

Shop Manual

930E-4

DUMP TRUCK

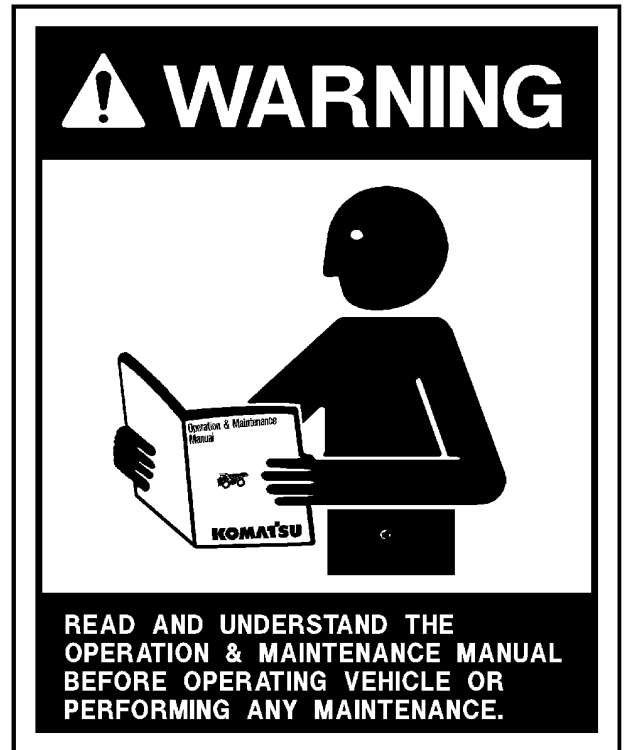
SERIAL NUMBERS **A31164 & UP**

KOMATSU®

This material is proprietary to Komatsu America Corp (KAC), and is not to be reproduced, used, or disclosed except in accordance with written authorization from KAC.

It is the policy of the Company to improve products whenever it is possible and practical to do so. The Company reserves the right to make changes or add improvements at any time without incurring any obligation to install such changes on products sold previously.

Because of continuous research and development, periodic revisions may be made to this publication. Customers should contact their local Komatsu distributor for information on the latest revision.



Unsafe use of this machine may cause serious injury or death. Operators and maintenance personnel must read and understand this manual before operating or maintaining this machine.

This manual should be kept in or near the machine for reference, and periodically reviewed by all personnel who will come into contact with it.

CALIFORNIA Proposition 65 Warning

Diesel engine exhaust, some of its constituents, and certain vehicle components contain or emit chemicals known to the State of California to cause cancer, birth defects or other reproductive harm.

CALIFORNIA Proposition 65 Warning

Battery posts, terminals and related accessories contain lead and lead compounds, chemicals known to the State of California to cause cancer and birth defects or other reproductive harm. Wash hands after handling.

CALIFORNIA Proposition 65 Warning

Mercury and mercury compounds are known to the State of California to cause developmental problems. This machine may be equipped with optional HID lamps which contain mercury. There is no risk of exposure unless the lamps are broken. However, the lamps must be reused, recycled or properly disposed of in accordance with Local, State and Federal Laws at the end of their useful lives.



NON-OEM PARTS IN CRITICAL SYSTEMS

For safety reasons, Komatsu America Corp. strongly recommends against the use of non-OEM replacement parts in critical systems of all Komatsu equipment. Critical systems include but are not limited to steering, braking and operator safety systems.

Replacement parts manufactured and supplied by unauthorized sources may not be designed, manufactured or assembled to Komatsu's design specifications; accordingly, use of such parts may compromise the safe operation of Komatsu products and place the operator and others in danger should the part fail.

Komatsu is also aware of repair companies that will rework or modify an OEM part for reuse in critical systems. Komatsu does not generally authorize such repairs or modifications for the same reasons as noted above.

Use of non-OEM parts places full responsibility for the safe performance of the Komatsu product on the supplier and user. Komatsu will not in any case accept responsibility for the failure or performance of non-OEM parts in its products, including any damages or personal injury resulting from such use.

FOREWORD

This manual is written for use by the operator and/or the service technician. It is designed to help these persons to become fully knowledgeable of the truck and all of its systems in order to keep it operating safely and efficiently. All operators and maintenance personnel should read and understand the information in this manual before operating the truck or performing maintenance and/or operational checks on the truck. All safety notices, warnings, and cautions should be understood and followed when operating the truck or performing repairs on the truck.

The first section covers component descriptions, truck specifications and safe work practices, as well as other general information. The major portion of the manual pertains to disassembly, service and reassembly. Each major serviceable area is dealt with individually. For example, the disassembly, service and reassembly of the radiator group is discussed as a unit. The same is true of the engine and engine accessories, and so on through the entire mechanical detail of the truck. Disassembly should be carried only as far as necessary to accomplish needed repairs.

The illustrations used in this manual are *typical* of the component shown and may not be an exact reproduction of what is found on the truck.

This manual shows dimensioning of U.S. standard and metric (SI) units throughout. All references to “right,” “left,” “front,” or “rear” are made with respect to the operator's normal seated position unless specifically stated otherwise.

When assembly instructions are provided without references to specific torque values, standard torque values should be used. Standard torque values are shown in torque charts in the General Information section of this manual. Specific torques, when provided in the text, are in bold face type, such as **135 N•m (100 ft lbs)**. All torque specifications have $\pm 10\%$ tolerance unless otherwise specified.

A product identification plate is located on the frame in front of the right side front wheel. It designates the Truck Model Number, Product Identification Number (vehicle serial number), and Maximum GVW (Gross Vehicle Weight) rating.

The KOMATSU truck model designation consists of three numbers and one letter (i.e. 930E).

The three numbers represent the basic truck model.

The letter “E” designates an Electrical wheel motor drive system.

The Product Identification Number (vehicle serial number) contains information which identifies several characteristics of this unit. For a more detailed explanation, see the end of Section A4.

The Gross Vehicle Weight (GVW) is what determines the load on the drive train, frame, tires, and other components. The vehicle design and application guidelines are sensitive to the maximum GVW.

GVW is *total weight*: **empty vehicle weight + fuel & lubricants + payload**.

To determine the *allowable payload*, fill all lubricants to the proper level and fill the fuel tank of an empty truck (which includes all accessories, body liners, tailgates, etc.), and then weigh the truck. Record this value and subtract it from the GVW. The result is the allowable payload.

NOTE: Accumulations of mud, frozen material, etc, become part of the GVW and reduces the allowable payload. To maximize payload and to keep from exceeding the maximum GVW rating, these accumulations should be removed as often as practical.

Exceeding the allowable payload will reduce the expected life of truck components.



This “ALERT” symbol is used with the signal words, “DANGER”, “WARNING”, and “CAUTION” in this manual to alert the reader to hazards arising from improper operating and maintenance practices.



“DANGER” identifies a specific potential hazard WHICH WILL RESULT IN EITHER INJURY OR DEATH if proper precautions are not taken.



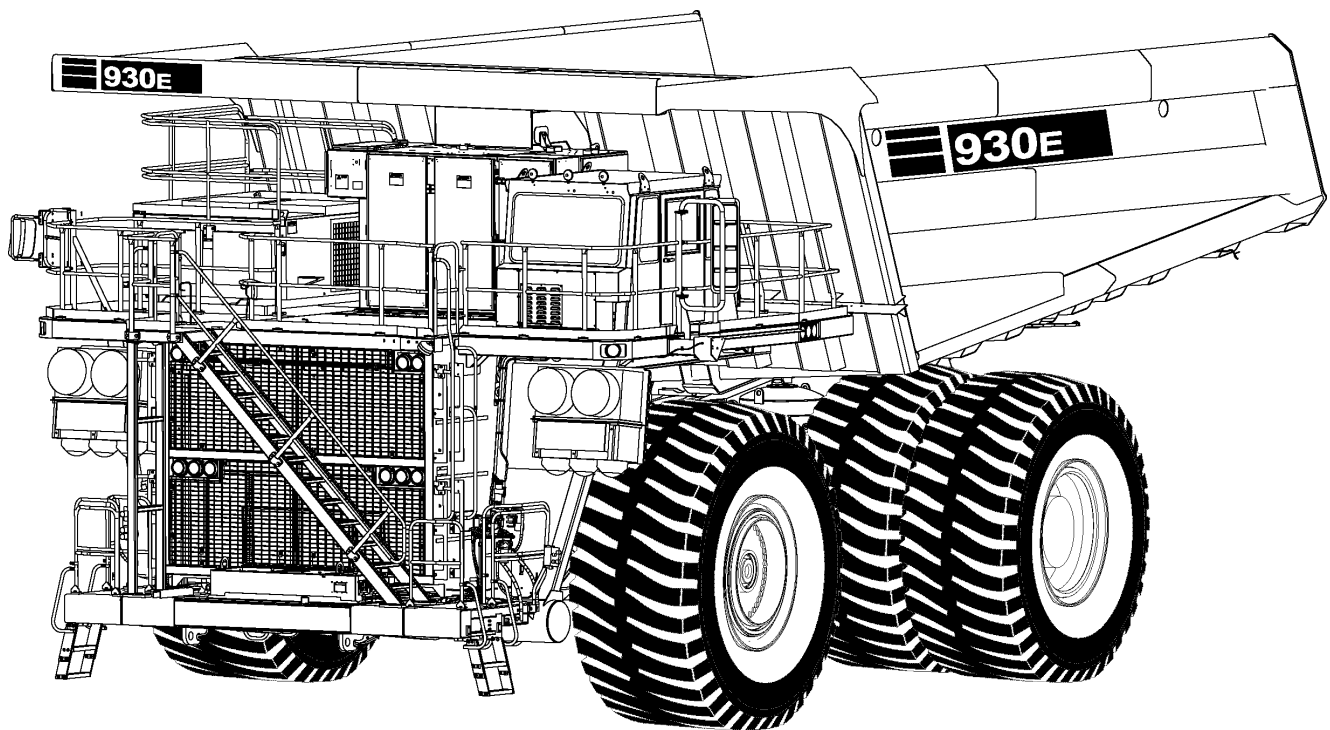
“WARNING” identifies a specific potential hazard WHICH MAY RESULT IN EITHER INJURY OR DEATH if proper precautions are not taken.



“CAUTION” is used for general reminders of proper safety practices OR to direct the reader’s attention to avoid unsafe or improper practices which may result in damage to the equipment.

TABLE OF CONTENTS

SUBJECT	SECTION
GENERAL INFORMATION	A
STRUCTURES	B
ENGINE	C
ELECTRICAL SYSTEM (24VDC NON-PROPULSION)	D
ELECTRIC PROPULSION SYSTEM	E
REAR AXLE, SPINDLES AND WHEELS	G
HYDRAIR® II SUSPENSIONS	H
BRAKE CIRCUIT	J
HYDRAULIC SYSTEM	L
OPTIONS AND SPECIAL TOOLS	M
OPERATOR CAB AND CONTROLS	N
LUBRICATION AND SERVICE	P
ALPHABETICAL INDEX	Q
SYSTEM SCHEMATICS	R



KOMATSU MODEL 930E-4 DUMP TRUCK

SECTION A GENERAL INFORMATION INDEX

MAJOR COMPONENTS & SPECIFICATIONS	A2
SAFETY AND OPERATING INSTRUCTIONS	A3
WARNINGS AND CAUTIONS.....	A4
TORQUE TABLES AND CONVERSION CHARTS	A5
STORAGE AND IDLE MACHINE PREPARATION	A7

NOTES

COMPONENTS & SPECIFICATIONS

MAJOR COMPONENT DESCRIPTIONS

Truck And Engine

The 930E-4 Dump Truck is an off-highway, rear dump truck with AC Electric Drive. The gross vehicle weight is 498 960 kg (1,100,000 lb). The engine is a Komatsu SSDA16V160 rated at 2 014 kW (2,700 HP).

Main Alternator

The diesel engine drives an in-line alternator at engine speed. The alternator produces AC current which is rectified to DC within the main control cabinet. The rectified DC power is converted back to AC by groups of devices called "inverters", which are also within the main control cabinet. Each inverter consists of six phase modules under the control of a gate driver power converter (GDPC). The two GDPCs control the operation of each phase module.

Each phase module contains paired positive and negative semiconductor switches referred to as insulated gate bipolar transistors (IGBT). The IGBTs cycle on and off at varying frequencies to create an AC power signal from the DC supply.

The AC power signal produced by each inverter is a variable-voltage, variable-frequency (VVVF) signal. Frequency and voltage are changed to suit the operating conditions.

Cooling air for the control/power group and wheel motors, as well as the alternator itself, is provided by dual fans mounted on the alternator shaft.

AC Induction Traction Motorized Wheels

The alternator output supplies electrical energy to the two wheel motors attached to the rear axle housing. The motorized wheels use three-phase AC induction motors with full-wave AC power.

The two wheel motors convert electrical energy back to mechanical energy through built-in gear trains within the wheel motor assembly. The direction of the wheel motors is controlled by a directional control lever located on the center console.

Suspension

Hydrair II® suspension cylinders located at each wheel provide a smooth and comfortable ride for the operator and dampens shock loads to the chassis during loading and operation.

Operator's Cab

The operator cab has been engineered for operator comfort and to allow for efficient and safe operation of the truck. The cab provides wide visibility with an integral 4-post ROPS/FOPS structure and an advanced analog operator environment. It includes a tinted safety-glass windshield and power-operated side windows, a deluxe interior with a fully adjustable seat with lumbar support, a fully adjustable/tilt steering wheel, controls mounted within easy reach of the operator, and an analog instrument panel which provides the operator with all instruments and gauges which are necessary to control and/or monitor the truck's operating systems.

Power Steering

The truck is equipped with a full time power steering system which provides positive steering control with minimum operator effort. The system includes nitrogen-charged accumulators which automatically provide emergency power if the steering hydraulic pressure is reduced below an established minimum.

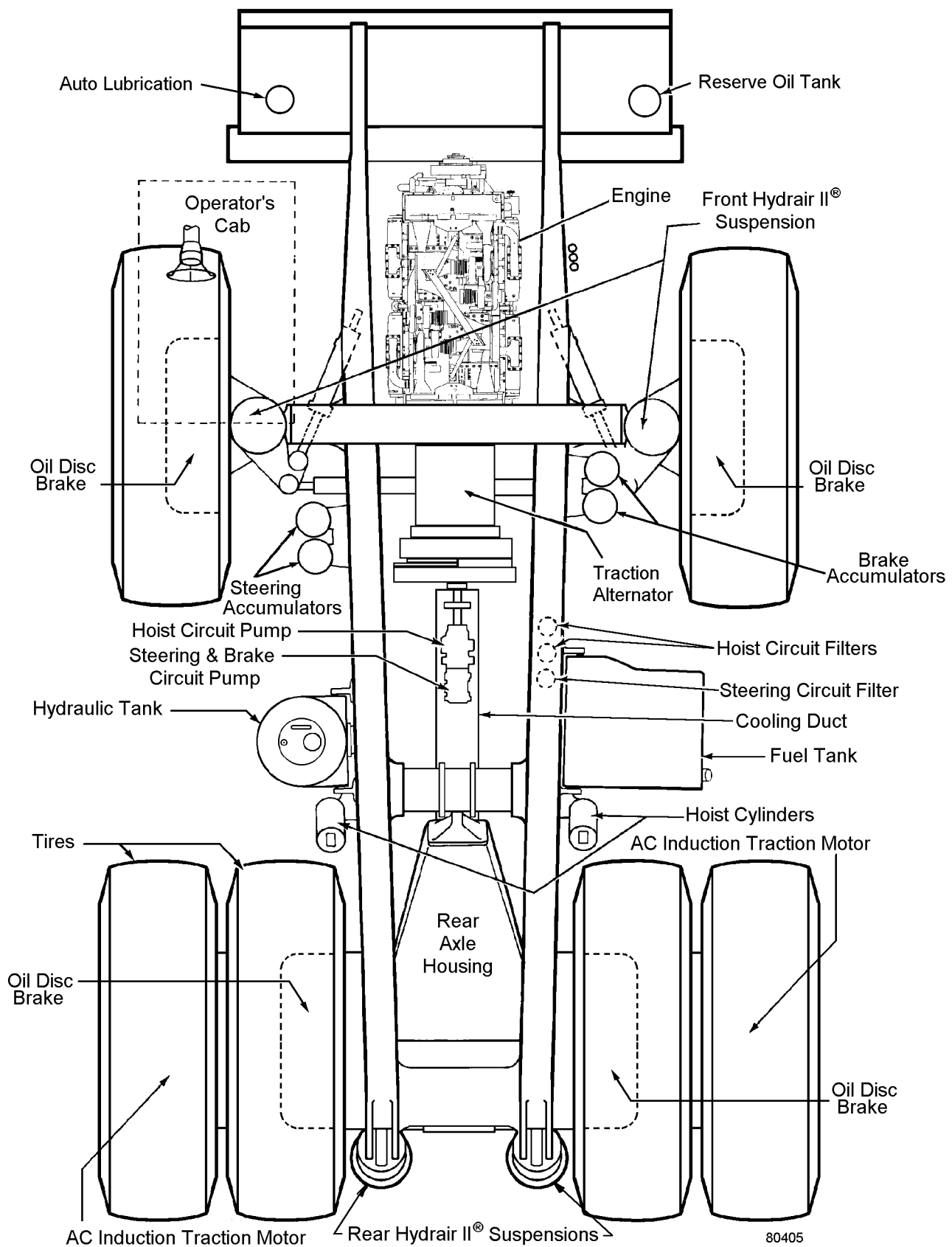
Dynamic Retarding

The dynamic retarding is used to slow the truck during normal operation or control speed coming down a grade. The dynamic retarding ability of the electric system is controlled by the operator through the activation of the retarder pedal (or by operating a lever on the steering wheel) in the operators cab and by setting the RSC (Retarder Speed Control). Dynamic retarding is automatically activated, if the truck speed goes to a preset overspeed setting.

Brake System

Service brakes at each wheel are oil-cooled multiple disc brakes applied by an all-hydraulic actuation system. Depressing the brake pedal actuates both front and rear brakes after first applying the retarder. All wheel brakes will be applied automatically if the brake system pressure decreases below a preset minimum.

The parking brake is a dry disc type, mounted inboard on each rear wheel motor, and is spring-applied and hydraulically-released with wheel speed application protection (will not apply with truck moving).



SPECIFICATIONS

These specifications are for the standard Komatsu 930E-4 Truck. Customer options may change this listing.

ENGINE

Komatsu SSDA16V160

No. of Cylinders	16
Operating Cycle	4-Stroke
Rated Brake HP	2 014 kW (2,700 HP) at 1900 RPM
Flywheel HP	1 902 kW (2,550 HP) at 1900 RPM
Weight (Wet)*	9 608 kg (21,182 lb)

* Weight does not include Radiator, Sub-frame, or Alternator.

AC ELECTRIC DRIVE SYSTEM

(AC/DC Current)

Alternator	General Electric GTA-41
Dual Impeller, In-Line Blower	340 m ³ / min (12,000 cfm)
Motorized Wheels	GDY106 AC Induction Traction Motors
Standard Gear Ratio*	32.62:1
Maximum Speed	64.5 kph (40 mph)

* Wheel motor application depends upon GVW, haul road grade and length, rolling resistance, and other parameters.
Komatsu and GE must analyze each job condition to ensure proper application.

DYNAMIC RETARDING

Electric Dynamic Retarding	Standard
Maximum Rating	4 026 kW (5,400 HP)
Continuous*	2 460 kW (3,300 HP)

* Continuously rated high-density blown grids with retard at engine idle and retard in reverse propulsion.

BATTERY ELECTRIC SYSTEM

Batteries	Four 8D, 12 volt wet batteries with disconnect switch
Cold Cranking Amps.	1450 CCA
Alternator	24 Volt, 140 Amp Output
Lighting	24 Volts
Starters (2)	24 Volts

SERVICE CAPACITIES

Crankcase (including lube oil filters)	280 liters (74 gallons)
Cooling System	594 liters (157 gallons)
Fuel	4 542 liters (1,200 gallons)
Hydraulic System	1 325 liters (350 gallons)
Wheel Motor Gear Box	95 liters (25 gallons) per wheel

HYDRAULIC SYSTEMS

Hoist and Brake Cooling Pump	Tandem Gear
Rating	931 L/min (246 gpm) at 1900 RPM and 17 237 kPa (2,500 psi)
Steering/Brake Pump	Pressure Compensated Piston
Rating	246 L/min (65 gpm) at 1900 RPM and 18 961 kPa (2,750 psi)
Relief Pressure - Hoist	17 237 kPa (2,500 psi)
Relief Pressure - Steering/Brake	27 579 kPa (4,000 psi)
Hoist Cylinders (2)	3-Stage Hydraulic
Tank (Vertical/Cylindrical)	Non-Pressurized
Tank Capacity	947 liters (250 gallons)
Filtration	In-line replaceable elements
Suction	Single, Full Flow, 100 Mesh
Hoist and Steering Filters (Dual In-Line, High Pressure)	Beta ₁₂ Rating =200

SERVICE BRAKES

All Hydraulic Actuation with Traction System Wheel Slip/Slide Control

Front and Rear Oil-Cooled Multiple Discs on each wheel

Total Friction Area / Brake 97 019 cm² (15,038 in²)

Maximum Apply Pressure. 17 238 kPa (2,500 psi)

STEERING

Twin hydraulic cylinders with accumulator assist to provide constant rate steering

Emergency power steering automatically provided by accumulators

Turning Circle (SAE). 29.7 m (97 ft. 7 in.)

TIRES

Radial Tires (standard). 53/80 R63

Rock Service, Deep Tread Tubeless

Rims patented Phase I New Generation™ rims

STANDARD DUMP BODY CAPACITIES AND DIMENSIONS

Capacity

Heaped @ 2:1 (SAE). 211 m³ (276 yd³)

Struck 171 m³ (224 yd³)

Width (Inside) 8.15 m (26 ft. 9 in.)

Depth 3.2 m (10 ft. 7 in.)

Loading Height 7.06 m (23 ft. 2 in.)

Dumping Angle 45°

NOTE: Optional capacity dump bodies are available.

WEIGHT DISTRIBUTION

Empty Vehicle

Front Axle (49.3%) 103 301 kg (227,738 lb)
Rear Axle (50.7 %) 107 518 kg (237,034 lb)
 Total (with 50% fuel) 210 819 kg (464,772 lb)

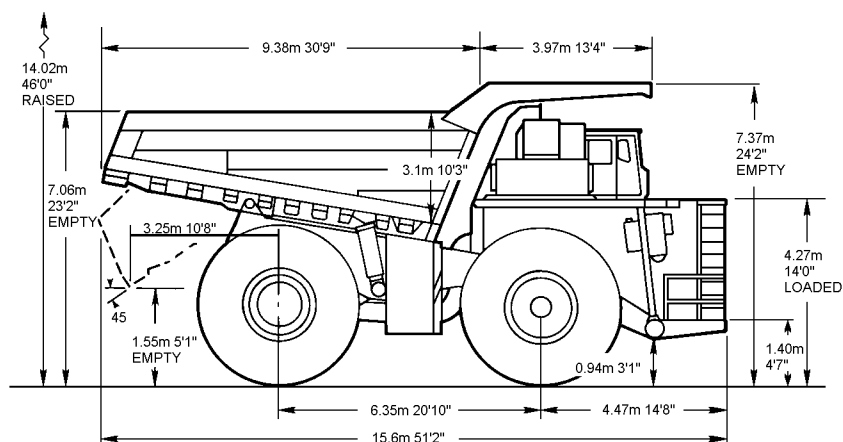
Standard Komatsu body 30 362 kg (66,936 lb)
 Standard tire weight 26 127 kg (57,600 lb)

Loaded Vehicle

Front Axle (33.0%) 165 554 kg (364,980 lb)
Rear Axle (67.0%) 336 124 kg (741,020 lb)
 Total 501 678 kg (1,106,000 lb)

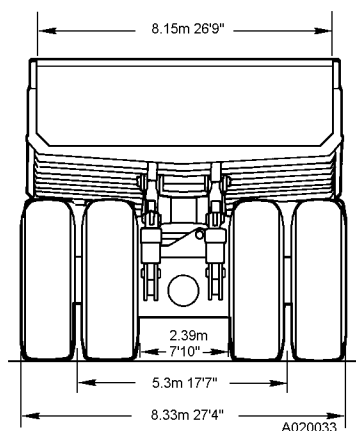
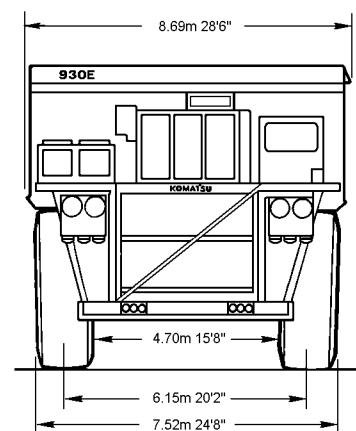
Nominal Payload*. 290 859 kg (641,228 lb)

* *Nominal payload* is defined within Komatsu America Corporation's payload policy documentation. Nominal payload must be adjusted if the weight of any customized body or tires vary from that of the standard Komatsu body and tires. Nominal payload must also be adjusted to take into account the additional weight of any custom/optional extras fitted to the truck which are not stated within the Standard Features list of the applicable specification sheet.



All Dimensions with 171/211 m³ 224/276 yd³ Body

BODIES	Struck		2:1 Heap		Loading Height	
	M ³	Yd ³	M ³	Yd ³	M	Feet
Standard	171	224	211	276	7.06	23'2"



A020033

NOTES

SAFETY

GENERAL

Safety records from most organizations will show that the greatest percentage of accidents are caused by unsafe acts performed by people. The remainder are caused by unsafe mechanical or physical conditions. Report all unsafe conditions to the proper authority.

The following safety rules are provided as a guide for the operator. However, local conditions and regulations may add many more to this list.



Read and follow all safety precautions. Failure to do so may result in serious injury or death.

Safety Rules

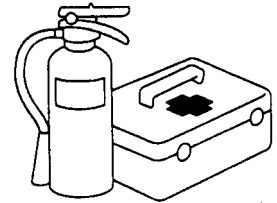
- Only trained and authorized personnel may operate and maintain the truck.
- Follow all safety rules, precautions and instructions when operating or performing maintenance on the truck.
- When working with another operator or a person on work site traffic duty, make sure that all personnel understand all hand signals that are to be used.

Safety Features

- Make sure that all guards and covers are in their proper position. Have any damaged guards and covers repaired. (See Operating Instructions - "Preparing For Operation".)
- Learn the proper use of safety features such as safety locks, safety pins, and seat belts. Use these safety features properly.
- Never remove any safety features. Always keep them in good operating condition.
- Improper use of safety features could result in serious bodily injury or death.
- Check the seat belt fabric, buckle and hardware for damage or wear. Replace any worn or damaged parts immediately.
- The seat belts must be replaced 5 years after seat belt manufacture, or after every 3 years of use, whichever comes first.

Fire Extinguisher And First Aid Kit

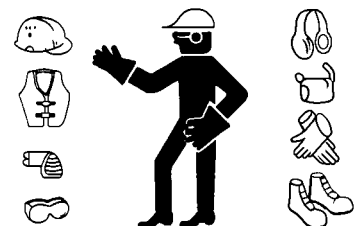
- Make sure that fire extinguishers are accessible and proper usage techniques are known.
- Provide a first aid kit at the storage point.
- Know what to do in the event of a fire.
- Keep the phone numbers of persons you should contact in case of an emergency on hand.



A0055070

Clothing And Personal Items

- Avoid loose clothing, jewelry, and loose long hair. They can catch on controls or in moving parts and cause serious injury or death. Also, never wear oily clothes as they are flammable.
- Wear a hard hat, safety glasses, safety shoes, mask and gloves when operating or maintaining a truck. Always wear safety goggles, hard hat and heavy gloves if your job involves scattering metal chips or minute materials--particularly when driving pins with a hammer or when cleaning air cleaner elements with compressed air. Also, ensure that the work area is free from other personnel during such tasks.



A0055010

Leaving The Operator's Seat

When preparing to leave the operator's seat, do not touch any control lever that is not locked. To prevent accidental operations from occurring, always perform the following:

- Move the directional control lever to PARK. **Do not use the wheel brake lock when the engine will be turned off.**
- Lower the dump body to the frame.
- Stop the engine. When exiting the truck, always lock compartments and take the keys with you. If the truck should suddenly move or move in an unexpected way, this may result in serious bodily injury or death.

Mounting And Dismounting

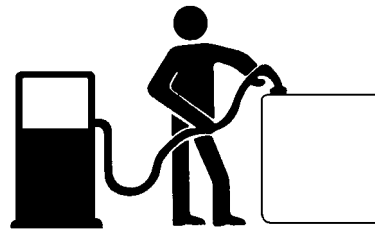
- Use the handrails and steps when getting on or off the truck.
- Never jump on or off the truck. Never climb on or off a truck while it is moving.
- When climbing on or off a truck, face the truck and use the hand-hold and steps.
- Never hold any control levers when getting on or off a truck.
- Always maintain three-point contact with the hand-holds and steps to ensure that you support yourself.
- When bringing tools into the operator's compartment, always pass them by hand or pull them up by rope.
- If there is any oil, grease, or mud on the hand-holds or steps, wipe them clean immediately. Always keep these components clean. Repair any damage and tighten any loose bolts.

Fire Prevention For Fuel And Oil

- Fuel, oil, and antifreeze can be ignited by a flame. Fuel is extremely flammable and can be hazardous. Keep flames away from flammable fluids.
- Keep oil and fuel in a designated location and do not allow unauthorized persons to enter.
- When refueling, stop the engine and do not smoke.
- Refueling and oiling should be done in well ventilated areas.
- Tighten all fuel and oil tank caps securely.



A0055020



A0055030



A0055040

Precautions With High Temperature Fluids

Immediately after truck operation, engine coolant, engine oil, and hydraulic oil are at high temperatures and are pressurized. If the cap is removed, the fluids are drained, the filters are replaced, etc., there is danger of serious burns. Allow heat and pressure to dissipate before performing such tasks and follow proper procedures as outlined in the service manual.



To prevent hot coolant from spraying:

1. Stop the engine.
2. Wait for the coolant temperature to decrease.
3. Depress the pressure release button on the cap to vent cooling system pressure.
4. Turn the radiator cap slowly to release the pressure before removing.

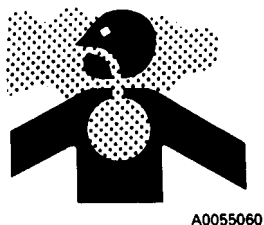
To prevent hot engine oil spray:

1. Stop the engine.
2. Wait for the oil temperature to cool down.
3. Turn the cap slowly to release the pressure before removing the cap.

Asbestos Dust Hazard Prevention

Asbestos dust is hazardous to your health when inhaled. If you handle materials containing asbestos fibers, follow the guidelines below:

- Never use compressed air for cleaning.
- Use water for cleaning to control dust.
- Operate the truck or perform tasks with the wind to your back whenever possible.
- Use an approved respirator when necessary.



Prevention Of Injury By Work Equipment

Never enter or put your hand, arm or any other part of your body between movable parts such as the dump body, chassis or cylinders. If the work equipment is operated, clearances will change and may lead to serious bodily injury or death.

Unauthorized Modification

Any modification made to this vehicle without authorization from Komatsu America Corp. can possibly create hazards.

Before making any modification, consult the authorized regional Komatsu America Corp. distributor. Komatsu will not be responsible for any injury or damage caused by any unauthorized modification.

Precautions When Using ROPS

The ROPS is intended to protect the operator if the truck should roll over. It is designed not only to support the load of the truck, but also to absorb the energy of the impact.

- The Rollover Protection Structure (ROPS) must be properly installed before the truck is operated.
- ROPS installed on equipment manufactured and designed by Komatsu America Corp. fulfills all of the regulations and standards for all countries. If it is modified or repaired without authorization from Komatsu, or if it is damaged when the truck rolls over, the strength of the structure will be compromised and will not be able to fulfill its intended purpose. Optimum strength of the structure can only be achieved if it is repaired or modified as specified by Komatsu.
- When modifying or repairing the ROPS, always consult your nearest Komatsu distributor.
- Even with the ROPS installed, the operator must always use the seat belt when operating the truck.

Precautions For Attachments

- When installing and using optional equipment, read the instruction manual for the attachment and the information related to attachments in this manual.
- Do not use attachments that are not authorized by Komatsu America Corp. or the authorized regional Komatsu distributor. Use of unauthorized attachments could create a safety problem and adversely affect the proper operation and useful life of the truck.
- Any injuries, accidents, and product failures resulting from the use of unauthorized attachments will not be the responsibility of Komatsu America Corp. or the authorized regional Komatsu distributor.

Precautions For Starting The Truck

Start the engine from the operator's seat only. Never attempt to start the engine by shorting across cranking motor terminals. This may cause a fire, or serious injury or death to anyone in truck's path.



PRECAUTIONS BEFORE OPERATION

Safety is thinking ahead. Prevention is the best safety program. Prevent a potential accident by knowing the employer's safety requirements and all necessary job site regulations. In addition, know the proper use and care of all the safety equipment on the truck. Only qualified operators or technicians should attempt to operate or maintain a Komatsu machine.

Safe practices start before the operator gets to the equipment.

Safety At The Worksite

- When walking to and from a truck, maintain a safe distance from all machines even when the operator is visible.
- Before starting the engine, thoroughly check the area for any unusual conditions that could be dangerous.
- Examine the road surface at the job site and determine the best and safest method of operation.
- Choose an area where the ground is as horizontal and firm as possible before performing the operation.
- If you need to operate on or near a public road, protect pedestrians and cars by designating a person for work site traffic duty or by installing fences around the work site.
- The operator must personally check the work area, the roads to be used, and the existence of obstacles before starting operations.
- Always determine the travel roads at the work site and maintain them so that it is always safe for the machines to travel.
- If travel through wet areas is necessary, check the depth and flow of water before crossing the shallow parts. Never drive through water that exceeds the permissible water depth.

Fire Prevention

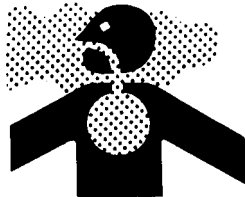
- Remove wood chips, leaves, paper, and other flammable items that have accumulated in the engine compartment. Failure to do so could result in a fire.
- Check the fuel, lubrication, and hydraulic systems for leaks. Repair any leaks. Clean any excess oil, fuel or other flammable fluids, and dispose of them properly.
- Make sure that a fire extinguisher is present and in proper working condition.
- Do not operate the truck near open flames.



A0055020

Ventilation In Enclosed Areas

If it is necessary to start the engine within an enclosed area, provide adequate ventilation. Inhaling exhaust fumes from the engine can kill.



A0055060

Preparing For Operation

- Always mount and dismount while facing the truck. Never attempt to mount or dismount the truck while it is in motion. Always use handrails and ladders when mounting or dismounting the truck.
- Check the deck areas for debris, loose hardware and tools. Check for people and objects that might be in the area.
- Become familiar with and use all protective equipment devices on the truck and ensure that these items (anti-skid material, grab bars, seat belts, etc.) are securely in place.

Mirrors, Windows And Lights

- Remove any dirt from the surface of the windshield, cab windows, mirrors and lights. Good visibility may prevent an accident.
- Adjust the rear view mirror to a position where the operator can see best from the operator's seat. If any glass or light is broken, replace it with a new part.
- Make sure that the headlights, work lights, and taillights are in proper working order. Make sure that the truck is equipped with the proper work lamps that are needed for the operating conditions.

In The Operator Cab (Before Starting The Engine)

- Do not leave tools or spare parts lying around. Do not allow trash to accumulate in the cab of the truck. Keep all unauthorized reading material out of the truck cab.
- Keep the cab floor, controls, steps and handrails free of oil, grease, snow and excess dirt.
- Read and understand the contents of this manual. Pay special attention to Section A3 pertaining to safety and operating instructions. Become thoroughly acquainted with all gauges, instruments and controls before attempting operation of the truck.
- Read and understand the **WARNING** and **CAUTION** decals in the operator's cab.
- Make sure that the steering wheel, horn, controls and pedals are free of any oil, grease or mud.
- Check the operation of the windshield wiper, condition of wiper blades, and the washer fluid reservoir level.
- Be familiar with all steering and brake system controls, warning devices, road speeds and loading capabilities before operating the truck.

Seat Belts

- On both driver and passenger seats, check the seat belt fabric, buckle, all belt retractors and hardware for damage or wear. Replace any worn or damaged parts immediately.
- Even if there are no signs of damage, replace both driver and passenger seat belts 5 years after seat belt manufacture, or every 3 years after start of use, whichever comes first. The passenger seat belt date of manufacture label is sewn into the seat belt near the buckle. The driver seat belt date of manufacture label is sewn into the shoulder harness belt, near the retractor end.

OPERATING THE TRUCK

When Starting The Engine

- Never attempt to start the engine by shorting across cranking motor terminals. This may cause a fire, or serious injury or death to anyone in truck's path.
- Never start the engine if a warning tag has been attached to the controls.
- When starting the engine, sound the horn as an alert.
- Start and operate the truck only while seated in the operator's seat.
- Do not allow any unauthorized persons in the operator's compartment or any other place on the truck.

General Truck Operation

- **Wear the seat belt at all times.**
- Only authorized persons are allowed to ride in the truck. Riders must be in the cab and belted in the passenger seat.
- Do not allow anyone to ride on the decks or steps of the truck.
- Do not allow anyone to get on or off the truck while it is in motion.
- Do not move the truck in or out of a building without a signal person present.
- Know and obey the hand signal communications between operator and spotter. When other machines and personnel are present, the operator should move in and out of buildings, loading areas, and through traffic under the direction of a signal person. Courtesy at all times is a safety precaution!
- Immediately report any adverse conditions at the haul road, pit or dump area that may cause an operating hazard.
- Check for flat tires periodically during a shift. If the truck has been operating on a "flat", do not park the truck inside a building until the tire cools. If the tire must be changed, do not stand in front of the rim and locking ring when inflating a tire mounted on the truck. Observers should not be permitted in the area and should be kept away from the side of such tires.



The tire and rim assembly may explode if subjected to excessive heat. Personnel should move to a remote or protected location if sensing excessively hot brakes, smelling burning rubber or observing evidence of fire near the tire and wheel area.

If the truck must be approached to extinguish a fire, those personnel should do so only while facing the tread area of the tire (front or back) unless protected by using large heavy equipment as a shield. Stay at least 50 ft. (15 m) from the tread of the tire.

In the event of fire in the tire and wheel area (including brake fires), stay away from the truck for at least 8 hours or until the tire and wheel are cool.

- Keep serviceable fire fighting equipment on hand. Report empty extinguishers for replacement or refilling.
- Always place the directional control lever in the PARK when the truck is parked and unattended. Do not leave the truck unattended while the engine is running.
- Park the truck a safe distance away from other vehicles as determined by the supervisor.
- Stay alert at all times! In the event of an emergency, be prepared to react quickly and avoid accidents. If an emergency arises, know where to get prompt assistance.

Ensuring Good Visibility

- When working in dark places, install work lamps and head lamps. Set up extra lighting in the work area if necessary.
- Discontinue operations if visibility is poor, such as in mist, snow, or rain. Wait for the weather to improve to allow the operation to be performed safely.

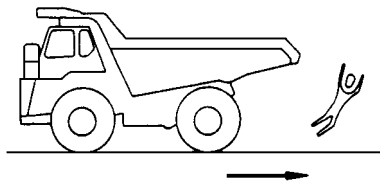
Traveling

- When traveling on rough ground, travel at low speeds. When changing direction, avoid turning suddenly.
- Lower the dump body and set the dump lever to the FLOAT position before traveling.
- If the engine stops while the truck is in motion, emergency steering and braking enable the truck to be steered and stopped. A fixed amount of reserve oil provides temporary steering and braking to briefly allow the truck to travel to a safe area. Apply the brakes immediately and stop the truck as quickly and safely as possible off of the haul road, if possible.

Traveling In Reverse

Before operating the truck:

- Sound the horn to warn people in the area. Make sure that the back-up horn also works properly.
- Check for personnel near the truck. Be particularly careful to check behind the truck.
- When necessary, designate a person to watch the area near the truck and signal the operator. This is particularly necessary when traveling in reverse.
- When operating in areas that may be hazardous or have poor visibility, designate a person to direct work site traffic.
- Do not allow any one to enter the line of travel of the truck. This rule must be strictly observed even with machines equipped with a back-up horn or rear view mirror.



Traveling On Slopes

- Traveling on slopes could result in the truck tipping over or slipping.
- Do not change direction on slopes. To ensure safety, drive to level ground before turning.
- Do not travel up and down on grass, fallen leaves, or wet steel plates. These materials may make the truck slip on even the slightest slope. Avoid traveling sideways, and always keep travel speed low.
- When traveling downhill, use the retarder to reduce speed. Do not turn the steering wheel suddenly. Do not use the foot brake except in an emergency.
- If the engine should stop on a slope, apply the service brakes fully and stop the truck. Move the directional control lever to PARK after the truck has stopped.

Operating On Snow Or Ice

- When working on snowy or icy roads, there is danger that the truck may slip to the side on even the slightest slope. Always travel slowly and avoid sudden starting, turning, or stopping in these conditions.
- Be extremely careful when clearing snow. The road shoulder and other objects are buried in the snow and cannot be seen.

Avoid Damage To Dump Body

Always be extremely cautious when working in tunnels, on bridges, under electric cables, or when entering a parking place or any other place where there are height limits. The dump body must be completely lowered before driving the truck.

Driving Near High Voltage Cables

Driving near high-voltage cables can cause electric shock. Always maintain safe distances between the truck and the electric cable as listed below.

Voltage	Minimum Safety Distance	
6.6 kV	3 m	10 ft.
33.0 kV	4 m	14 ft.
66.0 kV	5 m	17 ft.
154.0 kV	8 m	27 ft.
275.0 kV	10 m	33 ft.

The following actions are effective in preventing accidents while working near high voltages:

- Wear shoes with rubber or leather soles.
- Use a signalman to give warning if the truck approaches an electric cable.
- If the work equipment touches an electric cable, the operator should not leave the cab.
- When performing operations near high voltage cables, do not allow anyone to approach the truck.
- Check with the electrical maintenance department about the voltage of the cables before starting operations.

Working On Loose Ground

- Avoid operating the truck near cliffs, overhangs and deep ditches. If these areas collapse, the truck could fall or tip over and result in serious injury or death. Remember that ground surfaces in these areas may be weakened after heavy rain or blasting.
- Freshly laid soil and the soil near ditches is loose. It can collapse under the weight or vibration of the truck. Avoid these areas whenever possible.

When Dumping

- Before starting the dumping operation, make sure that there are no persons or objects behind the truck.
- Stop the truck in the desired location. Check again for persons or objects behind the truck. Give the determined signal, then slowly operate the dump body. If necessary, use blocks for the wheels or position a flagman.
- When dumping on slopes, truck stability is poor and there is danger of tipping over. Always use extreme care when performing such operations.
- Never travel with the dump body raised.

When Loading

- Make sure that the surrounding area is safe. Stop the truck in the correct loading position, then load the body uniformly.
- Do not leave the operator's seat during the loading operation.

Parking The Truck

- Choose a flat, level surface to park the truck. If the truck has to be parked on a slope, put blocks behind all the wheels to prevent truck movement.
- When parking on public roads, provide fences and signs, such as flags or lights, on the truck to warn pedestrians and other vehicles. Make sure that the truck, flags or lights do not obstruct traffic.
- Before leaving the truck, lower the dump body fully, move the directional control lever to PARK, stop the engine and lock everything. Always take the key with you.



Do not activate the wheel brake lock when the parking brake is activated. Bleed down of hydraulic pressure may occur, causing the truck to roll away.

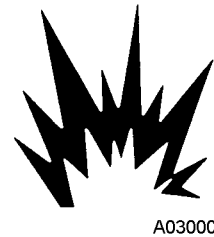
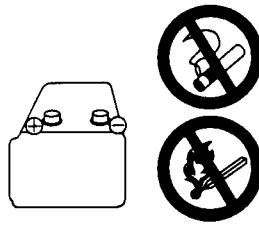
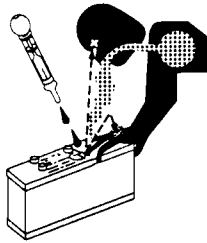
Towing

- Improper towing methods may lead to serious personal injury and/or damage.
- Use a towing device with ample strength for the weight of this truck.
- Never tow a truck on a slope.
- Inspect towing components, such as tow bars and couplings, for any signs of damage. Never use damaged or worn components to tow a disabled vehicle.
- Keep a safe distance from the trucks and towing apparatus while towing a vehicle.
- When connecting a truck that is to be towed, do not allow anyone to go between the tow vehicle and the disabled vehicle.
- Set the coupling of the truck being towed in a straight line with the towing portion of the tow truck, and secure it in position.
- For towing methods, refer to Operating Instructions, Towing.

WORKING NEAR BATTERIES

Battery Hazard Prevention

- Battery electrolyte contains sulfuric acid, which can quickly burn the skin and eat holes in clothing. If you spill acid on yourself, immediately flush the area with water.
- Battery acid can cause blindness if splashed into your eyes. If acid gets into your eyes, flush them immediately with large quantities of water and see a doctor at once.
- If you accidentally drink acid, drink a large quantity of water, milk, beaten eggs or vegetable oil. Call a doctor or poison prevention center immediately.
- Always wear safety glasses or goggles when working with batteries.
- Batteries generate hydrogen gas. Hydrogen gas is very explosive and can easily be ignited with a small spark or flame.
- Before working with batteries, stop the engine and turn the key switch to the OFF position.
- Avoid short-circuiting the battery terminals through accidental contact with metallic objects, such as tools, across the terminals.
- When removing or installing batteries, check which is the positive (+) terminal and the negative (-) terminal.
- Tighten the battery terminals securely. Loose terminals can generate sparks and lead to an explosion.
- Tighten battery caps securely.



A030005

Jump Starting With Booster Cables

- Always wear safety glasses or goggles when starting the truck with booster cables.
- When starting from another truck, do not allow the two trucks to touch.
- Perform the operation with two people: one person in the cab of the disabled machine, the other person working with the jumper cables.
- Ensure the parking brake is applied on both machines.
- Ensure the size of the booster cables and clips are suitable for the battery size. Inspect the cables and connectors for any damage or corrosion.
- Ensure the key switch and both battery disconnect switches on the disabled machine are in the OFF position.
- Connect the batteries in parallel: positive to positive and negative to negative.
- Connect the positive (+) cable from the good machine to the (+) on the disabled machine first. Then connect the ground cable from the negative (-) battery terminal on the good machine to the frame of the disabled machine, as far away as possible from the batteries. This will prevent a spark from possibly starting a battery fire.
- If the batteries are low, DO NOT attempt starting the machine with only one set of jumper cables installed. Install the second set of jumper cables in the same way as already described.
- For booster cable removal, disconnect the ground or negative (-) cable first, then the (24VDC +) cable last.
- If any tool touches between the positive (+) terminal and the chassis, it will cause sparks. Always be cautious when using tools near the battery.

Jump Starting With Receptacles

- Always wear safety glasses or goggles when starting the machine with booster cables.
- While jump starting with another machine, DO NOT allow the two machines to touch.
- Perform the operation with two people: one person in the cab of the disabled machine, the other person working with the jumper cables.
- Ensure the parking brake is applied on both machines.
- Inspect the cables and connectors for any damage or corrosion.
- Ensure the key switch and both battery disconnect switches on the disabled machine are in the OFF position.
- Connect the jumper cable to the receptacle on the good machine to the receptacle on the disabled machine.
- Allow time for the batteries to charge.

NOTE: The batteries will charge even with the battery disconnect switches are in the OFF position.

- If the batteries are low, DO NOT attempt starting the machine with only one set of jumper cables installed. Install the second set of jumper cables in the same way as already described.
- Turn the battery disconnect switches to the ON position and attempt starting.
- For booster cable removal, disconnect the cables from each machine.
- If any tool touches between the positive (+) terminal and the chassis, it will cause sparks. Always use caution when using tools near the battery.

BEFORE PERFORMING MAINTENANCE

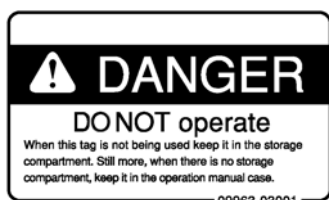
Stopping The Engine Before Service

- Before performing inspections or maintenance, stop the truck on firm flat ground, lower the dump body, move the directional control lever to PARK, and stop the engine.
- If the engine must be run during service, such as when cleaning the radiator, the directional control lever must be in PARK. Always perform this work with two people. One person must sit in the operator's seat to stop the engine if necessary. During these situations, never move any controls that are not related to the task at hand.
- When servicing the truck, do not touch any moving parts. Never wear loose clothing or jewelry.
- Put wheel blocks under the wheels to prevent truck movement.
- When performing service with the dump body raised, place the dump lever in the HOLD position and apply the lock (if equipped). Install the body-up safety pins or cable securely.

Warning Tag

Starting the engine or operating the controls while other personnel are performing maintenance on the truck can lead to serious injury and/or death. Always attach the warning tag to the control lever in the operator's cab to alert others that you are working on the truck. Attach additional warning tags around the truck, if necessary.

These tags are available from your Komatsu distributor. **Warning tag part number:** 09963-03001



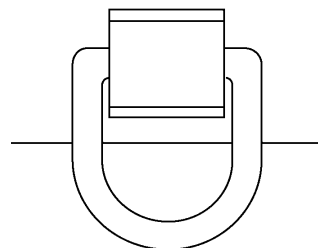
Proper Tools

Only use tools that are suited to the task. Using damaged, low quality, faulty or makeshift tools could cause personal injury.



Use of Tie-Off Anchor During Maintenance and Repair

While working at heights during assembly, maintenance or repair of the haul truck, workers should wear an appropriate fall protection harness and attach it to a tie-off anchor or tie-off point.



Komatsu anchor (58B-98-75190) is available for use with fall protection harnesses. Carefully read and understand the harness maker's instructions before using any fall protection harness.

NOTE: The anchor must not be used for lifting.

Securing The Dump Body

⚠ WARNING

To avoid serious personal injury or death, the body retention sling must be installed whenever personnel are required to perform maintenance on the truck while the dump body is in the raised position.

The Komatsu body-up safety sling can only be used with a Komatsu body. Non-OEM body may not accommodate the Komatsu body-up safety sling. The end user must ensure that a proper cable/sling is used.

1. Raise the body to its maximum height.
2. Install two shackles (2, Figure 3-1) and body retention sling (3) between rear body ear (1) and the axle housing.
3. Secure the shackle pins with cotter pins.
4. Move the hoist lever to the FLOAT position to slowly lower the body until the sling is supporting the full weight of the body. Then move the hoist lever to the HOLD position.
5. After service work is completed, reverse the installation steps to remove the sling.

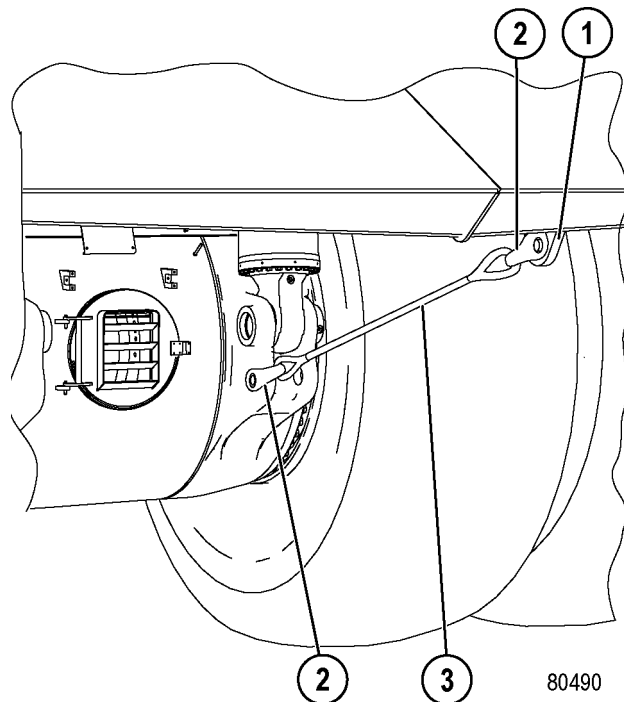


FIGURE 3-1. BODY RETENTION SLING
INSTALLATION

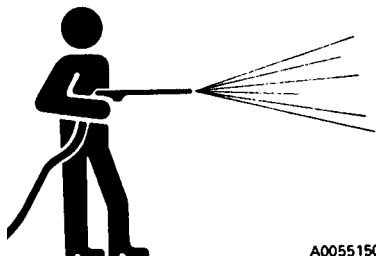
- | | |
|------------------|-------------------------|
| 1. Rear Body Ear | 3. Body Retention Sling |
| 2. Shackle & Pin | |

WHILE PERFORMING MAINTENANCE

NOTE: Only authorized personnel should service and repair the truck.

Keep The Truck Clean

- Spilled oil, grease, scattered tools, etc., can cause you to slip or trip. Always keep your truck clean and tidy.
- If water gets into the electrical system, there is danger that the truck may move unexpectedly and/or damage to components may occur. Do not use water or steam to clean any sensors, connectors or the inside of the operator's compartment.
- Use extreme care when washing the electrical control cabinet. Do not allow water to enter the control cabinet around the doors or vents. Do not allow any water to enter the cooling air inlet duct above the electrical control cabinet. If water enters the control cabinet through any opening or crevice, major damage to the electrical components is possible.
- Never spray water into the rear wheel electric motor covers. Damage to the wheel motor armatures may occur.
- Do not spray water into the retarding grids. Excess water in the retarding grids can cause a ground fault, which will prevent propulsion.



Attachments

Place attachments that have been removed from the truck in a safe place and manner to prevent them from falling.



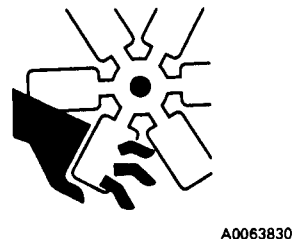
Working Under The Truck

- Always lower all movable work equipment to the ground or to their lowest position before performing service or repairs under the truck.
- Always block the tires of the truck securely.
- Never work under the truck if the truck is poorly supported.



Rotating Fan And Belts

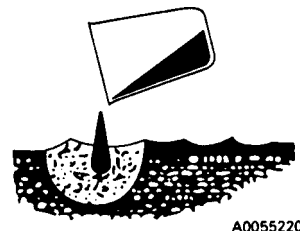
Stay away from all rotating parts such as the radiator fan and fan belts. Serious bodily injury may result from direct or indirect contact with rotating parts and flying objects.



Waste Materials

- Never dump oil into a sewer system, river, etc.
- Always put oil drained from your truck in appropriate containers. Never drain oil directly onto the ground.
- Obey appropriate laws and regulations when disposing of harmful objects such as oil, fuel, coolant, solvent, filters and batteries.
- The machine may be equipped with optional High Intensity Discharge (HID) lighting lamps which contain mercury. These lamps must be reused, recycled or properly disposed of in accordance with applicable local, state and federal laws.

INCORRECT

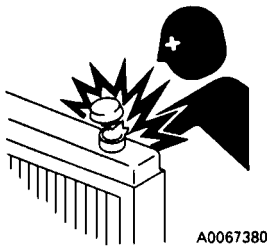


Adding Fuel Or Oil

- Spilled fuel and oil may cause slipping. Always clean up spills immediately.
- Always add fuel and oil in a well-ventilated area.
- When refueling, stop the engine and do not smoke.
- Tighten the cap of the fuel and oil fillers securely.
- Never use fuel to wash parts.

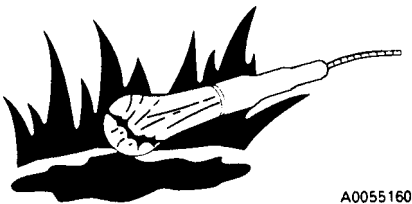
Radiator Coolant Level

If it is necessary to add coolant to the radiator, stop the engine and allow the engine and radiator to cool down before adding the coolant. Depress the pressure release button on the cap to vent cooling system pressure. Slowly loosen the cap to relieve any remaining pressure.



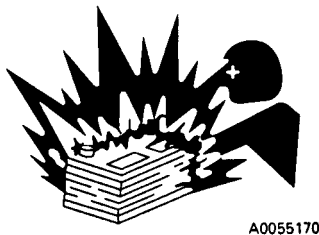
Use Of Lighting

When checking fuel, oil, coolant or battery electrolyte, always use lighting with anti-explosion specifications. If such lighting equipment is not used, there is danger of an explosion.



Precautions With The Battery

When repairing the electrical system or performing electrical welding, remove the negative (-) terminal of the battery to stop the flow of current.



Handling High Pressure Hoses

- Do not bend high pressure hoses or hit them with hard objects. Do not use any bent or cracked piping, tubes or hoses. They may burst during use.
- Always repair any loose or broken hoses. If fuel or oil leaks, it may result in a fire.

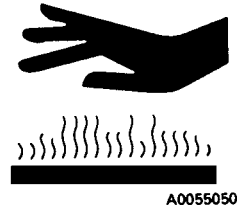
Precautions With High Pressure Oil

- Work equipment circuits are always under pressure. Do not add oil, drain oil or perform maintenance or inspections before completely releasing the internal pressure.
- Small, high-pressure pin hole leaks are extremely dangerous. The jet of high-pressure oil can pierce the skin and eyes. Always wear safety glasses and thick gloves. Use a piece of cardboard or a sheet of wood to check for oil leakage.
- If you are hit by a jet of high-pressure oil, consult a doctor immediately for medical attention.



Performing Maintenance Near High Temperature Or High Pressure

Immediately after stopping operation, engine coolant and operating oils are at high temperature and under high pressure. If the cap is removed, the oil or water is drained, or the filters are replaced under these conditions, it may result in burns or other injury. Wait for the temperature to cool and pressure to subside before performing the inspection and/or maintenance as outlined in the shop manual.



TIRES

Handling Tires

Rim and tire maintenance can be hazardous unless the correct procedures are followed by trained personnel. Improperly maintained or inflated tires can overheat and burst due to excessive pressure. Improper inflation can also result in cuts in the tire caused by sharp stones. Both of these conditions can lead to tire damage, serious personal injury, or even death. To safely maintain a tire, adhere to the following conditions:

- Before a tire is removed from a vehicle for tire repair, the valve core must be partially removed to allow deflation, and then the tire/rim assembly can be removed. During deflation, persons must stand outside of the potential trajectory of the locking ring of a multi-piece wheel rim.
- After the tire/rim assembly is installed on the vehicle, inflate the tires to their specified pressure. Abnormal heat is generated, particularly when the inflation pressure is too low.

NOTE: To prevent injury from the wheel rims during tire inflation, use one of the following:

- A wheel cage or other restraining device that will constrain all wheel rim components during an explosive separation of a multi-piece wheel rim, or during the sudden release of air.
- A stand-off inflation device which permits a person to stand outside of the potential trajectory of the wheel components.
- Use the specified tires.

The tire inflation pressure and permissible speeds, given in this manual, are general values. The actual values may differ, depending on the type of tire and the specific operating conditions. For details, please consult the tire manufacturer.

When the tires become overheated, a flammable gas is produced inside the tire which can ignite. It is particularly dangerous if the tires become overheated while the tires are pressurized. If the gas generated inside the tire ignites, the internal pressure will suddenly rise, and the tire will explode, resulting in danger and/or death to personnel in the area. Explosions differ from punctures or tire bursts because the destructive force of the explosion is extremely large. Therefore, the following operations are strictly prohibited when the tire is pressurized:

- Welding the rim
- Welding near the wheel or tire.
- Smoking or creating open flames



Tire Maintenance

If the proper procedure for performing maintenance or replacement of the wheel or tire is not used, the wheel or tire may burst, causing damage, serious injury, or even death. When performing such maintenance, consult your authorized regional Komatsu distributor, or the tire manufacturer.

Refer to the Society of Automotive Engineers (SAE), SAE J1337, Off-Road Rim Maintenance Procedures and Service Precautions, Section 4.2 for additional information on demounting the tires and rim assemblies. Also, refer to Section 4.4 of SAE J1337 for assembly and inflation recommendations.

The U.S. Department of Labor Mine Safety and Health Administration (MSHA) addresses tire repairs in its Title 30 Code of Federal Regulations, 30 CFR 57.14104.

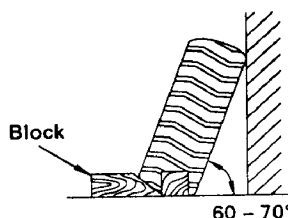


DO NOT stand in front of a rim and locking ring when inflating a tire mounted on the machine. Observers must not be permitted in the area.

DO NOT weld or heat the rim assembly with the tire mounted on the rim. Resulting gases inside the tire may ignite, causing explosion of the tire and rim.

Storing Tires After Removal

- As a basic rule, store the tires in a warehouse in which unauthorized persons cannot enter. If the tires are stored outside, always erect a fence around the tires and put up "No Entry" signs and other warning signs that even young children can understand.
- Stand the tire on level ground and block it securely so that it cannot roll or fall over.
- If the tire falls over, flee the area quickly. The tires for dump trucks are extremely heavy. Never attempt to hold or support the tire. Attempting to hold or support a tire may lead to serious injury.



A030008

- Mounted tires stored as spares must be inflated to the minimum inflation pressure necessary to keep the tire beads properly seated. Maximum inflation pressure of the stored tire must not exceed 15% of the tire's cold inflation pressure.

WHEN REPAIRS ARE NECESSARY

NOTE: Only qualified maintenance personnel who understand the systems being repaired should attempt repairs.

- Many components on the Komatsu truck are large and heavy. Ensure that lifting equipment - hoists, slings, chains, and lifting eyes - are of adequate capacity to handle the lift.
- Do not work under a suspended load. Do not work under a raised body unless body safety cables, props or pins are in place to hold the body in up position.
- Do not repair or service the truck while the engine is running, except when adjustments can only be made under such conditions. Keep a safe distance from moving parts.
- When servicing any air conditioning system with refrigerant, wear a face shield and cold resistant gloves for protection against freezing. Follow all current regulations for handling and recycling refrigerants.
- Follow package directions carefully when using cleaning solvents.
- If an auxiliary battery assist is needed, first use one cable to connect the 24V positive (+) post of the disabled truck batteries to the 24V positive (+) post of the auxiliary assist. Use a second cable to connect the 24V negative (-) post of the auxiliary assist battery to a **frame ground (-)** on the disabled truck away from the battery.
- Always disconnect the positive and negative battery cables of the vehicle before doing any welding on the unit. Failure to do so may seriously damage the battery and electrical equipment. Disconnect the battery charging alternator lead wire and isolate the electronic control components before making welding repairs. (It is not necessary to disconnect or remove any control circuit cards on electric drive dump trucks or any of the AID circuit control cards.)

- Always fasten the welding machine ground (-) lead to the piece being welded. The grounding clamp must be attached as near as possible to the weld area. Never allow welding current to pass through ball bearings, roller bearings, suspensions or hydraulic cylinders. Always avoid laying welding cables over or near the vehicle electrical harnesses. Welding voltage could be induced into the electrical harness and possibly cause damage to components.
- If the truck must be towed, use a rigid tow bar. Check the truck frame for a decal recommending special towing precautions. Also refer to Operating Instructions, Towing.
- Drain, clean, and ventilate fuel tanks and hydraulic tanks before making any welding repairs.



Any operating fluid, such as hydraulic oil, escaping under pressure can have sufficient force to enter a person's body by penetrating the skin. Serious injury and possibly death may result if proper medical treatment by a physician familiar with this injury is not received immediately.

- Relieve pressure in lines or hoses before making any disconnects.
- After adjustments or repairs, replace all shields, screens and clamps.
- Only a qualified operator or maintenance personnel who are also qualified in truck operation should move the truck under its own power in the repair facility or during road testing after repairs are complete.

SPECIAL PRECAUTIONS FOR WORKING ON AC DRIVE TRUCKS

Preliminary Procedures Before Welding or Performing Maintenance

Before welding or repairing an AC drive truck, maintenance personnel should notify a Komatsu service representative. Only qualified personnel, specifically trained for servicing the AC drive system, should perform this service.

If it is necessary to perform welding or repair to the truck without the field engineer present, the following procedures must be followed to ensure that the truck is safe for maintenance personnel to work on and to reduce the chance for damage to equipment.



Anytime the engine is running, do not open any of the cabinet doors or remove any covers. Do not use any of the power cables for hand holds or foot steps. Do not touch the retarding grid elements.



Before opening any cabinets or touching a retarding grid element or a power cable, the engine must be shutdown and the red drive system warning lights must not be illuminated.

Engine Shutdown Procedure Before Welding or Performing Maintenance

Normal operation of the AC drive system at shutdown should leave the system safe to service. However, in the event of a system failure, performing the following procedure before any maintenance activities will ensure that no hazardous voltages are present in the AC drive system.

1. Before shutting down the engine, verify the status of all the drive system warning lights on the overhead display panel. Use the lamp test switch to verify that all lamps are functioning properly.
If any of the red drive system warning lights remain on, do not attempt to open any cabinets, disconnect any cables, or reach inside the retarding grid cabinet without a trained drive system technician present, even if the engine is off. Only qualified personnel, specifically trained for servicing the AC drive system, should perform this service.
2. If all red drive system warning lights are off, follow the Normal Engine Shutdown Procedure on page A3-27.
3. After the engine has been stopped for at least five minutes, inspect the link voltage lights on the exterior of the main control cabinet and back wall of the operator's cab (DID panel). If all lights are off, it is safe to work on the retarding grids, wheel motors, alternator and related power cables.
4. Locate the GF cut-out switch in the access panel on the left side of the main control cabinet. Place the switch in the "alternator cutout" position. This will prevent the alternator from re-energizing and creating system voltage until the switch is returned to its former position.

5. Before welding on the truck, disconnect all electrical harnesses from the Engine Control System (ECS) inside the electrical cabinet behind the operator's cab. Also, disconnect the ground strap from the ECS.
6. **Do not weld on the rear of the control cabinet!** The metal panels on the back of the cabinet are part of capacitors and cannot be heated.
7. **Do not weld on the retarding grid exhaust louvers!** They are made of stainless steel. Some power cable panels throughout the truck are also made of aluminum or stainless steel. They must be repaired with the same material or the power cables may be damaged.
8. Power cables must be cleated in wood or other non-ferrous materials. Do not repair cable cleats by encircling the power cables with metal clamps or hardware. Always inspect power cable insulation before servicing the cables and returning the truck to service. Discard cables with broken insulation.
10. Power cables and wiring harnesses should be protected from weld spatter and heat.

Always fasten the welding machine ground (-) lead to the piece being welded. **The grounding clamp must be attached as near as possible** to the weld area.

Always avoid laying welding cables over or near the vehicle electrical harnesses. Welding voltage could be induced into the electrical harness and cause damage to components.

Before doing any welding on the truck, disconnect the battery charging alternator lead wire and isolate electronic control components.

Also, always disconnect the negative and positive battery cables of the vehicle. Failure to do so may seriously damage the battery and electrical equipment.

Never allow welding current to pass through ball bearings, roller bearings, suspensions or hydraulic cylinders.
11. If the red lights on the exterior of the control cabinet and/or the back wall of the operator's cab continue to be illuminated after following the above procedure, a fault has occurred.

Leave all cabinet doors in place. Do not touch the retarding grid elements. Do not disconnect any power cables or use them as hand or foot holds.

Notify your Komatsu service representative immediately. Only qualified personnel, specifically trained for servicing the AC drive system, should perform this service.
12. Replace all covers and doors. Place the GF cutout switch and battery disconnect switches in their original positions. Reconnect all harnesses prior to starting the truck.

Leave the drive system in the rest mode until the truck is to be moved.

CAPACITOR DISCHARGE SYSTEM

The control cabinets are equipped with two capacitor charge lights, one on the exterior of the cabinet and one in the interior. The capacitor charge lights, when off, indicate to service personnel that the drive system is safe to work on. Certain drive system failures, however, can result in a condition where one or more capacitors can remain in a charged state even though the capacitor charge lights are off.

Because a danger can still exist with the capacitor charge lights off, it is necessary to adhere to the following instructions before touching or servicing drive system components. Only authorized service personnel are allowed to service the drive system.

Refer to the shop manual for rules when servicing the drive system. Adhere to the proper procedures for disabling the drive system.

Necessary Tools

- 2000VDC meter kit (Komatsu p/n 58B-06-00800)
(Includes the parts for the volt meter assembly in Figure 3-2, the grounding sticks in Figure 3-3 and high voltage gloves)
- Additional Personal Protective Equipment (PPE) for working with 2000VDC meter, including safety shoes and safety glasses
- Multimeter for ground resistance measurement



FIGURE 3-2. VOLT METER ASSEMBLY

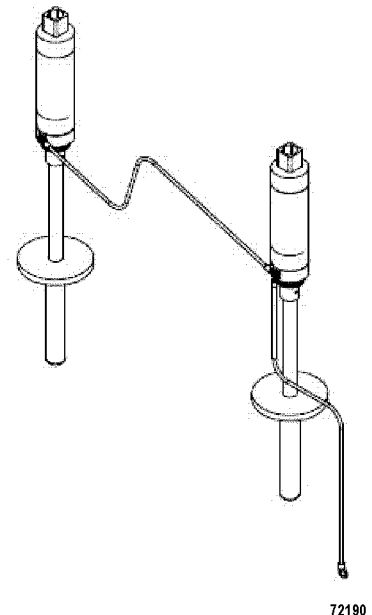


FIGURE 3-3. GROUNDING STICKS

Warnings And Cautions

All applicable local mine, government, and industry rules for working with high voltage must be followed. Required personal protective equipment, including but not limited to safety shoes, high voltage gloves, and safety glasses must be worn.

Safety cautions and warnings appear throughout the instructions. WARNING indicates the potential for personal injury and CAUTION indicates the potential for equipment damage. Read the following warnings prior to working on this drive system.

WARNING

Hazardous voltages are present in this equipment. Ensure that the Control Power Switch (CPS) is in the OFF position and that the Generator Field Contactor (GFCO) switch is in the CUT-OUT position before attempting any work on the drive system components. Check that Capacitor Charge Lights (CCL's) are not illuminated.

Use measurement and protective equipment rated for 2000 VDC minimum to verify that no voltage is present before touching any terminal.

Verify functionality of the measurement equipment using site-approved procedures both before and after performing control group measurements.

Failure to observe these precautions may result in death or serious personal injury.

WARNING

Verify that the Capacitor Charge Light (CCL) above the high voltage contactor area is not illuminated before opening the doors to the high voltage area or the high voltage contactor area.

Use measurement and protective equipment rated for 2000 VDC minimum to verify that no voltage is present before touching any terminal.

Verify functionality of the measurement equipment using site-approved procedures both before and after performing control group measurements.

Failure to observe these precautions may result in death or serious personal injury.

WARNING

Hazardous voltages are present in this equipment. Avoid touching any energized equipment when the door to the low voltage area is open.

Failure to do so may result in personal injury and equipment damage.

WARNING

Hazardous voltages may be present in this equipment even if the engine and capacitor charge lights are off.

Use measurement and protective equipment rated for 2000 VDC minimum to verify that no voltage is present before touching any terminal.

Verify functionality of the measurement equipment using site-approved procedures both before and after performing control group measurements.

Failure to observe these precautions may result in death or serious personal injury.

WARNING

Voltages in excess of 1500 VDC may be present. Any measurement and/or protective equipment used must be rated at 2000 VDC minimum.

Verify functionality of the measurement equipment using site-approved procedures both before and after performing control group measurements.

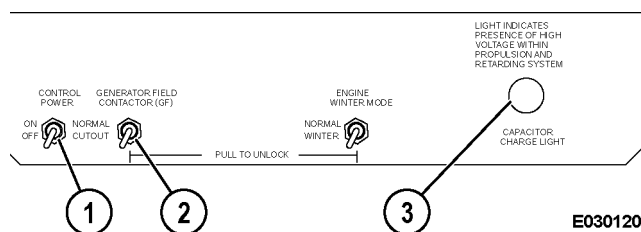
Failure to observe these precautions may result in death or serious personal injury.

Manual DC Link Capacitor Discharge Procedure

Follow any and all local and site specific procedures and requirements for working on off-highway mining equipment.

1. Verify that the engine is off and the parking brake is on.
2. Verify that the generator field contactor (GF) switch (2, Figure 3-4) in the low voltage area of the control cabinet is in the CUTOFF position.
3. Move the control power switch (1) to the ON position for a minimum of 30 seconds. Then, turn the control power switch to the OFF position.

(With control power on, an RP contactor closes and discharges the DC link through the retarding grids in less than 10 seconds.)

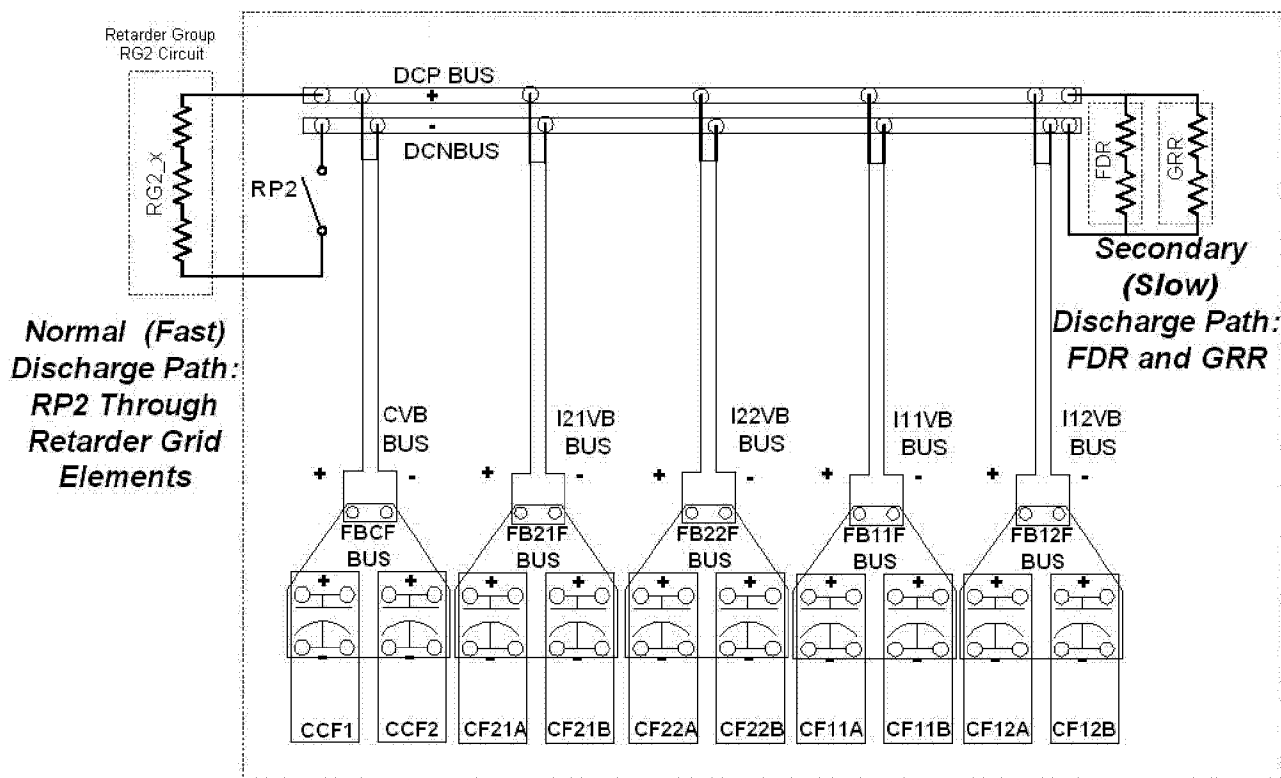


E030120

FIGURE 3-4. INFORMATION DISPLAY PANEL

- | | |
|-------------------------|---------------------------|
| 1. Control Power Switch | 3. Capacitor Charge Light |
| 2. GF Cutout Switch | |

In most control cabinets, RP2 is the normal discharge path. In groups containing an RP3 contactor, RP2 and RP3 are alternated as the normal discharge path. Refer to Figure 3-5.



83456

FIGURE 3-5. DISCHARGE PATHS

4. Observe both capacitor charge lights (CCL1, CCL2). CCL1 is on the outside of the contactor box. CCL2 is on the switch panel inside the low voltage area. The lights will stay illuminated as long as the voltage on the DC link is greater than 50 VDC. Refer to Figure 3-6.

⚠ WARNING

Hazardous voltages may be present in this equipment even if the engine and capacitor charge lights are off.

Use measurement and protective equipment rated for 2000 VDC minimum to verify that no voltage is present before touching any terminal.

⚠ WARNING

Verify functionality of the measurement equipment using site-approved procedures both before and after performing control group measurements.

Failure to observe these precautions may result in death or serious personal injury.

5. Use volt meter tester (58B-06-00770) to test the volt meter for proper operation. Voltage should read 100 VDC. If the voltmeter fails the test, repair or replace the volt meter. Do not proceed unless the volt meter is functioning properly.

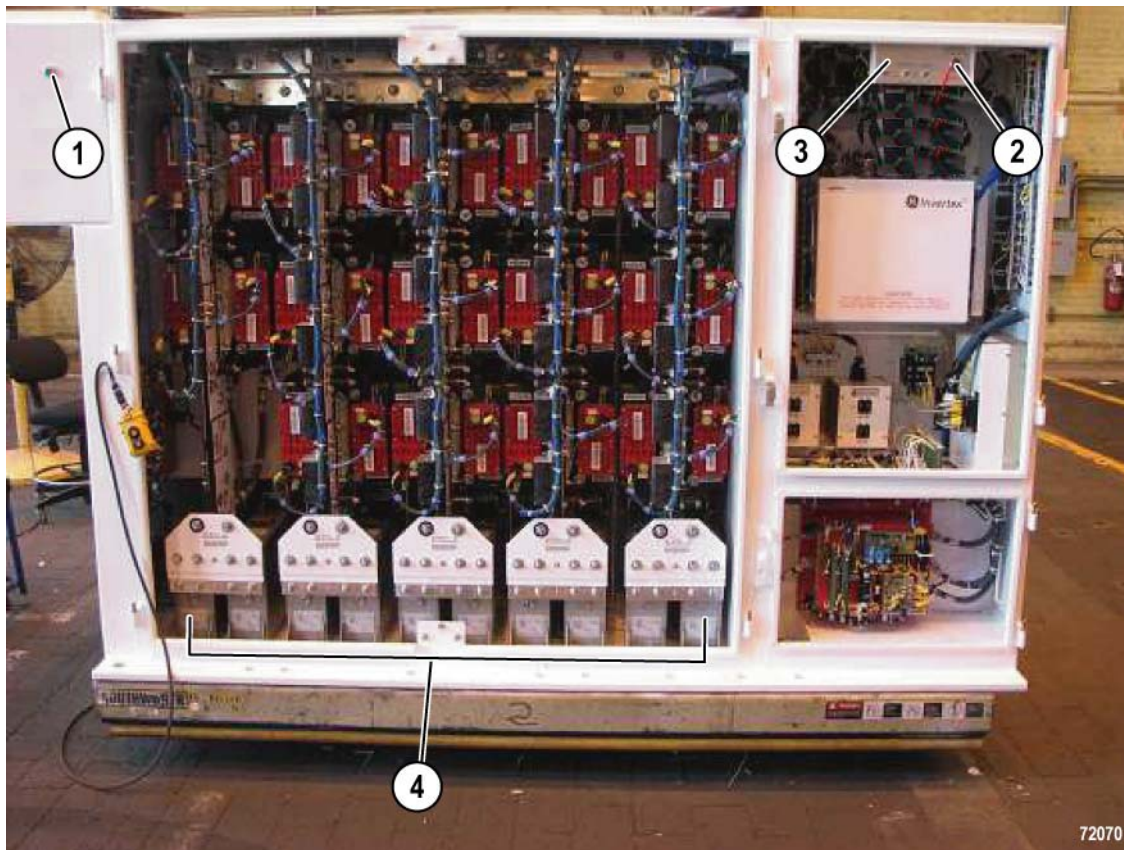


FIGURE 3-6. CAPACITOR CHARGE LIGHTS

- | | |
|---|------------------------------|
| 1. Exterior Capacitor Charge Light (CCL2) | 3. Information Display Panel |
| 2. Interior Capacitor Charge Light (CCL1) | 4. DC Link Capacitors |

Failure of the Discharge System

If the capacitor charge lights remain illuminated, a failure of the normal (fast) capacitor discharge system (RP discharge path) has likely occurred. The slow discharge resistors that are hard wired across the DC link should then discharge the capacitors. The slow discharge resistors will discharge the DC link to less than 1 VDC in under 7.5 minutes. If after 7.5 minutes, the capacitor charge lights are still illuminated, it must be assumed that the automatic discharge system is not working and that high voltage is present in the high voltage area.

1. Open the high voltage area doors.
2. Use a suitable high voltage meter (rated for at least 2000 VDC) and suitable protective equipment that are included in 2000VDC meter kit (58B-06-00800) to measure the voltage across each of the DC link capacitors. Place one meter lead on the positive (+) capacitor terminal, and the other meter lead on the negative (-) capacitor terminal, and observe the voltage. See Figure 3-7 and Figure 3-8. If the voltage is less than 1.0 V the capacitor is sufficiently discharged. Repeat on all DC Link capacitors. If the voltage of any capacitor is above 1.0 V, manually discharge the capacitor as described in "Manual Discharge of Capacitors".



FIGURE 3-7. DC METER ON CAPACITOR

72184



FIGURE 3-8. CAPACITOR TERMINAL POLARITY

72185

Manual Discharge of Capacitors

If the voltage is greater than 1V on any of the capacitors, the capacitor must be manually discharged. Use a suitable capacitor discharge device, such as the grounding sticks included in 2000VDC meter kit (58B-06-00800), to discharge the capacitors where needed per the following instructions:

1. With the control cabinet high voltage compartment doors closed, connect the grounding stick pair ground lead to ground. The most convenient location is the middle top bolt that attaches the door center post to the control cabinet frame.
2. Verify the ground connection and grounding stick resistance levels. Measure the resistance between each individual grounding stick tip to either the GND1 or GND2 ground block in the low voltage compartment of the control cabinet using a multimeter (Figure 3-9). Ensure that the resistance is within the manufacturer's specifications for the ground sticks. If the readings are not within specifications, repair or replace the grounding sticks. Do not proceed any further until the grounding sticks are within specifications.

For the grounding sticks included in 2000VDC meter kit (58B-06-00800), the valid range is 80 to 125 ohms per stick.

⚠ WARNING

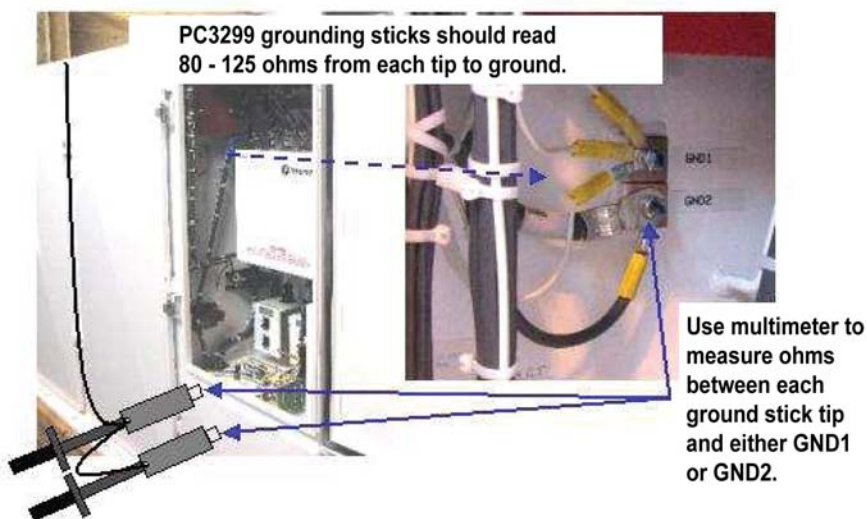
Hazardous voltages are present in this equipment. Avoid touching any energized equipment when the door to the low voltage area is open. Failure to do so may result in personal injury and equipment damage.

3. Discharge the relevant capacitors. Open the door(s) and place one grounding stick on one of the positive (+) capacitor terminals and the other on the diagonally located negative (-) terminal. Refer to Figure 3-8 and Figure 3-10. Attempt to minimize the time between application of the positive stick and the negative stick so that the current flow will be positive to negative rather than either to ground. Leave sticks in place until capacitors are discharged. Refer to Figure 3-10.

NOTE: On the capacitors located behind the door post, it is difficult to access diagonal terminals on the same capacitor. If using adjacent terminals, use care to keep the tips separated while discharging or else use a positive terminal on one capacitor and negative terminal of the adjacent capacitor on the same bus bar.

For the grounding sticks, the discharge times from 2000 volts are:

- 15 seconds (maximum) for 2 capacitors
- 75 seconds (maximum) for 10 capacitors



72185

FIGURE 3-9. GROUNDING STICK LEAD CONNECTION AND CHECK

⚠ WARNING

Voltages in excess of 1500 VDC may be present. Any measurement and/or protective equipment used must be rated at 2000 VDC minimum.

Verify functionality of the measurement equipment using site-approved procedures both before and after performing control group measurements.

Failure to observe these precautions may result in death or serious personal injury.

4. To ensure the discharge tool is working properly, verify the ground connection and ground stick resistance levels. Measure the resistance between each individual ground stick tip to either the GND1 or GND2 ground block in the low voltage compartment of the control cabinet using a multimeter (Figure 3-9). Ensure that the resistance is within the manufacturer's specifications for the ground sticks.

For the grounding sticks included in 2000VDC meter kit (58B-06-00800), the valid range is 80 to 125 ohms per stick.

5. Re-test for voltage on all capacitors as described in "Manual Discharge of Capacitors". The discharge procedure must be repeated on all capacitors that indicated 1 VDC or higher.
6. After it is confirmed that all capacitors are fully discharged, proceed to "Short Isolated Capacitor Terminals".



FIGURE 3-10. APPLICATION OF GROUNDING STICKS TO CAPACITOR TERMINALS

Short Isolated Capacitor Terminals

⚠ WARNING

Hazardous voltages may be present in this equipment even if the engine and capacitor charge lights are off.

Use measurement and protective equipment rated for 2000 VDC minimum to verify that no voltage is present before touching any terminal.

Verify functionality of the measurement equipment using site-approved procedures both before and after performing control group measurements.

Failure to observe these precautions may result in death or serious personal injury.

Any capacitor that is isolated from the DC link and confirmed discharged must have its terminals electrically shorted together to prevent static charge build up. Use bare wire to jumper all four terminals on the capacitor. See Figure 3-11. Proceed to troubleshoot and repair the control group to restore it to original functionality.

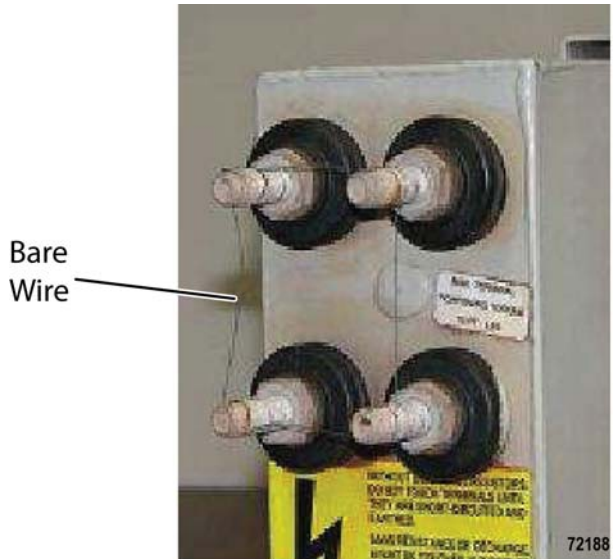


FIGURE 3-11. JUMPER ALL TERMINALS ON ISOLATED CAPACITOR

OPERATING INSTRUCTIONS

PREPARING FOR OPERATION

The safest trucks are those which have been properly prepared for operation. At the beginning of each shift, a careful check of the truck should be made before attempting to start the engine.

Safety is thinking ahead. Prevention is the best safety program. Prevent a potential accident by knowing the employer's safety requirements, all necessary job site regulations, and use and care of the safety equipment on the truck. Only qualified operators or technicians should attempt to operate or maintain a Komatsu truck.

Safe practices start before the operator gets to the equipment.

- Wear the proper clothing. Loose fitting clothing, unbuttoned sleeves and jackets, jewelry, etc., can catch on a protrusion and cause a potential hazard.
- Always use the personal safety equipment provided for the operator such as hard hats, safety shoes, safety glasses or goggles. There are some conditions when protective hearing devices should also be worn for operator safety.
- When walking to and from the truck, maintain a safe distance from all machines, even if the operator is visible.

WALK AROUND INSPECTION

At the beginning of each shift, a careful walk around inspection should be performed before attempting to start the engine. A walk around inspection is a systematic ground level inspection of the truck and its components to ensure that the truck is safe to operate before entering the operator's cab.

Start at the left front corner of the truck (see the illustration on the next page) and move in a counterclockwise direction. Move front-to-rear, across the rear, and continue forward up the opposite side of the truck to the original starting point.

If these steps are performed in sequence and are repeated from the same point and in the same direction before every shift, many potential problems may be avoided or scheduled for maintenance. Unscheduled downtime and loss of production can be reduced as a result.

Local work practices may prevent an operator from performing all tasks suggested here. To the extent permitted, the operator should follow this routine.



High voltage may be present on this truck! DO NOT open any electrical cabinet doors on this truck while the engine is running. Never climb on any power cables or use power cables for handholds or footholds unless the engine has been shut down and the system has been verified to be at "rest".

1. Start at left front of truck. Visually inspect all lights and safety equipment for damage. Make sure that the lenses are clean and unbroken.
2. Move in front of the left front tire. Inspect the hub and brake assemblies for leaks and any abnormal conditions.
3. Make sure that all suspension mounting hardware is secure. Inspect the mounting key area for evidence of wear. Make sure that the suspension extension (exposed piston rod) is correct and that there are no leaks.
4. Make sure that the anchor end of the steering cylinder is secure and properly greased.
5. With the engine stopped, check the engine oil level. Use the service light, if necessary.
6. Inspect the air conditioner belts for correct tension, obvious wear, and tracking. Make sure that the fan guard is secure and in good condition. When leaving this point, turn off the service light if used.
7. Move outside of the front wheel. Make sure that all of the mounting nuts/studs are in place and tight. Check the tires for cuts, damage, or bubbles. Check for evidence of incorrect tire inflation.
8. Move in behind the front wheel. Make sure that the steering cylinder is properly greased and the mounting hardware is tight. Check the suspension mounting hardware and suspension extension. Make sure that the suspension protective boot is in good condition. Inspect the hub and brakes for any unusual conditions. Check the entire area for leaks.

START HERE

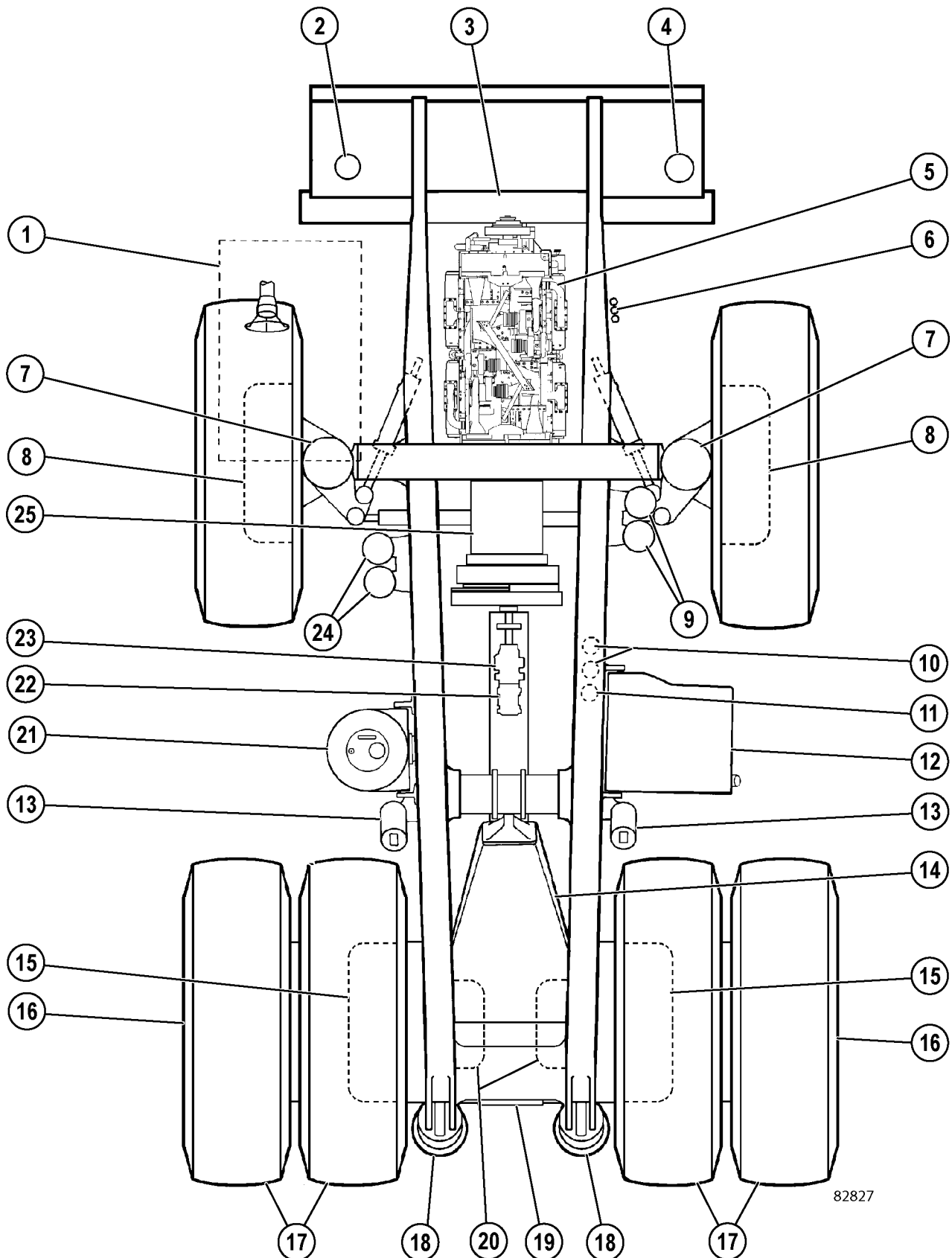


FIGURE 3-12. TOP VIEW

1. Operator Cab
2. Auto Lubrication
3. Radiator
4. Reserve Oil System
5. Engine
6. Fuel Filters
7. Suspension
8. Wet Disc Brake
9. Brake Accumulators
10. Hoist Filters
11. Steering Filter
12. Fuel Tank
13. Hoist Cylinder
14. Rear Axle Housing
15. Wet Disc Brake
16. Wheel Motor Transmission
17. Rear Tires
18. Rear Suspension
19. Rear Axle Hatch
20. AC Electric Motor
21. Hydraulic Tank
22. Steering Pump
23. Hoist Pump
24. Steering Accumulators
25. Alternator

9. Inspect the sight glass on the hydraulic tank. With the engine stopped and the body down, the hydraulic oil level must be visible in the center of the upper sight glass.
10. Verify that all hydraulic tank shut-off valves are locked in their fully open positions.
11. Move around the hydraulic tank and in front of the rear dual tires. Inspect the hoist cylinder for any damage and leaks. Make sure that both upper and lower hoist cylinder pins are secure and properly greased.
12. Look under the lower edge of the chassis to make sure that the flexible duct that carries the air from the main blower to the final drive housing is in good condition. Also, look up at the main hydraulic pumps to see if there is any leakage or any other unusual condition with the pumps or pump drive shafts.

13. Move around the left side dual tires. Make sure that all wheel nuts/studs are in place and tight. Inspect the wheel for any oil that would indicate brake leakage or wheel motor leakage.

Check the dual tires for cuts, damage, and bubbles. Verify that tire inflation is correct. If the truck has been run on a "flat", the tire must be cooled before moving the truck inside. Check for any rocks that might be lodged between the dual tires.

14. Inspect the left rear suspension for damage, correct rod exposure, and leaks. Ensure that the covers over the chrome piston rod are in good condition.
15. Open the rear axle hatch. Inspect for leaks around the parking brakes. Make sure the cooling air exhaust ductwork is intact and there are no obstructions.
16. While standing at the rear axle hatch, look up to check that the rear lights and backup alarms are in good condition. Look up at the panhard rod to check that it is properly greased. Also check both body hinge pins for proper greasing and any abnormal condition. Check the hoist limit switch and clear any mud or debris from the contacts.
17. Perform the same inspections on the right rear suspension.
18. Move around the right side dual tires. Inspect between the tires for rocks. Inspect the tires for cuts, damage and proper inflation.
19. Make sure that all wheel nuts/studs are in place and tight. Inspect the wheel for any oil that would indicate brake leakage or wheel motor leakage.
20. Move in front of right dual tires. Inspect the hoist cylinder in the same manner as the left side. Make sure the body-up limit switch is secure and in good condition. Remove any mud or dirt accumulation from the switch.
21. Move around the fuel tank. Make sure that the fuel gauge agrees with the fuel gauge in the cab. Inspect the mounting hardware for the fuel tank at the upper saddles and at the lower back side of the tank. Check the hoist filters for leaks.

22. Move in behind the right front wheel. Make sure that the steering cylinder is properly greased and the mounting hardware is secure. Check the suspension mounting hardware and suspension extension. Make sure that the suspension protective boot is in good condition. Inspect the hub and brakes for any unusual conditions. Check the entire area for leaks.
23. Move out and around the right front wheel. Make sure that all wheel nuts/studs are in place and tight.
24. Move behind the front of the right front wheel. Check the hub and brakes for leaks and any unusual conditions. Make sure that the steering cylinder is secure and properly greased. Inspect the engine compartment for any leaks and unusual conditions. Inspect the fan guard and belts. Check for any debris behind the radiator.
25. Inspect the auto lube system. See Section P3, Automatic Lubrication System, for more information about the auto lube system.
26. As you move in front of the radiator, remove any debris that is stuck in front of the radiator. Check for any coolant leaks. Inspect the headlights and fog lights.
27. Before climbing the ladder to the first level, make sure that the ground level engine shutdown switch is ON. Inspect the fire control actuator. Make sure that the safety pin and the plastic tie that prevents accidental actuation are in place and in good condition. Make sure that the battery disconnect switches are activated.
28. Always use handrails and ladders when mounting or dismounting the truck. Clean any foreign material such as ice, snow, oil or mud from the ladders and handrails.
29. If equipped with the retractable ladder system:
 - a. Visually inspect the ladder for mechanical damage. If movement is impaired in any way, the assembly must be repaired.
 - b. Visually inspect for cleanliness. Ensure that the ladder is dry and free of grease and oil.
 - c. Ensure that the correct oil level is maintained in the reservoir.
 - d. Raise and lower the ladder. Check for loose parts or any adverse noise conditions. Ensure that the movement alarm and both UP and DOWN LEDs operate correctly.
 - e. Check for any change in equipment appearance, especially that which will affect ladder system stability.



If the engine has been running, allow the coolant to cool before removing the fill cap or draining the radiator. Serious burns may result if skin comes in contact with hot coolant.

30. Use the coolant level sight gauge to check the coolant level in the radiator. If it is necessary to remove the radiator cap, make sure that the engine is off, then relieve the coolant pressure by *slowly* removing the radiator cap.
31. Inspect the retarding grid covers. Make sure that the latches are in place and secure. Inspect the main air inlet to ensure that it is clear. Make sure that all cabinet door latches are secure.
32. Move around to the rear of the cab. Open the doors of the brake cabinet and check inside the cabinet for leaks.
33. Clean the cab windows and mirrors. Clean out the cab floor, if necessary. Make sure that the steering wheel, operator controls, and pedals are free of any grease or mud.
34. Stow any personal gear in the cab so that it does not interfere with any operation of the truck. Dirt or trash buildup, specifically in the operator's cab, should be cleared. Do not carry tools or supplies in the cab or on the deck.
35. Adjust the seat and the steering wheel for the most operator comfort.
36. Before operating the truck, read about and understand the locations and functions of all operator controls listed in Section N5, Operator Cab Controls.
37. If equipped, raise the retractable ladder using the in-cab control panel.

RETRACTABLE LADDER SYSTEM (Optional)

Description

The retractable ladder system (RLS) is an electro-hydraulic ladder powered by the truck's 24VDC electrical system. The RLS provides a safe means to mount and dismount the truck under normal and emergency conditions. The RLS consists of two main control components, the in-cab control panel and the power pack that operates the ladder.

1. The in-cab control panel is a microprocessor that controls, monitors, stores and reports ladder system operational data.
2. The electro-hydraulic power pack is in a ground level stainless steel cabinet. It houses the main control hydraulics and electrical components that lower and raise the ladder.

The RLS also contains wiring harnesses, hydraulic hoses and an emergency down valve.

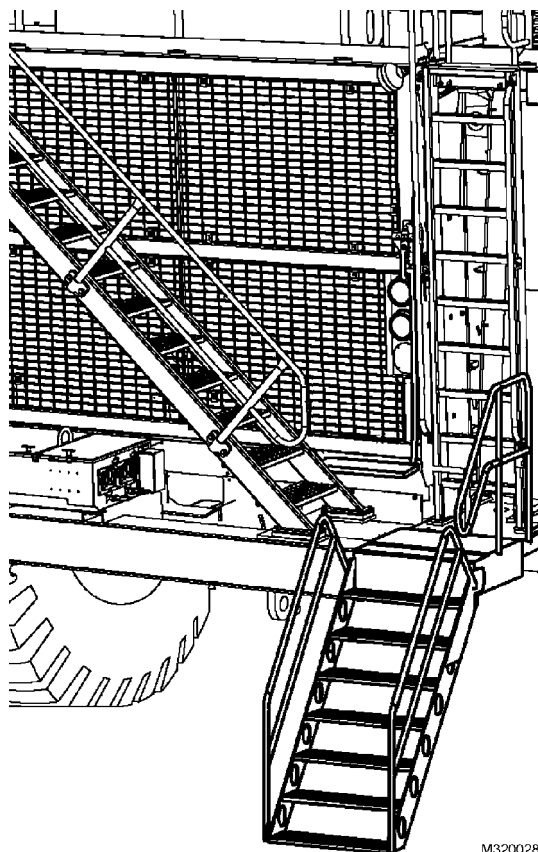


FIGURE 3-13. RETRACTABLE LADDER SYSTEM

Normal Operation

During normal operation, a person can lower or raise the RLS by using the:

- In-cab control panel
- Ground level control box located next to the ground level ladder light switch
- Control switches in the power pack.

NOTE: The master disconnect switch located on the battery disconnect box and the isolation switch in the power pack must both be in the ON position for the RLS to operate.

As an added safety measure, the RLS uses a parking brake interlock that requires the parking brake to be set before the ladder can be operated under normal conditions. The RLS will automatically raise the ladder if the operator moves the directional control lever out of PARK and fails to press the [UP] button on the in-cab control panel before attempting to drive the truck.

General Safety

The following safety procedures, at a minimum, must be followed to ensure safe operation and use of the RLS.

- DO NOT run or jump on the ladder.
- DO NOT overload the ladder. Use the ladder one person at a time.
- Hold onto the handrail when using the ladder.
- Always face the ladder when ascending or descending.
- DO NOT attempt to ride on the ladder while it is being raised or lowered or while the truck is in motion.
- Always visually check the ladder before use to ensure the unit has not been damaged.
- Ensure the ladder is in the fully down position before boarding.
- Keep hands and fingers away from pinch points while the ladder is in motion.
- Always check to ensure no personnel are on or in the immediate vicinity of the ladder while it is in motion.
- The ladder must be kept clean and free of moisture, grease and oil
- When in the truck's cab, always use the in-cab control panel to raise the ladder.
- Report defects to maintenance personnel immediately.

IN-CAB CONTROL PANEL

In-cab control panel (1, Figure 3-14) is located on the left side of the dash and contains a microprocessor that controls, displays, monitors, stores and reports ladder system operational data. The in-cab control panel provides real time position of the ladder to the operator.

The in-cab control panel features a digital display screen, command buttons ([UP], [DOWN], [ENTER], [EXIT], [LEFT], [RIGHT]) for operating the ladder and navigating through the various menu display screens, a USB key port and RS 232 port for downloading ladder system operational data to a laptop computer for troubleshooting. Refer to Figure 3-15.

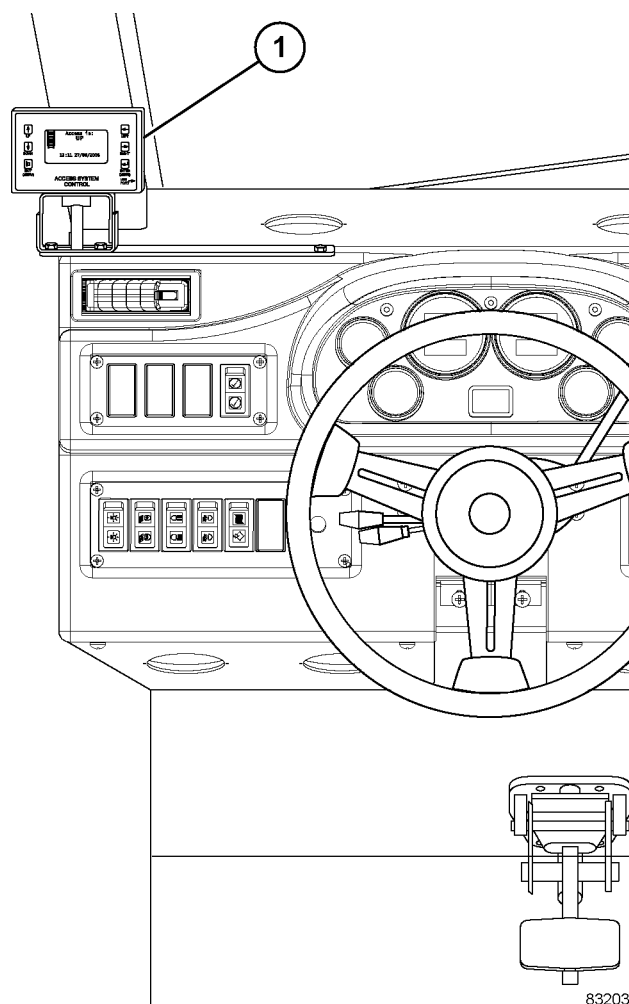


FIGURE 3-14. CAB CONTROLS
(OPERATOR VIEW)

1. In-Cab Control Panel

Digital Display Screen

In-cab control panel display screen (1, Figure 3-15) provides the user with system status, error/fault messages and a visual indicator of the ladder's position (lowered, raised, in motion).

Command Buttons

[UP] button (2) is a dual function control that is used to raise the ladder during normal operation and scroll through the menu displays. Pressing this button and holding it briefly will cause the ladder to raise. Pressing and releasing this button one time will scroll up one line of menu display, and holding it will result in continuous menu scrolling.

[DOWN] button (3) is a dual function control that is used to lower the ladder during normal operation and scroll through the menu displays. Pressing this button and holding it briefly will cause the ladder to lower. Pressing and releasing this button one time will scroll down one line of menu display, and holding it will result in continuous menu scrolling.

[DOWN] button (3) is a dual function control that is used to lower the ladder during normal operation and scroll through the menu displays. Pressing this button and holding it briefly will cause the ladder to lower. Pressing and releasing this button one time will scroll down one line of menu display, and holding it will result in continuous menu scrolling.

[LEFT] button (4) is for scrolling left when entering a password.

[RIGHT] button (5) is for scrolling right when entering a password.

[ENTER] button (6) is a multi-function button used for ladder system alarm acknowledgement and menu access when in Display Mode. This button is also used to confirm or accept changes shown on the display screen. Press this button to confirm or accept changes listed on the display screen.

[EXIT] button (7) is a multi-function button used for ladder system alarm acknowledgement and menu access when in Display Mode. This button is also used to exit from the menu screens. Press this button to exit from menu screens.

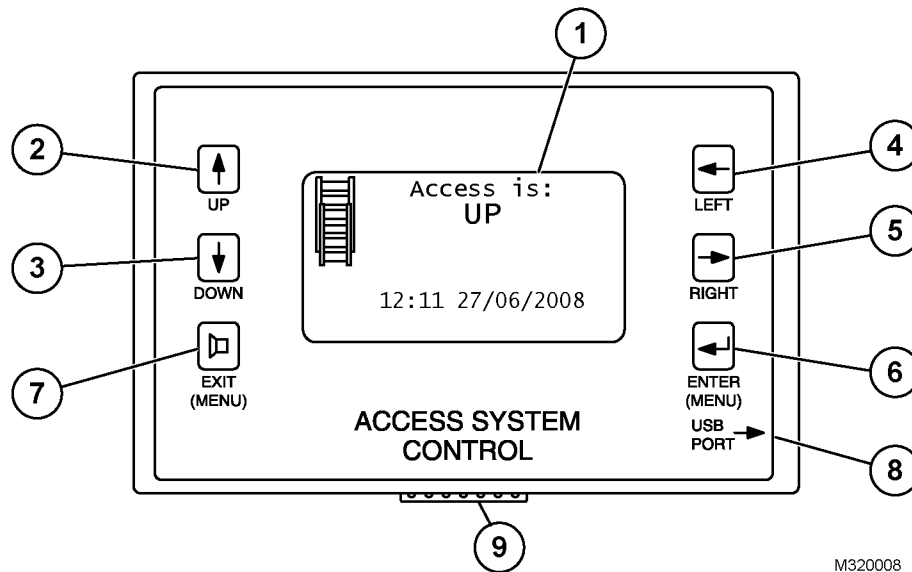


FIGURE 3-15. IN-CAB CONTROL PANEL (FRONT VIEW)

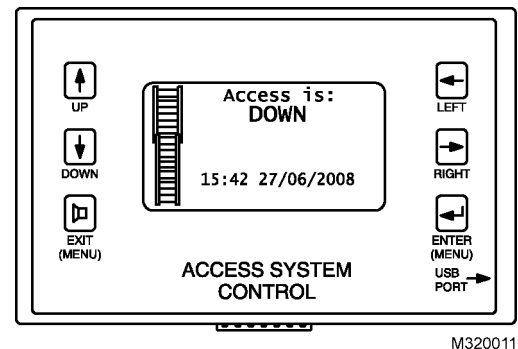
- | | | |
|-------------------|-------------------|------------------------|
| 1. Display Screen | 4. [LEFT] Button | 7. [EXIT] Button |
| 2. [UP] Button | 5. [RIGHT] Button | 8. USB Port |
| 3. [DOWN] Button | 6. [ENTER] Button | 9. 15-Pin Harness Plug |

USING THE IN-CAB CONTROL PANEL

While the ladder is in motion, the direction (up or down) can be changed by pressing the opposite direction ([UP] or [DOWN]) button on the control panel.

If the IN CAB LOCKOUT message is displayed on the control panel screen, then the RLS can only be operated from the ground level control box mounted next to the ground level ladder light switch. All other control panel functionality is still available including fault indication and audible alarms.

The ladder will typically be in the lowered position on a stationary truck. The in-cab control panel will indicate that the access is DOWN, as shown here:

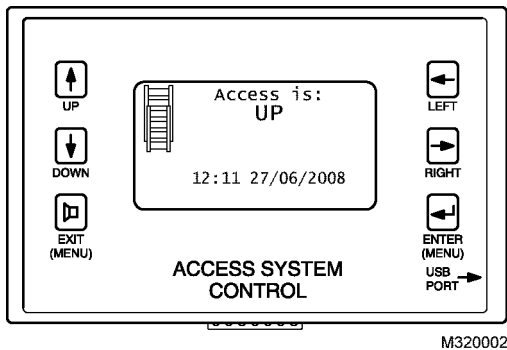


When in the operator's cab, always use the [UP] button on the control panel to raise the ladder instead of moving the directional control lever out of PARK. The automatic operation of the ladder when the parking brake is released is an emergency feature only. DO NOT move the directional control lever out of PARK to raise the ladder as part of normal operation.

Raising the Ladder

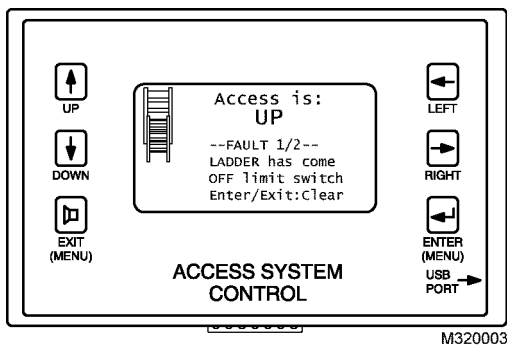
To raise the ladder, press and hold (temporarily) the [UP] button located on the in-cab control panel. The illustrated ladder on the screen will animate and begin to rise to the UP position and the UP arrow (on the control panel button) will flash. When the ladder is completely raised, the [UP] button will remain illuminated continuously.

When the ladder has reached its raised travel position and strikes the limit switch, the control panel will indicate that the access is UP, as shown here:



The RLS power pack will continue to operate for three more seconds to charge hydraulic accumulator. During this time, the UP arrow will continue to flash until the cycle is complete.

The RLS uses a limit switch to monitor ladder travel to the UP position. If the ladder fails to strike the limit switch or the limit switch fails to close, a fault alarm will activate after a short delay and the in-cab control panel will display this message, as shown here:



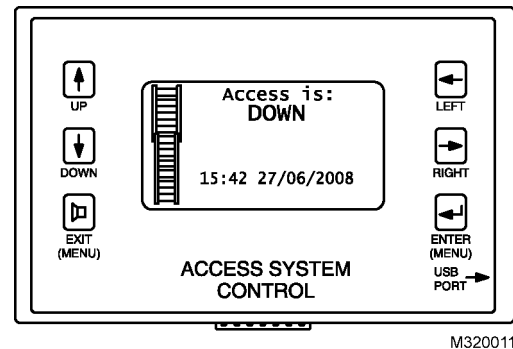
NOTE: The fault alarm must be acknowledged by pressing the [EXIT] or [ENTER] buttons on the control panel. Once the alarm has been acknowledged, the fault message will be replaced by a **FAULT IN SYSTEM** message. A defective limit switch will prevent normal RLS operation and must be corrected immediately.

Lowering the Ladder

NOTE: The directional control lever must be in **PARK** before the ladder can be lowered.

To lower the ladder, press and hold (temporarily) the [DOWN] button located on the in-cab control panel. The illustrated ladder on the screen will animate and begin to lower to the DOWN position and the [DOWN] button will flash.

When the ladder has reached its lowered travel position, the control panel will indicate that the access is DOWN, as shown here:



When the ladder is completely lowered, the [DOWN] button will remain illuminated continuously.

USING THE GROUND LEVEL CONTROL BOX

Ground level control box (1, Figure 3-16) is located next to the ground level ladder light switch. Toggle switch (2) lowers and raises the ladder.

NOTE: The parking brake must be applied before the ladder can be lowered. If toggle switch (2) is held in either position for more than ten seconds, a fault will be activated and will need to be acknowledged by pressing either the [EXIT] or [ENTER] buttons located on the in-cab control panel.

Raising the Ladder

To raise the ladder, push toggle switch (2) to the LADDER UP position and release. Ladder operation via this switch is the same as using the in-cab control panel. Any ladder movement will be shown on the in-cab control panel.

Lowering the Ladder

To lower the ladder, push toggle switch (2) to the LADDER DOWN position and release. Ladder operation via this switch is the same as using the in-cab control panel. Any ladder movement will be shown on the in-cab control panel.

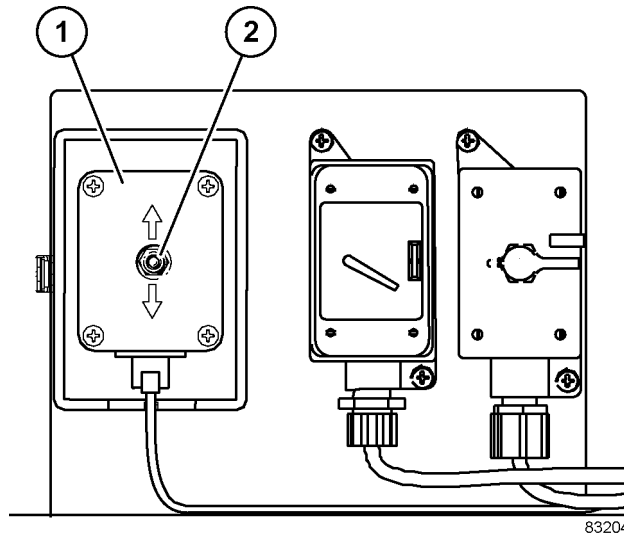


FIGURE 3-16. GROUND LEVEL CONTROL BOX

1. Ground Level Control Box
2. Toggle Switch

EMERGENCY OPERATION

In an emergency, the retractable ladder can be lowered by using emergency down valve (1, Figure 3-17) mounted on the grille above the left hand headlight assembly. The emergency down valve relieves ladder system hydraulic pressure and allows the ladder to smoothly lower to the ground.

To lower the ladder, rotate the handle on the emergency down valve clockwise. The ladder will lower smoothly until it reaches the ground.

To reset the ladder, rotate the handle counterclockwise to its original position and, with power restored to the power pack, press the [UP] button to raise the ladder.

NOTE: The handle on the emergency down valve must be in the original position before re-activating the ladder.

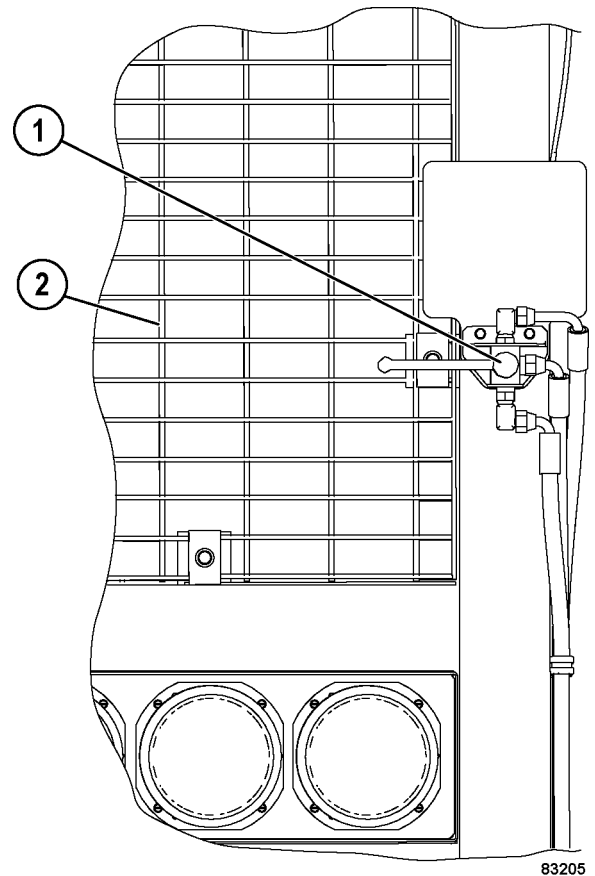


FIGURE 3-17. EMERGENCY DOWN VALVE

1. Emergency Down Valve
2. Grille

ENGINE START-UP



Never attempt to start the engine by shorting across the starter terminals. This may cause fire, or serious injury or death to anyone in the machine's path. Start the engine from the operator's seat only.

If the truck is equipped with an auxiliary cold weather heater system, do not attempt to start the engine while the heaters are in operation. Damage to the coolant heaters will result!



1. Make sure that all personnel are clear of the truck before starting the engine. Always sound the horn as a warning before moving any controls.
2. If the truck is in an enclosure, make sure that there is adequate ventilation before start-up. Exhaust fumes are dangerous!
3. Make sure that the directional control lever is in PARK before starting.
4. Place the rest switch in the ON position to put drive system in rest mode. For more information, refer to Section N5, Operator Cab And Controls.



Do not crank an electric starter for more than 30 seconds. Allow two minutes for cooling before attempting to start the engine again. Severe damage to the starter motors can result from overheating.

5. The key switch is a four-position switch (ACC, OFF, RUN, START). The ACC position is not currently used.
 - a. Turn the key switch one position clockwise to the RUN (not START) position. All electrical circuits except the start circuit are activated.
 - b. Turn the key switch fully clockwise to the START position and hold this position until the engine starts. The START position is spring-loaded and will return the switch to the RUN position when the key is released.

NOTE: This truck is equipped with an engine prelube system. With this feature, a noticeable time delay may occur before engine cranking begins as the engine lube oil passages are being filled and pressurized.

6. After the engine has started, place the rest switch in the OFF position to deactivate the rest mode. Refer to Section N5, Operator Cab And Controls for more information on the rest switch.

NOTE: In cold ambient conditions and when the engine is cold, the engine rpm will not increase above low idle speed until the engine controller determines it is safe to do so. This time delay will vary from 30 seconds to 11 minutes which allows the coolant and engine oil to warm up.

AFTER ENGINE HAS STARTED



Do not leave the truck unattended while the engine is running. Move the directional control lever to PARK and turn off the engine before leaving the truck.

Become thoroughly familiar with the steering, braking and emergency controls.

1. After the engine has been started and the low pressure warning systems are normal, test the steering by turning the steering wheel fully to the left and right.

If the steering system is not operating properly, turn the engine immediately. Determine the steering system problem and have it repaired before resuming operation.

2. Operate each of the truck's brake circuits at least twice before operating and moving the truck. These circuits include the service brake, parking brake, and brake lock. With the engine running and the hydraulic circuit fully charged, activate each circuit individually from the operator's cab.

If any application or release of any brake circuit appears sluggish or improper, or if warning alarms are activated, turn off the engine and notify maintenance personnel immediately. Do not operate the truck until the faulty brake circuit is fully operational.

3. Check the gauges, warning lights, and instruments before moving the truck to ensure proper system operation and proper instrument functioning. Pay special attention to the braking and steering hydraulic circuit warning lights. If warning lights come on, turn off the engine immediately and determine the cause.
4. Make sure that the headlights, worklights, and taillights are in proper working order. Check the operation of windshield wiper. Good visibility may help prevent an accident.

PRE-SHIFT BRAKE CHECK

NOTE: Komatsu recommends that operators perform static brake tests to verify that the braking systems are adequate at the beginning of each shift before operating the truck.

The static brake test allows the operator to check the service brake, parking brake and the dynamic retarder. The purpose of these tests is to verify the functionality of the service brake, parking brake and retarder systems at the time they are tested.

After performing each test, it is the operator's responsibility to determine if the truck passed each test and if the truck is safe for operation.

The order of performing the brake tests, (service brake, parking brake or retard system) does not matter. Each brake test is a separate test, where one brake system or all three can be tested at any time.

If an operator has questions during brake testing, refer to the drive system Diagnostic Information Display (DID) panel, located on the back wall of the cab for guidance.



If the truck fails any brake test, notify maintenance personnel immediately. Do not resume operation unless the truck passes all brake tests.

Events

The following events have been added to the drive system software to support the brake test feature.

- 645-1 Service Brake test Performed
- 645-2 Parking Brake Test Performed
- 645-3 Retard Test Performed
- 645-4 Brake Test Switch stuck closed

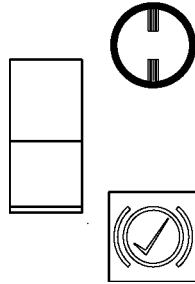
The Events Log will record when and which brake tests have been performed.

OPERATION

The static brake test utilizes a momentary switch and a check light located in the overhead display panel.

Brake Test Switch

The brake test switch is used to initiate a brake test. Press on the momentary switch to enter the brake test mode. If certain conditions are met, the operator can enter a brake test sequence.



If the brake check light does not illuminate immediately after pressing the brake test switch, there is most likely a problem with the setup. Refer to the setup conditions and take action to prepare the truck for a brake test.

If there is a problem with the truck setup, the DID panel will display the problem.

For example if the engine is off:

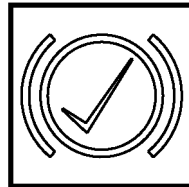
ERROR Entering Brake Test
Engine not running

If the Truck is loaded:

ERROR Entering Brake Test
Truck is NOT Empty

Brake Check Light

The amber light is used to indicate when the truck is in the brake test mode. When illuminated, a brake test is ready. When flashing, the brake test is at the validation point, or the retard system test is finished.



If all of the conditions are correct, except the brakes are not set correctly, an error message will be displayed. For example, if the service brake and parking brake are both applied together:

ERROR Entering Brake Test
Set Brakes for Test

Description

The operator can choose which brake test to perform, and will set the truck controls based on the settings in Table 1. The drive system will detect the position of the directional control lever, and will prepare for the appropriate test. The operator will then press the brake test switch.

If the brake check light is illuminated solid after pressing the brake test switch, the system is in brake test mode and is ready for the chosen test to be initiated by the operator. After testing, the operator will then determine if the truck passed the brake tests, and if it is safe for operation.

Brake Test Exit Criteria

Numerous conditions can occur which may interrupt a brake test, including the following:

- Any of the setup conditions becoming false
- Drive system fault which restricts the LINK or Propel mode
- Truck speed greater than 3.2 kph (2.0 mph)
- Drive system at torque level for more than 30 seconds
- Brake test requested, but not initiated by the operator within 60 seconds after pressing the brake test switch

Table 1: BRAKE TEST SETTINGS

Test Type	Wheel Brake Lock	Service Brake Pedal	Directional Control Lever
Service Brake	OFF	FULLY APPLIED	NEUTRAL
Parking Brake	OFF	RELEASED	PARK
Retard Test	OFF	RELEASED	PARK

PERFORMING THE BRAKE TESTS

Setup

Before performing any brake test, the following conditions must be met:

- Dump body empty

NOTE: The drive system will not enter any brake test if the truck is loaded.

- Truck located outside on a flat area, where truck movement is allowed.



Ensure the area around the truck is free of personnel and objects. Some truck movement could occur during brake testing.

The drive system is unable to determine if the truck is on a hill or in a parking ditch. Testing during these conditions will affect test results.

Before performing any brake test, the truck must be in the following state:

- Engine on (low idle)
- Drive system ready (Ready Mode)
- Dump body down
- No drive system warning lights ON (Can not be in LIMP mode.)
- Zero ground speed

Service Brake Test

1. Firmly depress the retard/brake pedal.
2. Place the directional control lever in the NEUTRAL position. Ensure the wheel brake lock switch is OFF.
3. Press the brake test switch and wait for the brake check light to be on solid.
4. Fully depress the retard/brake pedal. Ensure full brake application is reached. Failure to reach full brake application will affect the service brake test results.
5. The DID panel will display:

Service Brake Test READY
Press Accel Pedal to Start

6. With the retard/brake pedal still fully applied, fully depress the accelerator pedal. The drive system controller will enter propel mode and generate torque up to the service brake limit. Maintain full service brake pressure during the test.

NOTE: The drive system can only detect if the service brakes are applied. It can not detect the percentage of application. It is up to the operator to press hard enough on the service brake pedal to achieve a full brake application.

7. The DID panel will display:

Service Brake Test ACTIVE
Check Truck Movement when Light Flashes

8. Once torque has reached the limit for the service brake test, the brake check light will begin to flash. This is the indication for the operator to make a determination as to the status of the service brake system.

- **If the truck did not move:**
The service brake system passed the test.
- **If the truck moved during the test:**
The service brake system failed the test.

9. Release the accelerator pedal and the torque will be reduced, the test will stop and the brake check light will turn off.
10. Place the directional control lever in the PARK position. Release the service brake pedal.
11. If the truck failed the service brake test, notify maintenance personnel immediately.

WARNING

If the truck fails the service brake test, notify maintenance personnel immediately. Do not resume operation unless the truck passes all brake tests.

NOTE: If the operator partially or fully releases the accelerator pedal during the test, torque will be reduced and the brake check light will go back on solid when torque falls below the test set point. The operator can re-apply the accelerator pedal to increase torque and the brake check light will again flash when the torque is at the test limit.

If the test exits abnormally, or if the operator simply does not press the accelerator pedal far enough to achieve the torque level for the service brake test, the DID panel will display:

Brake Test ERROR
Test did NOT complete

Parking Brake Test

1. Place the directional control lever in the PARK position.
2. Press the brake test switch and wait for the brake check light to be on solid.
3. The DID panel will display:

Parking Brake or Retard Test READY
Press Accel or Retard Pedal to Start

4. Fully depress the accelerator pedal. The drive system controller will enter propel mode and generate torque up to the park brake limit.
5. The DID panel will display:

Park Brake Test ACTIVE
Check Truck Movement when Light Flashes

6. Once the torque has reached the limit for the parking brake test, the brake check light will begin to flash. This is the indication for the operator to make a determination as to the status of the parking brake system.

- **If the truck did not move:**

The park brake system passed the test.

- **If the truck moved during the test:**

The park brake test has failed. Release the accelerator pedal. If the truck starts to roll, apply the service brakes to hold the truck stationary. Notify maintenance personnel immediately.

CAUTION

If the truck moves at all during the parking brake test, immediately release the accelerator pedal to avoid serious damage to the parking brake.

WARNING

If the truck fails the parking brake test, notify maintenance personnel immediately. Do not resume operation unless the truck passes all brake tests.

7. When the operator releases the accelerator pedal, torque will be reduced, the test will stop and the brake check light will turn off.

Retard System Test

1. Place the directional control lever in the PARK position.
2. Press the brake test switch and wait for the brake check light to be on solid.
3. The DID panel will display:

Parking Brake or Retard Test READY
Press Accel or Retard Pedal to Start

4. Depress the retard/brake pedal far enough to engage retard, but not far enough to engage the service brakes. Depressing the retard/brake pedal too far will cancel the retarder test.

The drive system controller will ramp up the engine speed, close RP1, close RP2, close RP3 (if present), turn on the choppers, and test the retarding system. The system will verify current flow through each grid leg and the grid blower motor.

5. The DID panel will display the current status of the test.

Retard System Test ACTIVE
RP1 RP2 RP3 CHOP

(Elements are added as the test progresses.)

6. Upon successful completion of the test, the light will flash for 10 seconds indicating a successful test. The DID panel will display:

Retard System Test PASSED

or

Retard System Test FAILED or Incomplete

NOTE: If the brake check light never flashes, but turns off, the test has failed.



If the truck fails the retard system test, notify maintenance personnel immediately. Do not resume operation unless the truck passes all brake tests.

7. When the operator releases the retard/brake pedal, the test will stop and the brake check light will turn off.
8. If the retard system failed the test, notify maintenance personnel immediately. Do not resume operation unless the truck passes all brake tests.

EMERGENCY STEERING SYSTEM

Operation

This truck is equipped with an emergency steering system. This system is a backup in the event of loss of oil supply to the main steering system. The emergency steering system was designed to meet or exceed SAE J1511 and ISO 5010 standards.

If the low steering system pressure indicator light and alarm are activated, a failure in the hydraulic oil supply to the steering and brake system exists. When the alarm is activated, typically there is enough hydraulic pressure stored in the brake and steering accumulators to allow brief operation of the steering and brake functions. However, this oil supply is limited. Therefore, it is important to stop the truck as quickly and safely as possible after the alarm is first activated.

If the oil supply pressure drops to a predetermined level, the low brake pressure warning light will also illuminate. If the oil pressure continues to decrease, the brake auto-apply feature will activate the service brakes to stop the truck.

Pre-Operation Testing

NOTE: Komatsu recommends that operators perform this test to verify that the steering accumulator precharge pressure is adequate at the beginning of each shift before operating the truck.

1. Park the empty truck on flat, level ground. Lower the dump body onto the frame and stop the engine. Make sure that the key switch is in the OFF position.
2. Wait at least 90 seconds to verify that all hydraulic pressure has been relieved from the steering accumulators. Turn the steering wheel from stop to stop. If the front wheels do not move, there is no hydraulic pressure.
3. Check the hydraulic tank oil level. The oil level must be visible in the center of the upper sight glass and must not cover the entire upper sight glass. Add oil if necessary. **Do not overfill.**

4. Turn the key switch to the ON position, but do not start the engine.

a. *Steering system pressure:* Verify that the low steering pressure warning light is illuminated. If it is not illuminated, immediately notify maintenance personnel. Do not operate the truck until the problem is corrected.

b. *Steering accumulator precharge:* Verify that the low accumulator precharge warning light is not illuminated and the warning buzzer is not sounding. If the warning light is illuminated and the buzzer is sounding, immediately notify maintenance personnel. Do not operate the truck until the problem is corrected.

5. Start the engine and allow the steering accumulators to fully charge. Turn the steering wheel so that the front wheels are straight.

6. Check the hydraulic tank oil level while the engine is on.

a. If the oil level is visible in center of the lower sight glass and does not cover the entire lower sight glass, the steering accumulators are adequately charged. Proceed to Step 7.

b. If the oil level is below the lower sight glass, the steering accumulators are not adequately charged. Stop the engine and turn the key switch to the OFF position. Immediately notify maintenance personnel. Do not operate the truck until the problem is corrected.

7. If the steering accumulators are adequately charged, stop the engine by using the engine shutdown switch. Leave the key switch in the ON position. This allows the steering accumulators to retain their hydraulic charge. The low steering pressure warning light and the low accumulator precharge warning light should not illuminate.

8. Turn the steering wheel from stop to stop. The front wheels should turn fully to the left and to the right. Also, the low steering pressure warning light should illuminate and the warning buzzer should sound.

If the front wheels cannot be turned fully to the left and right, or if the warning light and buzzer do not activate, immediately notify maintenance personnel. Do not operate the truck until the problem is corrected.

If the truck passes this test, the emergency steering system is functioning properly.

MACHINE OPERATION SAFETY PRECAUTIONS

After the engine is started and all systems are functioning properly, the operator must follow all local safety rules to ensure safe machine operation.

WARNING

If any of the red warning lights come on or if any gauge reads in the red area during truck operation, a malfunction is indicated. Stop the truck as soon as possible and turn off the engine. Have the problem corrected before resuming truck operation.

CAUTION

The truck is equipped with “slip/slide” control. If this function becomes inoperative, operating the truck with stalled or free spinning wheel motors may cause serious damage to the wheel motors. If the truck does not begin to move within 10 seconds after depressing the throttle pedal with the directional control lever in a F or R position, release the throttle pedal and allow the wheels to regain traction before accelerating again.

- Always look to the rear before backing the truck. Watch for and obey the ground spotter's hand signals before making any reverse movements. Sound the warning horn (three blasts). The spotter should have a clear view of the entire area at the rear of the truck.
- Operate the truck only while properly seated with seat belt fastened. Keep hands and feet inside the cab while the truck is in operation.
- Do not allow unauthorized personnel to ride in or on the truck. Do not allow anyone to ride on the ladder of the truck.
- Check the gauges and instruments frequently during operation for proper readings.
- Observe all regulations pertaining to the job site's traffic pattern. Be alert to any unusual traffic pattern. Obey the spotter's signals.

- Match the truck speed to haul road conditions. Slow down the truck in congested areas. Keep a firm grip on the steering wheel at all times.
- Do not allow the engine to run at idle for extended periods of time.
- When the truck body is in the dump position, do not allow anyone beneath it unless the body-up retaining pin or cable is in place.



Do not use the brake lock or the secondary brake (if equipped) for parking. With the engine stopped, hydraulic pressure will bleed down, allowing the brakes to release.

- Check the brake lock performance periodically for safe loading and dump operation.
- Proceed slowly on rough terrain to avoid deep ruts or large obstacles. Avoid traveling close to soft edges or the edge of a fill area.
- Do not drive over unprotected power cables.
- Truck operation requires concentrated effort by the driver. Avoid distractions of any kind while operating the truck.

OPERATING ON THE HAUL ROAD

1. Always stay alert! If you are unfamiliar with the road, drive with extreme caution. The cab doors should remain closed at all times if the truck is in motion or unattended.
2. Obey all road signs. Operate the truck so it is under control at all times. Govern the truck speed by the road conditions, weather, and visibility. Report poor haul road conditions immediately. Muddy or icy roads, pot holes, or other obstructions can present hazards.
3. Initial propulsion with a loaded truck should begin from a level surface whenever possible. If there are circumstances where starting on a hill or grade cannot be avoided, refer to Starting On a Grade With a Loaded Truck on the next page.

4. When backing up the truck, give a back-up signal of three blasts on the horn. When starting forward, give two blasts on the horn. These signals must be given each time the truck is moved forward or backward.
5. Use extreme caution when approaching a haul road intersection. Maintain a safe distance from oncoming vehicles.
6. Maintain a safe distance when following another vehicle. Never follow another vehicle in the same lane closer than 15 m (50 ft). When driving on a downgrade, this distance should be at least 30 m (100 ft).
7. Do not stop or park on a haul road unless it is unavoidable. If you must stop, move the truck to a safe place, move the directional control lever to PARK and turn off the engine before leaving the cab. Chock the wheels securely and notify maintenance personnel for assistance.
8. Before driving up or down a grade, maintain a speed that will ensure safe driving and provide effective retarding under all conditions. Refer to Dynamic Retarding in Section N5, Operator Cab And Controls. Refer to the grade/speed chart in the operator's cab to determine the maximum safe truck speeds for descending various grades with a loaded truck.
9. When operating the truck in darkness, or when visibility is poor, do not move the truck unless all headlights, clearance lights, and tail lights are on. Do not back up the truck if the back-up horn or lights are inoperative. Always dim the headlights when meeting oncoming vehicles.
10. If the emergency steering light and/or low brake pressure warning light illuminate during operation, steer the truck immediately to a safe area away from other traffic, if possible, and stop the truck. Refer to Step 7.
11. Check the tires for proper inflation periodically during your shift. If the truck has been run on a "flat", or under-inflated tire, it must not be parked in a building until the tire cools.

STARTING ON A GRADE WITH A LOADED TRUCK

Initial propulsion with a loaded truck should begin from a level surface whenever possible. If the truck must be started on a hill or grade, use the following procedure:

1. Fully depress the service brake pedal. Do not use the retarder lever to hold the truck on the grade. With the service brakes fully applied, move the directional control lever to a drive position (FORWARD or REVERSE) and increase the engine rpm with the throttle pedal.
2. As the engine rpm approaches the maximum, and the operator senses the propulsion effort working against the brakes, release the brakes and let truck movement start. Completely release the service brake pedal. As the truck speed increases above 5 - 8 kph (3 - 5 mph), the propulsion system controller (PSC) will drop the propulsion if the retarder is still applied.

NOTE: Releasing and re-applying dynamic retarding during a hill start operation will result in loss of propulsion.

PASSING

1. Do not pass another truck on a hill or blind curve. Only use areas designated for passing.
2. Before passing, make sure that the road ahead is clear. If a disabled truck is blocking your lane, slow down and pass with extreme caution.

LOADING

1. Pull into the loading area with caution. Remain at a safe distance while the truck ahead is being loaded.
2. When approaching or leaving a loading area, watch out for other vehicles and personnel working in the area.
3. When pulling in under a loader or shovel, follow the spotter or shovel operator's signals. The truck operator may speed up loading operations by observing the location and loading cycle of the truck being loaded ahead, then follow a similar pattern.
4. During loading, the operator must stay in the cab with the engine running. Place the directional control lever in NEUTRAL and apply the brake lock.
5. When loaded, pull away from shovel as quickly as possible but with extreme caution.

Overload Speed Limit Function

NOTE: This feature can be enabled or disabled at the discretion of site management. The setting for payload trigger and speed limits are also adjustable by local site preference.

The overload speed limit function limits the speed of the truck if a payload limit setting has been exceeded. The overload signal becomes true (24 volts) when the PLMIII early estimate of payload exceeds the preset payload ton threshold.

The PLMIII early estimate of payload calculation occurs after 12 seconds of travel above 5 kph (3 mph) after being loaded. Once the overload signal becomes true, it stays true until the payload is dumped.

By default, the overload speed limit is disabled (option 0 in the configuration file). The configuration file can be set to:

Option 1 - Overload speed limit is imposed.

or

Option 2 - No propel restriction is imposed and logged. The truck cannot move under its own power when the overload signal becomes true.

The overload speed limit can be adjusted by modifying the configuration file. Use the Payload Service Software Lite Version (available for download at www.KomatsuAmerica.net) to set the overload threshold.

When option 1 is selected, and the overload signal is true, tractive effort is reduced to attempt to respect the speed limit setting. The retard effort will not be engaged to lower the speed of the truck. Once the truck speed matches that of the overload speed limit setting, limited propel efforts and retarding efforts will be used as necessary. This feature maintains the truck speed at the maximum of the overload speed limit setting.

DUMPING

Raising The Dump Body

1. Pull into the dump area with extreme caution. Make sure the area is clear of persons and obstructions, including overhead utility lines. Obey signals directed by the spotter, if present.
2. Avoid unstable areas. Stay a safe distance from the edge of the dump area.

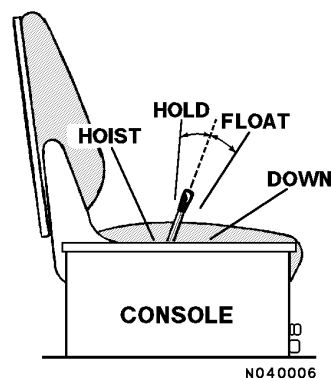
WARNING

To prevent tipping or rolling, position the truck on a solid, level surface before dumping. As the body raises, the truck center of gravity will move.

3. Carefully maneuver the truck into the dump position. When backing the truck into the dump position, use only the foot-operated brake pedal to stop and hold the truck. Do not rely on the wheel brake lock to stop the truck. This control is not modulated and applies the **rear service brakes only**.
4. When the truck is stopped and in the dump position, apply the brake lock and move the directional control lever to NEUTRAL.

WARNING

Dumping very large rocks (10% of payload or greater) or sticky material (loads that do not flow freely from the body) may allow the material to move too fast and cause the body to move rapidly and suddenly. This sudden movement may jolt the truck violently and cause possible injury to the operator and/or damage to the hoist cylinders, frame, and/or body hinge pins.



5. Pull the lever to the rear to actuate the hoist circuit. Releasing the lever anywhere during "hoist up" will hold the body at that position.
6. Raise the engine rpm to accelerate the hoist speed. If dumping very large rocks or sticky material as described in the WARNING above, slowly accelerate the engine to raise the body. When the material starts to move, release the hoist lever to the HOLD position. If the material does not continue moving and clear the body, repeat this procedure until the material has cleared the body.
7. Reduce the engine rpm as the last stage of the hoist cylinder begins to extend. Let the engine go to low idle as the last stage reaches half-extension.
8. Release the hoist lever as the last stage of the hoist cylinder reaches full extension.
9. After the material being dumped clears the body, lower the body to the frame.

NOTE: If the directional control lever is in REVERSE when the dump body is raised, reverse propel is inhibited. To deactivate, lower the dump body and move the directional control lever out of REVERSE.

Lowering The Dump Body (When dumping on flat ground)

It is very likely when dumping on flat ground that the dumped material will build up enough to prevent the body from lowering. In this case, the truck will have to be driven forward a short distance (just enough to clear the material) before the body can be lowered.

1. Move the directional control lever to FORWARD, release the brake lock, depress the override button, and drive just far enough forward for the body to clear the material.
2. Stop, move the directional control lever to NEUTRAL, and apply the brake lock.
3. Move the hoist lever forward to the DOWN position. Release the lever to place the hoist control valve in the FLOAT position, which allows the body to return to the frame.

If dumped material builds up at the rear of the body and the body cannot be lowered, then perform the following steps:

- a. Move the hoist lever back to the RAISE position to fully raise the dump body. Then release the hoist lever so it returns to the HOLD position.
- b. Move the directional control lever to FORWARD, release the brake lock, depress the override button, and drive forward to clear the material.
- c. Stop, move the directional control lever to NEUTRAL, apply the brake lock, and lower the body.

NOTE: Failure to hoist the body after making an unsuccessful attempt at lowering the body may result in the dump body suddenly lowering after the truck has pulled ahead of the material that was previously preventing the body from lowering.



Do not move the truck with the dump body raised except in an emergency. Failure to lower body before moving truck may cause damage to hoist cylinders, frame and/or body hinge pins.

4. With the body returned to frame, move the directional control lever to FORWARD, release the brake lock, and leave the dump area carefully.

Lowering The Dump Body (When dumping over a berm or into a crusher)

1. Move the hoist lever forward to the DOWN position. Release the lever to place the hoist control valve in the FLOAT position, which allows the body to return to the frame.

If dumped material builds up at the rear of the body and the body cannot be lowered, perform the following steps:

- a. Move hoist lever back to the HOIST position to fully raise the dump body. Release the hoist lever so it returns to the HOLD position.

NOTE: Do not drive forward if the tail of body will not clear the crusher wall in the fully raised position.

- b. Move the directional control lever to FORWARD, release the brake lock, depress the Override button, and drive forward to clear the material.
- c. Stop, move the directional control lever to NEUTRAL, apply the brake lock and lower the body.

NOTE: Failure to hoist the body after making an unsuccessful attempt at lowering the body may result in the dump body **suddenly lowering** after the truck has pulled ahead of the material that was previously preventing the body from lowering.



Do not move the truck with the dump body raised except in an emergency. Failure to lower the body before moving the truck may cause damage to the hoist cylinders, frame and/or body hinge pins.

2. With the body returned to the frame, move the directional control lever to FORWARD, release the brake lock, and leave the dump area carefully.

TOWING

⚠ WARNING

Before towing a truck, many factors must be carefully considered. Serious personal injury and/or significant property damage may result if important safety practices, procedures and preparation for moving heavy equipment are not observed.

A disabled machine may be towed after the following precautions have been taken.

- Do not tow the truck any faster than 8 kph (5 mph).
- Tow with a solid tow bar. DO NOT tow with a cable. Use a towing device with ample strength for the weight of this truck.
- Never tow a truck on a slope.
- Inspect towing components, such as tow bars and couplings, for any signs of damage. Never use damaged or worn components to tow a disabled vehicle.
- Keep a safe distance from the trucks and towing apparatus while towing a vehicle.
- When connecting a truck that is to be towed, do not allow anyone to go between the tow vehicle and the disabled vehicle.
- Set the coupling of the truck being towed in a straight line with the towing portion of the tow truck, and secure it in position.
- An operator is to remain in the cab of the towed vehicle at all times during the towing procedure.

Special Wiring Harness

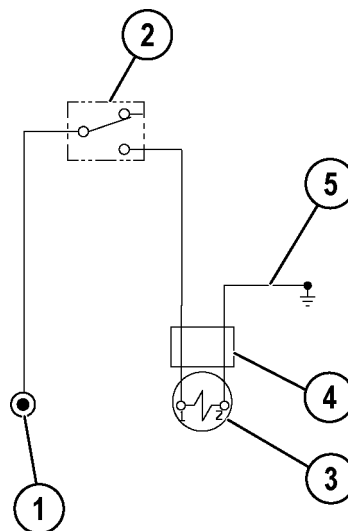
Before towing, a special wiring harness must be made in order to release the parking brake. The harness will require approximately 9 meters (30 ft) of #14 wire, one ON/OFF switch (capable of carrying 2 amps of current) and two ring terminals.

Refer to Figure 3-18 for a diagram of the special wiring harness circuit.

- Using a ring terminal, one end of the harness must connect to 24VDC terminal (1, Figure 3-18) in the auxiliary control cabinet to supply 24V to the parking brake solenoid.
- Switch (2) must be connected to the other end of the harness so the operator can use the switch while seated in the cab.
- The harness must be fitted with connector (4) to allow it to be plugged into parking brake solenoid (3) inside the brake cabinet. Ground wire (5) leading from this connector must be long enough to connect to a ground block using the second ring terminal.

Connector (4) consists of:

- one housing (PB8538)
- two sockets (08191-05430)
- one wedge (PB8540)



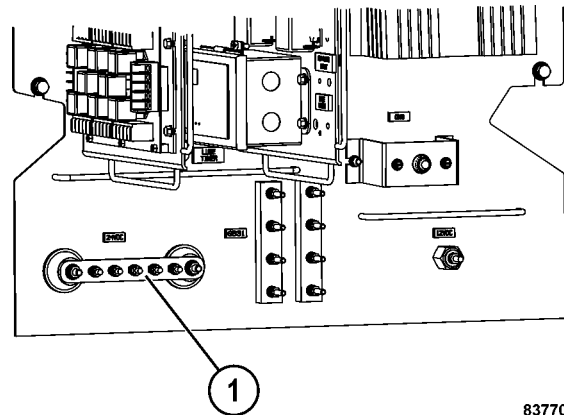
82834

FIGURE 3-18. PARKING BRAKE HARNESS CONNECTION DIAGRAM

- | | |
|---------------------------|----------------|
| 1. 24VDC Terminal | 4. Connector |
| 2. Switch (ON/OFF) | 5. Ground Wire |
| 3. Parking Brake Solenoid | |

1. Turn off the engine.
2. Block the wheels on the disabled truck to prevent movement while preparing the truck for towing and while attaching the tow bar.
3. Ensure that the towing vehicle has adequate capacity to both move and stop the towed truck under all conditions.
4. Ensure that the tow bar has adequate strength (approximately 1.5 times the empty vehicle weight of truck being towed). Install the tow bar between the two vehicles.
5. If necessary, install quick disconnect fittings to the bleeddown manifold to allow the hydraulic system to be operational. Install hydraulic connections for steering/braking between the tow vehicle and the disabled vehicle. An auxiliary power unit can also be used.
6. After the hydraulic connections are made, check the disabled vehicle braking and steering systems for normal operation.
7. If the truck is loaded, dump the entire load. Never pull or tow a loaded truck. Refer to "Disabled Truck Dumping Procedure".
8. The parking brakes must be released before towing. To release the parking brakes, use the following steps to install the special wiring harness.
 - a. Ensure that the ON/OFF switch is in the OFF position.
 - b. Connect one lead of the special wiring harness to a 24VDC terminal on bus bar (1, Figure 3-19) on the left side wall of the auxiliary control cabinet for the 24V supply power.
 - c. Disconnect the truck wiring harness from parking brake solenoid (2, Figure 3-20). Connect the special wiring harness to the parking brake solenoid. Connect the ground wire from the connector to a ground block in the right side of the brake cabinet.
 - d. Open the window and place the other end of the special wiring harness inside the cab so the operator can control the ON/OFF switch.
9. When ready to tow the disabled truck, remove the blocking from the wheels.
10. The operator in the disabled truck should now move the ON/OFF switch to the ON position. This will release the parking brakes.
11. Tow the disabled truck. Sudden movement may cause tow bar failure. Smooth, gradual truck movement is preferred. Do not tow the truck any faster than 8 kph (5 mph).

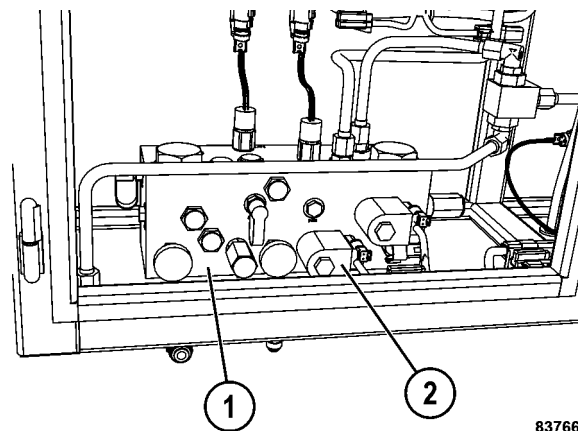
12. Minimize the tow angle at all times. **Never exceed 30 degrees.** The towed truck must be steered in the direction of the tow bar.
13. When the desired location has been reached, the operator in the towed vehicle must apply the service brakes, then move the switch to the OFF position. This will apply the parking brakes.
14. Block the wheels to prevent movement.
15. Turn off the engine in the towing vehicle.
16. If installed, disconnect the hydraulic hoses at the quick disconnect fittings on the bleeddown manifold.
17. Disconnect the special wiring harness from the truck. Connect the truck wiring harness to the parking brake solenoid.
18. Disconnect the tow bar.



83770

FIGURE 3-19. AUXILIARY CONTROL CABINET

1. 24VDC Bus Bar



83766

FIGURE 3-20. BRAKE CABINET

1. Brake Manifold

2. Parking Brake Solenoid (SV2)

SAFE PARKING PROCEDURES

Continue to use safety precautions when preparing for parking and engine shutdown.

If the equipment is being used in consecutive shifts, any questionable truck performance the operator may have noticed must be checked by maintenance personnel before the truck is released to another operator.

1. Park the truck on level ground, if possible. If the truck must be parked on a grade, position the truck at a right angle to the grade.
2. The directional control lever must be in PARK and chocks must be placed in front of and behind the wheels so the truck cannot roll. Each truck should be parked at a reasonable distance from one another.



Do not activate the wheel lock feature when the parking brake is activated. Bleed down of hydraulic pressure may occur, causing the truck to roll away.

3. Haul roads are not safe parking areas. In an emergency, pick the safest spot that is most visible to other machines in the area. If the truck becomes disabled where traffic is heavy, mark the truck with warning flags in daylight or flares at night.

NORMAL ENGINE SHUTDOWN PROCEDURE

Follow this procedure at each engine shutdown.

1. Stop the truck out of the way of other traffic on a level surface. Make sure that there are no overhead power lines or other obstructions in case the dump body must be raised.
2. Reduce the engine speed to idle.
3. Place the directional control lever in PARK. Make sure that the parking brake applied indicator light is illuminated.
4. Place the rest switch in the ON position to put the AC drive system in the rest mode. Make sure that the rest mode indicator light is illuminated.
5. Turn the key switch to the OFF position. A shutdown timing sequence will be activated automatically to allow the engine to cool down before it is turned off. The timing sequence may last up to three minutes.
If the engine does not shutdown with the key switch after the timing sequence is done, use the engine shutdown switch on the operator cab center console. Pull this switch up until the engine stops.
6. With the key switch OFF and engine stopped, wait at least 90 seconds. Make sure that the steering circuit is completely depressurized by turning the steering wheel back and forth several times. The front wheels should not turn when the hydraulic pressure is relieved. If the front wheels can still be turned, notify maintenance personnel.
7. Make sure that the link voltage light in the control cabinet is off. Notify maintenance personnel if a light remains illuminated for longer than five minutes after the engine is shut down.
8. Close and lock all windows. Remove the key from the key switch and lock the cab to prevent unauthorized truck operation. Dismount the truck properly. Put wheel chocks in place.

SUDDEN LOSS OF ENGINE POWER

If the engine suddenly stops, there is enough hydraulic pressure stored in the brake and steering accumulators to allow the operation of the steering and brake functions. However, this oil supply is limited, so it is important to stop the truck as quickly and safely as possible after the loss of engine power.

If the brake supply pressure drops to a pre-determined level, the low brake oil pressure warning light will illuminate and sonalarm will sound. If the brake pressure continues to decrease, the auto-apply feature will activate and the service brakes will apply automatically to stop the truck.

1. Bring the truck to a safe stop as quickly as possible by using the foot pedal to apply the service brakes. If possible, safely steer the truck to the side of the road while braking.

WARNING

Dynamic retarding will not be available. Do not use the service brakes for continuous retarding purposes.

2. Move the directional control lever to PARK as soon as the truck has stopped moving.
3. Turn the key switch to the OFF position and notify maintenance personnel immediately.
4. If traffic is heavy near the disabled machine, mark the truck with warning flags during daylight hours or use flares at night. Adhere to local regulations.

FUEL DEPLETION

The high pressure injection (HPI) fuel system uses fuel to adjust fuel delivery timing by creating a hydraulic link between the upper plunger and the timing plunger. Metered fuel is also used for lubricating the injector plunger and barrel. The maximum demand for metered fuel is required during high speed / low load conditions.

CAUTION

Operating the truck to fuel depletion forces the injector train into a no-follow condition. No fuel flow between the plungers may cause damage to the injectors and the overhead due to adhesive wear, resulting in costly repairs and unnecessary downtime.

WARNING

Allowing the Komatsu truck to operate until fuel depletion can lead to unsafe operating conditions possibly resulting in an uncontrollable vehicle and/or personal injury.

DISABLED TRUCK DUMPING PROCEDURE

It may be necessary to dump a load from the body of a truck if the hoist system is inoperable. The following instructions describe the use of a “good” truck to provide the hydraulic power required to raise the body of the “disabled” truck to dump the load.

In the following example, Figure 3-21 illustrates a typical hookup from the good truck. The disabled truck may be the same truck model or a different Komatsu truck model.

Hookup

1. Ensure that there is an adequate, clear area to dump the load from the body. When the good truck is in position, turn off the engine and allow the hydraulic system to bleed down completely before connecting the hoses.
2. With the good truck parked as close as possible to the disabled truck, attach a hose from **power up** quick disconnect (4, Figure 3-21) to the **power down** circuit of the disabled truck. The hose must be rated to withstand 17 000 kPa (2,500 psi) or greater pressure.

NOTE: The power down circuit will use a smaller diameter hose (tube) than the power up circuit.

3. Connect another hose from **power down** quick disconnect (3) to the **power up** circuit of the disabled truck.

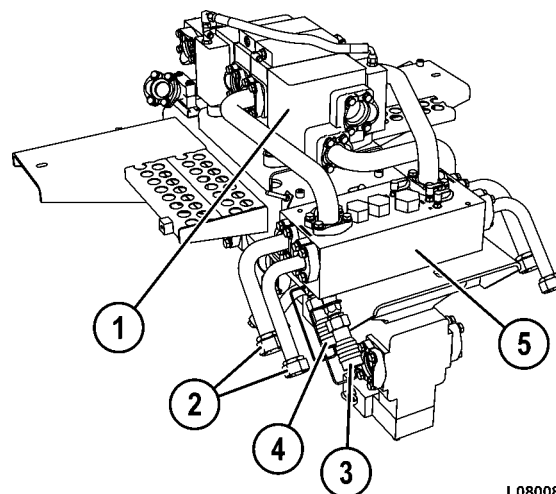
*NOTE: If both trucks are the same model, the hoses will be installed at the quick disconnects shown in Figure 3-21 and **will be crossed** when connected.*

Raising the Body

1. On the disabled truck, move the hoist control lever to power up and then release it to place the hoist pilot valve in the HOLD position. Leave it in this position during entire procedure.
2. Start the engine on the good truck. Place the hoist control lever in the POWER DOWN position and increase the engine rpm to high idle to dump the load from the disabled truck body. If the body of the disabled truck fails to raise, increase the good truck power down relief pressure as follows:
 - a. Turn off the engine and allow the hydraulic system to bleed down completely.
 - b. Remove the cap from the hoist pilot valve located in the hydraulic brake cabinet behind the cab. While counting the number of turns, slowly turn the relief valve adjustment screw clockwise until it bottoms. Repeat Step 2.

Lowering the Body

1. Place the hoist lever of the good truck in FLOAT to lower the body. If necessary, momentarily place the hoist control in POWER UP until the body is able to descend in FLOAT. Do not accelerate the engine.
2. After the body is lowered, shut down the truck, bleed the hydraulic system, and disconnect the hoses.
3. Reduce the power down relief valve pressure on the good truck to normal by turning the hoist pilot valve relief valve adjustment screw counterclockwise the same number of turns as required in Step 2b of “Raising the Body.”
4. Check the power down relief pressure. Refer to Section L10, Hydraulic Check-Out Procedure.
5. Check the hydraulic tank oil level.



L080087

FIGURE 3-21. PUMP MODULE, HOSE HOOKUP

1. Hoist Valve
2. Tubes to LH Hoist Cylinder
3. Power Down Quick Disconnect
4. Power Up Quick Disconnect
5. Overcenter Manifold

NOTES

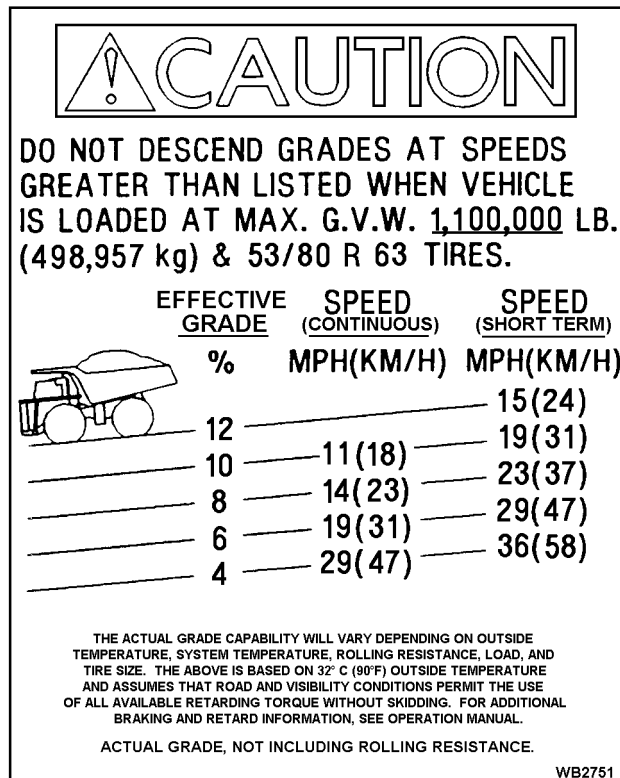
WARNINGS AND CAUTIONS

The following pages give an explanation of the warning, caution, and service instruction plates and decals attached to the truck. The plates and decals listed here are typical of this Komatsu model, but because of customer options, individual trucks may have plates and decals that are different from those shown here.

The plates and decals must be kept clean and legible. If any decal or plate becomes damaged or illegible, replace it with a new one.

GRADE/SPEED CHART

A grade/speed retard chart is located on the left front post of the operator's cab. It provides the recommended maximum speeds to be used when descending various grades with a loaded truck. Always refer to the decal in operator's cab. This decal may change with optional truck equipment such as wheel motor drive train ratios, retarding grids, tire sizes, etc.



KEY SWITCH

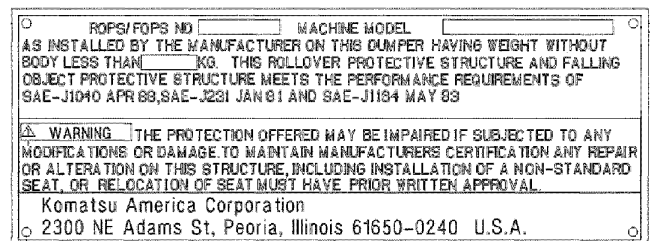
A warning decal is below the key switch located to the right of the steering column on the instrument panel. The warning stresses the importance of reading the operator's manual before operation.



ROPS/FOPS

A plate attached to the right rear corner of the cab states that the Rollover Protective Structure (ROPS) and Falling Object Protective Structure (FOPS) meet various SAE performance requirements.

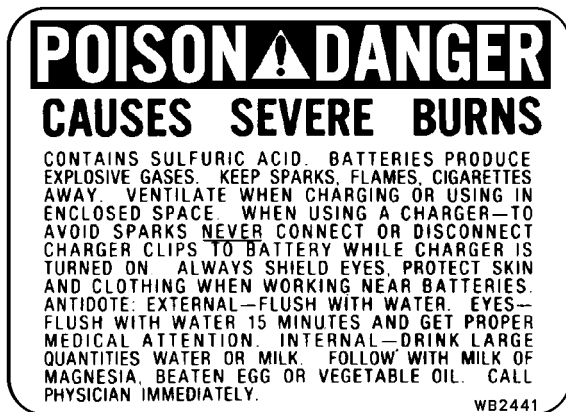
The plate also contains a WARNING about making modifications or repairs to the structure. Unauthorized modifications or repairs will void the certification.



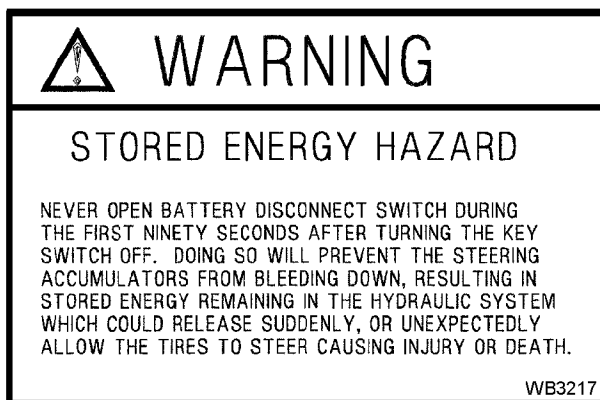
BATTERIES

A danger decal is located on the front exterior of the battery box. It stresses the need to keep from making any sparks near the battery. When another battery or 24VDC power source is used for auxiliary power, all switches must be OFF before making any connections. When connecting auxiliary power cables, positively maintain correct polarity. Connect the positive (+) posts together and then connect the negative (-) lead of the auxiliary power cable to a good frame ground. **Do not connect to the negative posts of the truck battery or near the battery box.** This hookup completes the circuit but minimizes the danger of sparks near the batteries.

Sulfuric acid is corrosive and toxic. Use proper safety gear, goggles, rubber gloves and a rubber apron when handling and servicing batteries. If sulfuric acid is swallowed, gets into your eyes or contacts your skin, get proper medical help immediately.



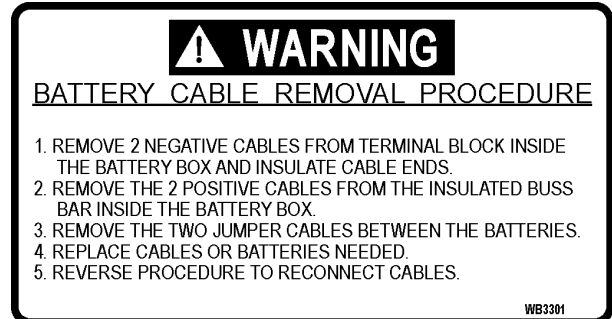
This decal is located on the front exterior of the battery box. It warns about the potential hazard of stored energy remaining in the hydraulic system if the battery disconnect switch is opened before the steering accumulators have bled down completely.



This decal is placed on the battery box near the battery disconnect switches to indicate that the battery system (24VDC) is a negative (-) ground system.



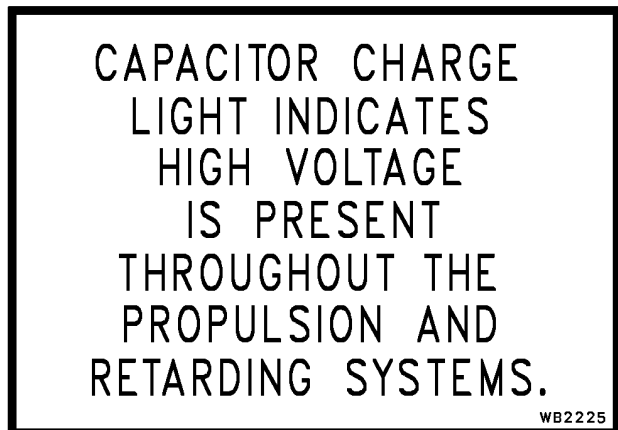
A decal that outlines the correct battery cable removal procedure is located on the front exterior of the battery box.



CAPACITORS

This decal is located on the front of the contactor box and on the inside of the left control cabinet door.

When any of the indicator lights are on, high voltage is present throughout the propulsion and retarding system. Extreme care should be exercised.



CRUSHING HAZARD

Warning decals are mounted on the frame in front of both front tires and to the rear of both front tires. They warn that the clearances change when the truck is steered and could cause serious injury.



Warning decals are attached to both the hydraulic tank and fuel tank to alert technicians not to work on the truck with the body in the raised position unless the body-up retention device (pins or cable) is in position.

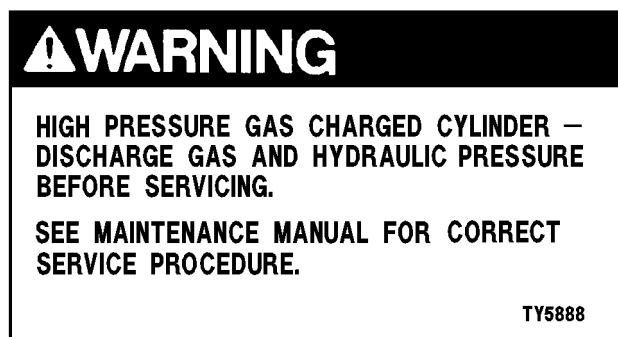


CYLINDER PRESSURE

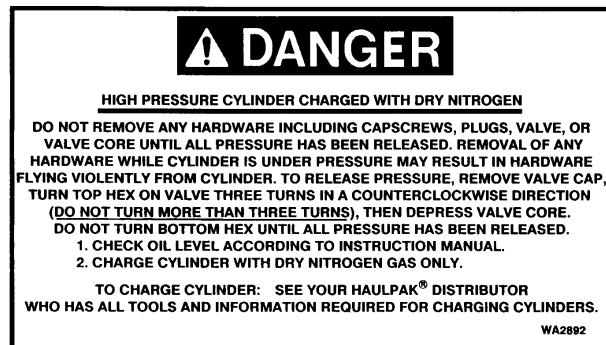
These danger decals are mounted on the outside of each frame rail to remind technicians to read the warning labels attached to the side of each of the accumulators (see below) before releasing internal nitrogen pressure or disconnecting any hydraulic lines or hardware. There are similar decals mounted on top of each of the accumulators (both steering and brake) with the same danger message.



These warning decals are mounted on the side of each of the accumulators (both steering and brake) to alert technicians to discharge all gas and hydraulic pressure, and to read the maintenance/service manual before performing any service.



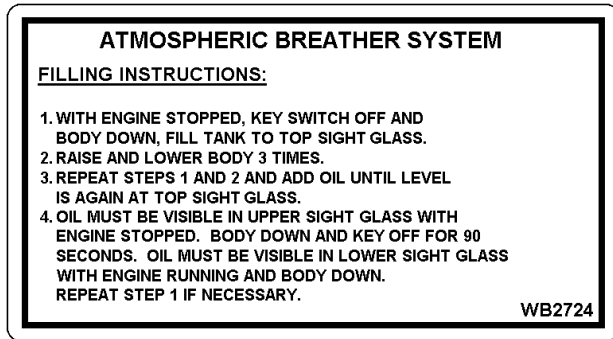
This danger decal is attached to all four suspensions. It contains instructions for releasing internal pressure before disconnecting any hardware. Serious injury can occur if these directions are not followed.



FILLING THE HYDRAULIC TANK

A decal mounted on the side of the hydraulic tank provides instructions for filling the hydraulic tank.

To lessen the chances of system contamination, keep the system open to the atmosphere only as long as absolutely necessary. Service the tank with clean Type C-4 hydraulic oil. All oil that is being put into the hydraulic tank should be filtered through 3 micron filters.



A caution decal is attached below the hydraulic tank oil level sight gauge. Check the oil level with the body down, engine stopped, and key switch OFF. Add oil per filling instructions if the oil level is below the top of the sight glass.

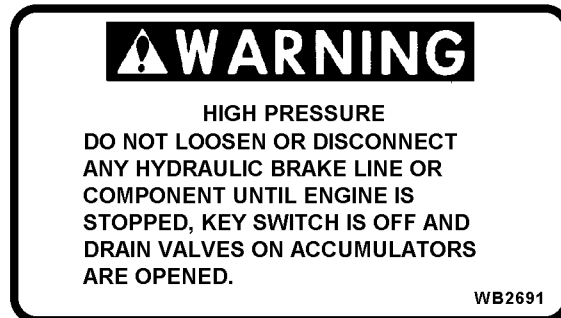


HYDRAULIC OIL PRESSURE

A warning plate is attached to the hydraulic tank to warn technicians that high pressure hydraulic oil is present during operation. When it is necessary to open the hydraulic system, be sure the engine is stopped and the key switch is OFF to bleed down hydraulic pressure. There is always a chance of residual pressure being present. Open fittings slowly to allow all pressure to bleed off before removing any connections.

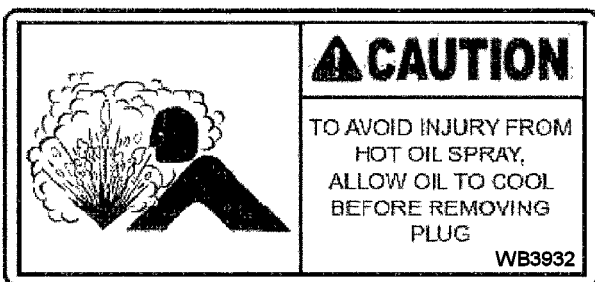


Two warning decals are located on the doors of the hydraulic brake cabinet to warn technicians not to disconnect any hydraulic brake lines until the machine has been properly shut down.



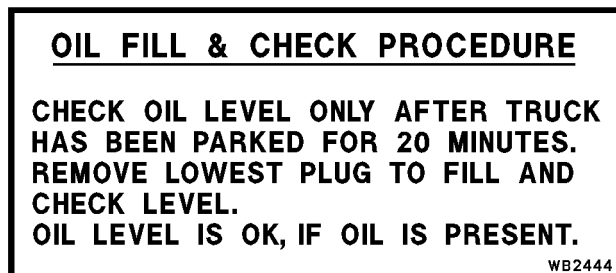
HOT OIL SPRAY

This decal is placed on both front wheel hubs to warn about hot oil inside the wheel hubs. Remove the oil level plug carefully to avoid injury.



WHEEL MOTOR OIL LEVEL

A wheel motor oil level decal is attached to the gear cover on both electric wheel motors. This decal stresses that the truck must be parked for 20 minutes prior to checking the oil level in order to get an accurate reading.



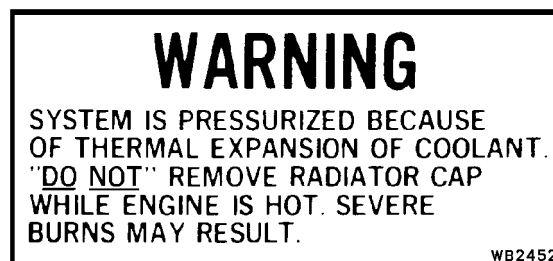
HOT EXHAUST

A caution decal is also attached to the door of the rear axle hatch and around the retarding grid cabinet to alert personnel that hot exhaust air is present and may cause injury.



RADIATOR

A warning decal is mounted on top of the radiator surge tank cover near the radiator cap. The engine cooling system is pressurized. **Always turn the key switch OFF and allow the engine to cool before removing the radiator cap.** Unless the pressure is first released, removing the radiator cap after the engine has been running for a time will result in hot coolant being expelled from the radiator. **Serious scalding and burning may result.**



EMERGENCY LADDER

This decal is located by the emergency ladder on both sides of the truck.



ACCUMULATOR DRAIN VALVES

Warning decals are applied to both brake accumulators located inside the hydraulic brake cabinet behind the operator cab. These decals remind servicing technicians to close the accumulator drain valves after they have been opened to bleed the brake pressure. It further warns not to overtighten the drain valves in order to prevent damage to the valve seat(s).



EMERGENCY DUMP PROCEDURE

A decal which provides the operator or technician with the emergency procedure for dumping a loaded, disabled truck by using a functional truck for hydraulic power is located on the frame near the left hoist cylinder.

Refer to Section L in the shop manual for additional information about using this procedure.

EMERGENCY DUMP PROCEDURE

1. CONNECT A HYDRAULIC POWER SUPPLY CAPABLE OF 2500 PSI (17.2 MPa) WHICH HAS A RESERVE CAPACITY EXCEEDING 80 GAL. (303 l) TO THE QUICK DISCONNECTS ON L.H. HOIST CYLINDER.
2. PLACE HOIST VALVE IN HOLD POSITION.
3. DUMP LOAD AND LOWER BODY USING CONTROL VALVE ON HYDRAULIC POWER SUPPLY UNIT.

WB2246

EMERGENCY TOWING PROCEDURE

A decal which provides the operator or technician with the emergency procedure for towing a disabled truck is located on the left frame rail near the steering accumulators.

WARNING

EMERGENCY TOWING PROCEDURE

1. ENGINE MUST BE STOPPED AND ACCUMULATOR(S) DISCHARGED.
2. EXTERNAL SUPPLY MUST BE ABLE TO MAINTAIN 3000 PSI AND HAVE A MIN. CAPACITY OF 20 GAL.
3. CONNECT EXTERNAL SUPPLY TO THE 'QD' SUPPLY AND 'QD' RETURN PORTS ON THE BLEEDOWN MANIFOLD.
4. CHECK OPERATION OF STEERING AND BRAKES.
5. PROCEED WITH TOWING OPERATION.

WB2443

WELDING

This caution decal is placed near the battery disconnect switches to alert service technicians to always disconnect the battery charging alternator lead wire before making welding repairs.

In addition, always isolate electronic control components and disconnect the positive and negative battery cables of the vehicle. Failure to do so may seriously damage the battery and electrical equipment.

CAUTION

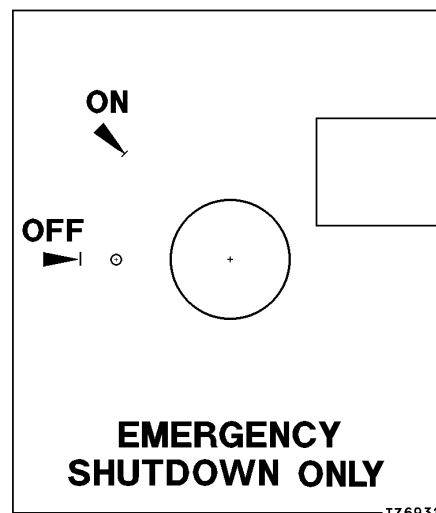
**PRIOR TO WELDING ON TRUCK
DISCONNECT LEAD WIRE ON
BATTERY CHARGING ALTERNATOR**

WB2442

Always fasten the welding machine ground (-) lead to the piece being welded. **The grounding clamp must be attached as near as possible to the weld area.** Never allow welding current to pass through ball bearings, roller bearings, suspensions, or hydraulic cylinders. Always avoid laying welding cables over or near the vehicle electrical harnesses. Welding voltage could be induced into the electrical harness and possibly cause damage to components.

EMERGENCY SHUTDOWN

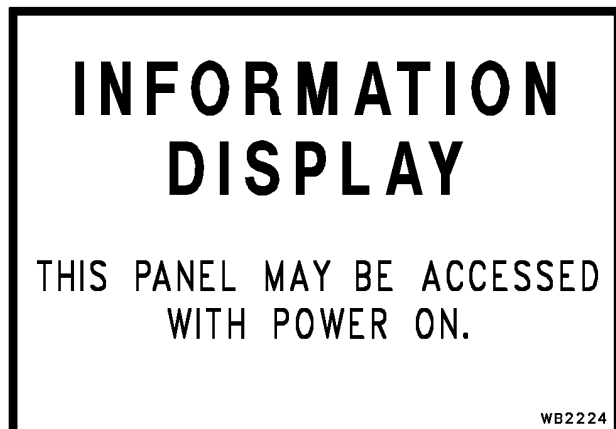
This decal is placed on the ground level engine shutdown switch, which may be mounted on either side of the front bumper. It specifies that this switch is for emergency shutdown only.



TZ6931

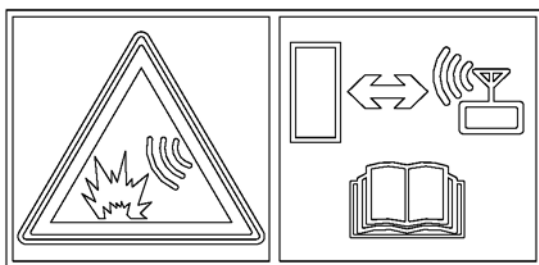
INFORMATION DISPLAY

This information decal is placed on the left door of the control cabinet.



WIRELESS SIGNALS

Wireless signals from the truck's VHMS system can interfere with other wireless signals in the area. This interference can cause a malfunction in a blast zone resulting in an unintended detonation. Know the locations of blast zones in the area and keep a safe distance to avoid unintentional blasts.



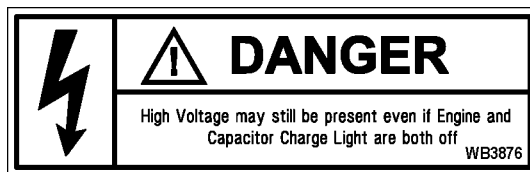
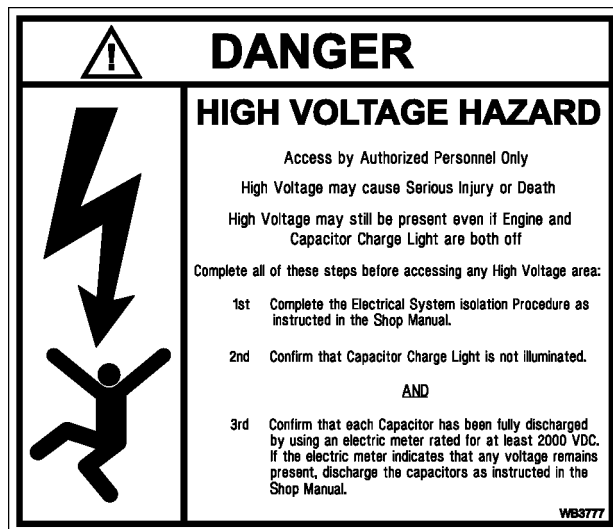
09845 - 00480

HIGH VOLTAGE

A high voltage danger decal is attached to the door of the rear hatch cover. **High voltage may be present!** Only authorized personnel should access this rear housing.



These danger decals are mounted on all the AC drive control housings and cabinets. **High voltage may be present!** Only authorized personnel should access this component.

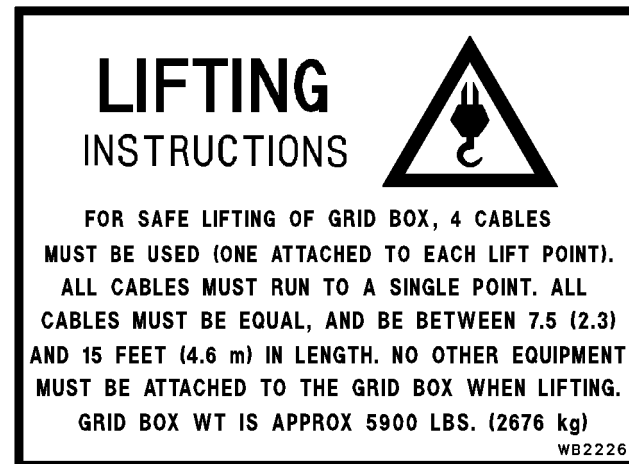
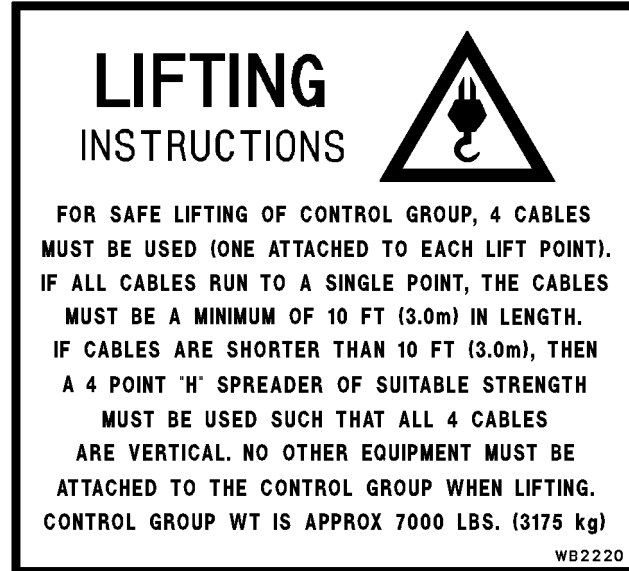
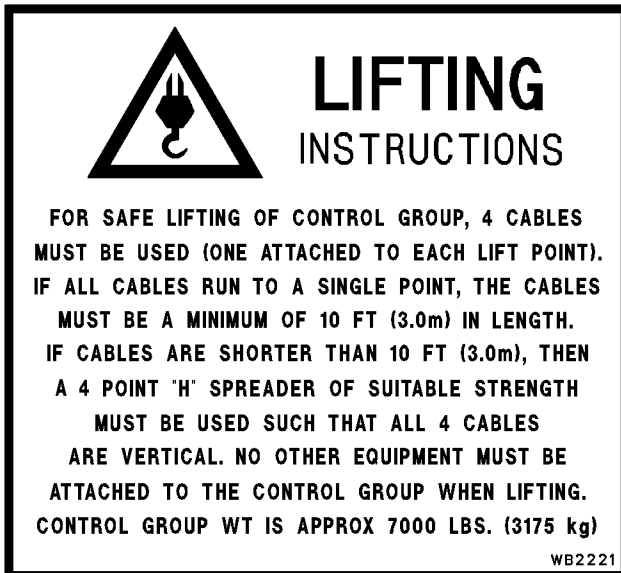
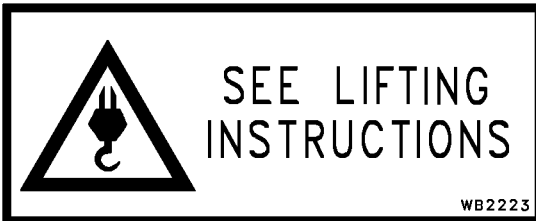
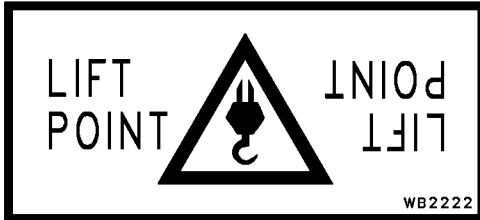


LIFTING INSTRUCTIONS

This page illustrates a variety of decals which are placed on deck mounted cabinets, housings, and structures that must be lifted in a specific manner and from specific points in order to safely move them.

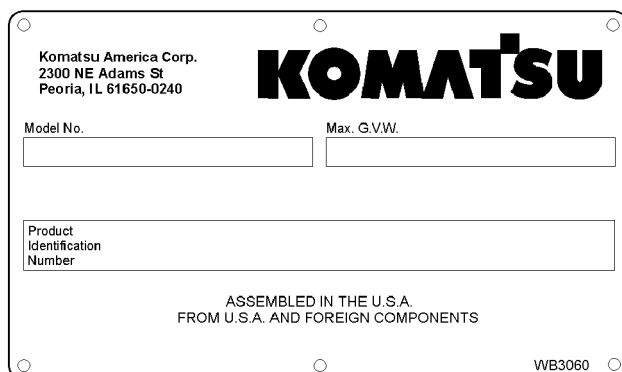
If any of these decals is damaged or defaced so it is no longer legible, it should be replaced immediately.

Maintenance personnel must follow these lifting instructions.



PRODUCT IDENTIFICATION PLATE

A product identification plate is located on the frame in front of the right side front wheel. It shows the Truck Model Number, Maximum GVW, and Product Identification Number (PIN).



The image shows a blank product identification plate template. It features the Komatsu logo at the top center. To the left of the logo is the text: "Komatsu America Corp. 2300 NE Adams St Peoria, IL 61650-0240". Below the logo are two input fields: "Model No." and "Max. G.V.W.". Below these fields is a larger input field labeled "Product Identification Number". At the bottom center, it says "ASSEMBLED IN THE U.S.A. FROM U.S.A. AND FOREIGN COMPONENTS". In the bottom right corner, the code "WB3060" is printed.

The Product Identification Number (PIN) consists of a total of 19 characters. The first and last characters are tamper preventive symbols (*). The remaining 17 alpha/numeric characters are used to identify the following five characteristics of the machine:

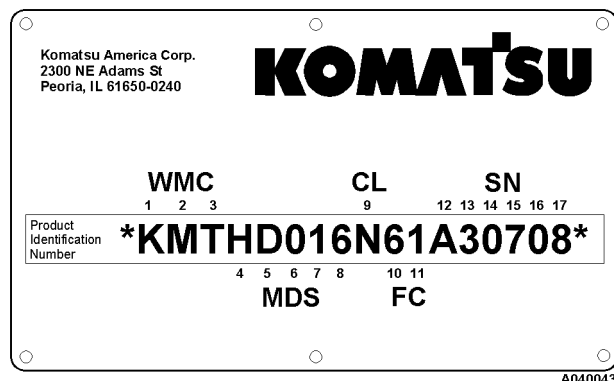
WMC - Character positions 1, 2 and 3 identify the Worldwide Manufacturer Code (WMC). The WMC designates the manufacturer of the product. Komatsu brand products are identified with the letters KMT.

MDS - Character positions 4, 5, 6, 7 and 8 identify the Machine Descriptor Section (MDS). The MDS code identifies general information regarding machine specifications. The MDS is a code for the machine type and model.

CL - Character position 9 identifies the Check Letter (CL). The CL is used to verify the accuracy of the individual PIN.

FC - Character positions 10 and 11 identify the Factory Code (FC). The FC identifies the Komatsu factory in charge of claims for the product. The FC for electric drive trucks is 61.

SN - Character positions 12, 13, 14, 15, 16 and 17 identify the Serial Number (SN). The SN is a unique sequential number.



The image shows a filled product identification plate example. It features the Komatsu logo at the top center. To the left of the logo is the text: "Komatsu America Corp. 2300 NE Adams St Peoria, IL 61650-0240". Below the logo are two input fields: "Model No." and "Max. G.V.W.". Below these fields is a larger input field labeled "Product Identification Number". The PIN is filled in as: *KMT HD016N61A30708*. Above the PIN, the characters are grouped into five sections: WMC (1, 2, 3), CL (9), SN (12, 13, 14, 15, 16, 17), MDS (4, 5, 6, 7, 8), and FC (10, 11). At the bottom center, it says "ASSEMBLED IN THE U.S.A. FROM U.S.A. AND FOREIGN COMPONENTS". In the bottom right corner, the code "A040043" is printed.

LUBRICATION CHART

The lubrication chart is mounted on the right hand side of the radiator grille structure. Refer to Section P2, Lubrication and Service, for more complete lubrication instructions.

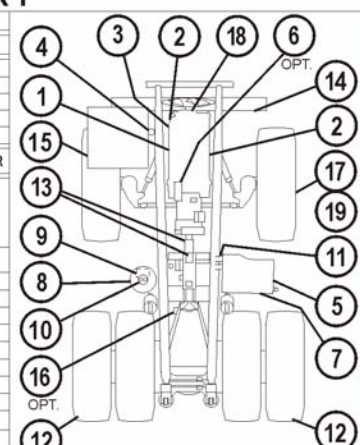
LUBRICATION CHART

LUBRICATION SPECIFICATIONS										
LUBE KEY	TYPE LUBRICANT									
A	ENGINE OIL - - - - SEE ENGINE MANUAL									
B	HYDRAULIC OIL - - - - SAE 10W C-4 - - - - AUXILIARY HEATERS REQUIRED BELOW -10°F (-23°C)									
C	HEAVY-DUTY SYNTHETIC GEAR OIL - - - - SEE DRIVE SYSTEM MANUAL									
D	MULTI-PURPOSE EXTREME PRESSURE GREASE - - - - NLGI NO.2 (5% MIN. MOLY-DISULFIDE)									
E	MULTI-PURPOSE GEAR OIL - - - - SAE 80W-90									

SYM	DESCRIPTION	PTS	LUBE KEY	10 HR	50 HR	100 HR	250 HR	500 HR	1000 HR	2000 HR	2500 HR
1	CRANKCASE OIL LEVEL	1	A	CHECK							
2	ENGINE LUBE OIL FILTERS										
3	FUEL FILTER										
4	FUEL SEPARATOR (DAVCO)										
5	FUEL TANK	1					DRAIN H ₂ O & SEDIMENT				
6	GE PREFILTER BLOWER	1	D				GREASE				
7	FUEL TANK BREATHER	1							CLEAN		
8	HYDRAULIC OIL LEVEL	1	B	CHECK					*CHANGE		
9	HYDRAULIC STRAINER	2							CLEAN		
10	HYDRAULIC TANK BREATHER	2							CHANGE		
11	HYDRAULIC FILTERS	3			**	**	**		CHANGE		
12	MOTORIZED WHEEL OIL LEVEL	2	C								
13	HYD. PUMP DRIVE SHAFT	2	D				GREASE				
14	CHASSIS LUBE LEVEL	1	D				GREASE				
15	SEAT SLIDES & STEER SHAFT	4	D						GREASE		
16	WHEEL MOTOR BLOWER	2	D				GREASE				
17	FRONT WHEEL BEARINGS ***	2	E				CHECK				CHANGE
18	FRONT TRUNION	1	D						GREASE		
19	MAGNETIC PLUG										
	FRONT WHL COVER ****	2					CHECK				

SEE ENGINE MANUAL

SEE DRIVE SYSTEM MANUAL



630E, 730E, 830E, & 930E

* 1000 HR INTERVAL CAN BE EXTENDED TO 2500 HR PROVIDED OIL SAMPLING AND ANALYSIS IS CONDUCTED EVERY 250 HR.

** ONE-TIME CHANGE AT 50, 100 AND 250 HR.

*** NOT APPLICABLE FOR 930E

**** APPLICABLE FOR 930E ONLY

WB2790

TORQUE TABLES AND CONVERSION CHARTS

This manual provides dual dimensioning for many specifications. Metric units are specified first, with U.S. standard units in parentheses. When torque values are not specified in the assembly instructions contained in this manual, use the standard torque value for the hardware being used. Standard value torque tables are contained in this chapter for metric and SAE hardware.

NOTE: This truck is assembled with both metric and SAE (U.S.) hardware. Reference the correct table when determining the proper torque value.



References throughout the manual to standard torques or other standard values will be to one of the following tables. Do not use standard values to replace specific torque values in assembly instructions.

For values not shown in any of the charts or tables, standard conversion factors for most commonly used measurements are provided in the following tables.

INDEX OF TABLES

TABLE	Page
TABLE 1 Standard Tightening Torque For Metric Cap screws and Nuts	12-2
TABLE 2 Standard Tightening Torque For SAE Grade 5 & 8 Cap Screws and Nuts	12-3
TABLE 3 Standard Tightening Torque 12-Pt, Grade 9, Cap Screws (SAE)	12-4
TABLE 4 Tightening Torque for T-Bolt Type Hose Clamps.	12-5
TABLE 5 Standard Tightening Torque For Split Flange Clamp Bolts	12-5
TABLE 6 Tightening Torque For Flared Tube And Hose Fittings	12-5
TABLE 7 JIC 37° Swivel Nuts Torque Chart (SAE)	12-6
TABLE 8 Pipe Thread Torque Chart (SAE)	12-6
TABLE 9 O-Ring Boss Torque Chart (SAE)	12-7
TABLE 10 O-Ring Face Seal Torque Chart (SAE)	12-7
TABLE 11 Common Conversions Multipliers - Metric -to- English	12-8
TABLE 12 Common Conversions Multipliers - English -to- Metric	12-9
TABLE 13 Temperature Conversions.	12-10

EFFECT OF SPECIAL LUBRICANTS On Fasteners and Standard Torque Values

The Komatsu engineering department does not recommend the use of special friction-reducing lubricants, such as Copper Coat, Never-Seez®, and other similar products, on the threads of standard fasteners where standard torque values are applied. The use of special friction-reducing lubricants will significantly alter the clamping force being applied to fasteners during the tightening process.

If special friction-reducing lubricants are used with the standard torque values listed in this chapter, excessive stress and possible breakage of the fasteners may result.

Where the torque tables specify “Lubricated Threads” for the standard torque values listed, these standard torque values are to be used with simple lithium base chassis grease (multi-purpose EP NLGI) or a rust-preventive grease (see below) on the threads and seats unless specified otherwise.

NOTE: Ensure the threads of fasteners and tapped holes are free of burrs and other imperfections before assembling.

Suggested* Sources for Rust Preventive Grease:

- American Anti-Rust Grease #3-X from Standard Oil Company (also American Oil Co.)
- Gulf NoRust #3 from Gulf Oil Company.
- Mobilarma 355, Product No. 66705 from Mobil Oil Corporation.
- Rust Ban 326 from Humble Oil Company.
- Rustolene B Grease from Sinclair Oil Co.
- Rust Preventive Grease - Code 312 from the Southwest Grease and Oil Company.

NOTE: This list represents the current engineering approved sources for use in Komatsu manufacture. It is not exclusive. Other products may meet the same specifications of this list.

STANDARD TIGHTENING TORQUES For Class 10.9 Cap screws & Class 10 Nuts

The following specifications apply to required assembly torques for all metric class 10.9 finished hexagon head cap screws and class 10 nuts.

- Cap screw threads and seats shall not be lubricated when assembled. These specifications are based on all cap screws, nuts, and hardened washers being phosphate and oil coated.

If zinc-plated hardware is used, each piece must be lubricated with simple lithium base chassis grease (multi-purpose EP NLGI) or a rust preventive grease (see list, this page) to achieve the same clamping forces provided below.

- Torques are calculated to give a clamping force of approximately 75% of proof load.
- The maximum torque tolerance shall be within $\pm 10\%$ of the torque value shown.
- In the following table under “Cap Screw Size”, the first number represents the shank diameter (mm). The second number represents threads per millimeter.

Example: M20 x 2.25

M20 = shank diameter (20 mm)

2.25 = thread pitch in millimeter



Cap Screw Size	Torque N•m	Torque ft lb	Torque kg•m
M6 x1	12	9	1.22
M8 x 1.25	30	22	3.06
M10 x 1.5	55	40	5.61
M12 x 1.75	95	70	9.69
M14 x 2	155	114	15.81
M16 x 2	240	177	24.48
M20 x 2.5	465	343	47.43
M24 x 3	800	590	81.6
M30 x 3.5	1600	1180	163.2
M36 x 4	2750	2028	280.5

STANDARD TIGHTENING TORQUES For SAE Grade 5 and Grade 8 Cap screws

The following specifications apply to required assembly torques for all grade 5 and grade 8 cap screws.

- Cap screw threads and seats shall be lubricated when assembled.

Unless instructions specifically recommend otherwise, these standard torque values are to be used with simple lithium base chassis grease (multi-purpose EP NLGI) or a rust- preventive grease (see list, previous page) on the threads.

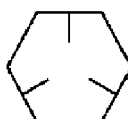
- Torques are calculated to give a clamping force of approximately 75% of proof load.

- The maximum torque tolerance shall be $\pm 10\%$ of the torque value shown.
- In the following table under Cap Screw Size, the first number represents the shank diameter (in.). The second number represents threads per inch.

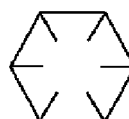
Example: 7/16 - 20

7/16 = shank diameter (7/16 inch (0.438 inch))

20 = threads per inch



GRADE 5



GRADE 8

TABLE 2.
Standard Tightening Torque
for SAE Hex Head Cap Screw And Nut Assembly

Cap Screw Size	Torque - Grade 5			Torque - Grade 8			Cap Screw Size	Torque - Grade 5			Torque - Grade 8		
	N·m	ft lb	kg·m	N·m	ft lb	kg·m		N·m	ft lb	kg·m	N·m	ft lb	kg·m
1/4-20	9.5	7	0.97	13.6	10	1.38	3/4-16	319	235	32.5	454	335	46.3
1/4-28	10.8	8	1.11	14.9	11	1.52	7/8-9	475	350	48.4	678	500	69.2
5/16-18	20.3	15	2.07	28	21	2.90	7/8-14	508	375	51.9	719	530	73.3
5/16-24	22	16	2.21	30	22	3.04	1.0-8	712	525	72.6	1017	750	103.7
3/8-16	34	25	3.46	47	35	4.84	1.0-12	759	560	77.4	1071	790	109.3
3/8-24	41	30	4.15	54	40	5.5	1.0-14	773	570	78.8	1085	800	110.6
7/16-14	54	40	5.5	79	58	8.0	1 1/8-7	881	650	89.9	1424	1050	145
7/16-20	61	45	6.2	84	62	8.57	1 1/8-12	949	700	96.8	1546	1140	158
1/2-13	88	65	9	122	90	12.4	1 1/4-7	1234	910	125.9	2007	1480	205
1/2-20	95	70	9.7	129	95	13.1	1 1/4-12	1322	975	134.8	2142	1580	219
9/16-12	122	90	12.4	169	125	17.3	1 3/8-6	1627	1200	166	2630	1940	268
9/16-18	129	95	13.1	183	135	18.7	1 3/8-12	1776	1310	181	2874	2120	293
5/8-11	169	125	17.3	237	175	24.2	1 1/2-6	2142	1580	219	3471	2560	354
5/8-18	183	135	18.7	258	190	26.2	1 1/2-12	2305	1700	235	3756	2770	383
3/4-10	298	220	30.4	420	310	42.8							

1 ft. lb. = 0.138 kg·m = 1.356 N·m

STANDARD TIGHTENING TORQUES

For SAE Grade 9 Cap screws

The following specifications apply to required assembly torques for all 12-point, grade 9 (170,000 psi minimum tensile), cap screws.

- Cap screw threads and seats shall be lubricated when assembled.

Unless instructions specifically recommend otherwise, these standard torque values are to be used with simple lithium base chassis grease (multi-purpose EP NLGI) or a rust- preventive grease (see list, this page) on the threads.

- Torques are calculated to give a clamping force of approximately 75% of proof load.
- The maximum torque tolerance shall be $\pm 10\%$ of the torque value shown.

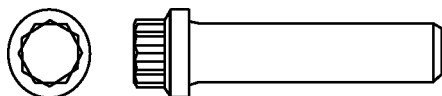


TABLE 3.
Standard Tightening Torque
for 12-Point, Grade 9 Cap Screws

Cap Screw Size*	Torque N·m	Torque ft lb	Torque kg·m
0.250 - 20	16	12	1.7
0.312 - 18	33	24	3.3
0.375 - 16	57	42	5.8
0.438 -14	95	70	9.7
0.500 -13	142	105	14.5
0.562 - 12	203	150	20.7
0.625 - 11	278	205	28.3
0.750 - 10	488	360	49.7
0.875 - 9	780	575	79.4
1.000 - 8	1166	860	119
1.000 - 12	1240	915	126
1.125 - 7	1670	1230	170
1.125 - 12	1800	1330	184
1.250 - 7	2325	1715	237
1.250 - 12	2495	1840	254
1.375 - 6	3080	2270	313
1.375 - 12	3355	2475	342
1.500 - 6	4040	2980	411
1.500 - 12	4375	3225	445

* Shank Diameter (in.) - Threads per inch

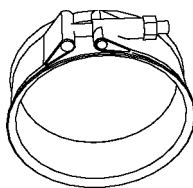


TABLE 4. Tightening Torque For T-Bolt Type Hose Clamp (SAE J1508 Type TB)			
Thread Size	Band Width	Newton meters (N·m)	Inch Pounds (in. lb)
0.25-28 UNF	19.05 mm (0.75 in.)	8.5 ± 0.6 N·m	75 ± 5 in lb

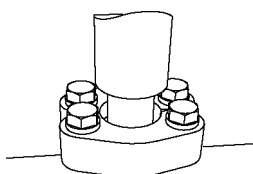


TABLE 5. Tightening Torque For Split Flange Clamp Bolts				
Cap Screw Thread Diameter (mm)	Width Across Flat (mm)	Newton meters (N·m) Tolerances ±10%	Foot Pounds (ft lb) Tolerances ±10%	Kilogram meters (kg·m) Tolerances ±10%
10	14	66	48	6.7
12	17	112	83	11.5
16	22	279	206	28.5

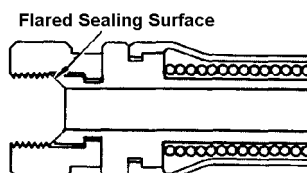


TABLE 6. Tightening Torque For Flared Tube And Hose Fittings				
Thread Diameter of Nut (mm)	Width Across Flat (mm)	Newton meters (N·m) Tolerances ±10%	Foot Pounds (ft lb) Tolerances ±10%	Kilogram meters (kg·m) Tolerances ±10%
14	19	25	18	2.5
18	24	50	36	5
22	27	80	58	8
24	32	140	101	14
30	36	175	130	18
33	41	195	145	20
36	46	245	180	25
42	55	295	215	30

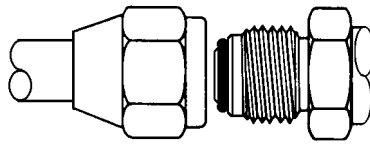


TABLE 7.
Torque Chart For JIC 37° Swivel Nuts
With Or Without O-ring Seals

Size Code	Tube Size (OD)	Threads UNF-2B	Newton meters (N·m)	Foot Pounds (ft lb)
– 2	0.125	0.312 – 24	5 ± 1	4 ± 1
– 3	0.188	0.375 – 24	11 ± 4	8 ± 3
– 4	0.250	0.438 – 20	16 ± 4	12 ± 3
– 5	0.312	0.500 – 20	20 ± 4	15 ± 3
– 6	0.375	0.562 – 18	24 ± 7	18 ± 5
– 8	0.500	0.750 – 16	40 ± 7	30 ± 5
– 10	0.625	0.875 – 14	54 ± 7	40 ± 5
– 12	0.750	1.062 – 12	75 ± 7	55 ± 5
– 14	0.875	1.188 – 12	88 ± 7	65 ± 5
– 16	1.000	1.312 – 12	108 ± 7	80 ± 5
– 20	1.250	1.625 – 12	136 ± 14	100 ± 10
– 24	1.500	1.875 – 12	163 ± 14	120 ± 10
– 32	2.000	2.500 – 12	312 ± 27	230 ± 20

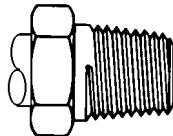


TABLE 8.
Torque Chart For
Pipe Thread Fittings

Size Code	Pipe Thread Size	With Sealant N·m	With Sealant ft lb	Without Sealant N·m	Without Sealant ft lb
– 2	0.125 – 27	20 ± 4	15 ± 3	27 ± 7	20 ± 5
– 4	0.250 – 18	27 ± 7	20 ± 5	34 ± 7	25 ± 5
– 6	0.375 – 18	34 ± 7	25 ± 5	48 ± 7	35 ± 5
– 8	0.500 – 14	48 ± 7	35 ± 5	61 ± 7	45 ± 5
– 12	0.750 – 14	61 ± 7	45 ± 5	75 ± 7	55 ± 5
– 16	1.000 – 11.50	75 ± 7	55 ± 5	88 ± 7	65 ± 5
– 20	1.250 – 11.50	95 ± 7	70 ± 5	108 ± 7	80 ± 5
– 24	1.500 – 11.50	108 ± 7	80 ± 5	129 ± 14	95 ± 10
– 32	2.000 – 11.50	129 ± 14	95 ± 10	163 ± 14	120 ± 10

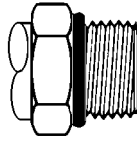


TABLE 9.
Torque Chart For
O-ring Boss Fittings

Size Code	Tube Size (OD)	Threads UNF-2B	Newton meters (N·m)	Foot Pounds (ft lb)
– 2	0.125	0.312 – 24	4 ± 3	4 ± 2
– 3	0.188	0.375 – 24	7 ± 3	5 ± 2
– 4	0.250	0.438 – 20	11 ± 4	8 ± 3
– 5	0.312	0.500 – 20	14 ± 4	10 ± 3
– 6	0.375	0.562 – 18	18 ± 4	13 ± 3
– 8	0.500	0.750 – 16	33 ± 7	24 ± 5
– 10	0.625	0.875 – 14	43 ± 7	32 ± 5
– 12	0.750	1.062 – 12	65 ± 7	48 ± 5
– 14	0.875	1.188 – 12	73 ± 7	54 ± 5
– 16	1.000	1.312 – 12	98 ± 7	72 ± 5
– 20	1.250	1.625 – 12	109 ± 7	80 ± 5
– 24	1.500	1.875 – 12	109 ± 7	80 ± 5
– 32	2.000	2.500 – 12	130 ± 14	96 ± 10

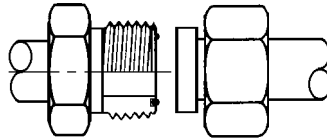


TABLE 10.
Torque Chart For
O-ring Face Seal Fittings

Size CodeE	Tube Size (O.D.)	Threads UNF-2B	Newton meters (N·m)	Foot Pounds (ft lb)
– 4	0.250	0.438 – 20	15 ± 1	11 ± 1
– 6	0.375	0.562 – 18	24 ± 3	18 ± 2
– 8	0.500	0.750 – 16	48 ± 5	35 ± 4
– 10	0.625	0.875 – 14	69 ± 7	51 ± 5
– 12	0.750	1.062 – 12	96 ± 10	71 ± 7
– 16	1.000	1.312 – 12	133 ± 8	98 ± 6
– 20	1.250	1.625 – 12	179 ± 10	132 ± 7
– 24	1.500	1.875 – 12	224 ± 20	165 ± 15

TABLE11. Common Conversion Multipliers Metric To English		
To Convert From	To	Multiply By
millimeter (mm)	inch (in.)	0.0394
centimeter (cm)	inch (in.)	0.3937
meter (m)	foot (ft)	3.2808
meter (m)	yard (yd)	1.0936
kilometer (km)	mile (mi)	0.6210
square centimeters (cm ²)	square inch (in. ²)	0.1550
square centimeters (cm ²)	square feet (ft ²)	0.001
cubic centimeters (cm ³)	cubic inch (in. ³)	0.061
liters (l)	cubic inch (in. ³)	61.02
cubic meters (m ³)	cubic feet (ft ³)	35.314
liters (l)	cubic feet (ft ³)	0.0353
grams (g)	ounce (oz)	0.0353
milliliter (ml)	fluid ounce (fl oz)	0.0338
kilogram (kg)	pound (mass)	2.2046
Newton (N)	pounds (lb)	0.2248
Newton meters (N·m)	kilogram meters (kg·m)	0.102
Newton meters (N·m)	foot pounds (ft lb)	0.7376
kilogram meters (kg·m)	foot pounds (ft lb)	7.2329
kilogram meters (kg·m)	Newton meters (N·m)	9.807
kilopascals (kPa)	pounds/square inch (psi)	0.1450
megapascals (MPa)	pounds/square inch (psi)	145.038
kilograms/cm ² (kg/cm ²)	pounds/square inch (psi)	14.2231
kilograms/cm ² (kg/cm ²)	kilopascals (kPa)	98.068
kilogram (kg)	short ton (tn)	0.0011
metric ton	short ton (tn)	1.1023
liters (l)	quart (qt)	1.0567
liters (l)	gallon (gal)	0.2642
Watts (W)	horsepower (hp)	0.00134
kilowatts (kW)	horsepower (hp)	1.3410

TABLE 12. Common Conversion Multipliers English to Metric		
To Convert From	To	Multiply By
inch (in.)	millimeter (mm)	25.40
inch (in.)	centimeter (cm)	2.54
foot (ft)	meter (m)	0.3048
yard (yd)	meter (m)	0.914
mile (mi)	kilometer (km)	1.61
square inch (in. ²)	square centimeters (cm ²)	6.45
square feet (ft ²)	square centimeters (cm ²)	929
cubic inches (in. ³)	cubic centimeters (cm ³)	16.39
cubic inches (in. ³)	liters (l)	0.016
cubic feet (ft ³)	cubic meters (m ³)	0.028
cubic feet (ft. ³)	liters (l)	28.3
ounce (oz)	kilogram (kg)	0.028
fluid ounce (fl oz)	milliliter (ml)	29.573
pound (lb)	kilogram (kg)	0.454
pound (lb)	Newton (N)	4.448
inch pounds (in. lb)	Newton meters (N·m)	0.113
foot pounds (ft lb)	Newton meters (N·m)	1.356
foot pounds (ft lb)	kilogram meters (kg·m)	0.138
kilogram meters (kg·m)	Newton meters (N·m)	9.807
pounds/square inch (psi)	kilopascals (kPa)	6.895
pounds/square inch (psi)	megapascals (MPa)	0.007
pounds/square inch (psi)	kilograms/square centimeter (kg/cm ²)	0.0704
short ton (tn)	kilogram (kg)	907.2
short ton (tn)	metric ton (t)	0.0907
quart (qt)	liters (l)	0.946
gallon (gal)	liters (l)	3.785
horsepower (hp)	Watts (w)	745.7
horsepower (hp)	kilowatts (kw)	0.745

TABLE 13. Temperature Conversions Formula: $F^{\circ} - 32 \div 1.8 = C^{\circ}$ or $C^{\circ} \times 1.8 + 32 = F^{\circ}$								
Celsius C°		Fahrenheit F°	Celsius C°		Fahrenheit F°	Celsius C°		Fahrenheit F°
121	250	482	63	145	293	4	40	104
118	245	473	60	140	284	2	35	95
116	240	464	57	135	275	- 1	30	86
113	235	455	54	130	266	- 4	25	77
110	230	446	52	125	257	- 7	20	68
107	225	437	49	120	248	- 9	15	59
104	220	428	46	115	239	- 12	10	50
102	215	419	43	110	230	- 15	5	41
99	210	410	41	105	221	- 18	0	32
96	205	401	38	100	212	- 21	- 5	23
93	200	392	35	95	203	- 23	- 10	14
91	195	383	32	90	194	- 26	- 15	5
88	190	374	29	85	185	- 29	- 20	- 4
85	185	365	27	80	176	- 32	- 25	- 13
82	180	356	24	75	167	- 34	- 30	- 22
79	175	347	21	70	158	- 37	- 35	- 31
77	170	338	18	65	149	- 40	- 40	- 40
74	165	329	15	60	140	- 43	- 45	- 49
71	160	320	13	55	131	- 46	- 50	- 58
68	155	311	10	50	122	- 48	- 55	- 67
66	150	302	7	45	113	- 51	- 60	- 76
NOTE: The numbers in the unmarked columns refer to temperature in either degrees Celsius (C°) or Fahrenheit (F°). Select a number in this unmarked column and read to the left to convert to degrees Celsius (C°) or read to the right to convert to degrees Fahrenheit (F°). If starting with a known temperature (either C° or F°), find that temperature in the marked column and read the converted temperature in the center, unmarked column.								

STORAGE AND IDLE MACHINE PREPARATION

There may be periods when it is necessary for a machine to be idle for an extended period of time. If properly prepared, a stored machine may promptly and safely be put back into operation. Improper preparation, or complete lack of preparation, can make the job of getting the vehicle back to operating status difficult.

The following information outlines the essential steps for preparing a unit for extended storage, and the necessary steps for bringing the unit back to operational status. Additional information is given to help restore machines that were not put into storage, but merely shut down and left idle for a long period of time.

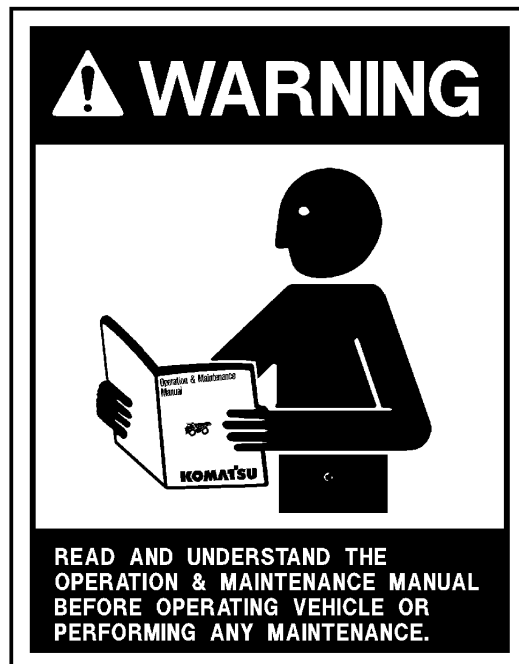
Much of this material is of a general nature since the environment (where the machine has been standing idle) will play a big part in its overall condition. A hot, humid climate will affect vehicle components much differently than a dry desert atmosphere or a cold arctic environment. These climatic aspects must be considered and appropriate actions taken when restoring a long term idle vehicle.

These instructions are not intended to be all inclusive, but are *minimum* guidelines. The final aim should always be to provide the operator with a safe, reliable, fully productive vehicle.

SHORT TERM IDLE PERIODS

There may be periods when a vehicle is idle from 30 to 60 days, but must be ready for use at any time. To most effectively handle this type of situation, follow the procedure below to prevent any deterioration from the beginning of vehicle idle time.

1. Keep the vehicle fully serviced.
2. Perform a visual check of the vehicle every week. Start and run the engine until it is at operating temperature. Move the vehicle around the yard for a few minutes to ensure that all internal gears and bearings are freshly lubricated.
3. Operate all hydraulic functions through their complete range to ensure that the cylinder rods and seals are fully lubricated.
4. Check and operate all systems.
5. Once a month, perform the 10 Hour service items shown in the Operation and Maintenance Manual. Keep batteries properly serviced.



PREPARATION FOR STORAGE

For long term idle periods, proper preparation will pay large dividends in time and money when future operation of the vehicle is scheduled.

1. The engine should be prepared for storage according to instructions found in the engine manufacturer's manual.
2. The vehicle should be in top operating condition with all problems corrected. The paint should be in good condition, with no rust or corrosion. All exposed, machined or unpainted surfaces should be coated with a good rust preventive grease.
3. After the vehicle has been parked in its storage location, all hydraulic cylinders, including Hydrair suspensions, should be retracted as much as possible (steering cylinders centered). Wipe the exposed portion of all cylinder rods clean and coat them (including the seals on the end of the barrel) with a good preservative grease.
4. If long term storage is anticipated, the vehicle should be blocked up with the tires clear of the ground to remove the vehicle weight from the tires. Lower the tire pressure to 103 - 172 kPa (15 - 25 psi). Completely cover the tires with tarpolins to minimize rubber oxidation and deterioration.
5. Clean the radiator. Refer to Section C, Cooling System, for the proper cleaning instructions.

WARNING

The cooling system is pressurized due to thermal expansion of coolant. Do not remove the radiator cap while the engine and coolant are hot. Severe burns may result.

6. The cooling system should be completely drained, chemically flushed, and refilled with a conditioned water/antifreeze solution suitable for the lowest temperature anticipated. Refer to Section P, Lubrication and Service, for the proper anti-freeze and conditioner concentrations. After refilling the system, always operate the engine until the thermostats open to circulate the solution through the cooling system.

NOTE: NEVER store a vehicle with a dry cooling system.

WARNING

Any operating fluid, such as hydraulic oil, escaping under pressure can have sufficient force to enter a person's body by penetrating the skin. Serious injury and possible death may result if proper medical treatment by a physician familiar with this injury is not received immediately.

7. New hydraulic filters should be installed and the hydraulic tank fully serviced with Type C-4 oil as specified in Section P, Lubrication and Service.
8. Disconnect the batteries, if possible. Batteries should be removed and stored in a battery shop or a cool dry location on wooden blocks. Do not store batteries on a concrete floor. Clean the battery compartment. Remove all corrosion and paint the compartment with an acid proof paint.
9. The wheel axle housings should be fully serviced with recommended lubricants. Seal all vents.
10. The exhaust openings and air cleaners should be covered tightly with moisture barrier paper and sealing tape.

11. All lubrication points (grease fittings) should be serviced with the recommended lubricants.
12. Relieve the tension from all drive belts. The engine manufacturer recommends insertion of heavy kraft paper between the belts and pulleys to prevent sticking.
13. All vandalism covers and locks should be in place and secured.
14. All cab windows should be closed, locked and sealed, and the cab door should be locked to prevent vandalism and weather effects.
15. The fuel tank should be completely drained of fuel, fogged with preservative lubricant (NOX-RUST MOTOR STOR., SAE10), and closed tightly. All fuel filters should be replaced.
16. If possible, all available service publications (vehicle and engine) and a current parts catalog should be packaged in a moisture proof package and placed in the cab to aid whomever will eventually place the unit back in operation.
17. Ensure that the water drain holes in the truck body are open.

REMOVAL FROM STORAGE

If the storage preparations were followed when placing the vehicle into storage, getting it back to operational status is a matter of reversing those steps.

NOTE: Before starting the job of restoring a vehicle to operation, obtain copies of the Operation and Maintenance Manual, Service Manual, Engine Manuals and Parts Book, if possible, and follow all special instructions regarding servicing the vehicle and its components.

In addition to removing the storage materials, the following actions should be taken:

1. Inspect the entire vehicle carefully for rust and corrosion. Correct as necessary.
2. Service the engine according to the engine manufacturer's Operation and Maintenance Manual.
3. Clean the radiator. Refer to Section C, Cooling System, for cleaning instructions.
4. The cooling system should be completely drained, chemically flushed, and refilled with a conditioned water/antifreeze solution suitable for the lowest temperature anticipated. Refer to Fluid Specifications in Section P, Lubrication and Service, for the proper anti-freeze and conditioner concentrations. After refilling the system, always operate the engine until the thermostats open to circulate the solution through the cooling system.
5. Thoroughly inspect all drive belts and hydraulic oil lines for damage, wear or deterioration. Replace any suspected lines. Do not take chances on possible ruptures or blow-outs.
6. New hydraulic filters should be installed and the hydraulic tank checked and serviced with Type C-4 oil as specified in Section P, Lubrication and Service.

7. Open the fuel tank drain valve to remove any moisture or sediment that may have accumulated while in storage. Close the drain valve and fill the fuel tank with diesel fuel.

DANGER

Never blend gasoline, gasohol and/or alcohol with diesel fuel. This creates an extreme fire hazard and, under certain conditions, may cause an explosion.

8. Ensure that all hydraulic controls, steering linkage and throttle linkage points are lubricated and operate freely before starting the engine.
9. All electrical connections must be clean and tight. Ensure that all ground straps and cables are secure.
10. Install fully charged batteries in the vehicle. Clean the connectors and connect the battery cables. The battery compartment must be free of corrosion. Secure the batteries with hold-downs.
11. Check all electrical cables for deterioration, cracks and other damage. Replace any defective cables.

DANGER

Air pressure must be released from tires with bad cuts or wear that extends into the plies before removal from the vehicle. Also, do not allow personnel to stand in removal path of tires.

12. Check all tires carefully for damage. Inflate the tires to the proper pressure.
13. If disconnected, reconnect the parking brake linkage.
14. Completely service the vehicle as recommended in Section P, Lubrication and Service, for both 10 and 100 Hour inspections.
15. Adjust all drive belts to the specified tension.

16. Refer to the Operation and Maintenance Manual for engine startup and vehicle checkout procedures. Check all hose and line connections for leaks when the engine is running.
17. Before moving the vehicle, cycle all hydraulic controls, steering controls, and braking systems to verify proper operation. Check all instruments and gauges to ensure that all systems are operational.
18. When all systems are operational and all known problems are corrected, road test the vehicle in a smooth, level, unobstructed area to check steering response, service brake efficiency, and hydraulic functions.
19. Recharge any fire protection equipment before the vehicle is returned to service.

RECONDITIONING AN IDLE VEHICLE

WARNING

Never attempt to operate a vehicle that has been standing idle for a long period until all systems which affect steering, brakes, engine, transmission and running gears have been completely reconditioned. An unsafe vehicle can cause serious injuries and/or major property damage.

Reconditioning a vehicle that has been subjected to long idle periods without being properly prepared for storage - merely shut down and left to the elements - may present a major expenditure of time and money when it is to be put into operation.

1. Remove all debris and thoroughly clean the vehicle before starting any inspection or maintenance.
2. Remove the batteries and move them to a battery shop for service and charging, or replace as necessary.

WARNING

Do not disassemble an inflated tire. Remove the valve core slowly and allow pressure to bleed off before attempting to remove the lockring. Also, eye protection must be worn during tire deflation to protect against any foreign object being projected into the eyes.

WARNING

Do not mix rim parts of different rim manufacturers. Rim parts may resemble those of a different manufacturer, but the required tolerances may be wrong. Use of mismatched rim parts is hazardous.

3. Inspect tires thoroughly for wear, cuts and cracks on the treads and side walls.
 - a. Any tire suspected of being unserviceable should be removed and thoroughly inspected before being inflated.
 - b. If the tires are removed, clean and inspect all wheel components. All rust and corrosion must be removed and parts repainted as needed before mounting the tires.
 - c. Mount and inflate tires as shown in Operation and Maintenance Manual. Follow all safety rules
4. Inspect the service brakes carefully. Before disabling the brake circuit, block all wheels to prevent possible movement of the vehicle.
 - a. All brake lines and connections must be clean and free of rust, corrosion and damage.
 - b. When reconditioning the braking system, the service brake hydraulic circuits should be checked out according to the instructions in Section J, Brake Circuit Check-Out Procedure.
5. The engine should be inspected and serviced according to the engine manufacturer's Operation And Maintenance Manuals.
 - a. Ensure that the exhaust is clear and clean. If water entry is suspected, disconnect the air tubes at the turbochargers to check for water before attempting to start the engine.
 - b. Replace the fuel filters and fill filter cans with fresh fuel for engine priming.

CAUTION

Have a new safety filter (secondary) element on hand before removing the old one. Do not keep the intake system open to the atmosphere any longer than absolutely necessary.

- c. Replace both the primary and safety filter (secondary) elements in the air cleaners. Check all intake lines between the air cleaners and the engine. All clamps must be tight. The plunger in the filter condition indicators must move freely.
- d. Inspect the tubes in the precleaner section of the air cleaner assembly. Use a light to inspect the tubes. The light should be visible through the end of the tube. If clogging is evident, the precleaner must be cleaned according to the instructions in Section C, Air Cleaners.
- e. Drain and flush the engine cooling system. Fill with coolant and inhibitors after checking all lines, hoses and connections. Refer to Section P, Lubrication and Service, for anti-freeze recommendations. Radiator cores must be clear of dirt and debris.



To prevent injuries, always release spring tension before replacing the fan belt.

- f. Check and tighten the engine fan drive belts. If necessary, install a new belt set.
- g. Check and tighten the engine mounts.
- 6. If fuel was left in the tanks, it must be removed. Do not attempt to use old diesel fuel.
 - a. With the tanks empty, remove the inspection plates and thoroughly check the interior of the tanks. Remove any sediment and contamination. If the fuel was contaminated, the lines should be disconnected and blown clear.
 - b. Check all fuel lines for deterioration or damage. Replace lines as necessary.
 - c. Install the inspection covers with new gaskets.
 - d. Fill the tanks with the specified diesel fuel.
 - e. Replace the fuel filters.

- 7. Drain the hydraulic tank. If the oil is not contaminated and is stored in clean containers, it may be reused if filtered through 3-micron filter elements when being pumped back into the tank. Do not attempt to use contaminated hydraulic oil, especially if water entry into the system is suspected.

NOTE: If filling is required, use clean hydraulic oil only. Refer to the Lubrication Chart in Section P, Lubrication and Service, for proper oil specifications.

- a. Replace the hydraulic filter elements and clean the suction strainer elements. While the suction strainers are removed, inspect and clean the inside of the tank thoroughly to remove all sediment and foreign material.
- b. Inspect all hydraulic lines for deterioration and damage. Replace any suspected lines. Do not risk possible hose ruptures or blow outs.
- c. Check all hydraulic components (pumps, valves and cylinders) for damage and corrosion. Secure all mountings and connections. The control valves must move freely with no binding.
- d. Check the exposed portions of all hydraulic cylinder rods for rust, pitting and corrosion. If the plating has deteriorated, the cylinder should be removed and overhauled or replaced. Pitted or scored plating will cause leakage at the cylinder seals.
- 8. Check the front wheel hub, final drive and wheel axle lubricant. If contamination is suspected, the oil should be drained completely and the component serviced with clean recommended lubricant. If major contamination is present, disassembly and overhaul will be necessary.
- 9. Check the parking brake. Since it is spring applied, the brake pads may be stuck tightly to the disc. It may be necessary to remove and overhaul the parking brake assembly.

10. Lubricate all grease fittings that are not part of the automatic lubrication system with the recommended lubricants. Pay special attention to the steering linkage connections. All pivot points must move freely with no binding.
11. Check the battery charging alternator for corrosion or deterioration. The alternator rotor must move freely with no binding or roughness. Inspect, install and properly tension the alternator drive belts.
12. Ensure that the steering cylinder ball joints link and hydraulic connections are secure.
13. Examine the Hydrair suspensions for damage.
 - a. Discharge the nitrogen from the suspensions as outlined in Section H, Hydrair II Suspensions.
 - b. Check the condition of the suspension oil and cylinder wipers. If the wipers are cracked or hardened, the suspension must be rebuilt. Recharge the suspension with new oil if the old oil is deteriorated.
 - c. Check the exposed chrome portions of cylinder for rust, pitting and corrosion. If the plating is deteriorated, the suspension should be removed and overhauled or replaced. Pitted or scored plating will rapidly cause leakage at the seals.
 - d. Recharge the suspensions as outlined in Section H, Hydrair II Suspensions.
14. If not done previously, install fully-charged batteries and ensure that the hook-up is correct.

ENGINE OPERATION

After all reconditioning procedures have been completed, a static check of engine operation, systems operations, and verification of the braking and steering functions must be done before the vehicle is moved.

WARNING

Before starting the engine, clear the immediate area of personnel and obstructions. Ensure that all tools and loose equipment have been removed from the vehicle. Always sound the horn as a warning before starting the engine or actuating any operational controls.

Never start the engine in a building unless the doors and windows are open and ventilation is adequate.

1. Turn the key switch ON. The warning lights for low brake and steering pressure should illuminate and the horn should sound. If it does not, check all components in the circuit and correct the problem before continuing.
2. Turn the key switch to the START position to crank the engine. Release the key when the engine starts. Watch the engine oil pressure gauge. If the pressure does not show within 10 - 15 seconds, turn off the engine and locate the problem.
3. While the engine is warming up, inspect the coolant, oil and fuel lines for leaks. Check the hydraulic pump and hydraulic lines for leaks.
4. When the engine is at operating temperature, check the operation of the throttle circuit. Acceleration should be smooth. Watch the gauges for any abnormal activity. Proper temperatures and pressures are shown in the Operation and Maintenance Manual.

After The Engine Has Started

Any machine which is unsafe and/or not in top operating condition should not be assigned to an operator for production use.

1. Become thoroughly familiar with the steering and emergency controls. Test the steering in the extreme right and left directions. If the steering system is not operating properly, turn off the engine immediately. Determine the steering system problem and have repairs made before resuming operation.
2. Operate each of the brake circuits at least twice prior to operating and moving the machine.
 - a. Activate each circuit individually using the controls in the cab with the engine running and the hydraulic circuit fully charged.
 - b. If any application or release of any brake circuit does not appear proper or if sluggishness is apparent, turn off the engine and notify maintenance personnel. Do not operate the machine until the brake circuit in question is fully operational.
3. Check all gauges, warning lights and instruments before moving the machine to ensure proper system operation and gauge function. Pay special attention to the braking and steering circuit warning lights. If any warning lights come on, turn off the engine immediately and determine the cause.
4. Cycle the hoist controls and steering several times to remove any trapped air. Complete the steering cycles in both directions to verify steering response, smoothness and reliability. Check the seals and lines for leaks.

5. After all known problems have been corrected, the vehicle is ready for a road test. This test should be performed only by a capable and experienced operator in a large open area where plenty of maneuvering room is available.

One of the road test items that should be performed is the repeated test of braking efficiency at progressively higher speeds. Start at slow speeds. Do not operate at higher speeds until the machine is determined to be operating safely.

6. When the vehicle is ready for production use, it should be visually rechecked and fully serviced according to Section P, Lubrication and Service.

A few of the conditions which might be encountered after a machine has been exposed to the elements for a long period of time include:

- increased corrosion and fungus growth on electrical components in humid/tropical areas.
- accelerated rust formation in humid climates.
- increased sand and dust infiltration in windy, dry dusty areas.
- deterioration of rubber products in extreme cold areas. Cables, hoses, O-rings, seals and tires may become weather checked and brittle.
- animal or bird's nests in unsealed openings.

ENGINE STORAGE

When an engine is to be stored or removed from operation for a period of time, special precautions should be taken to protect the interior and exterior of the engine and other parts from rust and corrosion.

All rust and corrosion must be completely removed from any exposed part before applying a rust preventive compound. Therefore, it is recommended that the engine be processed for storage as soon as possible after removal from operation.

The engine should be stored in a building which is dry and can be heated during the winter months. Moisture absorbing chemicals are commercially available for use when excessive moisture is prevalent in the storage area.

Temporary Storage (30 Days Or Less)

1. Drain the oil from the crankcase. Fill the crankcase to the proper level with the recommended viscosity and grade of new oil.
2. Fill the fuel tank with the recommended grade of fuel oil. Operate the engine for two minutes at 1200 rpm with no load. Turn off the engine. **Do not** drain the fuel system or the crankcase.
3. Check the air cleaner and service it, if necessary.
4. If freezing weather is expected during the storage period, add an ethylene glycol base antifreeze solution in accordance with the manufacturer's recommendations.
5. Clean the entire exterior of the engine (except the electrical system) with fuel oil and dry it with compressed air.



To prevent possible personal injury, wear adequate eye protection and do not exceed a compressed air pressure of 276 kPa (40 psi).

6. Seal all engine openings. The material used for this purpose must be waterproof, vaporproof and possess sufficient physical strength to resist puncture and damage from the expansion of entrapped air.

An engine prepared in this manner can be returned to service in a short period of time by removing the seals at the engine openings, checking the engine coolant, fuel oil, lubricating oil, transmission oil, and priming the raw water pump (if used).

Extended Storage (More Than 30 Days)

1. Drain the cooling system and flush with clean, softened water. Refill the system with clean, softened water and add a rust inhibitor.
2. Remove, check and recondition the injectors, if necessary, to ensure that they will be ready to operate when the engine is restored to service. Reinstall the injectors. Time them and adjust the exhaust valve clearance.
3. Circulate the coolant by operating the engine until the normal operating temperature of **71° - 85° C (160° - 185° F)** is reached.
4. Stop the engine.
5. Drain the oil from the crankcase, then reinstall and tighten the drain plug. Install new lubricating oil filter elements and gaskets.
6. Fill the crankcase to the proper level with a 30-weight preservative lubricating oil (MIL-L-21260C, Grade 2).

7. Drain the fuel tank. Refill with enough No. 1 diesel fuel or pure kerosene to allow the engine to operate for about ten minutes. If it is not convenient to drain the fuel tank, use a separate portable supply of the recommended fuel.

NOTE: If the engine is stored where condensation of water in the fuel tank may be a problem, add pure, waterless isopropyl alcohol (isopropanol) to the fuel at a ratio of one pint to 125 gallons(473 L) of fuel, or 0.010% by volume. Where biological contamination of fuel may be a problem, add a biocide, such as Biobor JF or an equivalent, to the fuel. When using a biocide, follow the manufacturer's concentration recommendations and all cautions and warnings.

8. Drain and disassemble the fuel filter and strainer. Discard the used elements and gaskets. Fill the cavity between the element and shell with No. 1 diesel fuel or pure kerosene, and install on the engine. If spin-on fuel filters and strainers are used, discard the used cartridges, fill the new ones with No. 1 diesel fuel or pure kerosene, and install on the engine.
10. Operate the engine for five minutes to circulate the clean fuel oil throughout the fuel system.
11. Service the air cleaner.
12. Turbocharger bearings are pressure lubricated through the external oil line leading from the engine cylinder block while the engine is operating. However, the turbocharger air inlet and turbine outlet connections should be sealed off with moisture resistant tape.
13. Apply a rust preventive compound to all exposed, non-painted surfaces.
14. Drain the cooling system.
15. Drain the preservative oil from the engine crankcase. Reinstall and tighten the drain plug.
16. Remove and clean the batteries and battery cables with a baking soda/water solution and rinse them with fresh water. Do not allow the soda solution to enter the battery. Add distilled water to the electrolyte, if necessary. Store the battery in a cool - **never below 0° C (32° F)** - dry place. Keep the battery fully charged and check the level and the specific gravity of the electrolyte regularly. Never set batteries on a concrete floor. Place them on wooded blocks.
17. Insert heavy paper strips between the pulleys and belts to prevent sticking.
18. Seal all engine openings, including the exhaust outlet, with moisture resistant tape. Use cardboard, plywood or metal covers where practical.
19. Clean and dry the exterior painted surfaces of the engine and spray with a suitable liquid automotive body wax, a synthetic resin varnish or a rust preventive compound.
20. Protect the engine with a weather-resistant tarpaulin and store it under cover, preferably in a dry building with temperatures above 0° C (32° F).



Outdoor storage of engines is not recommended. However, if an engine must be kept outdoors, follow the preparation and storage instructions above. Do not use plastic sheeting for outdoor storage. Plastic may be used for indoor storage. However, when used outdoors, moisture can condense on the inside of the plastic and cause ferrous metal surfaces to rust and/or pit aluminum surfaces. If a unit is stored outside for any extended period of time, severe corrosion damage can result.

21. Inspect the stored engine periodically. If there are any indications of rust or corrosion, corrective steps must be taken to prevent damage to the engine parts. Perform a complete inspection at the end of one year and apply additional treatment, as required.

RESTORING AN ENGINE AFTER EXTENDED STORAGE

1. Remove the covers and tape from all of the openings of the engine, fuel tank and electrical equipment. **Do not overlook the exhaust outlet or the intake system.**
2. Wash the exterior of the engine and flywheel with fuel oil to remove the rust preventive compound.
3. Remove the paper strips from between the pulleys and the belts.
4. Remove the drain plug and drain the preservative oil from the crankcase. Reinstall the drain plug. Then, refer to Lubrication and Service, Section P, and fill the crankcase with the recommended grade of lubricating oil to proper level using a pressure prelubricator.
5. Fill the fuel tank with the engine manufacturer's specified fuel.
6. Close all of the drain cocks and fill the engine cooling system with clean, softened water and a rust inhibitor. If the engine is to be exposed to freezing temperatures, fill the cooling system with an ethylene glycol base antifreeze solution. Refer to the coolant specifications in Section P, Lubrication and Service.
7. Install and connect the fully charged batteries.
8. Service the air cleaner as outlined in Section C, Air Cleaners.
10. Remove the covers from the turbocharger air inlet and turbine outlet connections.
11. After all of the preparations have been completed, start the engine. The small amount of rust preventive compound which remains in the fuel system will cause a smoky exhaust for a few minutes.

NOTE: Before subjecting the engine to a load or high speed, it is advisable to check the engine tune-up.

ELECTRIC DRIVE TRUCKS

These instructions provide the recommended procedures for protecting equipment from damage during both short term and long-term storage periods. Also included are instructions for placing this equipment into service after having been stored.

For the purposes of this instruction, a short term storage period is considered to be less than three months; a long-term storage period is considered to be three months or longer.

General Electric recommends a maximum storage period of three years, with these storage procedures being repeated after each year. After a storage period of three years or more, the motorized wheels should be removed and sent to an overhaul facility for teardown and inspection of seals and bearings. These should be replaced if necessary.

Periodic (every three months) inspections should be made to determine the lasting qualities of long-term storage protection measures. Such inspections will indicate the need for renewing protective measures when necessary to prevent equipment deterioration.

Proper storage of this equipment is vital to equipment life. Bearings, gears, and insulation may deteriorate unless adequate protective measures are taken to protect against the elements. For example, bearings and gears in the motorized wheel gear case are susceptible to the formation of rust, insulation in rotating electrical equipment can accumulate moisture, and bearings may become pitted.



Never apply any spray, coating, or other protective materials to areas not specifically recommended.

It is also important to note that these instructions cannot possibly anticipate every type of storage condition and, therefore, cannot prevent all equipment deterioration problems caused by inadequate storage. These instructions are not intended to be all inclusive, but are *minimum* guidelines for achieving the best possible equipment life and the lowest operating cost when the equipment is returned to service.

NOTE: Local conditions and/or experience may require additional procedures and/or additional storage precautions.

Storing A Truck That Is Operational

When a fully operational truck is being placed into storage for less than three months, the best protective measure which can be taken is to drive the truck once a week for at least 30 minutes. Prior to driving the truck, the rotating equipment should be Meggered and:

- If greater than 2 megohms, run normally.
- If less than 2 megohms, isolate the condition and correct it before running.

Driving the truck circulates oil in the gearcase to keep gears and bearings lubricated and free from rust. It also prevents deterioration of the brushes, commutators and slip rings.

When a fully operational truck is being placed into storage for three months or longer, and the truck cannot be operated weekly throughout the storage period as indicated above, perform the following:

1. Drain the oil from the gearcase and install rust preventive 4161 (a product of Van Straaten Chemical Co.) or equivalent. Fill per General Electric Motorized Wheel Service Manual.

2. Megger the wheels as indicated in the instructions above. Operate the truck for at least 30 minutes to ensure that the rust preventive compound has been thoroughly circulated throughout the gearcase. Stop the truck and drain the rust preventive compound.

NOTE: Do not run a loaded truck with rust preventive compound in the wheel motor gearcases.



Do not operate trucks without oil in the wheel motor gearcases.

3. Perform a megohmmeter test. Refer to the truck's Vehicle Test instructions for the correct procedure. Record the Megger readings for future reference. They will be helpful in determining if deterioration is being experienced when additional Megger tests are made as part of the periodic inspection.
4. Lift all brushes in the motorized wheels, blowers and the alternator. They must be removed from the brushholder. Disconnecting brush pigtails is not required.
5. Cover any open ductwork with screening material to prevent rodents from entering. Then tape over the screen to prevent the entry of water and dirt (allow breathing).
6. Examine all exposed machined surfaces for rust or dirt accumulation. Remove all contamination as necessary. Remove rust by using a fine abrasive paper. Old flushing compound can be removed with mineral spirits (GE-D5B8). Methanol should be used to remove all residue. When clean, coat with Tarp B rust preventive. Refer to General Electric Motorized Wheel Service Manual for specifications.
7. Loosen exciter drive belts (where applicable).
8. Open all switches in the control compartment.
9. Install a 500 watt heat source inside all control groups which house electronic control equipment. These heat sources are to be energized below 0° C (32° F) and de-energized above 5° C (41° F).
10. Install a 500 watt heat source inside the commutator chamber of both motorized wheels and inside the alternator slip ring chamber. This will minimize the accumulation of moisture. A hole in the bottom of the hubcap will accommodate the electrical cord for the heat source in the motorized wheels. These heat sources are to be energized continuously.
11. Seal the compartment doors with a weatherproof tape to prevent entry of rain, snow and dirt (allow breathing).

Storing A Truck That Is Not Operational

When a truck which is not fully operational is being stored for any period of time, perform the following:

1. Drain the oil from the gearcase and install rust preventive compound 4161 (or equivalent). Fill per General Electric Motorized Wheel Service Manual.
2. Jack up each side of the truck (one side at a time) enough to rotate the tires.
3. Connect a D-C welder as described in the Vehicle Test Instructions (Wheel Motor inst. 400A, arm & field in stress 900- 1000 rpm arm).
4. Rotate each motorized wheel (one at a time) for at least 30 minutes to ensure that the rust preventive compound has been thoroughly circulated throughout the gearcase. Disconnect the welder. Remove the jacks. Drain the gearcase.

5. If the truck is partially dismantled, pay close attention to ductwork, blower shrouds, etc., which may be exposed to weather conditions. These areas will require the same sealing measures as in Step 5 above which deals with protecting ductwork. Cover exposed blower housings to prevent entry of water and dirt.
6. Perform Steps 3 through 11 in "Storing A Truck That Is Operational" in this section.

Storing A Major Component

When storing a motorized wheel, alternator, blower or control group for any period of time, always store it inside a warm, climate-controlled environment. Do not attempt to store individual components where they would be exposed to inclement weather, climatic changes, high humidity and/or temperature extremes.

Periodic Inspections

It is important that periodic inspections (every three months) of stored equipment be performed to ensure the continued serviceability of all protective measures initially taken when the storage period began. The following items should be checked at each inspection interval:

1. Remove the weatherproof tape from the compartment doors and perform a Megger test as described in the Vehicle Test Instructions. Record the test results and compare them with the recorded Megger readings taken when storage first began, and those taken throughout the storage period. Remove all test equipment and close the compartment. Reseal the compartment doors with new weatherproof tape. If Megger readings indicate a deterioration of insulation quality (to below 2.0 megohms), more protection may need to be provided.
2. Check all other weatherproofing tape. Replace any that has become loose or is missing.
3. Check all heat sources. Replace or repair any that have become inoperative.
4. Check all machine surfaces that were coated with flushing compound when storage began. If the compound appears to be deteriorating, it must be cleaned off and renewed.

Placing Equipment Into Service After Storage

If a truck has been operated weekly throughout the storage period, perform a complete visual inspection of the motorized wheels, blowers, alternator and control compartments. Repair any defects that are found, then place the truck directly into service.

If the truck was not operated weekly throughout the storage period, perform the following procedures:

1. Remove all weatherproofing tape from control compartment doors and ductworks.
2. Remove all screening material from the ductwork.
3. Remove all heat sources from the motorized wheels, control compartments and the alternator.
4. Fill with recommended oil. Refer to the Motorized Wheel Service Manual for the type and amount oil to be used. This oil should be drained and new oil should be added after 500 hours of operation.
5. Clean all motorized wheel grease fittings in the axle box. Ensure that all grease lines are completely full of grease, then add the recommended amount of grease to all fittings.
6. Install brushes in the motorized wheels, blowers and alternator. Make sure that the brushes move freely in their carbonways and that they have enough length to serve until the truck's next inspection period. Install new brushes if necessary. Ensure that all brush pigtail screws are tight.

7. Perform a megohmmeter test. Refer to the truck's Vehicle Test Instructions for the correct procedure. If Megger readings are less than 2.0 megohms, the problem could be an accumulation of moisture in the motor or alternator. If this is the case, the faulty component will have to be isolated and dried out using procedures outlined in the G.E. Service Manual.
8. Perform a thorough inspection of the motorized wheels, alternator, blowers and control compartments. Look for:
 - Rust or dirt accumulation on machine surfaces
 - Damaged insulation
 - Any accumulation of moisture or debris, especially in the ductwork
 - Loose wiring and cables
 - Any rust on electrical connectors in the control compartment
 - Any loose cards in the card panels
9. Clean and make repairs as necessary.
10. Check the retarding grids and insulators for loose connections and dirt accumulation. Clean and tighten connections as necessary.
11. Where applicable, check exciter drive belts for cracks and deterioration. If acceptable, set the belt tension to specification.
12. Before starting the engine, turn on the control power. Check that the contactors and relays pick up and drop out normally.
13. Perform a startup procedure on the complete system to ensure maximum performance during service. Refer to the truck's Vehicle Test Instructions for the complete test procedure.

After all storage protection has been removed, the truck has been cleaned and inspected, all repairs have been made, the motorized wheel gearcase has been filled with new oil, the dirt seals have been completely purged with new grease, and the system has been completely checked, the truck can be placed into service. However, it is recommended that the truck be driven unloaded at a low speed of no more than 16 km/h (10 mph) for the first hour of operation.

NOTES

SECTION B STRUCTURES INDEX

STRUCTURAL COMPONENTS	B2-1
DUMP BODY.....	B3-1
FUEL TANK (S/N A31164 - A31494).....	B4-1
FUEL TANK (S/N A31496 & UP).....	B4-1

NOTES

SECTION B2
STRUCTURAL COMPONENTS
INDEX

PREPARATION	B2-3
DIAGONAL LADDER, GRILLE AND HOOD ASSEMBLY.....	B2-4
Removal	B2-4
Installation	B2-4
RH DECK AND RETARDING GRID	B2-5
Removal	B2-5
Installation	B2-5
LH DECK	B2-6
Removal	B2-6
Installation	B2-6
CENTER DECK	B2-7
Removal	B2-7
Installation	B2-7

NOTES

STRUCTURAL COMPONENTS

PREPARATION



- *Do not attempt to work in deck area until the body retention sling has been installed.*
- *Do not step on or use any power cable as a handhold when the engine is running.*
- *Do not open any electrical cabinet covers or touch the retarding grid elements until all shutdown procedures have been followed.*
- *All removal, repairs and installation of propulsion system electrical components and cables must be performed by an electrical maintenance technician who is properly trained to service the system.*
- *In the event of a propulsion system malfunction, a qualified technician should inspect the truck and verify that the propulsion system does not have dangerous voltage levels present before repairs are started.*

NOTE: Depending on optional equipment installed on the truck, additional steps may be required before the decks or other structures are removed.

Prior to removal or repair procedures, it may be necessary to remove the body to provide clearance for any lifting equipment. If body removal is not required, the body should be raised and the body retention sling installed at the rear of the truck. Refer to “Securing the Dump Body” in Section A3.

After the truck is parked for repairs, the truck must be shut down properly to ensure the safety of those working in the areas of the deck, control cabinets and retarding grids. Refer to “Normal Engine Shutdown Procedure” and “Capacitor Discharge System” in Section A3.

DIAGONAL LADDER, GRILLE AND HOOD ASSEMBLY

Removal

1. Attach an appropriate lifting device to the diagonal ladder. The combined weight of diagonal ladder (1, Figure 2-1), support (2) and the handrails is approximately 300 kg (660 lb).
2. Remove ladder mounting hardware (3).
3. Lift the structure off the truck. Collect shims (6).
4. Disconnect wiring harnesses and remove cable clamps as necessary to allow removal of the grille and hood assembly.
5. Attach an appropriate lifting device to hood and grille assembly (4). The weight of the grille and hood assembly is approximately 500 kg (1,100 lb).
6. Remove all hood mounting hardware (5) on both sides of the hood.
7. Loosen the radiator bumpers on both decks.

8. Lift the grille and hood assembly from the truck and move it to a work area.

Installation

1. Use an appropriate lifting device to lift grille and hood assembly (4, Figure 2-1) into position on the truck.
2. Align the mounting hardware holes with the brackets attached to the radiator assembly. Install hood mounting hardware (5). Tighten the capscrews to the standard torque.
3. Adjust and tighten both radiator bumpers.
4. Use an appropriate lifting device to lift diagonal ladder (1) and support (2) into position over the mounting pads on the front bumper. Align the mounting holes and install shims (6) as needed to level the structure.
5. Install ladder mounting hardware (3). Tighten the capscrews to the standard torque.
6. Connect all cables, wiring harnesses, hoses and clamps that were removed previously.

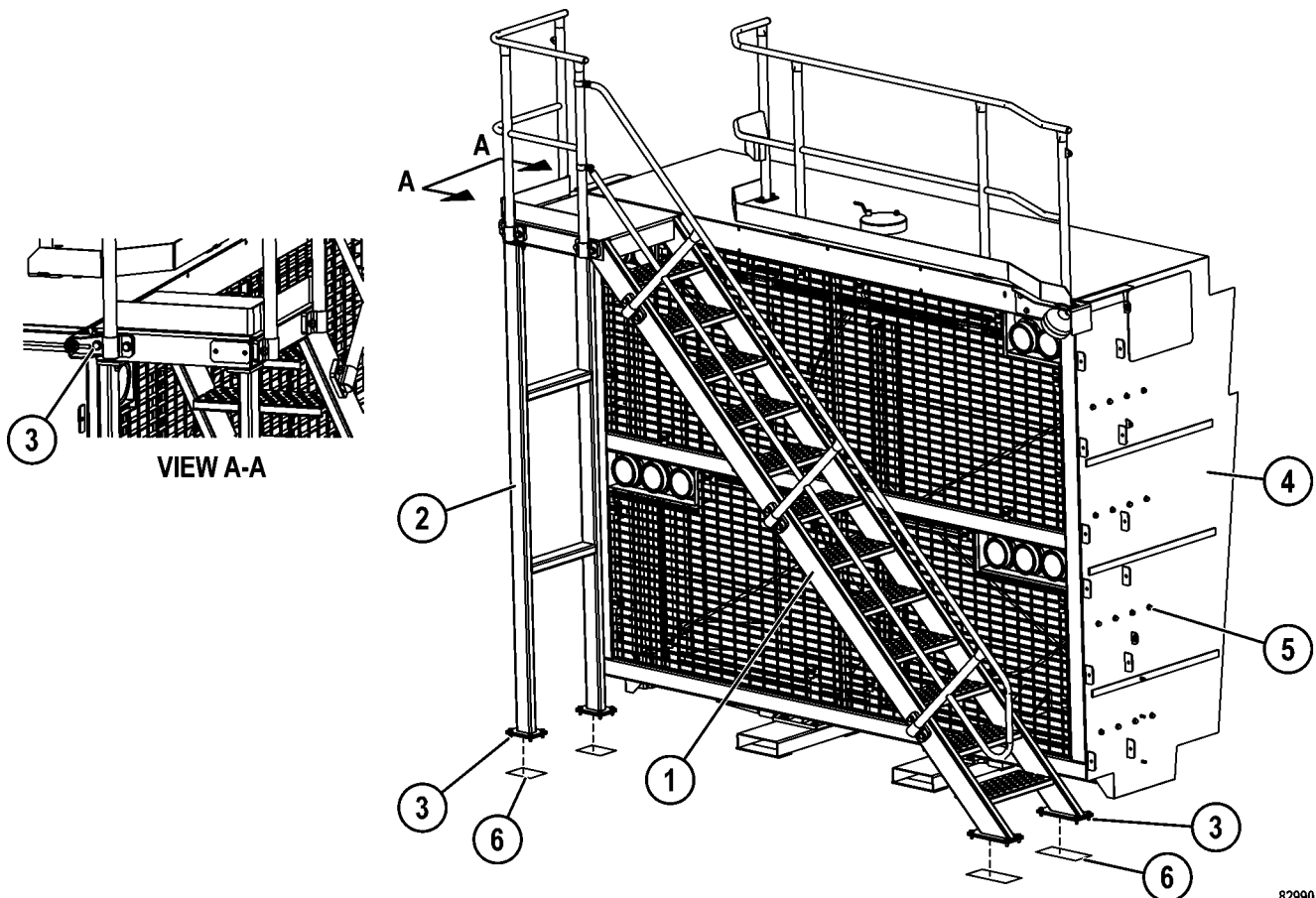


FIGURE 2-1. DIAGONAL LADDER, GRILLE & HOOD REMOVAL

82990

1. Diagonal Ladder
2. Support

3. Ladder Mounting Hardware
4. Grille and Hood Assembly

5. Hood Mounting Hardware
6. Shims

RH DECK AND RETARDING GRID

Removal

1. Remove the access covers from the retarding grid. Tag and disconnect all electrical leads in preparation for removal.

NOTE: Removal of the retarding grid is optional. If repairs to the retarding grid are required, refer to the applicable equipment manufacturer's publication for service and maintenance procedures.

2. Attach an appropriate lifting device to the lifting eyes on the retarding grid. The weight of the retarding grid (with handrails) is approximately 3 900 kg (8,600 lb).
3. Remove retarding grid mounting hardware (3, Figure 2-2) at six locations.
4. Lift the retarding grid clear of the deck and move it to a work area.
5. Disconnect deck lighting harness (4). Remove any wiring harnesses, cables or hoses as needed. The lighting harness and clamps do not require removal.

6. Install an appropriate lifting device to the lifting eyes at each corner of the deck and take up any slack. **Do not attach the lifting device to the handrails.** The weight of the RH deck (with handrails) is approximately 1 030 kg (2,270 lb).
7. Remove deck mounting hardware (2) and any shims at the deck support and the front upright.
8. Loosen the radiator bumpers on both decks.
9. Carefully lift deck structure (1) from the deck supports.

Installation

Repeat the removal procedure in reverse order to install the right deck and components. Clean all mount mating surfaces. Install shims with the deck mounting hardware, as needed, to keep the deck level. Tighten all hardware to standard torque.

Ensure that all wiring harnesses, clamps and hoses are undamaged and reinstalled securely. Replace any components as necessary.

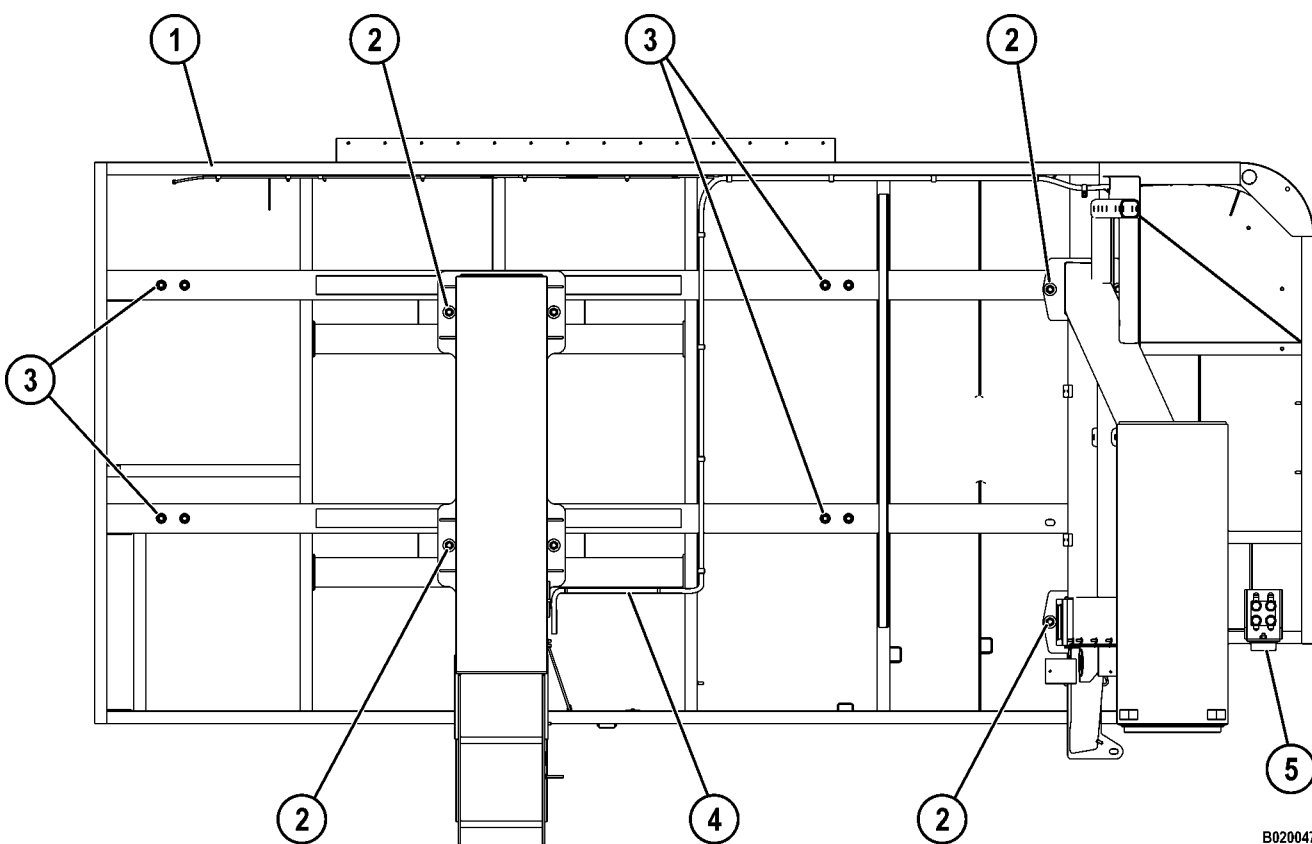


FIGURE 2-2. R.H. DECK STRUCTURE MOUNTING

1. Deck Structure
2. Deck Mounting Hardware

3. Retarding Grid Mounting Hardware

4. Deck Lighting Harness
5. Radiator Bumper

LH DECK

CAUTION

If air conditioning system components must be removed, refer to Section N4 for instructions on discharging the air conditioning system before disconnecting any air conditioning lines and servicing the air conditioning system after installation.

Removal

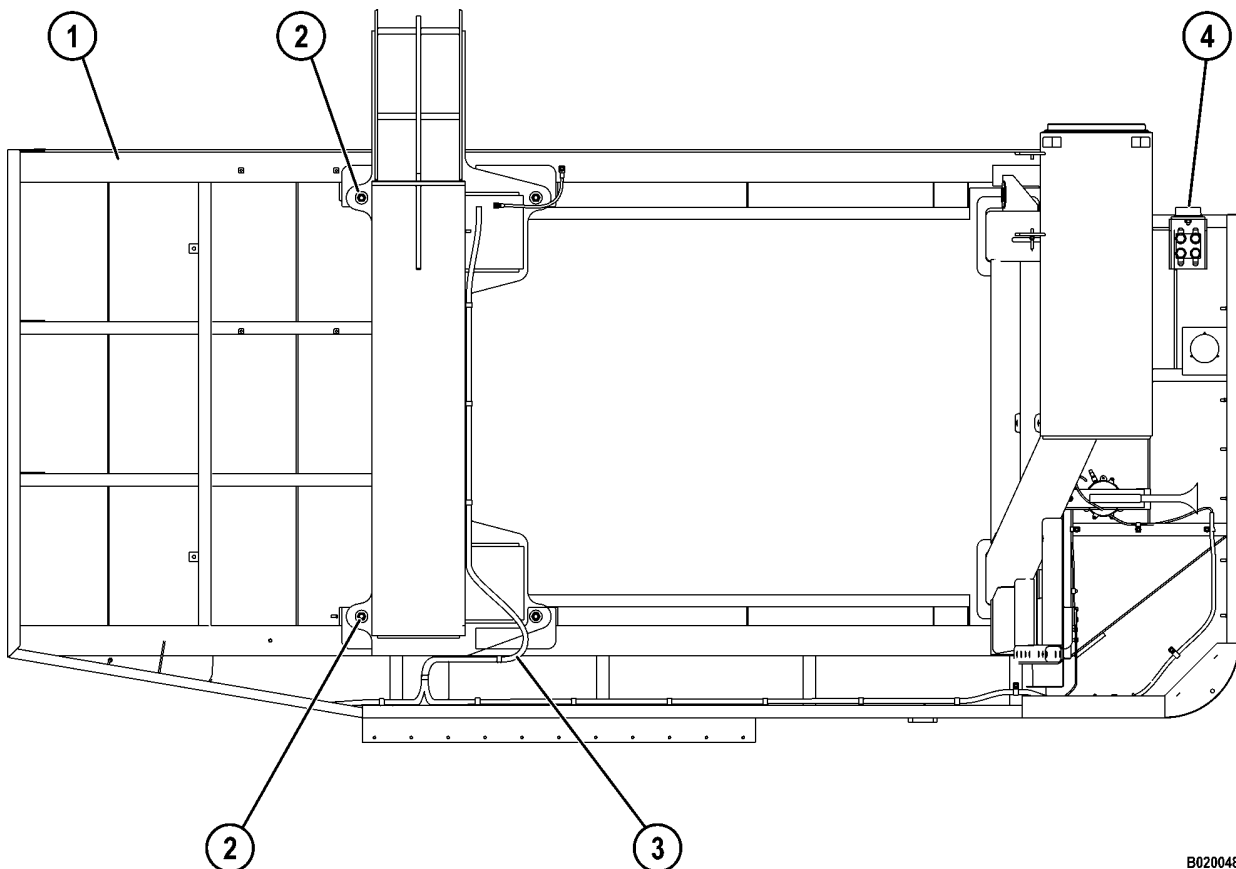
1. Remove the operator cab. Refer to Section N2, Truck Cab And Components, for cab removal and installation instructions.
2. Disconnect deck lighting harness (3, Figure 2-3). Remove any wiring harnesses, cables or hoses as needed. The lighting harness and clamps do not require removal.

3. Install an appropriate lifting device to the lifting eyes at each corner of the deck and take up any slack. **Do not attach the lifting device to the handrails.** The weight of the LH deck (with handrails) is approximately 820 kg (1,810 lb).
4. Remove deck mounting hardware (2) and any shims at the deck support and the front upright.
5. Loosen the radiator bumpers on both decks.
6. Carefully lift deck structure (1) from the deck supports.

Installation

Repeat the removal procedure in reverse order to install the left deck and components. Clean all mount mating surfaces. Install shims with the deck mounting hardware, as needed, to keep the deck level. Tighten all hardware to standard torque.

Ensure that all wiring harnesses, clamps and hoses are undamaged and reinstalled securely. Replace any components as necessary.



B020048

FIGURE 2-3. L.H. DECK STRUCTURE MOUNTING

- | | |
|---------------------------|--------------------------|
| 1. Deck Structure | 3. Deck Lighting Harness |
| 2. Deck Mounting Hardware | 4. Radiator Bumper |

CENTER DECK

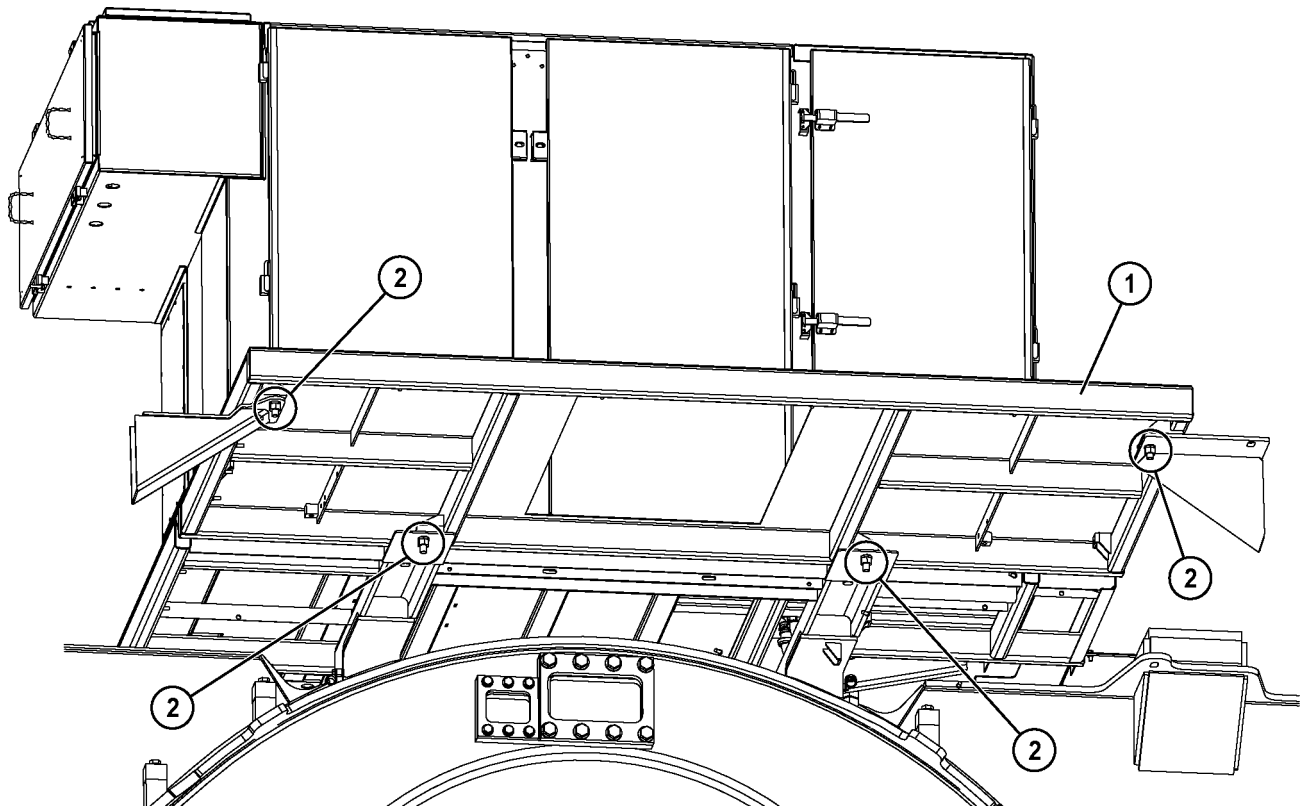
Removal

1. Install an appropriate lifting device to the deck and take up any slack. The weight of the center deck is approximately 285 kg (630 lb).
2. Remove deck mounting hardware (2, Figure 2-4) and any shims.
3. Ensure that all wiring harnesses, cables and hoses have been removed as necessary. Carefully lift center deck (1) from the truck.

Installation

Repeat the removal procedure in reverse order to install the center deck. Clean all mount mating surfaces. Install shims with the deck mounting hardware, as needed, to keep the deck level. Tighten all hardware to standard torque.

Ensure that all wiring harnesses, clamps and hoses are undamaged and reinstalled securely. Replace any components as necessary.



B020046

FIGURE 2-4. CENTER DECK MOUNTING

1. Center Deck

2. Deck Mounting Hardware

NOTES

SECTION B3 DUMP BODY INDEX

DUMP BODY	B3-3
Removal	B3-3
Installation	B3-4
BODY PADS	B3-5
Body Pad Shimming Procedure	B3-6
BODY GUIDE	B3-7
HOIST LIMIT SWITCH	B3-7
BODY-UP SWITCH	B3-7
BODY RETENTION SLING	B3-7
ROCK EJECTORS	B3-8
Inspection	B3-8

NOTES

DUMP BODY

DUMP BODY

⚠ WARNING

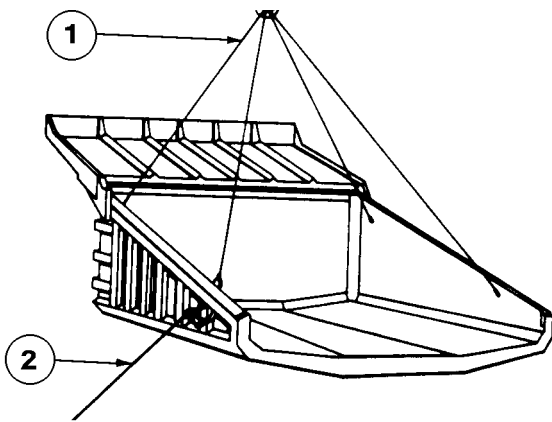
Inspect the condition and rating of all lifting devices, slings, chains, and cables. Refer to the manufacturer's manual for correct capacities and safety procedures when lifting components. Replace any questionable items.

Ensure that the lifting device is rated for at least a 45 ton capacity. Slings, chains, and cables used for lifting components must be rated to supply a safety factor of approximately 2X the weight being lifted. When in doubt as to the weight of components or any service procedure, contact the Komatsu area representative for further information.

Lifting eyes and hooks should be fabricated from the proper materials and rated to lift the load being placed on them.

Never stand beneath a suspended load. Use of guy ropes are recommended for guiding and positioning a suspended load.

Before raising or lifting the body, ensure there is adequate clearance between the body and overhead structures or electric power lines.



90909

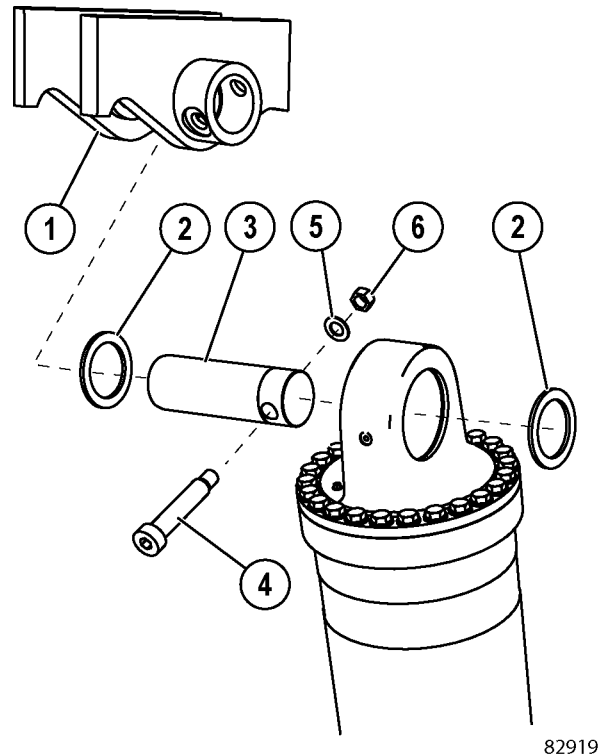
FIGURE 3-1. DUMP BODY REMOVAL

1. Lifting Cables 2. Guide Rope

Removal

1. Park truck on a hard, level surface and block all the wheels. Attach cables and a lifting device to the dump body and take up the slack as shown in Figure 3-1.
2. Remove the mud flaps from both sides of the body. Remove any electrical wiring and hoses that are attached to the body.
3. Attach chains around the upper end of both hoist cylinders to support them after the pivot pins are removed.
4. At the hoist cylinder upper mounting bracket, remove self-locking nut (6, Figure 3-2), flat washer (5) and shoulder bolt (4). Use a brass drift and hammer to drive pivot pin (3) from mounting bracket (1).

NOTE: Do not lose two spacers (2) between the cylinder bearing and mounting bracket (1).



82919

FIGURE 3-2. HOIST CYLINDER UPPER MOUNT

- | | |
|---------------------|---------------------|
| 1. Mounting Bracket | 4. Shoulder Bolt |
| 2. Spacer | 5. Flat Washer |
| 3. Pivot Pin | 6. Self-Locking Nut |

5. Remove self-locking nut (3, Figure 3-3), flat washer (2) and shoulder bolt (1) from both body pivots.
6. Attach a body pivot pin support fixture to the bracket on the underside of the dump body to aid in supporting the pin as it is removed.
7. Remove body pivot pin (4) far enough to allow shims (8) to drop out. Complete removal of pins is not necessary unless new pins are being installed.

NOTE: To prevent the body pivot pins from falling while removing the dump body, use chains to secure them to the truck.

8. Lift the dump body clear of the chassis and move it to a work area. Block up the body to prevent damage to the body guide.
9. Inspect bushings (5), (7), and (9), body ears (6), and the frame pivot for excessive wear or damage. Replace bushings and repair the pivot area as necessary.

Installation

1. Park the truck on a hard, level surface and block all the wheels. Attach cables and a lifting device to the dump body and take up the slack as shown in Figure 3-1. Lower the body over the truck frame and align the body ears with the frame pivot bores.
2. Install shims (8, Figure 3-3) in both body pivots, as required, to fill the outside gaps and center the body on the frame pivot.

NOTE: Do not install shims on the inside. A minimum of one shim is required at the outside end of both frame pivots.

3. Align the hole in body pivot pin (4) with the bolt hole in the pin retainer on the outer body pivot ear. Push the body pivot pin through the outer body pivot ear, the frame pivot, and the inner body pivot ear.
4. Install shoulder bolt (1) through the body pivot pin. Install flat washer (2) and self-locking nut (2). Tighten the nut and flat washer securely against the shoulder of the bolt.
5. Align the hoist cylinder bearing eye with the bore of hoist cylinder upper mounting bracket (1, Figure 3-2).
6. Align the bolt hole in pivot pin (3) with the bolt hole in mounting bracket (1). Install the pivot pin through the first spacer (2), then align the second spacer with the bore on the other side of hoist cylinder bearing eye. Finish installing the pin through second spacer (2).
7. Install shoulder bolt (4) through each pin.

NOTE: Do not use a flat washer under the head of the shoulder bolt. This will ensure that the shoulder bolt does not run out of threads when tightening.

8. Install flat washer (5) and self-locking nut (6) onto the threaded end of each shoulder nut bolt. Tighten the nuts and flat washers securely against the shoulder of the bolts.
9. Install all mud flaps, electrical wiring and hoses to the body.

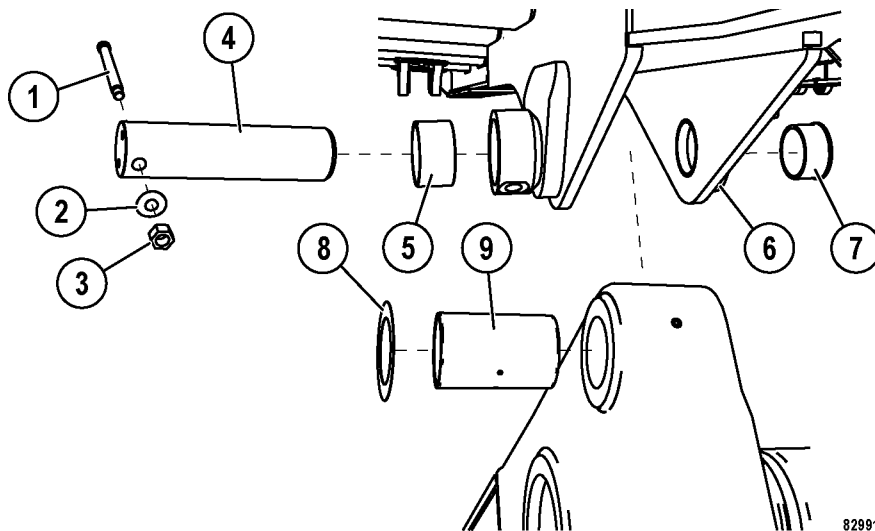


FIGURE 3-3. DUMP BODY PIVOT PIN

- | | | |
|---------------------|-------------------|-----------------------|
| 1. Shoulder Bolt | 4. Body Pivot Pin | 7. Pivot Bushing |
| 2. Flat Washer | 5. Pivot Bushing | 8. Shim |
| 3. Self-Locking Nut | 6. Body Pivot Ear | 9. Body Pivot Bushing |

BODY PADS

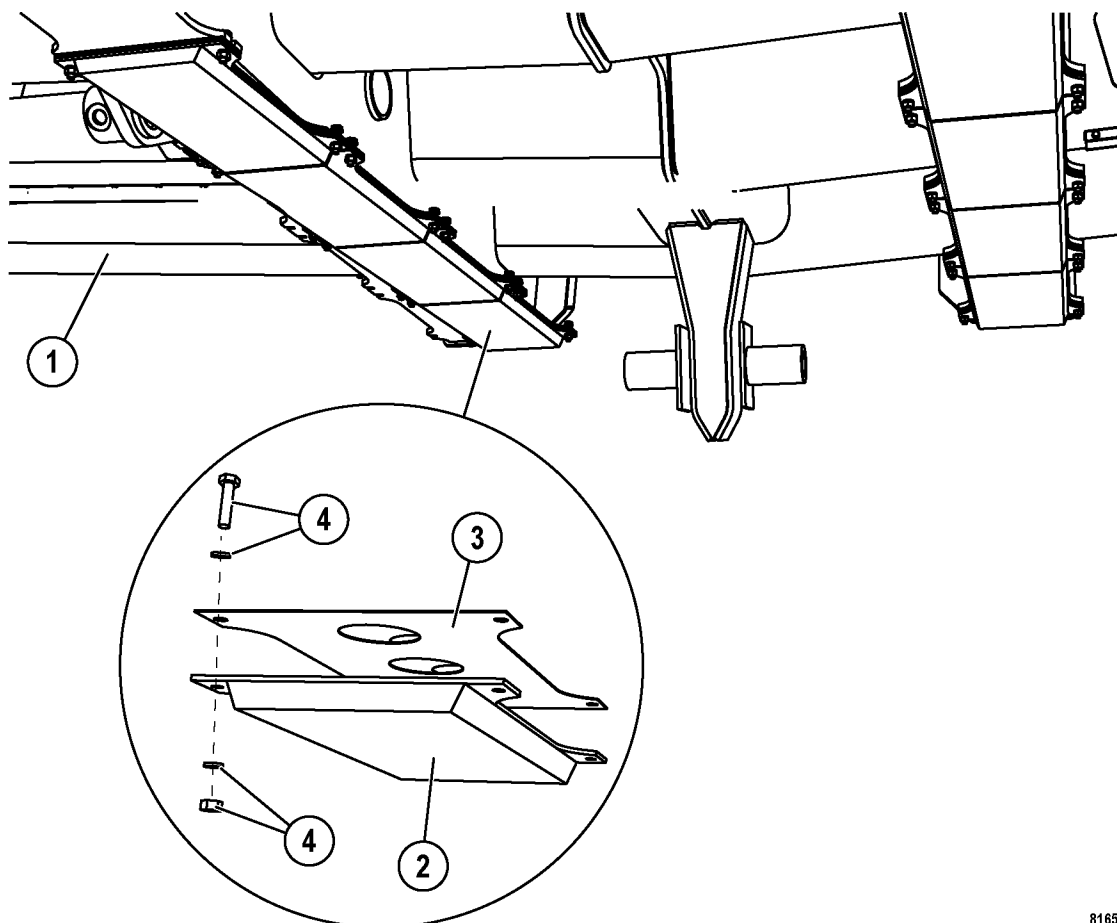
NOTE: It is not necessary to remove the dump body to replace the body pads. Body pads should be inspected during scheduled maintenance inspections and replaced if worn excessively.

1. Raise unloaded dump body (1, Figure 3-4) to a height that is sufficient to allow access to all of the body pads.

⚠ WARNING

To avoid serious personal injury or death, the body retention sling must be installed anytime personnel are required to perform maintenance on the truck with the dump body in the raised position.

2. Install the body retention sling. Refer to "Securing The Dump Body" in Section A3, General Safety & Operating Instructions, for the body retention sling installation procedure.
3. Place blocks between the body and frame. Secure the blocks in place.
4. Remove mounting hardware (4).
5. Remove body pad (2) and shims (3). Note the number of shims installed at each pad location. The rear pad on each side should have one less shim than the other pads.
6. Install new body pads with the same number of shims that were removed in step 3.
7. Install mounting hardware (4). Tighten the nuts to **88 N·m (65 ft lb)**.
8. Remove the blocks and body retention sling. Lower the dump body onto the frame.



81653

FIGURE 3-4. BODY PAD INSTALLATION

1. Dump Body
2. Body Pad

3. Shim
4. Mounting Hardware

Body Pad Shimming Procedure

Proper contact between the body pad and frame is required to ensure maximum pad life. All pads (except the rear pad on each side) should contact the frame with approximately equal compression of the rubber. A gap of approximately 1.9 mm (0.075 in.) is required at each rear pad. This can be accomplished by using one less shim at each rear pad.

If pad contact appears to be unequal, perform the following shimming procedure.

1. Park the truck on a flat, level surface.
2. Raise the unloaded dump body to a height that is sufficient to allow access to all of the body pads.

! WARNING

To avoid serious personal injury or death, the body retention sling must be installed anytime personnel are required to perform maintenance on the truck with the dump body in the raised position.

3. Install the body retention sling. Refer to "Securing The Dump Body" in Section A3, General Safety & Operating Instructions, for the body retention sling installation procedure.
4. Remove all of the body pads and shims.
5. Bolt one spacer block (XC2293) to each side of the body. Use the pad mounting holes closest to the front of the dump body.
6. Remove the body retention sling. Lower the dump body onto the spacer blocks.
7. Check the dump body's position and fit on the truck. If there is any interference when the body is on the spacer blocks, contact your local Komatsu distributor to resolve the issue.
8. With the body lowered, measure the distance from the frame rail to each pair of bolt holes at the dump body's pad mounting locations. There will be a total of eight measurements "A" per side. Refer to Figure 3-5.

9. Subtract the body pad thickness of 39.7 mm (1.56 in.) from each measurement that was recorded in Step 8. There will be a total of eight measurements "B" per side. Refer to Figure 3-5.

$$\text{"A"} - 39.7 \text{ mm (1.56 in.)} = \text{"B"}$$

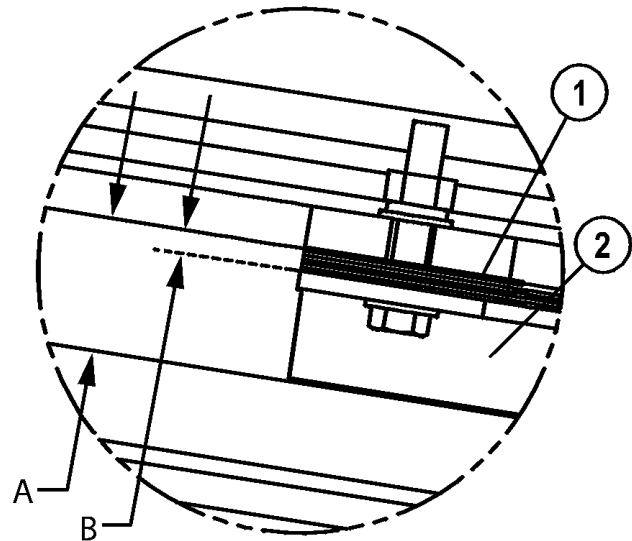
10. Divide each measurement "B" by the single shim thickness of 1.5 mm (0.06 in.) to determine the number of shims required for each mounting location.

$$\text{"B"} / 1.5 \text{ mm (0.06 in.)} = \text{number of shims}$$

11. Remove one shim from the calculation for the most rearward body pad on both sides.

NOTE: Using half shims is allowed if necessary. Half shims must be installed at the top of the stack.

NOTE: The frame rail and the body bolster do not have to be parallel.



72226

FIGURE 3-5. MEASUREMENT DETAILS

1. Shims

2. Body Pad

BODY GUIDE

1. Inspect the body guide wear points each time a body pad inspection is performed. Body guide (2, Figure 3-6) should be centered between wear plates (3) with a maximum gap of 4.8 mm (0.19 in.) at each side (new wear plates).
2. If the gap becomes excessive, install new parts.

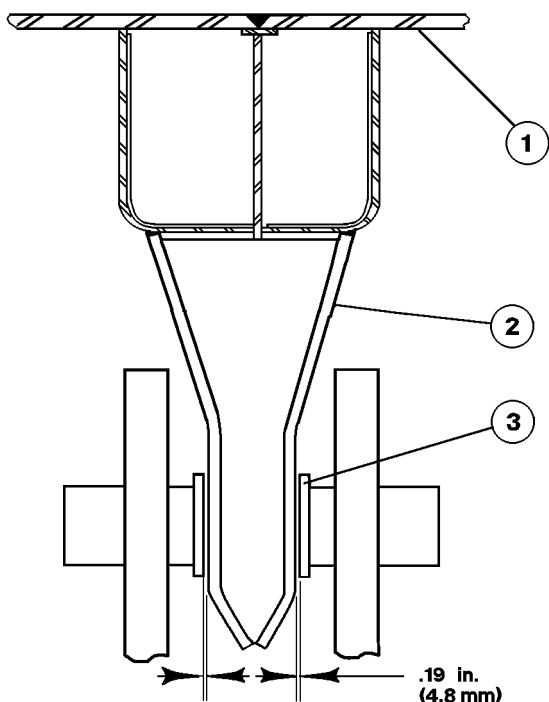


FIGURE 3-6. BODY GUIDE

- | | |
|---------------|---------------------------|
| 1. Dump Body | 3. Body Guide Wear Plates |
| 2. Body Guide | |

HOIST LIMIT SWITCH

Refer to Section D3, 24VDC Electrical System Components, for the hoist limit switch adjustment procedure.

BODY-UP SWITCH

Refer to Section D3, 24VDC Electrical System Components, for the body-up switch adjustment procedure.

BODY RETENTION SLING

⚠ WARNING

To avoid serious personal injury or death, the body retention sling must be installed anytime personnel are required to perform maintenance on the truck with the dump body in the raised position.

Refer to "Securing The Dump Body" in Section A3, General Safety & Operating Instructions, for the body retention sling installation procedure.

ROCK EJECTORS

Rock ejectors are placed between the rear dual wheels to keep rocks or other material from lodging between the tires. Failure to maintain the rock ejectors could allow debris to build up between the dual wheels and cause damage to the tires.

Inspection

1. The rock ejectors must be positioned on the center line between the rear tires within 6.0 mm (0.25 in.).
2. With the truck parked on a level surface, rock ejector arm (1, Figure 3-7) should be approximately 454.5 mm (17.9 in.) from wheel housing (2).

NOTE: With the rock ejector arm hanging vertical as shown in Figure 3-7, there must be no gap at stop block (3, Figure 3-8).

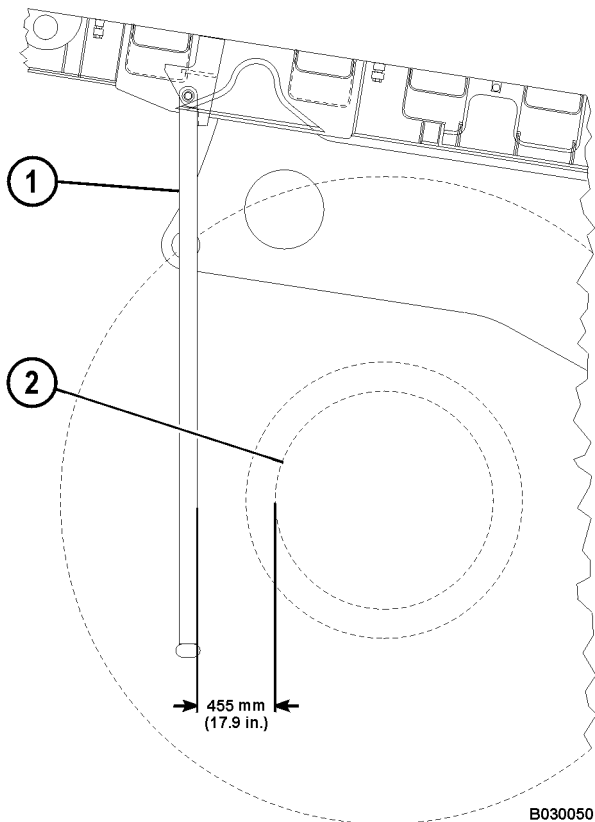


FIGURE 3-7. ROCK EJECTOR INSTALLATION

- | | |
|---------------------|------------------|
| 1. Rock Ejector Arm | 2. Wheel Housing |
|---------------------|------------------|

3. If the rock ejector arm becomes bent, it must be removed and straightened.
4. At each shift change, inspect mounting brackets (4, Figure 3-8), pins (2) and stop blocks (3) for wear and damage. Repair as necessary.

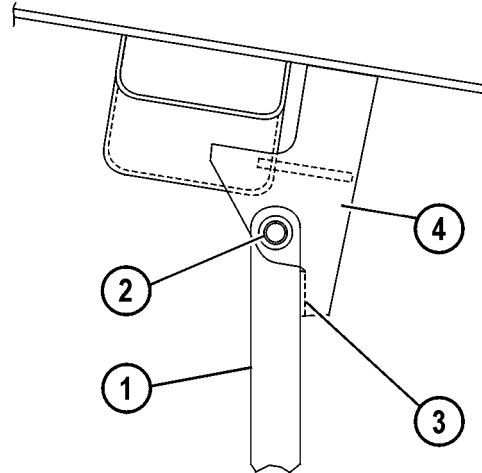


FIGURE 3-8. ROCK EJECTOR MOUNTING BRACKET

- | | |
|---------------------|---------------------|
| 1. Rock Ejector Arm | 3. Stop Block |
| 2. Pin | 4. Mounting Bracket |

SECTION B4

FUEL TANK

INDEX

FUEL TANK	B4-3
Removal	B4-3
Repair	B4-3
Cleaning	B4-3
Installation	B4-3
FUEL TANK VENT ASSEMBLY	B4-5
Disassembly	B4-5
Assembly	B4-5
FUEL GAUGE SENDER	B4-6
Removal	B4-6
Installation	B4-6
QUICK FILL FUEL RECEIVERS	B4-6

NOTES

FUEL TANK

FUEL TANK

Removal

1. Raise the truck body and install the body retention sling.
2. Loosen filler cap (5, Figure 4-1) and open the drain cock on the bottom of fuel tank (1) to drain the fuel from the tank into clean containers. Tighten the filler cap when the fuel is completely drained.
3. Disconnect the wire harness from hoist circuit filter assemblies (11) and steering circuit filter assembly (12). Remove ground wire (13).
4. If equipped, close the inline shutoff valves. Remove fuel return hose (6), fuel supply hose (7) and quick fill hose (10). Cap the hoses and tank fittings to prevent contamination.
5. Remove hoist circuit filter assemblies (11) and steering circuit filter assembly (12) from the fuel tank. Support the filter assemblies by placing a chain over the frame rail. It is not necessary to remove the hydraulic hoses.



The weight of the empty fuel tank is approximately 1 690 kg (3,725 lb). Use lifting devices with adequate capacity to remove the fuel tank.

6. Attach a lifting device to the lifting eyes on each side of the tank.
7. Remove capscrews (14), lockwashers (15), flat washers (16), large washers (17) and rubber mounts (16). Remove capscrews (2), lockwashers (3) and mounting caps (4).
8. Lift the fuel tank from the brackets and move it to a work area.

Repair

If a tank has been damaged and requires structural repair, perform such repairs before final cleaning.



If a tank is to be weld repaired, special precautions are necessary to prevent fire or explosion. Consult local authorities for safety regulations before proceeding.

Cleaning

The fuel tank has a drain cock and a cleaning port in the side that allow steam or solvent to be used for cleaning tanks that have accumulated foreign material.

It is not necessary to remove the tank from the truck for cleaning of sediment. However, rust and scale on the walls and baffles may require complete tank removal. This allows cleaning solutions to be in contact with all interior surfaces by rotating the tank in various positions.

Before a cleaning procedure of this type, the fuel gauge sender, vent assembly (8, Figure 4-1) and all hose connections must be removed and temporarily sealed. After cleaning is complete, the temporary plugs can be removed.

If the tank is to remain out of service, a small amount of light oil should be sprayed into the tank to prevent rust. Seal all openings for rust prevention.

Installation

1. Thoroughly clean the frame mounting brackets and the mounting hardware holes. Re-tap the threads if damaged.

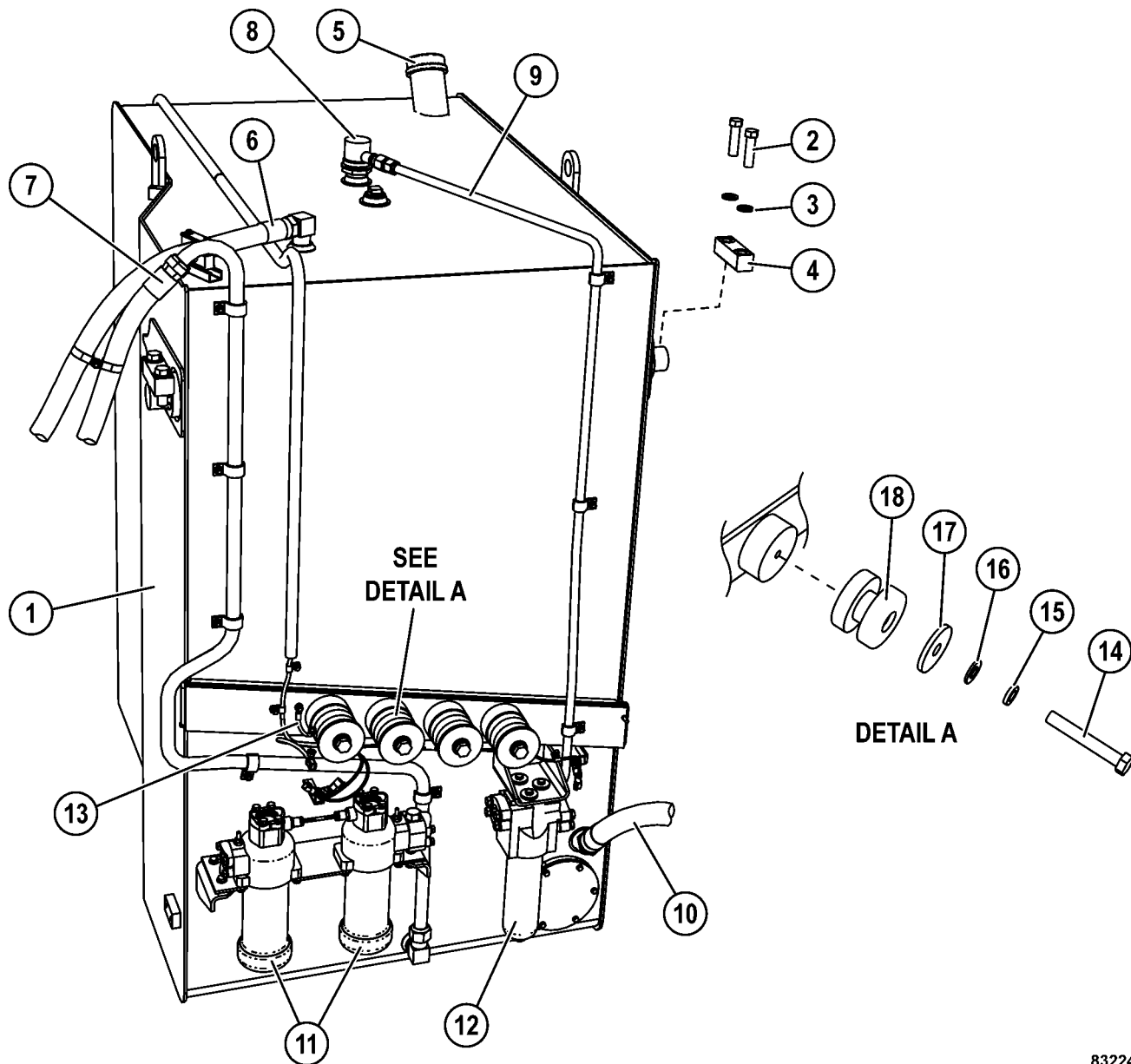


The weight of the empty fuel tank is approximately 1 690 kg (3,725 lb). Use lifting devices with adequate capacity to install the fuel tank.

2. Attach a lifting device to the lifting eyes on each side of the tank. Lift the fuel tank into position over the frame trunnion mounts and lower it into position.
3. Install mounting caps (4, Figure 4-1), lockwashers (3) and capscrews (2). Do not tighten the capscrews at this time.
4. Install rubber mounts (18), large washers (17), flat washers (16), lockwashers (15) and capscrews (14). Tighten the lower mounting capscrews to **800 ± 80 N·m (590 ± 59 ft lb)**.
5. Tighten the upper mounting capscrews to **712 ± 72 N·m (525 ± 53 ft lb)**.
6. Install hoist circuit filter assemblies (11) and steering circuit filter assembly (12) on the tank.

7. Connect fuel return hose (6) fuel supply hose (7) and quick fill hose (10). Open the inline shutoff valves, if equipped.

8. Install ground wire (13). Connect the wire harness to hoist circuit filter assemblies (11) and steering circuit filter assembly (12).



83224

FIGURE 4-1. FUEL TANK REMOVAL & INSTALLATION

- | | | |
|---------------------|--------------------------------------|------------------|
| 1. Fuel Tank | 7. Fuel Supply Hose | 13. Ground Wire |
| 2. Capscrew | 8. Vent Assembly | 14. Capscrew |
| 3. Flat Washer | 9. Overflow Tube | 15. Lockwasher |
| 4. Mounting Cap | 10. Quick Fill Hose | 16. Flat Washer |
| 5. Filler Cap | 11. Hoist Circuit Filter Assemblies | 17. Large Washer |
| 6. Fuel Return Hose | 12. Steering Circuit Filter Assembly | 18. Rubber Mount |

FUEL TANK VENT ASSEMBLY

NOTE: The relief pressure of the fuel tank vent assembly is 62 - 76 kPa (9 - 11 psi).

Disassembly

1. Remove fitting (1, Figure 4-2) from body (2).
2. Remove ball cage (8), solid ball (9) and float balls (10).
3. Unscrew end fitting (5) from body (2).
4. Remove stem (6) and valve spring (3).

Assembly

1. Clean and inspect all parts. If any parts are damaged, replace the entire assembly.
2. Place valve spring (3) into position in body (2).
3. Insert stem (6) into end fitting (5).
4. Screw end fitting (5) into body (2). Ensure the components are properly aligned and seated.
5. Install fitting (1) in body (2).
6. Insert the balls into ball cage (8) with solid ball (9) on top.
7. Insert the ball cage onto the stem. A minimum of two cage coils must be seated in the groove on the stem. Ensure the solid ball is able to seat properly on the stem. If not, adjust the cage accordingly.

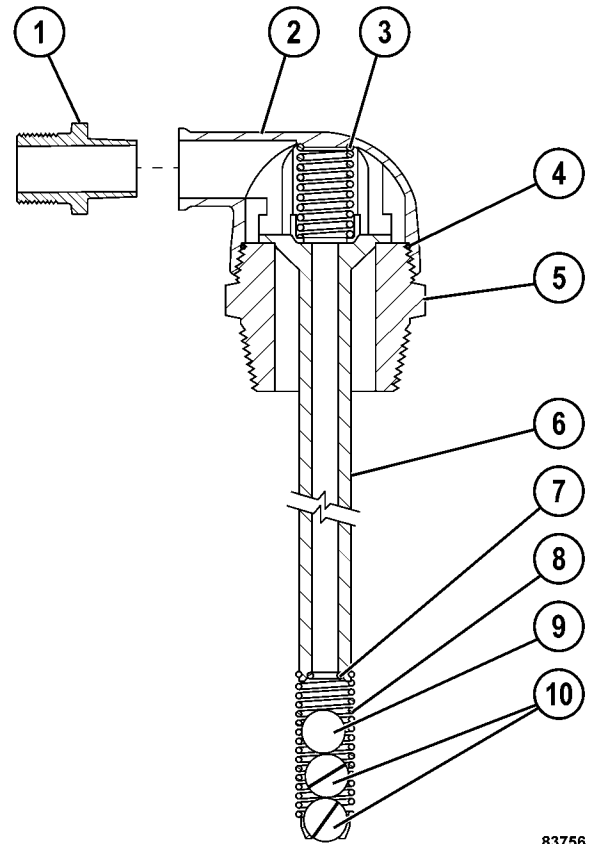


FIGURE 4-2. VENT ASSEMBLY (PB9887)

- | | |
|-----------------|----------------|
| 1. Fitting | 6. Stem |
| 2. Body | 7. O-Ring |
| 3. Valve Spring | 8. Ball Cage |
| 4. O-Ring | 9. Solid Ball |
| 5. End Fitting | 10. Float Ball |

FUEL GAUGE SENDER

Fuel gauge sender (1, Figure 4-3) provides an electrical signal to operate the fuel gauge on the instrument panel.

Removal

1. Drain the fuel below the level of the fuel gauge sender.
2. Disconnect the wires from terminals (3).
3. Remove sender mounting hardware (2). Carefully remove the sender and gasket.

Installation

1. Clean the mating surfaces. Install a new gasket.
2. Install fuel gauge sender (1, Figure 4-3) in the tank. Ensure the float is oriented properly and moves freely.
3. Install sender mounting hardware (2) and tighten the capscrews to the standard torque.
4. Connect the wires to terminals (3).
5. Fill the fuel tank and check for leaks.

QUICK FILL FUEL RECEIVERS

Fuel receiver assembly (4, Figure 4-3) is mounted on the front of the fuel tank. Fuel receiver assembly (5) is mounted to the bottom of the hydraulic tank. It is connected to the rear of the fuel tank by hose (6).

The quick fill fuel system is a pressurized system that is rated at a maximum of 568 l/m (150 gpm). When filling the tank, vent assembly (8, Figure 4-1) allows air to escape to prevent the tank from over-pressurizing. When the fuel level reaches the float balls in the vent, the balls rise with the fuel. When the fuel level reaches its maximum level, the float balls block air flow out of the vent, causing pressure to rise in the fuel tank. When pressure reaches 55 - 70 kPa (8 - 10 psi), the fuel nozzle should turn off.

If the nozzle fails to turn off and fuel continues to fill the tank, a relief valve in the vent will open at 79 kPa (11.5 psi) to prevent over-pressurization of the tank. If filling persists and the fuel reaches the top of the tank, fuel will flow out of the vent until the nozzle is turned off.

If fuel spills from the vent, or if the tank does not completely fill, check the vent to see whether the float balls are in place and overflow tube (9) is clean. If the vent is operating properly, the problem will most likely be in the fuel supply system.

NOTE: Keep the cap on each fuel receiver to prevent dirt buildup in valve area and nozzle grooves.

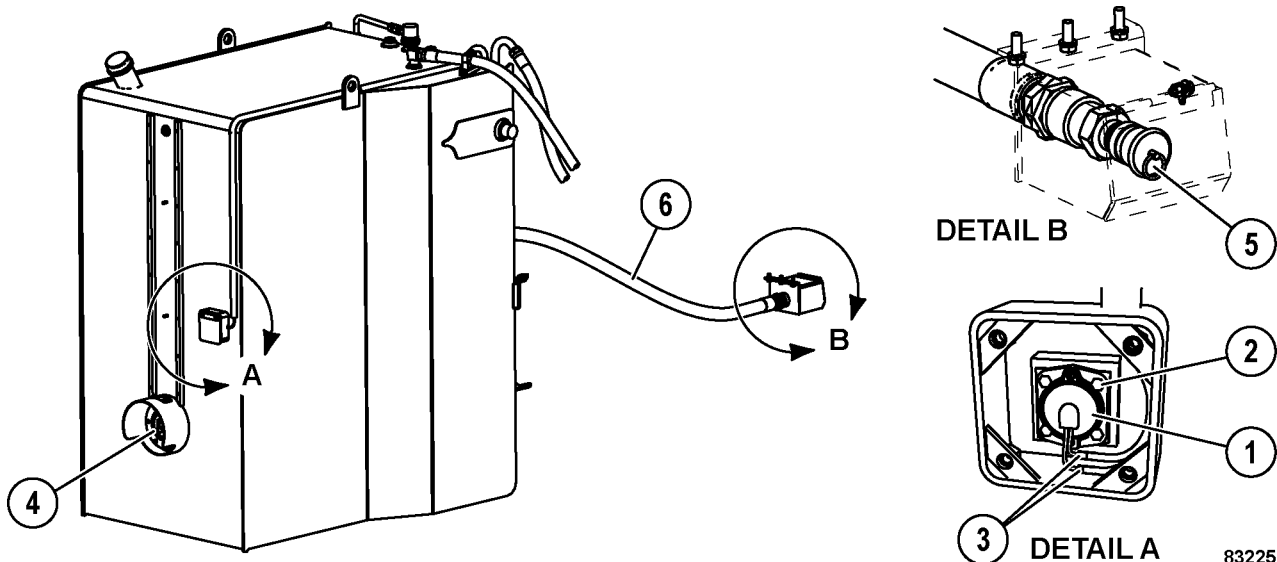


FIGURE 4-4. FUEL GAUGE SENDER & QUICK FILL FUEL RECEIVERS

1. Fuel Gauge Sender
2. Mounting Hardware
3. Terminals

4. RH Fuel Receiver Assembly
5. LH Fuel Receiver Assembly
6. Quick Fill Hose

SECTION B4

FUEL TANK

INDEX

FUEL TANK	B4-2
Removal	B4-2
Installation	B4-2
Repair	B4-4
Cleaning	B4-4
FUEL TANK BREATHER VALVE	B4-4
Disassembly	B4-4
Assembly	B4-4
FUEL GAUGE SENDER	B4-5
Removal	B4-5
Operation	B4-6
Specifications:	B4-6
QUICK FILL FUEL RECEIVERS	B4-7

FUEL TANK

Removal

1. Raise the truck body and install the body retention sling.
2. Drain sediment from the fuel tank and dispose of properly.
3. Loosen filler cap (5, Figure 4-1) and open the drain cock on the bottom of fuel tank (1) to drain the fuel from the tank into clean containers. Tighten the filler cap when the fuel is completely drained.
4. Disconnect the wire harness from hoist circuit filter assemblies (11) and steering circuit filter assembly (12). Remove ground wire (13).
5. If equipped, close the in-line shutoff valves. Remove fuel return hose (6), fuel supply hose (7) and quick fill hose (10). Cap the hoses and tank fittings to prevent contamination.
6. Remove hoist circuit filter assemblies (11) and steering circuit filter assembly (12) from the fuel tank. Support the filter assemblies by placing a chain over the frame rail. It is not necessary to remove the hydraulic hoses.

WARNING

The weight of the empty fuel tank is approximately 1 711 kg (3,772 lb). Use lifting devices with adequate capacity to remove the fuel tank.

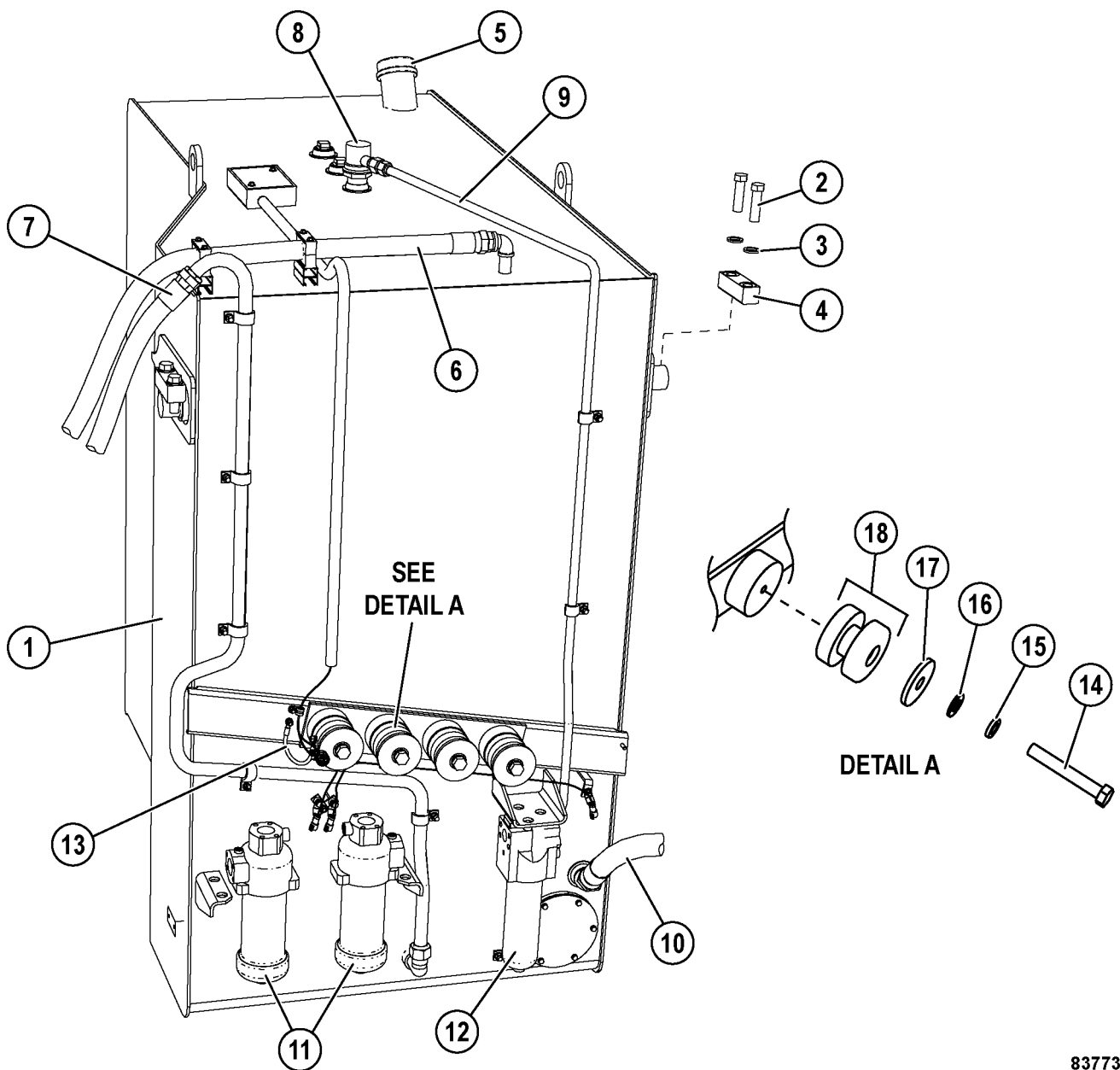
7. Attach a lifting device to the lifting eyes on each side of the tank.
8. Remove cap screws (14), lock washers (15), flat washers (16), large washers (17) and rubber mounts (18). Remove cap screws (2), flat washers (3) and mounting caps (4).
9. Lift the fuel tank from the brackets and move it to a work area.
10. Inspect rubber mounts (18) and replace if necessary.

Installation

WARNING

The weight of the empty fuel tank is approximately 1 711 kg (3,772 lb). Use lifting devices with adequate capacity to remove the fuel tank.

1. Thoroughly clean the frame mounting brackets and the mounting hardware holes. Re-tap the threads if damaged.
2. Attach a lifting device to the lifting eyes on each side of the tank. Lift the fuel tank into position over the frame trunnion mounts and lower it into position.
3. Install mounting caps (4, Figure 4-1), flat washers (3) and cap screws (2). Do not tighten the cap screws at this time.
4. Install rubber mounts (18), large washers (17), flat washers (16), lockwashers (15) and capscrews (14). Tighten the lower mounting cap screws to **800 ± 80 N·m (590 ± 59 ft lb)**.
5. Tighten upper mounting cap screws (2) to **712 ± 72 N·m (525 ± 53 ft lb)**.
6. Attach ground wire (17), and connect wire harness (13). Install wire harness clamps.
7. Attach fuel supply hose (7), return hose (6) and quick fill hose (10). If equipped, open the in line shutoff valves.
8. Attach hoist circuit filter assemblies (11) and steering circuit filter assembly (12) to the fuel tank.
9. Refill the fuel tank with clean fuel.



83773

FIGURE 4-1. FUEL TANK

- | | | |
|---------------------|--------------------------------------|------------------|
| 1. Fuel Tank | 7. Fuel Supply Hose | 13. Ground Wire |
| 2. Cap Screw | 8. Vent Assembly | 14. Cap Screw |
| 3. Flat Washer | 9. Overflow Tube | 15. Lock Washer |
| 4. Mounting Cap | 10. Quick Fill Hose | 16. Flat Washer |
| 5. Filler Cap | 11. Hoist Circuit Filter Assemblies | 17. Large Washer |
| 6. Fuel Return Hose | 12. Steering Circuit Filter Assembly | 18. Rubber Mount |

Repair

If a fuel tank has been damaged and requires structural repair, perform such repairs before final cleaning.

⚠ WARNING

If a tank is to be weld repaired, special precautions are necessary to prevent fire or explosion. Consult local authorities for safety regulations before proceeding.

Cleaning

The fuel tank is provided with a drain cock and a cleaning port in the side that allows steam or solvent to be utilized in cleaning tanks that have accumulated foreign material.

It is not necessary to remove the tank from the truck for cleaning of sediment. However, rust and scale on the walls and baffles may require complete tank removal. This allows cleaning solutions to be in contact with all interior surfaces by rotating the tank in various positions.

Prior to a cleaning procedure of this type, the vent assembly, fuel gauge sender and hose connections must be removed and temporarily sealed. After all scale, rust, and foreign material has been removed, the temporary plugs can be removed.

If the tank is to remain out of service, a small amount of light oil should be sprayed into the tank to prevent rust. Seal all openings for rust prevention.

FUEL TANK BREATHER VALVE

NOTE: The relief pressure of the fuel tank breather valve is 70 - 89 kPa (10 - 13 psi).

Disassembly

1. Remove fitting (1, Figure 4-2).
2. Remove ball cage (8), solid ball (9) and float balls (10).
3. Unscrew end fitting (5) from body (2).
4. Remove stem (6) and valve spring (3).

Assembly

1. Clean and inspect all parts. If any parts are damaged, replace the entire assembly.
2. Place valve spring (3) into position in body (2).
3. Insert stem (6) into end fitting (5).
4. Screw end fitting (5) into body (2). Ensure the components are properly aligned and seated.
5. Install fitting (1).
6. Insert the balls into ball cage (8) with solid ball (9) on top.
7. Insert the ball cage onto the stem. A minimum of two cage coils must be seated in the groove on the stem. Ensure the solid ball is able to seat properly on the stem. If not, adjust the cage accordingly.

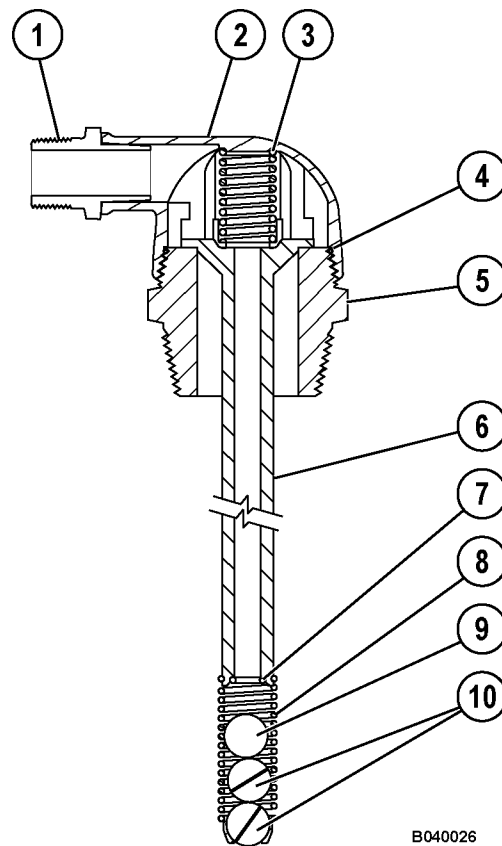


FIGURE 4-2. BREATHER VALVE

- | | |
|-----------------|----------------|
| 1. Fitting | 6. Stem |
| 2. Body | 7. O-Ring |
| 3. Valve Spring | 8. Ball Cage |
| 4. O-Ring | 9. Solid Ball |
| 5. End Fitting | 10. Float Ball |

FUEL GAUGE SENDER

Fuel gauge sender (1, Figure 4-3) is mounted on the top of the fuel tank and provides an electrical signal to operate the fuel level gauge on the instrument panel.

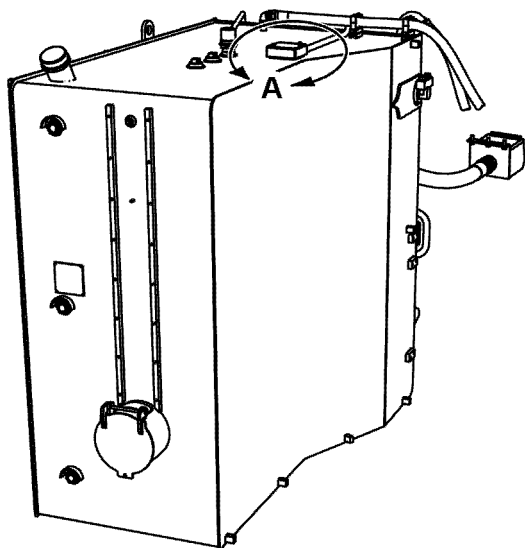
Removal

1. Raise the dump body all the way. Secure the dump body in the fully raised position using the body sling.



Never work under a raised body unless safety device(s) are in position to prevent the dump body from lowering.

2. Remove the fuel gauge cover from the top of the fuel tank.
3. Disconnect wiring harness (4, Figure 4-3) from connector (3) on the fuel gauge sender.
4. Remove fuel gauge sender mounting hardware (2). Carefully remove sender (1) and the gasket.

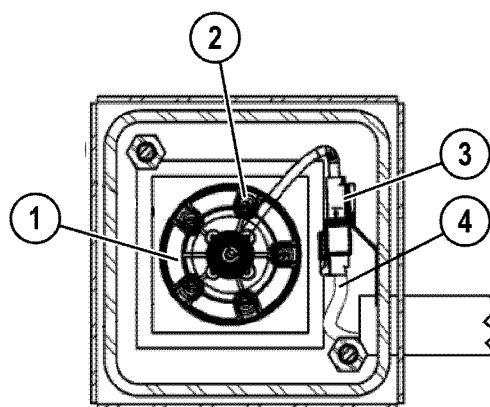


Installation



Never work under a raised body unless safety device(s) are in position to prevent the dump body from lowering.

1. Clean the fuel gauge sending mating surface on the fuel tank. If reinstalling the same fuel gauge sender, clean the mating surface on the sender and install a new gasket (2, Figure 4-4). (New fuel gauge senders are supplied with a new gasket.)
2. Install fuel gauge sender (1, Figure 4-3) into the tank. Ensure the guide on the bottom of the fuel sender is seated in the cup in the bottom of the fuel tank. If it is not seated in the cup, the mounting flange on the sender will not be level with the mounting surface on the fuel tank.
3. Install sender mounting hardware (2) and tighten the cap screws to the standard torque.
4. Connect wiring harness (4) to connector (3) attached to the sender.
5. Install the cover on the top of the sender.
6. Remove the body up sling and completely lower the dump body.



DETAIL A

84102

FIGURE 4-3. FUEL GAUGE SENDER

- | | |
|------------------------|-------------------|
| 1. Fuel Gauge Sender | 3. Wire Connector |
| 2. Mounting Cap Screws | 4. Wire Harness |

Operation

Float (4, Figure 4-4) moves up and down as the fuel level changes. The resistance value in the fuel gauge sender changes with the level of fuel in the tank.

The fuel gauge sender is equipped with one 30 ohm resistor (2, Figure 4-5) which is always connected to the sender circuit.

There are also 16 bi-stable reed switches (3) evenly spaced inside the fuel sender tube. Each reed switch is wired to a 13 ohm resistor (1). As the float moves up with the fuel level, the float will open the individual reed switches, and 13 ohms of resistance is added to the circuit. When the float is at the very top, all of the reed switches will be latched open.

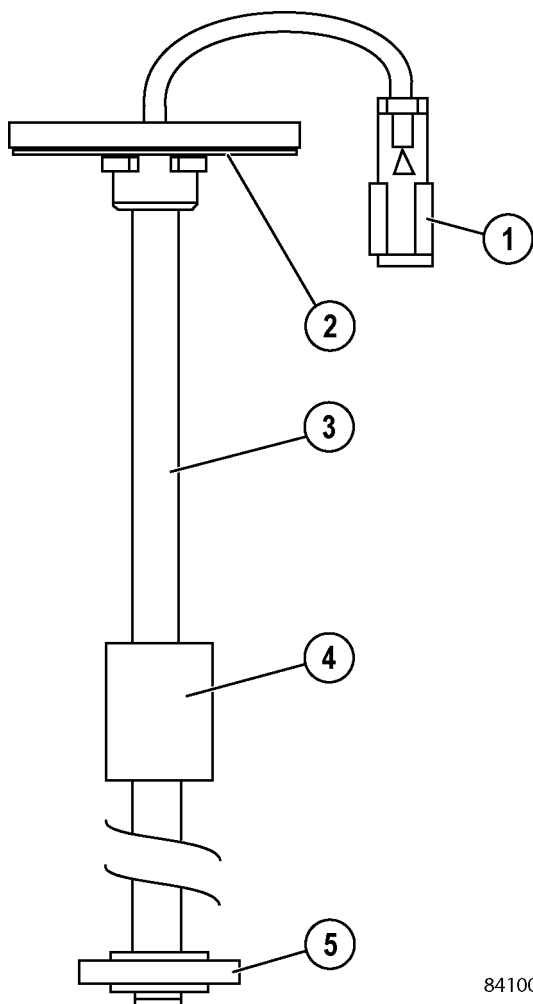


FIGURE 4-4. FUEL GAUGE SENDER

- | | |
|--------------|----------|
| 1. Connector | 4. Float |
| 2. Gasket | 5. Guide |
| 3. Tube | |

As the fuel level goes down, the float will move downward past the reed switches again. This will switch them to the closed position and it will reduce the resistance value by 13 ohms as each switch is closed.

Specifications:

Minimum Resistance: 30 ohms (float at bottom)

Maximum Resistance: 238 ohms (float at top)

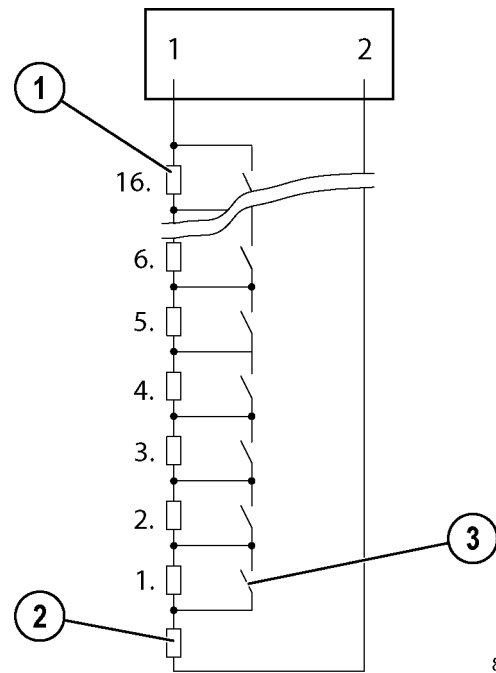


FIGURE 4-5. FUEL GAUGE SENDER SCHEMATIC

- | | |
|-----------------------|---------------------|
| 1. Resistors (13 Ohm) | 3. Reed Switch (NC) |
| 2. Resistor (30 Ohm) | |

QUICK FILL FUEL RECEIVERS

Fuel receiver assembly (4, Figure 4-6) is mounted on the RH face of the fuel tank. If equipped, fuel receiver assembly (6) is mounted to the bottom of the hydraulic tank. It is connected to the rear of the fuel tank by hose (2).

The quick fill fuel system is a pressurized system that is rated at a maximum of 568 l/m (150 gpm). When filling the tank, vent assembly (1) allows air to escape to prevent the tank from over-pressurizing. When the fuel level reaches the float balls in the vent, the balls rise with the fuel. When the fuel level reaches its maximum level, the float balls block air flow out of the vent, causing pressure to rise in the fuel tank. When pressure reaches 55 - 70 kPa (8 - 10 psi), the fuel nozzle should turn off.

If the nozzle fails to turn off and fuel continues to fill the tank, a relief valve in the vent will open at 79 kPa (11.5 psi) to prevent over-pressurization of the tank. If filling persists and the fuel reaches the top of the tank, fuel will flow out of the vent until the nozzle is turned off.

If fuel spills from the vent, or if the tank does not completely fill, check the vent to see whether the float balls are in place and overflow tube is clean. If the vent is operating properly, the problem will most likely be in the fuel supply system.

NOTE: Keep the cap on each fuel receiver to prevent dirt buildup in valve area and nozzle grooves.

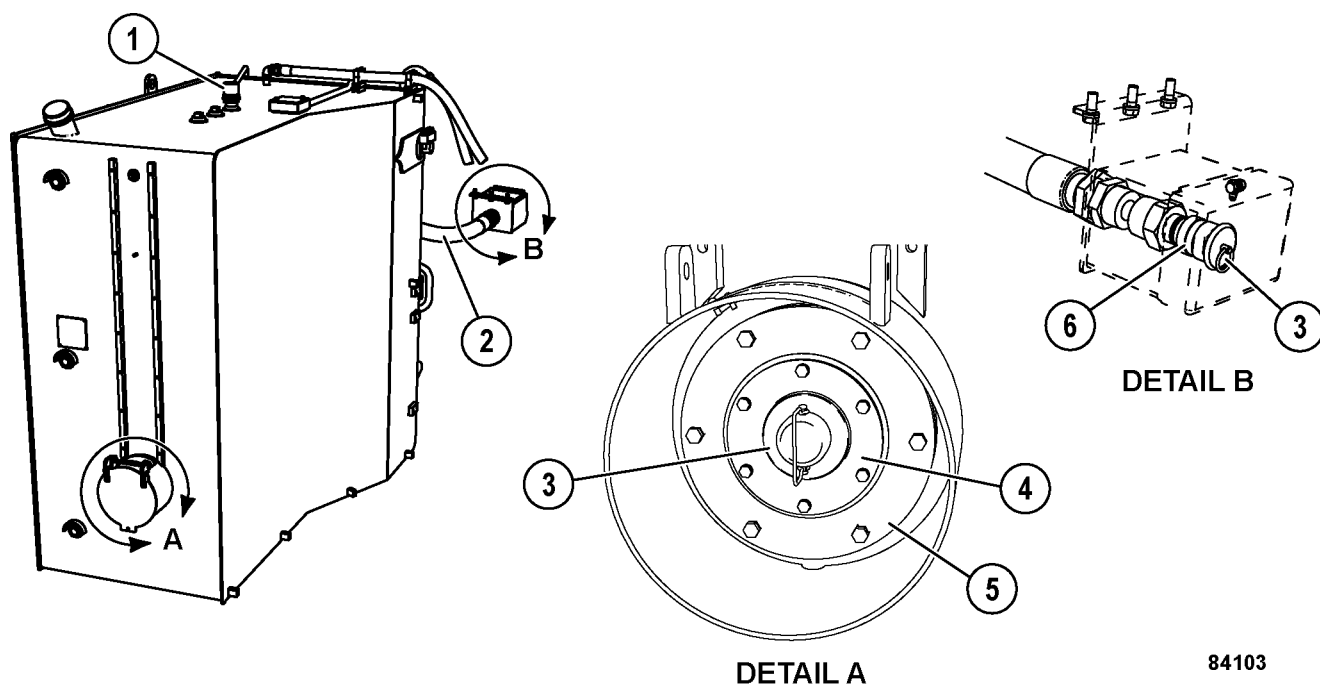


FIGURE 4-6. FUEL GAUGE SENDER & QUICK FILL FUEL

- 1. Vent Assembly
- 2. Quick Fill Hose

- 3. Fuel Receiver Cap
- 4. RH Fuel Receiver Assembly

- 5. Adapter Plate
- 6. LH Fuel Receiver Assembly

NOTES:

SECTION C ENGINE INDEX

POWER MODULE.....	C2-1
COOLING SYSTEM	C3-1
POWER TRAIN.....	C4-1
AIR CLEANERS	C5-1
FAN CLUTCH	C7-1

NOTES

SECTION C2

POWER MODULE

INDEX

POWER MODULE	C2-3
PREPARATION FOR REMOVAL	C2-3
REMOVAL.....	C2-8
INSTALLATION.....	C2-10
HOOKUP.....	C2-11

NOTES

POWER MODULE

The radiator, engine and alternator assemblies are mounted on a roller-equipped subframe which is contained within the truck's main frame. It is referred to as a "power module". This arrangement permits removal and installation of these components with a minimum amount of disconnect and by utilizing the unique "roll in/roll out" feature.

Although the instructions in this section are primarily based upon the "roll out" method for major component removal, the radiator and fan may be removed as separate items. Instructions for radiator and fan removal are included later in this section. If any optional equipment is installed on the truck, additional removal and installation steps that are not listed may be required.

The procedures outlined in this section of the manual are general instructions for power module removal and installation. It may be necessary to perform some procedures in a different order or use different methods for component removal and installation, depending on the lifting equipment available at the mine site. Prior to removal or repair procedures, it may be necessary to remove the body to provide clearance for lifting equipment to be used. If body removal is not required, raise the body and install the body retention sling at the rear of the truck.

Read and observe the following instructions before attempting any repairs.

WARNING

- ***Do not attempt to work in the deck area until body safety cables have been installed.***
- ***Do not step on or use any power cable as a handhold when the engine is running.***
- ***Do not open any control cabinet covers or touch the retarding grid elements until all shutdown procedures have been followed.***
- ***All removal, repairs and installation of propulsion system electrical components must be performed by an electrical maintenance technician properly trained to service the system.***
- ***In the event of a propulsion system malfunction, a qualified technician should inspect the truck and verify the propulsion system does not have dangerous voltage levels present before repairs are started.***

PREPARATION FOR REMOVAL

1. Properly park the truck and shutdown the engine. Refer to "Normal Engine Shutdown" in Section A3, General Safety & Operating Instructions.
2. To ensure that the link will not be energized during test and repair procedures, turn GF Cutout Switch (2, Figure 2-1) in the control cabinet to the CUTOOUT position by pulling the switch out and moving the switch downward as shown.

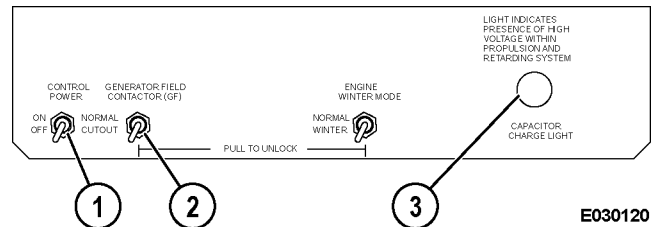


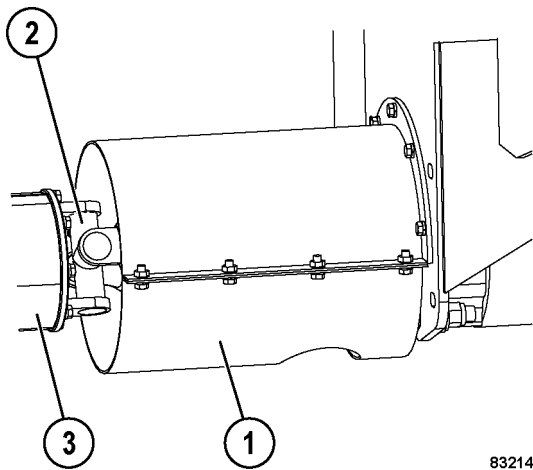
FIGURE 2-1. INFORMATION DISPLAY PANEL

1. Control Power Switch
2. GF Cutout Switch
3. Capacitor Charge Light

3. Open the battery disconnect switches.
4. Tag or mark all hydraulic lines, fuel lines and electrical connections to ensure correct hookup during installation of the power module. Plug all ports and cover all hose fittings and connections when disconnected to prevent contamination.

NOTE: It is not necessary to remove the radiator before removing the power module. However, the coolant must be drained and the piping connected to the brake cooling heat exchanger must be removed. If radiator removal is needed or only radiator repair is necessary, refer to Section C3, Cooling System.

5. If radiator removal is required, refer to Section B2, Structural Components, for hood and grille removal instructions. The power module may be removed with the hood installed if desired.

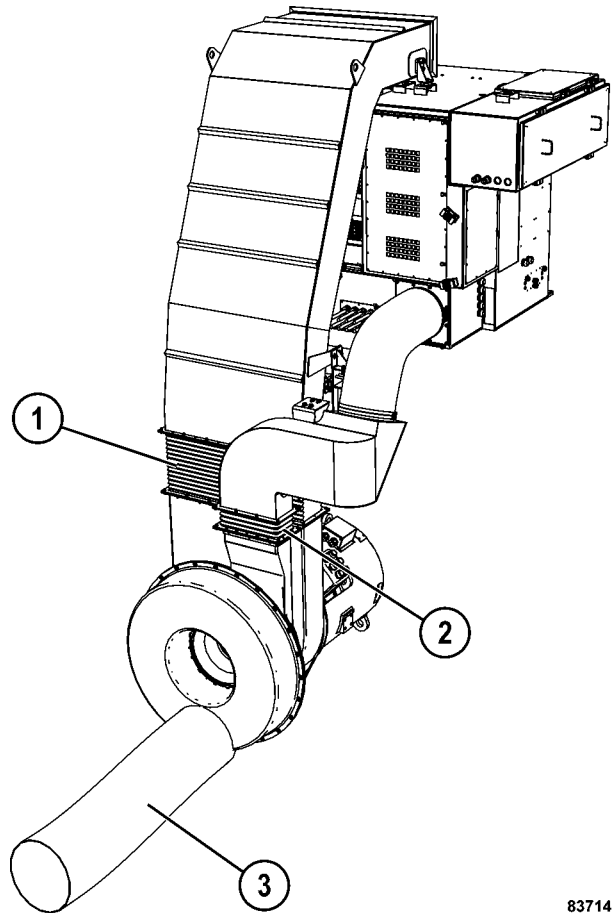


83214

FIGURE 2-2. PUMP DRIVE SHAFT REMOVAL

- | | |
|-----------------------------|---------------------|
| 1. Bottom Drive Shaft Cover | 2. Pump Drive Shaft |
| | 3. Alternator |

6. Remove bottom drive shaft cover (1, Figure 2-2). Disconnect pump drive shaft (2) at the flange mounted on the alternator (3).
7. Remove the alternator blower ducts from the alternator. Be careful not to tear the flexible neoprene material.
 - a. Disconnect cables and hoses as required.
 - b. Remove wheel motor blower transition (1, Figure 2-3) and the gasket.
 - c. Remove control cabinet blower transition flexible duct (2) and the gasket.
 - d. Remove drive axle blower hose (3).
 - e. Cover all openings to prevent foreign material from entering the system.

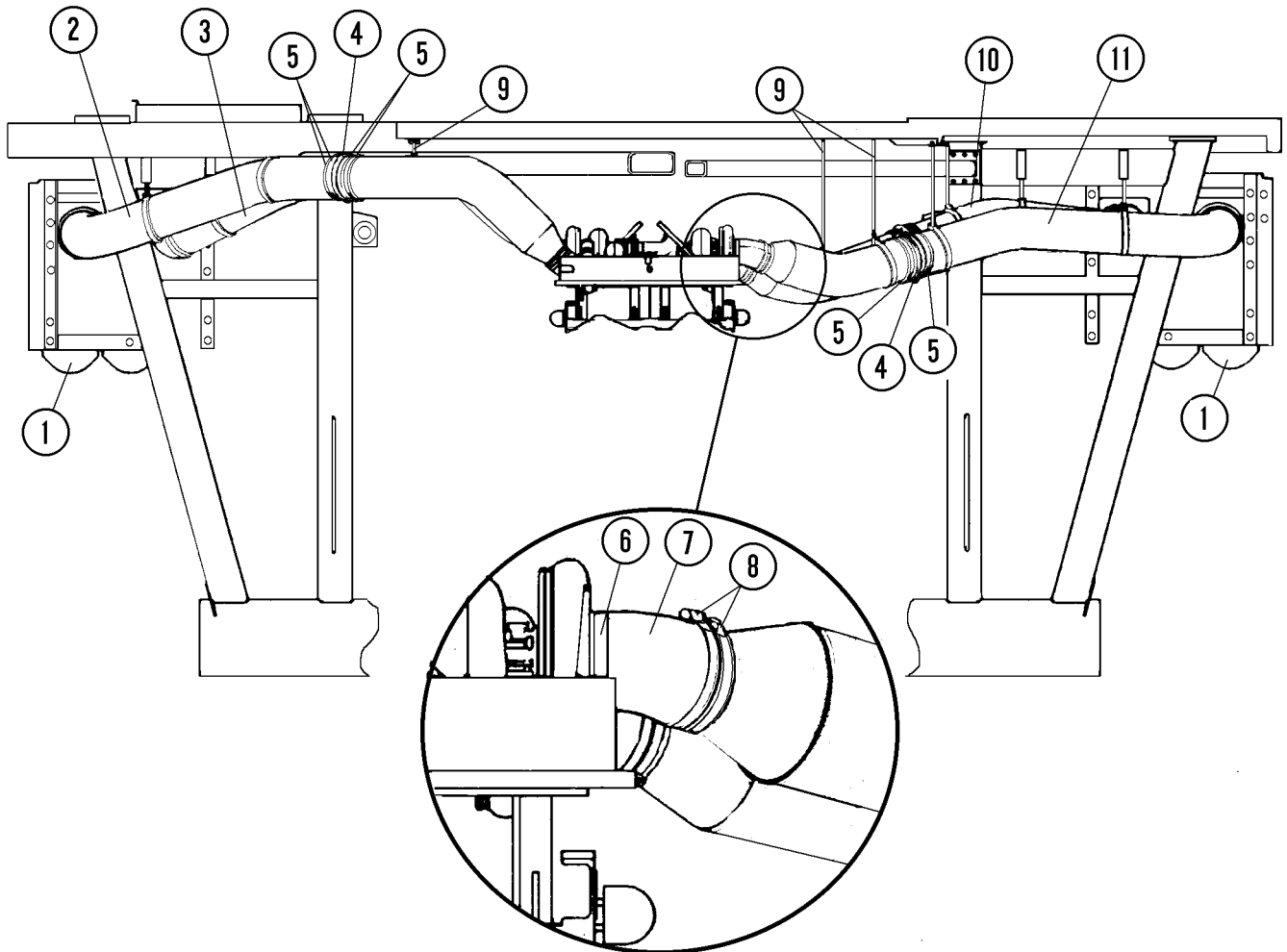


83714

FIGURE 2-3. ALTERNATOR BLOWER DUCTS

- | |
|--|
| 1. Wheel Motor Blower Transition |
| 2. Control Cabinet Blower Transition Flexible Duct |
| 3. Drive Axle Blower Hose |

8. Remove engine air intake duct support rods (9, Figure 2-4). Remove any hoses or electrical cables attached to center deck.
9. Remove the center deck. Refer to Section B2, Structural Components, for the removal procedure.
10. Disconnect the air cleaner restriction indicator nylon tubes at the inlet ducts.
11. Loosen T-bolt clamps (5) on hump hoses (4) between the four engine air inlet ducts and air cleaner outlet ducts.
12. Disconnect the inlet ducts at each of the four turbochargers. Remove the inlet ducts from truck. Cover the openings at the turbochargers.



Turbocharger Inlet Piping Detail

C020016

FIGURE 2-4. ENGINE AIR INLET PIPING

- | | |
|---------------------------|-----------------------------|
| 1. Air Cleaner Assemblies | 7. Reducer Elbow |
| 2. Left Rear Intake Duct | 8. T-Bolt Clamp |
| 3. Left Front Intake Duct | 9. Duct Support Rod |
| 4. Hump Hose | 10. Right Front Intake Duct |
| 5. T-Bolt Clamp | 11. Right Rear Intake Duct |
| 6. T-Bolt Clamp | |

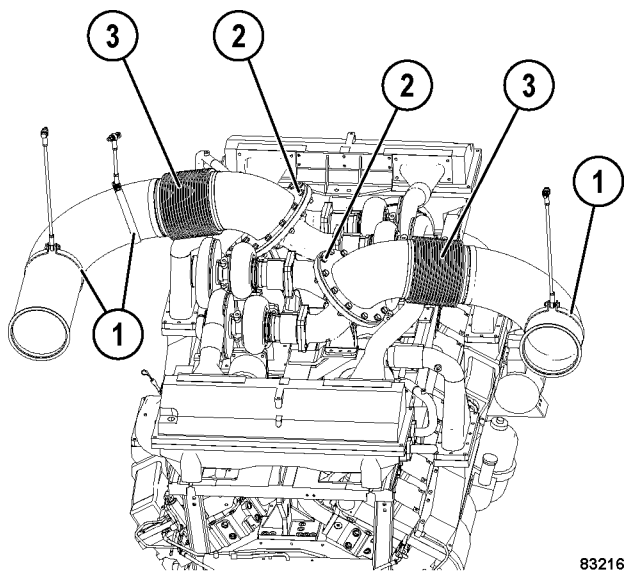


FIGURE 2-5. ENGINE EXHAUST PIPING

- 1. Clamps
- 2. Mounting Hardware
- 3. Exhaust Tubes

13. Remove clamps (1, Figure 2-5) from the hangers.
14. Remove mounting hardware (2) that retains exhaust tubes (3) to the turbocharger outlet flanges. Remove the exhaust tubes. Cover the turbocharger outlet openings on the engine.
15. Remove the alternator power cable protective cover. Disconnect all electrical cables, oil lines and fuel lines that would interfere with power module removal. Cover or plug all lines and their connections to prevent entrance of dirt or foreign material.
16. Close cab heater water shutoff valves (1, Figure 2-6), disconnect the water lines and drain the water from the heater core. Secure the water lines away from engine compartment to prevent interference with power module removal.

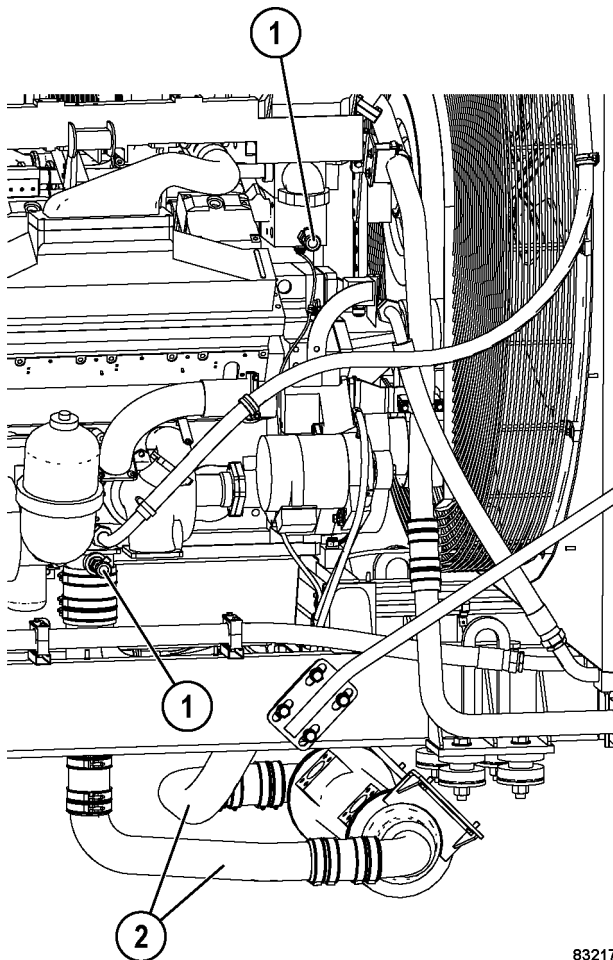


FIGURE 2-6. WATER SHUTOFF VALVES & HEAT EXCHANGER PIPING

- 1. Water Shutoff Valves
 - 2. Heat Exchanger Piping
17. Drain the engine coolant into clean containers for re-use after engine installation. Refer to Section P2, Lubrication & Service, for the coolant system capacity.
 18. Disconnect and remove heat exchanger piping (2, Figure 2-6) from the engine water pump and the radiator.

19. Remove radiator upper support rod (1, Figure 2-7) at each side of radiator.
20. Remove the diagonal ladder from truck. Refer to Section B2, Structural Components, for the removal procedure.
21. Refer to Section N4, Heater/Air Conditioner, for the procedures required to properly remove the refrigerant.

NOTE: The system contains HFC-134A refrigerant.



Federal regulations prohibit venting air conditioning system refrigerants into the atmosphere. An approved recovery/recycle station must be used to remove the refrigerant from the air conditioning system.

22. After the air conditioning system has been discharged, disconnect the refrigerant hoses that are routed to the cab at the compressor and the receiver/drier.

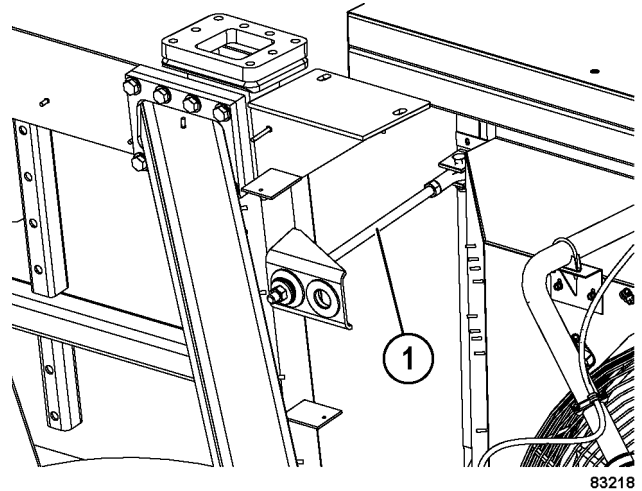


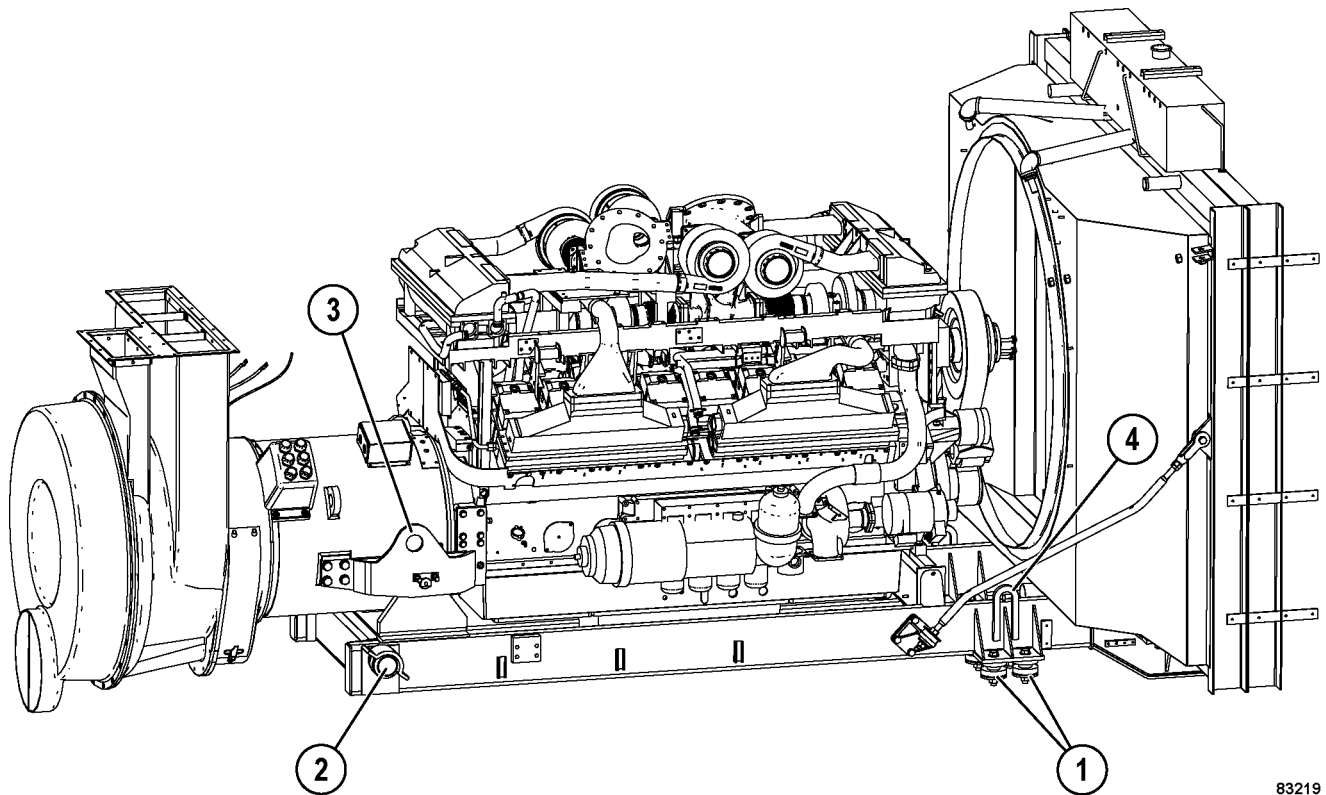
FIGURE 2-7. RADIATOR UPPER SUPPORT ROD

1. Radiator Upper Support Rod

REMOVAL

1. Remove the mounting hardware and front engine mounts (1, Figure 2-8) to the main frame on both sides of the engine.
2. Install a safety chain around the engine subframe cross member and the main frame to prevent the power module from rolling when the subframe rollers are installed.
3. Remove the capscrews and engine mounting caps that secure subframe mounting bushings (2) at the rear of the subframe.
4. Check the engine and the alternator to ensure that all cables, wires, hoses, tubing and linkages have been disconnected.
5. Place a jack under the rear of the power module. Raise the rear portion of engine subframe and install the subframe rollers (see Figure 2-9). Lower the rear portion of the subframe carefully until the rollers rest on the main frame guide rail.

NOTE: Subframe rollers are supplied in the truck tool group and can be installed in the storage position after use as shown in Figure 2-9.



83219

FIGURE 2-8. POWER MODULE REMOVAL & INSTALLATION

- | | |
|-------------------------------|---------------------------------|
| 1. Front Engine Mounts | 3. Engine/Alternator Cradle |
| 2. Subframe Mounting Bushings | 4. Front Subframe Lifting Point |

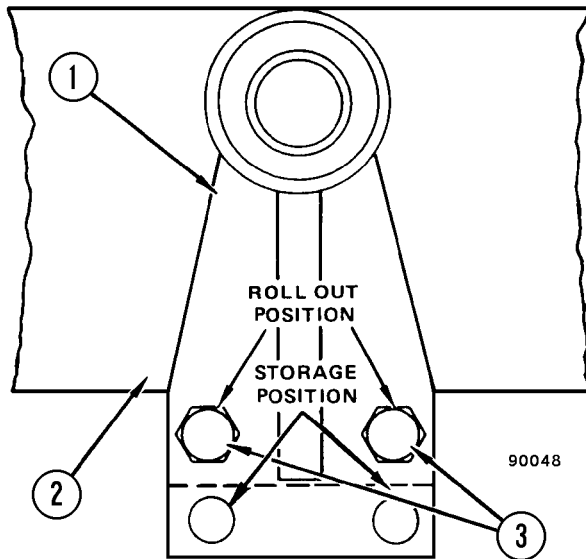


FIGURE 2-9. SUBFRAME ROLLERS

1. Roller Assembly
2. Subframe
3. Capscrews

⚠ WARNING

The complete power module, including the hood and grille, weighs approximately 17 000 kg (37,500 lb). Ensure that the lifting device to be used has adequate capacity.

Lift the power module only at the lifting points on the subframe and engine/alternator cradle structure. Refer to Figure 2-10.

6. Position the hoist to front subframe lifting points (4, Figure 2-8). Raise the engine subframe until the engine is on a level plane. Remove the safety chain.

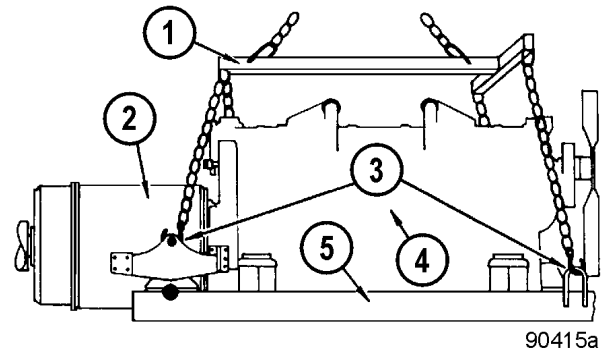


FIGURE 2-10. POWER MODULE LIFTING POINTS

1. Lifting Tool
2. Alternator
3. Lifting Points
4. Engine
5. Power Module Subframe

7. Roll the power module forward so that adequate clearance is provided in front of the control cabinet for the lifting device to be attached to engine/alternator cradles (3, Figure 2-8) and front subframe lifting points (4). Place stands or blocks under the front of the subframe and lower the hoist until the front of the subframe is supported. Install a safety chain to prevent the subframe from rolling.
8. Attach lifting tool (1, Figure 2-10) to the hoist and attach it to the engine/alternator cradles and the front subframe lifting points as shown in Figure 2-10. Remove the safety chain.
9. Raise the power module slightly to determine whether it is on an even plane. Move the power module straight out of the truck to a clean work area for disassembly.

For further disassembly of the engine, alternator and radiator, refer to the appropriate section of this manual.

INSTALLATION

1. Inspect the main frame guide rails. Remove any debris which would interfere with power module installation.
2. Clean the main frame rear support brackets. Apply a light film of soap solution to each subframe mounting bushings (2, Figure 2-8) located at the rear of the subframe.
3. Check the subframe rollers to ensure that they roll freely and are in the "roll out" position (see Figure 2-9).
4. Attach lifting tool (1, Figure 2-10) to the engine/alternator cradles and front subframe lifting points (see Figure 2-10).

WARNING

The complete power module, including the hood and grille, weighs approximately 17 000 kg (37,500 lb). Ensure that the lifting device to be used has adequate capacity.

5. Raise the power module and align the subframe rollers within the main frame guide rails.

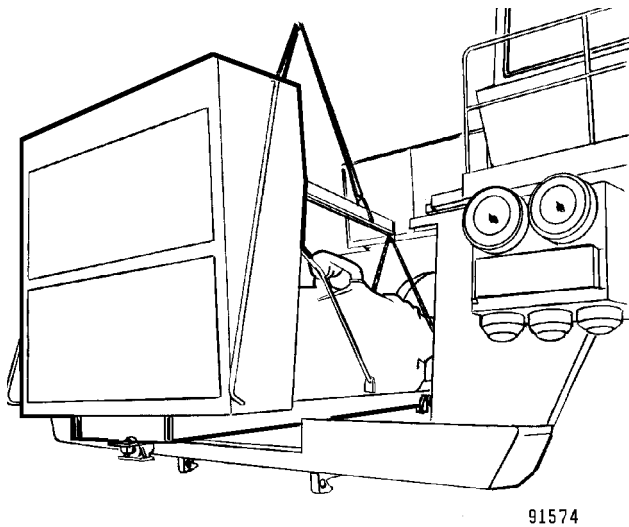


FIGURE 2-11. POWER MODULE INSTALLATION

6. Lower the power module to the subframe guide rails, relax the hoist slightly, and roll the power module into truck frame as far as possible before the lifting chains contact the control cabinet.
7. Place stands or blocks under the front of the subframe to support the assembly while repositioning the hoist.
8. Install a safety chain around the truck frame and the subframe to prevent the power module from rolling forward.
9. Place a small block behind each rear subframe roller to prevent rolling.
10. Lower the hoist to allow the subframe to rest on the stands and rollers. Remove the lifting tool.
11. Attach the hoist to the front lifting eyes on the subframe.
12. Remove the small blocks behind the subframe rollers. Remove the safety chain and slowly roll the power module into position over the main frame mounts. Lower the hoist until the front subframe mounts are aligned and seated on the front main frame mounts. Install the safety chain.
13. Place a jack under the rear of the subframe to support the power module. Raise the power module just enough to permit removal of the subframe rollers.
14. Lower the rear portion of the subframe until the subframe mounting bushings (2, Figure 2-8) are seated in the rear mounting brackets on the main frame of the truck.
15. After the subframe is seated in the frame mounts, the safety chain may be removed from the front subframe member.
16. Install front engine mounts (1, Figure 2-8). and the mounting hardware. Tighten the capscrews to **1017 ± 72 N·m (750 ± 53 ft lb)**.
17. Install the rear engine mounting caps and secure the caps with lubricated capscrews. Tighten the capscrews to **552 ± 20 N·m (407 ± 15 ft lb)**.

HOOKUP

1. Install all ground straps between the frame and subframe. Reconnect the electrical harnesses at the power module subframe connectors.
2. Install the diagonal ladder on the front platform. Refer to Section B2, Structural Components, for the installation procedure.
3. Install the center deck. Refer to Section B2, Structural Components, for the installation procedure.
4. Install the engine air intake duct supports. Install the intake ducts between the turbochargers and air cleaner outlet ducts (see Figure 2-4). Clamp them securely to ensure that a positive seal is made. Refer to Figures 2-4 and 2-12 for correct installation and alignment examples.
5. Install exhaust tubes (3, Figure 2-5).
6. Connect the cab heater inlet and outlet hoses, then open both water shutoff valves (1, Figure 2-6).
7. Install heat exchanger piping (2) to the heat exchanger, engine water pump and radiator.
8. Inspect the transition gaskets and replace them if damaged. Install wheel motor blower transition (1, Figure 2-3) and control cabinet blower transition (2). Install drive axle blower hose (3).
9. Connect pump drive shaft (2, Figure 2-2) to the alternator. Tighten the capscrews to the standard torque. Install bottom drive shaft guard (1).
10. Connect all remaining electrical, oil lines and fuel lines. Connect the air filter restriction indicator hoses.
11. Close the battery disconnect switches.
12. Connect the hoses routed from the cab to the receiver/drier and air conditioning compressor.
13. Refill the radiator and service the engine with appropriate fluids. Refer to Section P2, Lubrication & Service, for the service capacities.
14. Recharge the air conditioner system. Refer to Section N4, Heater/Air Conditioner.

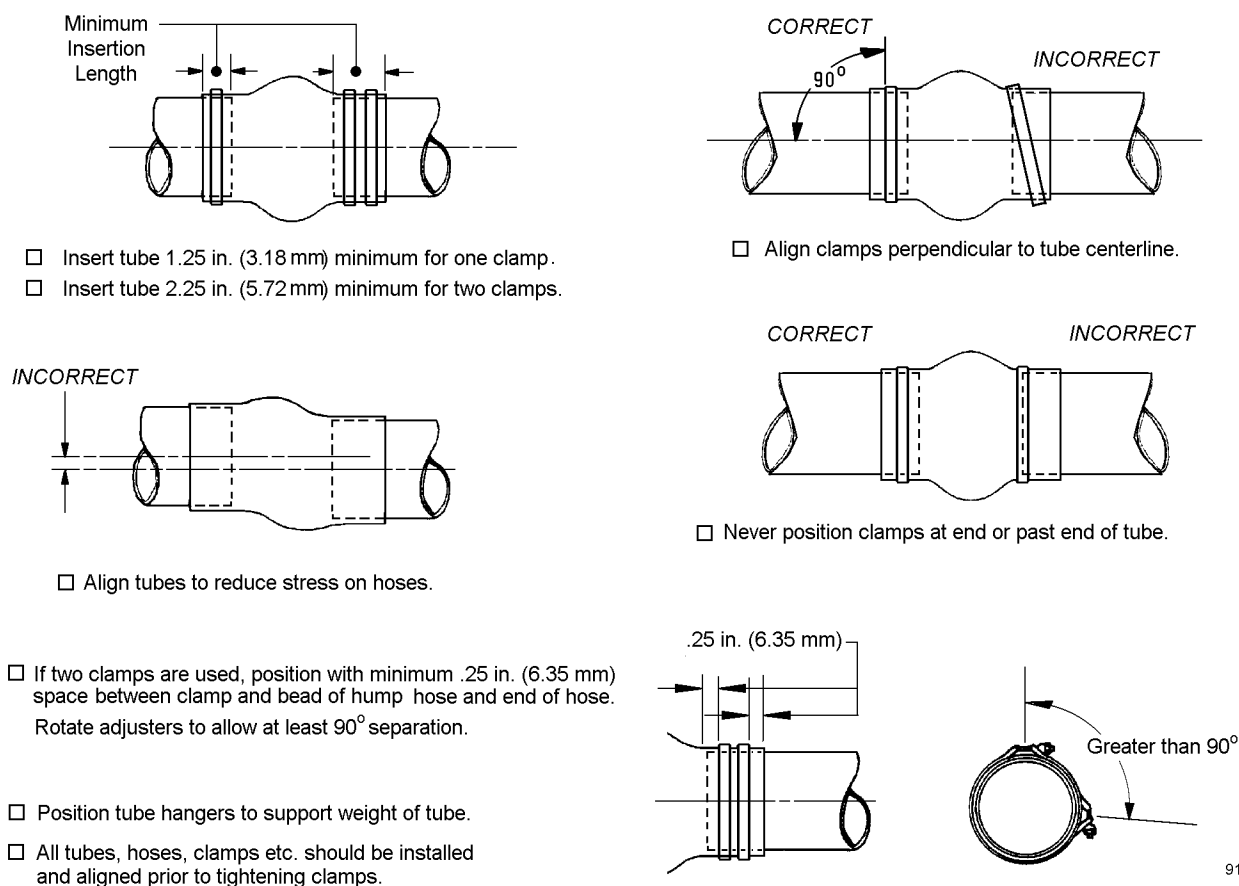


FIGURE 2-12. AIR INLET PIPING CONNECTIONS

NOTES

SECTION C3

COOLING SYSTEM

INDEX

RADIATOR	C3-3
Removal	C3-3
Installation	C3-5
Filling Procedure	C3-6
REPAIRING THE RADIATOR	C3-6
Internal Inspection	C3-6
External Cleaning	C3-6
Disassembly	C3-7
Cleaning and Inspection	C3-8
Assembly	C3-8
Pressure Testing	C3-9
COOLANT SYSTEM TROUBLESHOOTING	C3-9

NOTES

COOLING SYSTEM

RADIATOR

Removal

1. Drain the coolant into clean containers for possible reuse after engine installation. Refer to Section P, Lubrication and Service, for the cooling system capacity.
2. If the radiator is being removed without the complete power module, remove the grille and hood according to the removal instructions in Section B, Structural Components.



Federal regulations prohibit venting air conditioning system refrigerants into the atmosphere. An approved recovery/recycle station must be used to remove the refrigerant from the air conditioning system.

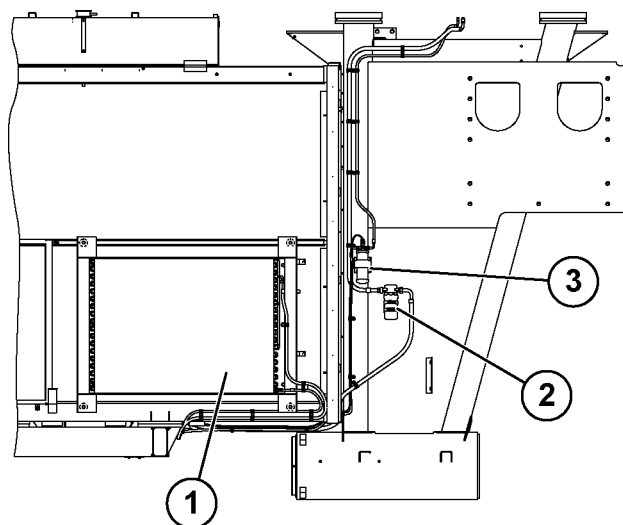


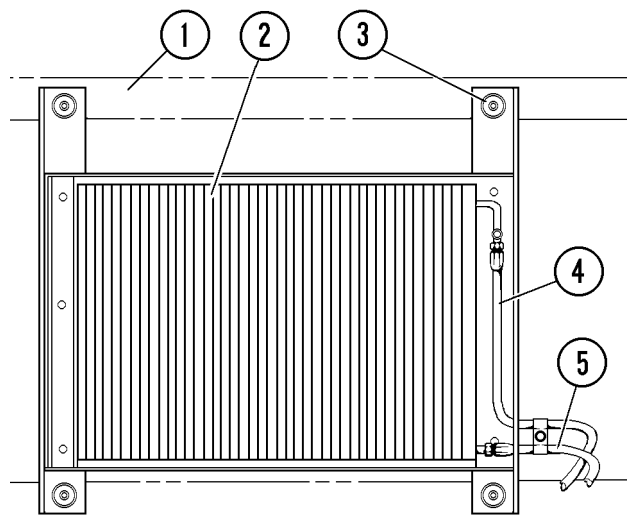
FIGURE 3-1. RECEIVER/DRYER LOCATION

- | | |
|----------------|-------------------|
| 1. Condenser | 3. Receiver/Drier |
| 2. Accumulator | |

3. Refer to Section N4, Heater / Air Conditioner, for the procedures required to properly remove the refrigerant from the air conditioning system.

NOTE: The system is charged with HFC-134A refrigerant.

- a. Remove the refrigerant hose clamps and receiver/drier (3, Figure 3-1) from the front left side of the radiator shroud. Disconnect the harness from the low pressure switch on the receiver/drier.
 - b. Remove the clamps and disconnect inlet hose (4, Figure 3-2) and outlet hose (5) at the condenser. Remove mounting hardware (3) and remove condenser (2) from radiator assembly (1).
 - c. Cap all hoses to prevent contamination. Remove any remaining clamps attaching the hoses and wiring to the radiator shroud. Reposition the hoses and wiring to allow removal of the radiator and shroud assembly.
4. Disconnect the lines at the fuel cooler. Remove all clamps that secure the lines to the radiator.



C030020

FIGURE 3-2. AIR CONDITIONER CONDENSER

- | | |
|----------------------|----------------|
| 1. Radiator Assembly | 4. Inlet Hose |
| 2. Condenser | 5. Outlet Hose |
| 3. Mounting Hardware | |

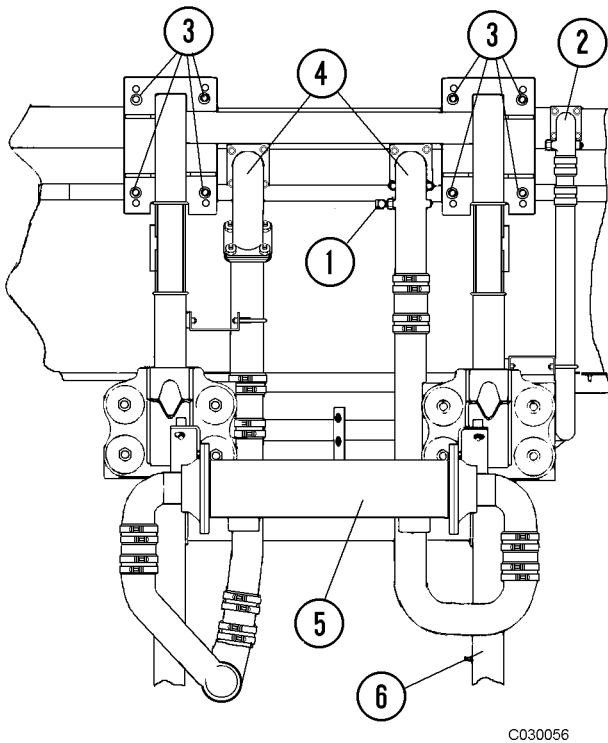


FIGURE 3-3. RADIATOR PIPING, BOTTOM VIEW

- | | |
|-------------------------------|------------------------|
| 1. Drain Cock | 4. Outlet Elbow (Rear) |
| 2. Outlet Elbow (Front) | 5. Heat Exchanger |
| 3. Radiator Mounting Hardware | 6. Engine Subframe |

5. Unclamp and separate all upper hoses between the radiator and the engine.
6. Remove outlet elbows (2, Figure 3-3) and (4) at the bottom tanks. Cap all coolant lines to prevent contamination.
7. Remove and cap the hoses from the radiator top tank and surge tank.
8. Remove the capscrews and lockwashers to free fan guards (1, Figure 3-4) and (2) from radiator shroud (3). The two halves of the fan guard may be disassembled and removed or the complete guard may be moved to the rear to clear the shroud during removal of the radiator.

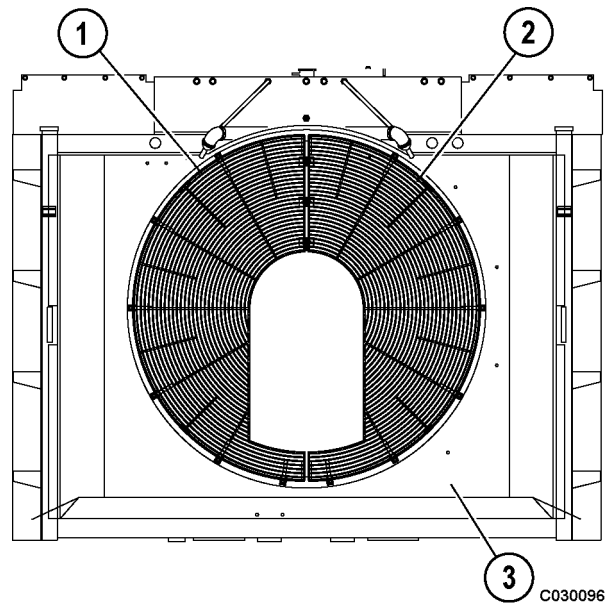


FIGURE 3-4. FAN GUARDS

- | | |
|-------------------|--------------------|
| 1. Fan Guard (LH) | 3. Radiator Shroud |
| 2. Fan Guard (RH) | |

9. Attach the hoist to the radiator and take up any slack.

NOTE: The radiator and shroud assembly weighs approximately 2600 kg (5730 lbs).

10. Remove upper support rods (2, Figure 3-5) and stabilizer bars (3), if equipped.
11. Remove radiator mounting hardware (4, Figure 3-3) that secures the radiator and to the power module subframe.
12. Loosen the radiator bumpers on both decks.
13. Ensure that all hoses and wiring harnesses have been removed. Lift the radiator slightly with the hoist and move the radiator forward until it is clear of the engine fan. Move the radiator to a work area for service.

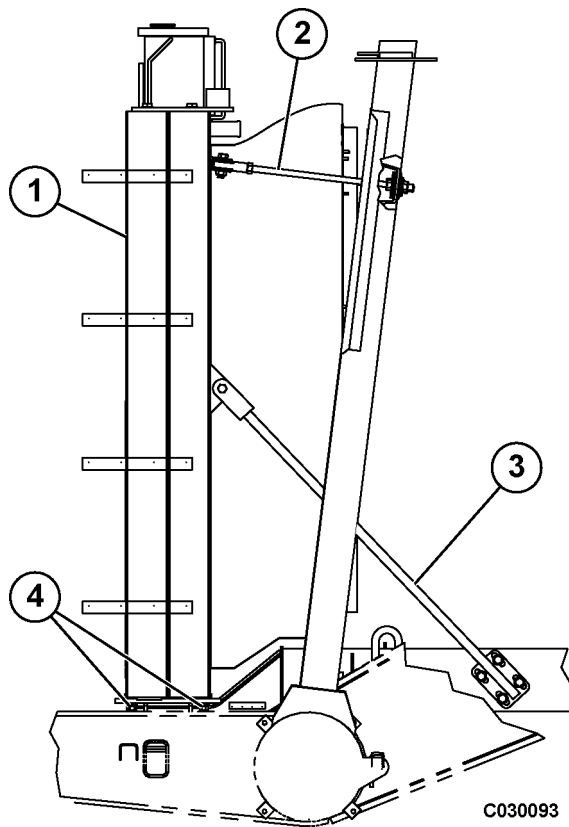


FIGURE 3-5. RADIATOR MOUNTING

- | | |
|----------------------|-------------------------------|
| 1. Radiator | 4. Radiator Mounting Hardware |
| 2. Upper Support Rod | |
| 3. Stabilizer Bar | |

Installation

1. Attach a hoist to the radiator and lift it into position on the power module subframe.
2. Install radiator mounting hardware (4, Figure 3-5) at the lower radiator mounting brackets. Do not tighten at this time.
3. Inspect the rubber bushings for upper support rods (2) and replace them if worn or damaged.
4. Install upper support rods (2) and the mounting hardware at both ends. Do not tighten at this time.
5. Adjust the upper support rods until the radiator is positioned perpendicular to the subframe within ± 3.0 mm (0.12 in.) measured at top of radiator. When the position is established, tighten the upper support rod locknuts and recheck the perpendicularity of the radiator.

6. Tighten radiator mounting hardware (4) at the lower radiator mounting brackets to the standard torque.
7. For added stability, stabilizer bars (3) may be installed. Install the stabilizer bars at the radiator brackets and subframe as shown, then tighten the mounting hardware to the standard torque.
8. Install fan guards (1, Figure 3-4) and (2) using the original mounting hardware.
10. Install the upper hoses between the radiator and the engine. Seat the hoses fully and tighten the clamps securely.
11. Install outlet elbows (2, Figure 3-3) and (4) with new gaskets.
12. Route the lines to the fuel cooler and clamp them in place. Attach the lines to the fuel cooler fittings.
13. Install the air conditioning system components.
 - a. Install condenser (2, Figure 3-2), hoses (4) and (5), and all clamps.
 - b. Install receiver/drier (3, Figure 3-1) and clamp the hoses. Attach the harness to the low pressure switch.
 - c. Clamp all hoses and wiring to the studs using the original clamps. Refer to Section N, Heater / Air Conditioner, for complete instructions on evacuating and recharging the air conditioning system.
14. Install the grille and hood according to the instructions in Section B, Structural Components. Make sure that both radiator bumpers are adjusted and secured.
15. Make sure that all coolant drains are closed, all pipe plugs are installed, and all hoses are connected securely. Service the cooling system with the proper mixture of antifreeze as recommended in Section P, Lubrication and Service. Check for leaks and correct as needed.
16. Start and operate the engine until the cooling system reaches normal operating temperature. Recheck for leaks during engine operation. Turn off the engine and correct any leaks.



The cooling system is pressurized due to thermal expansion of coolant. DO NOT remove the radiator cap while the engine and coolant are hot. Severe burns may result.

1. With engine and coolant at ambient temperature, remove the radiator cap.

Note: If coolant is added using the Wiggins quick fill system, the radiator cap must be removed before adding coolant.

2. Fill the radiator with the proper coolant mixture (as specified by the engine manufacturer) until coolant is visible in the sight gauge.
3. Install the radiator cap.
4. Operate the engine for five minutes, then check the coolant level.
5. If coolant is not visible in the sight gauge, repeat steps 1 through 4 of this procedure. Any excess coolant will be discharged through the vent hose after the engine reaches normal operating temperature.

NOTE: Engine coolant must always be visible in the sight gauge before truck operation.

REPAIRING THE RADIATOR

Internal Inspection

If desired, an internal inspection can be performed on the radiator before complete disassembly. The inspection involves removing tubes from the radiator core and cutting them open. This type of inspection can indicate overall radiator condition, as well as coolant and additive breakdown.

To perform this inspection, remove four random tubes from the air inlet side of the radiator. Remove tubes from both the top and bottom cores, and near each end of the radiator. Refer to Disassembly and Assembly in this section for the proper instructions for removing and installing tubes. Analyze any contaminant residue inside the tube to determine the cause of contamination. Flush the system before returning the truck to service. Contact your nearest L&M Radiator facility for further instructions or visit the L&M website at www.mesabi.com.

External Cleaning

Many radiator shops use a hot alkaline soap, caustic soda or chemical additives in their boil-out tanks, which can attack solders. These tanks are generally not recommended. **Before such tanks are used for cleaning, ensure that the cleaning solutions are not harmful to solder. Otherwise, damage to the radiator will result.** Completely rinse the cleaned tube or core in clean water after removing it from the boil-out tank.

As an alternative to boil-out tanks, radiators can be cleaned externally with a high pressure washer and soap. In most cases, it may be best to blow out any dry dirt with a high pressure air gun prior to washing the core with the high pressure washer.

Pressure washers should not exceed 8 275 kPa (1,200 psi). Unlike conventional cores, the spray nozzle can be used right up next to the core. Starting from the air exit side, place the high pressure washer nozzle next to the fins. Concentrate on a small area, slowly working from the top down. Spray straight into the core, not at an angle. Continue washing until the exit water is free of dirt. Repeat from the opposite side.

⚠ IMPORTANT ⚠

To aid in removal of the tubes, clean the radiator prior to disassembly. Heating the seals with hot water helps to loosen the grip on the tubes. Cleaning the radiator prior to disassembly also reduces the risk of internal contamination. After cleaning, spray lubricating oil at the top end of the tubes.

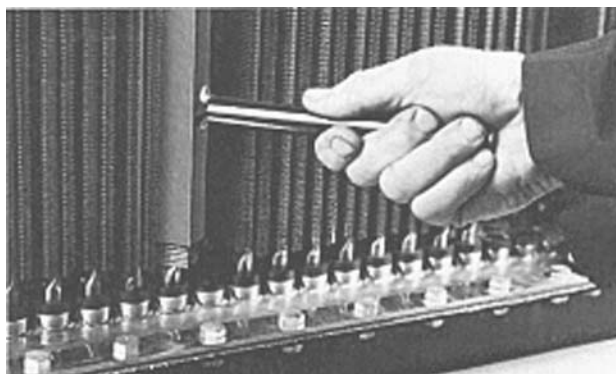


FIGURE 3-6. BREAKER TOOL (XA2307)

1. Start at the top row of tubes. Use the breaker tool (XA2307) to loosen the tube to be removed. When using the breaker tool, position it at the top or bottom of the tube. Never position it in the middle of the tube or damage may result. Use the breaker tool to lightly twist the tube back and forth within the seals to loosen the grip. Refer to Figure 3-6.

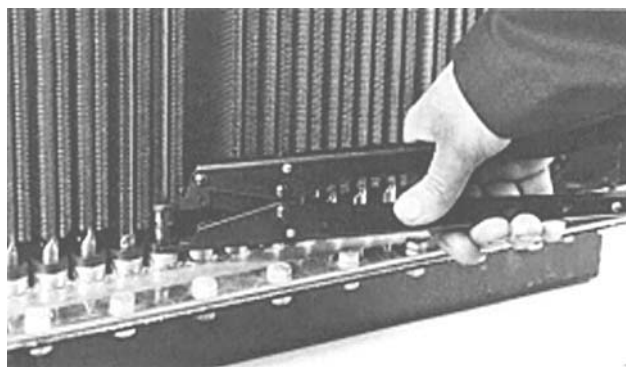


FIGURE 3-7. INSTALLATION TOOL (VJ6567)

2. After the tube is loose, position the installation tool (VJ6567) at the bottom of the tube to be removed. Refer to Figure 3-7. The upper jaw of the installation tool should be positioned just below the rectangular section of the tube. The bottom jaw should rest on the seal. Squeeze the installation tool just enough to allow the bottom of the tube to be removed from the bottom seal.

NOTE: To ease in the removal of tubes, use the breaker tool and installation tool simultaneously.

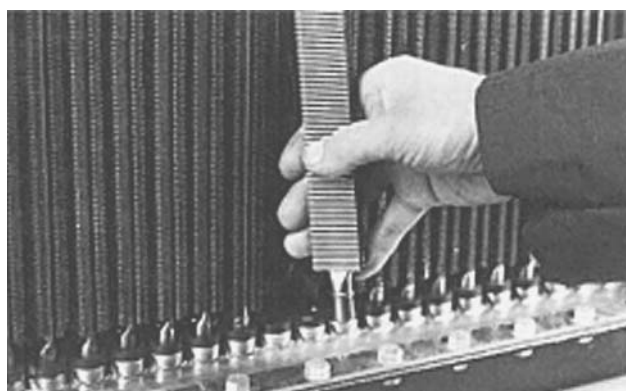


FIGURE 3-8. ANGLING TUBE DURING REMOVAL

3. Pull the tube from the top seal while simultaneously twisting the tube. Angle the tube only far enough to clear the radiator. Refer to Figure 3-8. Removing the tube at an excessive angle may cause damage to the tube.
4. Remove all the top tubes before removing the bottom tubes. After all of the tubes are removed, use pliers to remove the seals from the tanks. Discard all seals. New seals must be used for assembly.

Cleaning and Inspection

1. Use a drill with a 19 mm (3/4 in.) wire brush to remove any foreign material from the tube holes, then wipe the holes clean.
2. Clean the inside of the tanks and tubes. In most cases, just flushing the inside with soap and a high pressure hot water washer will be sufficient. If not, contact an L&M manufacturing facility for further instructions or visit the L&M website at www.mesabi.com.
3. Check for signs of internal blockage in the tubes and tanks. If desired, you may cut open tubes for inspection. If contamination is present, the tube should be analyzed. The radiator must be properly flushed of all contaminants and corrective action must be taken to prevent such contamination from occurring in the future. Refer to Internal Inspection in this section.
4. Buff the tube ends with a polishing wheel and a copper polishing compound. If any debris can not be removed by buffing, using an emery cloth, steel wool or a wire wheel with a wire size of 0.15 - 0.20 mm (0.006 - 0.008 in.) is acceptable. Be careful not to mar the tube ends.

Assembly

NOTE: For easier installation, soak the seals in hot water before installing.

1. Install new tube seals onto the bottom tank and the bottom side of the center tank. Do not install seals in the top core at this time. Seals for the top of the tubes do not have locking grooves; bottom tube seals do. Ensure the correct seals are installed in the proper position.

The seal holes must be dry during installation. Use a rubber mallet and a flat metal plate to lightly tap the seals into place. Using excessive force will drive the seals in too far. When installed properly, the seals should be slightly convex. Improperly installed seals are concave with a smaller diameter hole. Refer to Figure 3-9.

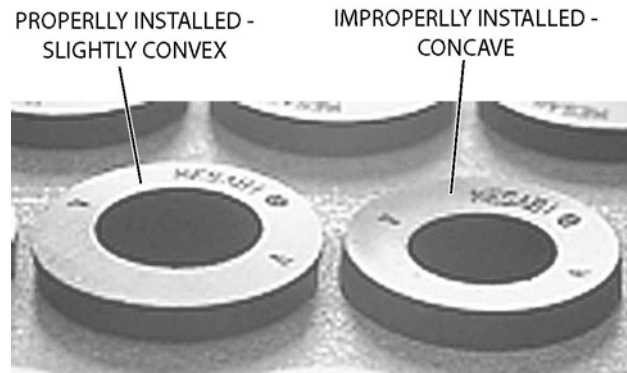


FIGURE 3-9. PROPER SEAL INSTALLATION

2. Use a 13 mm (1/2 in.) diameter brush to lubricate the seals with lube/release agent (XA2308).
3. Use a spray bottle to lubricate the tube ends with the lube/release agent.
4. When installing tubes, start at one end and work toward the center. After you reach the center, move to the opposite end, and again work toward the center. If any of the tubes are difficult to install, do not force the tube. Remove the tube and determine the problem. Possible causes may be:
 - adequate seal/tube lubrication
 - improperly installed seal
 - damaged seal or tube end
 - tube angle excessive during installation and/or tube not centered in seal.

Inspect the seals and tube ends for damage before trying to reinstall a tube. Replace as necessary.

5. Working from the front of the radiator (opposite of fan side), install the bottom row of tubes starting with the fan side row.

When installing the tubes, center the top of the tube in the top seal while angling the tube only as much as necessary. Twist the tube while applying upward force. Push the tube into the seal until enough clearance is available to install the bottom end of the tube into the bottom seal.

6. Center the bottom end of the tube in the bottom seal. Push the tube downward until the formed bead on the tube is seated inside the lock ring groove in the seal. If necessary, use the installation tool (VJ6567) to pull the tube downward into the seal. The tool has a hooking device on the end of one of the handles for aiding in installation. Refer to Figure 3-10.

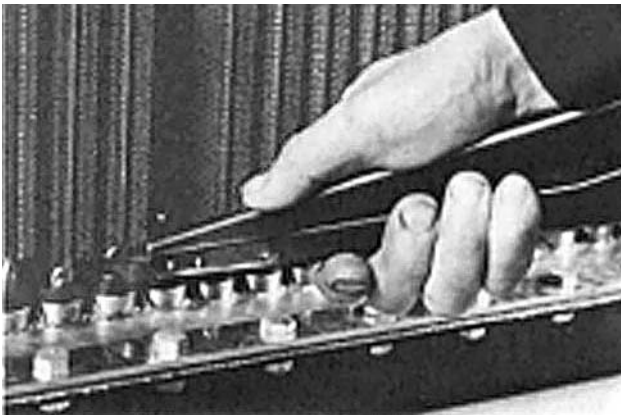


FIGURE 3-10. USING INSTALLATION TOOL TO INSTALL TUBE

7. Ensure that all tube beads are seated in their respective bottom seals. Align and straighten all tubes during the installation of each row to allow maximum air flow through the radiator.
8. Install tube stay ends. Install the felt air baffles behind the front and back rows while completing tube installation.

Pressure Testing

The radiator should be pressure tested at 103 kPa (15 psi) for 30 minutes. Various methods of pressure testing include the following:

- Pressurize the radiator and submerge into a test tank. Watch for leaks.
- Lay the front side of the radiator on the floor. Cap off ports, and fill the radiator with hot water. Pressurize the radiator and check for leaks.
- Cap off radiator ports. Install an air pressure gauge and pressurize to 103 kPa (15 psi). Remove the air source and monitor the pressure gauge.
- Pressurize the radiator with air, and spray sealed joints with soapy water.

Additional service information can be found on the L&M Radiator website at www.mesabi.com.

COOLANT SYSTEM TROUBLESHOOTING

If abnormal coolant temperatures are experienced, perform the following visual inspections and tests:

1. Check the coolant level and thoroughly inspect the system for leaks.
 - a. Check for proper coolant/antifreeze mixture.
 - b. Follow the recommendations of the engine manufacturer regarding use of cooling system additives.
2. Inspect the radiator fins for restrictions. Ensure the air flow through the radiator is not restricted by debris or bent radiator fins.
3. Inspect the fan blades for damage.
4. Check the radiator cap sealing surfaces.
5. If equipped with a fan clutch, refer to Section C7 for complete instructions for testing and repairs, if required.
6. Refer to the engine manufacturer's service manual for information about testing and replacing the cooling system thermostats.

NOTES

SECTION C4

POWER TRAIN

INDEX

ALTERNATOR REMOVAL	C4-3
ALTERNATOR INSTALLATION	C4-5
Measuring Procedure	C4-5
Joining the Alternator and Engine	C4-7
ENGINE	C4-8
Removal	C4-8
Service	C4-8
Installation	C4-8

NOTES

POWER TRAIN

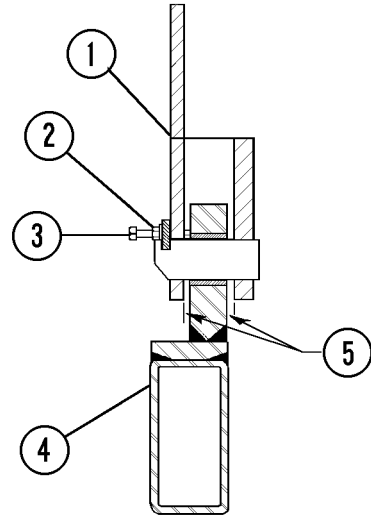
ALTERNATOR REMOVAL

The following instructions cover the removal of the main alternator from the engine after the power module has been removed from the truck.

⚠ WARNING

The alternator weighs approximately 3720 kg (8200 lbs). Ensure the lifting device to be used has adequate capacity. Lift the alternator only at the lifting eyes on the alternator.

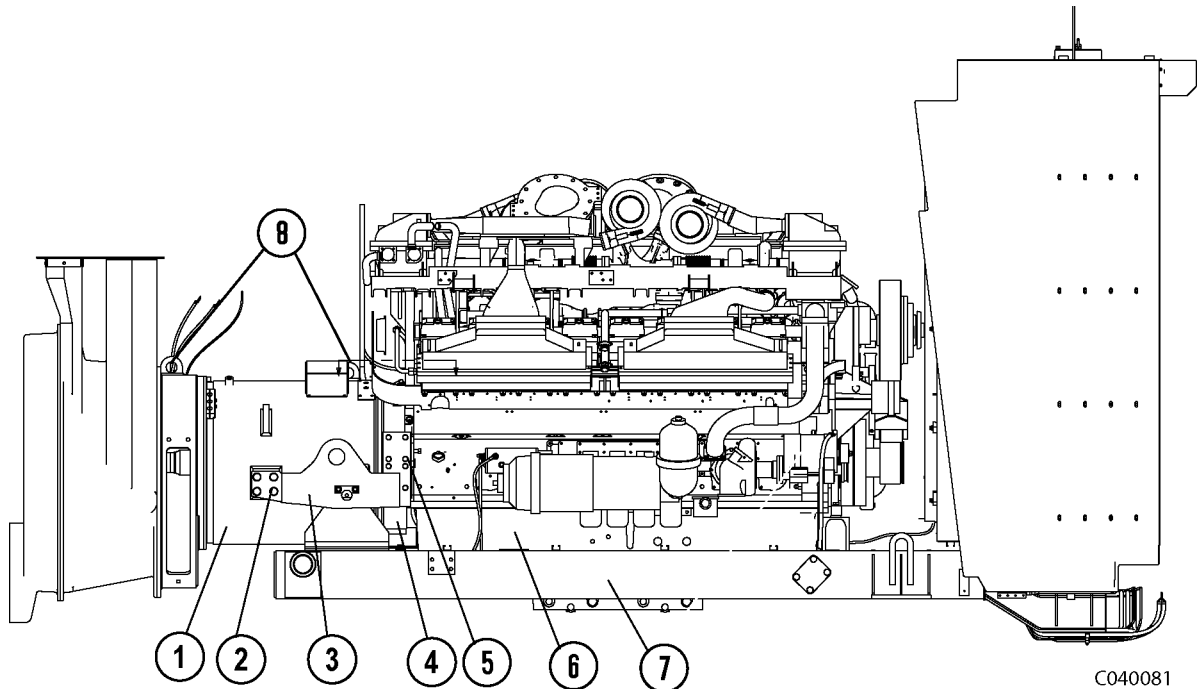
1. Attach a hoist with two lifting chains to both lifting eyes (8, Figure 4-2) on top of alternator (1).
2. Block up the rear of the engine.
3. Loosen cradle adjustment setscrews (3, Figure 4-1).
4. Loosen engine-to-cradle capscrews (5, Figure 4-2).



C040057

FIGURE 4-1. CRADLE STRUCTURE

- | | |
|------------------------|-------------|
| 1. Cradle Structure | 4. Subframe |
| 2. Jam Nut | 5. Gap |
| 3. Adjustment Setscrew | |



C040081

FIGURE 4-2. ENGINE AND ALTERNATOR

- | | | |
|----------------------|---------------------|----------------------------|
| 1. Alternator | 4. Flywheel Housing | 7. Subframe |
| 2. Mounting Hardware | 5. Capscrews | 8. Alternator Lifting Eyes |
| 3. Cradle Structure | 6. Engine | |

5. Remove the access cover at the front, right side of engine flywheel housing (4). Reach through the access openings and remove 12 capscrews (6, Figure 4-3) that join engine drive ring (7) to alternator rotor (8). Rotate the crankshaft to align each capscrew with the access hole.

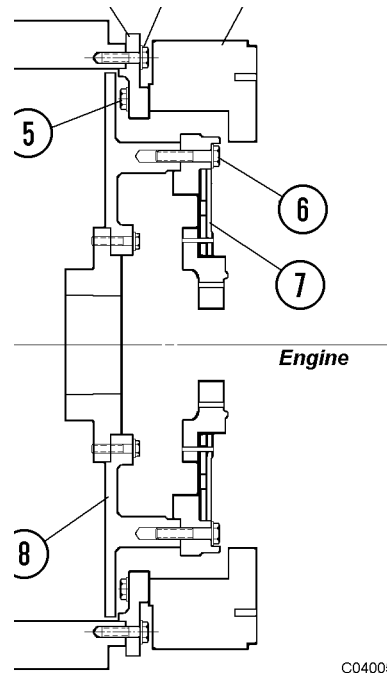
NOTE: Ensure that all 12 capscrews have been removed.

6. Remove 16 capscrews (3) that secure flywheel housing adapter (2) to alternator housing (1).

NOTE: The clearance between the head of capscrew (3) and flywheel housing (4) will not permit complete removal of the capscrews at all locations. Ensure that all of the capscrew threads are completely disengaged from alternator housing (1).

7. Take up any slack in the hoist. Remove mounting hardware (2, Figure 4-2) that secures the alternator to the cradle structures.
8. Keep the alternator as level as possible and move away from engine.
9. Note the location and quantity of the shims. Keep the shims for possible reuse during installation.

NOTE: For further disassembly instructions for the alternator, refer to the GE Service Manual.



C040058

FIGURE 4-3. ALTERNATOR TO ENGINE MOUNTING

- | | |
|-----------------------------|----------------------|
| 1. Alternator Housing | 5. Capscrew |
| 2. Flywheel Housing Adapter | 6. Capscrews |
| 3. Capscrews | 7. Engine Drive Ring |
| 4. Flywheel Housing | 8. Alternator Rotor |

ALTERNATOR INSTALLATION



The following instructions must be followed to ensure proper alignment and engine crankshaft endplay. Failure to follow these instructions can result in serious damage to the engine and/or alternator.

- Never pry on the engine crankshaft damper.
- Loosen or remove fan belts before measuring crankshaft endplay to ensure that the crankshaft moves easily and completely.
- When taking measurements, always take four equally spaced readings and average them.
- Always measure from mating surface to mating surface.
- A reference to crankshaft rotation - clockwise (CW) or counterclockwise (CCW) - is the direction of rotation when looking at the front (damper end) of engine.
- **Crankshaft end play: 0.13 - 0.38 mm (0.005 - 0.015 in.)**

SERVICE DATA - Eccentricity & Runout Limits	
Description	T.I.R
Maximum Eccentricity of Flywheel Housing Bore	0.66 mm (0.026 in.)
Maximum Face Runout of Flywheel Housing	0.25 mm (0.010 in.)
Maximum Eccentricity of Flywheel (Coupling Assembly)	0.18 mm (0.007 in.)
Maximum Axial Runout of Flywheel Face (Coupling Assembly)	0.25 mm (0.010 in.)

Measuring Procedure

1. Thoroughly clean the alternator housing mounting surface, rotor drive adapter mounting surface and flywheel housing adapter mounting surfaces.
2. Mount a magnetic base on the front of the engine and a dial indicator on the front of the crankshaft. Measure total crankshaft endplay.

Verify end play is within 0.13 - 0.38 mm (0.005 - 0.015 in.).

Total Crankshaft Endplay _____

3. Refer to Figure 4-4. Move the engine crankshaft to the rear of its end travel.
 - a. Carefully measure Dimension "C" at four locations, 90° apart:
 - 1st measurement: _____
 - 2nd measurement: _____
 - 3rd measurement: _____
 - 4th measurement: _____
 - Dimension "Cavg": _____
 - b. Add 1/2 (one-half) of Total Crankshaft Endplay from step 2 to Dimension "Cavg".
 - c. Record (Step 3a + Step 3b) as Measurement "C": _____

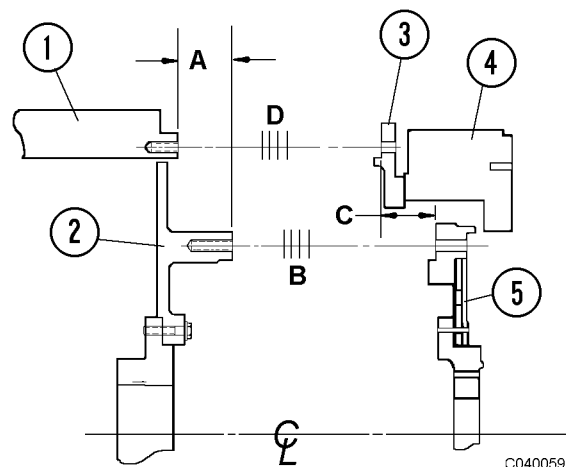


FIGURE 4-4. SHIM LOCATION

- | | |
|-----------------------------|---------------------|
| 1. Alternator Housing | "A" - Dimension "A" |
| 2. Alternator Rotor | "B" - Drive Shims |
| 3. Flywheel Housing Adapter | "C" - Dimension "C" |
| 4. Engine Flywheel Housing | "D" - Housing Shims |
| 5. Engine Drive Ring | |

4. Refer to Figure 4-5 to determine alternator endplay:

- a. Using flat steel bar (3) bolted rigidly to alternator rotor (2), install 5/8" - 11NC capscrew (4) finger tight at each end into alternator housing (1).
- b. Move alternator rotor (2) axially toward the rear (slip ring end) by alternately tightening capscrews (4) one-half turn at a time. **Do not exceed 16 N·m (12 ft lb) on each capscrew.** This establishes the maximum permissible rear travel for the alternator rotor.
- c. Alternately loosen capscrews (4) one turn at a time until all torque is released. Carefully remove steel bar (3).

Refer to Figure 4-4.

- d. Carefully measure Dimension "A" at four locations, 90° apart, and average the measurements. **Do not move the alternator rotor.**

1st measurement: _____
 2nd measurement: _____
 3rd measurement: _____
 4th measurement: _____
 Dimension "Aavg": _____

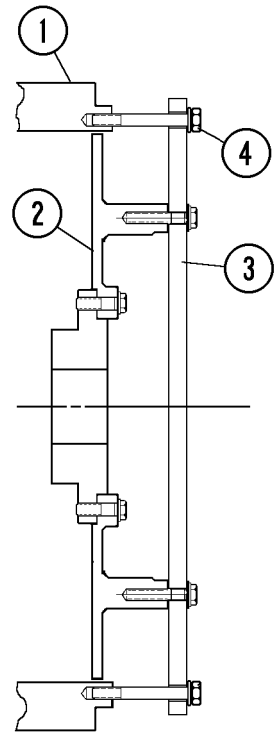
- e. Add 0.25 mm (0.010 in.) to Dimension "Aavg".

- f. Record (Step 4d + Step 4e) as Measurement "A": _____

5. To determine the correct shims to use, compare Measurement "C" (Step 3c) with Measurement "A" (Step 4f).

- a. If C is greater than A, subtract: $(C - A) = B$
 $B =$ _____ Shim pack thickness
 to be installed at location B, Figure 4-4.

Rotor-to-Drive Ring Location "B"	
Shim Part Number	Shim Thickness
EL3332	0.10 mm (0.004 in.)
EL3331	0.18 mm (0.007 in.)



C040078

FIGURE 4-5. ALTERNATOR END-PLAY

- | | |
|-----------------------|--------------|
| 1. Alternator Housing | 3. Steel Bar |
| 2. Alternator Rotor | 4. Capscrew |

- b. If A is greater than C, subtract: $(A - C) = D$
 $D =$ _____ Shim pack thickness
 to be installed at location D, Figure 4-4.

Alternator-to-Flywheel Housing Adapter Location "D"	
Shim Part Number	Shim Thickness
EJ1056	0.10 mm (0.004 in.)
EJ1055	0.18 mm (0.007 in.)

Joining the Alternator and Engine

WARNING

The alternator weighs approximately 3 720 kg (8,200 lb). Ensure the lifting device to be used has adequate capacity. Lift the alternator only at the lifting eyes.

1. Use two lifting eyes (8, Figure 4-2) provided on the alternator. The top front lifting eye should be equipped with some method of adjusting the alternator to keep it horizontal.
2. Carefully move the alternator into place and engage engine drive ring (6, Figure 4-6) into alternator rotor drive (7) using shims "B", if required. Refer to step 5a. in "Measuring Procedure".
3. Install flywheel housing adapter capscrews (2) into alternator housing (1). Tighten to **237 N·m (175 ft lb)**.
4. Install capscrews (5) through engine drive ring (6) into alternator rotor adapter (7). Rotate the crankshaft to access and align holes. Tighten capscrews (5) to **237 N·m (175 ft lb)**.
5. Install alternator-to-cradle mounting hardware (2, Figure 4-2). Tighten to **712 N·m (525 ft lb)**.
6. Tighten engine-to-cradle capscrews (5) to **465 N·m (345 ft lb)**.

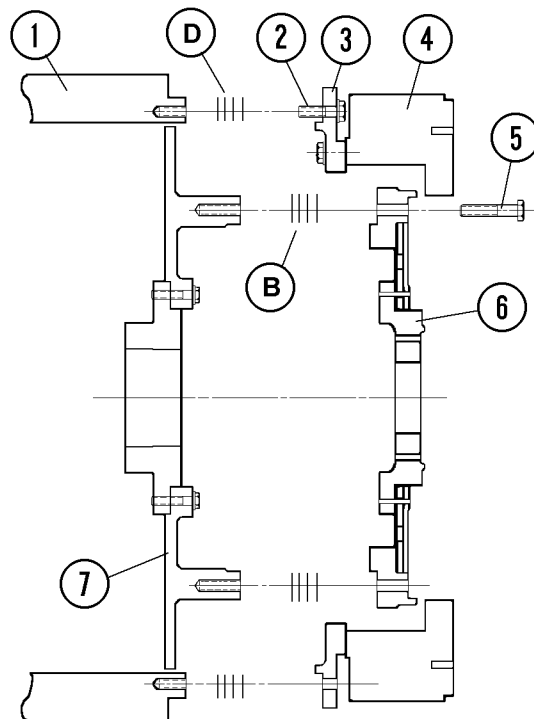
CAUTION

Never pry on the engine crankshaft damper. Severe engine damage can result.

7. Mount a magnetic base on the front of the engine and a dial indicator on the front of the crankshaft. Measure total crankshaft endplay.

Total Crankshaft Endplay: _____

8. Compare the value above to the measurement taken before the alternator was installed on the engine.



C040060

FIGURE 4-6. ALTERNATOR TO ENGINE MOUNTING

- | | |
|-----------------------------|-----------------------------|
| 1. Alternator Housing | 5. Capscrew |
| 2. Point Capscrew | 6. Engine Drive Ring |
| 3. Flywheel Housing Adapter | 7. Alternator Rotor Adapter |
| 4. Engine Flywheel Housing | "B" - Drive Shims |
| | "D" - Housing Shims |

The total engine crankshaft endplay (step 7) must equal the original measurement or 0.51 mm (0.020 in.), whichever is smaller.

If the endplay after the alternator and engine are assembled is less than 0.51 mm (0.020 in.), and less than the original engine crankshaft endplay, reshimming is required.

9. Rotate the crankshaft one full revolution and listen for any unusual noise caused by moving components contacting stationary parts.
10. Install the engine side cover, if removed. Install the lockwire on all alternator mounting capscrews.
11. Install the access cover on the flywheel housing.

ENGINE

Removal

NOTE: Refer to previous sections for removal procedures for the power module, alternator, and radiator assembly.



The engine weighs approximately 9 300 kg (20,515 lb). Ensure that the lifting device to be used has adequate capacity.

1. Disconnect any remaining wiring or hoses between engine (6, Figure 4-2) and subframe (7).
2. Remove the capscrews at the front engine mounts.
3. To ensure that the engine stays level while lifting, attach a spreader bar with lifting straps to the engine lifting eyes. Remove engine-to-cradle structure mounting capscrews (5).
4. Lift the engine from the subframe and move to clean work area for further disassembly.

Service

Complete instructions for the disassembly, assembly and maintenance of the engine and its components can be found in the engine manufacturer's service manual.

Installation

1. Align the engine with subframe (7, Figure 4-2) and install the capscrews at the front engine mounts.
2. Align and install capscrews (2) through the rear engine mounts and into cradle structure (3), but do not tighten at this time.
3. Install the alternator on the engine. Refer to Alternator Installation in this section.
4. Adjust setscrew (3, Figure 4-1) to equalize gap (5) between cradle structure (1) and subframe (4) at the left side and right side. Lock the setscrew with jam nut (2).

SECTION C5
AIR CLEANERS
INDEX

AIR CLEANERS C5-3

 OPERATION C5-3

 SERVICING THE AIR CLEANERS C5-3

 Replacing The Elements C5-4

 AIR CLEANER ASSEMBLY CLEANING C5-5

 Primary Element Cleaning C5-5

 Precleaner Section Cleaning C5-7

 AIR INTAKE TROUBLESHOOTING C5-8

NOTES

AIR CLEANERS

OPERATION

Air required by the diesel engine passes through the air cleaner assemblies mounted on each side of the radiator. These air cleaners discharge heavy particles of dust and dirt by centrifugal action and then remove finer particles by passing air through filter cartridges.

The engine demand for air creates a vacuum in the air cleaners and causes outside air to be drawn in through air inlets on the air cleaners. Dirty air entering here is drawn through a series of tubes that are designed to produce a cyclonic action. As the air passes through the outer portion of the tubes, a circular motion is set up causing dust and dirt particles to be thrown from the air stream into dust collectors (1, Figure 5-1). At the same time, the air stream turns and is directed up through the center of the tubes into the filter chamber. Here the air passes through the main filter element and safety filter element and out the clean air outlet to the engine's air intake system. The function of the safety filter is to increase overall reliability and engine protection.

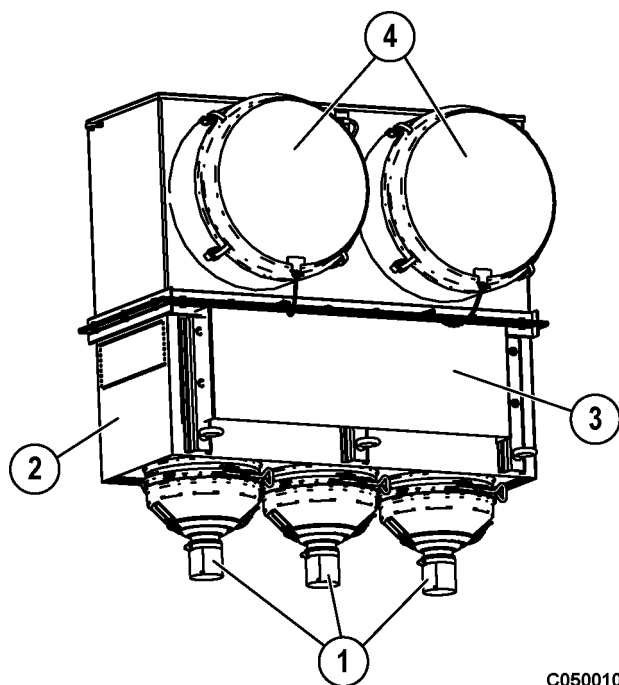


FIGURE 5-1. AIR CLEANERS

- | | |
|-----------------------|---------------------|
| 1. Dust Collectors | 3. Air Intake Cover |
| 2. Precleaner Section | 4. Element Covers |

SERVICING THE AIR CLEANERS

CAUTION

The engine must be turned off before servicing the air cleaner assemblies or opening the engine air intake system. Never start the engine with the filter elements removed. Serious engine damage can result.

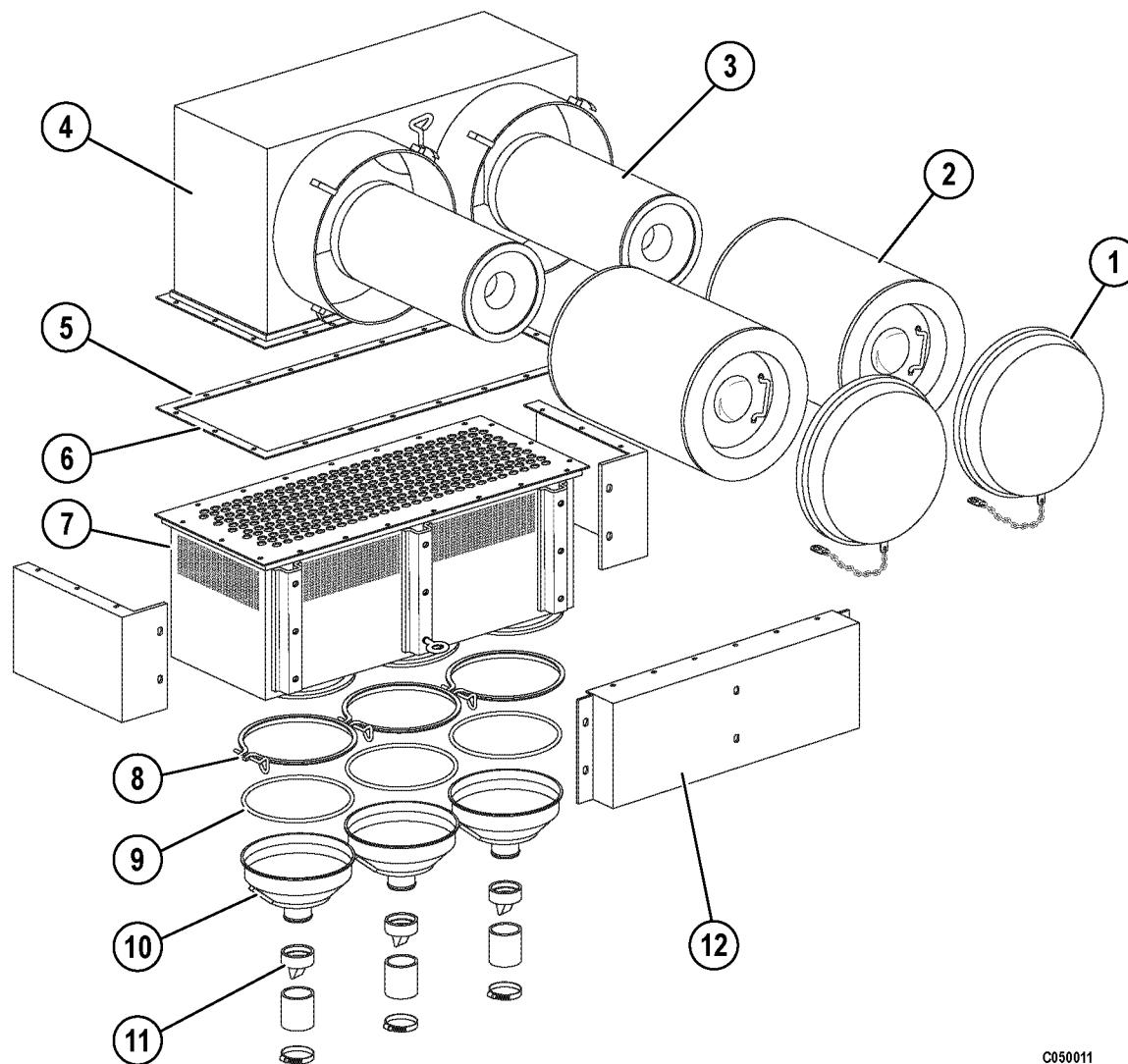
- Inspect and empty the dust cups at regular intervals. Daily inspection is recommended. Never allow the dust level to build up to the tube (precleaner) chamber.
- During operation or after the engine has been turned off, observe the air filter restriction gauges mounted on the overhead panel in the cab. When a gauge shows maximum restriction, filter service is required.
- Check all engine air inlet tubes, hoses and clamps. All connections must be air tight to prevent dirt from entering.
- Air cleaner housing fasteners and mountings must be tight.
- After the filters have been serviced, reset the air filter restriction gauges by pressing the reset button on the face of the gauge.

Replacing The Elements

NOTE: The function of the safety element is to increase overall reliability and engine protection. If the safety element indicator shows red, the element has become clogged and should be replaced with a new one.

1. Turn off the engine. Clean any dirt and dust from the area around the element cover.

2. Loosen the clips on element cover (1, Figure 5-2) and remove the element cover. Pull primary element (2) from air cleaner assembly (4).
3. Inspect the primary element carefully for damage, holes or breaks which might affect reuse of the element. If the element appears serviceable, proceed with the cleaning procedure. If defects are found in the element, replace the element.



C050011

FIGURE 5-2. AIR CLEANER ASSEMBLY

- | | | |
|-------------------------|--------------------------|----------------------|
| 1. Element Cover | 5. Gasket (long strips) | 9. O-Ring |
| 2. Primary Element | 6. Gasket (short strips) | 10. Dust Cup |
| 3. Safety Element | 7. Precleaner Section | 11. Dust Cup Valve |
| 4. Air Cleaner Assembly | 8. Clamp | 12. Air Intake Cover |

IMPORTANT

Have a new safety (secondary) filter element on hand before removing the used filter element. Do not keep the intake system open to the atmosphere any longer than necessary.

4. If the safety element must be replaced, remove and discard the safety element. **Do not** clean the damaged or dirty safety element.
5. Install the new safety element.
6. Install primary element (2) into the air cleaner. If the original element is being reused, ensure the sealing gasket is not damaged. The gasket must seal completely.
7. If open, close and latch dust cup valves (11) on the bottom of dust cups (10).

AIR CLEANER ASSEMBLY CLEANING

Primary Element Cleaning

IMPORTANT

Only the primary elements may be cleaned, and then only if they are structurally intact. Do not reuse an element that is damaged. Do not clean and reuse the safety elements. Replace them with new parts if necessary.

After inspection, determine the condition of the primary element. Choose either the washing method or compressed air method for cleaning the element. If the element is clogged with carbon, soot, oil and/or dust, the complete washing procedure will produce the best results.

Wash elements with water and detergent as follows:

1. Soak the element in a solution of detergent and water for at least 15 minutes. Rotate the element back and forth in the solution to loosen dirt deposits. **Do not soak elements for more than 24 hours.**
2. Rinse the element with a stream of fresh water in the opposite direction of normal air flow until rinse water runs clear. Maximum permissible water pressure is 276 kPa (40 psi). A complete and thorough rinse is essential.
3. Dry the element thoroughly. If drying is done with heated air, the maximum temperature must not exceed 60°C (140°F) and must be circulated continually. Do not use a light bulb to dry elements.

4. After cleaning, inspect the element thoroughly for the slightest ruptures and damaged gaskets. A good method for detecting paper ruptures is to place a light inside the filter element, as shown in Figure 5-3, and inspect the outer surface of the filter element. If holes or ruptures are found, do not reuse the element. Discard and replace with a new element.

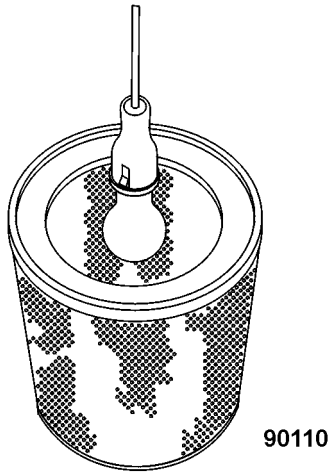


FIGURE 5-3. INSPECTING THE FILTER ELEMENT

Clean dust loaded elements with dry filtered compressed air as follows:

1. Maximum nozzle pressure must not exceed 207 kPa (30 psi). The distance from the nozzle to the surface of the filter element must be at least 25 mm (1 in.) to prevent damage to the filter material.
2. As shown in Figure 5-4, direct the stream of air from the nozzle against the inside of the filter element. This is the clean air side of the element and air flow should be opposite of normal air flow.
3. Move the air flow up and down vertically with the pleats in the filter material while slowly rotating the filter element.
4. When cleaning is complete, inspect the filter element as shown in Figure 5-3. If holes or ruptures are noted, discard the element and replace with a new element.

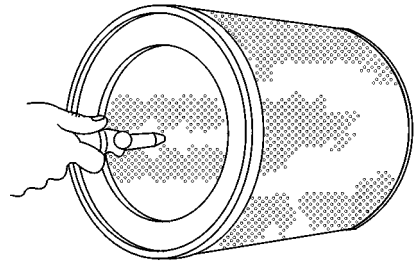


FIGURE 5-4. CLEANING THE FILTER ELEMENT WITH COMPRESSED AIR

Precleaner Section Cleaning

The tubes in precleaner section (7, Figure 5-2) should be cleaned at least once per year and at each engine overhaul. More frequent cleaning may be necessary depending upon operating conditions and the local environment.

To inspect the tubes in the precleaner section, remove the primary element. Do not remove the safety element. Loosen clamps (8) and remove dust cups (10) and O-rings (9). Use a light to inspect the tubes. All tubes should be clear and the light should be visible.

NOTE: Both the primary and safety elements must be installed in the air cleaner while Steps 1 and 2 are being accomplished to prevent any possibility of dirt being forced into the engine intake area.

Dust can be removed with a stiff fiber brush (see Figure 5-5). **Do not use a wire brush.** Dust may also be removed effectively using compressed air.

Heavy plugging of the tubes may require soaking and washing the entire precleaner section. Refer to the following procedure.

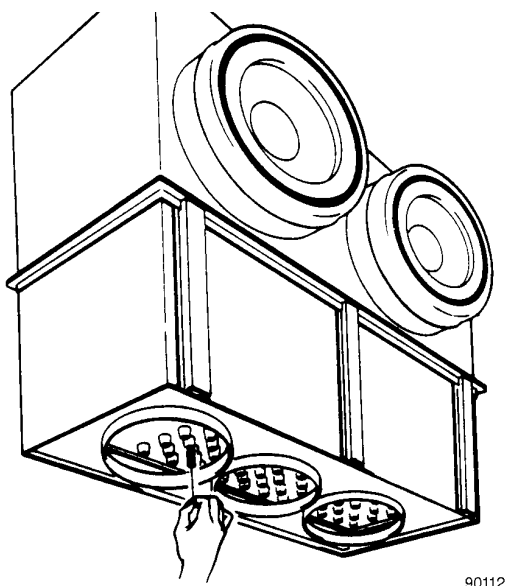


FIGURE 5-5. REMOVING DUST FROM THE TUBES

NOTE: The precleaner section may be separated from the air cleaner assembly without removing the entire air cleaner from the truck.

1. Remove air intake cover (12, Figure 5-2). Remove the mounting hardware that secures precleaner section (7) to air cleaner assembly (4). Remove the precleaner section. The safety element must remain in place to protect the engine intake.
2. Loosen clamps (8) and remove dust cups (10) and O-rings (9) from the precleaner section. Wash the dust cups with a water and liquid soap solution.
3. Submerge the precleaner section in a solution of Donaldson D-1400 and warm water (see Figure 5-6). Mix the solution according to the directions on the package. The tube section must be down. Soak for 30 minutes, then remove the precleaner section from the solution. Rinse thoroughly with fresh water and blow dry.

Severe plugging may require the use of an Oakite 202 and water solution instead. The solution should be 50% Oakite 202 and 50% fresh water.
4. Check precleaner gaskets carefully for any evidence of air leaks. Replace if necessary.
5. Install precleaner section (7) and gaskets (5) and (6) on air cleaner assembly (4). Install all mounting hardware that was removed.
6. Install dust cups (10) and O-rings (9) on the precleaner section. Secure the dust cups with clamps (8).

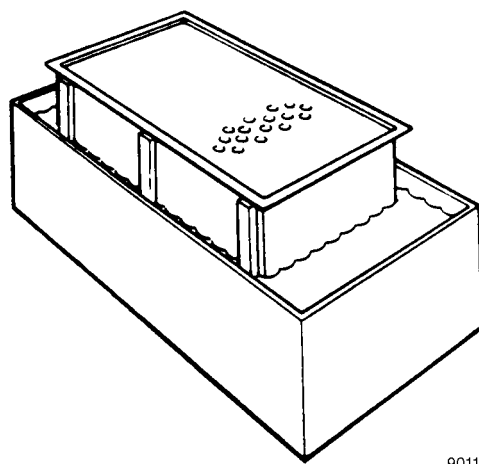


FIGURE 5-6. WASHING AND SOAKING THE PRECLEANER SECTION

AIR INTAKE TROUBLESHOOTING

To ensure maximum engine protection, ensure that all connections between the air cleaners and engine intake are tight and positively sealed. If air leaks are suspected, check the following:

1. All intake lines, tubes and hump hoses for breaks, cracks, holes, etc, which could allow an intake air leak.
2. Check all air cleaner gaskets for positive sealing.
3. Check the primary and safety elements for ruptures, holes or cracks.
4. Check air cleaner assembly for structural damage, cracks, breaks or other defects which could allow air leakage. Check all mounting hardware for tightness.

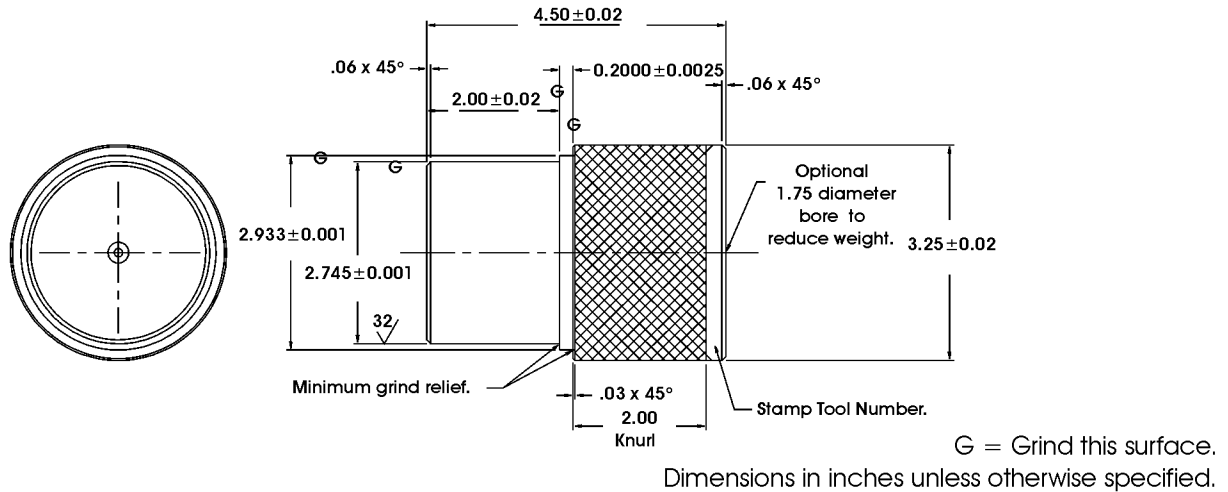
SECTION C7 FAN CLUTCH INDEX

REMOVAL & INSTALLATION TOOLING	C7-3
DISASSEMBLY	C7-6
CLEANING AND INSPECTION	C7-16
ASSEMBLY	C7-20
TEST PROCEDURE	C7-34

NOTES

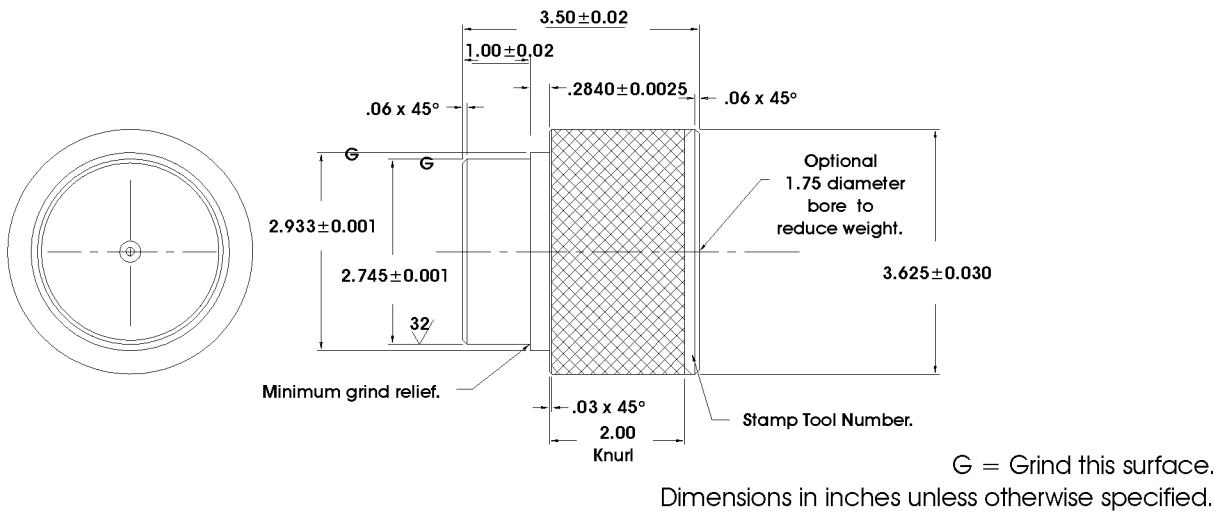
FAN CLUTCH

REMOVAL & INSTALLATION TOOLING



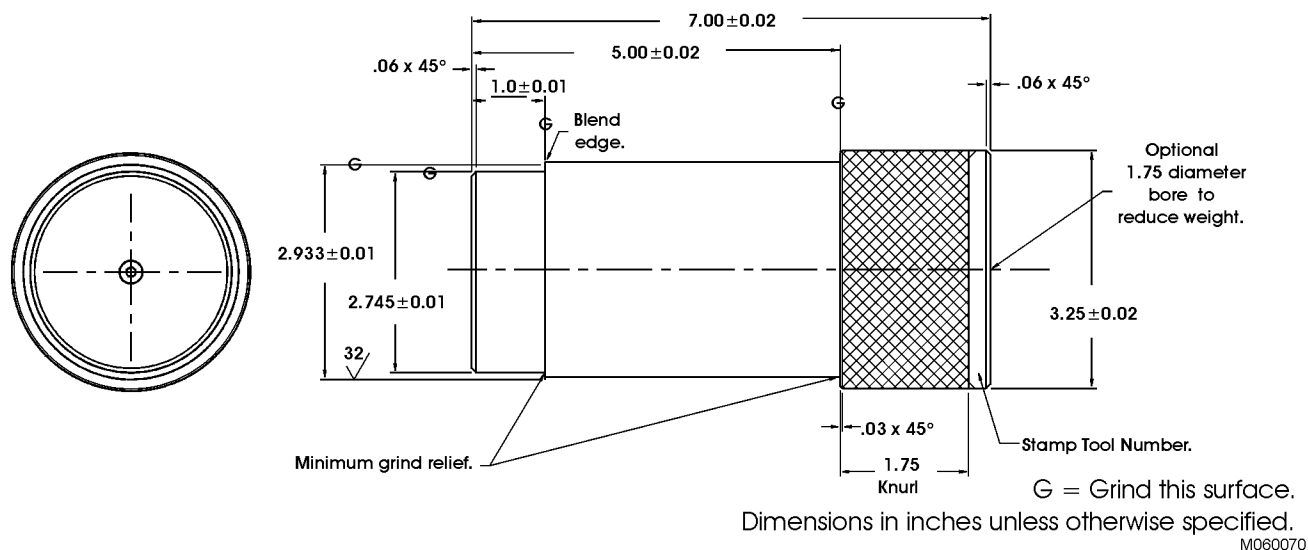
M060068

TOOL A - FRONT SLEEVE BEARING

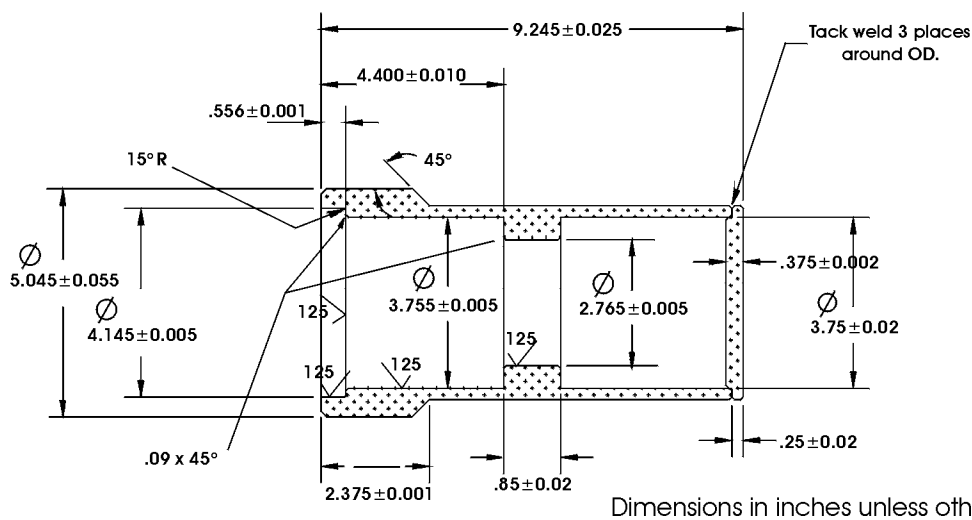


M060069

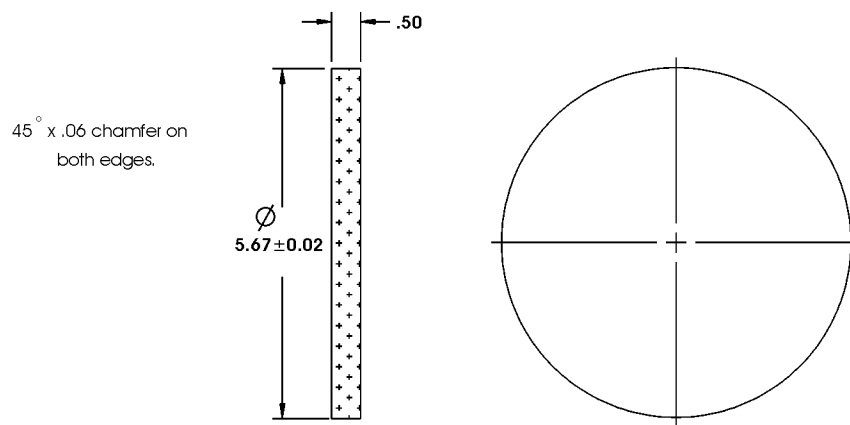
TOOL B - REAR SLEEVE BEARING



TOOL C - FRONT AND REAR SLEEVE BEARING REMOVER



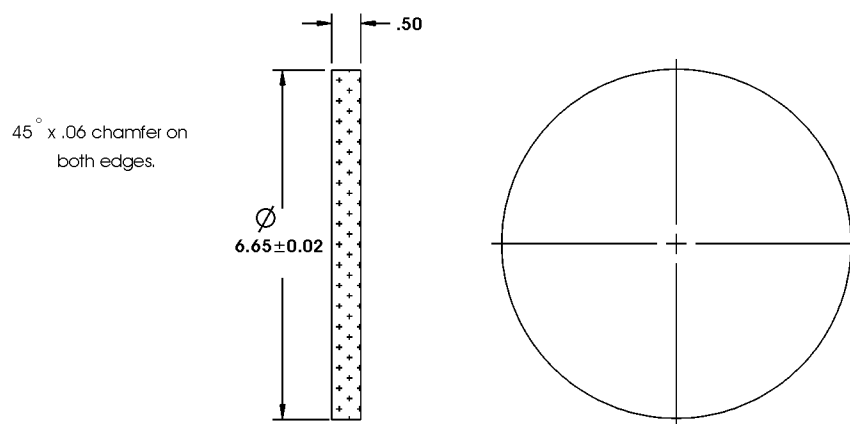
TOOL D - WEAR SLEEVE AND RETAINER/SEAL ASSEMBLY INSTALLER;
 BEARING REMOVER; ASSEMBLY PUSHER TOOL



Dimensions in inches unless otherwise specified.

M060072

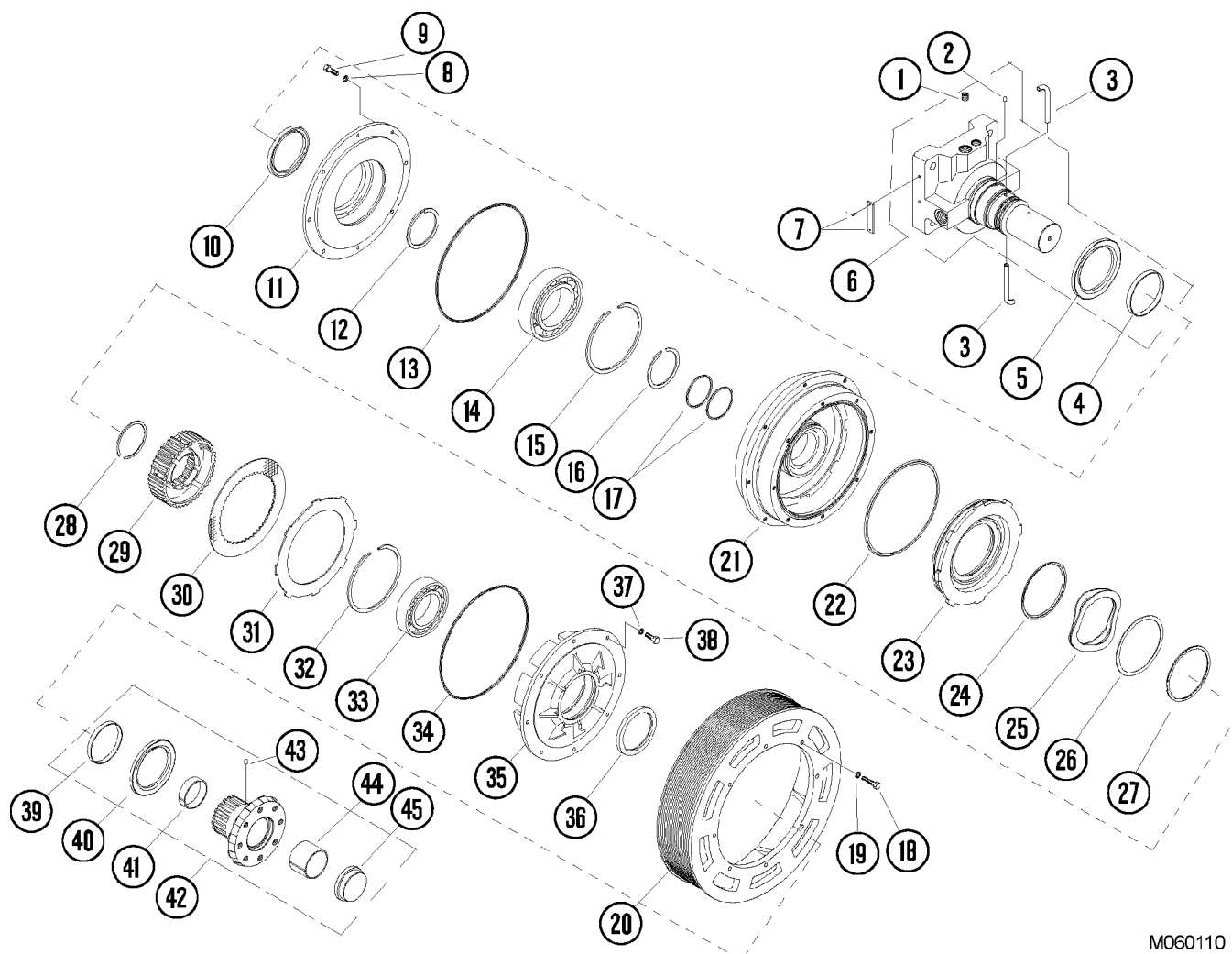
TOOL E - BEARING INSTALLER



M060073

TOOL F - BEARING INSTALLER

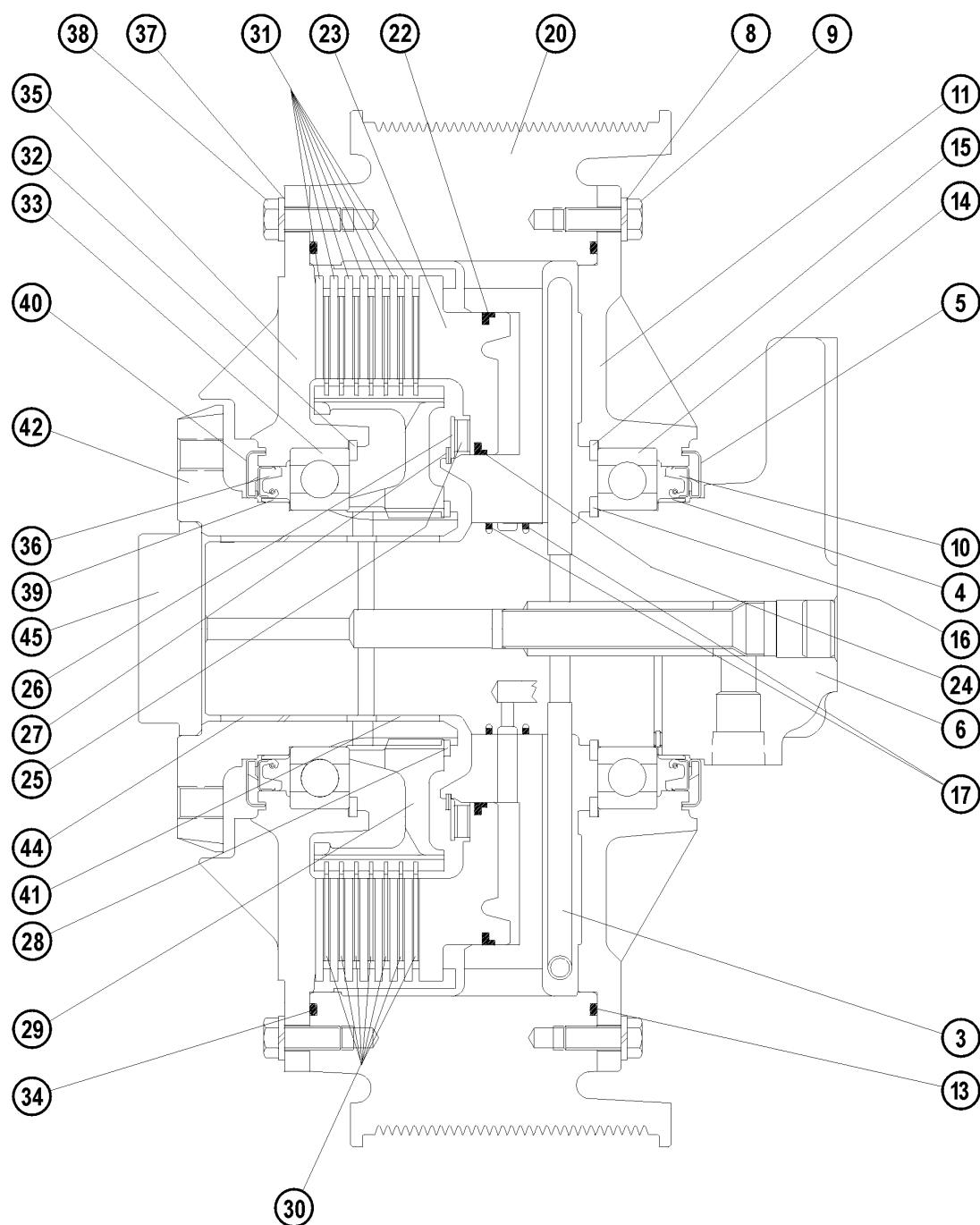
DISASSEMBLY



M060110

FIGURE 7-1. FAN CLUTCH EXPLODED VIEW

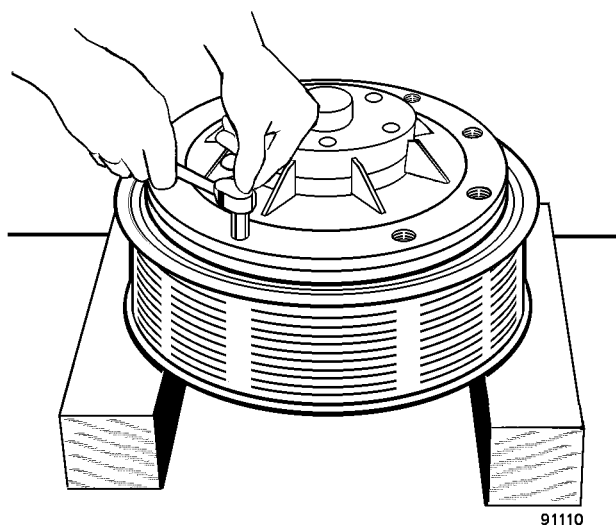
- | | | |
|--|---------------------------------|----------------------------------|
| 1. Orifice Fitting | 16. External Snap Ring (Spacer) | 32. Internal Snap Ring |
| 2. Dowel Pin (Rear) | 17. Seal Ring (Hook-Type) | 33. Main Bearing (Front) |
| 3. Pitot Tube | 18. Bolt | 34. O-Ring Seal |
| 4. Wear Sleeve | 19. Washer | 35. Bearing Retainer (Front) |
| 5. Retainer/Seal Assembly | 20. Pulley | 36. Oil Seal |
| 6. Shaft Assembly | 21. Pulley Adapter | 37. Washer |
| 7. Name Plate Kit | 22. Seal Ring (Large) | 38. Bolt |
| 8. Washer | 23. Piston | 39. Wear Sleeve |
| 9. Bolt | 24. Seal Ring (Small) | 40. Retainer/Seal Assembly |
| 10. Oil Seal | 25. Spring Washer | 41. Sleeve Bearing (Rear, Short) |
| 11. Bearing Retainer (Rear) | 26. Shim | 42. Fan Mounting Hub |
| 12. Bearing Spacer
(External Snap Ring) | 27. External Snap Ring | 43. Dowel Pin (Front) |
| 13. O-Ring Seal | 28. External Snap Ring | 44. Sleeve Bearing (Front, Long) |
| 14. Main Bearing (Rear) | 29. Clutch Hub | 45. End Cap |
| 15. Internal Snap Ring | 30. Facing Plate | |
| | 31. Steel Clutch Plate | |



M060075

FIGURE 7-1. FAN CLUTCH CUTAWAY (Typical)

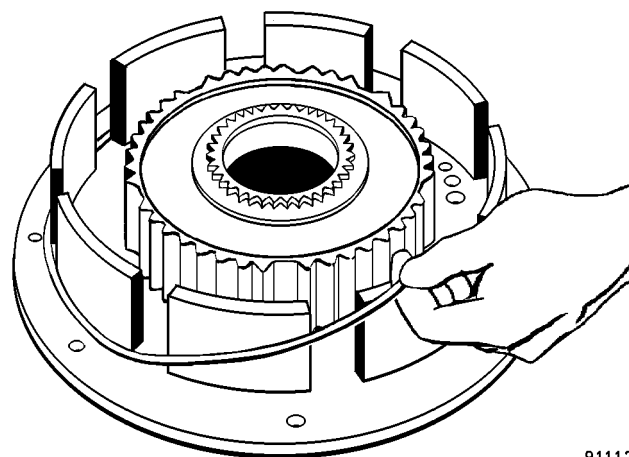
- | | | | |
|---------------------------|---------------------------|------------------------|----------------------------------|
| 3. Pitot Tube | 14. Main Bearing | 27. External Snap Ring | 37. Washer |
| 4. Wear Sleeve | 15. Internal Snap Ring | 28. External Snap Ring | 38. Bolt |
| 5. Retainer/Seal Assembly | 16. External Snap Ring | 29. Clutch Hub | 39. Wear Sleeve |
| 6. Shaft Assembly | 17. Seal Ring (Hook-Type) | 30. Facing Plate | 40. Retainer/Seal Assembly |
| 8. Washer | 20. Pulley | 31. Steel Clutch Plate | 41. Sleeve Bearing (Rear, Short) |
| 9. Bolt | 22. Seal Ring (Large) | 32. Internal Snap Ring | 42. Fan Mounting Hub |
| 10. Oil Seal | 23. Piston | 33. Main Bearing | 44. Sleeve Bearing (Front, Long) |
| 11. Bearing Retainer | 24. Seal Ring (Small) | 34. O-Ring Seal | 45. End Cap |
| 13. O-Ring Seal | 25. Spring Washer | 35. Bearing Retainer | |
| | 26. Shim | 36. Oil Seal | |



91110

FIGURE 7-2.

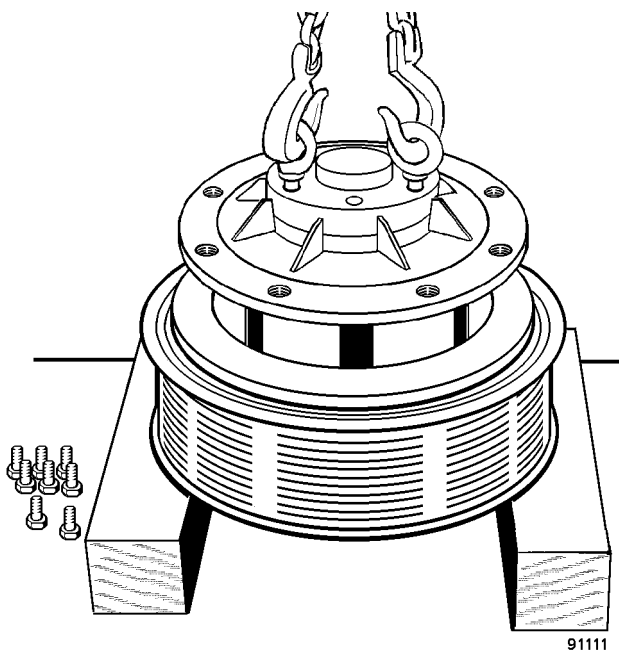
1. Support the fan clutch on a bench with fan mounting hub (42) facing upward. Support the assembly beneath the pulley. Remove bolts (38) and washers (37).



91112

FIGURE 7-4.

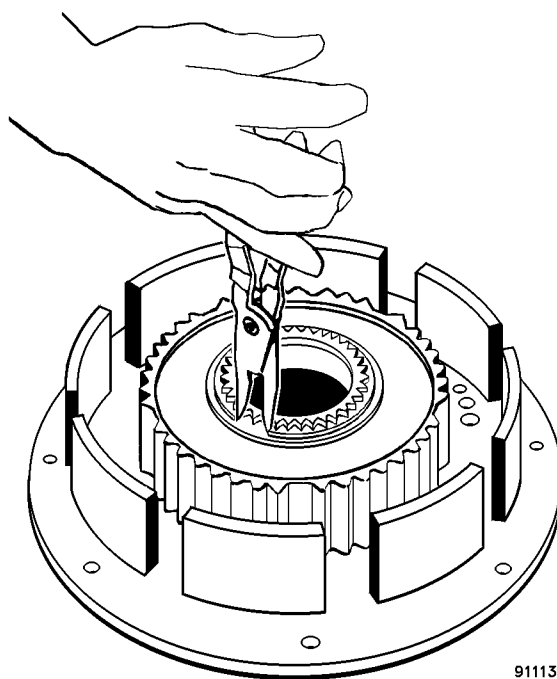
3. Remove O-ring seal (34).



91111

FIGURE 7-3.

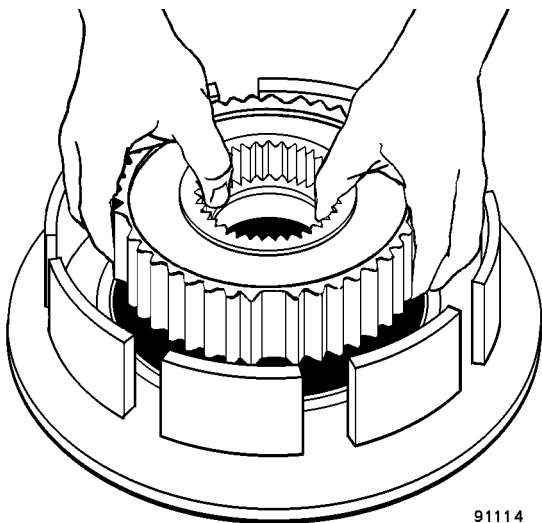
2. Install lifting eyes, and attach a hoist and chains to front bearing retainer (35). Use a small screwdriver to separate the front bearing retainer from pulley adapter (21), and set it aside on a bench.



91113

FIGURE 7-5.

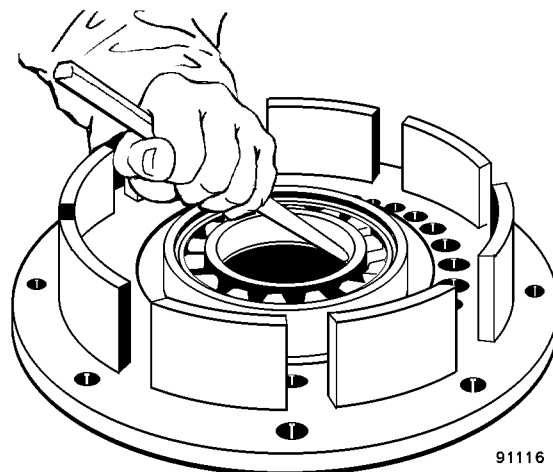
4. Position the bearing retainer and hub assembly on the bench with clutch hub (29) up. Remove external snap ring (28).



91114

FIGURE 7-6.

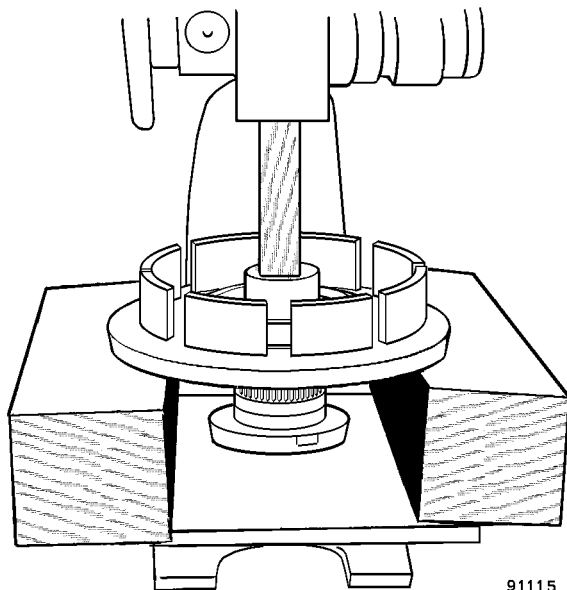
5. Remove clutch hub (29).



91116

FIGURE 7-8.

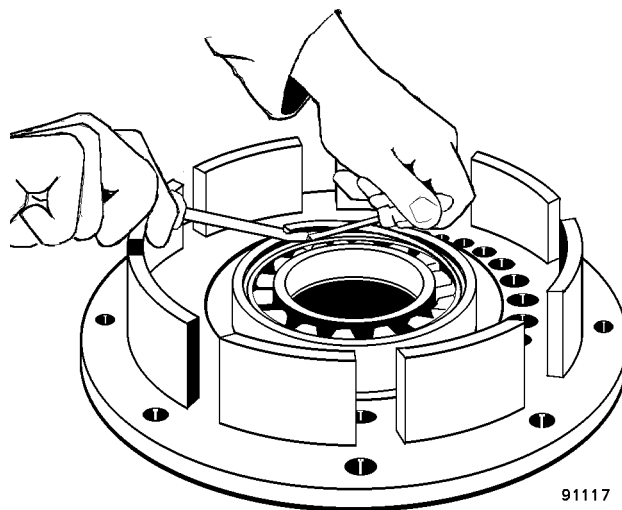
7. Remove front oil seal (36).



91115

FIGURE 7-7.

6. Position the sub-assembly beneath the ram of a press. Support the assembly beneath the bearing retainer as close as possible to fan mounting hub (42). Press the fan mounting hub out of the front bearing using tooling (B).



91117

FIGURE 7-9.

8. Remove internal snap ring (32).

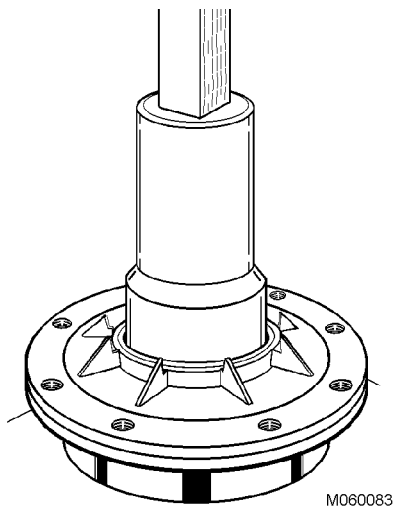


FIGURE 7-10.

9. Turn bearing retainer (35) over on the press bed. Press front bearing (33) out of the bearing retainer using tooling (D).

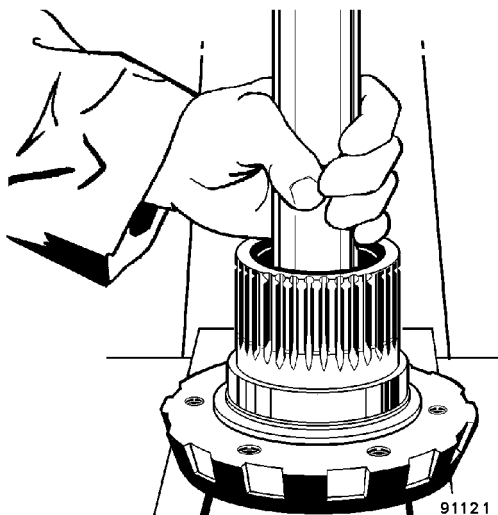


FIGURE 7-11.

10. Support beneath the fan mounting hub with end cap (45) down, but approximately 50 mm (2 in.) above the press bed. Using a solid steel bar or equivalent, press the end cap from the fan mounting hub.

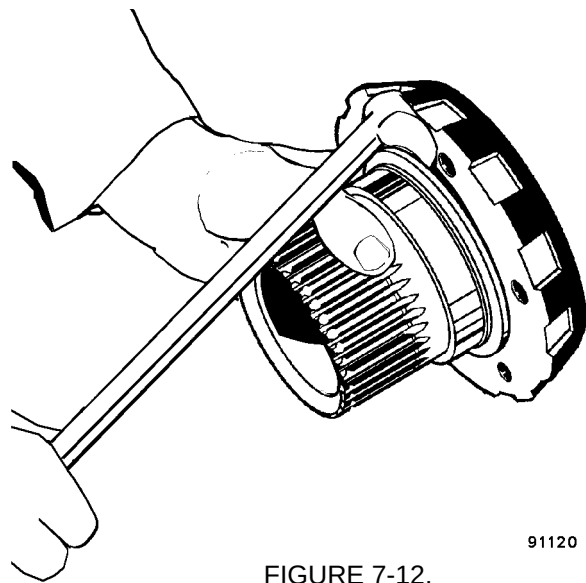


FIGURE 7-12.

11. Remove front retainer/seal assembly (40). Wedge a large chisel or other appropriate tool behind the retainer to force it off fan mounting hub (42).

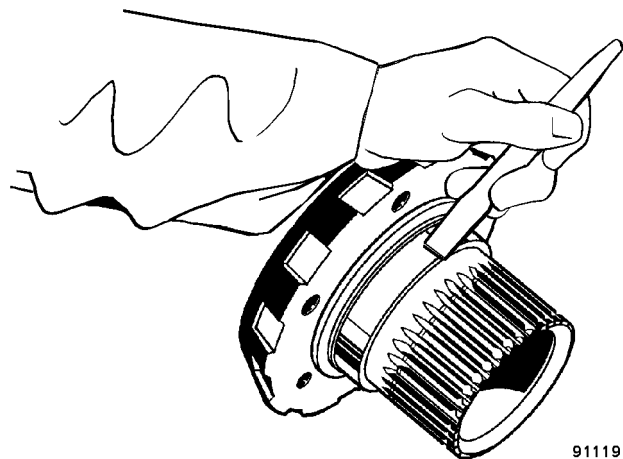


FIGURE 7-13.

Use a chisel to make three indentations in wear sleeve (39) in order to loosen the sleeve. The indentations should be approximately 120 degrees apart from one another. Remove the wear sleeve.

NOTE: Use caution when using the chisel. Do not cut through the sleeve. Damage to the shaft can cause future leaks.

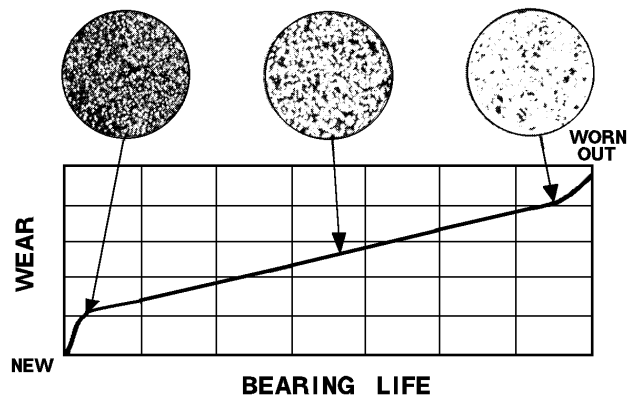


FIGURE 7-14.

12. Inspect sleeve bearing (44) and sleeve bearing (41). Compare the color of each bearing to the chart above. The lighter the appearance of the bearing, the more worn it is. If either bearing needs replacing, proceed to the next step. If the bearings are in good condition, skip the next step.

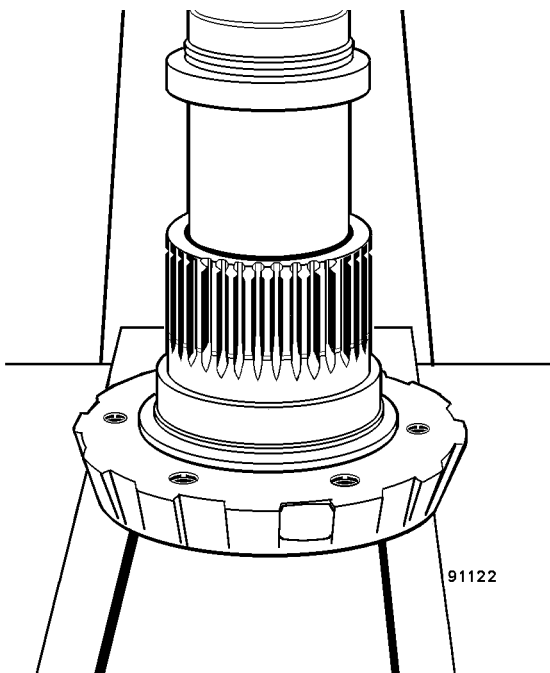


FIGURE 7-15.

13. Position tooling (C) against sleeve bearing (41). Press the front sleeve bearing downward to press it out of the fan mounting hub. Rear sleeve bearing (44) will be pressed out simultaneously.

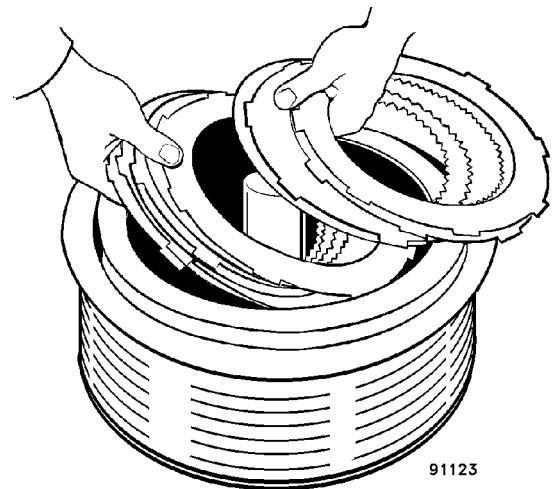


FIGURE 7-16.

14. Remove the stack of facing plates (30) and steel clutch plates (31) from inside the pulley.

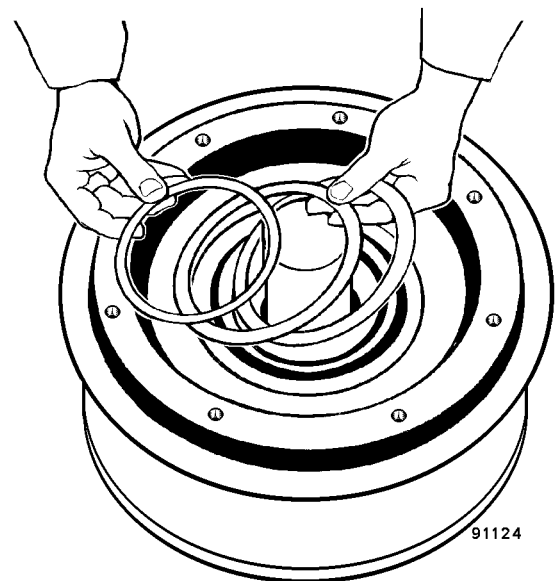
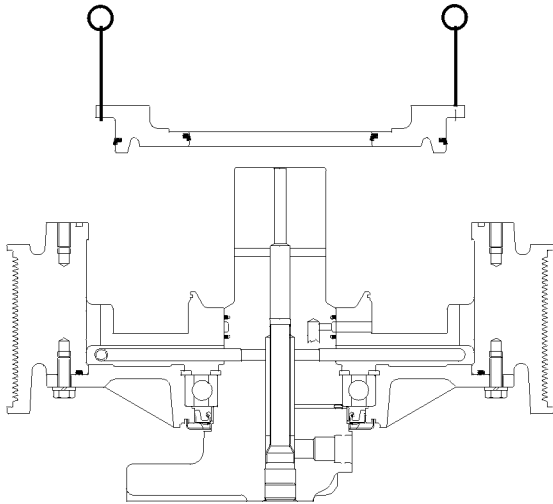


FIGURE 7-17.

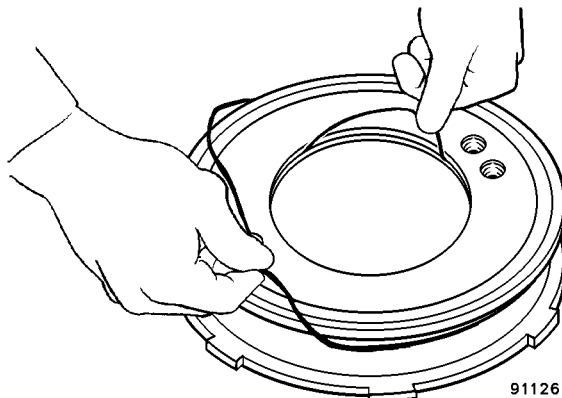
15. Remove external snap ring (27), shim (26), and spring washer (25).



M060077

FIGURE 7-18.

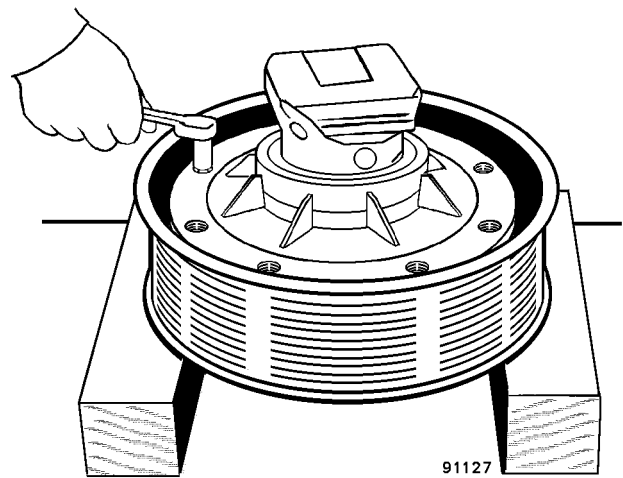
16. Attach wire lifting hooks to piston (23). Use the lifting hooks to pull the piston from pulley adapter (21).



91126

FIGURE 7-19.

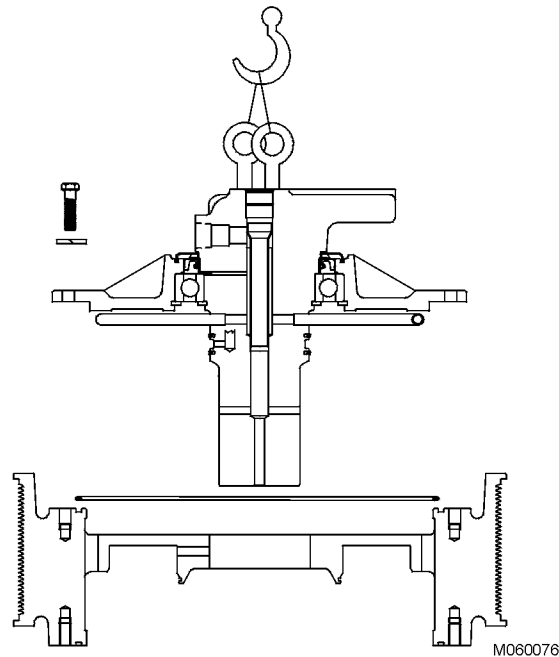
17. Remove seal rings (22) and (24) from the piston.



91127

FIGURE 7-20.

18. Support beneath the pulley to prevent it from dropping to the bench. Remove bolts (9) and lockwashers (8).



M060076

FIGURE 7-21.

19. Install lifting eyebolts to the shaft and bearing retainer assembly. Use a suitable lifting device to lift the assembly from the pulley. Remove O-ring seal (13).

NOTE: It may be necessary to use a soft rubber mallet to separate the shaft and bearing retainer from the pulley.

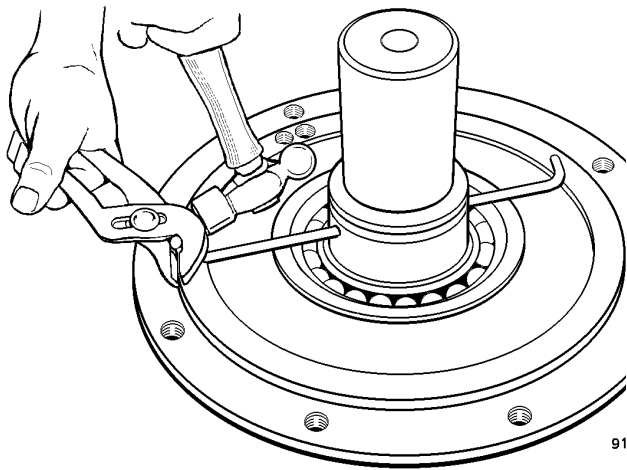


FIGURE 7-22.

20. Position the shaft as shown. Insert a phillips-head screwdriver into pitot tubes (3) to loosen and remove them from the shaft. Rotate the pitot tube until the sealant holding it tight is broken loose. Then grip the pitot tube with a pair of pliers and gently tap on the pliers to remove the pitot tubes from the hole in the shaft.

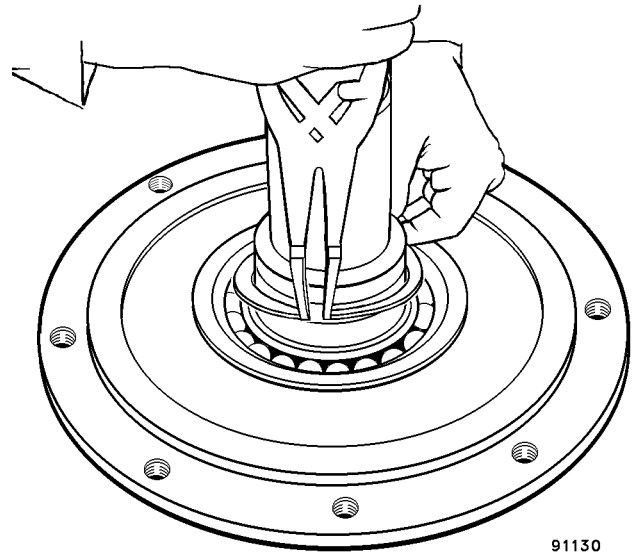


FIGURE 7-24.

22. Remove external snap ring (16).

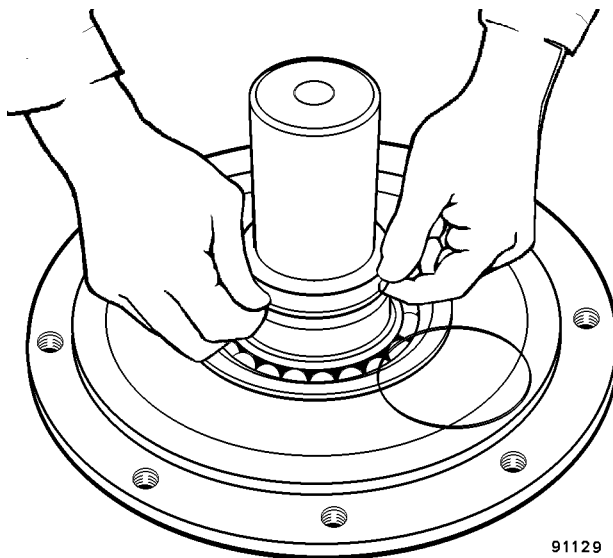


FIGURE 7-23.

21. Remove both seal rings (17).

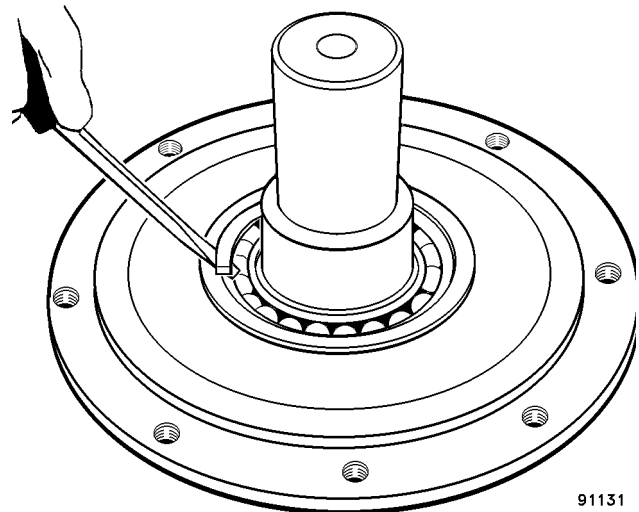


FIGURE 7-25.

23. Remove internal snap ring (15).

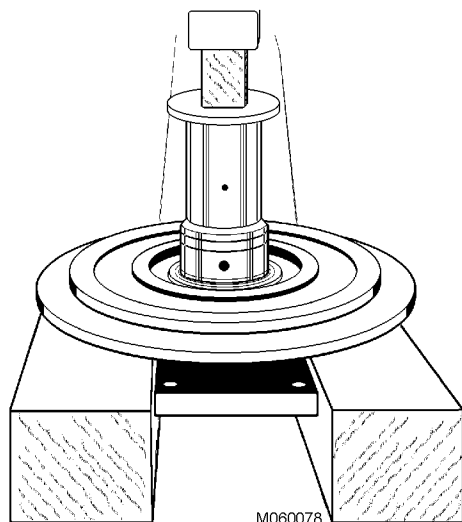


FIGURE 7-26.

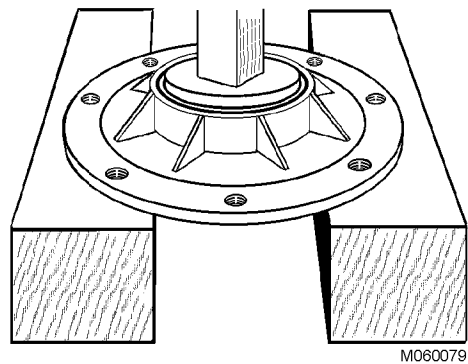


FIGURE 7-28.

24. Support the bearing retainer as close as possible to the bearing bore. Be careful not to damage the retainer/seal assembly. Press the shaft out of bearing (14) using tooling (E).

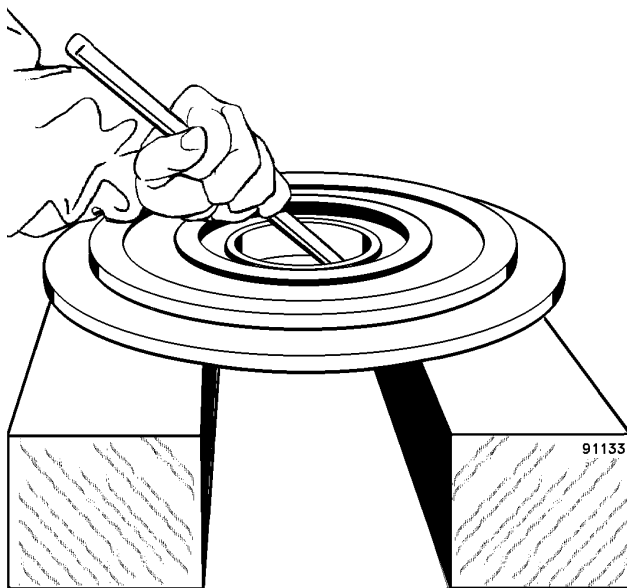


FIGURE 7-27.

25. Remove oil seal (10) from bearing retainer (11).

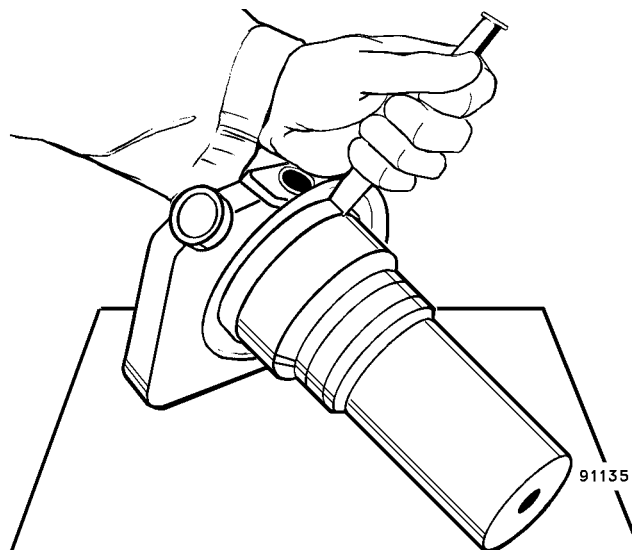


FIGURE 7-29.

27. Use a chisel to make three indentations in wear sleeve (4). The indentations should be approximately 120 degrees apart from one another. Remove the wear sleeve.

NOTE: Use caution when using the chisel. Do not cut through the sleeve. Damage to the shaft can cause future leaks.

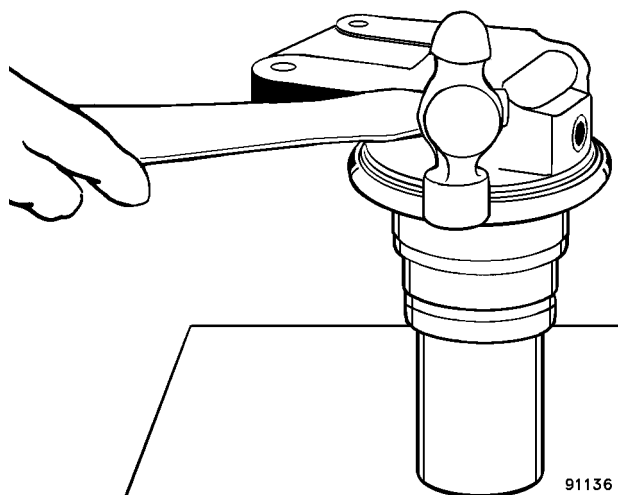


FIGURE 7-30.

28. Remove rear retainer/seal assembly (5). Drive the assembly off the shaft or wedge a large chisel or other appropriate tool behind the retainer to force it off.

CLEANING AND INSPECTION

Thoroughly clean all components before inspection.

Check each of the following components, and follow the guidelines for reuse:

- **Ball bearings** - Replace at time of rebuild.
- **Internal snap rings** - Must not be damaged or worn. Must be flat and have square edges at outer diameter.
- **External snap rings** - Must not be damaged or worn. Must be flat and have square edges at inner diameter.
- **Seal rings** - Replace during rebuild.
- **Oil seals** - Replace during rebuild.
- **Bolts and washers** - Reuse unless damaged or worn.
- **Retainer/Seal assemblies** - Replace if damaged or worn.
- **Wear sleeves** - Replace during rebuild.
- **Sleeve bearings** - Inspect color of surface. Refer to Figure 7-14.

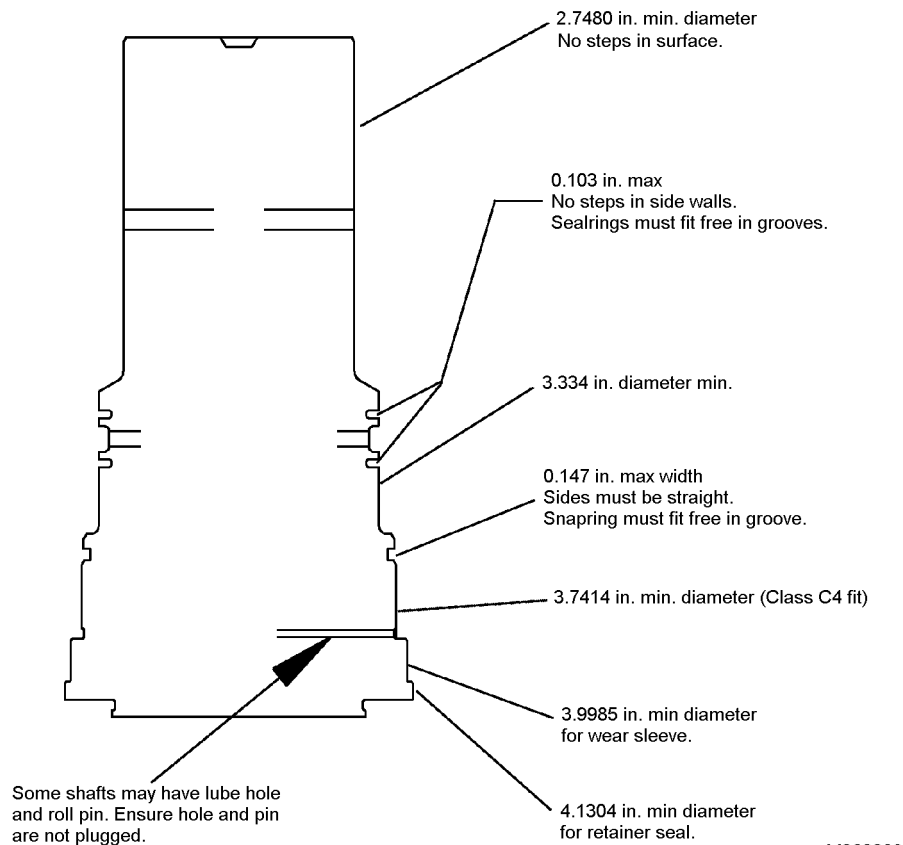


FIGURE 7-31. SHAFT ASSEMBLY WEAR DIMENSIONS

1. Check the shaft assembly for wear or damage. Refer to Figure 7-31 for dimensions.

NOTE: Some shafts were manufactured as two-piece assemblies. Do not attempt to separate the shaft assembly.

2. Inspect and clean the pitot tube holes in the shaft. Use a standard reamer (straight flute, 0.3770 in. diameter). Remove pipe plugs in the shaft for cleaning and reinstall using Loctite® Primer N and #242.

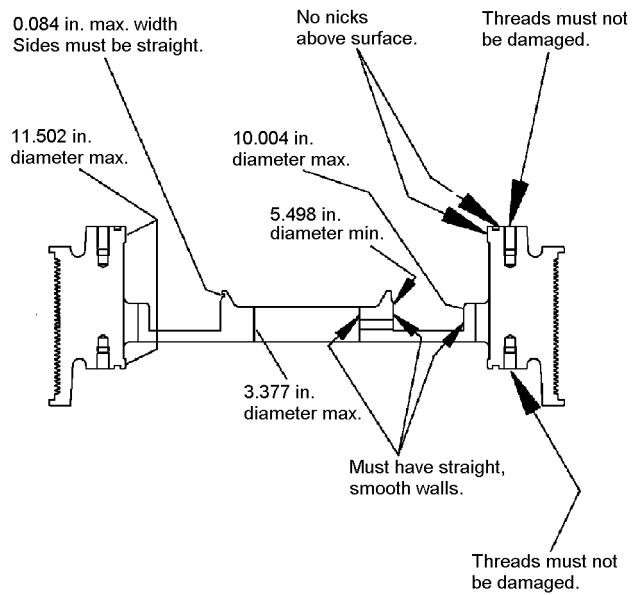


FIGURE 7-32.

M060092

3. Check pulley and adapter dimensions.

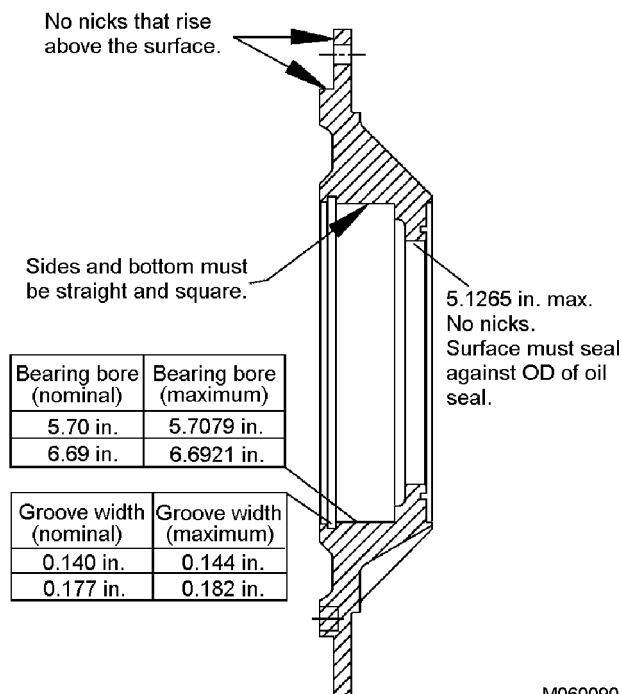


FIGURE 7-33.

M060090

4. Check rear bearing retainer (11) dimensions.

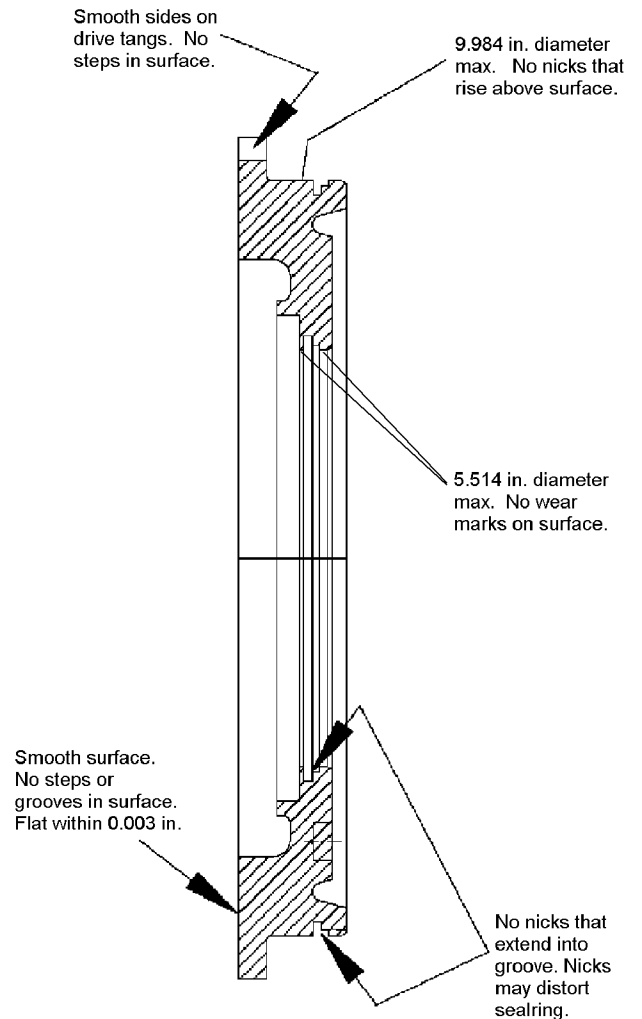
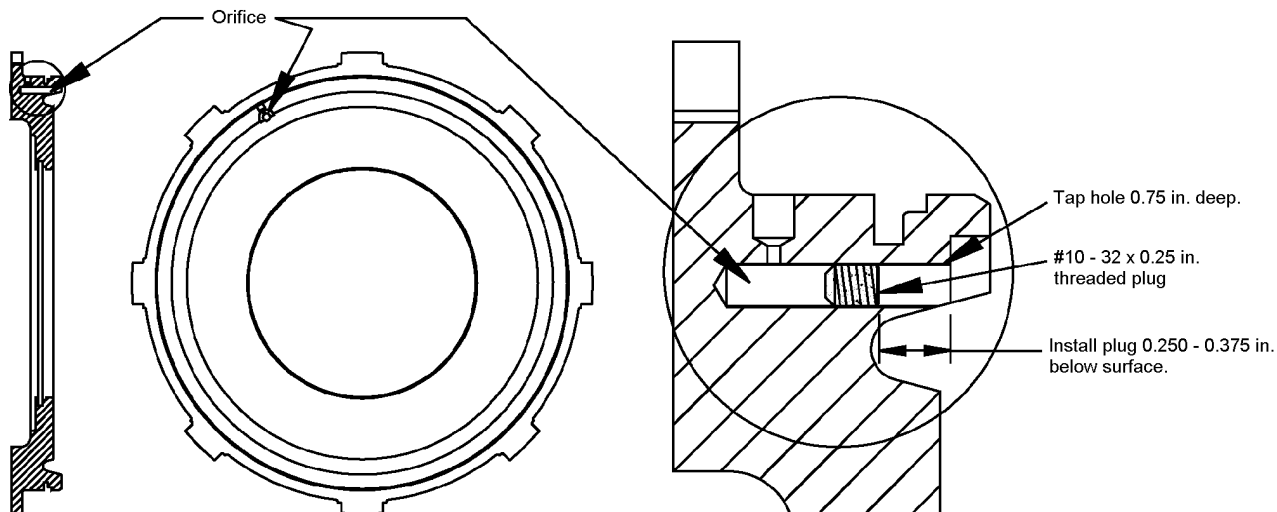


FIGURE 7-34.

M060091

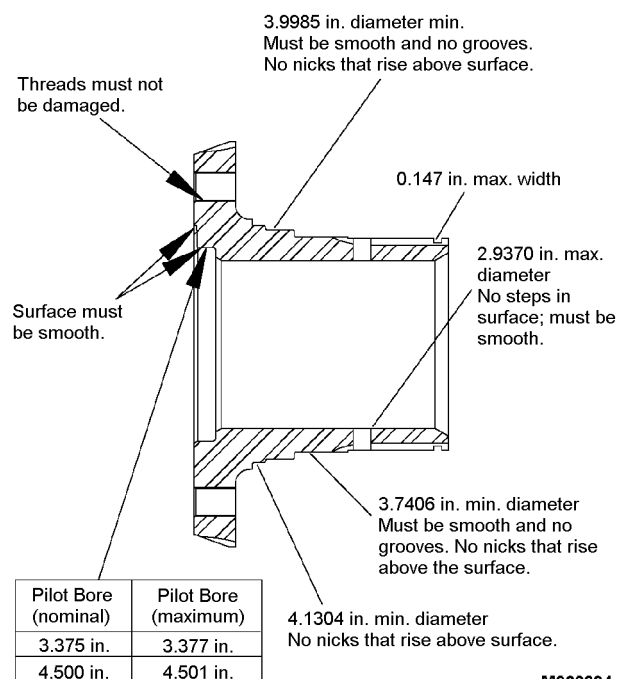
5. Check piston (23) dimensions.



M060088

FIGURE 7-35. PISTON REWORK
(For earlier pistons with the drilled orifice.)

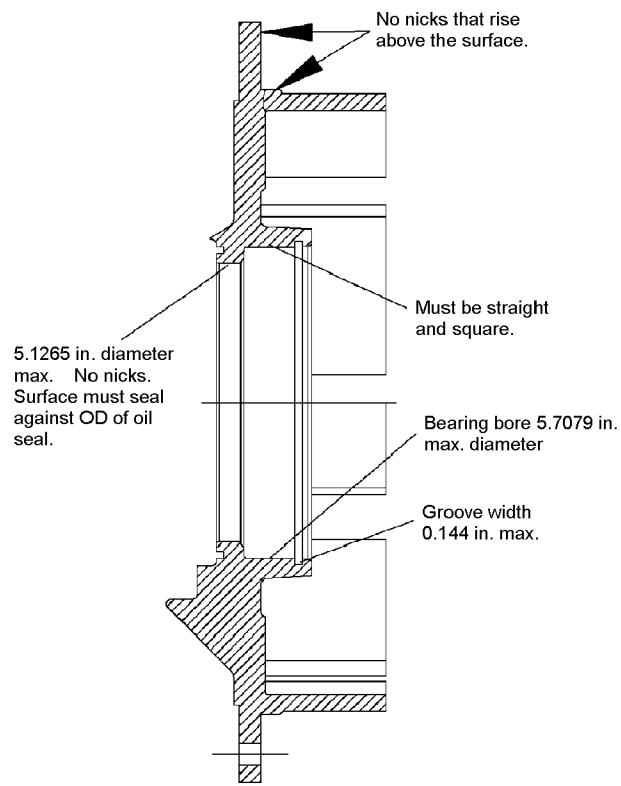
6. Check the piston for a drilled orifice at the inside face. If the piston contains the orifice, modify the piston as shown in Figure 7-35.
7. Inspect clutch hub (29) for wear. Wear marks that may be present on the teeth must not restrict plate movement. If they have smooth entry and exit ramps, the notches will not restrict plate movement and the clutch hub may be reused.
8. Check steel plates (31) for wear. The plates must be smooth and free of grooves or heat related damage. The plates are 3.07 mm (0.121 in.) minimum thickness when new and must be flat within 0.13 mm (0.005 in.).
9. Inspect facing plates (30). Minimum thickness for new facing plates is 2.77 mm (0.109 in.). Grooves are 0.15 mm (0.006 in.) deep. The plates must be flat within 0.13 mm (0.005 in.). Check the teeth for excessive wear. When new, the space between the teeth is approximately 7.11 mm (0.280 in.).



M060094

FIGURE 7-36.

10. Inspect fan mounting hub (42).



M060093

FIGURE 7-37.

11. Inspect front bearing retainer (35).
12. Inspect end cap (45) for any wear or raised nicks.

ASSEMBLY

NOTE: The fan clutch is reassembled using Loctite® (or equivalent) sealants. Follow manufacturer's recommendations regarding minimum cure time to prevent oil from washing the sealant from the sealing surfaces.

1. Place end cap (45) in a freezer or on dry ice to prepare for installation in the following steps.

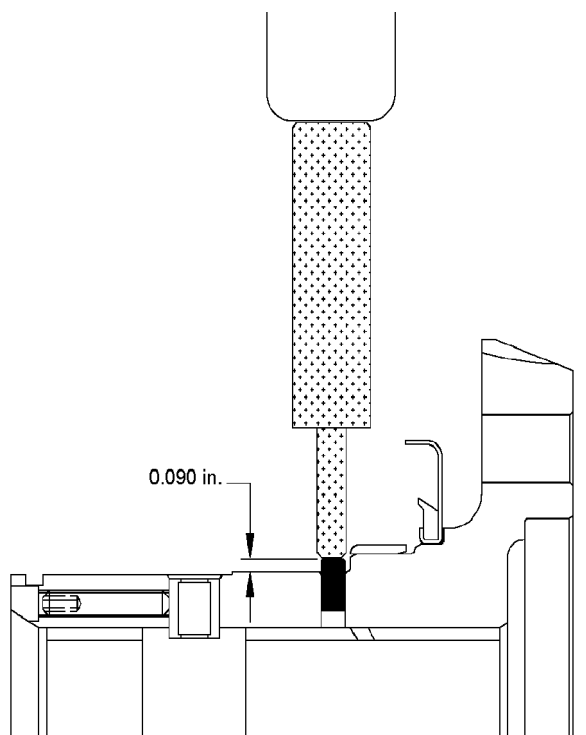


FIGURE 7-38.

2. If removed, install dowel pin (43) into fan mounting hub assembly (42). Refer to Figure 7-38. Press the pin into the hub, leaving 2.3 mm (0.090 in.) exposed.

If the shaft did not originally come with pinned bearings, install the dowel per instructions in Figures 7-38 and 7-39.

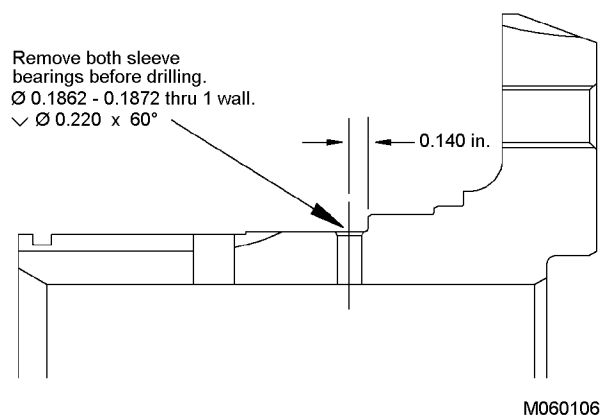


FIGURE 7-39.

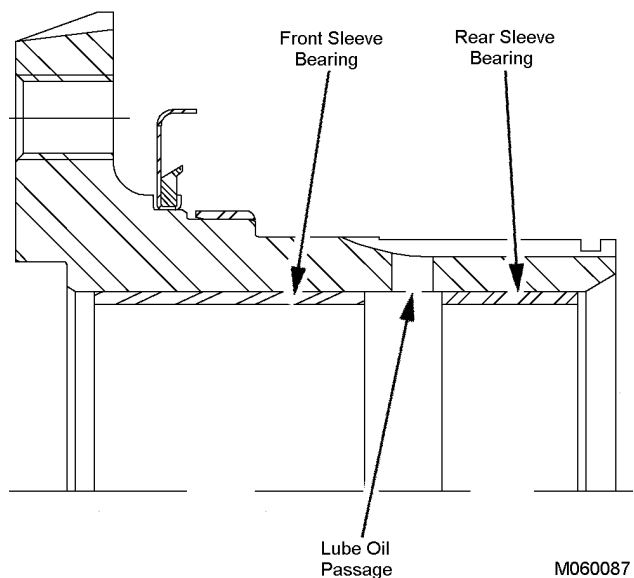


FIGURE 7-40.

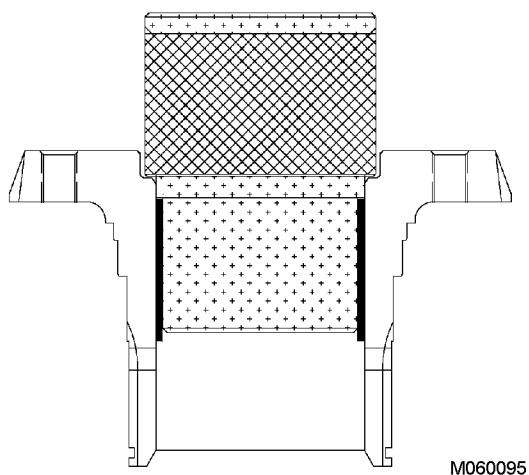


FIGURE 7-41.

3. Using tooling (A), press front (long) sleeve bearing (44) into the fan mounting hub until the tool contacts the shoulder of the hub. Ensure the correct bearing is installed. There are two sleeve bearings, and each one must be installed in the proper area of the hub to ensure the lube passage is not restricted. Refer to Figure 7-40.

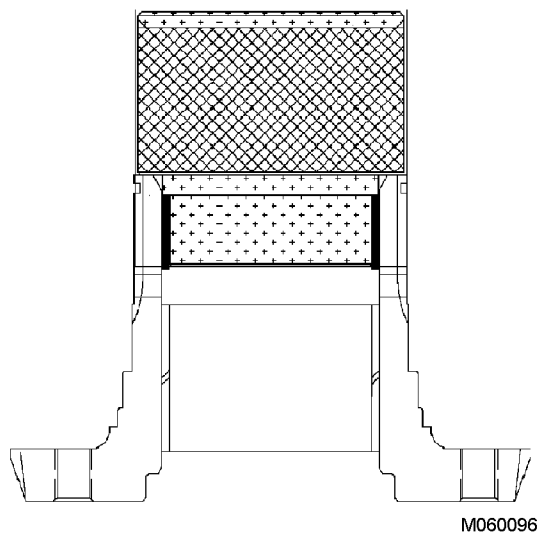
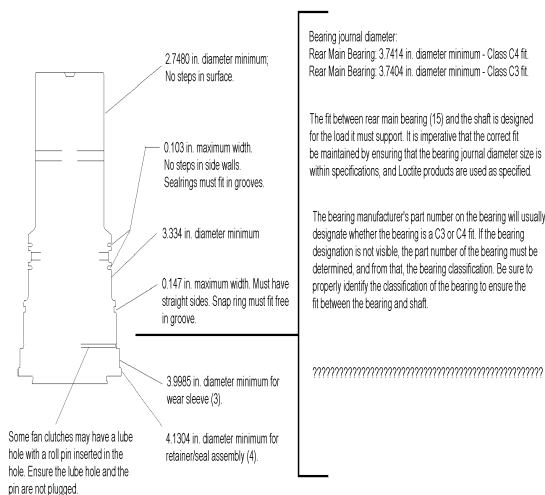


FIGURE 7-42.

4. Turn the hub over on the bed of the press. Using tooling (B), press rear sleeve bearing (41) into the fan mounting hub until the tool contacts the shoulder of the hub.



M060084

FIGURE 7-43.

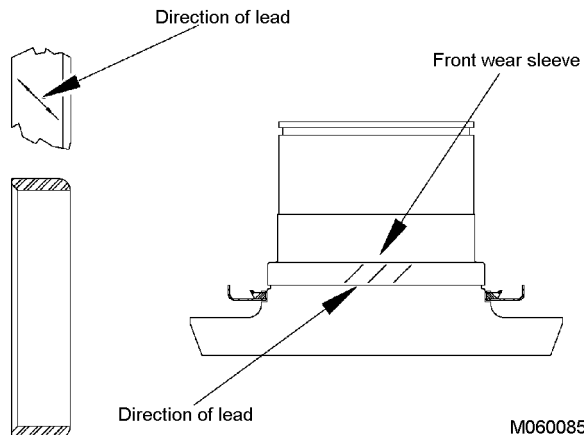
5. Press front retainer/seal assembly (40) onto the fan mounting hub (42) using tooling (D). The inner race of the retainer should be recessed 1.0 mm (0.040 in.) below the shoulder.

Check carefully to ensure that the retainer/seal assembly is installed straight and is not bent or damaged in any way which will cause interference between it and the bearing retainer after assembly.

⚠ IMPORTANT ⚠

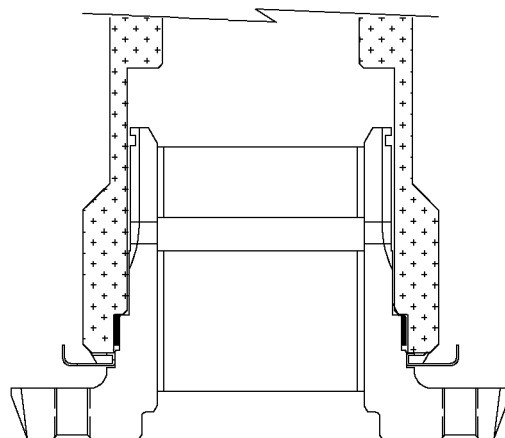
- **Front wear sleeve (39) is NOT interchangeable with rear (notched) wear sleeve (4). The inside diameter of the front wear sleeve is color coded red.**
- **Note the direction of the lead pattern on the sleeve. The wear sleeve must be installed with the pattern leading in the correct direction in order to prevent leakage from occurring.**
- **Use extreme care when handling the wear sleeve. The slightest nicks or scratches may cause leakage.**

NOTE: Some fan hubs may have a small hole on the wear sleeve mounting journal. This hole is not used and will be covered by the wear sleeve.



M060085

FIGURE 7-44.



M060099

FIGURE 7-45.

6. Coat the inside diameter of front wear sleeve (39) and the wear sleeve diameter of the shaft with Loctite Primer N and #242 (or equivalent). Using tooling (D), press the wear sleeve onto the shaft until it is flush with the shoulder.

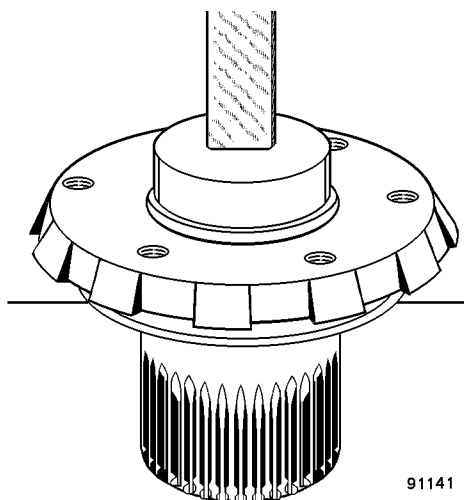


FIGURE 7-46.

7. Coat the bore of the fan mounting hub (42) with a thin coating of Loctite Primer N and #242. Remove frozen end cap (45) from the freezer. Do not apply Loctite to the end cap. Press the end cap into the hub until the cap bottoms out.

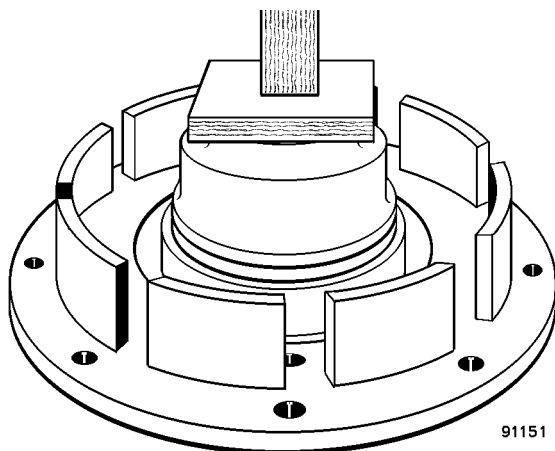


FIGURE 7-47.

8. Apply Loctite Primer N and #609 to the mating surfaces of front bearing (33) and front bearing retainer (35). Place the bearing into position on the retainer with the notch for the bearing pin facing downward.

Press the front bearing into the bearing retainer using tooling (E) or equivalent. Press **ONLY** on the outer race of the bearing until it seats at the bottom of the bore.

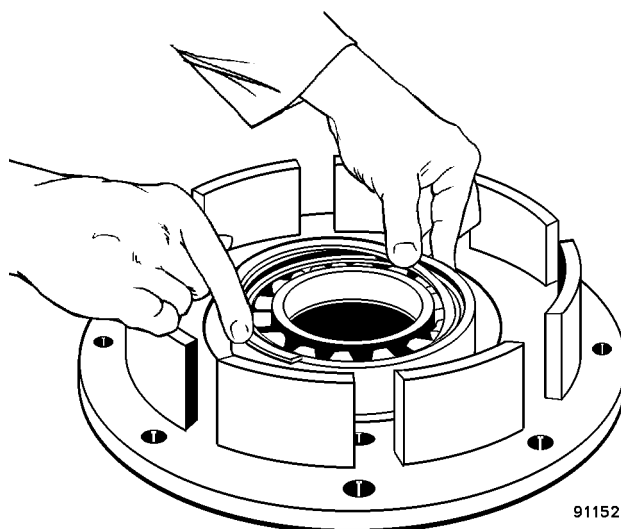


FIGURE 7-48.

9. Install internal snap ring (32).

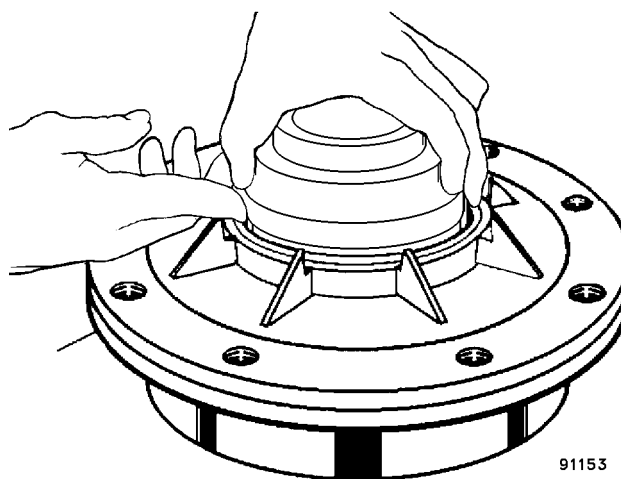
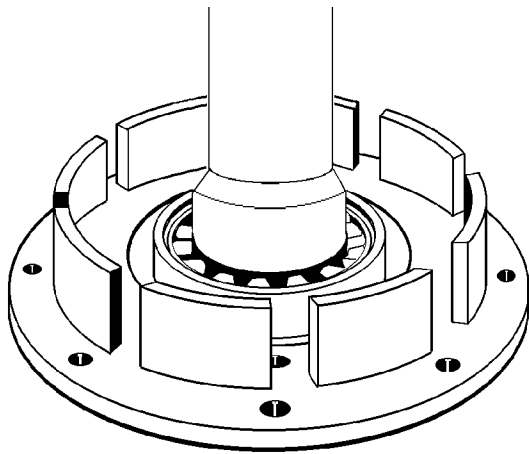


FIGURE 7-49.

10. Turn the retainer over on the press bed. Coat the outside diameter of front oil seal (36) and the mating surface on the bearing retainer with Loctite Primer N and #242 (or equivalent).

Use tooling (E) to press the oil seal into the front bearing retainer until it is flush with the front face. **Ensure that the lip of the seal is dry.** Wipe any excess Loctite from the seal area and remove any rubber strings from the seal.



M060100

FIGURE 7-50.

11. Coat the inside diameter of the bearing and the fan mounting hub bearing journal with Loctite Primer N and #609 (or equivalent). Place the front bearing retainer sub-assembly into position on the fan mounting hub. Ensure the notch in the bearing is aligned with the bearing dowel pin. Do not allow the seal lip to come in contact with the Loctite. Press the bearing onto the hub using tooling (D) until it contacts the wear sleeve.

Wipe any lubricant or sealer from the seal lip. The seal lip is teflon and must remain dry for proper sealing to occur.

Spin the bearing retainer at least 25 revolutions to ensure proper rotation of the bearing and to burnish the seal.

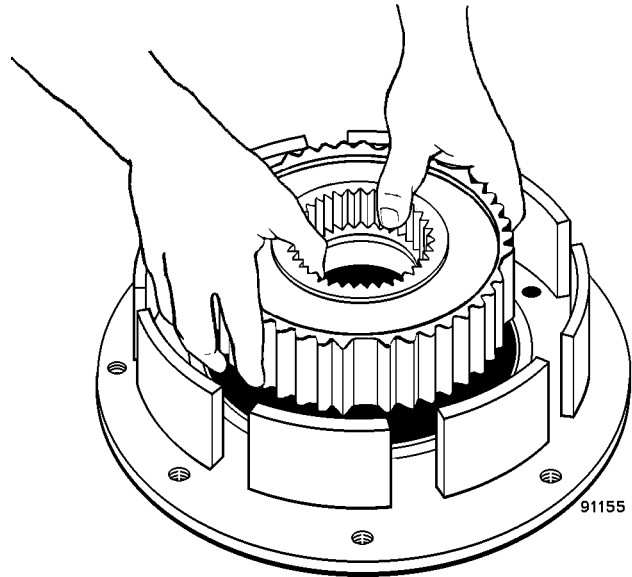


FIGURE 7-51.

12. Install clutch hub (29) on the fan mounting hub assembly (42) with the open end down. (No special timing is necessary.)

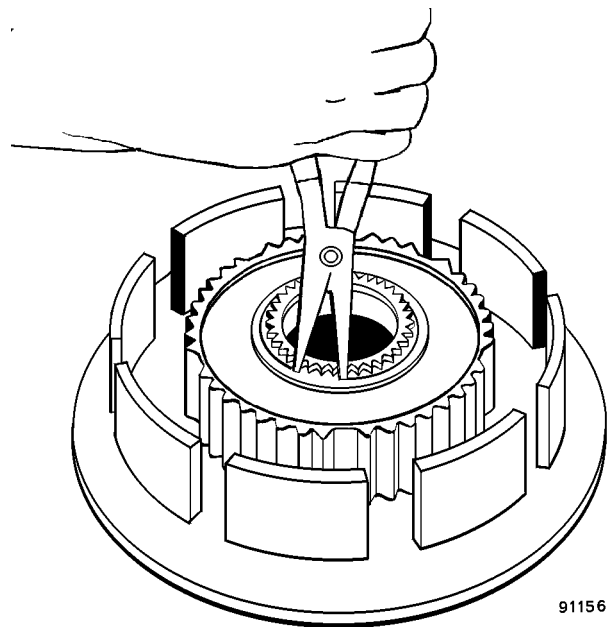


FIGURE 7-52.

13. Install external snap ring (28) to hold the clutch hub in place.

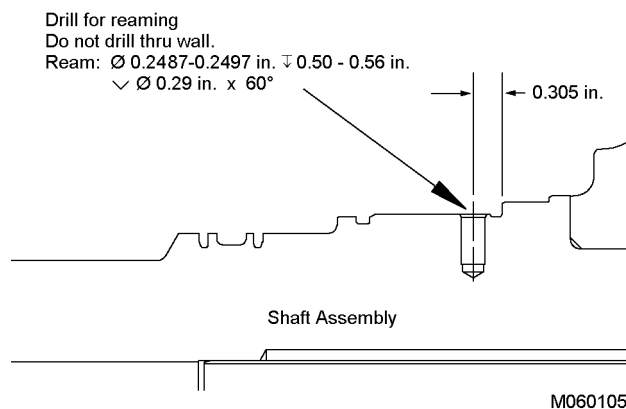


FIGURE 7-53.

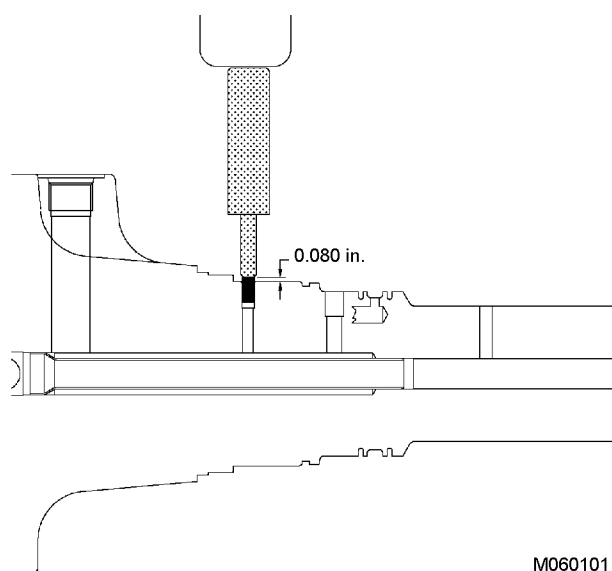


FIGURE 7-54.

14. If removed, install rear dowel pin (2) in shaft assembly (6). Press the pin until 2.0 mm (0.080 in.) is left exposed above the surface.

If the shaft did not originally come with pinned bearings, install the dowel per instructions in Figures 7-53 & 7-54.

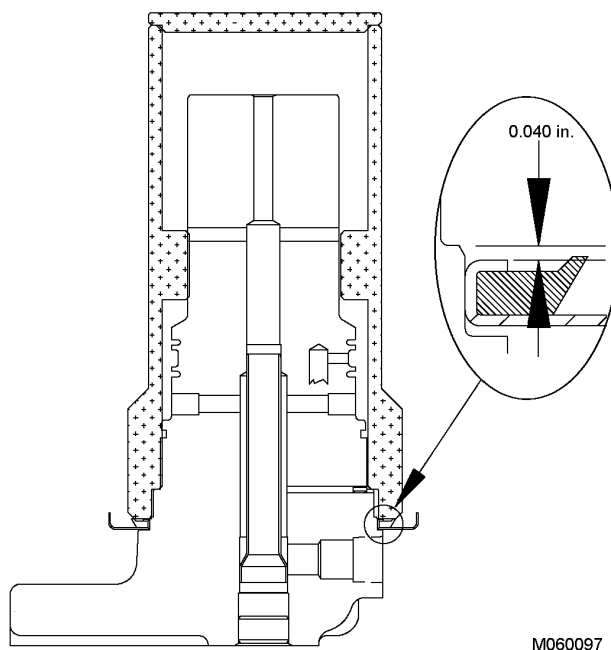


FIGURE 7-55.

15. Use tooling (D) to press rear retainer/seal assembly (5) onto shaft (6). The inner race of the retainer should be recessed 1.0 mm (0.040 in.) below the shoulder.

Check carefully to ensure that the retainer/seal assembly is installed straight and not bent or damaged in any way which will cause interference between it and the bearing retainer after assembly.

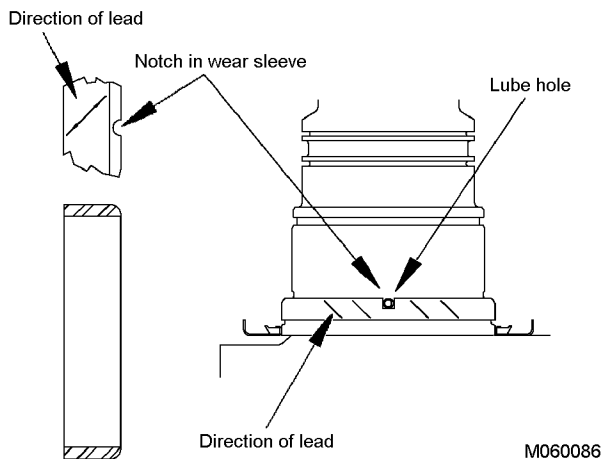


FIGURE 7-56.

⚠ IMPORTANT ⚠

- **Rear (notched) wear sleeve (4) is NOT interchangeable with front wear sleeve (39). The inside diameter of the rear wear sleeve is color coded blue.**
- **Note the direction of the lead pattern on the sleeve. The wear sleeve must be installed with the pattern leading in the correct direction in order to prevent leakage from occurring.**
- **Use extreme care when handling the wear sleeve. The slightest nicks or scratches may cause leakage.**

16. Coat the inside diameter of rear (notched) wear sleeve (4) and the wear sleeve diameter of the fan mounting hub with Loctite Primer N and #242 (or equivalent). Locate the sleeve so the notch in the sleeve will be aligned with the small lube hole in the shoulder. Using tooling (D), press the wear sleeve onto the fan mounting hub until it is flush with the shoulder.

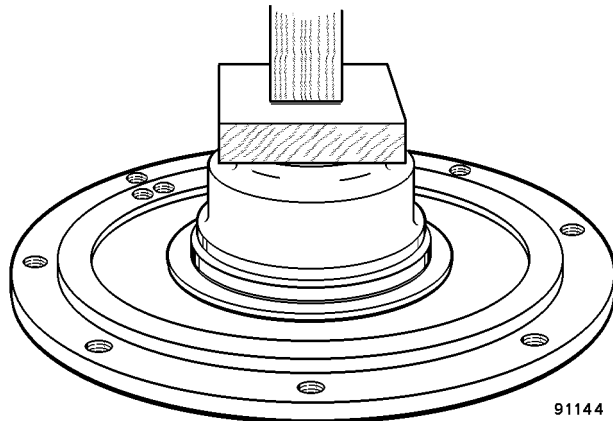


FIGURE 7-57.

17. Coat the outside diameter of rear bearing (14) and the mating surface of bearing retainer (11) with Loctite Primer N and #609 or equivalent.

The end of the bearing with the notch is installed first. Using tooling (F) or equivalent, press the bearing into the bearing retainer. Press **ONLY** on the outer race of the bearing until the bearing bottoms out in the bore.

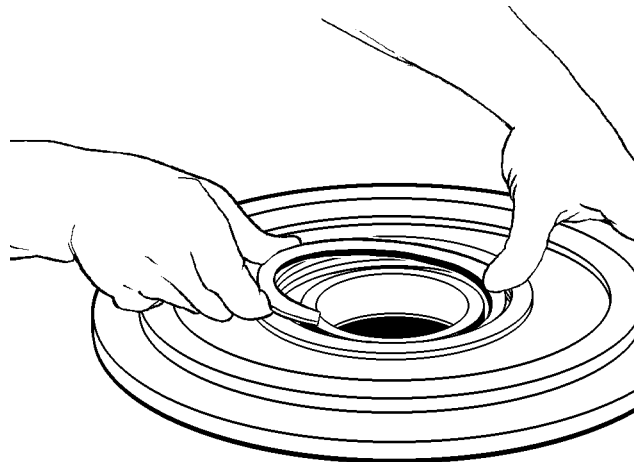


FIGURE 7-58.

18. Install internal snap ring (15).

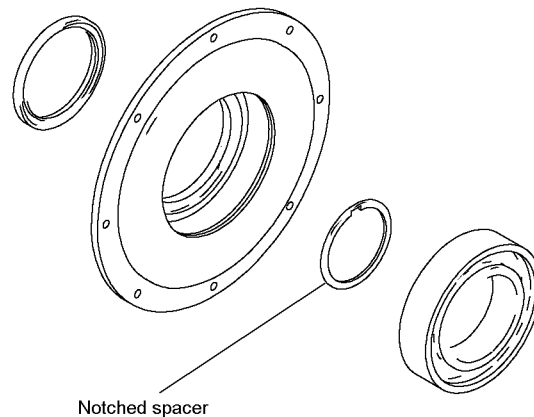
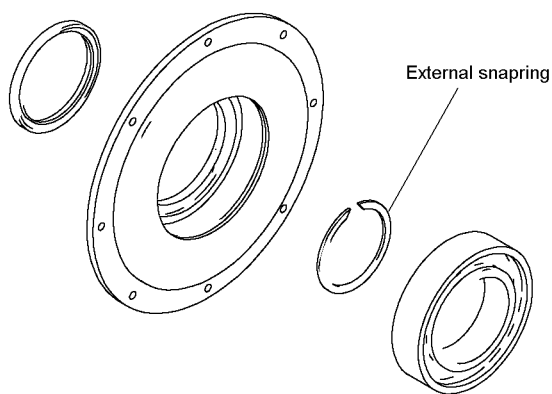


FIGURE 7-59.

M060107

19. Some fan clutches were assembled with an external snap ring that is used as a spacer between the bearing and the oil seal. Newer models were assembled using a notched spacer. If an external snap ring was used, place snap ring (12) on top of the bearing (oil seal side). If a notched spacer was used, the spacer will be installed in a later step. Proceed to the next step.

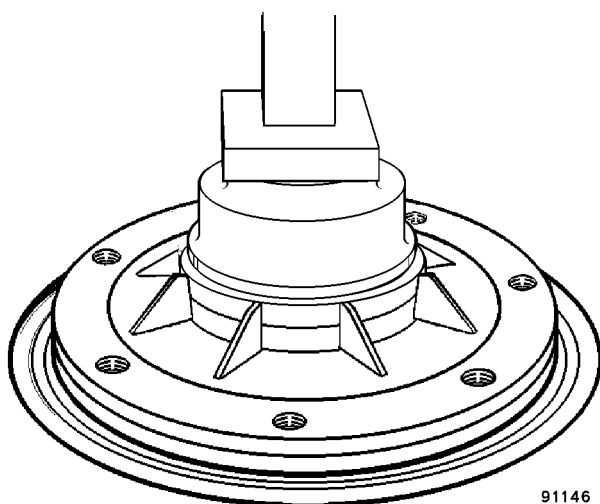


FIGURE 7-60.

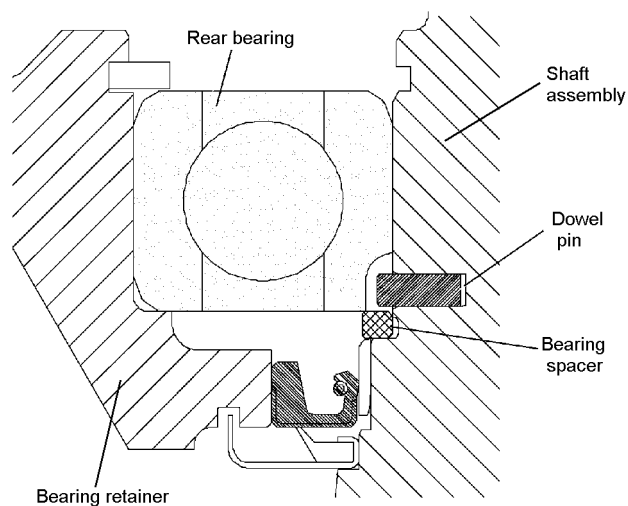


FIGURE 7-61.

M060108

20. Coat the outside diameter of rear oil seal (10) with Loctite Primer N and #242 (or equivalent). Use tooling (E) or an equivalent to install the oil seal in the rear bearing retainer until it is flush with the rear face.

Do not lubricate the seal. The seal is made of teflon and must be installed dry.

21. If a bearing spacer is used instead of a snap ring (as explained in Step 19), place the spacer into position in the groove on shaft assembly (6). Note the location of the spacer in Figure 7-61.

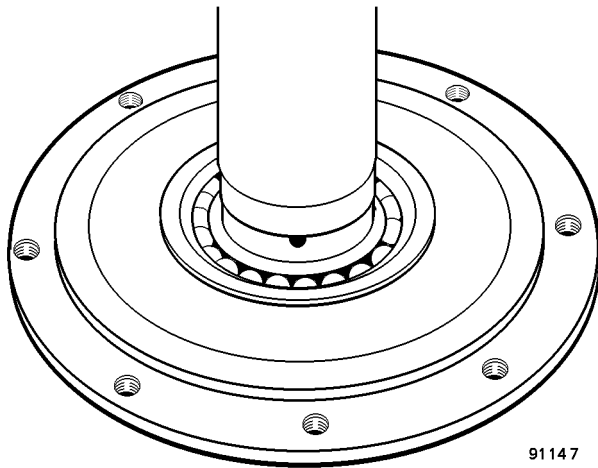


FIGURE 7-62.

22. Place the shaft sub-assembly on the press bed. Coat the inside diameter of the bearing and the bearing journal on the shaft with Loctite Primer N and #609 (or equivalent).

Carefully, lower the rear bearing retainer sub-assembly in place on the shaft. Do not allow the seal lip to come in contact with the Loctite. Ensure the notch in the bearing and the dowel pin are aligned. If external snap ring (16) was installed in the bearing retainer, ensure the opening is aligned with the dowel pin.

Press the bearing onto the shaft until it reaches the shoulder of the wear sleeve. Wipe any excess Loctite from the assembly.

Ensure the seal lip is dry. The seal must remain dry for proper sealing.

Spin the bearing retainer approximately 25 times to burnish the teflon seal on the wear sleeve. Check for abnormal sounds or other indications of contact between the retainer/seal assembly and the bearing retainer. If interference is found, remove the bearing retainer and eliminate the point of interference.

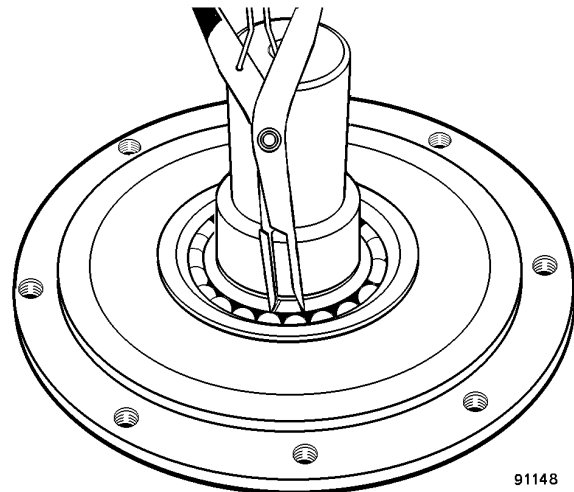


FIGURE 7-63.

23. Install external snap ring (16). Ensure that the snap ring is fully seated in the groove. It may be necessary to tap on the snap ring with a screwdriver to fully seat it.

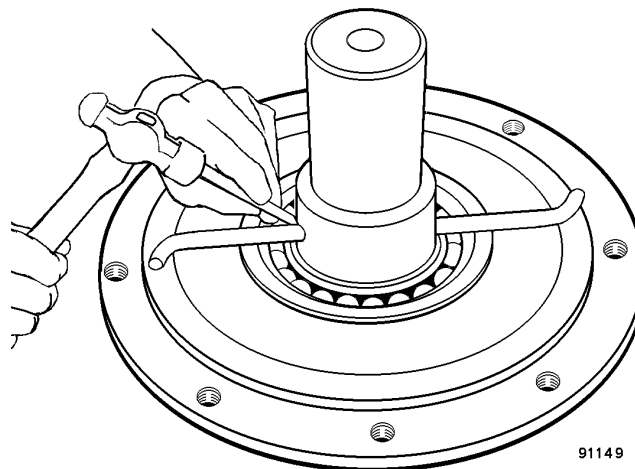


FIGURE 7-64.

24. Ensure that the pitot tube holes in the shaft are clean and free of burrs and staking material to allow the pitot tubes to fit into the holes and seat completely to the bottom. Apply a thin coating of Loctite Primer N and #609 (or equivalent) on the straight end of one pitot tube (3). Coat the tube to approximately 20 mm (0.75 in.) from the end.

Push the pitot tubes to the bottom of the hole. The outer end of the tube should be located well within the pulley-locating shoulder of the bearing retainer. Rotate the tube so the open, bent end faces in a counterclockwise direction and is *exactly* parallel to the surface of the bearing retainer. (A large phillips-head screwdriver inserted in the end of the tube can be used as an alignment gage).

Install the second pitot tube in the same manner as the first. Stake each pitot tube in three places (at the 9, 12, and 3 o'clock positions) to prevent the tubes from rotating in operation.

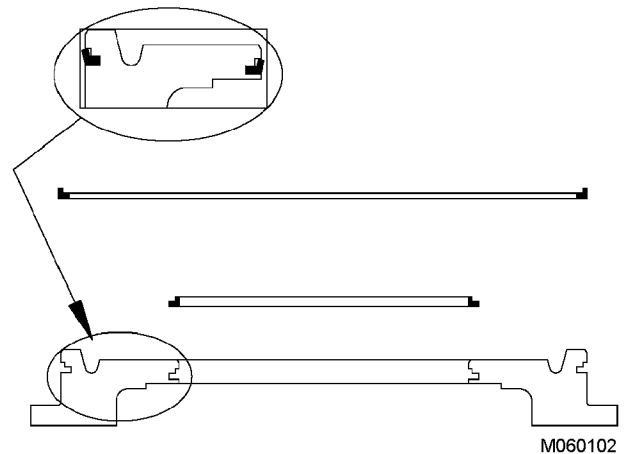


FIGURE 7-66.

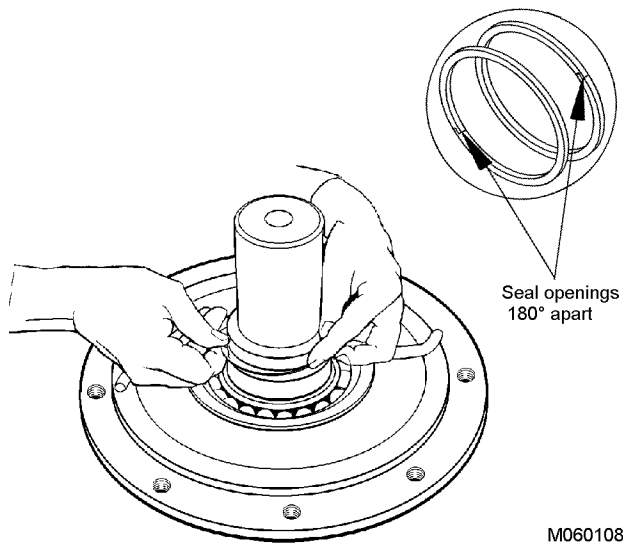


FIGURE 7-65.

25. Install both hook-type seal rings (17) in the grooves in the shaft. Rotate the rings so the slits in the rings are 180 degrees apart from one another.

26. Lubricate the seal ring grooves of piston (23) with an oil-soluble lubricant such as engine assembly grease. Install small seal ring (24) in the inside groove and large seal ring (22) in the outside groove. Refer to Figure 7-66 for proper orientation.



FIGURE 7-67.

27. Lubricate the external surfaces of seal rings (22) and (24) with an oil-soluble lubricant such as engine assembly grease. Also lubricate the seal mating surfaces in the pulley adapter.

⚠ IMPORTANT ⚠

Do not push the piston in place. Forcing the piston will usually cause the seal rings to be cut.

28. Carefully place the piston in the pulley. Without pressing down on the piston, rotate it slowly back and forth until it falls into place.

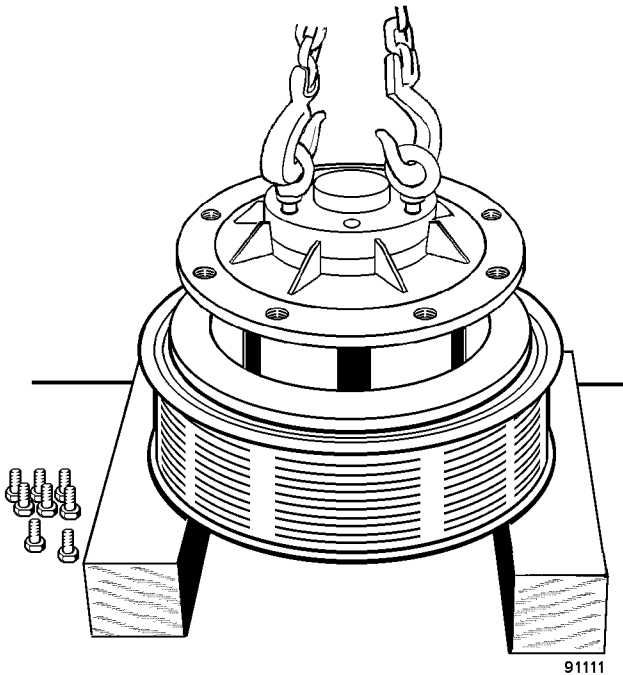


FIGURE 7-68.

29. Align the tangs of the piston for final assembly of the fan clutch. Lift the front bearing retainer sub-assembly in place on the pulley. While doing so, the slots of the front bearing retainer will engage the tangs of the piston, and the retainer will rest against the pulley.

Then, rotate the bearing retainer (and piston) until the bolt holes align in the bearing retainer and pulley. Carefully remove the bearing retainer sub-assembly.

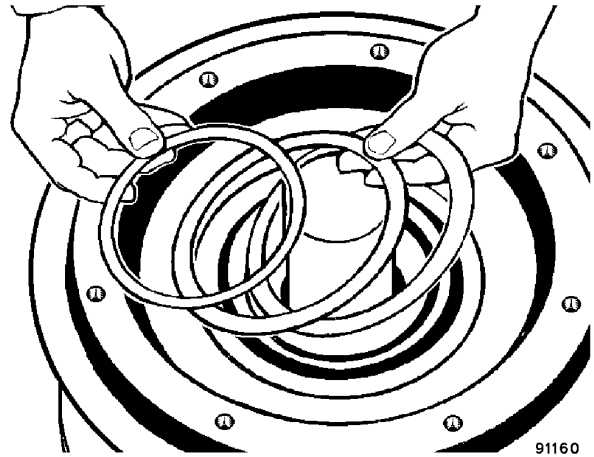


FIGURE 7-69.

30. Install spring washer (25), shim (26), and spirolock ring (27). It will be necessary to press downward to compress the spring washer while forcing the spirolock to properly seat in the groove. The shim must then be centered on the spring washer to prevent it from interfering with the movement of the piston.

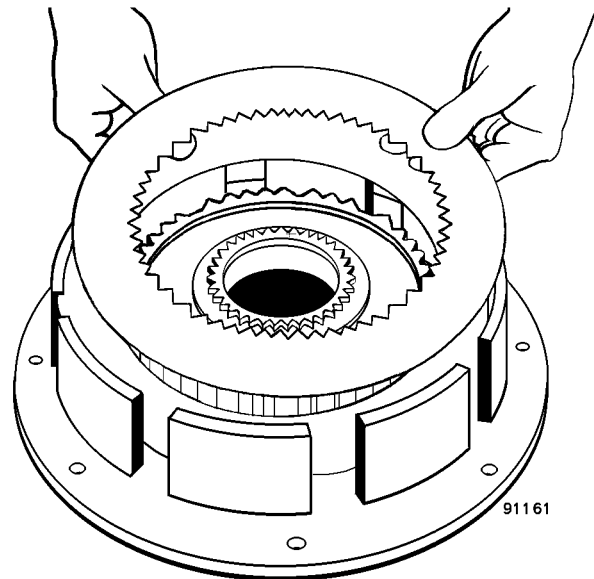
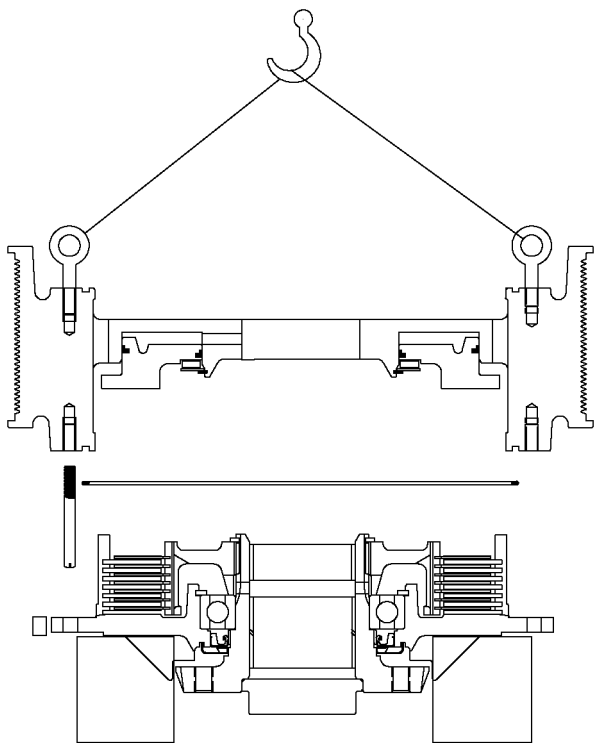


FIGURE 7-70.

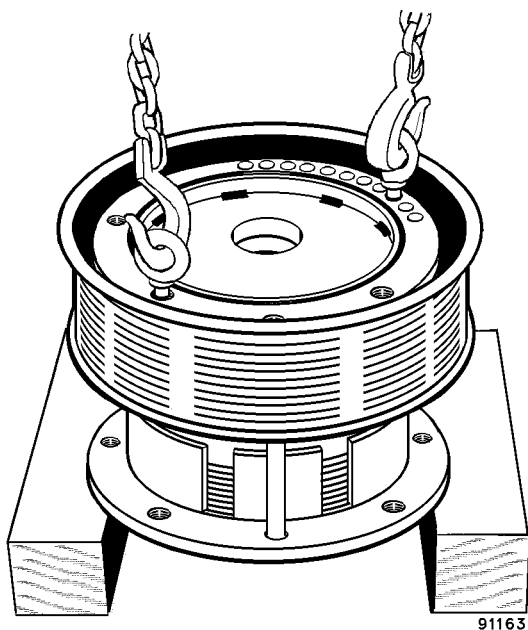
31. Place the front bearing retainer sub-assembly on the bench with the clutch hub up. Install one steel clutch plate (31) in place in the bearing retainer. Dip one facing plate (30) in new engine oil. Allow the excess oil to drain off, then place the facing plate on top of the steel plate.

Repeat this step until all 16 plates have been installed.



M060103

FIGURE 7-71.



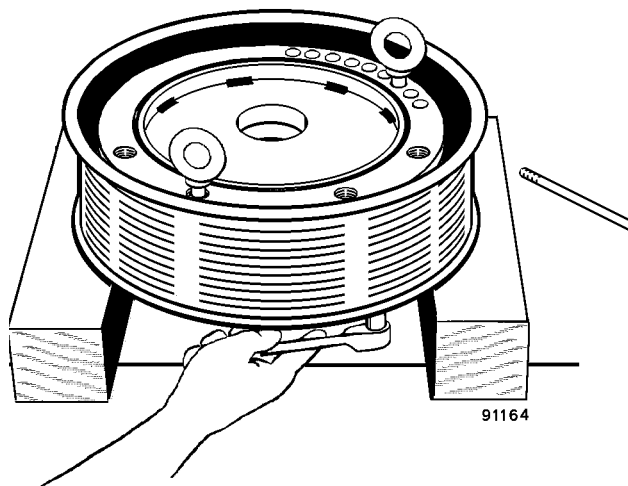
91163

FIGURE 7-72.

32. Turn the pulley adapter assembly over and install two lifting eyes 180 degrees apart. Install a guide bolt in one bolt hole of the pulley. Refer to Figure 7-71.

Coat front O-ring seal (34) with petroleum jelly or an oil-soluble grease. Place the seal in the groove in the pulley. The grease should secure the seal in the groove during installation.

Carefully lower the pulley. Ensure that the guide bolt is aligned with a bolt hole in the bearing retainer assembly and the O-ring seal is still securely in place. Lower the pulley until it rests on the front bearing retainer.



91164

FIGURE 7-73.

33. Install at least four bolts (38) with lockwashers (37) 90 degrees apart. Snug them down.

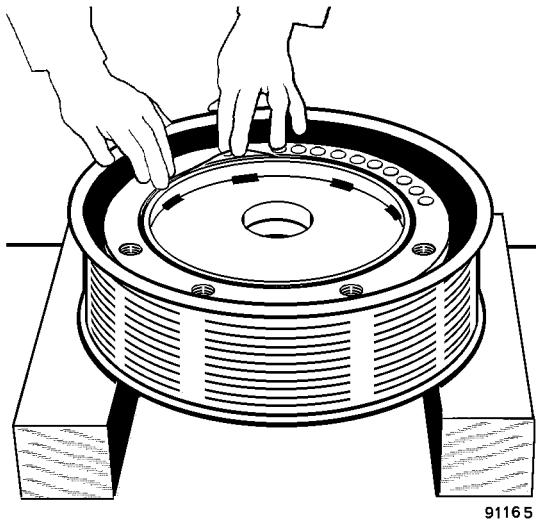


FIGURE 7-74.

34. Lubricate O-ring seal (13) with petroleum jelly or an oil-soluble grease and install it in the pulley groove.

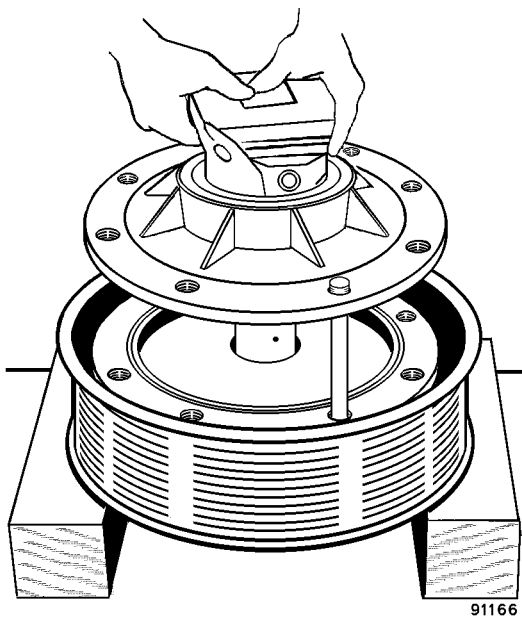


FIGURE 7-75.

35. Lubricate hook-type seal rings (17) on the shaft assembly. Carefully lower the shaft sub-assembly into the pulley bore and onto the pulley until the retainer rests on the pulley.

Use caution when lowering. Damage to the sleeve bearings may result if the shaft is cocked during installation.

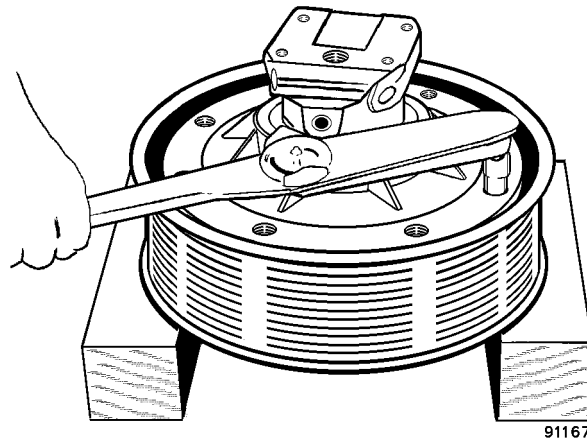


FIGURE 7-76.

36. Install bolts (9) with lockwashers (8). Tighten each bolt to **49 - 58 N•m (36 - 43 ft lbs)**.

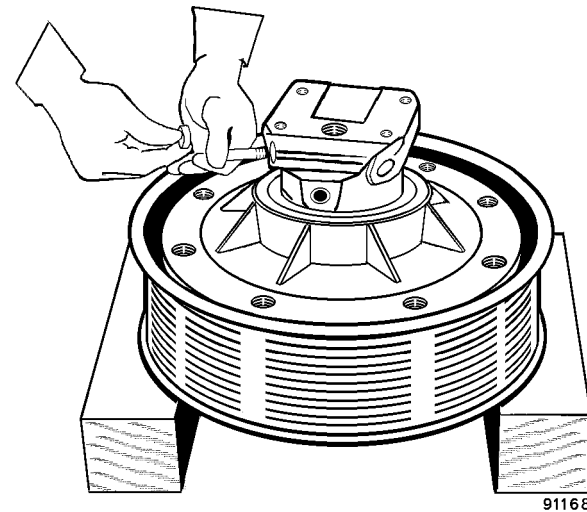


FIGURE 7-77.

37. If removed, install orifice fitting (1) in the "oil in" port of the bracket.

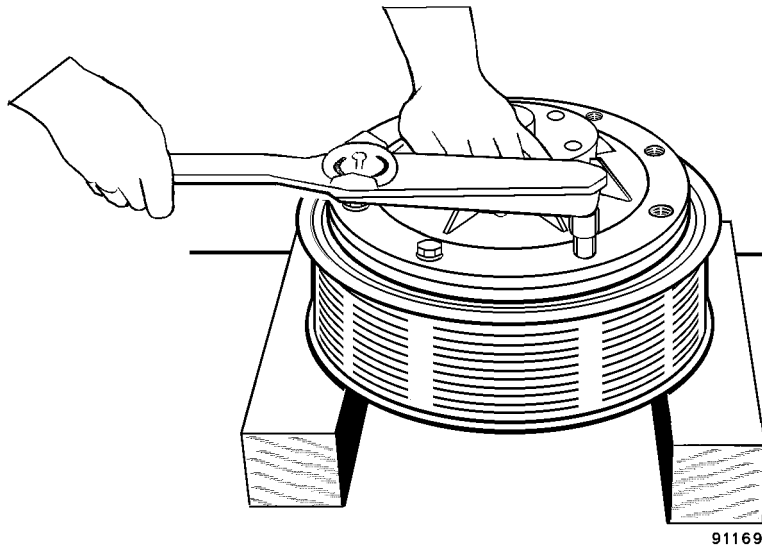
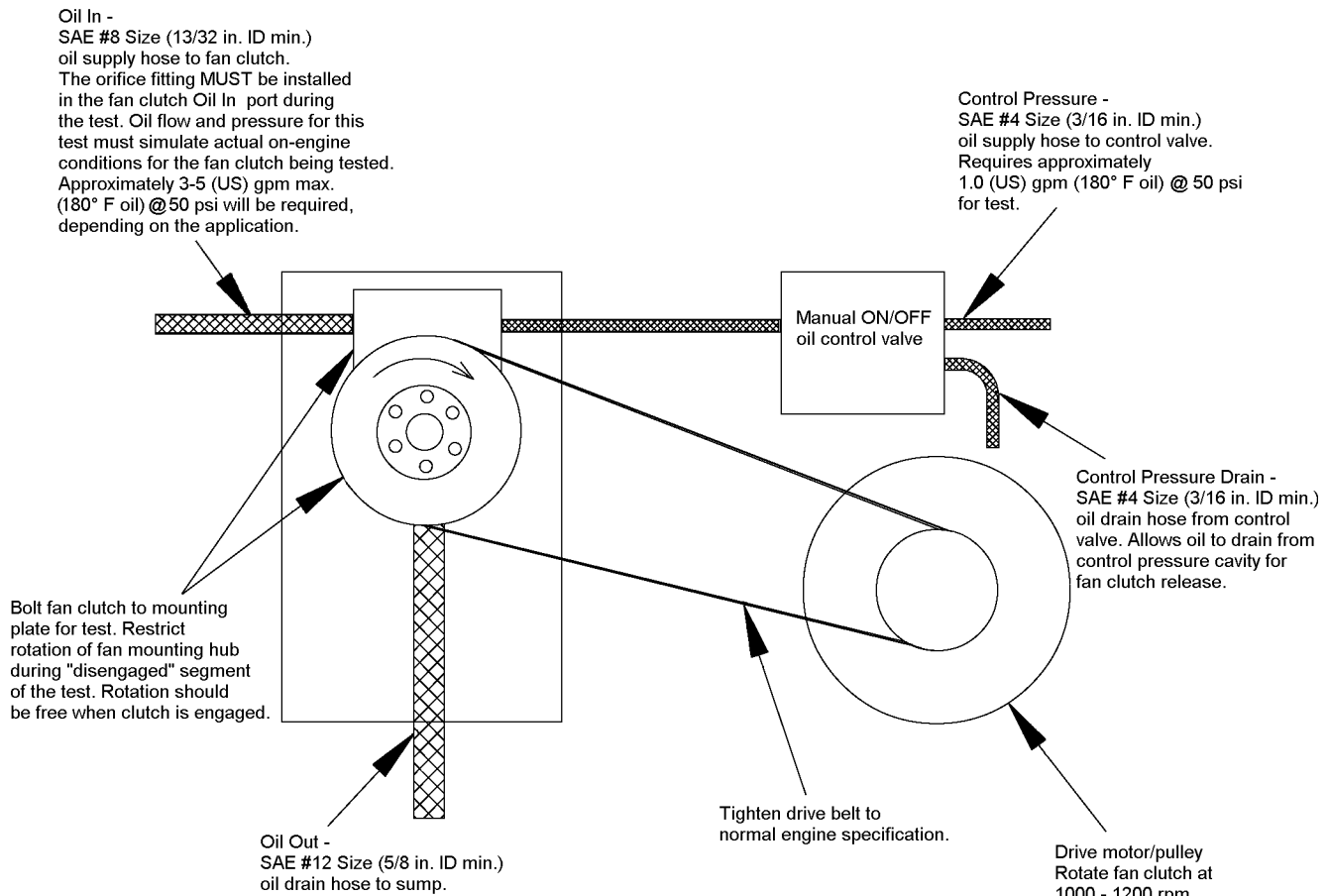


FIGURE 7-78.

38. Turn the assembly over on the bench. Install remaining bolts (38) and lockwashers (37). Tighten each bolt to **49 - 58 N•m (36 - 43 ft lbs)**.

TEST PROCEDURE



M060104

1. The fan clutch should be fully locked up with 275 kPa (40 psi) oil pressure supplied at the control pressure port.
2. Operate the fan clutch with 82° C (180° F) oil supplied to the "oil in" port for 2 hours. Manually engage and disengage the clutch during the test to operate seals in both modes. Restrict the fan mounting hub rotation while the clutch is disengaged, but ensure that the fan mounting hub is allowed to rotate freely while the clutch is engaged.

⚠ IMPORTANT ⚠

The fan clutch rotation causes the pitot tubes to pump lubricating oil from inside the fan clutch, maintaining low internal oil pressure. If lubricating oil is supplied to the fan clutch before it is rotating in the proper direction, internal pressures will become excessive, causing the oil seals to leak.

SECTION D

ELECTRICAL SYSTEM (24VDC NON-PROPULSION)

INDEX

24VDC ELECTRICAL SYSTEMS	D2-1
24VDC ELECTRICAL SYSTEM COMPONENTS	D3-1
KOMTRAX PLUS	D11-1
INTERFACE MODULE	D12-1
INTERFACE MODULE AND KOMTRAX PLUS TROUBLESHOOTING	D13-1
KOMTRAX PLUS FORMS	D14-1

NOTE: Electrical system wiring hookup and electrical schematics are located in Section R of this manual.



DANGEROUS VOLTAGE LEVELS ARE PRESENT WHEN THE TRUCK IS RUNNING AND CONTINUE TO EXIST AFTER SHUTDOWN IF THE REQUIRED SHUTDOWN PROCEDURES ARE NOT FOLLOWED. Before attempting repairs or working near propulsion system components, the following precautions and truck shutdown procedure must be followed:

- **DO NOT** step on or use any power cable as a handhold.
- Never open any electrical cabinet covers or touch the retarding grid elements. Additional procedures are required before it is safe to do so. Refer to Section E for additional propulsion system safety checks to be performed by a technician trained to service the system.
- ALL removal, repairs and installation of propulsion system electrical components, cables etc. must be performed by an electrical maintenance technician properly trained to service the system.
- In the event of a propulsion system malfunction, a qualified technician should inspect the truck and verify the propulsion system does not have dangerous voltage levels present before repairs are started.
- Prior to welding on the truck, maintenance personnel should attempt to notify the Komatsu Factory Representative. The welding ground electrode should be attached as close as possible to the area to be welded. Never weld on the rear of the electrical control cabinet or the retard grid exhaust air louvers.

After the truck is parked in position for the repairs, the truck must be shut down properly to ensure the safety of anyone working in the areas of the deck, electrical cabinet, traction motors, and retarding grids. The following procedure will ensure that the electrical system is properly discharged before repairs are begun.

TRUCK SHUTDOWN PROCEDURE

1. Reduce the engine speed to idle. Place the directional control lever in PARK. Make sure that the parking brake applied indicator light is illuminated.
2. Place the rest switch in the ON position to put the AC drive system in the rest mode. Make sure that the rest mode indicator light is illuminated.
3. Turn the key switch to the OFF position. A timing sequence will be activated automatically to allow the engine to cool down before it is turned off.

If the engine does not shut down with the key switch after the timing sequence is done, use the engine shutdown switch on the operator cab center console. Pull this switch up until the engine stops.

4. With the key switch OFF and the engine stopped, wait at least 90 seconds. Make sure that the steering circuit is completely depressurized by turning the steering wheel back and forth several times. The front wheels should not turn when the hydraulic pressure is relieved. If the front wheels can still be turned, notify maintenance personnel.
5. Make sure that the link voltage light in the control cabinet is off. Notify maintenance personnel if the light remains illuminated for longer than five minutes after the engine is shut down.
6. Locate the GF cutout switch in the access panel on the left side of the main control cabinet. Place the switch in the CUTOFF position. This will prevent the alternator from re-energizing and creating system voltage until the switch is returned to its former position.

SECTION D2

24VDC ELECTRICAL SUPPLY SYSTEM

INDEX

24VDC ELECTRIC SUPPLY SYSTEM	D2-3
ELECTRICAL SYSTEM DESCRIPTION	D2-3
BATTERIES	D2-3
Maintenance and Service	D2-3
Troubleshooting	D2-3
BATTERY SUPPLY SYSTEM	D2-5
24VDC Battery Charging Alternator	D2-5
Batteries	D2-5
Engine Start Relays	D2-5
24VDC Auxiliary Battery Receptacles	D2-5
Battery Disconnect Switches	D2-5
24VDC to 12VDC Converter	D2-6
24VDC ELECTRIC CRANKING MOTOR SYSTEM (WITH PRELUBE)	D2-6
Operation	D2-6
Pressure Switch	D2-6
Check Valve	D2-7
Timer Solenoid	D2-7
MAINTENANCE	D2-7
Prelube System Operation Checks	D2-7
Check Valve	D2-7
Timer Solenoid	D2-7
TROUBLESHOOTING PRELUBE CRANKING MOTOR CIRCUIT	D2-8

24VDC ELECTRIC START SYSTEM..... D2-10

 CRANKING MOTORS D2-10

 Operation..... D2-10

 Removal D2-10

 Installation D2-10

 CRANKING MOTOR TROUBLESHOOTING..... D2-11

 Preliminary Inspection..... D2-11

 No-Load Test..... D2-11

 Interpreting Results of Tests D2-12

 Disassembly D2-12

 Cleaning and Inspection D2-13

 Armature Servicing D2-15

 Field Coil Checks..... D2-15

 Field Coil Removal D2-15

 SOLENOID CHECKS D2-16

 Test D2-16

 Assembly..... D2-17

 Bearing Replacement:..... D2-17

 Motor Assembly: D2-17

 Pinion Clearance D2-18

 MAGNETIC SWITCH..... D2-18

 Removal D2-18

 Installation D2-18

 Coil Test..... D2-19

24VDC ELECTRIC SUPPLY SYSTEM

ELECTRICAL SYSTEM DESCRIPTION

The truck uses a 24VDC electrical system which supplies power for engine starting circuits and most non-propulsion electrical components. The 24VDC engine starting circuit is supplied by four heavy duty, type 8D, 12-volt storage batteries. Several components require 12VDC and are supplied by circuits tapped off the starting batteries.

The batteries are of the lead-acid type, each containing six 2-volt cells. With the engine off, power is supplied by the batteries. During engine cranking, power is supplied by the four engine cranking batteries only. When the engine is running, power is supplied by a high capacity alternator that is driven by the engine.

DANGER

Lead-acid storage batteries contain sulfuric acid which, if handled improperly, may cause serious burns on skin or other serious injuries to personnel. Wear protective gloves, aprons and eye protection when handling and servicing lead-acid storage batteries. See the precautions in Section A of this manual to ensure proper handling of batteries and accidents involving sulfuric acid.

During operation, the storage batteries function as an electrochemical device that converts chemical energy into the electrical energy that is required for operating the accessories when the engine is off.

BATTERIES

Maintenance and Service

The electrolyte level of each cell should be checked at the interval specified in Section P, Lubrication and Service. Add water if necessary. The proper level to maintain is 10 to 13 mm (3/8 to 1/2 in.) above the plates. To ensure maximum battery life, use only distilled water or other types of water recommended by the battery manufacturer. After adding water in freezing weather, operate the engine for at least 30 minutes to thoroughly mix the electrolyte.

DANGER

DO NOT SMOKE or allow flame around a dead battery or during the recharging process. The expelled gas from a dead cell is extremely explosive.

Excessive consumption of water indicates leakage or overcharging. Normal water usage for a unit operating eight hours per day is about 30 to 60 cm³ (1 to 2 oz.) per cell per month. For heavy duty operation (24 hours per day), normal consumption should run about 30 to 60 cm³ (1 to 2 oz.) per cell per week. Any appreciable increase over these figures should be considered a danger signal.

Troubleshooting

Two most common problems that occur in the charging system are undercharging and overcharging of the truck's batteries.

An undercharged battery is incapable of providing sufficient power to the truck's electrical system.

Some possible causes for an undercharged battery are:

- Sulfated battery plates
- Loose or corroded battery connections
- Defective wire in electrical system
- Loose alternator drive belt
- Defective alternator

Overcharging, which causes overheating, is first indicated by excessive use of water. If allowed to continue, the cell covers will push up at the positive ends and, in extreme cases, the battery container will become distorted and cracked.

Leakage can be detected by continual wetness of the battery or excessive corrosion of the terminals, battery carrier and surrounding area. (A slight amount of corrosion is normal in lead-acid batteries). Inspect the case, covers and sealing compound for holes, cracks and other signs of leakage. Check the battery hold down connections to ensure that the tension is not great enough to crack the battery or loose enough to allow vibration to open the seams. A leaking battery must be replaced.

To remove corrosion, clean the battery with a solution of ordinary baking soda and a stiff, non-wire brush and flush with clean water. Make sure that none of the soda solution is allowed to enter the battery cells. Make sure that the terminals are clean and tight. Clean terminals are very important in a voltage regulated system. Corrosion creates resistance in the charging circuit, which causes undercharging and gradual starvation of the battery.

NOTE: When washing batteries, make sure that the cell caps are tight to prevent cleaning solution from entering the cells.

Addition of acid will be necessary if considerable electrolyte has been lost through spillage. Before adding acid, make sure that the battery is fully charged by putting the battery on charge and taking hourly specific gravity readings on each cell. When all the cells are gassing freely and three successive hourly readings show no rise in specific gravity, the battery is considered charged. Additional acid may now be added. Continue charging for another hour and check specific gravity again. Repeat the above procedure until all cells indicate a specific gravity of 1.260 - 1.265 corrected to 27°C (80°F).

NOTE: Use 1.400 strength sulfuric acid when making specific gravity adjustments. Acid of higher strength will attack the plates and separators before it has a chance to diffuse into the solution.

If the temperature of the electrolyte is not reasonably close to 27°C (80°F) when the specific gravity is taken, temperature should be corrected to 27°C (80°F) as follows:

- For every 5°C (10°F) below 27°C (80°F), 0.004 should be SUBTRACTED from the specific gravity reading.
- For every 5°C (10°F) above 27°C (80°F), 0.004 should be ADDED to the reading.

Idle batteries should not be allowed to stand unattended. If equipment is to stand unused for more than two weeks, the batteries should be removed and placed in a cool, dry place where they may be checked periodically and charged when necessary. Remember, all lead-acid batteries discharge slowly when not in use. This self-discharge takes place even though the battery is not connected in a circuit, and it is more pronounced in warm weather than in cold weather.

The rate of self-discharge of a battery kept at 38°C (100°F) is about six times that of a battery kept at 10°F (50°F), and self-discharge of a battery kept at 27°C (80°F) is about four times that one at 10°F (50°F). Over a 30 day period, the average self-discharge runs about 0.002 specific gravity per day at 27°C (80°F).

To offset the results of self-discharge, idle batteries should receive a booster charge (not a quick charge) at least once every 30 days. Batteries allowed to stand for long periods in a discharged condition are attacked by a crystallization of the lead sulfate on the plates. Such batteries are called "sulfated" and are, in the majority of cases, irreparably damaged. In less severe cases, the sulfated battery may be restored to limited service by prolonged charging at a low rate (approximately 1/2 normal rate).

An undercharged battery is extremely susceptible to freezing when allowed to stand in cold weather.

The electrolyte of a battery in various stages of charge will start to freeze at temperatures indicated in the table below.

The temperatures in the table below indicate the points at which the first ice crystals appear. Lower temperatures must be reached for a solid freeze. Solid freezing of the electrolyte may crack the battery case and damage the positive plates. As will be noted, a charged battery is in no danger of freezing. Therefore, a battery should be kept charged, especially during winter weather.

SPECIFIC GRAVITY Corrected to 27°C (80°F)	FREEZING TEMPERATURE
1.280	-70°C (-90°F)
1.250	-54°C (-60°F)
1.200	-27°C (-16°F)
1.150	-15°C (+5°F)
1.100	-7°C (+19°F)

BATTERY SUPPLY SYSTEM

24VDC Battery Charging Alternator

The battery charging alternator is a 26-Volt (140 Amp) alternator. See the engine service manual for more information.

Batteries

Four type 8D batteries (1, Figure 2-1) for the 24VDC engine start circuit are located in the battery box in the center of the front platform.

For access to the batteries, open the hinged cover by turning the cover latches counterclockwise until released. Lifting eyes are attached to each end of the battery box so that the entire battery box assembly can be removed, if necessary.

Engine Start Relays

Engine start relays (2) receive the signal to begin cranking from the start relay located on relay board RB6. When the engine start relays are activated, a current is provided to each of the cranking motors to engage the drives and begin cranking the engine, eliminating the need for magnetic switches.

24VDC Auxiliary Battery Receptacles

Two receptacles (3) are provided to attach battery charger leads for charging the batteries. These receptacles can also be used for connecting external batteries to aid engine starting during cold weather.

When external batteries are used, they should be of the same type (8D) as the batteries installed on the truck. Two pairs of batteries should be used. Each pair should be connected in series to provide 24VDC, with one pair connected to the each receptacle.

Battery Disconnect Switches

Battery disconnect switches provide a convenient method of disconnecting the truck batteries from the truck electrical circuits without having to remove any battery cables. Starter disconnect switch (4) opens the starter battery circuit only, preventing engine startup while still allowing battery power to the 24VDC control system circuits. Master disconnect switch (5) disconnects the 24VDC system batteries.

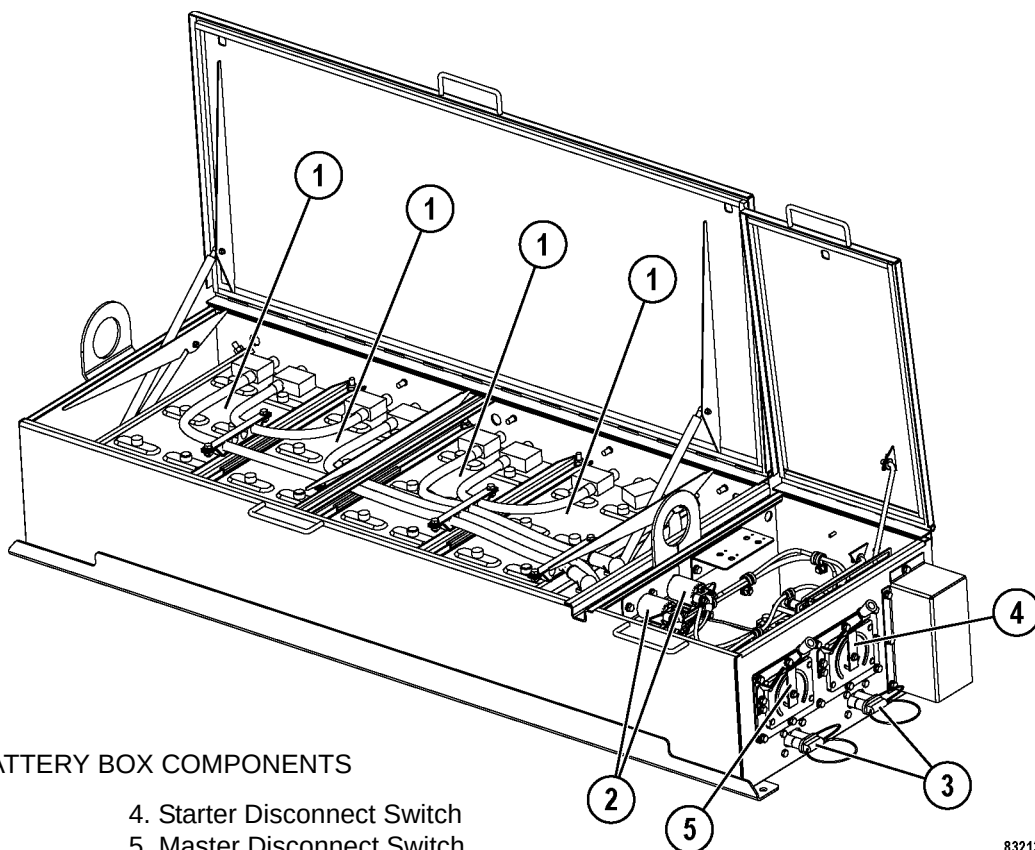


FIGURE 2-1. BATTERY BOX COMPONENTS

- | | |
|----------------------------------|------------------------------|
| 1. Batteries | 4. Starter Disconnect Switch |
| 2. Engine Start Relay | 5. Master Disconnect Switch |
| 3. Auxiliary Battery Receptacles | |

83213

24VDC to 12VDC Converter

24VDC to 12VDC converter (1, Figure 2-2) is used to convert the 24 volt battery system voltage to 12 volts for various truck components such as the radio/CD player, cab power windows, and the auxiliary power receptacles in the cab.

The converter is powered by the engine start circuit batteries. Converter output circuits are protected by 50 amp circuit breaker (2).

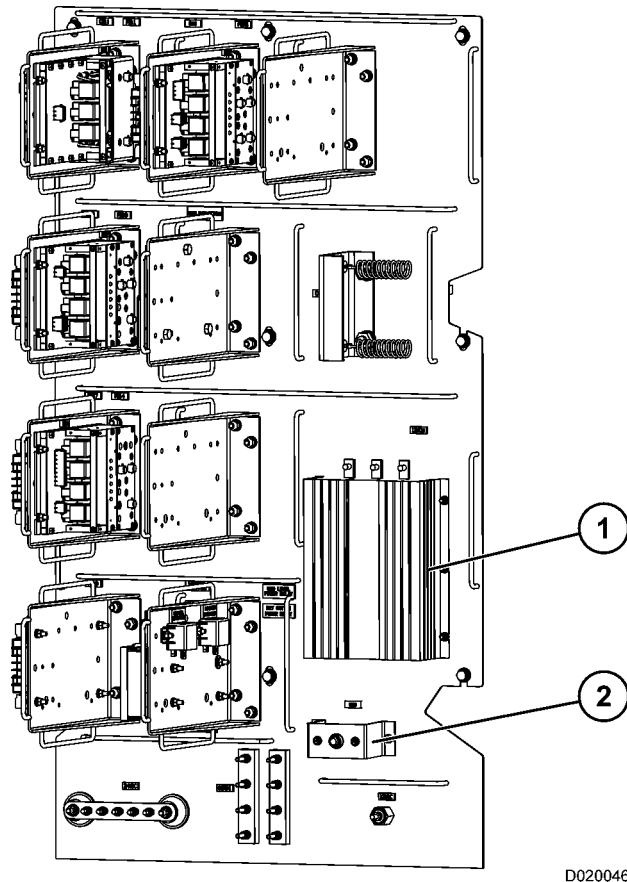


FIGURE 2-2. AUXILIARY CONTROL CABINET
(LEFT WALL)

1. 24VDC to 12VDC Converter
2. 50 Amp Circuit Breaker

24VDC ELECTRIC CRANKING MOTOR SYSTEM (WITH PRELUBE)

The engine includes an engine pre-lubrication system designed to reduce wear due to dry starts.

The prelube system automatically, safely and quickly fills filters and all oil passages prior to cranking at each engine startup. In addition, the system prevents startup if no oil is present in the engine.

The prelube system includes:

- Remote mounted 24VDC powered pump
- Timer solenoid
- Oil pressure switch
- Oil suction line
- Oil outlet line
- Check valve
- Electrical harness

Operation

The prelube system is activated when the operator turns the key switch and holds it in the START position. This allows the current to flow to prelube timer solenoid (2, Figure 2-2). When this timer solenoid is activated, current flows to the prelube motor, driving the prelube pump, but does not allow the cranking motor motors to engage the cranking motor pinion gears at this time. The prelube pump supplies oil from the engine oil pan to fill the engine oil filters and oil passages prior to cranking.

When the pressure in the engine cam oil rifle reaches 17.2 kPa (2.5 psi), the circuit to the prelube timer solenoid is opened. After a three second delay, the current is supplied to the cranking motor solenoids. The cranking motor motors will then be activated and the pinion gears will be engaged into the flywheel ring gear. Normal cranking will now occur with sufficient lubrication to protect the engine bearings and other components.

Pressure Switch

Pressure switch (3) is a 17.2 kPa (2.5 psi), normally closed (N.C.) switch, located so that it can sense oil pressure after the engine oil has passed through the filters. (Normally, this location is the cam cover at the rear of the engine block.)

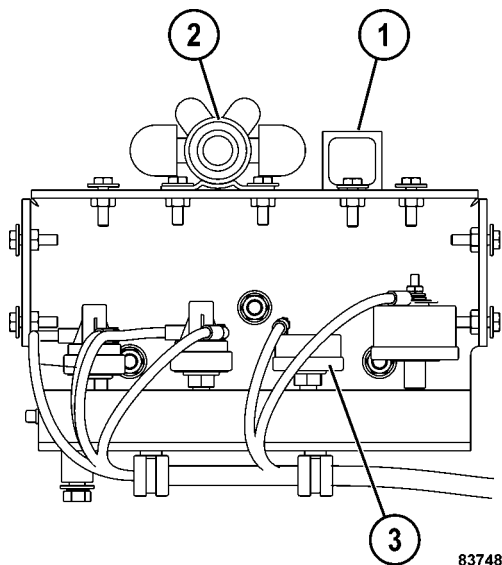


FIGURE 2-2. ENGINE OIL PRESSURE MANIFOLD

- | | |
|-------------------------|----------------------------|
| 1. Engine Prelube Relay | 3. Prelube Pressure Switch |
| 2. Timer Solenoid | |

Check Valve

The oil pressure supply hose will have a check valve installed between the prelube pump and the engine. The check valve prevents the passage of oil from the engine back through the prelube pump to the pan after the engine is started. Check valve leakage back to the prelube pump will cause extensive damage to the pump.

Timer Solenoid

Timer solenoid (2) controls the prelubrication cycle. Current is supplied to the timer through the key switch. The ground path is completed by the normally closed pressure switch (3).

When the switch opens, current is redirected to the engine cranking motor solenoids for engine cranking.



DO NOT attempt to jump start the truck using the terminals on the timer solenoid. Internal damage to the timer will result.

MAINTENANCE

Prelube system maintenance must be performed annually or at 5000 hour intervals as described below.

Prelube System Operation Checks

Verify system operates according to the two phases of operation as listed in *"Troubleshooting Prelube Cranking Motor Circuit"* on the following page. If a problem exists, refer to the list of problems and possible causes for troubleshooting system components.

If system is operating properly, continue with the inspection of component parts below:

Check Valve

Verify that no internal leakage exists in the check valve when the engine is running. Check valve leakage back to the prelube pump will cause extensive damage to the pump.

If check valve replacement is required, ensure the valve is installed with the arrow pointed toward the engine, and NOT toward the pump.

Timer Solenoid

Inspect the timer solenoid for physical damage and to verify wiring is in good condition.

TROUBLESHOOTING PRELUBE CRANKING MOTOR CIRCUIT

Two distinct phases are involved in a complete prelubrication cycle. The two phases are:

1. *Prelubrication Phase*- Begins when the key switch is held in the START position. A circuit is provided to ground through the normally closed pressure switch. The circuit is interrupted upon opening of the pressure switch when the prelube pressure reaches 17.2 kPa (2.5 psi).
2. *Delay and Crank Phase*- Begins when the pressure switch opens. A three second delay precedes the crank mode.

Problem	Probable Cause
• Cranking motor prelubricates only. Does not delay or crank.	Indicates oil pressure is not sufficient to open the pressure switch. a. No oil or low oil in engine. The pump can not build sufficient pressure to open switch. b. Pump failure. c. Pressure switch has failed (closed) and is grounding circuit. d. Oil pressure switch wire chafed and shorting to block.
• Cranking motor prelubricates continuously regardless of key switch position.	Indicates prelube timer solenoid contacts have welded. a. Low voltage can cause relay failure. b. Jump starting of the vehicle with a voltage that is higher than was designed for the system, can cause solenoid contacts to weld.
• Cranking motor delays and cranks. No prelubrication mode.	If an operator indicates the ignition is totally dead, ensure the key is being held in the crank position for 3 to 4 seconds. If the engine cranks after a short delay, this indicates that a ground connection to the pressure switch has been broken. Without a ground path, the prelubrication unit will proceed to delay and crank. a. Check the wire to the pressure switch. If the wire is removed or cut, replace it. b. Check the ground strap to engine block. If the ground strap is missing the block is not grounded. c. Check the pressure switch for an open circuit. Remove the wire, then check for an open circuit between the switch terminal and the switch base. If open, replace pressure switch.
• Starting circuit is irregular when in crank mode.	a. Check for low or dead batteries. b. Check alternator output. c. Check ground connection at "G" terminal of cranking motor bendix solenoid. d. Check for defective cranking motor safety relays. e. If everything checks OK, replace batteries. <i>NOTE: Maximum allowable voltage drop is - 2 volts for cranking motor control circuit.</i>

Problem	Probable Cause
• Cranking motor has very long prelubrication cycle.	Except for severe cold weather starts, the prelube cycle must not exceed 45 seconds. a. Low oil pressure. b. Ensure oil of the proper viscosity is being used in respect to outside temperature. (Refer to engine manufacturer's specifications). c. Check for suction side air leaks, loose connections, cracked fittings, pump casting, or hose kinks and blockage. d. Check the oil pressure switch for the correct location. Be certain that it has not been moved into a metered oil flow, as in a bypass filter or governor assembly.
• Cranking motor has no prelubrication, no delay and no crank.	If the cranking motor is totally inoperative and no prelubrication, no delay and crank, this indicates a possible failure of the prelubrication timer solenoid. Remove the wire from the pressure switch (ground wire) and activate the key switch for several seconds. a. If the cranking motor delays - then cranks, the Prelube Timer Solenoid is bad. Replace the timer solenoid assembly. b. If the cranking motor is still inoperative, check the truck cranking motor switch circuit. Ensure proper voltage is available to the Prelube Timer Solenoid when the key is activated.
• Cranking motor prelubricates, delays, then does not crank.	Indication is either a timer failure, or a cranking motor problem. a. Place a jumper wire to the cranking motor solenoid "S" post. If the engine starts to crank, replace the Prelube Timer Solenoid. b. If the engine fails to crank when the "S" post is energized with voltage, check out cranking motor bendix solenoid and cranking motor pinion drive.

24VDC ELECTRIC START SYSTEM

CRANKING MOTORS

Operation

Heavy duty batteries supply 24VDC to each of the two cranking motors through magnetic switches activated by the key switch on the instrument panel.

NOTE: With a prelube system installed, there is a delay between the time the key switch is moved to the START position and the cranking motors actuate.

When the key switch is placed in the START position, the magnetic switches close, connecting the motor solenoid "S" terminals to the batteries. When the solenoid windings are energized, plunger (56, Figure 2-5) is pulled in, moving cranking motor drive assembly (71) forward in the nose housing to engage the engine flywheel ring gear. Also, when the solenoid plunger is pulled in, the main solenoid contacts close to provide current to the motor armature and cranking takes place. When the engine starts, an overrunning clutch in the drive assembly protects the armature from excessive speed until the key switch is released. When the key switch is released, a return spring causes the drive pinion to disengage.

After the engine is running, a normally closed pressure switch senses engine oil pressure and opens the electrical circuit to prevent actuation of the motor(s) after the engine has started.

Removal

1. Disconnect battery power:
 - a. Open the master battery disconnect switch to remove power from the system.
 - b. Disconnect the negative (-) battery cables first.
 - c. Disconnect the battery positive (+) battery cables last.
2. Mark wires and cables and remove them from the terminals on cranking motor (2, Figure 2-3) and solenoid (3).
3. Remove cranking motor cap screws (1).
4. Remove the cranking motor assembly from flywheel housing.

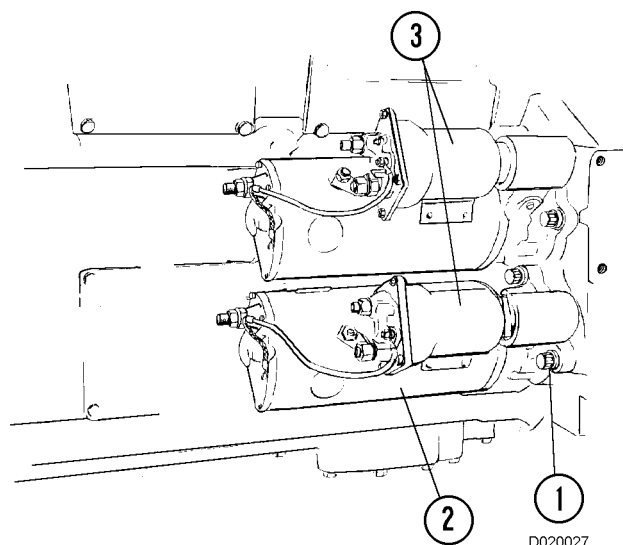


FIGURE 2-3. CRANKING MOTORS

- | | |
|-------------------|-------------|
| 1. Cap Screws | 3. Solenoid |
| 2. Cranking Motor | |

Installation

1. Align cranking motor (2, Figure 2-3) housing with the flywheel housing adaptor mounting holes and slide the cranking motor into position.
2. Insert cranking motor cap screws (1).
3. Connect marked wires and cables to the cranking motor and solenoid terminals.
4. Install in the following sequence:
 - a. Connect the battery positive (+) cables first.
 - b. Connect the battery negative (-) cables.
5. Close the master battery disconnect switch.

CRANKING MOTOR TROUBLESHOOTING

If the cranking system is not functioning properly, check the following to determine which part of the system is at fault:

- ❑ Batteries -- Verify the condition of the batteries, cables, connections and charging circuit.
- ❑ Wiring -- Inspect all wiring for damage or loose connections at the key switch, magnetic switches, solenoids and cranking motor(s). Clean, repair or tighten as required.

If the above inspection indicates the cranking motor to be the cause of the problem, remove the motor and perform the following tests prior to disassembly to determine the condition of the motor and solenoid and repairs required.

Preliminary Inspection

1. Check the cranking motor to ensure that the armature turns freely.
 - a. Insert a flat blade screwdriver through the opening in the nose housing.
 - b. Pry the pinion gear to ensure that the armature can be rotated.
2. If the armature does not turn freely, the cranking motor must be disassembled immediately.
3. If the armature can be rotated, perform the No-Load Test before disassembly.

No-Load Test

Refer to Figure 2-4 for the following test setup.

! IMPORTANT !

Make sure that the master battery disconnect switch is open before connections or disconnections are made during the following procedures.

1. Setup the motor for test as follows:
 - a. Connect a voltmeter from the motor terminal to the motor frame.
 - b. Use an RPM indicator to measure armature speed.
 - c. Connect a carbon pile across one battery to limit battery voltage to 20VDC.

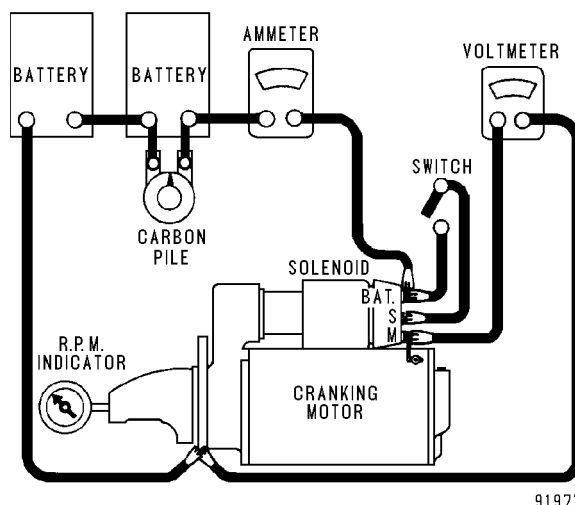


FIGURE 2-4. NO-LOAD TEST CIRCUIT

! WARNING

DO NOT apply voltages in excess of 20 volts. Excessive voltage may cause the armature to throw windings.

- d. Connect the motor and an ammeter in series with two fully charged 12 volt batteries.
- e. Connect a switch in the open position from the solenoid battery terminal to the solenoid switch terminal.
2. Close the switch and compare the RPM, current, and voltage reading to the following specifications:
 - ❑ **RPM:** 5500 Minimum to 7500 Maximum
 - ❑ **AMPS:** 95 Minimum to 120 Maximum
 - ❑ **VOLTS:** 20VDC

Interpreting Results of Tests

1. Rated current draw and no-load speed indicates normal condition of the cranking motor.
2. Low free speed and high current draw indicates:
 - a. Too much friction; tight, dirty, or worn bearings, bent armature shaft or loose pole shoes allowing armature to drag.
 - b. Shorted armature. This can be further checked on a growler after disassembly.
 - c. Grounded armature or fields. Check Further after disassembly.
3. Failure to operate with high current draw indicates:
 - a. A direct ground in the terminal or fields.
 - b. "Frozen" bearings (are determined by turning the armature by hand).
4. Failure to operate with no current draw indicates:
 - a. Open field circuit. This can be checked after disassembly by inspecting internal connections and tracing circuit with a test lamp.
 - b. Open armature coils. Inspect the commutator for badly burned bars after disassembly.
 - c. Broken brush springs, worn brushes, high insulation between the commutator bars or other causes which would prevent good contact between the brushes and commutator.
5. Low no-load speed and low current draw indicates:
 - a. High internal resistance due to poor connections, defective leads, dirty commutator and causes listed under Number 4.
6. High free speed and high current draw indicates shorted fields. If shorted fields are suspected, replace the field coil assembly and check for improved performance.

Disassembly

The cranking motor must be disassembled only as far as necessary to repair or replace defective parts.

1. Note the relative position of the solenoid (53, Figure 2-5), lever housing (78), nose housing (69), and C.E. frame (1) so the motor can be reassembled in the same manner.
2. Disconnect field coil connector (42) from solenoid motor terminal, and lead from solenoid ground terminal.
3. Remove the brush inspection plug (52), and brush lead screws (15).
4. Remove the attaching bolts (34) and separate the commutator end frame (1) from the field frame (35).
5. Separate the nose housing (69) and field frame (35) from lever housing (78) by removing attaching bolts (70).
6. Remove armature (45) and drive assembly (71) from lever housing (78).
7. Separate solenoid (53) from lever housing by pulling apart.

Cleaning and Inspection

1. Drive (71, Figure 2-5), armature (45) and fields (46) must not be cleaned in any degreasing tank or with grease dissolving solvents. These will dissolve the lubricant in the drive and damage the insulation in the armature and field coils.
2. All parts except the drive must be cleaned with mineral spirits and a clean cloth.
3. If the commutator is dirty, it may be cleaned with No. 00 sandpaper.

NOTE: DO NOT use an emery cloth to clean the commutator.

4. Inspect brushes (13) for wear.
 - a. If worn excessively when compared with a new brush, they must be replaced.
 - b. Ensure that the brush holders (10) are clean and the brushes are not binding in the holders.
 - c. The full brush surface must ride on the commutator. Check by hand to insure that brush springs (16) are giving firm contact between brushes (13) and commutator.
 - d. If springs (16) are distorted or discolored, they must be replaced.

FIGURE 2-5 CRANKING MOTOR ASSEMBLY

- | | |
|----------------------------|----------------------|
| 1. C.E. Frame | 52. Inspection Plug |
| 2. Washers | 53. Solenoid Housing |
| 3. O-Rings | 54. Lockwasher |
| 4. Insulator | 55. Screw |
| 5. Support Plate | 56. Plunger |
| 6. Brush Plate Insulator | 57. Washer |
| 7. Washers | 58. Boot |
| 8. Plate & Stud | 59. Washer |
| 9. Plate | 60. Spring |
| 10. Brush Holder | 61. Retainer |
| 11. Lockwasher | 62. Snap Ring |
| 12. Screw | 63. Shift Lever |
| 13. Brush (12 required) | 64. Nut |
| 14. Lockwasher | 65. O-Ring |
| 15. Screw | 66. O-Ring |
| 16. Brush Spring | 67. Snap Ring |
| 17. Screw | 68. Lever Shaft |
| 18. Screw | 69. Drive Housing |
| 19. Screw | 70. Screw |
| 20. Lockwashers | 71. Drive Assembly |
| 21. Plate | 72. Gasket |
| 22. Brush Holder Insulator | 73. Plug |
| 23. Screw | 74. Gasket |
| 24. Lockwasher | 75. Brake Washer |
| 25. Washer | 76. Screw |
| 26. O-Ring | 77. Lockwasher |
| 27. Bushing | 78. Lever Housing |
| 28. Insulator | 79. Washer |
| 29. Washer | 80. O-Ring |
| 30. Lockwasher | |
| 31. Nut | |
| 32. Nut | |
| 33. Lockwasher | |
| 34. Screw | |
| 35. Field Frame | |
| 36. Stud Terminal | |
| 37. Bushing | |
| 38. Gasket | |
| 39. Washers | |
| 40. Washers | |
| 41. Nut | |
| 42. Connector | |
| 43. Lockwasher | |
| 44. Nut | |
| 45. Armature | |
| 46. Field Coil | |
| 47. Shoe | |
| 48. Insulator | |
| 49. Screw | |
| 50. Washer | |
| 51. O-ring | |



3/12 D02042

Armature Servicing

If the armature commutator is worn, dirty, out of round, or has high insulation, the armature (45, Figure 2-5) must be put on a lathe and the commutator turned down. The insulation must then be undercut 0.79 mm (0.031 in.) wide and 0.79 mm (0.031 in.) deep, and the slots cleaned out to remove any trace of dirt or copper dust. As a final step in this procedure, the commutator must be sanded lightly with No. 00 sandpaper to remove any burrs left as a result of the undercutting procedure.

Check the armature for opens, short circuits and grounds as follows:

1. Opens are usually caused by excessively long cranking periods. The most likely place for an open to occur is at the commutator riser bars. Inspect the points where the conductors are joined to the commutator bars for loose connections. Poor connections cause arcing and burning of the commutator as the cranking motor is used. If the bars are not too badly burned, repair can often be effected by resoldering or welding the leads in the riser bars (using rosin flux), and turning down the commutator in a lathe to remove the burned material. The insulation must then be undercut.
2. Short circuits in the armature are located by use of a growler. When the armature is revolved in the growler with a steel strip such as a hacksaw blade held above it, the blade will vibrate above the area of the armature core in which the short circuit is located. Shorts between bars are sometimes produced by brush dust or copper between the bars. These shorts can be eliminated by cleaning out the slots.

3. Grounds in the armature can be detected by the use of a 110-volt test lamp and test points. If the lamp lights when one test point is placed on the commutator with the other point on the core or shaft, the armature is grounded. Grounds occur as a result of insulation failure which is often brought about by overheating of the cranking motor produced by excessively long cranking periods or by accumulation of brush dust between the commutator bars and the steel commutator ring.

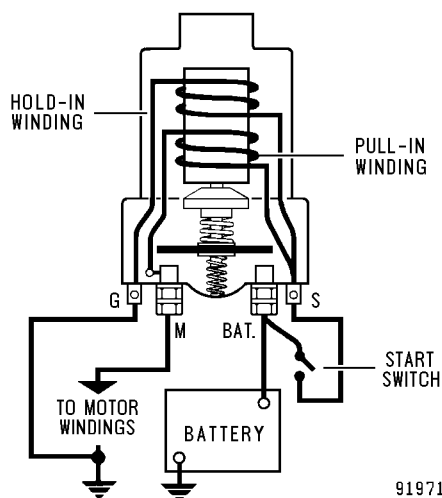
Field Coil Checks

Field coils (46, Figure 2-5) can be checked for grounds and opens by using a test lamp.

1. **Grounds** - The ground connections must be disconnected during this check. Connect one lead of the 110-volt test lamp to field frame (35) and the other lead to field connector (42). If the lamp lights, at least one field coil is grounded and must be repaired or replaced.
2. **Opens** - Connect test lamp leads to ends of field coils (46). If lamp does not light, the field coils are open.

Field Coil Removal

Field coils can be removed from the field frame assembly by using a pole shoe screwdriver. A pole shoe spreader must also be used to prevent distortion of the field frame. Careful installation of the field coils is necessary to prevent shorting or grounding of the field coils as the pole shoes are tightened into place. Where the pole shoe has a long lip on one side and a short lip on the other, the long lip must be assembled in the direction of armature rotation so it becomes the trailing (not leading) edge of the pole shoe.



91971

FIGURE 2-6. SIMPLIFIED SOLENOID CIRCUIT

SOLENOID CHECKS

A basic solenoid circuit is shown in Figure 2-6. Solenoids can be checked electrically using the following procedure.

Test

1. With all leads disconnected from the solenoid, make test connections as shown to the solenoid, switch terminal and to the second switch terminal "G", to check the hold-in winding (Figure 2-7).
2. Use the carbon pile to decrease the battery voltage to 20 volts. Close the switch and read current.
 - ❑ The ammeter must read 6.8 amps maximum.
3. To check the pull-in winding, connect from the solenoid switch terminal "S" to the solenoid motor "M" or "MTR" terminal (Figure 2-8).

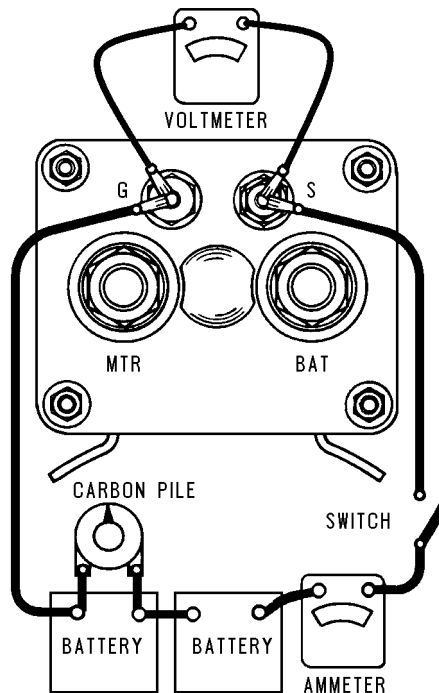
⚠ WARNING

To prevent overheating, DO NOT leave the pull-in winding energized more than 15 seconds. The current draw will decrease as the winding temperature increases.

4. Use the carbon pile to decrease the battery voltage to 5 volts. Close the switch and read current.
 - ❑ The ammeter must read 9.0 to 11.5 amps.

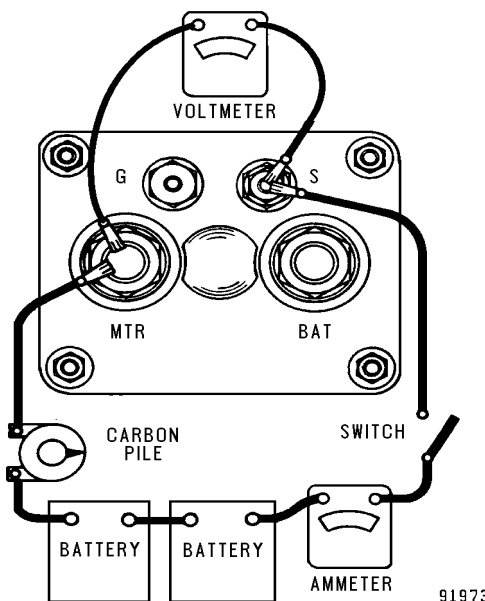
NOTE: High readings indicate a shorted winding. Low readings indicate excessive resistance.

5. To check for grounds, move battery lead from "G" (Figure 2-7) and from "MTR" (Figure 2-8) to the solenoid case. Ammeter must read zero. If not, the winding is grounded.



91972

FIGURE 2-7. SOLENOID HOLD-IN WINDING TEST



91973

FIGURE 2-8. SOLENOID PULL-IN WINDING TEST

Assembly

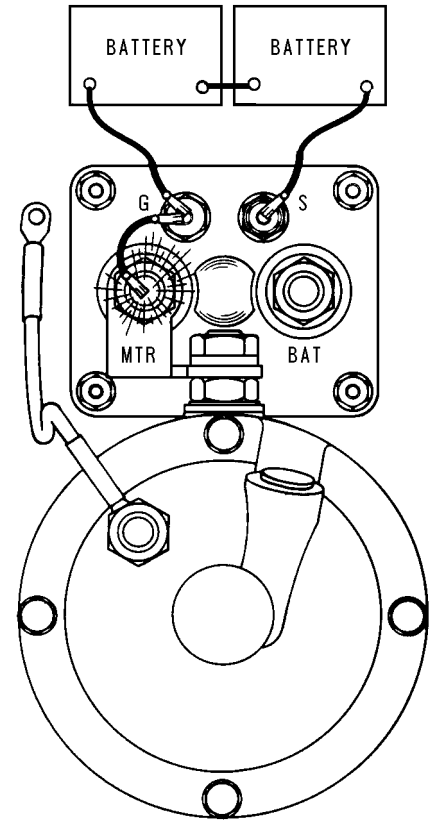
Lubricate all bearings, wicks and oil reservoirs with SAE No. 20 oil during assembly.

Bearing Replacement:

1. If any of the bronze bearings are to be replaced, dip each bearing in SAE No. 20 oil before pressing into place.
2. Install a wick, soaked in oil, prior to installing the bearings.
3. DO NOT attempt to drill or ream sintered bearings. These bearings are supplied to size. If drilled or reamed, the I.D. will be too large and the bearing pores will seal over.
4. DO NOT cross-drill bearings. Because the bearing is so highly porous, oil from the wick touching the outside bearing surface will bleed through and provide adequate lubrication.
5. The middle bearing is a support bearing used to prevent armature deflection during cranking. The clearance between this bearing and the armature shaft is large compared to the end frame bearings.

Motor Assembly:

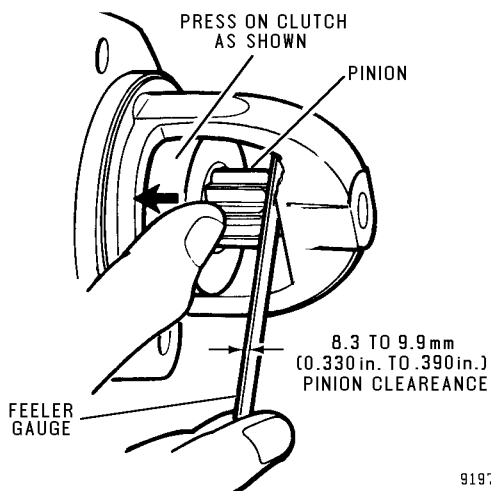
1. Install the end frame (with brushes) onto the field frame as follows:
 - a. Insert armature (45, Figure 2-5) into field frame (35). Pull the armature out of the field frame just far enough to permit the brushes to be placed over the commutator.
 - b. Place end frame (1) on the armature shaft. Slide the end frame and armature into place against the field frame.
 - c. Insert screws (34) and washers (33) and tighten securely.
2. Assemble lever (63) into lever housing (78) if removed.
3. Place washer (79) on the armature shaft and install new O-ring (80). Position drive assembly (71) in lever (63) in the lever housing. Apply a light coat of lubricant (Delco Remy Part No. 1960954) on washer (75) and install it over the armature shaft. Align the lever housing with the field frame and slide the assembly over the armature shaft. Secure it with screws (76) and washers (77).
4. Assemble and install the solenoid assembly through the lever housing and attach it to the field frame. Install nut (64), but do not tighten at this time. Install brush inspection plugs (52).



91975

FIGURE 2-9. PINION CLEARANCE CHECK CIRCUIT

5. Using new gasket (72), install drive housing (69) and secure it with screws (70).
6. Assemble field coil connector (42) to the solenoid.
7. Adjust the pinion clearance per instructions on the following page.
8. After the pinion clearance has been adjusted, install gasket (74) and plug (73).



91976

FIGURE 2-10. CHECKING PINION CLEARANCE

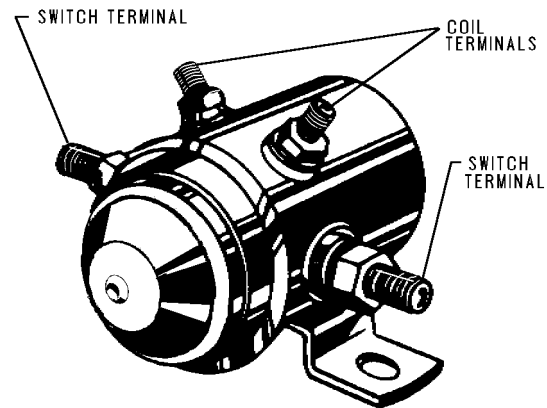
Pinion Clearance

To adjust the pinion clearance, follow the steps listed below.

1. Make connections as shown in Figure 2-9.
2. Momentarily flash a jumper lead from terminal "G" to terminal "MTR". The drive will now shift into cranking position and remain so until the batteries are disconnected.
3. Push the pinion or drive back towards the commutator end to eliminate slack movement.
4. The distance between the drive pinion and housing must be between 8.3 mm to 9.9 mm (0.330 to 0.390 in.) as shown in Figure 2-10.
5. Adjust clearance by turning shaft nut (64, Figure 2-5).

MAGNETIC SWITCH

The magnetic switch is a sealed unit and not repairable.



92052

FIGURE 2-11. MAGNETIC SWITCH ASSEMBLY

Removal

1. Remove battery power as described in Cranking Motor Removal.
2. Disconnect the cables from the switch terminals and the wires from coil terminals (Figure 2-11).

NOTE: If the magnetic switch being removed has a diode across the coil terminals, mark the leads prior to removal to ensure correct polarity during installation.

3. Remove the mounting cap screws and washers. Remove the switch from the mounting bracket.
4. The switch coil circuit can be tested as described on the next page.

Installation

1. Attach the magnetic switch to the mounting bracket using the cap screws and lockwashers removed previously.
2. Inspect the cables and switch terminals. Clean as required and install the cables.
3. Install the diode across the coil terminals. Ensure that diode polarity is correct. Attach the wires from the truck harness to the coil terminals (See Figure 2-4).
4. Connect battery power as described in Cranking Motor Installation.

Coil Test

1. Using an ohmmeter, measure the coil resistance across the coil terminals.
 - a. The coil must read approximately 28Ω at 22.2°C (72°F).
 - b. If the ohmmeter reads ∞ , the coil is open and the switch must be replaced.
 - c. If the ohmmeter reads 0Ω , the coil is shorted and the switch must be replaced.
2. Place one of the ohmmeter probes on a coil terminal and another on the switch mounting bracket. If the meter displays any resistance reading, the coil is grounded and the switch must be replaced.
3. The ohmmeter must display when the probes are placed across the switch terminals.

NOTE: The switch terminals will show continuity when 24VDC is applied to the coil terminals. However, high resistance across the internal switch contacts due to arcing, etc, could prevent the switch from delivering adequate current to the cranking motor. If the coil tests are satisfactory but the switch is still suspect, it must be replaced with a new part.

NOTES

SECTION D3 **24VDC ELECTRICAL SYSTEM COMPONENTS** **INDEX**

BRAKE WARNING BUZZER	D3-3
AUXILIARY CONTROL CABINET COMPONENTS	D3-3
Power Distribution Terminals	D3-3
Control Power Relay	D3-3
Auto Lube Timer	D3-3
Ground Level Power Relay	D3-3
Diode Board - DB1	D3-5
Fuse Blocks	D3-6
Alarm Indicating Device (AID) System	D3-6
RELAY BOARDS	D3-9
Relay Boards RB1, RB3, RB4, RB5	D3-9
Relay Boards RB6, RB7, RB8, RB9	D3-11
Relay Board Functions	D3-12
BODY-UP SWITCH	D3-14
Operation	D3-14
Adjustment	D3-14
Service	D3-14
HOIST LIMIT SWITCH	D3-15
Operation	D3-15
Adjustment	D3-15
Service	D3-15
FUSE BLOCKS AND CIRCUIT BREAKERS	D3-16

NOTES

24 VDC ELECTRICAL SYSTEM COMPONENTS



Do not attempt repairs until the truck is properly shut down. Dangerous voltage levels are present in the propulsion system while the engine is running and for a period of time after shutdown. Refer to the Index in Section D for additional warnings and truck shutdown procedure.

BRAKE WARNING BUZZER

The brake warning buzzer provides an audible alarm for the operator if a malfunction occurs in the hydraulic service brake system. This buzzer is located inside the radio module in the overhead panel. Refer to Section J for additional details.

AUXILIARY CONTROL CABINET COMPONENTS

The following 24VDC electrical system components are located in the auxiliary control cabinet, which is mounted on the left side of the main control cabinet behind the cab. The auxiliary control cabinet houses various components for the 24VDC circuits, engine related devices, and terminal strips that connect truck wiring harnesses with the main control cabinet and cab.

The following information describes the components in the auxiliary control cabinet and their operation. Additional detailed information for operation and troubleshooting procedures not included below can be found in Section E, Electrical Propulsion System, the engine manufacturer's service publications, and the appropriate GE publications. Use the electrical schematics in Section R when troubleshooting problems with the following 24VDC electrical system components.

Power Distribution Terminals

24VDC terminal (1, Figure 3-1) and 12VDC terminal (2) are mounted on the left wall of the cabinet. These terminals distribute battery voltage and 12VDC for devices requiring reduced voltage. The 24VDC terminal is a convenient test point for measuring battery voltage during troubleshooting procedures.

Control Power Relay

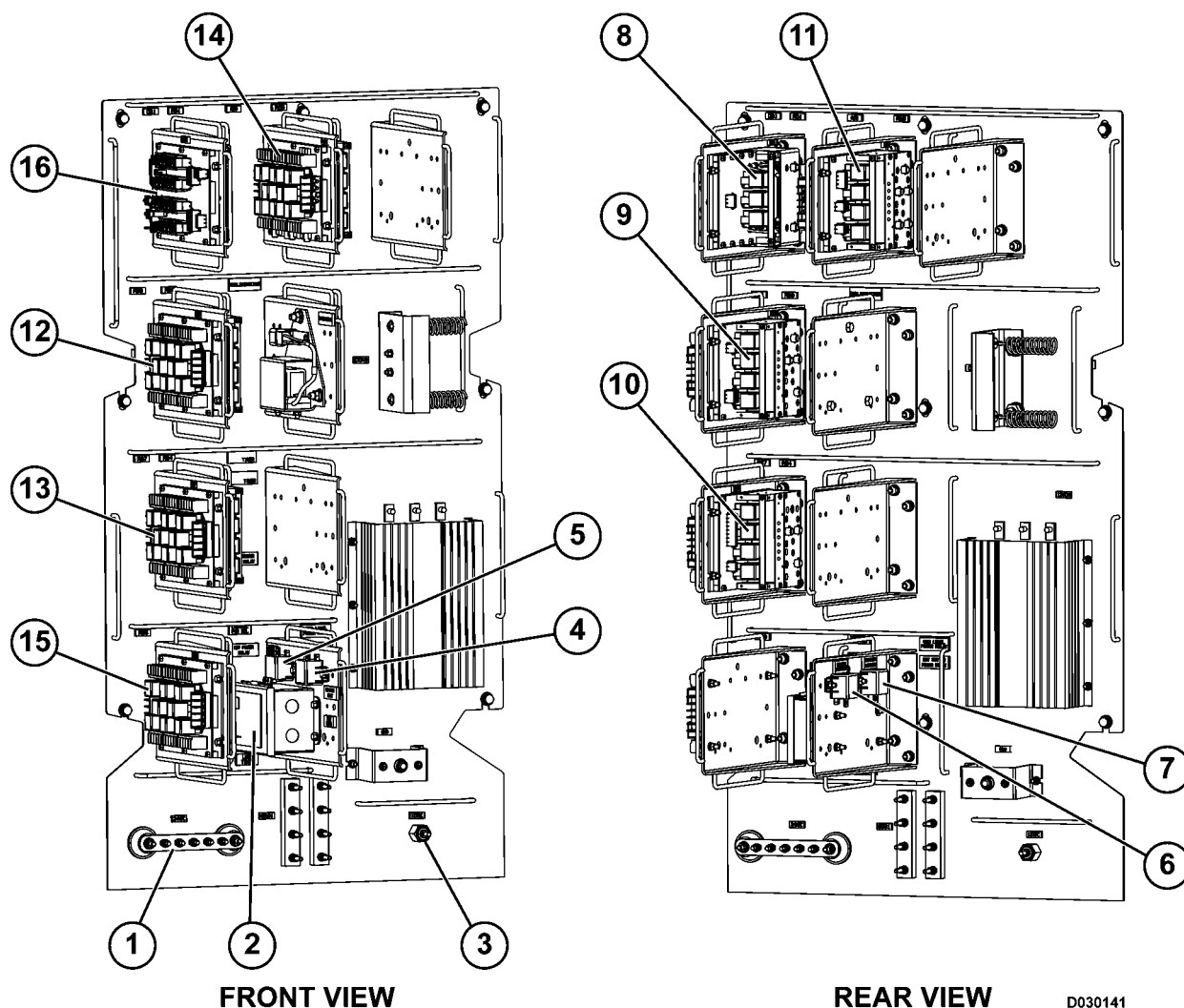
Control power relay (5) is energized when the control power switch, located in the main control cabinet, is turned ON. This relay isolates the GE control power from the truck circuits and provides power to non-propulsion system 24VDC components.

Auto Lube Timer

The automatic lubrication system lubrication interval is controlled by auto lube timer (6). Lubrication cycle frequency can be adjusted by removing the timer cover and selecting one of five different timing intervals available. System "on" time is automatically determined by the timer and is not adjustable. Refer to Section P for additional automatic lubrication system details.

Ground Level Power Relay

When the ground level shutdown switch is OFF, ground level power relay (6) is energized and the truck can operate. When the switch is ON (activated), the engine will stop and the ground level power relay is de-energized, interrupting power to the payload meter system and the timed engine shutdown sequence.



D030141

FIGURE 3-1. AUXILIARY CONTROL CABINET - LEFT WALL

- | | | |
|-----------------------------|---------------------------|-----------------------|
| 1. 24VDC Terminal | 7. Key Switch Power Relay | 13. Relay Board - RB7 |
| 2. Auto Lube Timer | 8. Relay Board - RB1 | 14. Relay Board - RB8 |
| 3. 12VDC Terminal | 9. Relay Board - RB3 | 15. Relay Board - RB9 |
| 4. Control Power Relay | 10. Relay Board - RB4 | 16. Diode Board - DB1 |
| 5. 12V Power Relay | 11. Relay Board - RB5 | |
| 6. Ground Level Power Relay | 12. Relay Board - RB6 | |

NOTE: For more information about relay boards RB1 through RB9, refer to Relay Boards later in this section.

Diode Board - DB1

Diode board (15, Figure 3-1) contains 24 replaceable diodes that are mounted on a plug-in connector for easy replacement. Some of the diodes are used in the coil circuit of various relays to suppress the resultant coil voltage spike when power is removed from the circuit, preventing damage to other circuit components (lamp filaments, etc.). Other diodes are used to control the flow of current in a circuit as required. Resistors or diodes may also be installed in sockets P7 through P12.

Refer to the electrical schematic in Section R of this manual for specific circuits.

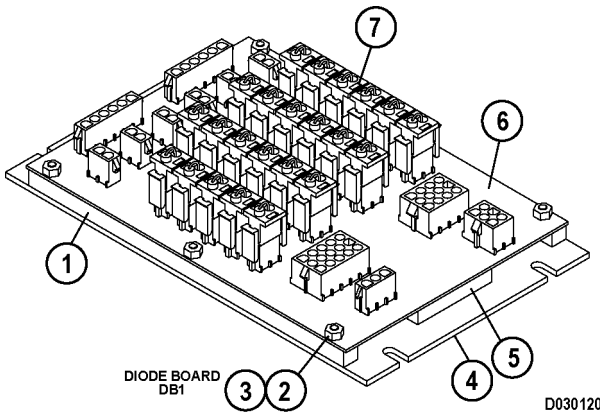


FIGURE 3-2. DIODE BOARD

- | | |
|-------------------|---------------|
| 1. Mounting Rail | 5. Foam Block |
| 2. Screw | 6. Board |
| 3. Nut | 7. Diode |
| 4. Mounting Plate | |

If a diode failure is suspected, remove and check the diode as follows:

1. Grasp the diode connector, compressing the locking "ears" while pulling the connector off the board. Note the connector "key" used to ensure correct polarity.

NOTE: Some digital multimeters are designed to test diodes. If this type is used, follow the manufacturer's instructions for proper test.

2. An analog ohmmeter can be used to test the diode as follows:
 - a. Place the meter on the "X100" scale.
 - b. With the red meter lead (+) on the banded end of the diode and the black lead (-) on the other diode lead, the meter should read between 1000 and 2000 ohms.
 - c. Reverse the meter leads and read infinite resistance.
3. If no resistance is read on the meter, the diode is open and must be replaced.
4. If the meter reads zero ohms, the diode is shorted and must be replaced.
5. Orient the diode assembly for proper polarity ("key" noted in step 1.) and insert connector until locked in position on mating receptacle.

Fuse Blocks

Four fuse blocks (1, Figure 3-3) and four fuse holders (2) contain fuses that protect various circuits on the truck. Always replace a blown fuse with a new one of the same rating. For a listing of fuse sizes and circuits, refer to the Fuse Blocks charts at the end of this section.

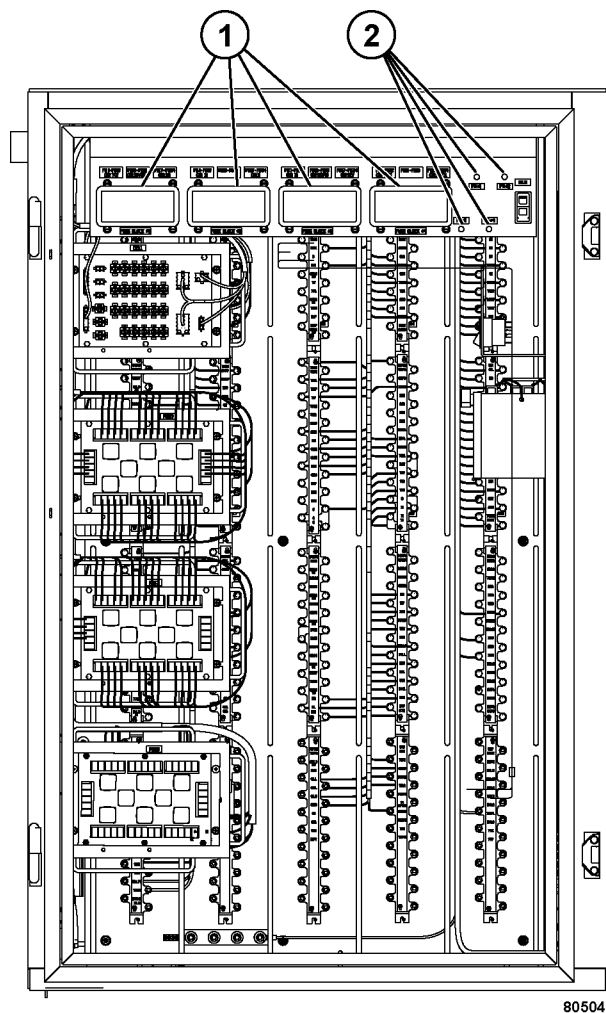


FIGURE 3-3. AUXILIARY CONTROL CABINET - FRONT VIEW

1. Fuse Blocks 2. Fuse Holders

Alarm Indicating Device (AID) System

Alarm indicating device (AID) module (1, Figure 3-4) is connected to the electrical accessories circuits to provide the operator with a warning indication of a malfunction. This system consists of up to eight printed circuit cards located in the auxiliary control cabinet. The actual quantity of cards will depend on the that are options installed on the truck.

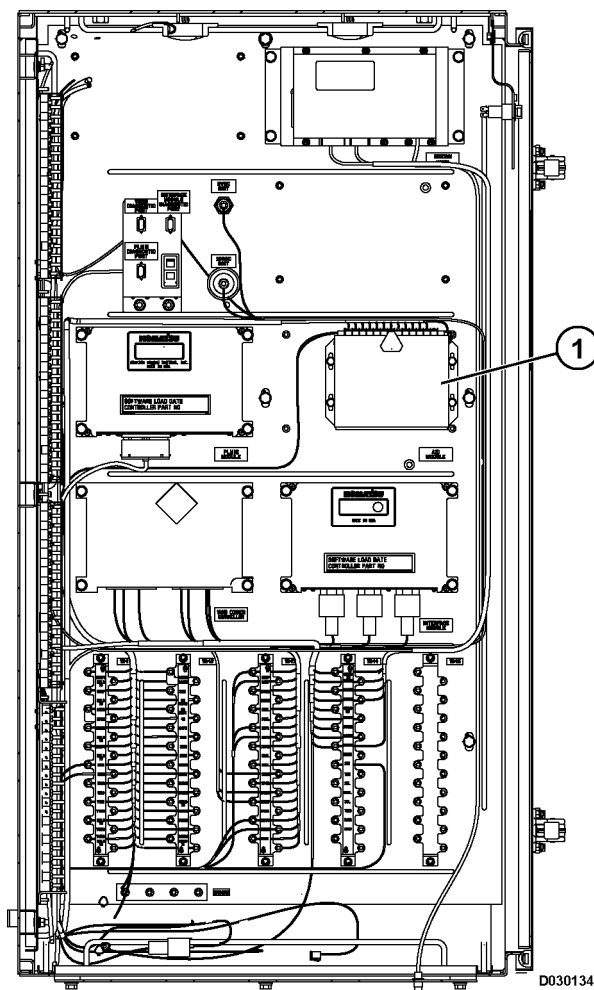


FIGURE 3-4. AUXILIARY CONTROL CABINET - RIGHT WALL

1. AID Module

The AID system enables the indicating lights to be flashing or constant. The AID also has the capability of operating an audible alarm along with the light. The eight printed circuit cards are:

- Diode Matrix (With Sound) Card (Slot 1)
- Diode Matrix (Without Sound) Card (Slot 2)
- Hot Switch Inverter Card (Slot 3)
- Not Used (Slot 4)
- Temperature Card (Slot 5) (Optional)
- Oil Level Card (Slot 6) (Optional)
- Temperature and Latch Card (Slot 7)
- Coolant Level and Flasher Card (Slot 8)

NOTE: Each card is identified with a number which corresponds to a mating number on the housing. If any cards are removed, make sure that the card numbers correspond with the housing numbers during installation. See Figure 3-5.

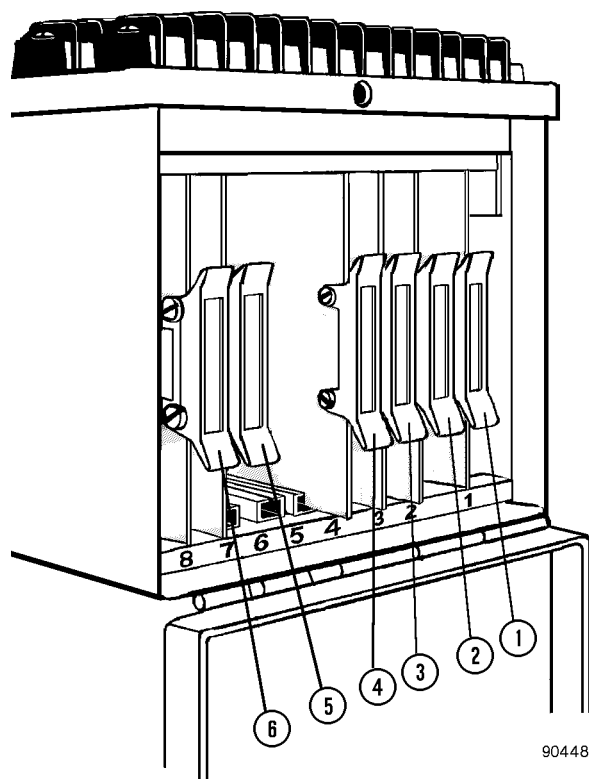


FIGURE 3-5. AID SYSTEM CARD ENCLOSURE

1. Diode Matrix With Sound
2. Diode Matrix Without Sound
3. Hot Switch Inverter
4. Not Used
5. Temperature & Latch
6. Coolant Level & Flasher

The following briefly describe each card and its function. Refer to Section R, Schematics, for the circuit components described below.

Diode Matrix (With Sound)

The diode matrix with sound card works very much like the other diode matrix card, except that it contains extra diodes to activate the alarm horn in addition to the flasher. The circuits connected to terminals A1 through A8 operate in the same manner.

All of the card circuits are connected to the lamp test switch on the overhead display area. In normal operation, these circuits are open and not functional. When the operator pushes the lamp test switch, it activates all the indicator circuits by grounding them. This is used to verify that all lamps are functional.

Diode Matrix (Without Sound)

The diode matrix without sound card consists of a series of diodes capable of working with eight different indicator circuits. The indicator light can be a flashing light by connecting it to the 12F circuit or a steady light by connecting it to the 12M circuit. In addition, some of the indicator light circuits are routed through a dimmer module to allow the operator to vary the intensity of the lamps. These lamps are fed by circuits 12FD (flashing) and 12MD (steady).

When an indicator circuit is not activated, there is no ground circuit for the bulb. When the indicator detecting switch activates the circuit, it grounds the lamp and the flasher circuit through the diodes. Any circuits connected to terminals C1 through C8 will operate in the same manner. The alarm horn is not activated by this card.

Hot Switch Inverter

The hot switch inverter card is used to operate and test the service brake indicator light. In normal conditions Q4 transistor is off and the Indicator Light is off. When the stoplight switch is activated, 24 volts is sent to pin "E" of the hot switch inverter card. Transistor Q4 is turned on by this voltage and, in turn, grounds the service brake indicator light. There is no alarm horn operation with this card.

A second circuit on this card is used to operate and test the retard speed control indicator light. When RSC is turned OFF, transistor Q7 is off and the indicator light is off. When RSC is turned on, 24 volts is sent to pin "J" of the card. This voltage turns on Q7, grounding the indicator light circuit.

Temperature and Latch

The temperature and latch card has two circuits to operate two different indicator lights. The temperature circuit is controlled by a coolant temperature sensor which decreases electrical resistance as its temperature increases. It will have a resistance of approximately 1000 ohms at 85°C (185°F) and 500 ohms at 121°C (250°F). The normal setting is 96°C (204°F).

When the temperature is low and the resistance is high, Q1 is off and no high temperature indication occurs. When the coolant temperature is excessive, resistance decreases to a point where Q1 will turn on and ground the flasher through D8, the alarm horn through D12, and the high temperature light through terminal D8. R14 can adjust the temperature (resistance) at which the circuit is activated.

NOTE: Some electronic engine controls monitor coolant temperature. If the engine controls monitor the circuit, a 2K ohm resistor is installed to replace the temperature sensor and disable the AID system circuit.

The latch circuit monitors the accumulator precharge pressure switches. When one of the pressure switches closes, Q5, which supplies power to the gate of SCR Q7, will be turned off. With Q7 turned on, Q9 will supply the ground path to turn on the low accumulator precharge indicator light and sound the alarm horn. The indicator light is connected to 12F and will flash off and on. The SCR will remain on until power is removed from the card by turning the key switch OFF.

Coolant Level and Flasher

The coolant level and flasher card contains two separate circuits. The flasher circuit at the top of the card has Q12 transistor biased to be saturated when no malfunction is present, resulting in 24 volt positive output on pin "H" of the card and on wire 12F. When an indicating circuit is activated, the ground side of the circuit connected to card pin "K" is grounded. Q12 will turn off initially and then after a delay, adjusted by R20, will turn on and off to give the intermittent 24 volt output.

The other half of the circuitry on the coolant level and flasher card operates the coolant level light. The water level probe connected to terminal B11 grounds the 31L circuit when the coolant in the radiator is above the probe position. The coolant saturates the probe and electrically grounds the circuit. When the circuit is grounded, Q6 transistor is off, resulting in no indication. When the coolant level drops below the probe, 31L is no longer grounded and Q6 turns on to ground the flasher through D5, the coolant level light through terminal D11, and the alarm horn through D6. The light and alarm horn will operate intermittently as their 24 volt supply is from circuit 12F, the flasher output.

NOTE: Some electronic engine controls monitor coolant level. If the engine controls monitor the circuit, a 2K ohm resistor is installed to replace the probe and disable the AID system circuit.

RELAY BOARDS

The auxiliary control cabinet contains eight relay boards to provide control for many of the 24VDC circuits. Two types of boards are used. One type of board contains circuit breakers in addition to 24VDC relays and a PC board for special functions. The second type of board contains relays only.

All relays are interchangeable. The circuit breakers are also interchangeable as long as the circuit breaker capacity is the same.



Do not interchange or replace any circuit breaker with one of a different capacity than specified for the circuit. Serious damage or a fire may result if the wrong capacity breaker is used.

The relay boards are identified as follows:

Relay Board	Functions
RB1	Clearance/Turn Signal Lights
RB3	Stop, Retard, Backup Lights
RB4	Parking Brake, Horn, Body-Up
RB5	Headlights (standard)
RB6	Headlights (HID), Backup Horn, Auto Lube Solenoid & Timer, Ether Start, Engine Interlock, Bleeddown Power
RB7	Brake Lock, IM Warning, Timed Engine Shutdown
RB8	PLMIII Lights, Shutter Control, TCI
RB9	Start Circuit, Low Steering Pressure, Engine Running/Selector Switch, Parking Brake OFF, Key Switch START, Rear Backup Lights

Refer to Figure 3-1 for the location of each relay board. Refer to the Circuit Breakers chart at the end of this section for electrical circuit identification numbers.

Relay Boards RB1, RB3, RB4, RB5

Each relay board of this type is equipped with four green lights (9, Figure 3-6) and one red “breaker open” light (7). Each relay board has a fifth green (8) light that has a different function on each board.

Four green lights (9) are labeled K1, K2, K3, or K4. These lights will be on only when that particular control circuit has been switched ON and the relay coil is being energized. The light will not turn on if the relay board does not receive the 24 volt signal to turn on a component.

If illuminated, red “breaker open” light (7) indicates that a circuit breaker on that relay board is in the OFF position. A light on the overhead display panel will also illuminate, informing the operator that a circuit breaker is in the OFF position. The red “breaker open” light will turn ON whenever there is a voltage difference across the two terminals of a circuit breaker.

If a control switch has been turned ON and a green (K) light is on, but that component is not operating, check the following on the relay board for that circuit:

- ☐ If a circuit breaker light is on, press all the circuit breakers to make sure that they are all on. There is no visual indication as to which circuit breaker has been tripped. Check the operation of the component. If it trips again, check the wiring or component for the cause of the overload.
- ☐ The contacts inside the relay may not be closing, or the contacts may be open, preventing an electrical connection. Swap relays and check again. Replace defective relays. Relays may take one minute to trip and 30 seconds before they can be reset.
- ☐ Check the wiring and all of the connections between the relay board and the component for an open circuit.
- ☐ The component may be defective. Replace the component.
- ☐ There is a poor ground at the component. Repair the ground connection.

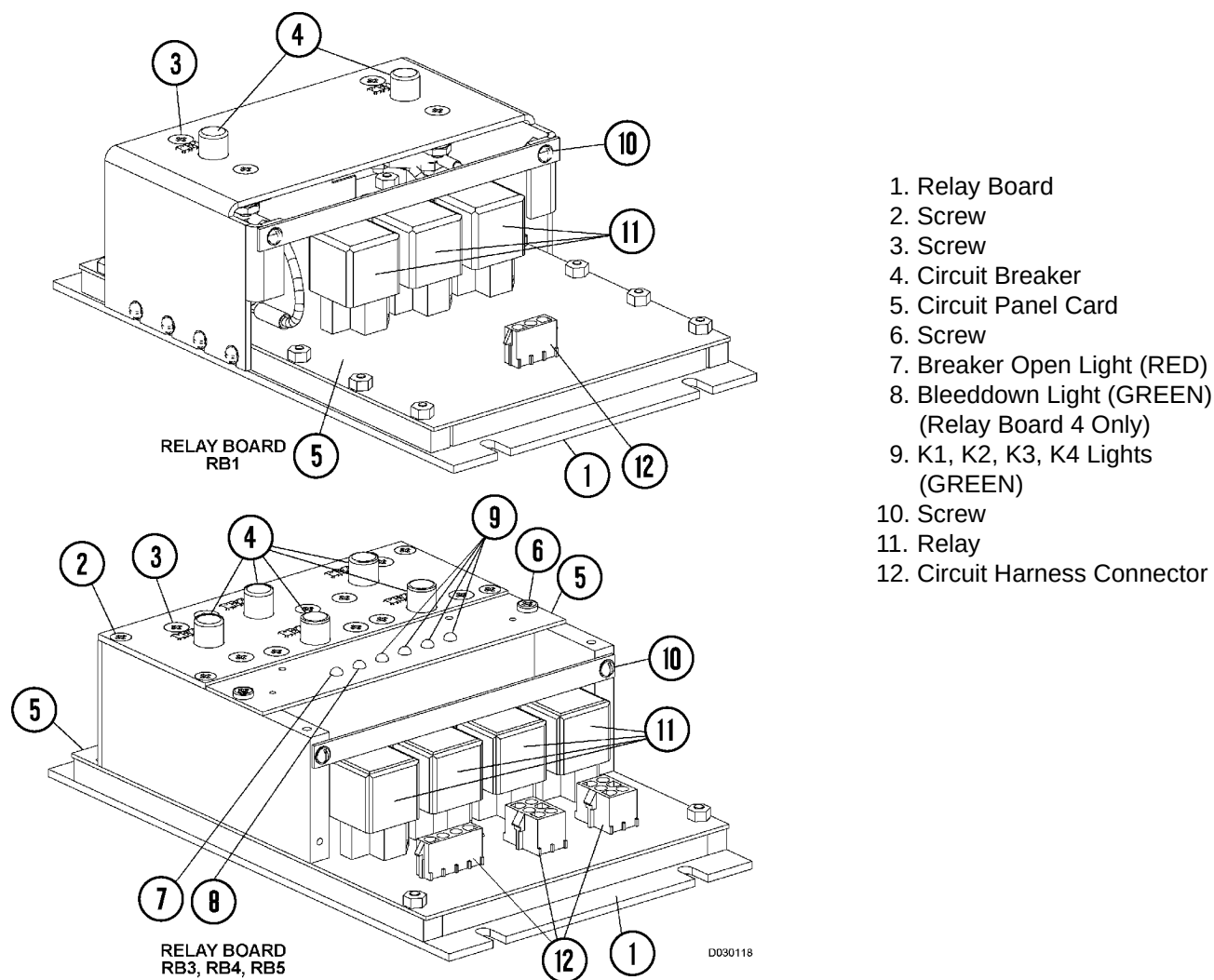


FIGURE 3-6. TYPICAL RELAY BOARDS - RB1, RB3, RB4, RB5

To replace a relay:

NOTE: The relays are labelled to identify the applicable circuits and components. Also, refer to the Fuse Blocks charts at the end of this section.

1. Remove one screw (10, Figure 3-6) holding the crossbar in place and loosen the other screw.
2. Swing the crossbar away.
3. Gently wiggle and pull outward to remove relay (11).
4. Line up the tabs and install a new relay.
5. Place the crossbar in its original position and install screw (10). Tighten both screws.

To replace a circuit breaker:

NOTE: Always replace a circuit breaker with one of the same amperage capacity as the one being removed.

1. Activate the battery disconnect switches.
2. Unplug all wiring harnesses from the relay board. Remove the four relay board mounting screws. Remove the relay board from the truck.
3. Remove four hold down screws (2, Figure 3-6) (one in each corner) in the circuit breaker cover plate. Remove two screws (6) and card (5).
4. Remove the nuts on the wire terminal leads on the circuit breaker to be replaced. Remove mounting screws on circuit breaker to be replaced.
5. Lift out circuit breaker. Retain flat washers from wire terminals.
6. Install new circuit breaker of the same capacity rating as the one removed. Install one nut and two flat washers for each wire connection to the new circuit breaker.
7. Install cover plate and all screws removed during disassembly.
8. Carefully install card (5) with screws (6).
9. Install relay board in truck and connect all wiring harnesses.

To replace a circuit panel card:

NOTE: DO NOT remove the small screws that hold the cover plate to the circuit panel. Replace circuit panel as a complete assembly.

1. Place battery disconnect switches in the OFF position.
2. Remove the two mounting screws (6, Figure 3-6) and carefully remove the circuit panel card from the relay board.
3. Line up the new circuit panel in slots and with the socket on the relay board and install carefully.
4. Install two mounting screws (6).

Relay Board RB6, RB7, RB8, RB9

Relay boards 6, 7, 8 and 9 (Figure 3-7) do not contain circuit breakers or modular cards. Additional circuits may be added by utilizing a spare relay socket as described below:

The control circuit for the relays are the “+” and “-” terminals:

- “+” terminal is for positive voltage.
- “-” terminal is for grounding of the control circuit.
- Either circuit can be switched “open” or “closed” to control the position of the relay.

The terminals of the switched circuit from the relay contacts are labeled as follows:

- NC - Normally Closed
 - COM - Common
 - NO - Normally Open
- ☐ COM terminal is for the voltage source (protected by a circuit breaker) coming into the relay which will supply the electrical power for the component being controlled.
 - ☐ NC terminal is connected (through the relay) to the “COM” terminal when the relay is not energized (when the control circuit terminals “+” & “-” are not activated).
 - ☐ NO terminal is connected (through the relay) to the “COM” terminal when the relay is energized (by the control circuits “+” & “-” being energized).

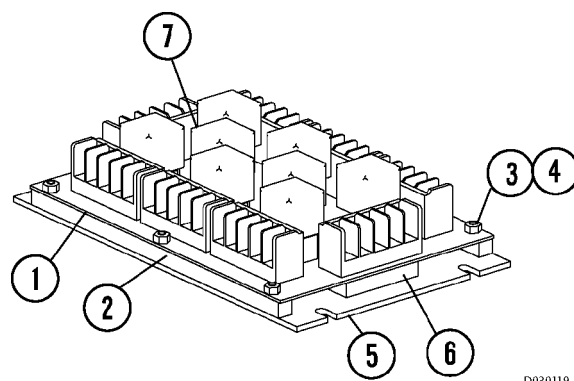


FIGURE 3-7. AUXILIARY RELAY BOARD - RB6, RB7, RB8, RB9

- | | |
|------------------|-------------------|
| 1. Circuit Board | 5. Mounting Plate |
| 2. Mounting Rail | 6. Foam Block |
| 3. Screw | 7. Relay |
| 4. Nut | |

Relay Board Functions

The components and functions of each relay board are described below.

Relay Board 1 (RB1)

- 1 - Flasher Power Light (green): This light will be illuminated when the turn signals or hazard lights are activated.
 - ☐ 1 light will be illuminated during right turn signal operation
 - ☐ 2 light will be illuminated during left turn signal operation
 - ☐ 3 light will be illuminated when clearance lights are activated.
 - ☐ 4 light will be flashing when the turn signals or hazard lights are in operation.

NOTE: If circuit breakers CB13 and CB15 are in the OFF position, no warning will be noticed until the clearance light switch is turned ON.

- 1 - Flasher Module card
- 2 - 12.5 amp circuit breakers (CB13, CB15)
- 4 - Relays
 - ☐ Left Turn Light Relay (K1)
 - ☐ Right Turn Light Relay (K2)
 - ☐ Clearance Lights Relay (K3)
 - ☐ Flasher Relay (K4)

Relay Board 3 (RB3)

- 1 - Light Module Display card
- 1 - Rev Light (green): This light is illuminated whenever the directional control lever is in the REVERSE position and the key switch is in the ON position.
- 4 - 12.5 amp circuit breakers (CB16, CB17, CB18, CB19)
- 4 - Relays
 - ☐ Manual Backup Lights Relay (K1)
 - ☐ Stop Lights Relay (K2)
 - ☐ Retard Lights Relay (K3)

Relay Board 4 (RB4)

- 1 - Steering Pressure Bleeddown Timer Module card
- 1 - Bleeddown Light (green): This light is illuminated when the bleeddown solenoid is being energized. The bleeddown timer will energize the solenoid for 90 seconds after the key switch is turned OFF.
- 3 - 12.5 amp circuit breakers (CB20, CB21, CB22)
- 4 - Relays
 - ☐ Parking Brake Failure Relay (K1)
 - ☐ Spare Relay (K2)
 - ☐ Horn Relay (K3)
 - ☐ Body-Up Relay (K4)

Relay Board 5 (RB5)

- 1 - Light Display Module card
- 1 - Lights Control Light (green): This light is illuminated when 24 volts is being supplied to the battery terminal of the light switch.
- 5 - 12.5 amp circuit breakers (CB23, CB24, CB25, CB26, CB27)
- 4 - Relays
 - ☐ Left Low Beam Relay (K1)
 - ☐ Right Low Beam Relay (K2)
 - ☐ Left High Beam Relay (K3)
 - ☐ Right High Beam Relay (K4)

Relay Board 6 (RB6)

The following relays are installed on RB6:

- ☐ Backup Horn Relay (K1)
- ☐ Engine Run/Ignition Relay (K2)
- ☐ Auto Lube Solenoid Relay (K3)
- ☐ Bleeddown Power Supply Relay (K4)
- ☐ Ether Start Relay (K5)
- ☐ Spare Relay (K6)
- ☐ HID Headlights Relay (K7)
- ☐ Auto Lube Timer Relay (K8)

Relay Board 7 (RB7)

The following relays are installed on RB7:

- ☐ Brake Lock Limitation Relay (K1)
- ☐ Brake Lock Limitation Relay (K2)
- ☐ Brake Lock Limitation Relay (K3)
- ☐ Brake Lock Limitation Relay (K4)
- ☐ IM Warning Relay (K5)
- ☐ Timed Engine Shutdown Light Relay (K6)
- ☐ Brake Lock Limitation Relay (K7)
- ☐ Timed Engine Shutdown Relay (K8)

Relay Board 8 (RB8)

The following relays are installed on RB8:

- ☐ 70% Load Signal to PSC Relay (K1)
- ☐ Full Load Signal to PSC Relay (K2)
- ☐ A/C Shutter Control Relay (K3)
- ☐ PLMIII Lights Relay (K4)
- ☐ Shutter Control Relay (K5)
- ☐ PLMIII Red Lights Relay (K6)
- ☐ PLMIII Amber Lights Relay (K7)
- ☐ PLMIII Green Lights Relay (K8)

Relay Board 9 (RB9)

The following relays are installed on RB9:

- ☐ Start Circuit Relay (K1)
- ☐ Engine Running/Selector Switch Relay (K2)
- ☐ Low Steering Pressure Power Relay (K3)
- ☐ Parking Brake OFF Relay (K4)
- ☐ Key Switch Start Activation Relay (K5)
- ☐ Low Steering Pressure Ground Relay (K6)
- ☐ Spare Relay (K7)
- ☐ Rear Backup Lights Relay (K8)

BODY-UP SWITCH

The body-up switch is located inside the right frame rail near the front of the body. It must be adjusted to specifications to ensure that the proper electrical signal is obtained when the body is raised or lowered. The body-up switch is designed to prevent propulsion in REVERSE when the dump body is not resting on the frame rails. The switch also prevents forward propulsion with the body up unless the override button is depressed and held.

Operation

When the body is resting on the frame, actuator arm (4, Figure 3-8) or magnet assembly (6) causes the electrical contacts in magnetically-operated proximity switch (3) to close. When the body is raised, the actuator arm or magnet assembly moves away from the switch, opening the contacts. The electrical signal is sent to the control system and the body-up relay.

The switch must be properly adjusted at all times. Improper adjustment or loose mounting bolts may cause false signals or damage to the switch assembly.

Adjustment

Before adjusting the body-up switch, inspect the body pads for wear and damage. Replace the body pads if required. The body must be resting on the frame in the normal body down position when adjustments are made.

1. Loosen capscrews (2) and adjust switch mounting bracket (1) to achieve an air gap (dimension "A") of 45 mm (1.77 in.) between proximity switch (3) and magnet assembly (4). Tighten the capscrews after the adjustment.
2. Vertical set up (dimension "B") should be 98.2 mm (3.86 in.). If necessary, bracket (5) can be repositioned and welded back onto the body to obtain this dimension.

Service

Keep the sensing area clean and free of metallic dust and other debris that may damage or inhibit operation of the switch. If the switch is damaged or not functioning, the switch must be replaced.

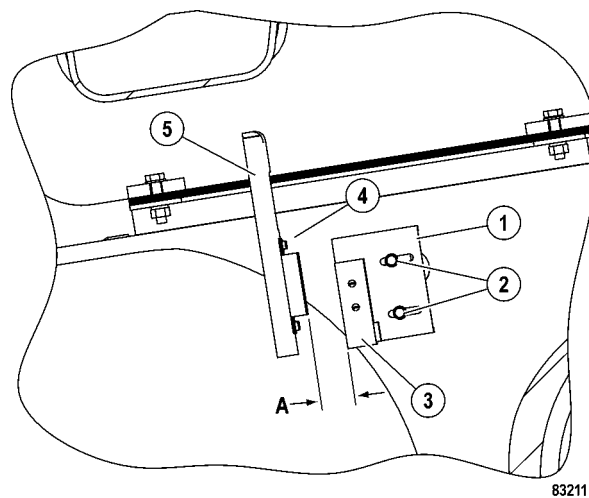


FIGURE 3-8. BODY-UP SWITCH

- | | |
|----------------------------|--------------------|
| 1. Switch Mounting Bracket | 4. Magnet Assembly |
| 2. Adjustment Capscrews | 5. Bracket |
| 3. Proximity Switch | |

HOIST LIMIT SWITCH

The hoist limit switch is located inside the right frame rail above the rear suspension, near the body pivot pin. The hoist limit switch is designed to stop the hoist cylinders before they reach full extension, preventing possible damage to the dump body or hoist cylinders.

Operation

When the hoist cylinders approach maximum stroke and the body pivots on the pins, actuator arm (3, Figure 3-9) or magnet assembly (6) moves close enough to magnetically-operated proximity switch (5) to close the electrical contacts. When the contacts close, an electrical signal is sent to the hoist limit solenoid valve, located in the hydraulic brake cabinet, to prevent further oil flow to the hoist cylinders.

The switch must be properly adjusted at all times. Improper adjustment or loose mounting bolts may cause false signals or damage to the switch assembly.

Adjustment

1. Raise the body so that the hoist cylinders are within 152 mm (6 in.) of maximum travel.
2. Adjust the hoist limit switch to achieve an air gap (dimension "A") of 45 mm (1.77 in.) between the sensing area and magnet assembly (6). Tighten the capscrews.

Service

Keep the sensing area clean and free of metallic dust and other debris that may damage or inhibit operation of the switch. If the switch is damaged or not functioning, the switch must be replaced.

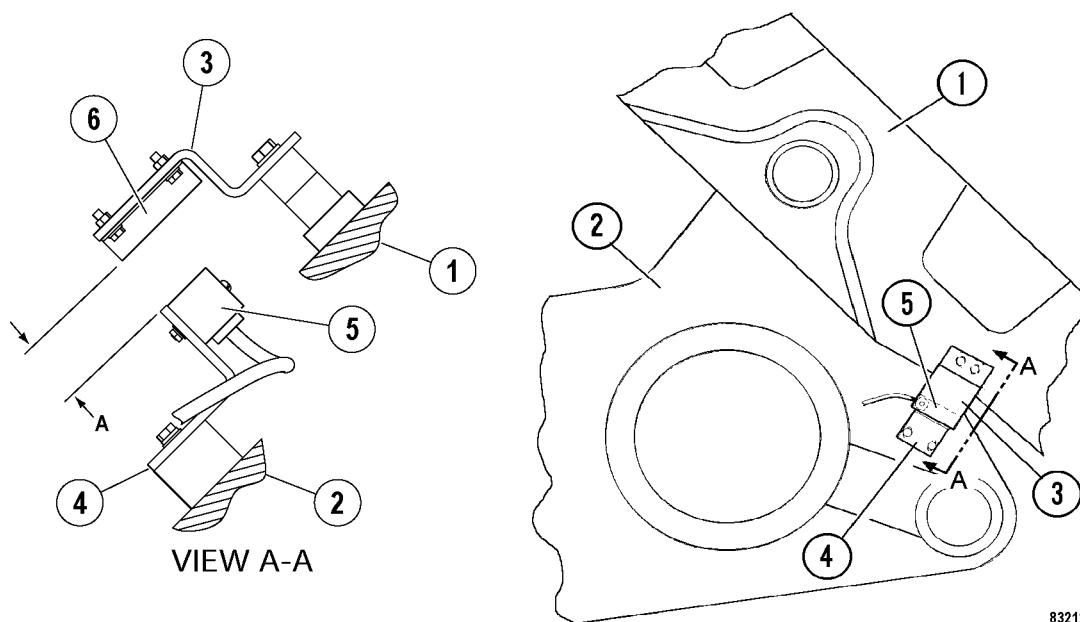


FIGURE 3-9. HOIST LIMIT SWITCH

- | | |
|-----------------|--------------------------|
| 1. Body | 4. Switch Mounting Plate |
| 2. Frame | 5. Hoist Limit Switch |
| 3. Actuator Arm | 6. Magnet Assembly |

FUSE BLOCKS AND CIRCUIT BREAKERS

FUSE BLOCK #1

Location	Amps	Devices protected	Circuit
1	15	A/C, Heater Blower Motor	12H
2	15	Windshield Washer / Wiper	63
3	5	Instrument Panel Gauges	712G
4	10	Key Switch Power	712P
5	10	Hoist Limit Switch	712H
6	15	Turn Signal / Clearance Lights	712T
7	10	Engine Options	712E
8	10	AID Module and Indicator Lights	12M
9	5	Engine Start Failure	712SF
10	10	Engine Shutters	712R
11	10	Auxiliary Control Cabinet Dome Lights	712A
13	20	Radio (Communication) - 12V	12VREG
14	10	Radio (Entertainment) - 12V	65
17	15	Ground Level Power	11GP
18	15	Payload Meter Lights	39J
19	5	Payload Meter Module	39G

FUSE BLOCK #2

Location	Amps	Devices protected	Circuit
1	15	Engine Service Lights	11SL
2	15	Cab Dome, Fog, Ladder Lights, Beacon, Hour Meter	11L
3	15	Hazard Lights	46
4	10	Interface Module	11INT
5	10	KOMTRAX Plus & Orbcomm Controllers Power	85
6	20	Modular Mining Supply	11M
7	15	Display Module	11DISP
8	10	Interface Module Power 2	11IM2
9	15	Reserve Oil System Pump	11ORS
10	15	Reserve Oil System Control Module	11RCNT
11	20	Hydraulic Bleed Down Power	11BD
12	20	Heated Mirrors	11HM
13	10	Key Switch Power	11KS
17	20	Engine Battery Power	11E1
18	20	Engine Battery Power	11E2
19	20	Engine Battery Power	11E3
20	20	Engine Battery Power	11E4
21	20	Ladder Lights	11DSL
22	20	Radio (Communication) - 24V	11ER
23	5	Fire Suppression	11FS

FUSE BLOCK #3

Location	Amps	Devices protected	Circuit
1	15	Cab Drive System Power	71P
2	10	Auto Lube Pump Power	68ES
3	15	Interface Module (GE Power)	71IM
4	20	Operator Seat Power	71OS
5	20	Rear Backup Lights	71RB
17	10	12V Auxiliary Power Outlets	67C
18	20	R.H. Cab Window Switch	67R
19	20	L.H. Cab Window Switch	67P

FUSE BLOCK #4

Location	Amps	Devices protected	Circuit
1	10	Brake Circuits	71BC
2	5	Payload Meter Controller	712PL
3	5	Interface Module	87
4	10	KOMTRAX Plus Controller	71VHM
5	5	Modular Mining Hub	712MM
6	5	Display Module	86
7	10	Hydraulic Bleed Down Power	71BD
8	10	Switch LED Power	71LS
9	1	Selector Switch Power	71SS
10	10	Headlight Switch	11D
17	5	Gauge Voltage	15V
18	15	Pedal Voltage	15PV

FUSE HOLDERS

Location	Amps	Devices protected	Circuit
FH #1	1	Left Rear Wheel Speed Sensor	15LRW
FH #2	1	Right Rear Wheel Speed Sensor	15RRW
FH #3	1	Left Front Wheel Speed Sensor	15LWS
FH #4	1	Right Front Wheel Speed Sensor	15RWS

The following two fuses are located in the fusible link between the prelube timer solenoid and #2 cranking motor.

Location	Amps	Devices protected	Circuit
Fusable link	150 each	Cranking Motors	11ST

The following circuit breakers are located on the relay boards on the left inside wall of the auxiliary control cabinet.

Location	Amps	Devices protected	Circuit
RB1 - CB13	12.5	Turn Signals / Clearance Lights	11CL
RB1 - CB14	12.5	Turn Signal Flasher	11Z
RB1 - CB15	12.5	Tail Lights	41T
RB3 - CB11	12.5	Backup Lights and Horn	79A
RB3 - CB16	12.5	Retard Lights	44D
RB3 - CB17	12.5	Manual Backup Lights	47B
RB3 - CB18	12.5	Stop Lights	44A
RB3 - CB19	12.5	Backup Lights and Horn	79A
RB4 - CB20	5	Parking Brake Failure Relay	439E
RB4 - CB21	12.5	Service Lights, Forward Horn	11A
RB4 - CB22	5	Engine Control Power	23D
RB5 - CB23	12.5	Headlights, Left Low Beam	11DL
RB5 - CB24	12.5	Headlights, Right Low Beam	11DR
RB5 - CB25	12.5	Headlights, Left High Beam	11HL
RB5 - CB26	12.5	Headlights, Right High Beam	11HR
RB5 - CB27	12.5	Headlights and Dash Lights	11D

The following circuit breaker is located on the left wall in the auxiliary control cabinet.

Location	Amps	Devices protected	Circuit
CB60	50	24V to 12V Converter	11BS

SECTION D11

KOMTRAX PLUS

INDEX

KOMTRAX PLUS BASIC FEATURES	D11-3
Gather Data	D11-3
Convert and Record Data	D11-3
Communicate Data to Off-Board Systems	D11-4
USING KOMTRAX PLUS	D11-5
Turning KOMTRAX PLUS ON	D11-5
Normal KOMTRAX PLUS Operation	D11-5
Turning KOMTRAX PLUS OFF	D11-6
Downloading from the KOMTRAX Plus Controller	D11-6
KOMTRAX PLUS DATA ITEMS	D11-7
Fault Codes	D11-7
Machine History	D11-7
KOMTRAX Plus History	D11-7
Snapshots	D11-8
Manual Snapshots	D11-8
Trends	D11-10
Histogram (Load Map) Data	D11-10
Haul Cycle Data	D11-11
Alarm and Snapshot Triggers	D11-12
Satellite Features	D11-12
KOMTRAX PLUS DIAGNOSTIC FEATURES	D11-14
Fault History	D11-14
LED Digits	D11-14
KOMTRAX PLUS CONTROLLER	D11-14
Removal	D11-15
Installation	D11-15
KOMTRAX PLUS SOFTWARE	D11-16
VHMS Tool Box Installation	D11-16
VHMS Setting Tool Installation	D11-16
KOMTRAX PLUS INITIALIZATION PROCEDURE	D11-17
Controller Setup Procedure	D11-17
Snapshot Procedure	D11-20
Download Procedure	D11-21
FTP Upload Procedure	D11-22
Initialization Forms	D11-23

WHEN REPLACING A KOMTRAX PLUS CONTROLLER	D11-24
KOMTRAX PLUS CONTROLLER CHECKOUT	D11-29
Necessary Equipment	D11-29
Preliminary	D11-29
KOMTRAX Plus Controller Checkout Procedure	D11-30
ORBCOMM CONTROLLER	D11-32
Removal	D11-32
Installation	D11-32

KOMTRAX PLUS

KOMTRAX PLUS BASIC FEATURES

The center of KOMTRAX Plus is the KOMTRAX Plus controller, which gathers data about the operation of the truck from sensors and other controllers installed on the truck. Refer to Figure 11-1 for an overview of the KOMTRAX Plus components.

For instructions on how to use KOMTRAX Plus software programs, refer to KOMTRAX Plus Software elsewhere in this section. For error codes, checkout and troubleshooting procedures, refer to KOMTRAX Plus Troubleshooting and Checkout Procedures elsewhere in this section.

Gather Data

The KOMTRAX Plus controller gathers data from three sources. Real-time and alarm data from each controller is gathered continually. In addition, haul cycle summary data from the PLM III is requested by the controller one time per day.

The system performs three primary functions:

1. Gathers data from on-board sources:
 - PLM III controller
 - Interface Module (IM)
 - Engine controllers
2. Converts data into usable formats and records it into permanent memory.
3. Communicates data to off-board systems:
 - Satellite (Orbcomm)
 - PC download

NOTE: The electric drive system does provide a limited number of faults to the interface module. Refer to KOMTRAX Plus Troubleshooting later in this section for a complete listing of fault codes that are generated by the drive system.

Convert and Record Data

KOMTRAX Plus controller (2, Figure 11-1) processes data received from external controllers and stores the following data in internal memory:

- Fault codes from the engine, Interface Module, and PLM III
- Snapshots of data when specific fault codes occur
- Trends of specific engine and chassis parameters
- Load map and other measures of engine and chassis usage
- Haul cycle summary information, including payload, distance traveled, and travel times

In addition to data gathered from external controllers, KOMTRAX Plus records information about the vehicle and KOMTRAX Plus usage, including:

- Key ON and engine ON record
- KOMTRAX Plus configuration changes.

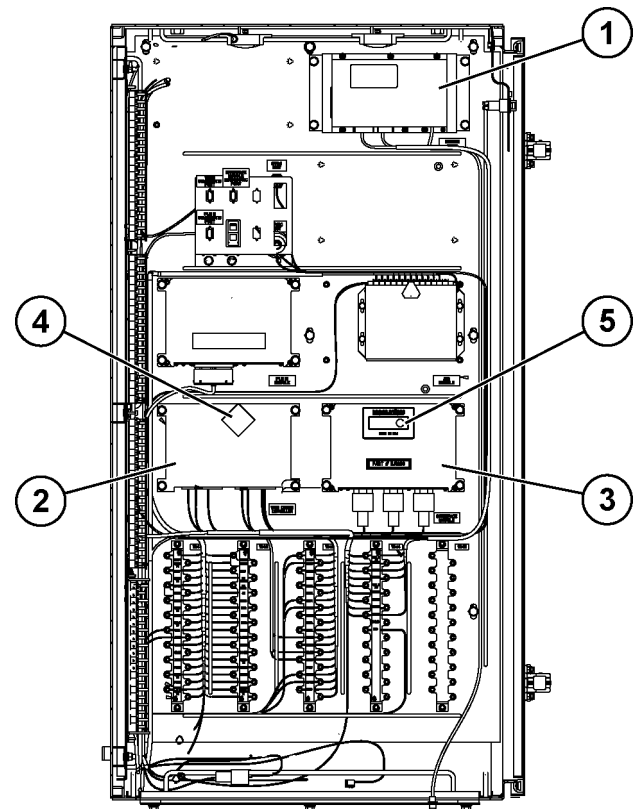
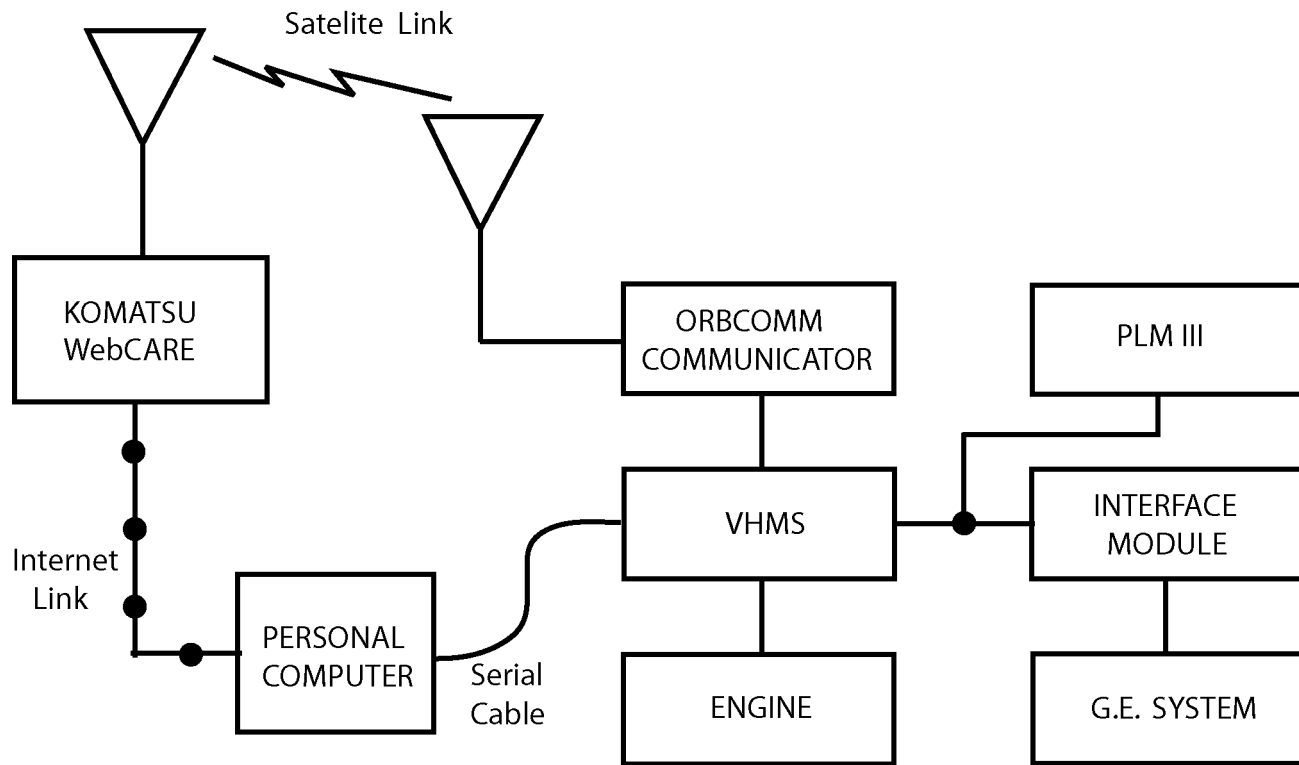


FIGURE 11-1. KOMTRAX PLUS COMPONENTS

- | | |
|----------------------------|---------------------|
| 1. Orbcomm Controller | 3. Interface Module |
| 2. KOMTRAX Plus Controller | 4. Red LED Digits |
| | 5. Green LED Light |



D110023

FIGURE 11-2. KOMTRAX PLUS COMPONENT FLOW CHART

Communicate Data to Off-Board Systems

KOMTRAX Plus has two methods to communicate data to off-board systems:

- Via satellite to the WebCARE database
- Download to a laptop PC running the *VHMS Technical Analysis Tool Box* software

Communication to the satellite (using Orbcomm) occurs automatically, but only sends critical data items. Orbcomm controller (1, Figure 11-1) is located inside the auxiliary control cabinet. Orbcomm antenna (1, Figure 11-3) is mounted on the front left corner of the cab by magnetic mount (2).

NOTE: The Orbcomm controller installed at the factory on all new trucks may not be approved for use in certain countries of the world. Local regulation may prohibit the use of the Orbcomm controller/satellite communicator. The controller and antenna may be disconnected and/or removed.

Communication to a laptop PC occurs whenever a user connects a laptop PC to the KOMTRAX Plus controller and requests a data download. All KOMTRAX Plus data is available for download to a laptop PC. Once downloaded to a laptop PC, the information is then sent to Komatsu via FTP. This data is then compiled at the Komatsu computer server. Based on this information, the local Komatsu distributor will suggest improvements and provide information aimed at reducing machine repair costs and downtime.

In order to collect all the necessary machine data, a preventive maintenance (PM) snapshot needs to be recorded every 500 hours of operation. The snapshot and other data is then downloaded into a laptop PC. This data is to be sent to Komatsu via the FTP program which is a part of the *VHMS Technical Analysis Tool Box* program. Refer to the check-out procedure for more detailed information regarding a PM snapshot.

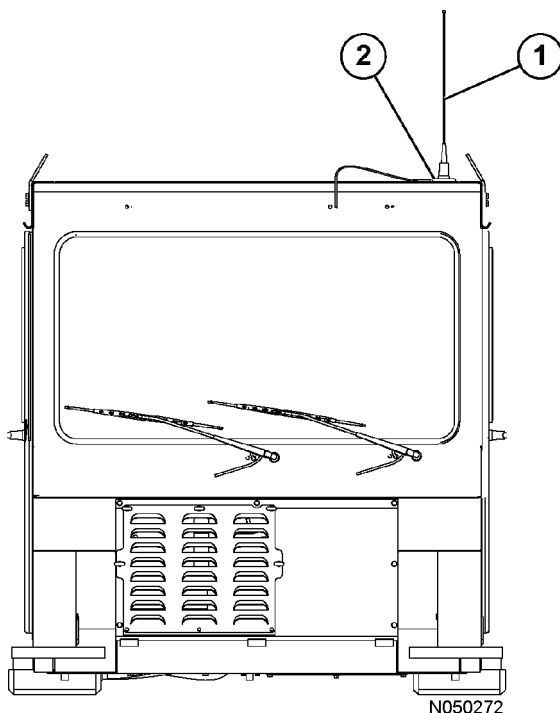


FIGURE 11-3. ORBCOMM ANTENNA

1. Orbcomm Antenna 2. Magnetic Mount

USING KOMTRAX PLUS

The primary tool for configuring, downloading, and viewing KOMTRAX Plus data is the *VHMS Technical Analysis Tool Box* software. Use of this software requires:

- A laptop PC running Windows 95/98/2000/ME/XP operating system.
- A serial cable to connect the laptop PC to the KOMTRAX Plus controller.

Refer to the *VHMS Technical Analysis Tool Box* instruction manual for additional information about using this software.

NOTE: It is recommended that the engine be OFF when downloading or configuring the KOMTRAX Plus controller.

Turning KOMTRAX Plus ON

The KOMTRAX Plus controller is turned on by the truck key switch (circuit 712). Immediately after receiving input from the key switch signal, the controller begins its power-up initialization sequence. This sequence takes about three seconds, during which time red LED digits (4, Figure 11-1) on the top of the controller will display a circular sequence of flashing LED segments. The controller will not support a connection from a laptop PC or a manual snapshot during this initialization time.

The KOMTRAX Plus controller is connected directly to the battery circuit, which provides a constant 24 volt signal from the truck batteries. However, the controller has the ability to turn itself off and will do so automatically within three minutes after the key switch is turned off.

CAUTION

The battery disconnect switch, located at the truck battery box, will remove 24 volt power from the KOMTRAX Plus controller and cause the controller to lose all data gathered since the key switch was last turned on. DO NOT disconnect the batteries until the controller has completed its shutdown operations and has turned off its LED digits.

Normal KOMTRAX Plus Operation

The red LED digits on the top of the KOMTRAX Plus controller indicate the current condition of the system. The possible conditions are shown in Table 1.

Table 1: KOMTRAX Plus Status	
LED DISPLAY	DESCRIPTION
Flashing LED segments in circular sequence	Power-on initialization
Numeric display, counting 00 - 99 at rate of 10 numbers per second	Normal Operation
Flashing Fault Codes	Normal operation, but a fault code is active

NOTE: Only a limited number of fault codes are displayed on the LED display. Most fault conditions are recorded internally in the KOMTRAX Plus controller, but are not indicated on the LED digits.

Turning KOMTRAX Plus OFF

The KOMTRAX Plus controller is connected directly to the truck batteries, but will remain in normal operation only if the truck key switch input (circuit 712) is on. When the controller senses that the truck key switch has been turned off, it finishes its internal processing and then saves recent data into permanent memory. This process can take up to three minutes.

If 24 volt power is removed from the KOMTRAX Plus controller before it has time to save data to permanent memory, data loss or corruption may occur.

The controller will turn off the red LED digits when it is off.



Do not remove 24 volt power from the KOMTRAX Plus controller unless the red LED digits on the controller are off!

Downloading from the KOMTRAX Plus Controller

Downloading data requires a laptop PC running Windows 95/98/2000/ME/XP operating system, the *VHMS Technical Analysis Tool Box* software, and a serial cable to connect the laptop PC to the KOMTRAX Plus controller. Refer to the *VHMS Technical Analysis Tool Box* instruction manual for additional information about using this software.

When a download to a laptop PC is performed, certain files are generated to store data. A listing of the file types and data is shown in Table 2.

KOMTRAX Plus diagnostic port (2, Figure 11-4), located on the DID panel at the rear of the operator cab, is used to download from the KOMTRAX Plus controller.

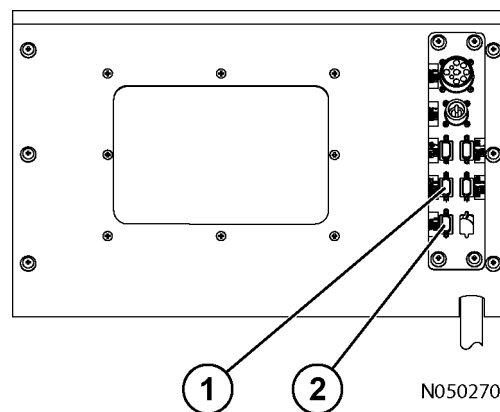


FIGURE 11-4. DIAGNOSTIC PORTS

1. IM Diagnostic Port
2. KOMTRAX Plus Diagnostic Port

Table 2: File Types of Download Data		
File Name	Data Type	Description
cyc_int0	Cycle Interval	Changes in engine speed
csvdata_3f.csv	Temporary Brake Load Map	
Fault0.csv	Fault History	Records all faults
index00.csv	Index	Lists all common data files
loadm1.csv	Temporary Load Map	
mcn_his0.csv	Machine History	Key On, Key Off
m_area0.csv	Running Area Map	Records engine operation distribution
m_drct0.csv	Running Direction Map	Records engine performance movement
snap00.csv	Snapshot	Records snapshot data over time period
vhmshis0.csv	KOMTRAX Plus History	Records changes to KOMTRAX Plus
*.k	Zipped File	Contains all data files

KOMTRAX PLUS DATA ITEMS

Fault Codes

The KOMTRAX Plus controller maintains a history of the most recent 600 fault codes. For each fault code, the controller records the following information:

- Fault Code Number
- SMR (service meter reading) when the fault occurred
- Time/Date when the fault occurred
- SMR (service meter reading) when the fault cleared
- Time/Date when the fault cleared

If a fault occurs more than once within 30 minutes, the KOMTRAX Plus controller will only maintain a single fault entry, but it will count the number of times the fault occurred and was cleared. This feature prevents an intermittent fault that occurs repeatedly from filling up the fault memory.

Serious fault conditions will be sent to WebCARE via the Orbcomm satellite network (if equipped) as well as being recorded in permanent memory. Some fault codes are configured to generate a snapshot when they occur. Refer to Table 6 for detailed information showing which fault codes will send data to WebCARE and which ones trigger a snapshot.

Machine History

The KOMTRAX Plus controller maintains a history of the most recent 400 Key ON and Engine ON conditions.

KOMTRAX Plus History

The KOMTRAX Plus controller maintains a history of the most recent 400 KOMTRAX Plus configuration changes. The controller will record a history entry each time one of the following configuration changes occurs:

- Changing the date or time of the KOMTRAX Plus controller
- Changing the Orbcomm satellite settings
- Performing a KOMTRAX Plus memory clear operation

Snapshots

A snapshot is a time history of real-time data that is recorded before and after the instant that a fault code occurs. The KOMTRAX Plus controller is continually recording real-time data for various engine data items. This allows the controller to record data for the time period before and after a fault code occurred.

Only certain fault codes generate snapshots. When a snapshot enabled fault code occurs, the KOMTRAX Plus controller will record data for 330 seconds (5.5 minutes) before the fault to 120 seconds (2 minutes) after the fault.

In order to conserve storage memory, the KOMTRAX Plus controller records snapshot data at two different sample rates. Each data item is recorded at a rate of one sample every 10 seconds up until 30 seconds prior to the fault occurrence. Each data item is then recorded at a rate of one sample per second from 30 seconds prior to 120 seconds after the fault occurrence.

If a snapshot enabled fault condition occurs more than one time, the KOMTRAX Plus controller will record the snapshot for the first (earliest) fault occurrence. The only exception is the manual snapshot button, in which case the controller will record the latest (most recent) snapshot.

Refer to Table 3 for all the items that are recorded in each snapshot.

Manual Snapshots

A manual snapshot is taken by pressing the data store button (1, Figure 11-5), located at the rear of the center console. When the 7.5 minute snapshot is being recorded by the KOMTRAX Plus controller, snapshot in progress light (2) will be illuminated. During the first five minutes, the LED will be on continuously. During the next two minutes, the LED will flash slowly. During the last 30 seconds, the LED will flash rapidly.

Manual snapshots are used to record current machine data that can then be downloaded and stored on a laptop PC. These snapshots can be used to observe current conditions on a machine. Over time, these snapshots can be compared and trends can be monitored. During the snapshot recording period, the machine should be driven over a variety of conditions so useful data can be collected.

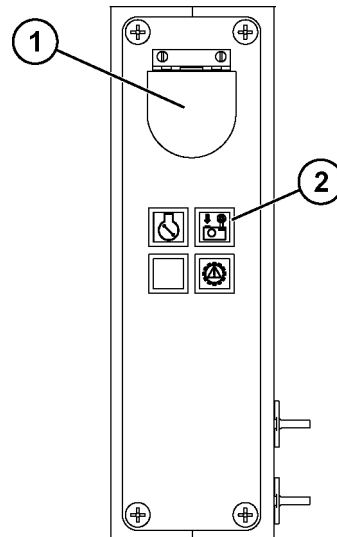


FIGURE 11-5. REAR OF CENTER CONSOLE

1. Data Store Button
2. Snapshot In Progress Light

Table 3: Snapshot Data	
Data Item	Data Source
Engine Coolant Temperature	Cummins QUANTUM Controller
Engine Oil Pressure	Cummins QUANTUM Controller
Accelerator Position%	Cummins QUANTUM Controller
Engine Speed	Cummins QUANTUM Controller
Exhaust Gas Temperature (Left Front)	Cummins CENSE Controller
Exhaust Gas Temperature (Left Rear)	Cummins CENSE Controller
Exhaust Gas Temperature (Right Front)	Cummins CENSE Controller
Exhaust Gas Temperature (Right Rear)	Cummins CENSE Controller
Engine Oil Temperature	Cummins CENSE Controller
Fuel Rate	Cummins QUANTUM Controller
Boost Pressure	Cummins QUANTUM Controller
Blow-by Pressure	Cummins QUANTUM Controller
Vehicle Speed	PLM III
Sprung Weight	PLM III
haul cycle State	PLM III
Brake Pressure	Interface Module
Hoist Pressure 1	Interface Module
Hoist Pressure 2	Interface Module
Steering Pressure	Interface Module
Front Left Brake Oil Temperature	Interface Module
Front Right Brake Oil Temperature	Interface Module
Rear Left Brake Oil Temperature	Interface Module
Rear Right Brake Oil Temperature	Interface Module
Ambient Temperature	Interface Module

Trends

The KOMTRAX Plus controller develops trends by monitoring real-time data and reducing the data into 20 hour statistical values. For each trended data item, the controller can determine the maximum value, minimum value, and average value during the preceding 20 hour period. Table 4 shows the type of statistical data recorded for each item.

NOTE: Trend data is only collected when the engine is running.

Histogram (Load Map) Data

The KOMTRAX Plus controller develops histograms by sampling data every 100ms while the engine is running. The data is presented as a two dimensional histogram showing time-at-level for various combinations of the two input data items.

The KOMTRAX Plus controller maintains an engine speed vs. fuel rate histogram called the Engine Load Map, and a brake pressure vs. speed histogram.

The Engine Load Map histogram shows time-at-level for specific engine speed and fuel rate combinations.

The Brake Pressure vs. Speed histogram shows time-at-level for specific brake pressure and vehicle speed combinations.

Two engine load maps are maintained in the KOMTRAX Plus controller. The Permanent Load Map contains load map data for the life of the engine. The Temporary Load Map contains load map data since the most recent memory clear action.

Although the engine data is sampled every 100ms internally, the histograms are only updated every two hours.

Table 4: Trend Data				
Data Item	Data Source	MAX	AVG	MIN
Engine Coolant Temperature	QUANTUM Controller	X		X
Engine Oil Pressure	QUANTUM Controller	X		X
Engine Speed	QUANTUM Controller	X		
Atmospheric Pressure	QUANTUM Controller		X	
Exhaust Gas Temperature	CENSE Controller	X		
Engine Oil Temperature	CENSE Controller	X		
Fuel Rate	QUANTUM Controller		X	
Boost Pressure	QUANTUM Controller	X		
Blow-by Pressure	QUANTUM Controller	X		
Brake Pressure	Interface Module	X	X	
Hoist Pressure 1	Interface Module	X		
Hoist Pressure 2	Interface Module	X		
Steering Pressure	Interface Module	X		
Front Left Brake Oil Temperature	Interface Module	X	X	
Front Right Brake Oil Temperature	Interface Module	X	X	
Rear Left Brake Oil Temperature	Interface Module	X	X	
Rear Right Brake Oil Temperature	Interface Module	X	X	
Ambient Temperature	Interface Module	X	X	X

Haul Cycle Data

The KOMTRAX Plus controller downloads haul cycle data from Payload Meter III one time every 24 hours, at a time specified by the *VHMS Setting Tool* software. The data consists of a summary report of all haul cycles completed in the past 24 hours. The summary data items are listed in Table 5.

After receiving the haul cycle summary data from Payload Meter III, the KOMTRAX Plus controller immediately attempts to send the data to WebCARE via the Orbcomm satellite (if equipped). The haul cycle summary data is also stored in controller internal memory. The controller maintains a record of the payload summary data from the past 100 daily transmissions to Orbcomm.

NOTE: The haul cycle summary statistics exclude haul cycles that the Payload Meter III controller has marked as 'not trusted'. The total number of haul cycles that occurred during the summary period, but were excluded from the summary, are indicated in the 'Total Excluded Cycles' field. See the Payload Meter III coverage in Section M, Options, for more information on excluded cycles.

Table 5: Haul Cycle Data	
Summary Data Item	Description
Summary Start Time	Start time of first haul cycle in summary
Summary End Time	Start time of last haul cycle in summary
Total Cycles	Total haul cycles included in this summary
Total Excluded Cycles	Total haul cycles occurring during summary period, but excluded from the statistics
Average Carried Load	Average Gross Payload
Standard Deviation of Carried Load	Standard Deviation of Gross Payload
Number of Loads Over Rated	Number of haul cycles with carried load > rated payload for this truck.
Number of Loads Over 110%	Number of haul cycles with carried load > 110% of rated payload for this truck.
Number of Loads over 120%	Number of haul cycles with carried load > 120% of rated payload for this truck.
Maximum Carried Load	Maximum carried load during this summary
Maximum Speed EMPTY	Maximum truck speed while truck was empty
Average Speed EMPTY	Average truck speed while truck was empty
Maximum Speed LOADED	Maximum truck speed while truck was loaded
Average Speed LOADED	Average truck speed while truck was loaded
Maximum Sprung Load	Maximum instantaneous sprung weight recorded during this summary
Average Maximum Sprung Load	Average of all 'Maximum Sprung Load' values recorded in each haul cycle
Maximum Frame Torque	Maximum instantaneous frame torque recorded during this summary
Average Maximum Frame Torque	Average of all 'Maximum Frame Torque' values recorded in each haul cycle.
Right Front Tire TKPH	Total tire ton kilometer per hour recorded for the right front tire.
Left Front Tire TKPH	Total tire ton kilometer per hour recorded for the left front tire
Rear Tires TKPH	Total tire ton kilometer per hour recorded for the rear tires
Relative Application Severity	Total frame damage recorded during this summary
Reserved_1	Future Use
Reserved_2	Future Use
Reserved_3	Future Use

Alarm and Snapshot Triggers

Serious fault conditions will be sent to WebCARE via the Orbcomm satellite network(if equipped), as well as being recorded in permanent memory. Some fault codes are configured to generate a snapshot when they occur.

Table 6 shows which fault codes trigger a snapshot and which fault codes will be sent to WebCARE via satellite.

Satellite Features

The KOMTRAX Plus controller sends data to WebCARE via the Orbcomm satellite network (if equipped) in the following conditions:

- A fault code occurs that has been configured for transmission via Orbcomm.
- A periodic event occurs, such as reception of daily PLM III summary data or a 20 hour trend.
- A remote request for data is received via the satellite network.

Table 6: Alarm and Snapshot Triggers

Fault Code	Fault Description	Source	Sent via Orbcomm	Snapshot Trigger
#A018	RR Flat Cylinder Warning	PLM III	X	
#A019	LR Flat Cylinder Warning	PLM III	X	
#A101	Pump Filter Switches	IM	X	
#A107	Propel System Caution	IM	X	
#A108	Propel System Temp Caution	IM	X	
#A109	Propel System Reduced Level	IM	X	
#A115	Low Steering Precharge	IM	X	
#A124	No Propel / Retard	IM	X	
#A125	No Propel	IM	X	
#A126	Hydraulic Tank Level	IM	X	
#A127	IM Sensor +5V Low	IM	X	
#A128	IM Sensor +5V High	IM	X	
#A190	Auto Lube Switch	IM	X	
#A194	High FL Brake Oil Temp	IM	X	X
#A195	High FR Brake Oil Temp	IM	X	X
#A196	High RL Brake Oil Temp	IM	X	X
#A197	High RR Brake Oil Temp	IM	X	X
MFA0	Manual Trigger	Manual	X	X
C00115	Speed Signal Lost	Engine	X	X
C00135	Oil Pressure Circuit Failed High	Engine	X	X

Table 6: Alarm and Snapshot Triggers (continued)				
Fault Code	Fault Description	Source	Sent via Orbcomm	Snapshot Trigger
C00143	Low Oil Pressure	Engine	X	X
C00151	High Coolant Temperature	Engine	X	X
C00155	High IMT LBF	Engine	X	X
C00158	High IMT LBR	Engine	X	X
C00162	High IMT RBF	Engine	X	X
C00165	High IMT RBR	Engine	X	X
C00214	High Oil Temperature	Engine	X	X
C00219	Remote Oil Level Low	Engine	X	X
C00233	Low Coolant Pressure	Engine	X	X
C00234	Engine Overspeed	Engine	X	X
C00235	Low Coolant Level	Engine	X	X
C00261	High Fuel Temperature	Engine	X	X
C00292	OEM Temp out of Range	Engine	X	X
C00293	OEM Temp Failed High	Engine	X	
C00294	OEM Temp Failed Low	Engine	X	
C00296	OEM Pressure Out of Range	Engine	X	X
C00297	OEM Pressure Failed High	Engine	X	
C00298	OEM Pressure Failed Low	Engine	X	
C00473	Remote Oil Level Signal Invalid	Engine	X	X
C00555	High Blow-by Pressure	Engine	X	X
C00639	Intake Air Leak LBR	Engine	X	X
C00641	High Exh Temp #1 LB	Engine		X
C00642	High Exh Temp #2 LB	Engine		X
C00643	High Exh Temp #3 LB	Engine		X
C00644	High Exh Temp #4 LB	Engine		X
C00645	High Exh Temp #5 LB	Engine		X
C00646	High Exh Temp #6 LB	Engine		X
C00647	High Exh Temp #7 LB	Engine		X
C00648	High Exh Temp #8 LB	Engine		X
C00651	High Exh Temp #1 RB	Engine		X
C00652	High Exh Temp #2 RB	Engine		X
C00653	High Exh Temp #3 RB	Engine		X
C00654	High Exh Temp #4 RB	Engine		X
C00655	High Exh Temp #5 RB	Engine		X
C00656	High Exh Temp #6 RB	Engine		X
C00657	High Exh Temp #7 RB	Engine		X
C00658	High Exh Temp #8 RB	Engine		X

KOMTRAX PLUS DIAGNOSTIC FEATURES

KOMTRAX Plus provides several basic data items that are useful for troubleshooting failures in the KOMTRAX Plus itself.

Fault History

The fault history recorded in the KOMTRAX Plus controller can help identify failures within the system and in the communications network to the engine controllers, interface module, or PLM III. For a complete listing of all the error codes, refer to the KOMTRAX Plus Troubleshooting and Checkout Procedures in this section.

LED Digits

The KOMTRAX Plus controller indicates some system errors or communication errors on two red LED digits (2, Figure 11-6) on the controller. Error codes are flashed as a two-part sequence. If no errors are occurring, the LED digits count from 00 - 99 continuously at a rate of 10 numbers per second. For a complete listing of all the error codes, refer to the KOMTRAX Plus Troubleshooting and Checkout Procedures in this section.

The KOMTRAX Plus controller also has two red LED lights (10 and 11, Figure 11-6).

Light (10) PLM III communication

- OFF - no communication with the PLM III controller
- ON - is communication with the PLM III controller

Light (11) Orbcomm

- OFF - no communication with Orbcomm controller
- ON - communication with Orbcomm controller
- FLASHING - satellite in view

KOMTRAX PLUS CONTROLLER

The KOMTRAX Plus controller collects and stores signals from sensors and data from other controllers. It also gives commands for transmitting the accumulated data through the communications system. The controller operates on 20VDC - 30VDC.

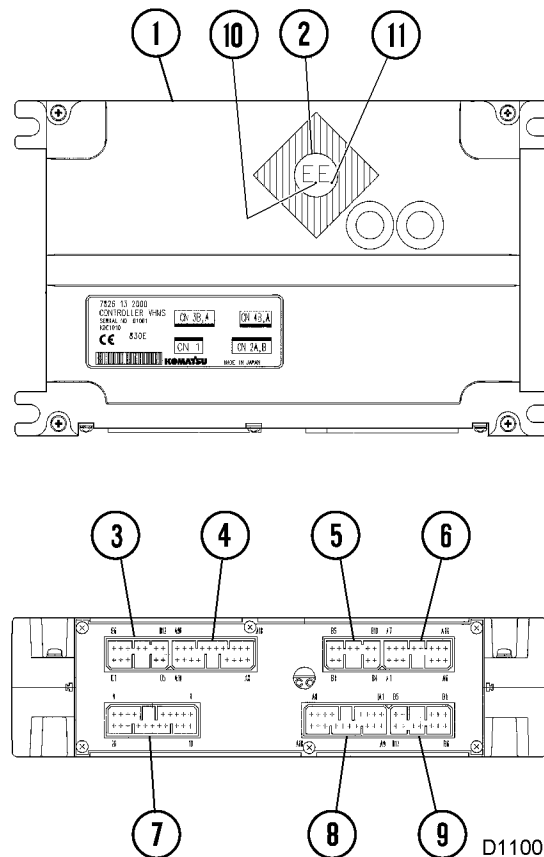


FIGURE 11-6. KOMTRAX PLUS CONTROLLER

- | | |
|----------------------------|-------------------|
| 1. KOMTRAX Plus Controller | 6. Connector CN4A |
| 2. LED Digit Display | 7. Connector CN1 |
| 3. Connector CN3B | 8. Connector CN2A |
| 4. Connector CN3A | 9. Connector CN2B |
| 5. Connector CN4B | 10. PLM III Light |
| | 11. Orbcomm Light |

Removal

If the KOMTRAX Plus controller has to be replaced, the following steps must be performed in order to maintain accurate information after the controller has been replaced. If the new controller is not set up correctly (like the one being removed), the data in the controller and at WebCARE may not be usable.

Some steps will require using a laptop PC and the *VHMS Setting Tool* software or the *VHMS Technical Analysis Tool Box* software. For more detailed instructions on performing these steps with a laptop PC and software, refer to "KOMTRAX Plus Software" later in this section.

During the controller replacement process, two data downloads will have to be taken (one before, one after) and sent to WebCARE. Also, a KOMTRAX Plus initialization form will have to be filled out and sent to Komatsu North America as shown on the form.

1. With the key switch OFF, connect a laptop PC to the controller using the serial cable.
2. Using a laptop PC and the *VHMS Technical Analysis Tool Box* software, perform a complete data download from the controller. Refer to "When Replacing a KOMTRAX Plus Controller" later in this section.
3. Save this data so it can be sent to WebCARE at a later time when a connection to the internet is available.
4. Using the *VHMS Setting Tool* software, enter the Service ID and choose the "Save/Load" function.
5. From the File menu, select "Save".
6. Capture a screen shot ("Alt" and "Print Screen" keys at the same time) of the Save Confirmation window, paste it into a Microsoft Word document and save it.
7. Click the "OK" button to save the settings.
8. Exit the *VHMS Setting Tool* program.
9. Turn the key switch OFF.
10. Wait three minutes, then disconnect battery power.
11. After the two LED lights are off, disconnect the wiring harnesses and remove the controller.

Installation

1. Install the new controller and connect the wiring harnesses to it. Connect the laptop PC to the controller with the serial cable.
2. Connect battery power. Turn the key switch ON, but do not start the engine.
3. With the *VHMS Setting Tool* software, enter the Service ID and choose the "Save/Load" function.
4. From the file menu, select "Load".
5. Capture a screen shot ("Alt" and "Print Screen" keys at the same time) of the Save Confirmation window, paste it into a Microsoft Word document and save it.
6. Click the [OK] button to load the settings.
7. Click the [Apply] button to reset the controller, then click the [OK] and [Yes] buttons to confirm. Then select the [Close] button.
8. Fill out a KOMTRAX Plus initialization form and send it to Komatsu as instructed on the form.
9. Exit the *VHMS Setting Tool* program.
10. Turn the key switch OFF and wait three minutes.
11. Turn the key switch ON. Wait three minutes and watch for any error messages on the controller LED lights that might indicate a problem in the system.
12. If there are no error messages, continue to Step 13. If there are error messages, refer to the KOMTRAX Plus Troubleshooting and Checkout procedures elsewhere in this section.
13. Using a laptop PC and the *VHMS Technical Analysis Tool Box* software, perform a complete data download from the controller.
14. Confirm the download data is good by using the *VHMS Technical Analysis Tool Box* software. Make sure the settings are correctly applied by looking at the date, time, SMR, etc.
15. Turn the key switch OFF. Disconnect the laptop PC from the controller.
16. Use internet access available to the laptop PC to send the download data set that was taken before the controller was removed from the truck to WebCARE. Use the FTP feature built into the *VHMS Technical Analysis Tool Box* program to send the files.
17. Use the FTP program to send the download data set that was taken after the new controller was installed to WebCARE.

KOMTRAX PLUS SOFTWARE

To work with KOMTRAX Plus, several special tools and software programs are required. Refer to Tables 1 and 2 for detailed information on KOMTRAX Plus software and tools.

NOTE: Be aware that the software and data files are updated periodically. Check with the local Komatsu distributor for the latest software versions.

Table 1: KOMTRAX Plus Software

Name	Description	Source
VHMS Technical Analysis Tool Box	To maintain KOMTRAX Plus	Komatsu Distributor
VHMS Setting Tool	To initialize KOMTRAX Plus	Komatsu Distributor
PDM	Payload Meter III Data Manager	Komatsu Distributor
PLM III *	PLM III Controller Software to work with KOMTRAX Plus	Komatsu Distributor

*NOTE: PLMIII ships with original software version **EJ0575-5**. Update to the current version.*

Table 2: KOMTRAX Plus Tools

Name	Description	Source
Laptop PC	200 MHz or higher 64 MB RAM or more Serial or USB Port CD/DVD-Rom drive Windows operating system	Purchase Locally
Serial cable (RS232 port)	Male DB9 connector at one end Female DB9 connector at other end	Purchase Locally
Adaptor	USB port to RS232 serial port (If the laptop PC does not have an RS232 port, this adaptor is required.)	Purchase Locally

VHMS Tool Box Installation

1. Insert the CD. The *VHMS Technical Analysis Tool Box* software will begin installing automatically.
2. Accept the recommended defaults and finish installing *VHMS Technical Analysis Tool Box*.
3. Double-click on the new icon on the desktop, *VHMS Technical Analysis Tool Box*.
4. Initialize the software by inserting the Set Up Disk.
5. Enter the User Name. The User Name is **user**.
6. Enter the Password. The Password you entered the first time will be your Password from then forward, unless you change it.
7. *VHMS Technical Analysis Tool Box* is installed.

VHMS Setting Tool Installation

1. Insert the CD. If the *VHMS Technical Analysis Tool Box* software begins installing automatically, select the [Cancel] button to stop the installation process.
2. Open My Computer.
3. Right-click on the CD drive and select Open.
4. Open the Setting Tool folder.
5. Double-click on the Setup.exe file.
6. Accept the recommended defaults and finish installing *VHMS Setting Tool*.

PDM Software Installation

Insert the CD. The *Payload Data Manager (PDM)* software will begin installing automatically.

If this does not happen, the software can be installed by running the Setup.exe program on the CD ROM.

KOMTRAX PLUS INITIALIZATION PROCEDURE

When a new truck is being assembled, there are several procedures to perform in order to initialize KOMTRAX Plus. Following the procedures will ensure a smooth initialization process which should not take longer than an hour to complete. To ensure the initialization process has been completed properly, check off each item on the list below as it is done. It is important to complete the entire procedure at one time. Submitting a data download with a date and SMR that does not match the KOMTRAX Plus initialization form will not allow the system to be initialized.

NOTE: The interface module must be fully operational before initializing the KOMTRAX Plus controller.

The initialization procedure consists of the following:

- ☐ Controller Setup Procedure
- ☐ Snapshot Procedure
- ☐ Download Procedure
- ☐ FTP Upload Procedure
- ☐ Initialization Forms

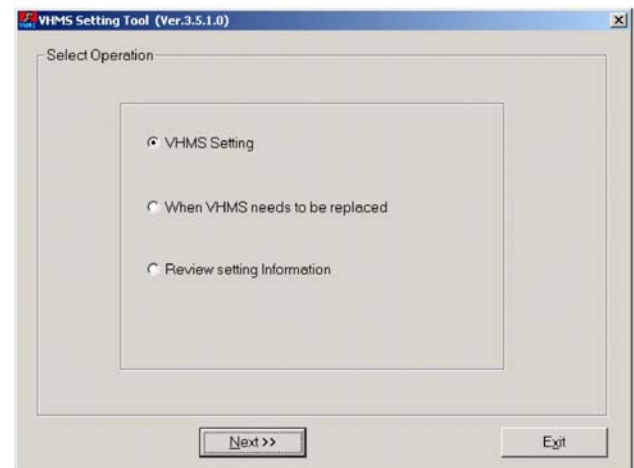
CONTROLLER SETUP PROCEDURE

VHMS Setting Tool software program

1. Start the *VHMS Setting Tool* software program. There will be three choices to choose from.
 - Use the [VHMS Setting] function to initialize a truck or change a truck's settings.
 - Use the [When VHMS needs to be replaced] function when replacing a truck's KOMTRAX Plus controller.
 - Use the [Review setting information] function when only needing to view a truck's settings.

Select Operation

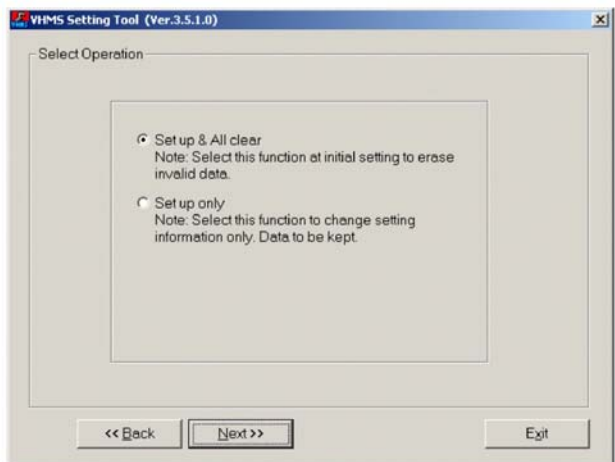
2. Select VHMS Setting, then click [Next].



D120031

KOMTRAX Plus Setting Function

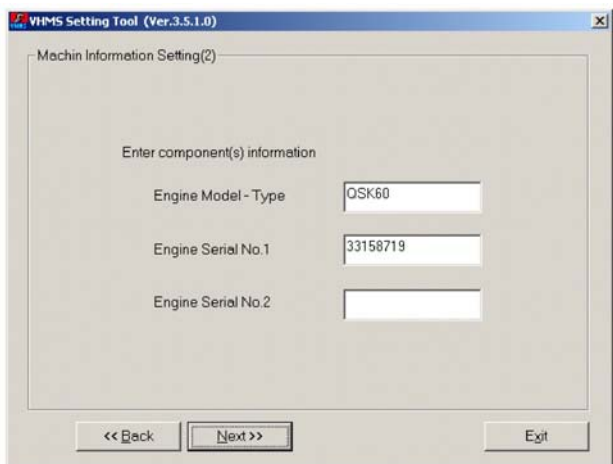
3. Select Set up & All clear if initializing a truck, then click [Next].



D120032

Machine Information Setting(2)

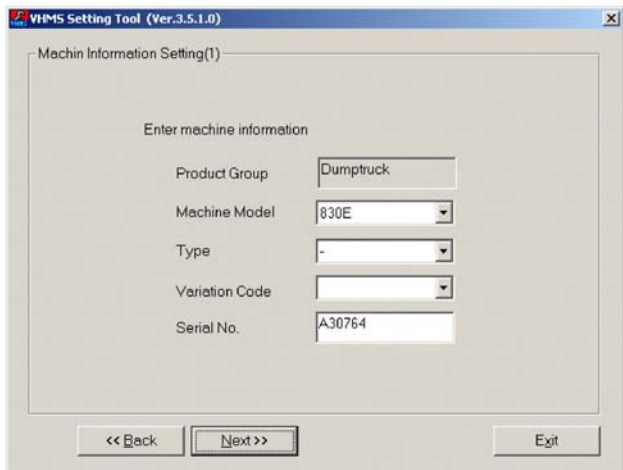
5. Verify that the Machine Information Settings are correct. If not, enter the correct settings. Then click [Next].



D120039

Machine Information Setting(1)

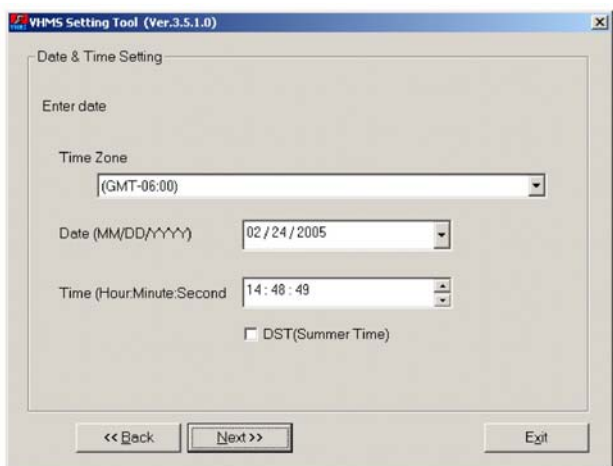
4. Verify that the Machine Information Settings are correct. If not, enter the correct settings. Then click [Next].



D120038

Date & Time Setting

6. Enter the correct Time Zone, Date and Time. Check [DST (Summer Time)] if the truck's location uses Daylight Savings Time. Then click [Next].



D120034

GCC Setting

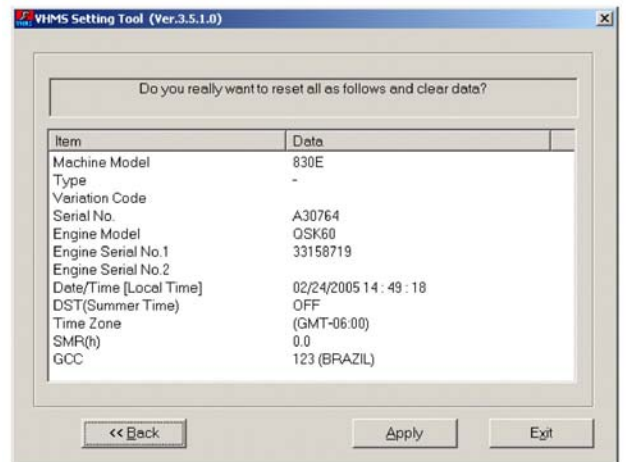
7. Choose the correct GCC code. The GCC code tells trucks that are equipped with Orbcomm which satellite ground station to use. Then click [Next].



D120040

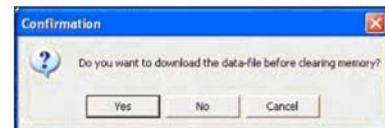
Setting Summary

8. Verify that all the setting information is correct and click [Apply].



D120041

9. Click [YES].



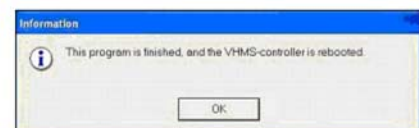
D120042

10. Click [OK].



D120043

11. Click [OK]. The VHMS Setting Tool program will close.



D120044

SNAPSHOT PROCEDURE

A snapshot through KOMTRAX Plus records important data about different systems on the truck. Take snapshots on a periodic schedule and store them as part of the truck history. These snapshots can then be compared and trends can be analyzed to predict future repairs.

A single snapshot records truck data for 7.5 minutes.

NOTE: On AC drive trucks, a laptop PC should also be connected to the GE drive system to allow for maximum horsepower check of the engine during the snapshot recording process.

1. Allow the truck to run until it is at normal operating temperatures.
2. Press and hold the GE data store switch for three seconds, then release it. The white data store in progress LED should illuminate.
3. While the manual snapshot is being taken, operate the truck.
 - a. Operate the engine at high and low idle.
 - b. Raise the dump body to the full dump position.
 - c. Lower the dump body to the frame, then hold it in the power down position momentarily.
 - d. Turn the steering wheel to full left, then full right against the stops momentarily.
 - e. Travel forward to maximum speed and apply the brakes hard.
 - f. Travel in reverse.
 - g. Perform a horsepower check using a laptop PC connected to the GE drive system.
4. The white LED will begin flashing slowly after five minutes has elapsed, then flash rapidly during the last 30 seconds.
5. Wait until the LED has finished flashing. After one more minute, turn the key switch OFF to stop the engine. Verify that the KOMTRAX Plus controller red LED display is off.
6. Use *VHMS Technical Analysis Tool Box* program to download the snapshot data into a laptop PC. Use the FTP feature to send the download data to WebCARE.

DOWNLOAD PROCEDURE

NOTE: Always verify a full download has been taken before disconnecting the laptop PC from the truck.

1. Turn the key switch to the OFF position to stop the engine.
2. Turn the key switch to the ON position, but DO NOT start the engine.
3. Allow the KOMTRAX Plus controller to start up. This should take about one minute. Verify the red LED display starts counting up.
4. Attach the KOMTRAX Plus serial cable to the truck's KOMTRAX Plus port and the other end to the laptop PC's serial port.
5. Double-click on the *VHMS Technical Analysis Tool Box* icon on the laptop PC's desktop.
6. Enter the appropriate User Name and Password and click the [OK] button.
7. Double-click on the [Download] icon.
8. Select the COM port in the Port No. drop-down box and click the [Connection] button.
9. Verify that the date and time is correct for current local date and time. Also verify that the displayed service meter hours are equal to the value entered previously.
10. If this is the first time this laptop PC has been connected to the truck, you will need to download its definition file by clicking the [OK] button.
11. Verify that a manual snapshot (MFAO) has been recorded. The display should show an item named "Snapshot" with the code MFAO and text "Manual Trigger".
12. On the Download screen, click the [Select All] button. All items will become checked.
13. Click the [Download] button. The download may take one to ten minutes. Generally, if there are several snapshots in the download items, the download will take longer.
14. Click the [OK] button to complete the download.
15. Verify that the "Download Completed" message is displayed. Click on [Exit].

16. Select the [Machine History] option from the list on the left side of the screen.
17. Verify that the key ON/OFF and engine ON/OFF records are recorded correctly.
18. Exit any open windows on the laptop PC.
19. Verify that a full download has been taken. Refer to Location of Downloaded Files on Computer for more detailed instructions.
20. Disconnect the KOMTRAX Plus serial cable from the laptop PC and from the truck.
21. Turn the key switch to OFF.

Location of Downloaded Files

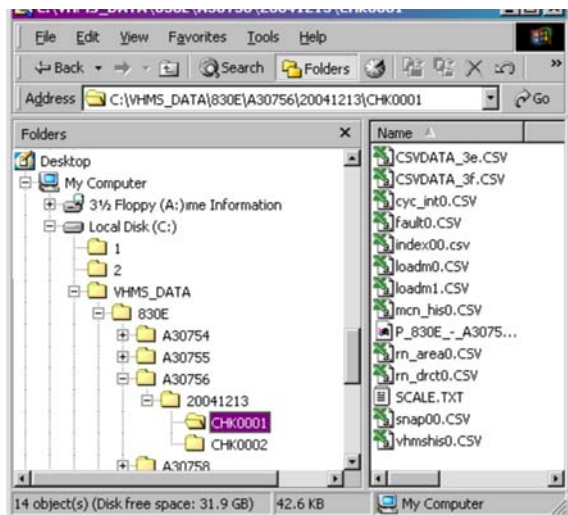
When a download using *VHMS Technical Analysis Tool Box* is performed, several files are downloaded onto the computer. They are organized in a specific way so that they can be used by *VHMS Technical Analysis Tool Box* at a later time. This structure is created automatically when the computer is used to perform the download from the KOMTRAX Plus controller. The situation may arise where the files need to be sent to someone, or someone gives these files to you.

1. Open Windows Explorer by right-clicking on the Start button and choosing Explore.
2. In the left frame, the computer's file structure will be displayed. The right frame will show the details for the folder that is highlighted in the left frame.
3. In the left frame, navigate to the download files.

The basic path is as follows:

- Desktop
- My Computer
- Local Disk (C:)
- VHMS_Data
- Model
- Serial Number
- Date
- Check Number

NOTE: The Date folder is named in the format YYYYMMDD.



D120064

The screenshot shows the location the KOMTRAX Plus download files on a computer. The Check Number folder is named in the format CHK000#. Each time a download is taken, it is placed in one of these folders. The first download will be in the CHK0001 folder. If a second download is taken on the same day, will be in the CHK0002 folder, etc.

Once the appropriate folder is selected, the contents will be shown in the right frame. These files can then be e-mailed or copied to a disk.

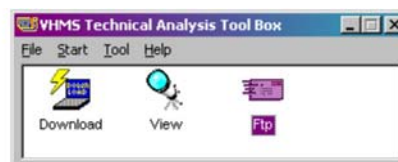
If someone provides KOMTRAX Plus download files through e-mail or on a disk, the same folder organization must be created in order to view them in *VHMS Technical Analysis Tool Box*.

FTP UPLOAD PROCEDURE

After downloading, the KOMTRAX Plus data resides on the laptop PC that performed the download. At this point, it can be reviewed and analyzed using *VHMS Technical Analysis Tool Box* on this laptop PC only. In order to make this data available to others, it must be sent to an online database named WebCARE. Once the data has been uploaded via FTP to WebCARE, it is accessible to anyone with an internet connection and an ID and password.

VHMS Technical Analysis Tool Box is used to perform the FTP upload. Perform an FTP upload as soon as the person who performed the download can obtain an internet connection. All downloads should be uploaded to WebCARE.

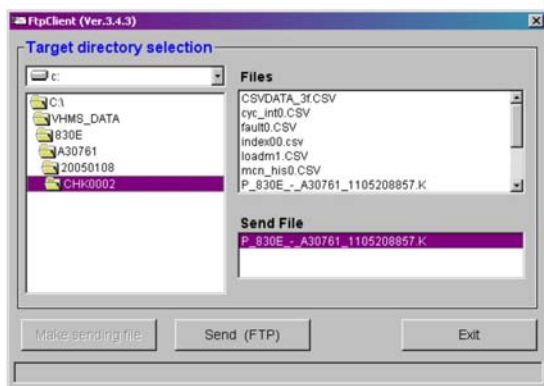
1. Double-click on the VHMS Technical Analysis Tool Box icon on the laptop PC's desktop.
2. Enter the appropriate user name and password and click the [OK] button.
3. Double-click the [FTP] icon.



D120046

4. At the FTP Client Login window, enter the FTP User ID and Password.
User ID = komatsu
Password = vhms

5. The target directory should be set to the laptop PC's hard drive (usually drive C:\).
 - a. Double-click the VHMS_Data folder to drop down the model folders.
 - b. Double-click the appropriate model folder to drop down the serial number folders.
 - c. Double-click the appropriate serial number folder to drop down the date folders.
 - d. Double-click the appropriate date folder to drop down the check number folders.
 - e. Double-click the appropriate check number folder to display its contents in the files window.

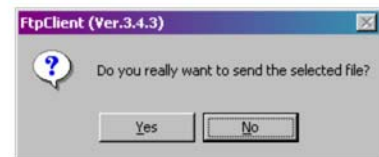


D120047

6. Some models will automatically create a sending file during the download process. Others need to have the sending file created at this time. A sending file is just a compressed version of all the other downloaded files. If there is already a sending file in the Send File window, you do not need to perform this step. If there is not a sending file in the Send File window, click the [Make Sending File] button.

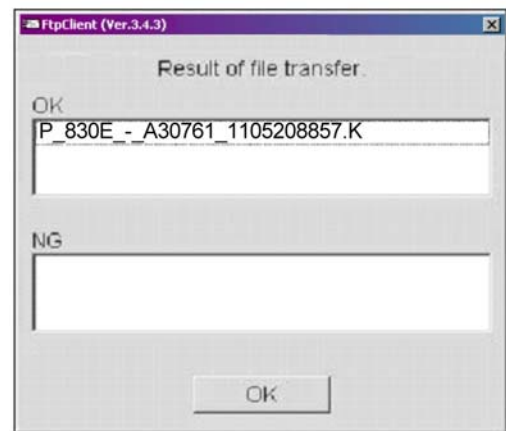
NOTE: The compressed sending file will look similar to this file name, and will always end with a ".K".
P_930E_-_A30462_1105208857.K

7. After selecting the correct file to send, click the [Send (FTP)] button.
8. Click the [Yes] button to verify that you want to upload the data to WebCARE.



D120048

9. If the sending file was uploaded successfully, the file will appear in the OK window. If the sending file was not uploaded successfully, the file will appear in the NG (No Good) window. Make sure that the laptop PC has an internet connection.



D120049

10. Click the [OK] button, then the [Exit] button. Close all other open windows.

INITIALIZATION FORMS

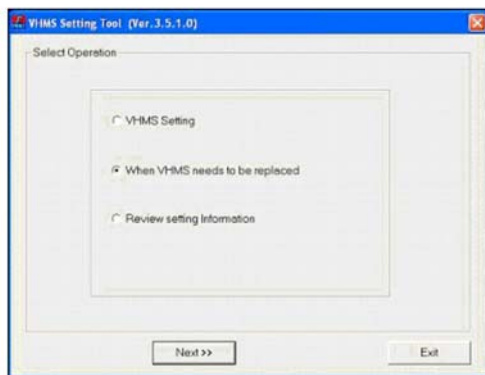
Complete the initialization check list and initialization forms found later in this section. Send the initialization form to Komatsu.

Initialization is now complete.

WHEN REPLACING A KOMTRAX PLUS CONTROLLER

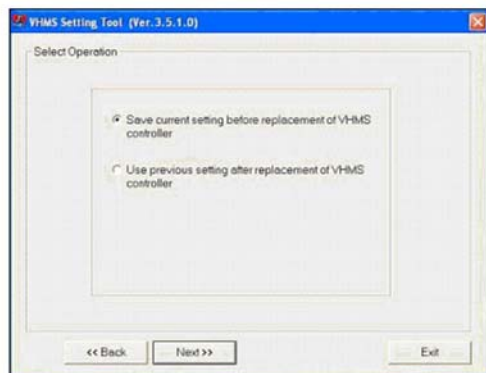
Refer to the KOMTRAX Plus controller removal and installation instructions earlier in this section to replace a KOMTRAX Plus controller. Follow the steps below when using the *VHMS Setting Tool* software to save the data and settings so they can be transferred from the old controller to the new controller.

1. Select the [When VHMS Needs To Be Replaced] function.



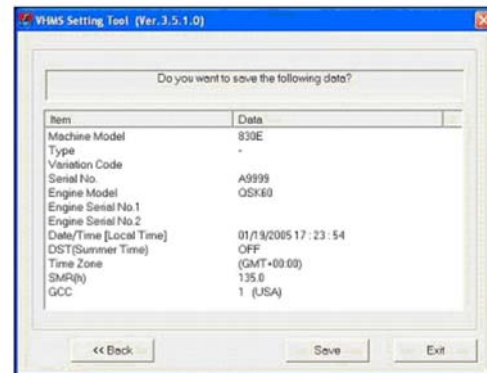
D120050

2. Select the [Save current setting before replacement of VHMS controller] function.



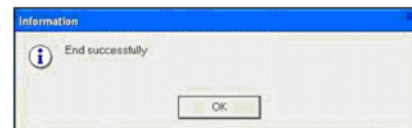
D120051

3. Click the [Save] button.



D120052

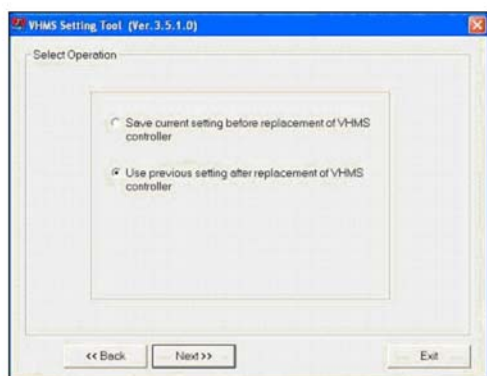
4. Click the [OK] button.



D120053

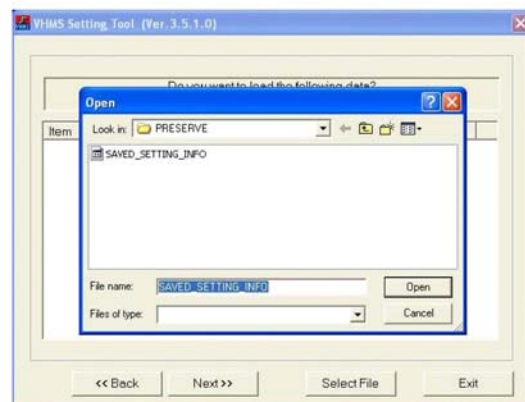
5. Replace the KOMTRAX Plus controller.

6. Select the [Use previous setting after replacement of VHMS controller] function.



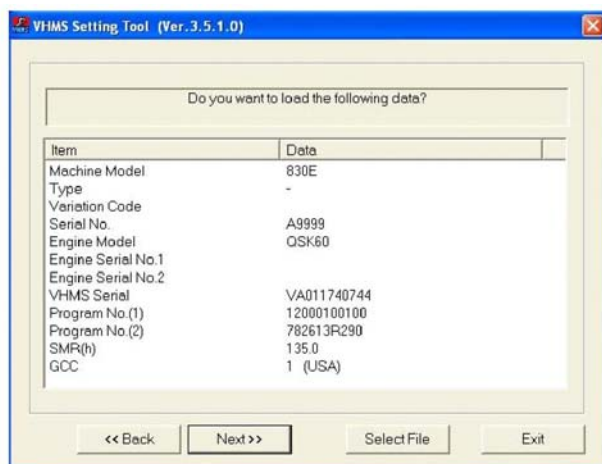
D120054

8. If the correct data is not showing, click the [Select File] button and choose the correct data. Then click the [Next] button.



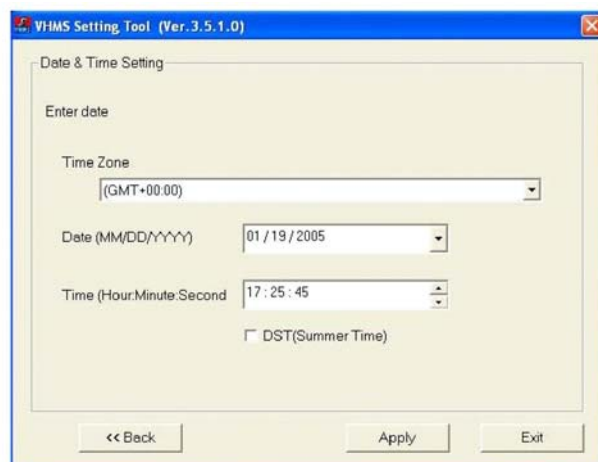
D120056

7. Verify that the data showing is the data to be loaded and then click the [Next] button.



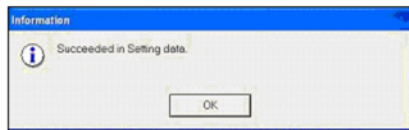
D120055

9. Enter the correct Time Zone, Date and Time information. Check [DST (Summer Time)] if the truck's location uses Daylight Savings Time. Click the [Apply] button.



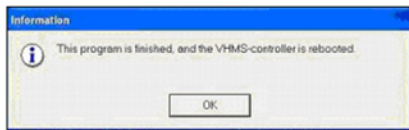
D120057

10. Click the [OK] button.



D120058

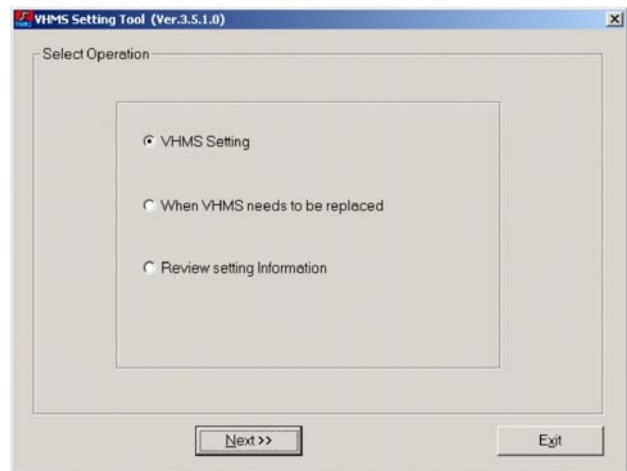
11. Click the [OK] button. The *Setting Tool Program* will close.



D120059

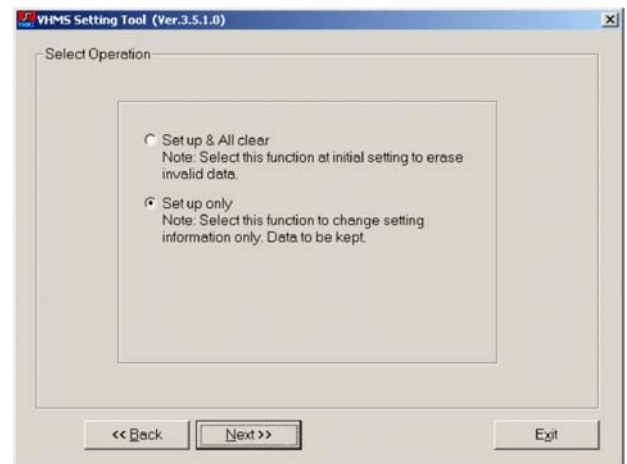
Setting Date & Time, Satellite & Payload Meter

1. Select the [VHMS Setting] function, then click the [Next] button.



D120031

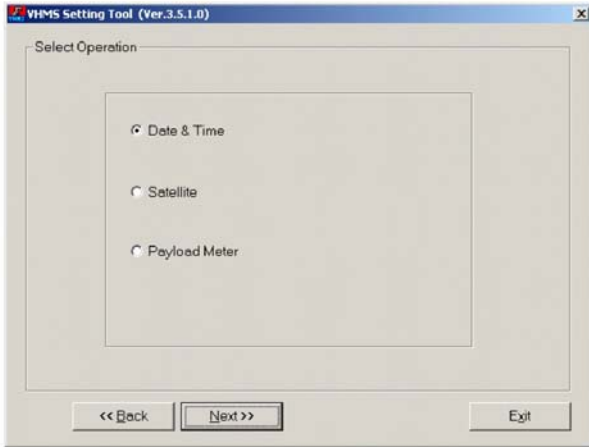
2. Select the [Set up only] function, then click the [Next] button.



D120033

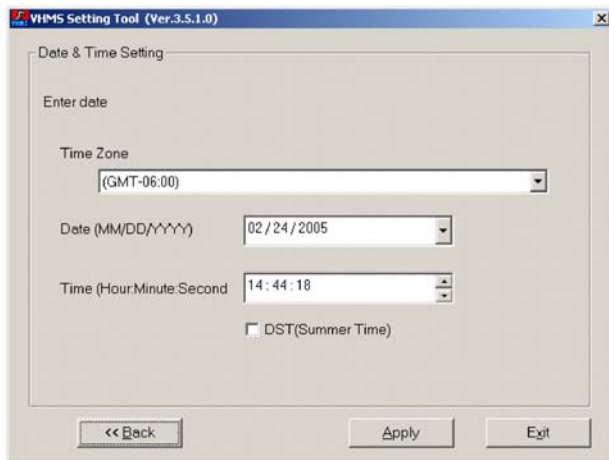
3. After selecting one of the following choices, click the [Next] button.

- Date & Time
- Satellite
- Payload Meter



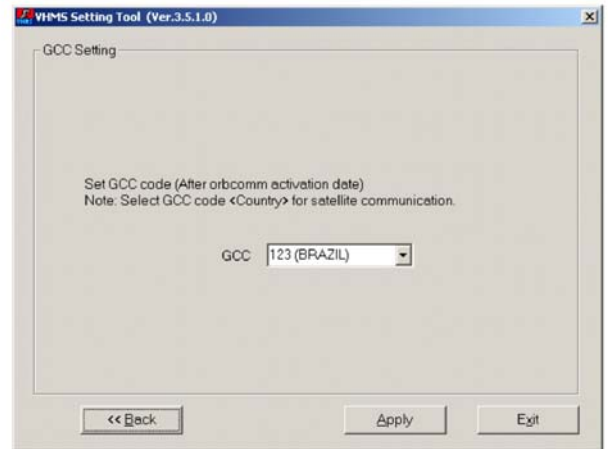
D120036

a. Date & Time: Should be set to current date and time. If not correct, set the correct Time Zone, Date and Time to current time zone, date and time. Select [DST Summer Time]] if it applies. Click the [Apply] button.



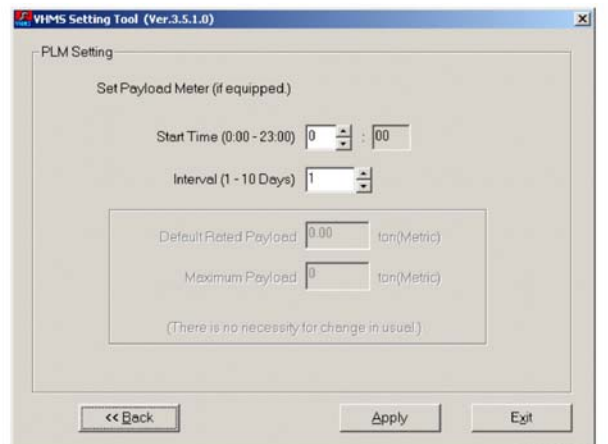
D120037

b. Satellite: Select the correct country location from the drop-down menu, then click the [Apply] button to change the setting.



D120035

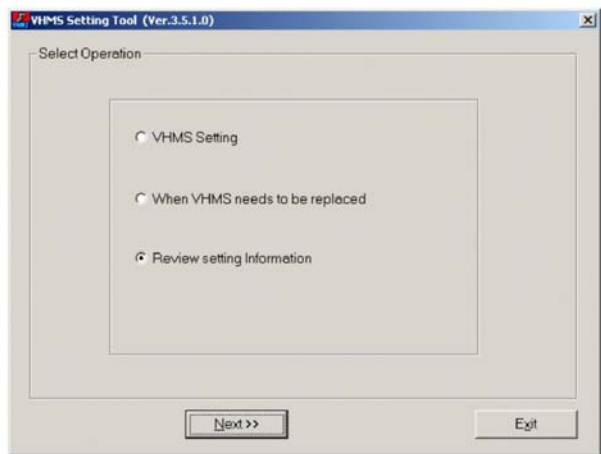
c. Payload Meter: Set Start Time to "0", and Interval to 1. Then click the [Apply] button to save the setting.



D120062

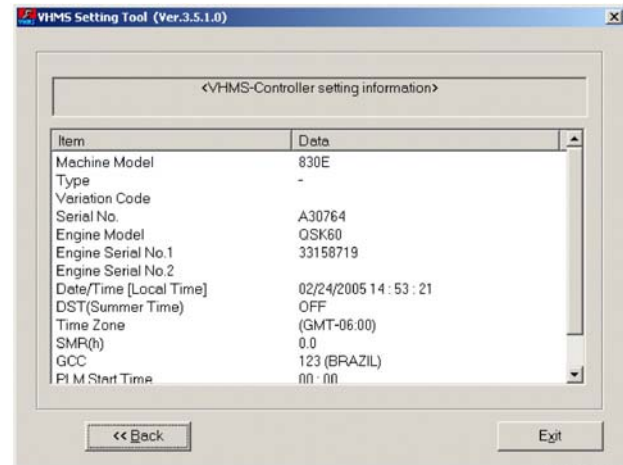
Review Setting Information

1. Select the [Review setting information] function and then click the [Next] button.



D120063

2. Review the settings for accuracy. If something is not correct, click the [Back] button, select the appropriate category and reset the information to the correct settings. If everything is correct, click the [Exit] button.



D120060

3. Click the [Yes] button to close the *Setting Tool Program*.



D120061

KOMTRAX PLUS CONTROLLER CHECKOUT

Necessary Equipment

- Checkout procedure
- System schematic
- Laptop personal computer (PC)
- *VHMS Technical Analysis Tool Box* software
- *VHMS Setting Tool* software
- Tera Term Pro software
- Serial cable (RS232)
(male DB9 connector on one end, female connector on the other end)

NOTE: Before performing this procedure, the interface module must be fully functional and the KOMTRAX Plus controller must be initialized and fully functional.

Preliminary

1. Turn the key switch to the OFF position to stop the engine. Verify that the seven segment LED display on the KOMTRAX Plus controller is off.
2. Turn the key switch to the ON position, but DO NOT start the engine.
3. Allow the KOMTRAX Plus controller to boot up. Watch the red, two digit LED display on the controller to show a circular sequence of seven flashing segments on each digit. After a short time the two digit display should start counting up from 00 - 99 at a rate of ten numbers per second.
4. Attach the KOMTRAX Plus serial cable to the truck's KOMTRAX Plus diagnostic port (2, Figure 13-3) and the other end to the laptop PC's serial port.
5. Double-click on the *VHMS Technical Analysis Tool Box* icon on the laptop PC's desktop.
6. Enter the appropriate User Name and Password and click the [OK] button.
7. Check for any active fault codes. If any are found, these circuits should be analyzed to determine the cause of the fault and they must be repaired before continuing.

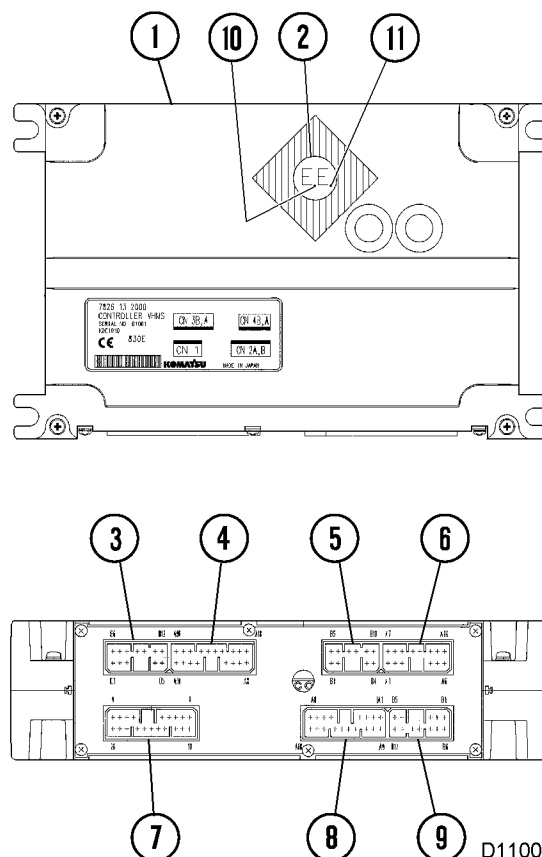
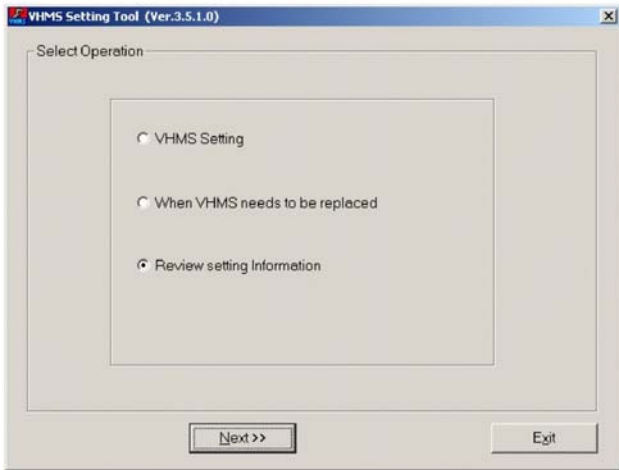


FIGURE 13-7. KOMTRAX PLUS CONTROLLER

- | | |
|----------------------------|-------------------|
| 1. KOMTRAX PLUS Controller | 6. Connector CN4A |
| 2. LED Display | 7. Connector CN1 |
| 3. Connector CN3B | 8. Connector CN2A |
| 4. Connector CN3A | 9. Connector CN2B |
| 5. Connector CN4B | 10. PLM III Light |
| | 11. Orbcomm Light |

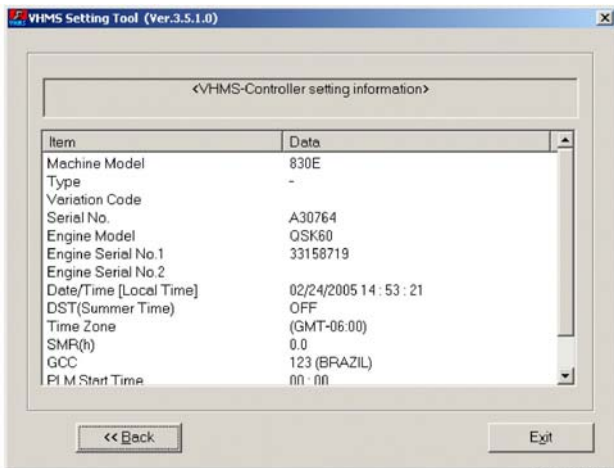
8. Start the *VHMS Setting Tool* program by clicking on the icon on the laptop PC screen.
9. Select the [Review setting information] function and then click the [Next] button.



D120063

10. Review the settings for accuracy.

- If everything is correct, click the [Exit] button. The checkout procedure is complete.
- If a setting is not correct, click the [Back] button, select the appropriate category and reset the information to the correct settings. Then proceed to the next step.



D120060

11. If any one of the following settings was changed, a new KOMTRAX Plus initialization form must be filled out and submitted to Komatsu America Service Systems Support Team.

- KOMTRAX Plus controller replaced
- Engine or alternator replaced
- Adjusted time or time zone

12. Select [Apply] and exit the *VHMS Setting Tool* program. Click [YES] when prompted to reset the controller.
13. E-mail or fax the completed KOMTRAX Plus initialization form to Komatsu America Service Systems Support Team.

KOMTRAX Plus Controller Checkout Procedure

1. Connect the serial cable from the PC to the serial port of the KOMTRAX Plus controller.
2. Start the serial communications software (Tera Term).
3. Setup the serial communications software by selecting the appropriate serial COM port, and baud rate equal to 19200.
4. After completing the setup, wait for 5 seconds then while holding the CTRL key, type "VHMS" (Notice that nothing will display on the screen while typing).
5. After "VHMS" has been typed, some text followed by a prompt > will be displayed. This confirms that proper communication between the PC and KOMTRAX Plus has been established.
6. At the prompt >, type "ver". Something similar to the following will be displayed:

>ver

VHMS OS Ver 1.6.5.1 Mar 01 2004 16:37:25

>

7. At the prompt >, type "dispvhmsinf". Information similar to the following will be displayed:

>dispvhmsinf

---- MACHINE INFORMATION -----

PRODUCT GROUP: Dumptruck

MACHINE_MODEL: 930E-

MACHINE_SERIAL:

ENG_MODEL: QSK60

ENG_SERIAL_NO1:

ENG_SERIAL_NO2:

PRG_NO1: 12000100100

PRG_NO2: 782613R290

---- DEVICES -----

PLC NO CONNECTION

PLM23 Disabled

PLM3 CONNECTED

---- Condition -----

SMR: 90.0 H

DATE 04-10-25 TIME14:44:24

TIMEZONE: 0.0 H SUMMERTIME 0

----Controller Info -----

PartNumber: 0000000000

Serial No.: 000000

Compo Name: KDE1010

SilkyID: VA011740744

>

NOTE: Use the results of step 6 and 7 to confirm that the correct software is installed in the KOMTRAX Plus controller.

8. The KOMTRAX Plus controller also has two red LED lights (10 and 11, Figure 13-6). Verify the connection status and repair any problems.

Light (10) PLM III communication

- OFF - No communication with the PLM III controller. Troubleshoot and repair the connection.
- ON - Communication with the PLM III controller is good.

Light (11) Orbcomm

- OFF - No communication with Orbcomm controller. Troubleshoot and repair the connection.
- ON - Communication with Orbcomm controller is good.
- FLASHING - Satellite signal is established. This may take as long as 15 minutes to occur.

ORBCOMM CONTROLLER (if equipped)

The Orbcmm controller receives data from the KOMTRAX Plus controller and sends this data through the antenna to the Komatsu computer center.

NOTE: The Orbcmm controller may not be approved for use in certain countries of the world. Local regulation may prohibit the use of the Orbcmm controller/satellite communicator. If equipped, the controller and antenna may be disconnected and/or removed from the truck.

Removal

1. Turn the key switch OFF. Disconnect battery power by using the battery disconnect switches.
2. Disconnect the wire harnesses from the Orbcmm controller.
3. Remove the Orbcmm controller.

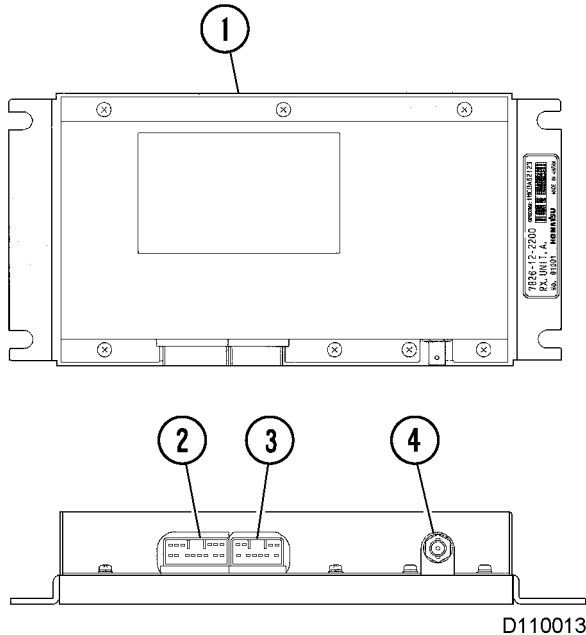


FIGURE 11-8. ORBCOMM CONTROLLER

- | | |
|----------------------|----------------------|
| 1. Orbcmm Controller | 3. Connector CN1B |
| 2. Connector CN1A | 4. Antenna Connector |

Installation

1. Install the Orbcmm controller. Connect the wire harnesses to the controller.
2. Turn the key switch ON, but do not start engine. Wait three minutes and watch for any error messages on the KOMTRAX Plus controller LED lights that might indicate a problem with the Orbcmm controller or communications to the controller.
3. If there are no error messages, turn the key switch OFF. If there are error messages, refer to the KOMTRAX Plus Troubleshooting and Checkout Procedures elsewhere in this section.
4. Fill out the KOMTRAX Plus initialization form and send it to Komatsu as instructed on the form. Failure to submit the form to Komatsu will prevent machine data from being sent to the Komatsu computer center.

NOTE: The new controller should come with a special Orbcmm Terminal Activation form that includes space to list the failed controller serial number and new controller serial number. Komatsu must have this information to maintain accurate data.

5. It may take up to two weeks for Komatsu to activate the new Orbcmm controller. During this time, a manual download of data must be taken one time each week using a laptop PC. This data must then be sent to WebCARE using the FTP feature in *VHMS Technical Analysis Tool Box* program. Keep downloading data and sending it to WebCARE one time each week until the new Orbcmm controller has been activated.

Komatsu will notify the person who performed the controller replacement by e-mail when the new controller has been activated and no more manual downloads will have to be performed.

SECTION D12

INTERFACE MODULE

INDEX

INTERFACE MODULE	D12-3
Removal	D12-3
Installation	D12-3
SENSORS	D12-4
Temperature Sensors	D12-4
Pressure Sensors	D12-4
INTERFACE MODULE SOFTWARE	D12-5
Flashburn Software Installation	D12-5
Interface Module Application Code Installation	D12-6
Interface Module Realtime Data Monitor Software	D12-6
INTERFACE MODULE CHECKOUT	D12-7
Necessary Equipment	D12-7
Preliminary	D12-8
Check Digital Inputs To The Interface Module	D12-9
Check Analog Inputs To The Interface Module	D12-12
Check Serial Interfaces To The Interface Module	D12-12
Check Outputs From The Interface Module	D12-13

NOTES

INTERFACE MODULE

The interface module (IM) (1, Figure 12-1) collects data from various sensors and sends this information to the KOMTRAX Plus controller through the main wiring harness.

If a new interface module is purchased, the operating system (software) has to be installed into the new interface module. To install the operating system, a laptop PC must be connected to the IM diagnostic port (1, Figure 12-5). Two software programs are required to install the software: the operating system and the program to perform the installation of the software (flashburn).

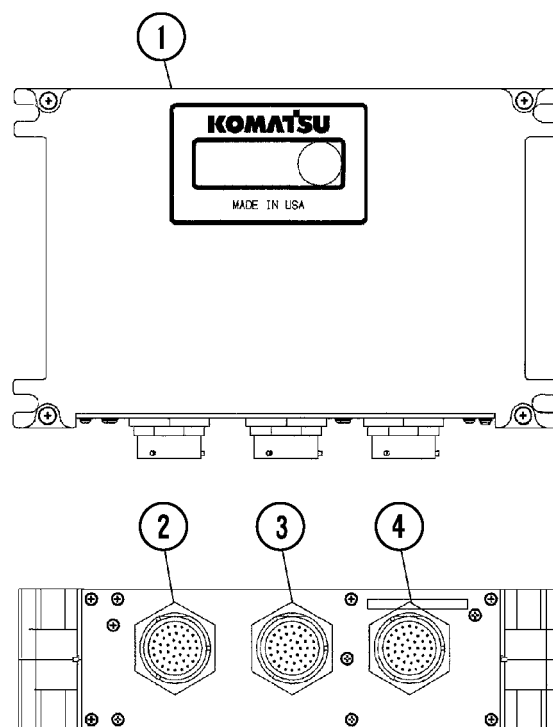
Removal

1. Turn the key switch OFF. Wait three minutes to allow the KOMTRAX Plus controller to process and store data.
2. Disconnect the battery using the battery disconnect switch.
3. Disconnect the wiring harnesses from the interface module.
4. Remove the mounting hardware and remove the interface module.

Installation

1. Install the interface module. Attach all wire harnesses to the interface module.
2. Refer to the KOMTRAX Plus Software instructions to install the flashburn program on a laptop PC.
3. Connect the laptop PC to IM diagnostic port (1, Figure 12-5).
4. Turn the key switch ON, but do not start the engine.
5. Run the flashburn program to install the application code into the interface module. Make sure the correct application code is installed for the model and serial number range (if any) of truck that is being serviced. After the application code has been installed, proceed with to Step 6.

6. Turn the key switch OFF and wait one minute.
7. Turn the key switch ON, but do not start the engine. Wait three minutes and watch for any error messages on the KOMTRAX Plus controller LED lights that might indicate a problem in the system.
8. If there are no error messages, turn the key switch OFF. If there are error messages, refer to the KOMTRAX Plus & Interface Module Troubleshooting and Error Codes elsewhere in this section.



D110014

FIGURE 12-1. INTERFACE MODULE

- | | |
|---------------------|------------------|
| 1. Interface Module | 3. Connector IM2 |
| 2. Connector IM1 | 4. Connector IM3 |

SENSORS

Temperature Sensors

Temperature sensors (Figure 12-2) monitor the ambient air temperature and the hydraulic oil temperature. An ambient air temperature sensor is located on the left side of the air blower inlet duct for the traction alternator. The hydraulic oil temperature sensors are located at each wheel to measure the oil temperature as it leaves each brake assembly.

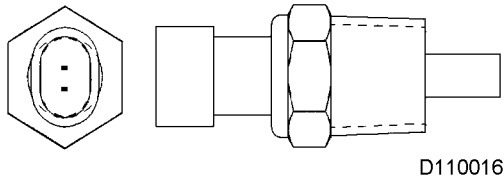


FIGURE 12-2. TEMPERATURE SENSOR

Pressure Sensors

Four pressure sensors (Figure 12-3) have been added to the truck to monitor various hydraulic circuits. The four circuits are:

- both inlets to the hoist valve
- steering supply circuit
- front brake apply circuit

The hoist pressure sensors are both located right at the inlet of the hoist valve. The front brake apply pressure sensor is located in the brake circuit junction block in the hydraulic cabinet behind the cab. The steering pressure sensor is located on the bleeddown manifold in the port labeled "TP2".

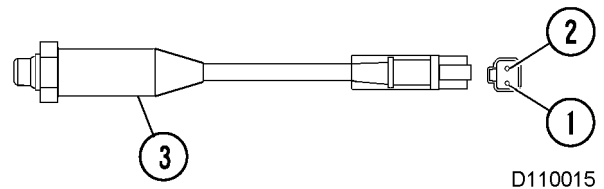


FIGURE 12-3. PRESSURE SENSOR

- | | |
|-------------------------|-----------|
| 1. Pin 1, Input (Brown) | 3. Sensor |
| 2. Pin 2, Signal (Red) | |

INTERFACE MODULE SOFTWARE

To work with the interface module, several special tools and software programs are required. Refer to Tables 1 and 2 for detailed information on software and tools.

Refer to the following pages for detailed information on how to perform certain procedures using specific software.

The data files, application code and flashburn software are only required if the interface module is being replaced. Replacement interface modules from Komatsu do not have any software installed in them.

NOTE: Be aware that the software and data files are updated periodically. Check with the local Komatsu distributor for the latest software versions.

Table 1: Interface Module Software		
Name	Description	Source
Interface Module Realtime Data Monitor Software	Use to watch inputs and outputs in the interface module	Komatsu distributor
Flashburn program	To install application code in interface module	Komatsu distributor
Application code	Application code for interface module	Komatsu distributor

Table 2: Interface Module Tools		
Name	Description	Source
Laptop computer	200 MHz or higher 64 MB RAM or more Serial or USB Port CD/DVD-Rom drive Windows operating system	Purchase locally
Serial cable (RS232 port)	Male DB9 connector at one end Female DB9 connector at other end	Purchase locally
Adaptor	USB port to RS232 serial port (If the laptop computer does not have an RS232 port, this adaptor is required.)	Purchase locally

Flashburn Program Installation

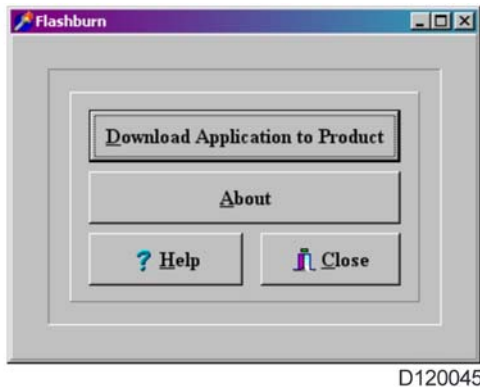
The *Flashburn* program is used to install the application code into the interface module controller.

1. Save the *Flashburn* program file to a folder on a local hard drive (such as C:\temp) on the laptop computer.
2. Double-click on the *Flashburn* program file so it will extract the file. Chose a folder on a local hard drive to save the file into (such as C:\temp).
3. Inside that directory, double click on Setup.exe to install the *Flashburn* program.
4. Follow the on-screen prompts to install the program.

Interface Module Application Code Installation

The application code is truck specific software that is installed into the interface module. Application code is installed using the *Flashburn* program.

1. Using a laptop PC, save the application code files to a folder on a local hard drive (such as C:\temp).
2. Double-click on the correct application code file so it will extract the file. Chose a folder on a local hard drive to save the file into (such as C:\temp).
3. Using a serial cable, connect the laptop PC to the IM-Diag connector located near the interface module.
4. Start the *Flashburn* program.
5. Select [Download Application to Product].



6. Make sure that the power to the interface module is off. Then click [Next].
7. Select the correct COM port. Then click [Next].
8. Select the correct ".KMS" file. Then click [Next].

Flashburn will now install the application code into the interface module.

Interface Module Realtime Data Monitor Software

The Interface Module Realtime Data Monitor Software is used to display the data going into and out of the interface module.

To install:

1. Copy the file onto the laptop PC hard drive.
2. Double-click on the file and follow the screen prompts to install the software.

To use:

1. Start the *Interface Module Realtime Monitor* program.
2. Click on the [Select Serial Port] menu item. Select the correct communication port. It will usually be Com1.
3. Click on the [Start/Stop] menu item and choose [Start].
4. Click on the [Units] menu to select the desired units to display the information.

INTERFACE MODULE CHECKOUT

⚠ IMPORTANT ⚠

If a new truck with KOMTRAX Plus is being assembled, or a new KOMTRAX Plus system has just been installed, refer to the KOMTRAX Plus System for instructions regarding the initialization procedure. The initialization procedure and form must be completed before the truck can be put into service.

Necessary Equipment

- System schematic
- Laptop PC
- *Interface Module Real Time Data Monitor* software
- Serial cable (RS232)
- Jumper wire 77 mm (3 in.) or longer
- Volt Meter
- 300 to 332 ohm resistor
- 3/8 in. nut driver

NOTE: The interface module must already have the application code installed.

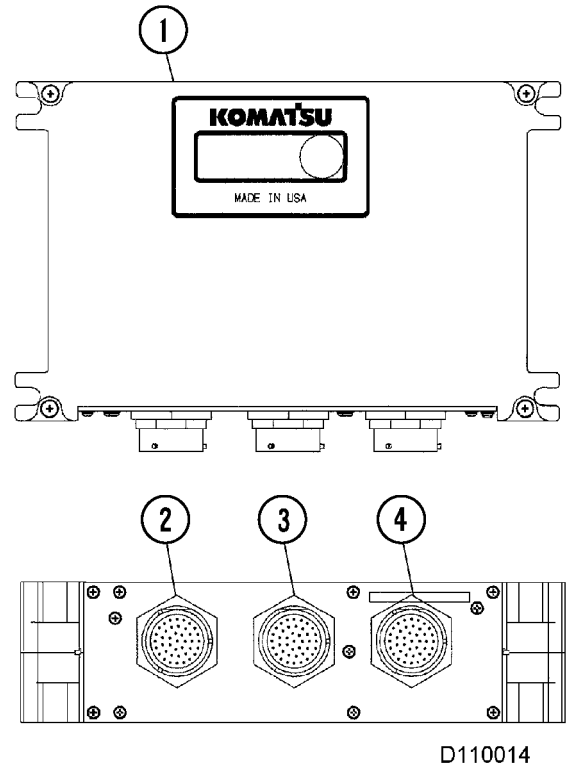


FIGURE 12-4. INTERFACE MODULE

- | | |
|---------------------|------------------|
| 1. Interface Module | 3. Connector IM2 |
| 2. Connector IM1 | 4. Connector IM3 |

Preliminary

1. Turn the key switch to the OFF position to stop the engine.
2. Turn the key switch to the ON position, but DO NOT start the engine.
3. Allow the KOMTRAX Plus controller to start up. This should take about one minute. Verify the red LED display starts counting up.
4. Attach the KOMTRAX Plus serial cable to the machine's KOMTRAX Plus diagnostic port (2, Figure 12-5), and the other end to the laptop PC's serial port.

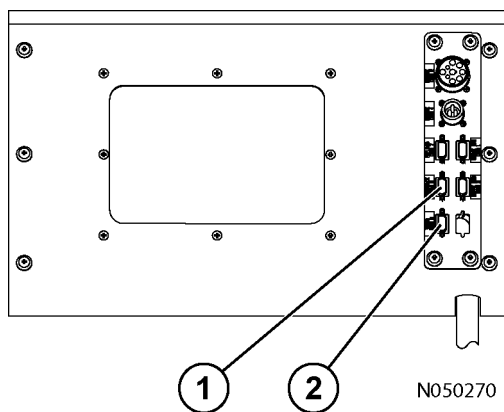


FIGURE 12-5. DIAGNOSTIC PORTS

1. IM Diagnostic Port
2. KOMTRAX Plus Diagnostic Port

5. Double-click on the *VHMS Technical Analysis Tool Box* icon on the computer's desktop.
6. Enter the appropriate User Name and Password and click the [OK] button.

7. Check for fault codes associated with the interface module.
 - a. Perform a KOMTRAX Plus download with the *VHMS Technical Analysis Tool Box* program. Refer to KOMTRAX Plus Download for detailed instructions on performing a download.
 - b. In the download data, view the fault history and confirm that there are no fault codes associated with the interface module. If any are found, these circuits should be analyzed to determine the cause of the fault and repaired.
 - c. Confirm that there are no fault codes associated with the communications between PLM III, engine controller, interface module, drive system controller or the Orbcomm controller. If any are found, these circuits should be analyzed to determine the cause of the fault and repaired.
8. Attach the KOMTRAX Plus serial cable to the machine's IM diagnostic port (1, Figure 12-5), and the other end to the laptop PC's serial port.
9. Start the *Interface Module Real Time Data Monitor* program by double-clicking on the shortcut. The program begins with a blank window. On the menu bar, there are five items: Select Serial Port, Start/Stop, Logging, Screenshot, and Units.
10. Click on [Select Serial Port] in the menu bar. Select the correct communication port. It will usually be Com1.
11. Click on [Start/Stop] in the menu bar and select Start.
12. The program should display data as shown in Figures 12-6 and 12-7.

NOTE: If any fault codes associated with the interface module are found, these circuits should be analyzed to determine the cause of the fault, and they should be repaired.

Check Digital Inputs To The Interface Module

1. Hydraulic Tank Level (IM2-k) - short wire 34LL to ground at TB35-N momentarily and confirm state change (one to zero).
2. No Propel / Retard - (IM2-n) short wire 75-6P to ground at TB26-C momentarily and confirm state change (one to zero).
3. Reduced Retard - (IM2-r) short wire 76LR to ground at TB28-D momentarily and confirm state change (one to zero).
4. Propel System Temp Caution - (IM3-A) short wire 34TW to ground at TB26-B momentarily and confirm state change (one to zero).
5. Lamp Test (IM2-R) - actuate lamp test switch and confirm state change (zero to one).
6. Low Steering Precharge (IM2-W) - short wire 33KL to ground at TB44-P momentarily and confirm state change (one to zero).

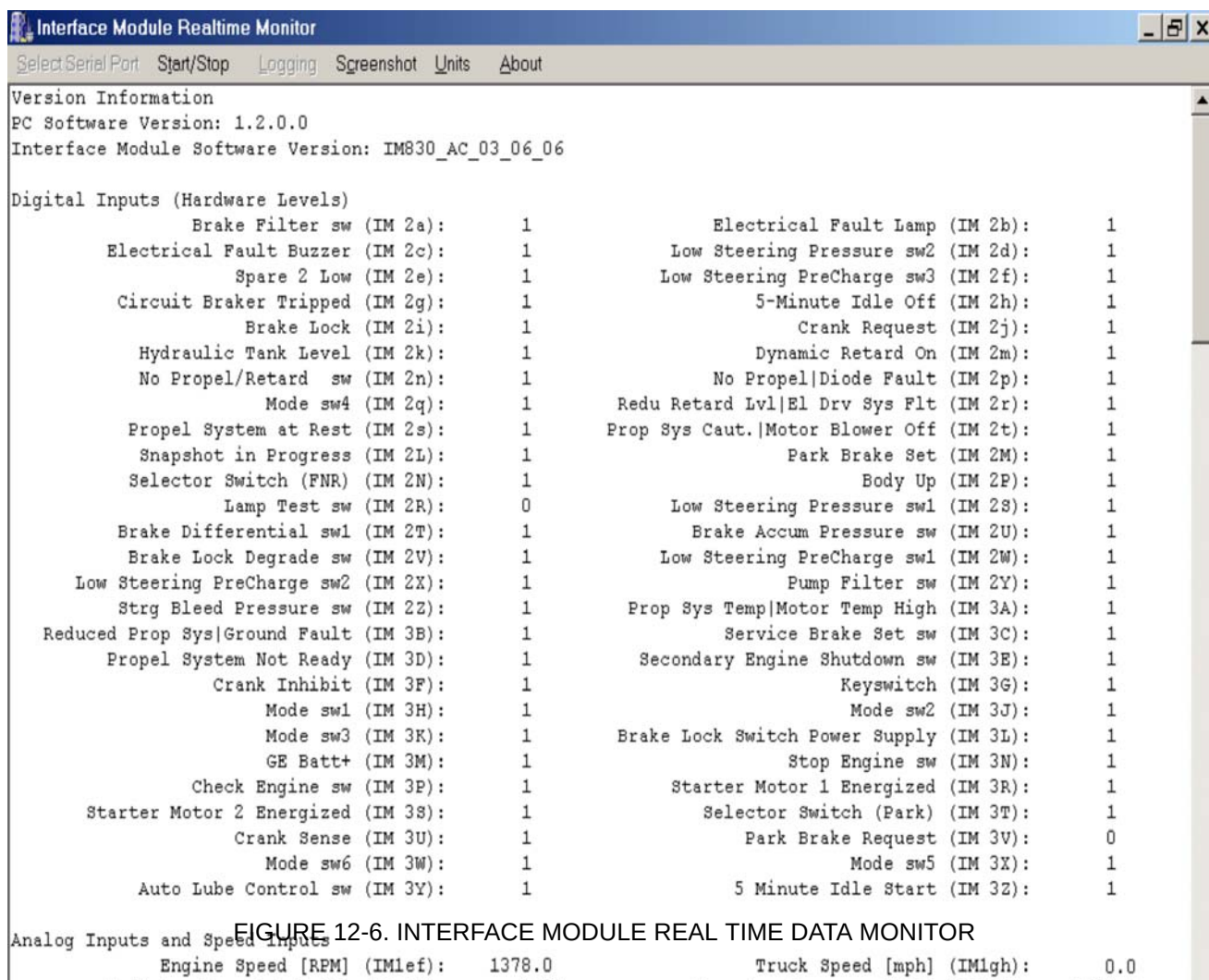


FIGURE 12-6. INTERFACE MODULE REAL TIME DATA MONITOR

7. Pump Filter Switches (IM2-Y) - short wire 39L to ground at TB44-N momentarily and confirm state change (one to zero).
8. No Propel (IM2-p) - short wire 75NP to ground at TB25-P momentarily and confirm state change (one to zero).
9. Propel System Caution (IM2-t) - short wire 79W to ground at TB26-D momentarily and confirm state change (one to zero).
10. Reduced Propel System (IM3-B) - short wire 72LP to ground at TB25-W momentarily and confirm state change (one to zero).
11. Park Brake Set (IM2-M) - disconnect park brake pressure switch in brake cabinet at CN240 momentarily and confirm state change toggles continually (zero to one).

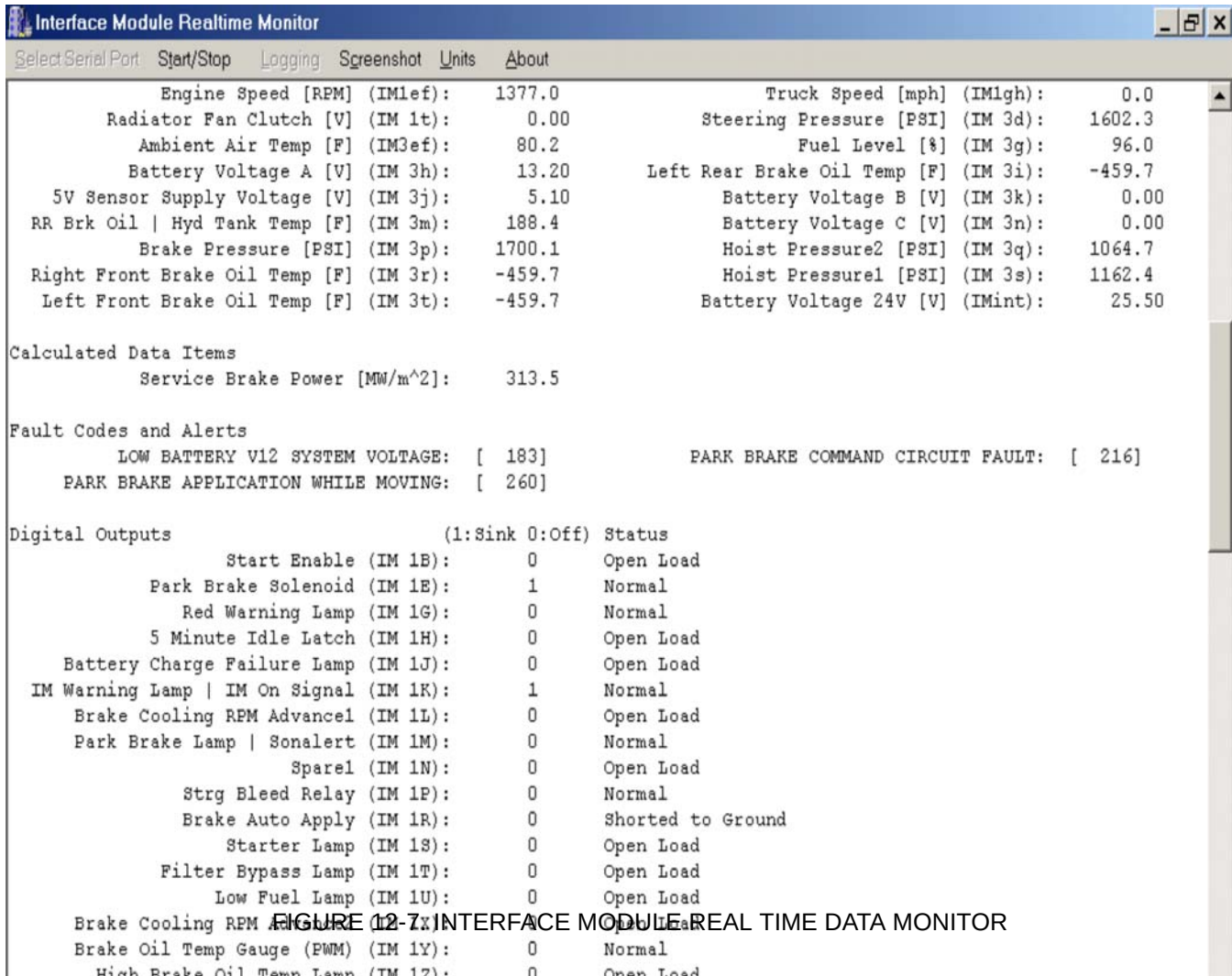


FIGURE 12-7 INTERFACE MODULE REAL TIME DATA MONITOR

12. Park Brake Request (IM3-V) - Short the engine oil pressure switch wire circuit 36 on TB26-L to ground. Move shift lever from neutral to park position and confirm state change (one to zero). Remove the ground from TB26-L.
13. Auto Lube Switch (IM3-Y) - short wire 68LLP1 to ground at TB24-T momentarily and confirm state change (one to zero).
14. GE Batt + (IM3-M) -- confirm this is a one.
15. Starter Motor 1 Energized (IM3-R) - Disconnect wire 11SM1 from cranking motor to TB29-K at TB29-K. Momentarily short TB29-K to 24V and confirm state change (zero to one). Reconnect disconnected wire.
16. Starter Motor 2 Energized (IM3-S) - Disconnect wire 11SM2 from cranking motor to TB29-G at TB29-G. Momentarily short TB29-G to 24V and confirm state change (zero to one). Reconnect disconnected wire.
17. Crank Sense (IM3-U) -Open the start battery disconnect switch so that there is no battery voltage to the starters. Momentarily short TB32-M to 24V and confirm state change (zero to one). After removing 24V short from TB32-M, close the start battery disconnect switch.
18. Selector Switch (Park) (IM3-T) - Place shifter into park position and confirm 1 state then shift into neutral and confirm 0 state. Return shifter to park position.
19. Selector Switch (FNR) (IM2-N) - Place shifter into park position and confirm 0 state then shift into neutral and confirm 1 state. Return shifter to park position.
20. Steering Bleed Pressure Switch (IM2-Z) - Disconnect the steering bleed down pressure switch and confirm state change (zero to one). Reconnect the switch.
21. Brake Lock Switch Power Supply (IM3-L) - Use GE DID to simulate a vehicle speed of 2 kph. Confirm state change (zero to one). Leave vehicle speed at 2 kph until completion of step 24.
22. Brake Lock (IM2-i) -Actuate brake lock switch and confirm state change (zero to one). Turn off brake lock switch. (The Brake Lock Switch Power Supply test, item 21, must be completed before this test can be successfully completed.) Reset vehicle speed to zero.
23. Service Brake Set Switch (IM3-C) - Short wire 44R at TB26-X to 24 volts momentarily and confirm state change (zero to one).
24. Engine Shutdown (IM3-F) - Disconnect wire 21ISL from the engine ECM at TB36-W leaving wire 21ISL to IM connected at TB36-W. Momentarily short TB36-W to 24V and confirm state change (zero to one). Reconnect disconnected wire.
25. Secondary Engine Shutdown Switch (IM3-E) - Actuate the Secondary Engine Shutdown switch and confirm state change (zero to one).

Check Analog Inputs To The Interface Module

NOTE: Instead of using a resistor in place of a sensor for verifying pressure readings, a calibrated pressure gauge can be installed in the hydraulic circuit to compare system pressures with the pressures displayed in the Interface Module Real Time Data Monitor program.

Verify that the used analog inputs are in the range of the values listed below.

1. Truck Speed [kph] (IM1-g, h): Use GE DID to simulate vehicle speed and confirm reported speed matches vehicle speed set using GE DID +/- 2 kph.
2. Steering Pressure [kPa] (IM3-d): Disconnect steering pressure sensor and confirm fault A204, Steering Pressure Sensor Low, is active. Reconnect sensor.
3. Ambient Air Temp [C] (IM3-e): confirm reported temperature matches ambient temperature within 3 C.
4. Fuel Level [%] (IM3-g): confirm reported % level matches actual fuel level in tank +/- 5%.
5. Battery Voltage A [V] (IM3-h): confirm reported voltage is +/- 1 volt of actual measured 12 volt battery voltage.
6. Brake Pressure [kPa] (IM3-p): Disconnect service brake pressure sensor located in brake cabinet (reference circuit 33SP) and confirm fault A205, Brake Pressure Sensor Low, is active. Reconnect sensor.
7. Right Rear Brake Oil Temp [C] (IM3-m): Disconnect right rear brake oil temp sensor and confirm fault A167, Hydraulic Oil Temp - Right Rear Sensor Low, is active. Reconnect sensor.
8. Left Rear Brake Oil Temp [C] (IM3-i): Disconnect left rear brake oil temp sensor and confirm fault A166, Hydraulic Oil Temp - Left Rear Sensor Low, is active. Reconnect sensor.
9. Right Front Brake Oil Temp [C] (IM3-r): Disconnect right front brake oil temp sensor and confirm fault A169, Hydraulic Oil Temp - Right Front Sensor Low, is active. Reconnect sensor.
10. Left Front Brake Oil Temp [C] (IM3-t): Disconnect left front brake oil temp sensor and confirm fault A168, Hydraulic Oil Temp - Left Front Sensor Low, is active. Reconnect sensor.
11. Hoist Pressure 2 [kPa] (IM3-q): Short wire 33HP2 to ground at TB41-J momentarily and confirm fault A203, Hoist Pressure 2 Sensor Low, is active.
12. Hoist Pressure 1 [kPa] (IM3-s): Short wire 33HP1 to ground at TB41-A momentarily and confirm fault A202, Hoist Pressure 1 Sensor Low, is active.
13. Battery Voltage 24V [V] (IMint): confirm reported voltage is +/- 1 volt of actual measured battery voltage.

Check Serial Interfaces To The Interface Module

1. Disconnect and isolate all circuits 90MMT at TB33-B and confirm that fault A276 becomes active after ten seconds.
2. Reconnect all circuits 90MMT at TB33-B and confirm that fault A276 clears.
3. Confirm that faults A184, A257, and A276 are not active.

Check Outputs From The Interface Module

NOTE: Before performing these next steps, the key switch must be turned off for at least 7 minutes to allow the IM to completely shutdown. Confirm that the IM has shutdown by verifying that the green LED on the IM controller has stopped flashing. While performing the following IM output checks, ensure that no output short circuit fault codes are reported by the IM Realtime Data Monitor software.

1. Short the engine oil pressure switch wire circuit 36 to ground on TB26-L.
 - a. Turn the key switch ON and shift into NEUTRAL. Confirm that park brake solenoid is energized by verifying that coil is magnetized.
 - b. Use the GE DID panel to set the truck speed to a speed above 1 kph. Shift into PARK. Confirm that the park brake solenoid remains energized.
 - c. Reduce the truck speed to 0 kph. Confirm that the auto apply solenoid energizes. After one second, confirm that the park brake solenoid de-energizes. After another 0.5 second, confirm that the auto apply solenoid de-energizes.
 - d. Remove the ground from TB26-L.
2. Actuate the lamp check switch and confirm that the Battery Charger Failure lamp energizes.
3. Actuate the lamp check switch and confirm that the IM Warning lamp energizes.
4. Actuate the lamp check switch and confirm that the Engine Start Fail lamp energizes.
5. If the Low Fuel lamp is already illuminated, confirm that it cycles off/on three times and that the buzzer sounds three times within 15 minutes.

Alternately, if the Low Fuel lamp is illuminated, completely power down the IM, then power it up again. Verify that the lamp cycles on/off and the buzzer sounds three times.

Otherwise, if the Low Fuel lamp is not illuminated, disconnect and isolate all circuits 32G and TB32-D. Confirm that the Low Fuel lamp cycles on/off twice then stays on and the buzzer sounds three times. Reconnect all circuits 32G and TB32-D.

6. Disconnect the park brake pressure switch. Actuate the lamp check switch and confirm that the Park Brake lamp energizes. Reconnect the pressure switch.
7. Actuate the lamp check switch and confirm that the Hydraulic Filter lamp energizes.
8. Actuate the lamp check switch and confirm that the Brake Oil Temp lamp energizes.
9. Confirm that the voltage on circuit 33ES1 at TB35-B is approximately 0 volts.
10. Confirm that the voltage on circuit 33ES2 at TB35-C is approximately 0 volts.
11. Check the Brake Oil Temperature gauge by placing a 316 ohm resistor with a range of 300 to 332 ohms between circuit 5VIM on TB42-L and 34BT3 on TB24-G. Verify that the gauge needle moves clockwise. Do not remove the resistor yet.
12. Confirm that the voltage on circuit 33ES1 at TB35-B is approximately battery volts.
13. Confirm that the voltage on circuit 33ES2 at TB35-C is approximately battery volts. Remove the resistor between circuits 5VIM and 34BT3.
14. Crank Enable Output: Disconnect circuit 21A from the prelube timer.
 - a. Shift into PARK and confirm that circuit 21A on TB25-D is 24 volts while cranking.
 - b. Shift into NEUTRAL and confirm that circuit 21A on TB25-D is 0 volts while cranking.
 - c. Reconnect circuit 21A to the prelube timer.
15. Steering Bleeddown Solenoid: Confirm that the steering bleeddown solenoid is de-energized. Turn the key switch OFF and confirm that the steering bleeddown solenoid is energized by verifying that coil is magnetized.

NOTES

SECTION D13

INTERFACE MODULE AND KOMTRAX PLUS TROUBLESHOOTING

INDEX

GENERAL	D13-3
TRUBLESHOOTING	D13-4
Communications Networks	D13-4
Coaxial Cable	D13-4
FAULT CODES	D13-4
Fault History	D13-4
KOMTRAX Plus LED Display Fault Codes	D13-5
Chassis Fault Codes	D13-6
Engine Fault Codes	D13-10
FAULT TREE ANALYSIS	D13-16
Unable to connect to KOMTRAX Plus from laptop PC	D13-16
Flashing Error Code N4-23 (PLM III Communications Fault)	D13-17
Flashing Error Code N4-22 (Engine Communications Fault)	D13-18
No Data Received By WebCARE	D13-19
Coaxial Cable Troubleshooting	D13-20

NOTES

INTERFACE MODULE AND KOMTRAX PLUS TROUBLESHOOTING

GENERAL

The center of KOMTRAX Plus is the KOMTRAX Plus controller, which gathers data about the operation of the truck from sensors and other controllers installed on the truck. Refer to Figure 13-1 for an overview of the KOMTRAX Plus system components.

For instructions on how to use KOMTRAX Plus software programs, refer to KOMTRAX Plus Software elsewhere in this section.

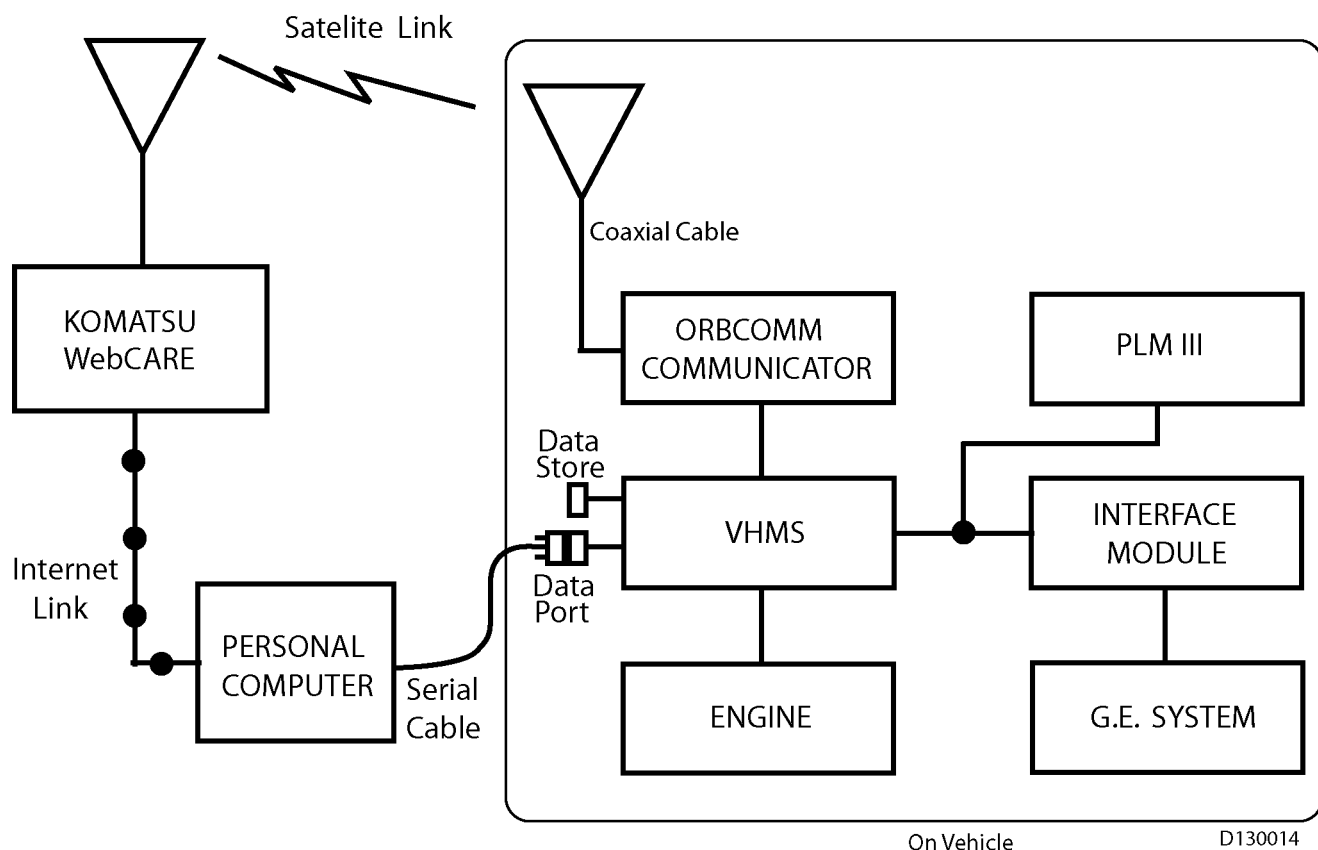


FIGURE 13-1. KOMTRAX PLUS SYSTEM

TROUBLESHOOTING

KOMTRAX Plus basically consists of five communications networks connected to the KOMTRAX Plus and Orbcomm controllers. Figure 13-1 shows the KOMTRAX Plus block diagram.

Communications Networks

Each RS232 network uses three wires: transmit, receive, and ground. Both transmit and receive are voltage signals, referenced individually to the ground wire. The shield for the cable is grounded at one end only.

Each CAN network uses two wires: CAN_High & CAN_Low. The communications signal is a voltage differential measured between CAN_High and CAN_Low. The cable shields are connected at each module through a high pass filter and grounded at one point only on the truck. Both ends of each network have termination resistors.

Coaxial Cable

The coaxial cable carries the Radio Frequency (RF) communications signal between the Orbcomm modem and the antenna. The coaxial cable consists of an inner conductor and an outer shield (connected to the connector shell) that are separated by a non-conductive dielectric material.

In an RF application such as KOMTRAX Plus, the communications signal sent over coaxial cable is very susceptible to changes in the cable. Physical damage, as well as contaminants such as water, may affect the ability of the cable to properly transmit the RF signal. Bending the coaxial cable into a small loop may also damage the inner conductor.

Effective troubleshooting of RF communications systems can be complex and cannot always be reduced to a simple check of electrical resistance. However, a few basic troubleshooting procedures may be helpful in identifying common problems. The following steps can help identify a failed coaxial cable. Repair or replace the cable if any of the following is true:

1. The center conductor is broken. There are more than two ohms of resistance when measuring from one end of the coaxial cable to the other.
2. The outer shield is broken.
3. There is an electrical connection between the center conductor and the outer shield. There are less than two megohms of resistance when measuring from the center conductor to the outer shield.

FAULT CODES

Fault History

The fault history recorded in the KOMTRAX Plus controller can help identify a failure within KOMTRAX Plus and in the communications network to the engine, interface module and PLMIII. KOMTRAX Plus provides the following fault codes.

Table 1: Fault History	
Fault Code	Description
DBB0KK	Source Voltage Error
DBB3KK	Abnormality in VBAT Voltage (VBAT <10V)
DBBRKR	Can-net System (J1939)
DBB0KQ	KOMTRAX Plus Connector Mismatch
DAW0KR	IM Stopped Real Time Data
7P70Kr	Too Much Payload Data For Requested Period
7P70KR	PLMIII Stopped Real Time Data
9843KM	Truck Frame Number Changed
MFA0	Manual Trigger

KOMTRAX Plus LED Display Fault Codes

The KOMTRAX Plus controller also indicates some faults on the two red LED digits on the top of the controller. Fault codes are flashed as a two part sequence, as shown in the table below.

When no communication errors are occurring, the KOMTRAX Plus controller LED digits count from 00 - 99 continuously at a rate of ten numbers per second.

Table 2: KOMTRAX Plus LED Display Error Codes		
Fault Code	Fault Condition	LED Display
M101	Truck Frame Number Changed	Alternates 'n1' and '01'
M801	Can-net System (J1939)	Alternates 'n8' and '01'
M804	Can-net System (RPC)	Alternates 'n8' and '04'
M806	IM Stopped Real Time Data	Alternates 'n8' and '06'
M807	Too Much Payload Data For Requested Period	Alternates 'n8' and '07'
M808	PLMIII Stopped Real Time Data	Alternates 'n8' and '08'
M809	Can-net System (QUANTUM)	Alternates 'n8' and '09'
M80A	Can-net System (CENSE)	Alternates 'n8' and '0A'
M901	Source Voltage Error	Alternates 'n9' and '01'
M902	24V Source System Error	Alternates 'n9' and '02'
M903	12V Source System Error	Alternates 'n9' and '03'
M904	5V Source System Error	Alternates 'n9' and '04'
M905	Abnormality in VBAT Voltage (VBAT <10V)	Alternates 'n9' and '05'
M990	Ethernet Power Short	Alternates 'n9' and '90'
MC10	MEMORY CLEAR: Failure History	Alternates 'nc' and '10'
MC31	MEMORY CLEAR: (Load Map)	Alternates 'nc' and '31'
MC40	MEMORY CLEAR: (Trend Analysis)	Alternates 'nc' and '40'
MC60	MEMORY CLEAR: (Snap Shot)	Alternates 'nc' and '60'
MC91	MEMORY CLEAR: (Maintenance History)	Alternates 'nc' and '90'
ME01	Change Service Meter	Alternates 'ne' and '01'
ME02	Change Calendar	Alternates 'ne' and '02'
ME03	Orbcomm Settings	Alternates 'ne' and '03'
ME04	Other Settings	Alternates 'ne' and '04'
ME05	MEMORY CLEAR: All	Alternates 'ne' and '05'
ME06	Initialized	Alternates 'ne' and '06'
MF11	KOMTRAX Plus Connector Mismatch	Alternates 'nf' and '11'
MFA0	Manual Trigger	Alternates 'nf' and 'A0'

Chassis Fault Codes

Fault codes generated from the truck chassis, PLM III or GE are shown in Table 3.

NOTE: Not of the fault codes that are listed may be used for this truck.

Table 3: Chassis Fault Codes				
Fault Code	Fault Description	Source	Sent via Orbcomm	Snapshot Trigger
#A1	LF Pressure Sensor Signal High	PLMIII		
#A2	LF Pressure Sensor Signal Low	PLMIII		
#A3	RF Pressure Sensor Signal High	PLMIII		
#A4	RF Pressure Sensor Signal Low	PLMIII		
#A5	LR Pressure Sensor Signal High	PLMIII		
#A6	LR Pressure Sensor Signal Low	PLMIII		
#A7	RR Pressure Sensor Signal High	PLMIII		
#A8	RR Pressure Sensor Signal Low	PLMIII		
#A9	Inclinometer Sensor Signal High	PLMIII		
#A10	Inclinometer Sensor Signal Low	PLMIII		
#A13	Body Up Switch Failure	PLMIII		
#A14	Internal Checksum Failure	PLMIII		
#A16	Internal Memory Write Failure	PLMIII		
#A17	Internal Memory Read Failure	PLMIII		
#A18	RR Flat Cylinder Warning	PLMIII		
#A19	LR Flat Cylinder Warning	PLMIII		
#A20	Date/Time Change	PLMIII		
#A21	Manual Tare Reset	PLMIII		
#A22	Alarm Carry Back	PLMIII		
#A26	User Switch Select Failure	PLMIII		
#A27	User Switch Clear Failure	PLMIII		
#A101	Pump Filter Switches - Pressure High	IM		
#A103	Hydraulic Tank Temp Sensor Low	IM		
#A104	Hydraulic Tank Temp Sensor High	IM		
#A105	Fuel Level Sensor Low	IM		
#A107	Propel System Caution	IM/GE	X	
#A108	Propel System Temp Caution	IM/GE	X	
#A109	Propel System Reduced Level	IM/GE	X	
#A115	Low Steering Precharge	IM	X	
#A123	Reduced Retard Level	IM/GE		
#A124	No Propel/Retard	IM/GE	X	
#A125	No Propel	IM/GE	X	
#A126	Hydraulic Tank Level Low	IM	X	

Table 3: Chassis Fault Codes (continued)				
Fault Code	Fault Description	Source	Sent via Orbcomm	Snapshot Trigger
#A127	IM Sensor +5V Low	IM	X	
#A128	IM Sensor +5V High	IM	X	
#A139	Low Fuel	IM		
#A145	Brake Cooling RPM Advance 1	IM		
#A146	Brake Cooling RPM Advance 2	IM		
#A152	Starter Failure	IM		
#A153	Low Battery Voltage - Engine Running	IM		
#A154	High Battery Charge Voltage	IM		
#A155	Low Battery Charge Voltage	IM		
#A158	Fuel Level Sensor High	IM		
#A159	Battery Voltage, 12-Volt, System Sensing Low	IM		
#A164	Battery Voltage, 12-Volt, System Sensing High	IM		
#A166	Hydraulic Oil Temp - LR Sensor Low	IM		
#A167	Hydraulic Oil Temp - RR Sensor Low	IM		
#A168	Hydraulic Oil Temp - LF Sensor Low	IM		
#A169	Hydraulic Oil Temp - RF Sensor Low	IM		
#A170	Hydraulic Oil Temp - LR Sensor High	IM		
#A171	Hydraulic Oil Temp - RR Sensor High	IM		
#A172	Hydraulic Oil Temp - LF Sensor High	IM		
#A173	Hydraulic Oil Temp - RF Sensor High	IM		
#A182	System Battery 12V High	IM		
#A183	System Battery 12V Low	IM		
#A184	J1939 Not Connected	IM		
#A190	Auto Lube Switch	IM	X	
#A194	High LF Brake Oil Temp	IM	X	X
#A195	High RF Brake Oil Temp	IM	X	X
#A196	High LR Brake Oil Temp	IM	X	X
#A197	High RR Brake Oil Temp	IM	X	X
#A198	Hoist Pressure 1 Sensor High	IM		
#A199	Hoist Pressure 2 Sensor High	IM		
#A200	Steering Pressure Sensor High	IM		
#A201	Brake Pressure Sensor High	IM		
#A202	Hoist Pressure 1 Sensor Low	IM		
#A203	Hoist Pressure 2 Sensor Low	IM		
#A204	Steering Pressure Sensor Low	IM		
#A205	Brake Pressure Sensor Low	IM		

Table 3: Chassis Fault Codes (continued)				
Fault Code	Fault Description	Source	Sent via Orbcomm	Snapshot Trigger
#A206	Ambient Temperature Sensor High	IM		
#A207	Ambient Temperature Sensor Low	IM		
#A212	Bad Truck Speed Signal	IM		
#A213	Park Brake Not Set When Expected	IM		
#A214	Park Brake Not Released When Expected	IM		
#A215	Brake Auto Apply Circuit Fault	IM		
#A216	Park Brake Command Circuit Fault	IM		
#A223	Excessive Cranking	IM		
#A230	Park Brake Request While Moving	IM		
#A236	Steering Bleed Fault	IM		
#A240	IM Key Switch Power Lost	IM		
#A249	Red Warning Lamp Short	IM		
#A250	Low Battery Voltage - Engine Off	IM		
#A252	Start Enable Circuit Fault	IM		
#A253	Steering Bleed Circuit Not Open While Running	IM		
#A256	Red Warning Lamp Open	IM		
#A257	Payload CAN/RPC Not Connected	IM		
#A258	Steering Accumulator Bleed Pressure Switch Fault	IM		
#A260	Park Brake Failure - Application While Moving	IM		
#A262	Steering Bleed Circuit Open During Shutdown	IM		
#A263	Steering Bleed Circuit Short to Ground	IM		
#A264	Park Brake Relay Circuit Defective	IM		
#A265	Service Brake Failure	IM		
#A270	Brake Lock Switch Power Supply Not On When Required	IM		
#A272	Brake Lock Switch Power Supply Not Off When Required	IM		
#A273	Pump Filter Circuit Fault	IM		
#A274	Brake Malfunction	IM		
#A275	Starter Stuck ON	IM		
#A276	Drive System Data Link Not Connected	IM		
#A282	Excessive Cranking Count Limit	IM		
#A283	Engine Shutdown Delay Aborted by Lack of Park Brake	IM		
#A284	Engine Shutdown Delay Aborted by Secondary Shutdown Switch	IM		
#A285	Park Brake Not Set When Keyswitch Turned Off	IM		

Table 3: Chassis Fault Codes (continued)				
Fault Code	Fault Description	Source	Sent via Orbcomm	Snapshot Trigger
#A286	Shutdown Delay Relay Circuit Fault	IM		
#A292	Shutdown Delay Relay Fault	IM		
#A303	Shifter Defective	IM		
#A306	Red Lamp Circuit Fault	IM		
#A309	No Brakes Applied When Expected	IM		
#A350	Output Overload 1B	IM		
#A351	Output Overload 1E	IM		
#A352	Output Overload 1H	IM		
#A353	Output Overload 1J	IM		
#A354	Output Overload 1K	IM		
#A355	Output Overload 1L	IM		
#A356	Output Overload 1M	IM		
#A357	Output Overload 1N	IM		
#A358	Output Overload 1P	IM		
#A359	Output Overload 1R	IM		
#A360	Output Overload 1S	IM		
#A361	Output Overload 1T	IM		
#A362	Output Overload 1U	IM		
#A363	Output Overload 1V	IM		
#A364	Output Overload 1Y	IM		
#A365	Output Overload 1Z	IM		

Engine Fault Codes

Table 4: Engine Fault Codes				
Fault Code	Fault Description	Source	Sent via Orbcomm	Snapshot Trigger
C112	Timing Fueling Flow Mismatch	Engine		
C113	Timing Actuator Circuit Shorted	Engine		
C115	Speed Signal Lost	Engine	X	X
C116	Timing Rail Pressure Ckt Failed High	Engine		
C117	Timing Rail Pressure Ckt Failed Low	Engine		
C118	Fuel Pump Pressure Ckt Failed High	Engine		
C119	Fuel Pump Pressure Ckt Failed Low	Engine		
C121	One Engine Speed Signal Lost	Engine		
C122	LB Boost Ckt Failed High	Engine		
C123	LB Boost Ckt Failed Low	Engine		
C124	High Boost LB	Engine		
C125	Low Boost LB	Engine		
C126	High Boost RB	Engine		
C127	Low Boost RB	Engine		
C128	RB Boost Ckt Failed High	Engine		
C129	RB Boost Ckt Failed Low	Engine		
C131	Throttle Ckt Failed High	Engine		
C132	Throttle Ckt Failed Low	Engine		
C133	PTO Circuit Shorted High	Engine		
C134	PTO Circuit Shorted Low	Engine		
C135	Oil Pressure Circuit Failed High	Engine	X	X
C136	Pre Filter Oil Press Ckt Failed High	Engine		
C137	Pre Filter Oil Press Ckt Failed Low	Engine		
C141	Oil Press Ckt Failed Low	Engine		
C143	Low Oil Pressure	Engine	X	X
C144	Coolant Temp Ckt Failed High	Engine		
C145	Coolant Temp Ckt Failed Low	Engine		
C147	Freq Throttle OOR High	Engine		
C151	High Coolant Temperature	Engine	X	X
C153	LBF IMT Ckt Failed High	Engine		
C154	LBF IMT Ckt Failed Low	Engine		
C155	High IMT LBF	Engine	X	X
C156	LBR IMT Ckt Failed High	Engine		
C157	LBR IMT Ckt Failed Low	Engine		
C158	High IMT LBR	Engine	X	X

Table 4: Engine Fault Codes (continued)				
Fault Code	Fault Description	Source	Sent via Orbcomm	Snapshot Trigger
C159	RBF IMT Ckt Failed High	Engine		
C161	RBF IMT Ckt Failed Low	Engine		
C162	High IMT RBF	Engine	X	X
C163	RBR IMT Ckt Failed High	Engine		
C164	RBR IMT Ckt Failed Low	Engine		
C165	High IMT RBR	Engine	X	X
C212	Oil Temp Ckt Failed High	Engine		
C213	Oil Temp Ckt Failed Low	Engine		
C214	High Oil Temperature	Engine	X	X
C219	Remote Oil Level Low	Engine	X	X
C221	Ambient Air Press Failed High	Engine		
C222	Ambient Air Press Failed Low	Engine		
C223	CORS Burn Valve Open Circuit	Engine		
C225	CORS Makeup Valve Open Circuit	Engine		
C231	Coolant Press Ckt Failed High	Engine		
C232	Coolant Press Ckt Failed Low	Engine		
C233	Low Coolant Pressure	Engine	X	X
C234	Engine Overspeed	Engine	X	X
C235	Low Coolant Level	Engine	X	X
C237	Multi Unit Sync Error	Engine		
C252	Oil Level Signal Invalid	Engine		
C253	Oil Level Low	Engine		
C254	FSOV Open Circuit	Engine		
C259	FSOV Mech Stuck Open	Engine		
C261	High Fuel Temperature	Engine	X	X
C263	Fuel Temp Ckt Failed High	Engine		
C265	Fuel Temp Ckt Failed Low	Engine		
C292	OEM Temp out of Range	Engine	X	X
C293	OEM Temp Failed High	Engine	X	
C294	OEM Temp Failed Low	Engine	X	
C296	OEM Pressure Out of Range	Engine	X	X
C297	OEM Pressure Failed High	Engine	X	
C298	OEM Pressure Failed Low	Engine	X	
C299	Hot Shutdown	Engine		
C316	Fuel Pump Open Circuit	Engine		

Table 4: Engine Fault Codes (continued)

Fault Code	Fault Description	Source	Sent via Orbcomm	Snapshot Trigger
C318	Fuel Pump Mech Stuck	Engine		
C343	ECM Hardware Issue	Engine		
C346	ECM Software / Hardware Failure	Engine		
C349	Output Shaft Speed Above Normal	Engine		
C384	Ether Solenoid Ckt Failed	Engine		
C422	Coolant Level Signal Invalid	Engine		
C423	Timing Press Incorrect	Engine		
C426	J1939 Broadcast Data Missing	Engine		
C427	J1939 Datalink Can Not Transmit	Engine		
C431	Idle Validation Invalid	Engine		
C432	Idle Validation Invalid	Engine		
C441	Low Battery Voltage	Engine		
C442	High Battery Voltage	Engine		
C451	Rail Press Ckt Failed High	Engine		
C452	Rail Press Ckt Failed Low	Engine		
C455	Rail Actuator Open Ckt	Engine		
C467	Desired Timing Not Achieved	Engine		
C468	Desired Rail Press Not Achieved	Engine		
C473	Remote Oil Level Signal Invalid	Engine	X	X
C487	Ether Bottle Empty	Engine		
C489	AXG Speed Low Error	Engine		
C514	Rail Actuator Mech Stuck	Engine		
C524	Alt Droop SW Val Fault	Engine		
C527	Dual Output A Shorted High or Open	Engine		
C528	Alt Torque SW Val Fault	Engine		
C529	Dual Output B Shorted High or Open	Engine		
C553	Rail Press OOR High	Engine		
C554	Rail Press Incorrect	Engine		
C555	High Blow-by Pressure	Engine	X	X
C611	Engine Hot Shutdown	Engine		
C612	High Oil Filter Rest	Engine		
C616	High Turbo Comp Inlet Temp LBR	Engine		
C621	Low Power #1 LB	Engine		
C622	Low Power #2 LB	Engine		
C623	Low Power #3 LB	Engine		

Table 4: Engine Fault Codes (continued)				
Fault Code	Fault Description	Source	Sent via Orbcomm	Snapshot Trigger
C624	Low Power #4 LB	Engine		
C625	Low Power #5 LB	Engine		
C626	Low Power #6 LB	Engine		
C627	Low Power #7 LB	Engine		
C628	Low Power #8 LB	Engine		
C631	Low Power #1 RB	Engine		
C632	Low Power #2 RB	Engine		
C633	Low Power #3 RB	Engine		
C634	Low Power #4 RB	Engine		
C635	Low Power #5 RB	Engine		
C636	Low Power #6 RB	Engine		
C637	Low Power #7 RB	Engine		
C638	Low Power #8 RB	Engine		
C639	Intake Air Leak LBR	Engine	X	X
C641	High Exh Temp #1 LB	Engine		X
C642	High Exh Temp #2 LB	Engine		X
C643	High Exh Temp #3 LB	Engine		X
C644	High Exh Temp #4 LB	Engine		X
C645	High Exh Temp #5 LB	Engine		X
C646	High Exh Temp #6 LB	Engine		X
C647	High Exh Temp #7 LB	Engine		X
C648	High Exh Temp #8 LB	Engine		X
C649	Change Lubricating Oil and Filter	Engine		
C651	High Exh Temp #1 RB	Engine		X
C652	High Exh Temp #2 RB	Engine		X
C653	High Exh Temp #3 RB	Engine		X
C654	High Exh Temp #4 RB	Engine		X
C655	High Exh Temp #5 RB	Engine		X
C656	High Exh Temp #6 RB	Engine		X
C657	High Exh Temp #7 RB	Engine		X
C658	High Exh Temp #8 RB	Engine		X
C661	High Power #1 LB	Engine		
C662	High Power #2 LB	Engine		
C663	High Power #3 LB	Engine		
C664	High Power #4 LB	Engine		

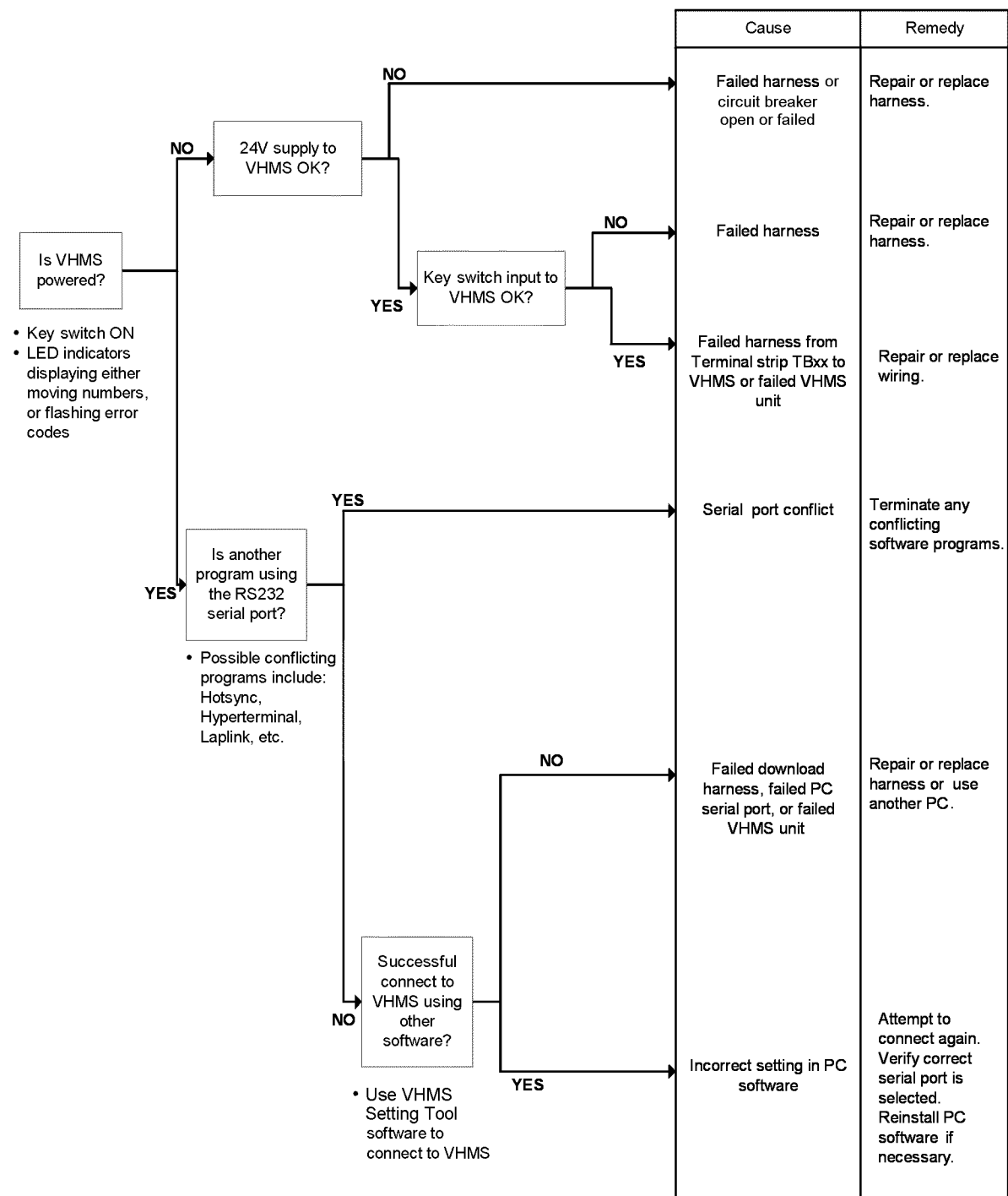
Table 4: Engine Fault Codes (continued)

Fault Code	Fault Description	Source	Sent via Orbcomm	Snapshot Trigger
C665	High Power #5 LB	Engine		
C666	High Power #6 LB	Engine		
C667	High Power #7 LB	Engine		
C668	High Power #8 LB	Engine		
C671	Exh Temp Ckt Failed Low #1 LB	Engine		
C672	Exh Temp Ckt Failed Low #2 LB	Engine		
C673	Exh Temp Ckt Failed Low #3 LB	Engine		
C674	Exh Temp Ckt Failed Low #4 LB	Engine		
C675	Exh Temp Ckt Failed Low #5 LB	Engine		
C676	Exh Temp Ckt Failed Low #6 LB	Engine		
C677	Exh Temp Ckt Failed Low #7 LB	Engine		
C678	Exh Temp Ckt Failed Low #8 LB	Engine		
C694	LBR Turbo Comp Inlet Temp Sensor Ckt Failed High	Engine		
C695	LBR Turbo Comp Inlet Temp Sensor Ckt Failed Low	Engine		
C711	High Power #1 RB	Engine		
C712	High Power #2 RB	Engine		
C713	High Power #3 RB	Engine		
C714	High Power #4 RB	Engine		
C715	High Power #5 RB	Engine		
C716	High Power #6 RB	Engine		
C717	High Power #7 RB	Engine		
C718	High Power #8 RB	Engine		
C719	Blowby Press Ckt Failed High	Engine		
C721	Exh Temp Ckt Failed Low #1 RB	Engine		
C722	Exh Temp Ckt Failed Low #2 RB	Engine		
C723	Exh Temp Ckt Failed Low #3 RB	Engine		
C724	Exh Temp Ckt Failed Low #4 RB	Engine		
C725	Exh Temp Ckt Failed Low #5 RB	Engine		
C726	Exh Temp Ckt Failed Low #6 RB	Engine		
C727	Exh Temp Ckt Failed Low #7 RB	Engine		
C728	Exh Temp Ckt Failed Low #8 RB	Engine		
C729	Blowby Press Ckt Failed Low	Engine		
C753	Cam Sync Error	Engine		
C777	Ambient Derate Error	Engine		

Table 4: Engine Fault Codes (continued)				
Fault Code	Fault Description	Source	Sent via Orbcomm	Snapshot Trigger
C2144	High Exh Temp #9 LB	Engine		
C2145	High Exh Temp #9 RB	Engine		
C2146	Exh Temp Ckt Failed Low #9 LB	Engine		
C2147	Exh Temp Ckt Failed Low #9 RB	Engine		
C2148	High Power #9 LB	Engine		
C2149	High Power #9 RB	Engine		
C2151	Low Power #9 LB	Engine		
C2152	Low Power #9 RB	Engine		
C2154	Post Oil Filter Press Ckt Failed High	Engine		
C2155	Post Oil Filter Press Ckt Failed Low	Engine		
C2157	Rapid Rise in LBR IMT	Engine		
C2158	Rapid Rise in RBF IMT	Engine		
C2159	Rapid Rise in RBR IMT	Engine		
C2241	High IMT LBM	Engine		
C2242	LBM IMT Ckt Failed High	Engine		
C2243	LBM IMT Ckt Failed Low	Engine		
C2244	Rapid Rise in LBM IMT	Engine		
C2245	High IMT RBM	Engine		
C2246	RBM IMT Ckt Failed High	Engine		
C2247	RBM IMT Ckt Failed Low	Engine		
C2248	Rapid Rise in RBM IMT	Engine		

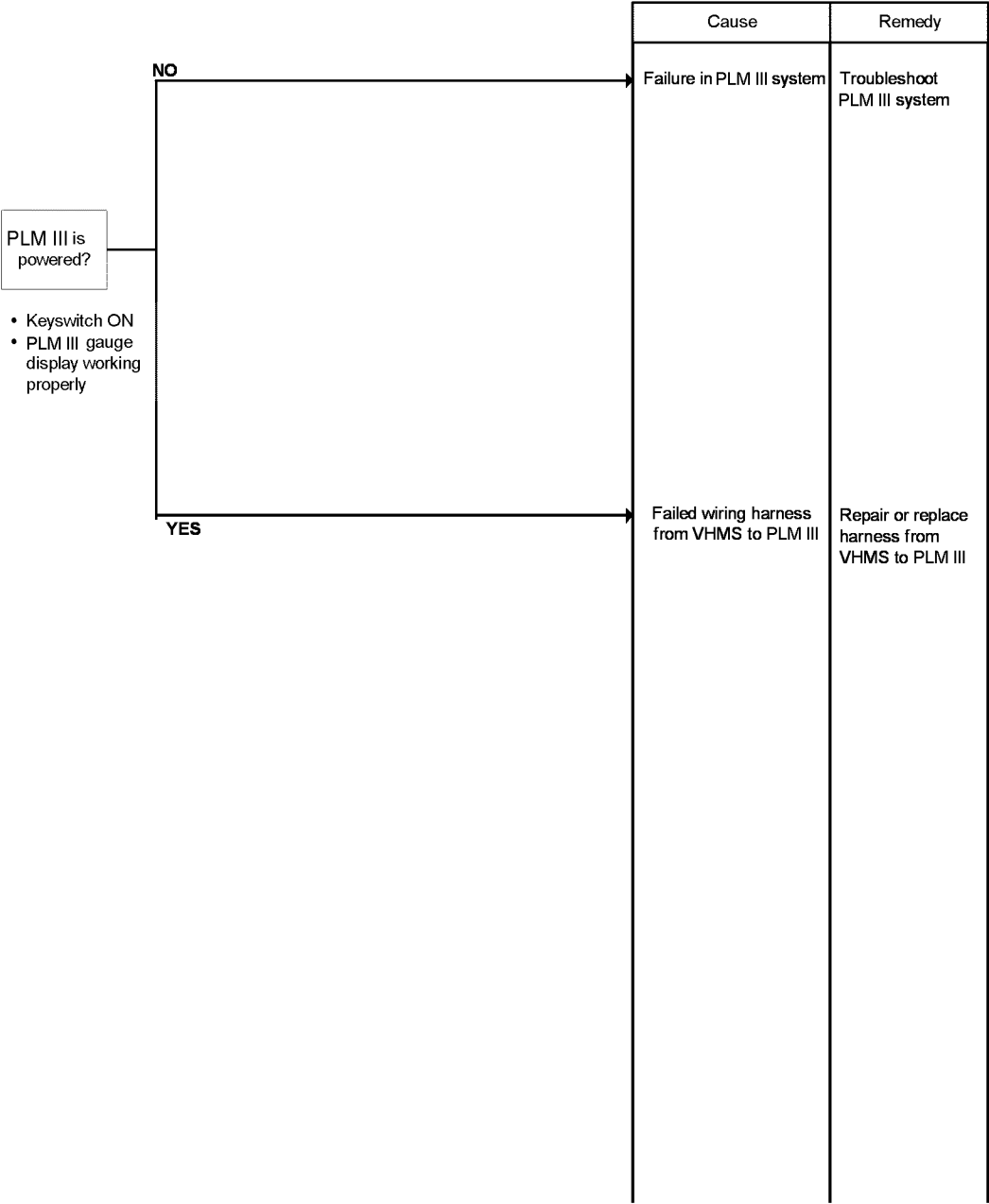
FAULT TREE ANALYSIS

Unable to connect to KOMTRAX Plus from laptop
PC



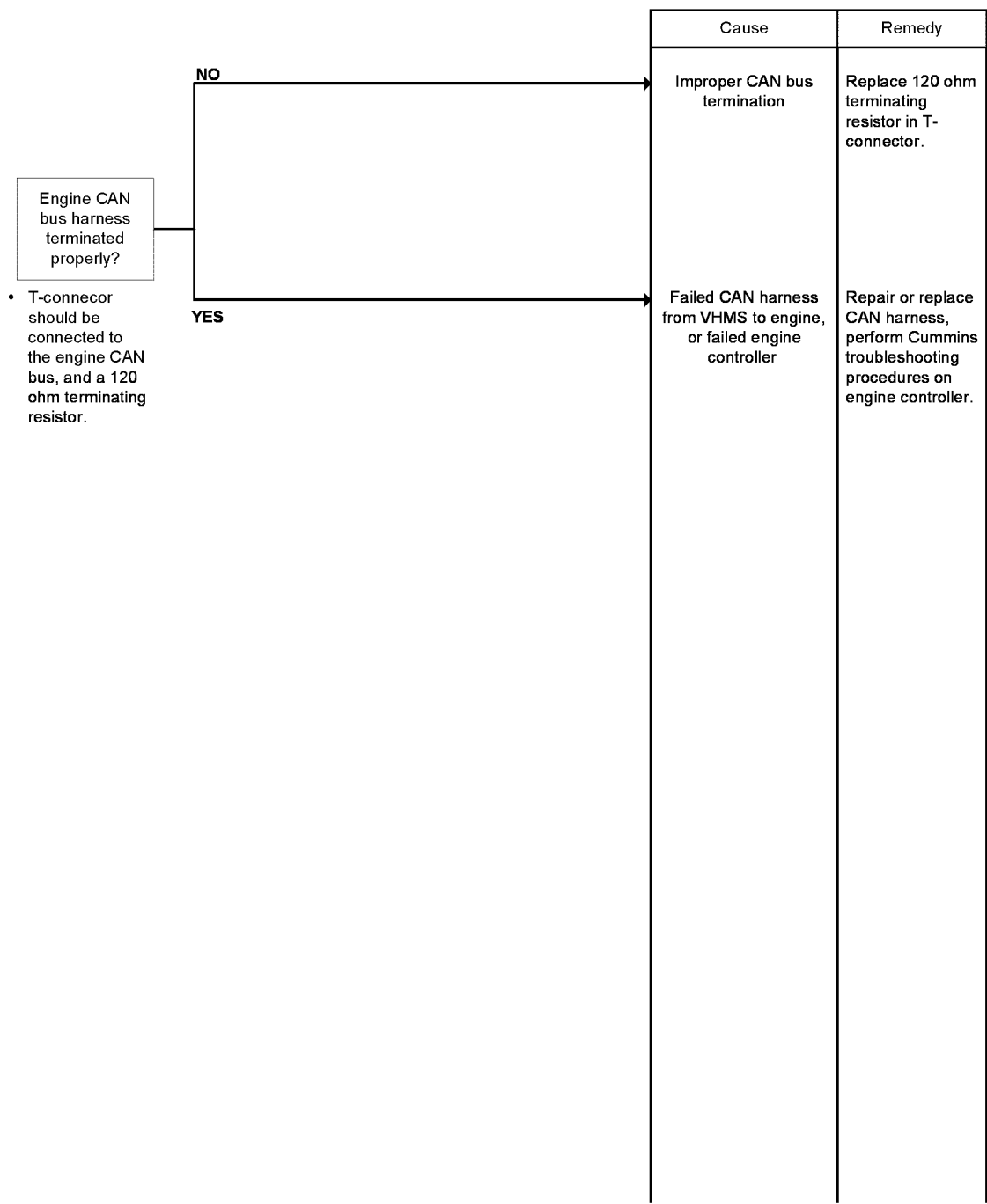
D130009

Flashing Error Code N4-23 (PLM III Communications Fault)



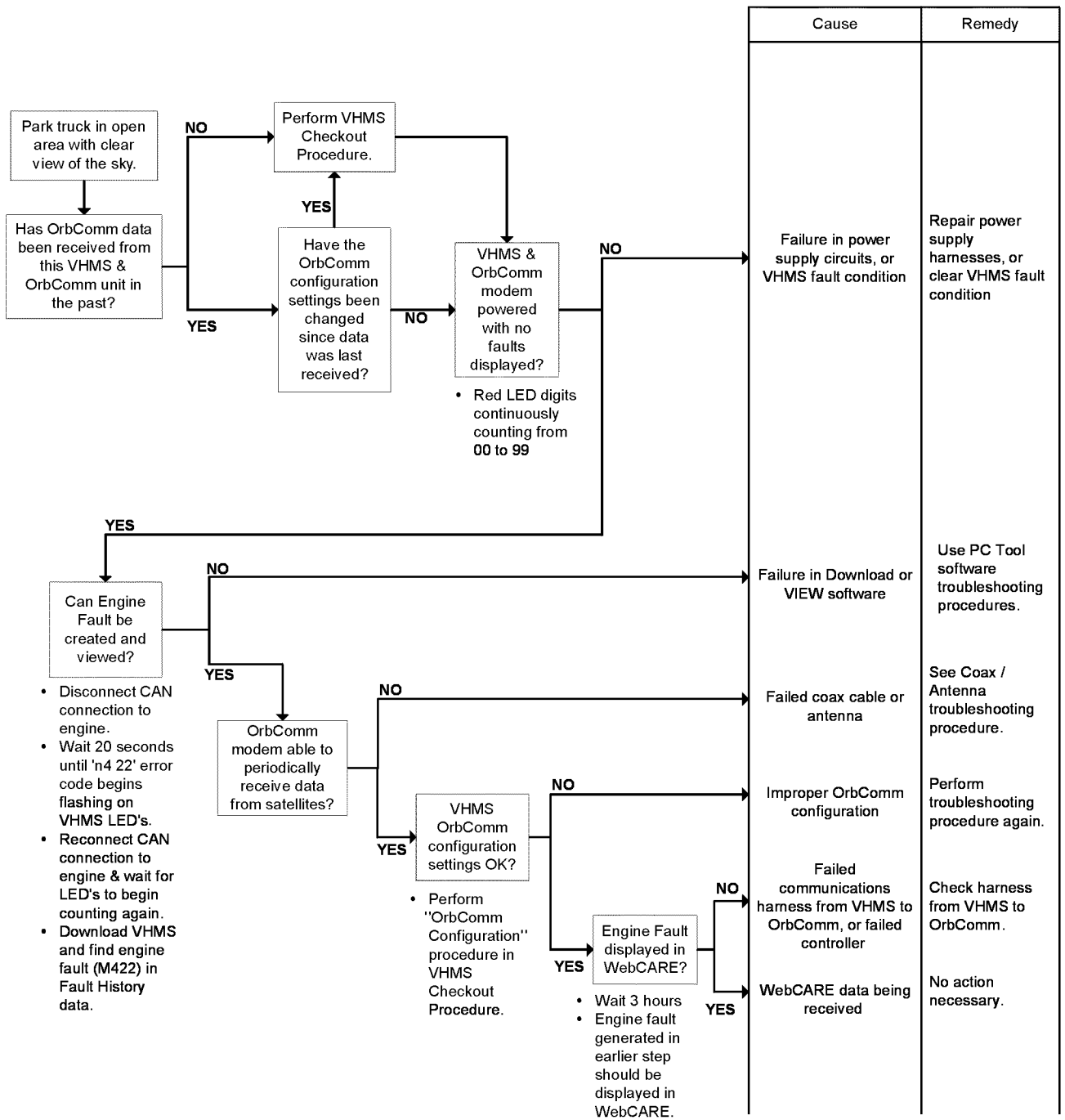
D130010

Flashing Error Code N4-22 (Engine Communications Fault)



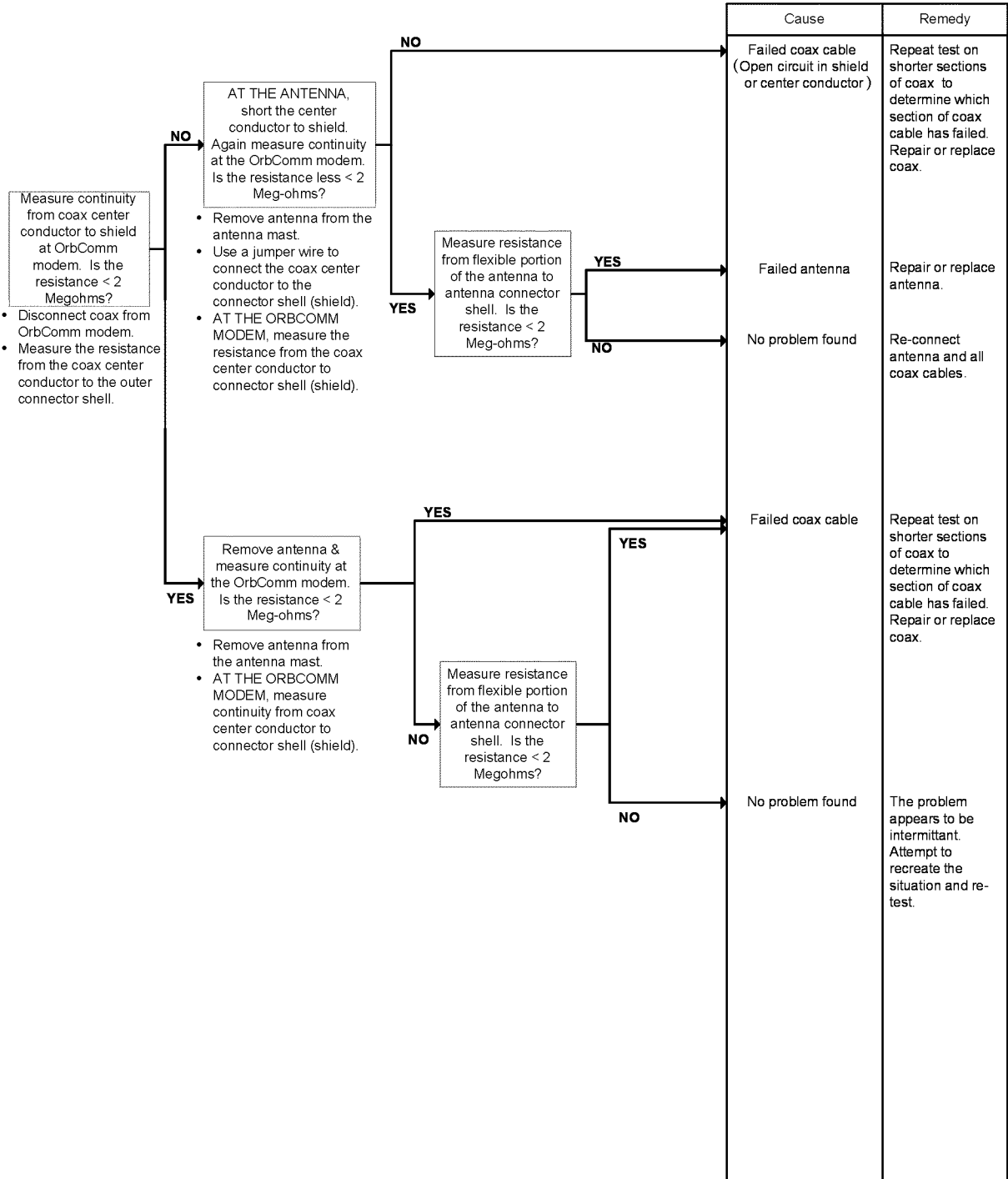
D130011

No Data Received By WebCARE



D130012

Coaxial Cable Troubleshooting



D130013

SECTION D14

KOMTRAX PLUS FORMS

INDEX

KOMTRAX PLUS FORMS	D14-3
KOMTRAX PLUS INITIALIZATION CHECK LIST	D14-3
KOMTRAX PLUS DATA DOWNLOAD	D14-4
KOMTRAX PLUS INITIALIZATION CHECK LIST	D14-5
KOMTRAX PLUS INITIALIZATION FORM	D14-7

NOTES

KOMTRAX PLUS FORMS

The preferred method to submit this form is in electronic format. This check list and initialization form are available in PDF format, where the information can be typed into the form. The form should then be saved using the model, serial number and "KOMTRAX Plus Initialization" to create the file name.

Example: 930E-3SE-A30300-KOMTRAX_Plus Initialization.pdf.

The following instructions will help ensure an accurate and complete form.

When a new machine equipped with KOMTRAX Plus has been assembled, there are several procedures to perform in order to initialize the KOMTRAX Plus system. Following the procedures in the order listed will ensure a smooth initialization process which should not take longer than one hour to complete. Check off each item on the list below as it is done. It is important to complete the entire procedure at one time. Submitting the download data with a date and service meter reading (SMR) that do not match the rest of the forms will not allow the system to be initialized.

1. ___ KOMTRAX Plus Initialization Check List
2. ___ KOMTRAX Plus Data Download
3. ___ KOMTRAX Plus Initialization Form

Orbcomm terminal activation can take up to two weeks. For this reason, it is important to complete these forms and submit them as early as possible after new machine assembly.

KOMTRAX PLUS INITIALIZATION CHECK LIST

This form is used as a check list during the initialization process. Fill in all information. All questions should be answered with a YES. If not, determine the cause and repair as required.

Each machine model will have a different KOMTRAX Plus Initialization Check List. Use the correct form for the model of machine being setup.

1. With the key switch OFF, record all of the data for item 1.
2. Using a serial cable, connect a laptop PC to the KOMTRAX Plus controller.
3. Turn the key switch ON, check operation of the LED lights.
4. Start the *VHMS Setting Tool* program.
 - a. At the Select Operation screen, select the "VHMS Setting" option, then click [Next].
 - b. Select the "Set up & All clear" option, then click [Next].
5. At the Machine Information Setting (1) screen:
 - a. Is Product Group correct?
 - b. Select the correct Machine Model.
 - c. Select the correct Machine Type.
 - d. Select the correct Machine Variation Code.
 - e. Enter the machine serial number.
 - f. Click [Next].
6. At the Machine Information Setting(2) screen:
 - a. Verify information is correct.
 - b. Enter engine serial number.
 - c. Then click [Next].
7. At the Date & Time Setting screen:
 - a. Select the correct time zone.
 - b. Enter the correct date.
 - c. Enter the correct time.
 - d. Select Daylight Savings Time (DST) if necessary.
 - e. Click [Next].
8. At the GCC Setting screen:
 - a. Select the correct country setting.
 - b. Click [Next].

9. At the Verification screen:

- a. Ensure that all information is correct.
- b. Click on [Apply].
- c. At the confirmation screen, select [Yes].
- d. Select [OK].
- e. Select [OK] to close the program.

10. Start the *VHMS Setting Tool* program.

- a. Select "VHMS Setting", then click [Next].
- b. Select "Setup only", then click [Next].
- c. Select "Payload Meter", then click [Next].
- d. Set Start Time to "0".
- e. Set Interval to "1".
- f. Click [Apply].
- g. Click [Exit].

11. Perform a manual snapshot.

- a. With the engine running, press the GE data store switch and hold it for three seconds. The white data store in progress LED should illuminate.
- b. While the manual snapshot is in process, operate the machine if possible. The snapshot lasts for 7 1/2 minutes.

12. After the "data store in progress" LED has been off for one minute, turn the key switch OFF. Wait three minutes before turning the key switch ON.

KOMTRAX PLUS DATA DOWNLOAD

A manual snapshot must be performed before downloading any data. For new machines, this should have been performed during the KOMTRAX Plus Initialization Check List procedure.

1. Perform a KOMTRAX Plus download. For more detailed information on how to perform a download, refer Section D11 in this shop manual.
2. Start the *VHMS Technical Analysis Tool Box* program. Use the view feature to look at the data and verify the settings are correct, the SMR is correct, the manual snapshot is recorded in fault history, and the engine ON/OFF is stored in machine history file.
3. E-mail the downloaded data files to Komatsu America Service Systems Support Team at ServicePrograms@KomatsuNA.com. Refer to Location of Download Files for more detailed instructions on locating the files.

KOMTRAX PLUS INITIALIZATION CHECK LIST

(Page 1 of 2)

FOR: 730E, 830E, 930E & 960E DUMP TRUCKS

Date of set-up	(MM/DD/YY) / /
Distributor and Branch	
Person performing initialization	

Item No.	To be checked when	Check Item	Results	
			Yes	No
1.	Key switch OFF	Machine Model Number		
		Machine Serial Number		
		Service Meter Reading		
		Engine Serial Number		
		Alternator Serial Number		
		KOMTRAX Plus Serial Number		
		Orbcomm Serial Number		
2.	Connect PC to KOMTRAX Plus controller	Are they properly connected?		
3.	Key switch ON	Check operation of controller LED (after segment rotation, display to count-up).		
4.	Start <i>VHMS Setting Tool</i> program	Select "VHMS Setting", then "Set up & All clear".		
5.	Initial setup of KOMTRAX Plus controller Machine Information Setting(1)	Is Product Group correct? (Dump truck)		
		Is Machine Model correct? (ex. 930E)		
		Is Type correct? (ex. -2)		
		Is Variation Code correct? (ex. SE)		
		Is Serial Number correct?		
6.	Machine Information Setting (2)	Is Engine Model - Type correct?		
		Is Engine Serial Number correct?		
7.	Date & Time Setting	Is Time Zone correct?		
		Is Date correct?		
		Is Time correct?		
		Is DST (daylight saving time) correct?		
8.	GCC Setting	Is correct GCC code selected for location?		
9.	Setting Data	Verify Setting Data is Correct.		
10.	Setting of Payload Meter	Set PLM time $\pm$ 2 minutes of KOMTRAX Plus time.		
		Start Time (set to 0)		
		Interval (set to 1)		

KOMTRAX PLUS INITIALIZATION CHECK LIST (Continued)

(Page 2 of 2)

FOR: 730E, 830E, 930E & 960E DUMP TRUCKS

Item No.	To be checked when	Check Item	Result	
			Yes	No
11.	With engine running, perform quick PM with manual snapshot switch.	While recording data, the white LED should be illuminated, indicating snapshot is in recording stage.		
12.	Key switch OFF	Red LED turns off?		
KOMTRAX PLUS DATA DOWNLOAD				
1.	Download data to laptop PC	What time did download start (use wrist watch)?		
		Select all files, and is download complete?		
		Is download start time correct?		
2.	Download Data Check	Settings correct?		
		SMR correct?		
		Manual snapshot recorded and no data missing?		
		Manual snapshot data recorded in fault history, key switch ON/OFF and engine on/off records are saved in machine history file?		
3.	Send download data to Komatsu	Send download data to KAC Service Systems Support at ServicePrograms@komatsuNA.com		

KOMTRAX PLUS INITIALIZATION FORM

This form must be completed and submitted at:

- New machine delivery
- KOMTRAX Plus controller replacement
- Orbcomm controller replacement
- Engine or alternator replacement

Customer Information

- Enter the customer information. All fields are required.

Distributor Information

- Enter the distributor information. All fields are required.
- All distributors are required to have one contact person who is responsible for coordinating KOMTRAX Plus, Payload Meter and Fleet Manager activities for all branches.

Machine Information

- Enter machine information. All fields are required.
- KOMTRAX Plus controller and Orbcomm controller part numbers and serial numbers can be found on a sticker on each controller. Verify that this matches the information displayed in the *VHMS Setting Tool* and download.

KOMTRAX Plus Setting Tool Information

- Enter the setting date. This should be the date when the first data download was taken and the *VHMS Setting Tool* program was first used.
- Enter the setting time. This should be the time shown in the first data download. Verify that it is the correct time.
- Enter the Greenwich Mean Time (GMT) for the location the machine will be working.
- Check whether the location where the machine will be working uses Daylight Savings Time (DST).
- Enter the service meter reading (SMR) at time of the first download.
- Enter the GCC Code. This setting tells the Orbcomm unit which satellite network to communicate with. Select the correct location from the drop down menu list.
- Enter the Orbcomm activation date. In the Orbcomm Activation Date field, enter a date at least two weeks ahead of today's date

Reason for Form Submittal

Check the reason for submitting the KOMTRAX Plus initialization form.

KOMTRAX PLUS INITIALIZATION FORM

NOTE: This form is available in electronic "fill-in" format, which is preferred. If an electronic form is needed, send request to ServicePrograms@KomatsuNA.com. After filling out the form, save the file using the Model Type, Serial Number and "KOMTRAX Plus Initialization" in the file name. (Example: **930E-3SE-A30300-KOMTRAX_Plus Initialization.pdf**),

1. E-mail the completed form to the Service Systems Support Team at ServicePrograms@KomatsuNA.com.
2. Attach the KOMTRAX Plus download files and a copy of the completed machine-specific KOMTRAX Plus Initialization Check List. The E-mail subject line should include the Model-Type, Serial Number, and "KOMTRAX_Plus Initialization".
(Example: **Subject: 930E-3SE-A30300-KOMTRAX_Plus Initialization**)

Customer Information	
Company Name	
Site Name	
Customer Employee Contact	
Mailing Address	
Phone Number	
Fax Number	
E-mail	
Distributor Information	
Distributor Name	
Distributor Service System Support Administrator Name and E-mail	
Distributor Branch	
Distributor Branch Employee Contact and E-mail	
Distributor 4 + 2 Code	
Machine Information	
Machine Model - Type	
Machine Serial Number	
Customer Unit Number	
Engine Serial Number	
Transmission / Alternator Serial Number	
KOMTRAX Plus Controller Part Number	
KOMTRAX Plus Controller Serial Number	
Orbcomm Controller Part Number	
Orbcomm Controller Serial Number	
Setting Tool Information	
Setting Date (MM:DD:YYYY)	
Setting Time (HH:MM:SS)	
GMT (Time Zone)	
Daylight Savings Time (DST) (Yes/No)	
Service Meter Reading (SMR)	
GCC code (Orbcomm satellite)	
Orbcomm Activation Date	
Reason for Form Submittal (Check One)	
Factory Installed KOMTRAX Plus Initialization	
Retrofitted KOMTRAX Plus Initialization	
KOMTRAX Plus Controller Replacement	
Major Component (Engine/Transmission Replacement)	
Customer or Distributor Change	
Setting Tool Information Change	

ELECTRIC PROPULSION SYSTEM COMPONENTS	E2-1
AC DRIVE SYSTEM ELECTRICAL CHECKOUT PROCEDURE	E3-1
TROUBLESHOOTING FAULT CODES (A001-A146)	E5-1
TROUBLESHOOTING FAULT CODES (A152-A216)	E5-1
TROUBLESHOOTING FAULT CODES (A223-A292)	E5-1
TROUBLESHOOTING FAULT CODES (A303-A365)	E5-1

E1-1



DANGEROUS VOLTAGE LEVELS ARE PRESENT WHEN THE TRUCK IS RUNNING AND CONTINUE TO EXIST AFTER SHUTDOWN IF THE REQUIRED SHUTDOWN PROCEDURES ARE NOT FOLLOWED.

Before attempting repairs or working near propulsion system components, the following precautions and truck shutdown procedure must be followed:

- **DO NOT** step on or use any power cable as a handhold when the engine is running.
- **NEVER** open any electrical control cabinet covers or touch the retarding grid elements. Additional procedures are required before it is safe to do so. Refer to “SAFETY”, Section A3, for additional propulsion system safety checks to be performed by a technician trained to service the system.
- **ALL** removal, repairs and installation of propulsion system electrical components, cables, etc, must be performed by an electrical maintenance technician properly trained to service the system.
- **IN THE EVENT OF A PROPULSION SYSTEM MALFUNCTION**, a qualified technician should inspect the truck and verify the propulsion system does not have dangerous voltage levels present before repairs are started.
- **THE LINK VOLTAGE LIGHTS MUST NOT BE ILLUMINATED WHEN TEST OR REPAIRS ARE INITIATED.** It requires approximately 5 minutes after the truck is shut down before the link voltage has dissipated.
- **BEFORE WELDING ON THE TRUCK**, disconnect the ECM harnesses. In the PSC and TCI enclosures, pull cards forward far enough to disconnect the card connector from the backplane connector. Disconnect the battery charging alternator lead wire and open the battery disconnect switches. The welding ground electrode should be attached as close as possible to the area to be welded. **NEVER** weld on the rear of the electrical control cabinet or the retard grid exhaust air louvers. Avoid laying welding cables across or near truck wiring harnesses or power cables. Voltages can be induced in adjacent cables, damaging electrical components.

SECTION E2

ELECTRIC PROPULSION SYSTEM COMPONENTS

INDEX

ELECTRICAL PROPULSION SYSTEM COMPONENTS	E2-3
GENERAL SYSTEM DESCRIPTION	E2-3
SYSTEM COMPONENTS	E2-5
Propulsion System Controller (PSC)	E2-5
Truck Control Interface (TCI)	E2-6
Diagnostic Information Display (DID) Panel	E2-7
DID Panel Event Codes	E2-7
PSC SOFTWARE FUNCTIONS	E2-26
Input Processing	E2-26
State Machine	E2-26
DC Link State	E2-29
Engine Control	E2-30
ALTERNATOR FIELD CONTROL	E2-30
Desired Three-Phase Voltage	E2-30
Desired DC Link Voltage	E2-30
Self-Load	E2-31
Propel Torque Control	E2-31
Retard Torque Control	E2-31
Wheel Slide Control	E2-32
Resistor Grid Control	E2-32
Chopper Voltage Control	E2-32

EVENT DETECTION AND PROCESSING	E2-32
Power-On Tests	E2-32
Initiated Tests	E2-33
Periodic Tests	E2-33
EVENT RESTRICTIONS	E2-33
EVENT LOGGING AND STORAGE	E2-33
Event History Buffer	E2-34
Data Packs	E2-34
To Record and Save a Data Pack to a Disk	E2-34
Event Reset	E2-35
SERIAL DATA COMMUNICATIONS	E2-35
PSC - TCI Communications Processing	E2-35
PSC - PTU Communications Processing	E2-35
Inverter Communications Processing	E2-36
OUTPUT PROCESSING	E2-36
ABNORMAL CONDITIONS/OVERRIDING FUNCTIONS	E2-36
Fast Start	E2-36
Engine Shutdown/Engine Not Running	E2-36
Limp Home Mode	E2-37
PROPULSION SYSTEM COMPONENT ABBREVIATIONS & LOCATIONS	E2-38
ELECTRONIC ACCELERATOR AND RETARD PEDALS	E2-46
Removal	E2-46
Installation	E2-46
Disassembly	E2-46
Assembly	E2-46

ELECTRICAL PROPULSION SYSTEM COMPONENTS

The following information provides a brief description of system operation and major components of the AC propulsion system. Refer to the appropriate GE publication for detailed information and theory of operation.

A list of commonly used propulsion system component abbreviations is listed in Table V at the end of this Section. Figures 2-3 through 2-11 illustrate the physical location of these components where applicable.

GENERAL SYSTEM DESCRIPTION

The AC drive system consists of the following major components:

- Alternator (coupled to a diesel engine)
- In-line Cooling Blower
- Gate Drive Power Converters
- Rectifier Diode Modules
- AC Power Inverters
- AC Induction Traction Motors

The alternator supplies three phase AC power for the gate driver power converters and rectifier diode modules. The rectifier diode modules convert the AC power to DC power, then supply that DC power to two AC power inverters via the DC link. Each AC power inverter inverts the rectified DC voltage, delivering variable voltage, variable frequency power to each of the AC induction traction motors.

NOTE: Refer to Figure 2-1 for the following description.

The two AC induction traction motors, each with its own inverter, are connected in parallel across the rectified output of the alternator. The inverters change the rectified voltage back to AC by turning on and off (*chopping*) the applied DC voltage.

The output AC voltage and frequency are controlled to produce optimum slip and efficiency in the traction motors. At low speeds, the rectified alternator output (DC link or DC bus) voltage is chopped with patterns called pulse width modulation (PWM) inverter operation. At higher speeds, the DC link voltage is applied to the traction motors using square wave inverter operation. The voltage of the DC link is dependent upon the Propulsion System Controller (PSC) and engine RPM during propulsion. The DC link voltage will vary between 600 and 1600 volts.

The alternator field is supplied from a tertiary winding on the alternator and is controlled by a silicon controlled rectifier (SCR) bridge. A starting boost circuit initially energizes the alternator from the truck batteries until the flux builds up enough to sustain excitation.

Cooling air for the alternator, control cabinet and traction motors is supplied by a dual in-line fan assembly mounted on the rear of the alternator. This blower provides cooling air to the traction motors, propulsion inverters, dynamic retarding choppers, and control system.

A resistor grid package is used to dissipate power from the traction motors (operating as generators) when in dynamic retarding mode. The total retard power produced by the traction motors is controlled by the two motor inverters. The amount of retard power dissipated by the grid package is controlled by an IGBT chopper circuit and stage-controlled contactors.

The PSC, which is mounted in the main control cabinet, determines the optimum engine operating speeds based on what the operator requests, propulsion system requirements, and efficient fuel usage. Interfaces between the PSC and the truck brake system allow the PSC to provide proper retarding, braking and wheel slide control.

The PSC interfaces with the Truck Control Interface (TCI), which is mounted in the same card rack as the PSC. System status and control signals are transmitted and received between these two components to access real time data and event information that is stored in the PSC. This data is displayed on the Diagnostic Information Display (DID) panel located in the cab behind the operator's seat.

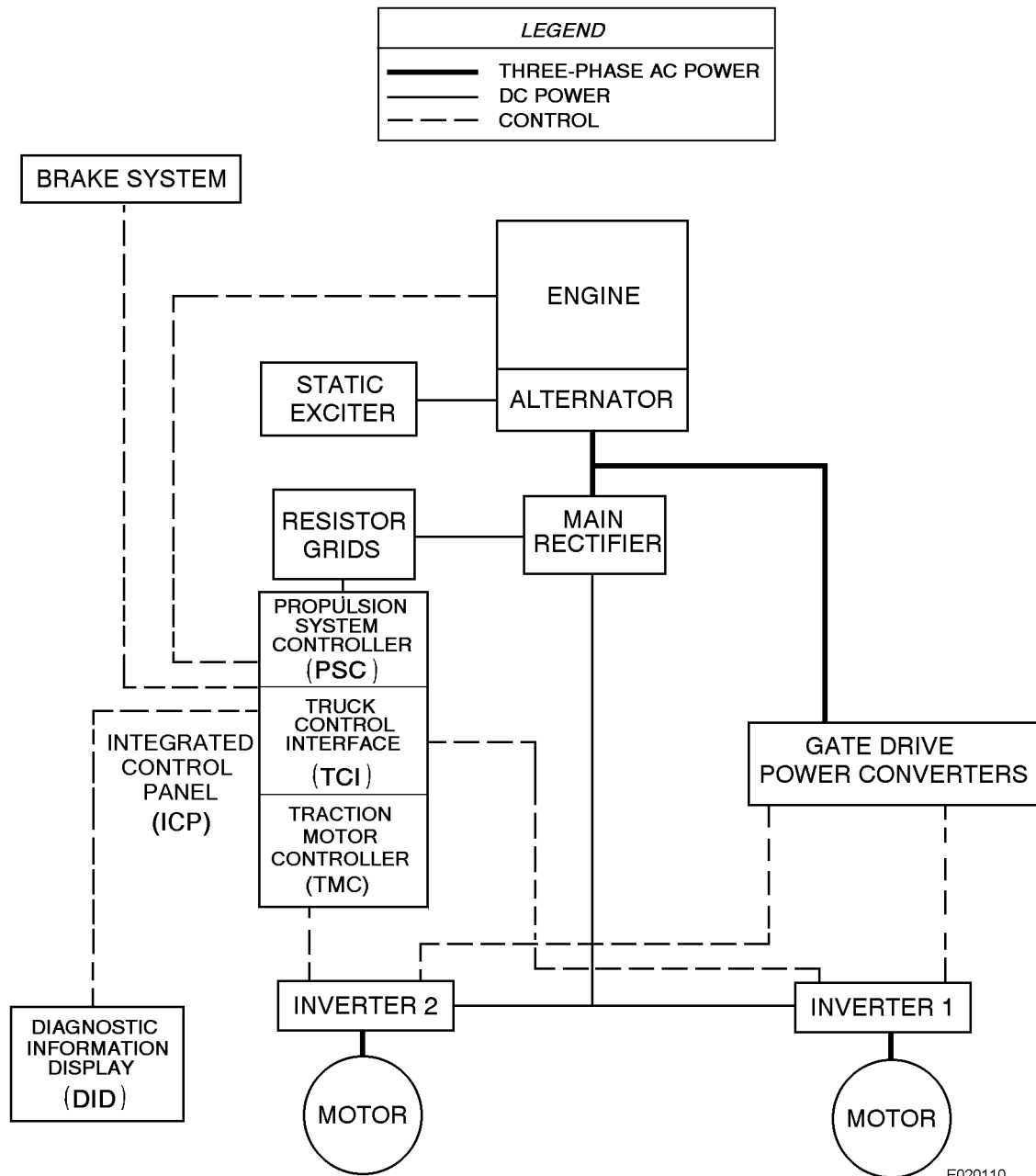


FIGURE 2-1. PROPULSION SYSTEM DIAGRAM

SYSTEM COMPONENTS

The ICP (Integrated Control Panel) consists of three major components: the PSC (Propulsion System Controller), the TCI (Truck Control Interface) and the TMC (Traction Motor Controller).

Propulsion System Controller (PSC)

The PSC is the main controller for the AC drive system. The ICP panel receives input signals from speed sensors mounted on the alternator and traction motors, voltage and current feedback signals from various control devices, and status/command inputs from the TCI. Using these inputs, the PSC controls the two inverters, retarding circuits, relays, contactors, and other external devices to provide the following functions:

- Propulsion and wheel slip control
- Retarding and wheel slide control
- Engine speed control
- Event detection
- Initialization of the necessary operating restrictions, including the shut down of the truck if a serious system fault (event) is detected. If the fault is not serious, an indicator lamp alerting the operator to the problem will illuminate. All event data is recorded for future review by maintenance personnel.
- Log event data
- Store statistical data of the history of various component and system function operations.
- Communicate with the TCI to exchange propulsion system status and control data (event data, statistical data, etc.) and to receive required truck systems status data.
- Communicate with the TCI to exchange propulsion Portable Test Unit (PTU) data (propulsion real time, history, diagnostic, and parameter data such as software code, etc.)
- Drive the operator cab status and warning lamps.

The PSC contains the following internal, removable printed circuit boards and two fiber optic boards:

System CPU Card: Provides serial communications and control functions, RS232 communications to PTU, and microprocessor controls for internal panel circuits.

Digital I/O Card: Receives digital inputs and feedback information from various propulsion and control system components. Digital outputs drive propulsion system contactors, relays and provide equipment enable commands.

System Analog I/O Card: Receives engine, voltage and current signals for the main alternator, link voltage and current, retard pedal input, and retard lever input. Controls retard effort, engine speed request, and AFSE firing pulses.

Inverter 1 & 2 CPU and I/O Cards (2 ea.): Receives motor speed signals, link voltage, phase voltages, and phase currents for microprocessor control for inverters 1 and 2. Controls IGBT phase modules through the fiber optic assembly. Phase module status is returned via a separate fiber optic assembly.

Fiber Optic Assembly: Provides electrical isolation for control and feedback signals for phase modules and chopper modules.

Truck Control Interface (TCI)

The TCI is the main interface between the truck systems/devices and service personnel. This panel is used in conjunction with the DID panel.

The TCI panel provides the following functions:

- Communicates with the PSC to exchange propulsion control system status and control data and to provide the PSC with truck systems status data.
- Communicates with the DID panel to exchange PSC and/or TCI diagnostic and parameter data.
- Communicates with a PTU to exchange TCI data.
- Communicates with a Modular Mining Dispatch System to exchange truck status data.
- Monitors engine control system, payload information, ambient and propulsion system temperature, operator control inputs, etc.
- Controls the engine start sequence.
- Provides signals to activate many of the cab mounted warning lamps and gauges. Controls the parking brake solenoid.
- Processes the front wheel speed signals for the PSC and speedometer.

The TCI contains the following internal, removable printed circuit boards:

CPU Card: Provides high speed communications to PSC and RS232 serial communication with the PTU.

Analog I/O Card: Provides RS232 serial communications with the DID and an optional Modular Mining Dispatch System. Receives signals for front wheel speed, motor cooling and barometric air pressures, accelerator, retard speed setting, payload, ambient and hydraulic oil temperature, and engine cranking voltage. Outputs drive the cab mounted temperature gauges.

Digital I/O Card: Receives operator control, engine and body-up signals. Provides engine start controls and drives the cab mounted indicator/warning lamps.

Diagnostic Information Display (DID) Panel

The DID panel (Figure 2-2) is located in the cab behind the operator's seat. The display provides service personnel with a means of communicating with the TCI.

The panel has two display lines. Each line is 40 characters long. The top line is the "message" line and is used by the TCI to inform service personnel of the truck systems and components status.

The bottom display line provides information in addition to the top line or relates to the keypad, displaying possible selection options and display functions. The keypad, located below the display lines, is used by service personnel to direct the activity of the TCI.

The display provides service and status information on the various truck systems and the propulsion system by displaying system status information or fault codes, as well as a description of the system status or a problem on the top display line. Information on the second display line may change to indicate which functions are available by pressing keys [F1] through [F5].

The DID panel can also be used to perform the self-load test.

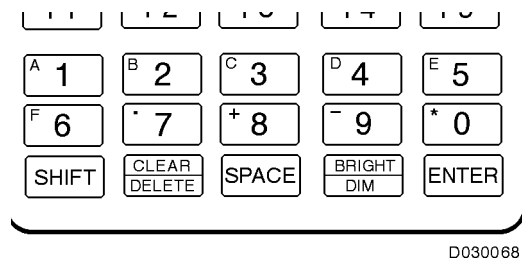


FIGURE 2-2. DIAGNOSTIC INFORMATION DISPLAY

NOTE: In addition to displaying English text, the DID panel can also display messages in Russian or Spanish.

DID Panel Event Codes

The tables on the following pages list the possible event codes which may be displayed on the DID panel when accessed. Table 1 describes restrictions to operation of the propulsion and retarding systems when a fault occurs for a particular code listed in Tables 2, 3 and 4.

Event codes numbered 000 through 099 are applicable to the PSC and are listed in Table 2. Codes numbered 100 through 199 are applicable to Inverter 1, and codes numbered 200 through 299 are applicable to Inverter 2. These are listed in Table 3. Codes numbered 600 through 699 are applicable to the TCI and are listed in Table 4. The codes listed in the Tables are applicable to Release 21 software.

Table 1: Event Restrictions	
RESTRICTION	DEFINITION
No Power	NO RETARD (red) light illuminates. No retarding allowed. No propulsion allowed. No power on the link.
No Propel	NO PROPEL (red) light illuminates. No propulsion allowed. Retarding allowed. Link power allowed.
Speed Limit	PROPEL SYSTEM CAUTION<170> (amber) light illuminates. Propel, retard and DC link power still allowed. Speed limited to 10 MPH (16 KPH).
INV1 Disable	Prohibits system from enabling inverter #1 drive signal.
INV2 Disable	Prohibits system from enabling inverter #2 drive signal.
Engine Speed/ RP1	Raises engine speed to account for a possible stuck RP contactor. Closes RP1.
SYS Event	No restrictions. Event is for information purposes only.

Table 2: DID PANEL FAULT CODES
(Codes Received from PSC)

EVENT NUMBER		EVENT DESCRIPTION	EVENT RESTRICTION	DETECTION INFORMATION
000		NO FAULT	None	Displayed when all faults have been reset
002		GROUND FAULT	No power	A ground fault has been detected: For voltage < 1000 V, detection threshold is 166 mA For voltage >= 1000 V, detection threshold ramps from 166 mA at 1000 V down to approximately 70 mA at approximately 1500 V.
003		FAILED DIODE	No power	Failed diode(s) in main rectifier
004		GFCO OPEN and not in REST	None	GF Cutout Switch is open with the system not in REST.
005		DRIVE SYSTEM OVERTEMP	No Propel	Temperature exceeds a limit for a sufficient time.
	:01	auxiliary phase control		
	:02	auxiliary inverter		
	:03	afse		
	:04	alternator		
	:05	left stator		
	:06	left rotor		
	:07	right stator		
	:08	right rotor		
	:09	chopper IGBT		
	:10	chopper diode		
	:11	left IGBT module		
	:12	left diode		
	:13	right IGBT module		
	:14	right diode		
	:15	rectifier diode		
006		BOTH INVERTERS COMMUNICATION FAILED	No power	Lost communication with both inverters
008		DC LINK OVERVOLTAGE	No power	DC link voltage exceeds limit for a sufficient time.
	:01	not in retard		Occurs while not in retard, exceeds propel voltage limit
	:02	in retard		Occurs while in retard, exceeds retard voltage limit
	:03	instantaneous		Occurs instantaneously in propel or retard, exceeds link voltage limit
009		ALT FIELD OVERCURRENT	No power	Alternator field current exceeds limit.
	:01	normal		Exceeds current limit over time
	:02	instantaneous		Exceeds current limit with no persistence
	:03	persistent		With persistence due to low engine speed
011		RETARD LEVER BAD	None	Incorrect input from retard lever
	:01	voltage too high		
	:02	voltage too low		
012		RETARD PEDAL BAD	None	Incorrect input from retard pedal
	:01	voltage too high		
	:02	voltage too low		
013		LINKV TEST FAILED	No power	Incorrect link volts

Table 2: DID PANEL FAULT CODES
(Codes Received from PSC)

EVENT NUMBER		EVENT DESCRIPTION	EVENT RESTRICTION	DETECTION INFORMATION
014		ANALOG SENSOR FAULT	Speed limit	Incorrect input from a sensor
	:01	alt field amps		
	:02	link amps		
	:03	load box amps		
	:04	3 phase alt volts		
	:05	alt field volts		
	:10	PSC link volts		
	:11	inv1 link volts		
	:12	inv2 link volts		
	:13	A2D ground		
	:14	A2D gain		
	:15	fault current		
	:16	ATOC		
	:21	grid blower 1 amps		
	:22	grid blower 2 amps		
015		ANALOG SENSOR FAULT (restrictive)	Speed limit	Incorrect input from a sensor
	:02	link amps		
016		PSC CPU CARD (FB147)	No power	Problem has occurred in the system CPU card.
	:01	task_1		Failed to initialize
	:02	task_2		
	:03	task_3		
	:04	task_4		
	:05	task_5		
	:06	task_6		
	:07	maintenance task		
	:09	flash CRC		Flash CRC computation did not match expected value.
	:10	BRAM CRC		CRC on BRAM does not match expected value.
	:11	excess timeouts		On power up, excessive timeouts occurred.
	:12	invalid pointers (data pack corrupted)		On power up, the status of data in BBRAM is invalid.
017		DIGITAL I/O CARD FAULT (FB104)	No power	System CPU cannot communicate with digital I/O card.
018		ANALOG I/O CARD FAULT (FB173)	No power	System CPU cannot communicate with analog I/O card.
	:01	analog card no response		Card missing
	:02	analog card timeout		Read timeout
019		RIDING RETARD PEDAL	SYS Event	Brake pedal applied while truck speed is >5 mph
020		LO SPEED HI TORQUE TIMEOUT	No propel	Torque limit exceeded

Table 2: DID PANEL FAULT CODES
(Codes Received from PSC)

EVENT NUMBER		EVENT DESCRIPTION	EVENT RESTRICTION	DETECTION INFORMATION
021		TCI COMM. FAULT	No propel	PSC received no serial data from TCI over period of time.
	:01	Message missing		
	:02	Bad tick		
	:03	Bad CRC		
	:04	Overflow		
	:05	Bad start		
	:06	Bad stop		
022		PERSISTENT TCI COMM FAULT	No power	No serial data received from TCI and truck is stopped for 10 seconds.
023		TERTIARY OVERCURRENT	No propel	Current in alternator field tertiary winding exceeds limit over time.
024		PSC CONFIG FILE INCORRECT	No power	Incorrect or missing PSC configuration file
	:01	no file		No configuration file selected
	:02	bad CRC		
	:03	wrong version		Wrong configuration file version
	:04	overspeeds incorrect		Incorrect overspeed values
025		AUX INVERTER FAULT	No power	Auxiliary blower system fault
	:01	not ok or no speed feedback		Auxiliary speed feedback indicates no or incorrect blower speed.
	:02	numerous shutdowns		Auxiliary OK goes low twice when speed command is greater than running speed.
026		CAPACITOR OVERPRESSURE	No power	Excessive filter cap pressure
	:01	INV1	No power	INV1 capacitor
	:02	INV2	No power	INV2 capacitor
027		PSC PANEL CONNECTOR	No power	A panel connector B, C, or D is not properly connected.
	:01	CNFB		
	:02	CNI/CNX (3500 HP, 150 TON)		
	:03	Aux blower connector		
030		GF CONTACTOR	Speed limit	GF command/feedback don't agree.
031		BATTERY BOOST CIRCUIT	Speed limit	
	:01	GFR failed to open		GFR command/ feedback don't agree.
	:02	GFR failed to close		
	:03	SCR3 failed		
032		RP CONTACTOR	Speed limit & engine speed/RP	RP command/ feedback don't agree.
	:01	RP1		
	:02	RP2		
	:03	RP3		
033		RETARD CIRCUIT	Speed limit & engine speed/RP	
035		ESS INPUT	Speed limit	Engine speed sensor is out of range.

Table 2: DID PANEL FAULT CODES
(Codes Received from PSC)

EVENT NUMBER		EVENT DESCRIPTION	EVENT RESTRICTION	DETECTION INFORMATION
036		GY19 GRID BLOWER FAILURE	No power	A grid blower has failed.
	:01	blower 1 stall		
	:02	blower 2 stall		
	:03	blower 1 open		
	:04	blower 2 open		
	:05	blower 1 & 2 delta too large		
037		COMPUTER POWER SUPPLY	Speed limit	
	:01	VOLTS 5 POS		+5V power supply is out of limits.
	:02	VOLTS 15 POS		+15V power supply is out of limits.
	:03	VOLTS 15 NEG		-15V power supply is out of limits.
040		VOLTS 24 POS		+24V power supply is out of limits.
041		VOLTS 24 NEG		-24V power supply is out of limits.
042		DIRECTION SELECTED IN LOAD BOX MODE	No propel	Selector switch moved to FORWARD or REVERSE during self load.
043		DRIVE SYSTEM BATTERY LOW	Speed limit	Battery volts are below limit.
044		DRIVE SYSTEM BATTERY HIGH	None	Battery volts are above limit.
045		CHOPPER OPEN CIRCUIT	Speed limit	Open circuit in a chopper
	:01	chopper 1		Open circuit in chopper 1
	:02	chopper 2		Open circuit in chopper 2
046		RETARD SHORT CIRCUIT	Speed limit & engine speed	Failure during chopper self test. Link voltage decayed too quickly when AFSE command set low, prior to starting test.
047		ENGINE STALL	No power	An engine stall condition has occurred.
048		SHORTED DC LINK	No power	DC link short detected at startup.
051		TACH LEFT REAR	INV1 disable	Input from M1 sensor is out of tolerance.
	:01	zero output with truck moving		Zero output from sensor with front wheels moving, brake released.
	:02	high output with truck stopped		High output from sensor with all other wheel speeds at zero.
052		TACH RIGHT REAR	INV2 disable	Input from M2 sensor is out of tolerance.
	:01	zero output with truck moving		Zero output from sensor with front wheels moving, brake released.
	:02	high output with truck stopped		High output from sensor with all other wheel speeds at zero.
053		TACH LEFT FRONT	SYS Event	Input from left front wheel sensor is out of tolerance.
	:01	zero output with truck moving		Zero output from sensor with rear wheels moving, brake released.
	:02	high output with truck stopped		High output from sensor with all other wheel speeds at zero.
054		TACH RIGHT FRONT	SYS Event	Input from right front wheel sensor is out of tolerance.
	:01	zero output with truck moving		Zero output from sensor with rear wheels moving, brake released.
	:02	high output with truck stopped		High output from sensor with all other wheel speeds at zero.

Table 2: DID PANEL FAULT CODES
(Codes Received from PSC)

EVENT NUMBER		EVENT DESCRIPTION	EVENT RESTRICTION	DETECTION INFORMATION
055		FRONT WHEEL TACHS	SYS Event	
056		INVERTER SW VERSION	SYS Event	Incorrect version of Inverter Software is installed.
	:01	Inverter # 1		
	:02	Inverter # 2		
061		MOTOR OVERSPEED	SYS Event	Truck is over the motor overspeed limit.
063		ENGINE LOAD SIGNAL	SYS Event	Engine load out of range.
	:01	below minimum		
	:02	above maximum		
	:03			PWM signal failed low.
	:04			PWM signal failed high.
	:05			PWM signal failed incorrect period.
065		TEMP INPUT RANGE CHECK	Speed limit	An analog input is outside the design range of valid values.
	:01	aux pc temp sensor		Auxiliary phase controller temperature sensor
	:02	aux inv temp sensor		Auxiliary inverter temperature sensor
	:03	afse temp sensor		AFSE temperature sensor
	:04	alternator temp		Temperature is out of range.
	:05	left stator temp		
	:06	left rotor temp		
	:07	right stator temp		
	:08	right rotor temp		
	:09	chopper IGBT temp		
	:10	chopper diode temp		
	:11	left IGBT module temp		
	:12	left diode temp		
	:13	right IGBT module temp		
	:14	right diode temp		
	:15	rectifier diode temp		
070		LINK CAPACITANCE LEVEL LOW	SYS Event	Link capacitance level is low, but OK.
071		LINK CAPACITANCE LEVEL TOO LOW	Speed Limit	Link capacitance level is too low.
072		GROUND FAULT CIRCUIT	Speed Limit	Ground fault detection circuit
074		INV1 COMM FAILED	INV1 Disable	
	:01	No communication Inverter #1		
	:02	Inverter #1 customer option bit		
075		INV2 COMM FAILED	INV2 Disable	
	:01	No communication Inverter #2		
	:02	Inverter #2 customer option bit		

Table 2: DID PANEL FAULT CODES
(Codes Received from PSC)

EVENT NUMBER		EVENT DESCRIPTION	EVENT RESTRICTION	DETECTION INFORMATION
076		FB173 CARD	No power	FB173 card failure
	:01	speed FPGA DL		
	:02	speed FPGA run		
	:03	ALT FPGA DL		
	:04	Microcontroller		
	:05	slow task		
	:06	med task		
	:07	fast task		
	:08	FD task		
	:09	Alternator 3 phase volts bad		
	:10	alt FPGA timeout		
077		INVERTER FAILED VI TEST	No power	Inverter failed during test.
078		Inverter Background Communication Failure	Sys Event	A failure in the inverter background communication was detected.
084		CONTROL POWER SWITCH OFF	SYS Event	Control power switch is turned off while truck is moving.
085		AUX COOLING	SYS Event	A fault has occurred in the auxiliary blower operation.
	:02	aux rpmfb input		Rpm of Aux Blower out of range.
	:03	aux rpm feedback		Rpm feedback does not match rpm command.
	:04	abnormal shutdown		A fault occurred during shutdown
087		HP LOW	SYS Event	Horsepower adjust is at negative limit for 30 seconds.
088		HP LIMIT	SYS Event	Horsepower limit exceeded while in propulsion.
089		ENGINE SPEED DOES NOT MATCH COMMAND	SYS Event	Engine speed feedback does not match commanded speed.
	:02	RPM does not match command		
091		INVERTER 1 CUTOUT	SYS Event	
092		INVERTER 2 CUTOUT	SYS Event	
094		ILLEGAL LIMP REQUEST	SYS Event	A "limp mode" request is received while truck is moving.
095		BAD BRAM BATTERY	SYS Event	BRAM battery voltage is low.
096		UNEXPECTED PSC CPU RESET	SYS Event	PSC CPU reset without request.
098		DATA STORE	SYS Event	PTU data store command

Table 3: DID PANEL FAULT CODES
(Codes Received from Inverter 1 & 2)

EVENT NUMBER	EVENT DESCRIPTION	EVENT RESTRICTION	DETECTION INFORMATION
100/200	INVERTER CPU CARD (FB172)	INV1 (INV2) off	
:23	pat fail out 100		Pattern had bad A, B, C output 100%.
:29	no extvi TIC		Extrapolation interrupt not running
:30	no vector TIC		Vector interrupt not running
:31	no I TIC TIC		I TIC interrupt not running
:32	NMI occurred		Non-maskable interrupt occurred.
:34	no background TIC		Background not running
:35	PGA not programmed		PGA could not be programmed.
:38	PGA init failed		PGA initialization failed.
:39	PGA DP failed		PGA D/P did not initialize.
:40	par not found		Parameter not found
:41	multiple par		Parameter multiply defined
:48	no cam TIC		Cam ISR not running
:49	no peak samp TIC		Peak sample ISR not running
101/201	INVERTER CPU CARD (NR)	INV1 (INV2) off	
:01	Aup cmd not off		Phase A up command not off
:02	Adn cmd not off		Phase A down command not off
:03	Bup cmd not off		Phase B up command not off
:04	Bdn cmd not off		Phase B down command not off
:05	Cup cmd not off		Phase C up command not off
:06	Cdn cmd not off		Phase C down command not off
:07	Aup cmd not on		Phase A up command not on
:08	Adn cmd not on		Phase A down command not on
:09	Bup cmd not on		Phase B up command not on
:10	Bdn cmd not on		Phase B down command not on
:11	Cup cmd not on		Phase C up command not on
:12	Cdn cmd not on		Phase C down command not on
:13	no chopper TIC1		Chopper 1 interrupt not running
:14	no chopper TIC2		Chopper 2 interrupt not running
:16	inv CPU reset		Inverter CPU was reset.

Table 3: DID PANEL FAULT CODES
(Codes Received from Inverter 1 & 2)

EVENT NUMBER		EVENT DESCRIPTION	EVENT RESTRICTION	DETECTION INFORMATION
102/202		INV I/O CARD (FB172)	INV1 (INV2) off	
	:05	gnd not ok		Logic ground not OK
	:08	no IO card		Could not access I/O card
	:09	eoc not working		A/D conversion did not work.
	:10	DB no brake		DB on too long while not braking
	:11	ptf A signal		Phase A overcurrent signal too long
	:12	ptf B signal		Phase B overcurrent signal too long
	:13	ptf C signal		Phase C overcurrent signal too long
	:14	IC zero not ok		Current IC not zero at start up
	:15	IC not ok		C phase current too high
	:16	ptl not ok		Protective turn off circuit not OK
	:17	cur measure not ok		Phase A and B currents do not match.
103/203		INV I/O CARD (NR)	None	
	:01	chop 1 cmd not off		Chopper 1 command not off
	:02	chop 2 cmd not off		Chopper 2 command not off
	:03	chop 1 cmd not on		Chopper 1 command not on
	:04	chop 2 cmd not on		Chopper 2 command not on
	:05	volt scale A flt		Scale A volts out of range 70%, 100%
	:06	volt scale B flt		Scale B volts out of range 70%, 100%
	:07	link V scale flt		Link V scale out of range 70%, 100%
	:08	current scale A flt		Scale A current out of range 70%, 100%
	:09	current scale B flt		Scale B current out of range 70%, 100%
	:10	input V scale flt		Input V scale out of range 70%, 100%
	:11	V test VCO high		High frequency on VCO Vtest channel
	:12	V test VCO low		Low frequency on VCO Vtest channel
	:13	IA VCO hi		High frequency on IA channel
	:14	IB VCO hi		High frequency on IB channel
	:15	link V VCO hi		High frequency on VCO link filter V channel
	:16	infilV VCO hi		High frequency on VCO in filter V channel
	:17	IA too high		IA current too positive
	:18	IA too low		IA current too negative
	:19	IB too high		IB current too positive
	:20	IB too low		IB current too negative
	:21	link V too hi		Link voltage too positive
	:22	infilV too hi		Input filter voltage too positive
	:23	DB chop VCO hi		High frequency on VCO DB chopper channel

Table 3: DID PANEL FAULT CODES
(Codes Received from Inverter 1 & 2)

EVENT NUMBER	EVENT DESCRIPTION	EVENT RESTRICTION	DETECTION INFORMATION
103/203	INV I/O CARD (NR)	None	
:24	DB chopV too hi		DB chopper voltage too positive
:25	VA VCO hi		High frequency on VCO VA channel
:26	VB VCO hi		High frequency on VCO VB channel
:27	VC VCO hi		High frequency on VCO VC channel
:28	VA volts too hi		VA voltage too positive
:29	VB volts too hi		VB voltage too positive
:30	volt scale C flt		Scale C volts out of range 70%, 120%
:31	VC volts too hi		VC voltage too positive
104/204	FIBER OPTIC CARD	INV1 (INV2) off	
:01	fo ps low		Fiber optic power supply monitor
:02	fo card disable		Fiber optic card disabled
:03	fo card enable		Fiber optic card enabled and no dir
105/205	POWER SUPPLY CARD	INV1 (INV2) off	
:01	P5V not ok		+5 volt not in tolerance
:02	P15V not ok		+15 volt not in tolerance
:03	N15V not ok		-15 volt not in tolerance
:06	P24V not ok		+24 volt not in tolerance
:07	N24V not ok		-24 volt not in tolerance
106/206	DC WIRING	INV1 (INV2) off	
:01	DC pwr conn open		DC power connection is open.
:02	link V phase V mismatch		Link and phase voltage are mismatched.
107/207	GDPS FAILURE	SYS Event INV1 (INV2) off	
:01	gate dr ps off		No power to gate drive power supply or it failed
:02	gate dr ps off S		No power to gate drive power supply or it failed with enable/DC volts
:03	multiple IGBT not off S		Multiple IGBTs not off with enable/DC volts
109/209	LINK VOLTS SENSOR	INV1 (INV2) off	
:01	linkV sensor flt		Link voltage sensor failed
111/211	INPUT VOLTS SENSOR	INV1 (INV2) off	
:01	Vfil not ok		Filter voltage outside limits

Table 3: DID PANEL FAULT CODES
(Codes Received from Inverter 1 & 2)

EVENT NUMBER		EVENT DESCRIPTION	EVENT RESTRICTION	DETECTION INFORMATION
113/213		INVERTER, GENERAL	INV1 (INV2) off	
	:01	Aup cur hi		Phase A current out too high
	:02	Adn cur hi		Phase A current in too high
	:03	Bup cur hi		Phase B current out too high
	:04	Bdn cur hi		Phase B current in too high
	:05	Cup cur hi		Phase C current out too high
	:06	Cdn cur hi		Phase B current in too high
	:07	Aup cur lo		Phase A current out too low
	:08	Adn cur lo		Phase A current in too low
	:09	Bup cur lo		Phase B current out too low
	:10	Bdn cur lo		Phase B current in too low
	:11	Cup cur lo		Phase C current out too low
	:12	Cdn cur lo		Phase C current in too low
	:13	A zero cur hi		Phase A current out not zero
	:15	B zero cur hi		Phase B current out not zero
	:17	A volt hi Adn		Phase A volt too high while phase A down on
	:18	A volt lo Aup		Phase A volt too low while phase A up on
	:19	A volt hi Bdn		Phase A volt too high while phase B down on
	:20	A volt lo Bup		Phase A volt too low while phase B up on
	:21	A volt hi Cdn		Phase A volt too high while phase C down on
	:22	A volt lo Cup		Phase A volt too low while phase C up on
	:23	B volt hi Adn		Phase B volt too high while phase A down on
	:24	B volt lo Aup		Phase B volt too low while phase A up on
	:25	B volt hi Bdn		Phase B volt too high while phase B down on
	:26	B volt lo Bup		Phase B volt too low while phase B up on
	:27	B volt hi Cdn		Phase B volt too high while phase C down on
	:28	B volt lo Cup		Phase B volt too low while phase C up on
	:29	C volt hi Adn		Phase C volt too high while phase A down on
	:30	C volt lo Aup		Phase C volt too low while phase A up on
	:31	C volt hi Bdn		Phase C volt too high while phase B down on
	:32	C volt lo Bup		Phase C volt too low while phase B up on
	:33	C volt hi Cdn		Phase C volt too high while phase C down on
	:34	C volt lo Cup		Phase C volt too low while phase C up on
	:35	Aup fault cur		Phase A fault current when phase A up on
	:36	Adn fault cur		Phase A fault current when phase A down on
	:37	Bup fault cur		Phase B fault current when phase B up on
	:38	Bdn fault cur		Phase B fault current when phase B down on
	:39	Cup fault cur		Phase C fault current when phase C up on
	:40	Cdn fault cur		Phase C fault current when phase C down on
	:48	A volt hi off		Phase A voltage high with all IGBTs off
	:49	A volt lo off		Phase A voltage low with all IGBTs off

Table 3: DID PANEL FAULT CODES
(Codes Received from Inverter 1 & 2)

EVENT NUMBER		EVENT DESCRIPTION	EVENT RESTRICTION	DETECTION INFORMATION
113/213		INVERTER, GENERAL	INV1 (INV2) off	
	:50	B volt hi off		Phase B voltage high with all IGBTs off
	:51	B volt lo off		Phase B voltage high with all IGBTs off
	:52	C volt hi off		Phase C voltage high with all IGBTs off
	:53	C volt lo off		Phase C voltage high with all IGBTs off
	:54	phase short pos		Possible phase to DC+ short
	:55	phase short neg		Possible phase to DC- short
	:60	linkV too hi PTL		Link volts above PTL
	:70	Aph neg I low		Phase A negative current low (unbalance)
	:71	Bph neg I low		Phase B negative current low (unbalance)
	:72	Cph neg I low		Phase C negative current low (unbalance)
	:73	Aph neg I hi		Phase A negative current high (unbalance)
	:74	Bph neg I hi		Phase B negative current high (unbalance)
	:75	Cph neg I hi		Phase C negative current high (unbalance)
	:76	Aph pos I low		Phase A positive current low (unbalance)
	:77	Bph pos I low		Phase B positive current low (unbalance)
	:78	Cph pos I low		Phase C positive current low (unbalance)
	:79	Aph pos I hi		Phase A positive current high (unbalance)
	:80	Bph pos I hi		Phase B positive current high (unbalance)
	:81	Cph pos I hi		Phase C positive current high (unbalance)
	:82	no current w run		No current while running
114/214		INVERTER, GENERAL (NR)	None	
	:22	IA VCO lo		Low frequency on IA channel
	:24	IB VCO lo		Low frequency on IB channel
	:26	linkV VCO lo		Low frequency on VCO link filter V channel
	:28	infilV VCO lo		Low frequency on VCO in filter V channel
	:38	LinkV too lo		Link voltage too negative
	:40	infilV too lo		Input filter voltage too positive
	:46	DB chop VCO lo		Low frequency on VCO DB chopper channel
	:48	DB chopV too lo		DB chopper voltage too negative
	:50	VA VCO lo		Low frequency on VCO VA channel
	:52	VB VCO lo		Low frequency on VCO VB channel
	:54	VC VCO lo		Low frequency on VCO VC channel
	:56	VA volts too lo		VA voltage too negative
	:58	VB volts too lo		VB voltage too negative
	:61	VC volts too lo		VC voltage too negative

Table 3: DID PANEL FAULT CODES
(Codes Received from Inverter 1 & 2)

EVENT NUMBER		EVENT DESCRIPTION	EVENT RESTRICTION	DETECTION INFORMATION
119/219		INVERTER, PHASE A-	INV1 (INV2) off	
	:01	alarm AN		Phase A negative IGBT did not turn off.
	:02	Adn fb not off		Phase A down feedback is not off.
	:03	phase A modl neg		Phase A negative module failed.
	:04	hold AN		Phase A positive and negative IGBTs are on (negative turn on).
	:05	Adn fb not on		Phase A down feedback is not on.
	:06	Adn IGBT not on		Phase A negative IGBT did not turn on.
	:07	IGBT_PS_AN		IGBT protective shutoff
120/220		INVERTER, PHASE A- (NR)	None	
	:02	Adn temp short		Phase A down thermistor short
	:03	Adn temp open		Phase A down thermistor open
	:04	Adn temp warm		Phase A down thermistor warm
	:05	Adn temp hot		Phase A down thermistor hot
	:06	Adn fb not off S		Phase A down not off with enable/DC volts
121/221		INVERTER, PHASE A CURR	INV1 (INV2) off	
	:01	I sensor ph A		Phase A current sensor failed.
	:02	IA zero not ok		Current IA not zero at startup
	:03	IA not ok		Phase A current too high
	:04	I snsr ph A open		Phase A current sensor open
	:05	I snsr ph A short		Phase A current sensor short
123/223		INVERTER, PHASE A VOLTS	INV1 (INV2) off	
	:01	V sensor phase A		Phase A voltage sensor failed.
	:02	VA not ok		Phase A voltage too high
125/225		INVERTER, PHASE B+/B-	INV1 (INV2) off	
	:01	alarm B		Phase B IGBT did not turn off
	:02	PTF B		Overcurrent on phase B
	:03	IGBT_SAT_BP		IGBT saturated
	:04	IGBT_SAT_BP		IGBT saturated
126/226		INVERTER, PHASE B+	INV1 (INV2) off	
	:01	alarm BP		Phase B IGBT did not turn off.
	:02	Bup fb not off		Phase B up feedback is not off.
	:03	phase B modl pos		Phase B positive module failed.
	:04	hold BP		Phase B positive and negative IGBTs are on (positive turn on).
	:05	Bup fb not on		Phase B feedback is not on.
	:06	Bup IGBT not on		Phase B positive IGBT did not turn on.

Table 3: DID PANEL FAULT CODES
(Codes Received from Inverter 1 & 2)

EVENT NUMBER	EVENT DESCRIPTION	EVENT RESTRICTION	DETECTION INFORMATION
127/227	INVERTER, PHASE B+	None	
:02	Bup temp short		Phase B up thermistor short
:03	Bup temp open		Phase B up thermistor open
:04	Bup temp warm		Phase B up thermistor warm
:05	Bup temp hot		Phase B up thermistor hot
:06	Bup fb not off S		Phase B up not off with enable/DC volts
128/228	INVERTER, PHASE B-	INV1 (INV2) off	
:01	alarm BN		Phase B negative IGBT did not turn off.
:02	Bdn fb not off		Phase B down feedback is not off.
:03	phase B modl neg		Phase B negative module failed.
:04	hold BN		Phase B positive and negative IGBTs are on (negative turn on).
:05	Bdn fb not on		Phase B down feedback is not on.
:06	Bdn IGBT not on		Phase B negative IGBT did not turn on.
:07	IGBT_PS_BN		IGBT protective shutoff
129/229	INVERTER, PHASE B- (NR)	None	
:02	Bdn temp short		Phase B down thermistor short
:03	Bdn temp open		Phase B down thermistor open
:04	Bdn temp warm		Phase B down thermistor warm
:05	Bdn temp hot		Phase B down thermistor hot
:06	Bdn fb not off S		Phase B down not off with enable/DC volts
130/230	INVERTER, PHASE B CURR	INV1 (INV2) off	
:01	I sensor ph B		Phase B current sensor failed.
:02	IB zero not ok		Current IB not zero at startup
:03	IB not ok		Phase B current too high
:04	I snsr ph B open		Phase B current sensor open
:05	I sensr ph B short		Phase B current sensor short
132/232	INVERTER, PHASE B VOLTS	INV1 (INV2) off	
:01	V sensor phase B		Phase B voltage sensor failed.
:02	VB not ok		Phase B voltage too high
134/234	INVERTER, PHASE C+/C-	INV1 (INV2) off	
:01	alarm C		Phase C IGBT did not turn off.
:02	PTF C		Overcurrent on phase C
:04	IGBT_SAT_CP		IGBT saturated
:05	IGBT_SAT_CN		IGBT saturated

Table 3: DID PANEL FAULT CODES
(Codes Received from Inverter 1 & 2)

EVENT NUMBER		EVENT DESCRIPTION	EVENT RESTRICTION	DETECTION INFORMATION
135/235		INVERTER, PHASE C+/C-	INV1 (INV2) off	
	:01	alarm CP		Phase C positive IGBT did not turn off.
	:02	Cup fb not off		Phase C up feedback is not off.
	:03	phase C modl pos		Phase C positive module failed.
	:04	hold CP		Phase C positive and negative IGBTs are on (positive turn on).
	:05	Cup fb not on		Phase C up feedback is not on.
	:06	Cup IGBT not on		Phase C positive IGBT did not turn on.
	:07	IGBT_PS_CP		IGBT protective shutoff
136/236		INVERTER, PHASE C+	None	
	:02	Cup temp short		Phase C up thermistor short
	:03	Cup temp open		Phase C up thermistor open
	:04	Cup temp warm		Phase C up thermistor warm
	:05	Cup temp hot		Phase C up thermistor hot
	:06	Cup fb not off S		Phase C up not off with enable/DC volts
137/237		INVERTER, PHASE C-	INV1 (INV2) off	
	:01	alarm CN		Phase C negative IGBT did not turn off.
	:02	Cdn fb not off		Phase C down feedback is not off.
	:03	phase C modl neg		Phase C negative module failed.
	:04	hold CN		Phase C positive and negative IGBTs are on (negative turn on).
	:05	Cdn fb not on		Phase C down feedback is not on.
	:06	Cdn IGBT not on		Phase C negative IGBT did not turn on.
	:07	IGBT_PS_CN		IGBT protective shutoff
138/238		INVERTER, PHASE C- (NR)	None	
	:02	Cdn temp short		Phase C down thermistor short
	:03	Cdn temp open		Phase C down thermistor open
	:04	Cdn temp warm		Phase C down thermistor warm
	:05	Cdn temp hot		Phase C down thermistor hot
	:06	Cdn fb not off S		Phase C down not off with enable/DC volts
141/241		INVERTER, PHASE C VOLTS	INV1 (INV2) off	
	:01	V sensor phase C		Phase C voltage sensor failed.
	:02	VC not ok		Phase C voltage too high
143/243		INVERTER, TACH 1 (NR)	INV1 (INV2) off	
	:01	tach1 rate hi		Tach 1 high rate of change
	:02	tach1 no input		Tach 1 no frequency input
	:03	TACH_INTERMIT		
144/244		INVERTER, TACH 1 (NR)	None	
	:01	tach1 one channel		Tach 1 single channel operation
145/245		INVERTER, TACH 2	None	
	:01	tach2 high rate		Tach 2 high rate of change
	:02	tach2 no input		Tach 2 no frequency input

Table 3: DID PANEL FAULT CODES
(Codes Received from Inverter 1 & 2)

EVENT NUMBER	EVENT DESCRIPTION	EVENT RESTRICTION	DETECTION INFORMATION
146/246	INVERTER, TACH 2 (NR)	None	
:01	tach2 one channel		Tach 2 single channel operation
148/248	INVERTER, CHOPPER 1 (NR)	None	
:01	chop1 fb not off		Chopper 1 feedback is not off.
:02	chop1 fb not on		Chopper 1 feedback is not on.
:03	chopA temp short		ChopA thermistor short
:04	chopA temp open		ChopA thermistor open
:05	chopA temp warm		ChopA thermistor warm
:06	chopA temp hot		ChopA thermistor hot
:07	DB1 fb not off S		Chopper 1 not off with DC volts
150/250	INVERTER, CHOPPER 2 (NR)	None	
:01	chop2 fb not off		Chopper 2 feedback is not off.
:02	chop2 fb not on		Chopper 2 feedback is not on.
:03	chop B temp short		Chop B thermistor short
:04	chop B temp open		Chop B thermistor open
:05	chop B temp warm		Chop B thermistor warm
:06	chop B temp hot		Chop B thermistor hot
:07	DB2 fb not off S		Chopper 2 not off with DC volts
151/251	MISCELLANEOUS	INV1 (INV2) off	
:01	tach differential		Too much speed difference
153/253	INVERTER, MOTOR	INV1 (INV2) off	
:01	motor open		Motor connection open
:02	motor short		Motor connection short
154/254	INVERTER MOTOR FAULTS (NR)	None	
:01	rotor temp hi		Motor rotor temperature is high.
:02	stator temp hi		Motor stator temperature is high.
155/255	INVERTER, SECOND LOAD	None	
:01	second load open		Second load connection open
175/275	INV 1 GENERIC EVENT	None	Inverter shutdown with no event code
176/276	INV 1 GENERIC EVENT	INV1 (INV2) off	

Table 4: DID PANEL FAULT CODES
(Codes Received from TCI)

EVENT NUMBER		EVENT DESCRIPTION	EVENT RESTRICTION	DETECTION INFORMATION
601		TCI FB144 CPU CARD	No propel	TCI CPU card problem
	:01	10ms task failed to init		
	:02	20ms task failed to init		
	:03	50ms task failed to init		
	:04	100ms task failed to init		
	:05	200ms task failed to init		
	:06	flt manager task		
	:07	flash CRC		Flash CRC computation did not match expected value.
	:09	main task failed to init		Upon power-up, excessive bus timeouts occurred.
	:10	excess timeouts		
	:11	BBRAM bad		
	:12	BBRAM CRC		CRC on BBRAM did not match expected value.
602		FB104 DIGITAL I/O CARD FAULT	No propel	Internal TCI self-test detected a digital I/O card problem.
603		FB160 ANALOG I/O CARD FAULT	No propel	Internal TCI self-test detected an analog I/O card problem.
604		PSC FAULT	Speed limit	Lost RS422 communication with PSC.
	:01	missing message		
	:02	bad tick		
	:03	bad CRC		
	:04	FIFO overflow		
	:05	bad start bit		
	:06	bad stop bit		
605		AUX BLOWER COMM. FAULT	None	Lost RS422 communication with auxiliary blower controller while auxiliary blower is in failure mode and DC link is not energized.
607		POSITIVE 5 VOLTS	Speed limit	+5V power supply out of limits
608		POSITIVE 15 VOLTS	Speed limit	+15V power supply out of limits
609		NEGATIVE 15 VOLTS	Speed limit	-15V power supply out of limits
610		POT REFERENCE	Speed limit	Pot reference (10.8V) out of limits
611		FREQUENCY INPUT	None	Front wheel speed input out of range
	:01	left front wheel speed		Left front wheel sensor out of range
	:02	right front wheel speed		Right front wheel sensor out of range
613		ANALOG INPUT	Speed limit	Signal is outside the design range of valid values.
	:01	A2D gnd		
	:02	A2D gainchk		
614		BATTERY SEPARATE CONTACTOR FAILURE	SYS Event	Signal is outside the design range of valid values.
	:01	Battery Separate Failure		Voltage difference greater than 3V
	:02	crank batt > cntrl batt		
	:03	cntrl batt > crank batt		
616		DIRECTION MISMATCH	No propel	Simultaneous FORWARD and REVERSE commands were received.

Table 4: DID PANEL FAULT CODES
(Codes Received from TCI)

EVENT NUMBER		EVENT DESCRIPTION	EVENT RESTRICTION	DETECTION INFORMATION
617		ENGINE START REQUEST DENIED	SYS Event	
	:01	engine warn while cranking		Engine warning occurs after engine crank command is given.
	:02	engine kill while cranking		Engine kill input occurs while engine crank command is active.
619		ENGINE WARNING RECEIVED	No propel	Engine controller sends caution signal, rpm above low idle.
620		ENGINE KILL WHILE VEHICLE MOVING	No propel	Engine shutdown switch is activated while truck is moving.
622		PARK BRAKE FAULT	No propel	Error in parking brake operation has occurred.
	:01	command/response failure		Park brake command and feedback don't agree.
	:02	set above maximum speed		Parking brake set feedback is received while truck is moving.
623		HYDRAULIC BRAKE FLUID	SYS Event	Hydraulic brake oil temperature has exceeded the limit.
	:01	tank		
	:02	left front outlet		
	:03	right front outlet		
	:04	left rear outlet		
	:05	right rear outlet		
624		BODY UP AND PAYLOAD INDICATION	Speed Limit	Full payload and body up signal are received at the same time.
625		Extended Battery Reconnect Time	None	Excessive time since battery separate and battery reconnection
628		CONNECTED BATTERY VOLTS	SYS Event	One of the connected batteries' volts are incorrect with engine speed above low idle.
	:01	control battery low		Control battery voltage below minimum limit (20)
	:02	control battery high		Control battery voltage above maximum limit (32)
	:03	crank battery low		Cranking battery voltage below minimum limit (20)
	:04	crank battery high		Cranking battery voltage above maximum limit (32)
629		BAROMETRIC PRESSURE SIGNAL	SYS Event	Barometric pressure signal is outside operational limits.
	:01	low		Voltage is below minimum operational limit.
	:02	high		Voltage is above maximum operational limit.
630		MOTOR BLOWER PRESSURE	Speed Limit	Motor inlet and outlet pressure signal is outside operational limits.
	:01	no cooling air		No voltage signal feedback
	:02	low voltage		Voltage feedback is below minimum operational limit.
	:03	high voltage		Voltage is above maximum operational limit.
	:04	sensor reversed		
631		AMBIENT TEMPERATURE	SYS Event	Ambient temperature signal is outside operational limits.
	:02	high		Voltage is above maximum operational limit.
632		TCI CONFIGURATION DATA	No propel	Problem with TCI configuration file
	:01	no file loaded	No propel	No configuration file is loaded.
	:02	bad CRC	No propel	
	:03	version incorrect	No propel	Wrong version of file is loaded.

Table 4: DID PANEL FAULT CODES
(Codes Received from TCI)

EVENT NUMBER		EVENT DESCRIPTION	EVENT RESTRICTION	DETECTION INFORMATION
633		BGRAM CORRUPTED	SYS Event	Battery backed RAM has failed.
634		TRUCK OVERLOADED - RESTRICTIVE	NO PROPEL	The over-payload signal is on, operation restricted.
635		TRUCK OVERLOADED - NON-RESTRICTIVE	SYS Event	The over-payload signal is on, propulsion allowed.
636		AUX INVERTER	SYS Event	An auxiliary blower control failure has occurred.
	:01	buss volts low		Low DC bus was detected during powerup.
	:02	buss volts high		High DC bus was detected during powerup.
	:03	overcurrent		Overcurrent condition was detected during operation.
	:04	battery loss		Loss of blower control battery voltage has occurred.
	:05	high dc buss when running		High DC bus voltage was detected during operation.
	:06	high dc buss after pc powerup		High DC bus voltage was detected after phase controller powerup.
	:07	Low dc buss after pc powerup		Low DC bus voltage was detected after phase controller powerup.
	:08	high dc buss when running		High DC bus voltage was detected during operation.
	:09	overcurrent after pc powerup, current overload		Overcurrent condition was detected after phase controller power up.
	:10	current overload		Sustained current overload exists.
	:11	low dc buss overcurrent		Overcurrent due to low DC bus voltage
	:12	low dc buss current overload		Sustained current overload due to low DC bus voltage
	:13	gate drive trip		IGBT protection circuit detected an overload.
	:14	no input voltage		Zero input voltage was detected.
638		ENGINE CRANKING TIMEOUT	SYS Event	Engine is cranking longer than allowed.
639		ENGINE START REQUEST WHILE RUNNING	SYS Event	Engine start request signal occurred while engine RPM greater than 600 RPM, and longer than 3 seconds.
640		ACCEL PEDAL TOO HIGH	No Propel	Accelerator pedal voltage is high.
641		ACCEL PEDAL TOO LOW	SYS Event	Accelerator pedal voltage is low.
696		UNEXPECTED TCI CPU RESET	SYS Event	TCI CPU reset without request.
698		DATA STORE	SYS Event	A data snapshot has been manually initiated.

PSC SOFTWARE FUNCTIONS

The operation of the AC drive system is regulated by a software program which resides in the propulsion system control panel's memory. The software program also contains instructions to test and fault isolate the system.

This section describes the PSC software program and its functions without regard to hardware.

Input Processing

This function reads in all external inputs for use by the PSC. The input processing function performs any signal conditioning that is required and computes the required derived inputs.

State Machine

As part of the total software package, a particular group of regulatory software commands is included called a "state machine". The state machine controls the various functions of truck operation.

The software implements the state machine by keeping track of which state the truck is in and which state the truck is allowed to move into if the operator requests a different mode of operation. Each software state is defined as follows:

Startup/Shutdown State: The purpose of this state is to ensure the system is in a desired known state upon startup or shutdown. This is an unpowered state.

NOTE: "Powered" and "unpowered" refer to the state of the DC link. 600 volts or more equals "powered", 50 volts or less equals "unpowered".

Test State: The purpose of this state is to provide an environment for the verification of system functionality. The test state will support a variety of activities, including:

- Waiting for the engine to start (if needed).
- Automatic testing on initial system startup or following rest state.
- Application of power to the DC link.
- Externally initiated testing to clear a fault, set temporary variables, or for maintenance purposes.

NOTE: The test state may be either powered or unpowered at a given point in time, depending on which activities are being performed.

Ready State: This is the default powered state. The system will be in this state whenever the engine and control system are ready to provide power, but none is requested.

Ready state is also the state where the DC link is discharged in preparation for shutdown, rest, or in reaction to certain event conditions. Therefore, the ready state should not be considered strictly a powered state (as are propel and retard).

Rest State: The purpose of this state is to conserve fuel while the truck idles for an extended period of time. The rest state also provides an environment where maintenance personnel can control the engine without causing power to be applied to the DC link. The rest state is an unpowered state.

Propel State: The purpose of this state is to provide the power system configuration and overall environment for engine-powered propulsion. This is a powered state. The system will not be allowed to maintain the propel state without sufficient power on the DC link.

Retard State: This state provides the power system configuration and overall environment for retard functions, where energy from vehicle movement is dissipated in the retarding grid resistors in an effort to slow the truck. The retard state is a powered state.

Transitions between states under normal operational conditions (no failures, etc.) are described as follows:

Transition to Startup/Shutdown State

(for Startup): The system will transition to the Startup/Shutdown state for the purpose of “startup” whenever execution control is initially transferred to the application program (after application of power, system reset, etc).

Transition to Startup/Shutdown State

(for Shutdown): The system will transition to the Startup/Shutdown state for the purpose of “shutdown” from the Test, Rest, Ready, or Startup/Shutdown (if previously entered for the purpose of startup) state if all of the following conditions are true:

- System power is removed, or the control power switch or key switch is turned off.
- The truck is not moving.
- There is essentially no voltage on the DC link.
- Any testing in progress is complete.

NOTE: Testing in progress does not have to be successful, but for the purpose of ensuring an orderly shutdown it must be complete before the current state is exited.

Transition from Startup/Shutdown State to Test State: This transition will occur automatically once initialization is complete (functions performed while in Startup/Shutdown state for the purpose of startup have been completed).

Transition from Test State to Ready State: This transition will occur upon completion of any required testing if the TCI Rest state request is not active and there is sufficient voltage on the DC link.

Transition to Rest State: This transition will occur automatically from the Test or Ready state if a request for Rest state is received from the TCI and all of the following conditions are true:

- Any testing in progress is complete.
- The system temperatures are cool enough to allow the Rest state (function of IGBT phase module, chopper module, and motor temperatures).
- The AFSE panel is disconnected and there is essentially no voltage on the DC link.
- The truck is not moving.

Transition from Ready State to Test State: This transition will occur if the truck is not moving and a request for testing is received.

Transition from Ready State to Propel State: This transition will occur if all of the following conditions are true:

- The accel pedal is pressed.
- A direction has been chosen (the truck is either in FORWARD or REVERSE).
- There is sufficient voltage on the DC link.
- At least one of the following conditions is true:
 - a. The retard pedal or lever is not pressed or is pressed such that an insignificant amount of retarding effort is requested.
 - b. Truck speed is such that retard is not allowed.
 - c. Truck speed is less than the motor overspeed limit.
 - d. The TCI accel inhibit is not active.

Transition from Ready State to Retard State: This transition will occur if truck speed is such that retard is allowed and at least one of the following conditions exists:

- Truck speed is greater than or equal to motor overspeed limit. Overspeed will not be engaged such that it prevents the truck from propelling at 40 mph (64 kph).
- The retard pedal or lever is pressed such that a significant amount of retarding effort is requested.
- All of the following conditions are true:
 - a. Retard speed control is selected.
 - b. Truck speed exceeds the set retard speed, or the truck is accelerating such that the truck speed will soon exceed the set retard speed if no action is taken.
 - c. The accel pedal is not pressed and/or the truck is configured such that accelerator pedal signal does not override retard speed control.

Transition from Rest State to Test State: This transition will occur upon release of the TCI rest request.

NOTE: A transition directly from Rest state to Ready state is not allowed because the system is essentially off and should be brought back on-line and checked out before Ready state is entered.

Transition from Propel State to Ready State: This transition will occur if all of the following conditions exist:

- The accelerator pedal is not pressed.
- The retard pedal or lever is not pressed or is pressed such that an insignificant amount of retarding effort is requested.
- Truck speed is less than the motor overspeed limit.
- At least one of the following conditions is true:
 - a. Retard speed control is not selected.
 - b. Truck speed is below the set retard speed, and acceleration is such that no retard effort is (currently) required to maintain this condition.

Transition from Propel State to Retard State: This transition will occur if at least one of the following conditions exists:

- Truck speed is such that retard is allowed, and the retard pedal or lever is pressed such that a significant amount of retarding effort is requested.
- Truck speed exceeds the motor speed limit. Overspeed will not be engaged such that it prevents the truck from propelling at 40 mph (64 kph).
- All of the following conditions are true:
 - a. Retard speed control is selected.
 - b. Truck speed exceeds the set retard speed, or the truck is accelerating such that the truck speed will soon exceed the set retard speed if no action is taken.
 - c. The truck is configured such that accelerator pedal signal does not override the retard speed control.

Transition from Retard State to Ready State: This transition will occur if all of the following conditions exist:

- Overspeed is not active.
- At least one of the following conditions is true:
 - a. The retard pedal or lever is not pressed or is pressed such that an insignificant amount of retarding effort is requested.
 - b. Truck speed is such that retarding is not allowed.
- At least one of the following conditions is true:
 - a. Retard speed control is not selected.
 - b. Truck speed is low enough such that retard speed control is not active.
 - c. The accelerator pedal is pressed, and the truck is configured such that the accelerator pedal overrides the retard speed control. This allows the configuration constant to determine if pressing on the accelerator pedal kicks the truck out of retard, even if the retard speed control is still active.
- The retard torque control logic exit sequence is complete.

DC Link State

Power is provided to the inverters and motors via the DC link. The DC link has two associated states: *powered* and *unpowered*. The following defines the conditions necessary to establish each state, as well as the transitional conditions between the two states:

Powering the DC Link: The PSC software will attempt to power the DC link (command the system configuration defined below) if all of the following conditions are true:

- Event restrictions do not prohibit power on the DC link.
- The system is in Test state and any initiated testing is complete.
- The engine is running.
- The gate drive power converters have been enabled.
- Neither inverter is requesting that a low voltage test be run.

In attempting to power the DC link, the PSC software will establish the necessary system configuration as follows:

1. GF is closed and GFR is picked up.
2. AFSE is enabled.
3. Alternator field reference is commanded so that the desired DC link voltage or three-phase voltage is maintained.
4. RP contactors open.
5. Chopper turn-on voltage is set above 600 volts.

NOTE: Before the AFSE is allowed to output firing pulses, the RP contactors will be commanded to open and the GF contactor will be verified to be closed. The AFSE will not output firing pulses if it is disabled, if GFR is dropped out, or if the alternator reference signal is 0.

De-Powering the DC Link: The PSC software will attempt to de-power the DC link (command the system configuration defined below) if the system is in Test or Ready state and any of the following conditions are true:

- Event restrictions prohibit power on the DC link.
- The system is preparing to transition to Startup/Shutdown state for the purpose of shutdown (all the non-link-related conditions for Startup/Shutdown state have been satisfied).
- The system is preparing to transition to Rest state (all the non-link-related conditions for Rest state have been satisfied).
- The engine is being shut down.

To accomplish this, the PSC software will establish the necessary system configuration as follows:

1. Alternator field reference is set to zero.
2. AFSE is disabled.
3. GF is open and GFR is dropped out.
4. Chopper turn-on voltage is set below 600 volts.
5. RP2 is closed.

Engine Control

This software function generates the engine speed command. The engine electronic fuel control is responsible for maintaining that speed.

The desired engine speed is determined according to the system state:

Propel State: The engine speed is commanded such that the engine supplies only as much horsepower as is required to achieve the desired torque.

All Other States: The engine speed is a direct function of the accelerator pedal. Additional constraints on the engine speed command are as follows:

- If the truck is in NEUTRAL, the commanded engine speed at full scale accelerator pedal will be the engine's high idle. If the truck is not in NEUTRAL, the maximum commanded engine speed will be the engine's rated horsepower rpm. This allows faster hoisting of the truck bed, if desired.
- During retard state the engine speed command will not be increased to support the DC link when retard is being ramped out at low truck speeds. However, engine speed may be increased if needed to support the DC link during normal retard when wheel slides are occurring.

The following constraints are applied to generating the engine speed command during all operating states:

- The engine speed command will always be greater or equal to the minimum idle signal. The TCI can request that the engine speed command be increased by setting minimum idle.
- The engine speed command will be increased if more alternator cooling is needed.

ALTERNATOR FIELD CONTROL

The alternator is controlled by controlling the alternator field reference sent to the AFSE panel.

The desired alternator output voltage is dependent on the system state. The PSC will command an alternator field reference such that the desired DC link voltage or three-phase voltage is maintained.

Desired Three-Phase Voltage

During all powered states, the three-phase line-to-line voltage will not be allowed to drop below 444 volts. This is the minimum voltage needed to supply the gate drive power converters.

During all powered states except retard, the DC link voltage will represent the rectified three-phase voltage. In this case, as long as the DC link voltage is above 600 volts, the three-phase voltage will be adequate.

During retard, the DC link voltage is not necessarily related to the three-phase voltage since the motors will be powering the DC link and reverse biasing the rectification diodes. In this case, the control ensures that the minimum three-phase voltage is maintained.

Desired DC Link Voltage

The desired link voltage is controlled by the alternator during all powered states except retard. The desired voltage is based on:

1. During propel, the desired DC link voltage will be adjusted based on motor speed and horsepower commanded to the inverters.
2. During retard, the DC link voltage may rise above the rectified three-phase voltage. When this occurs, the DC link voltage is controlled by the retard torque command, grid resistor command and chopper start. If conditions occur which prevent the motors from producing power to support the resistor grids, the alternator may be required to supply some power. In this case, the alternator field control will maintain at least 600 volts on the DC link.
3. During all powered states, the DC link voltage will not be allowed to drop below 600 volts.
4. While the DC link is being powered up, the voltage will be controlled to the levels necessary to support the inverter self-tests.

Self-Load

During self-load, the alternator provides power to the resistor grids. The rectifying diodes will be forward biased, and DC link voltage will be controlled by the alternator. The alternator field control will be based on the following:

- The DC link voltage will not be allowed to drop below 600 volts.
- One mode of self-load will require the alternator output to be controlled to maintain a set desired horsepower dissipation in the resistor grids.
- Another mode of self-load will require the alternator output to be controlled to maintain a set desired link voltage between 600 and 1500 volts.

Propel Torque Control

This software function commands the appropriate motor torque to the inverters during propel. The torque command is primarily a function of the accel pedal position and is limited by the physical constraints of the system.

Each wheel torque is computed independently because the wheels may be operating at different speeds. Each torque command is adjusted to account for the following constraints:

- **Speed Override**

The propulsion system will attempt to limit truck speed to the design envelope of the wheel motors. The torque command will be modulated as the truck speed approaches the motor overspeed limit so that this limit is not exceeded if possible. Note, however, that steady state operation is kept as close to the overspeed limit as possible without exceeding it.
- **Motor Torque Limits**

The torque command will be constrained to the operating envelope of the inverters and the traction motors. The maximum torque that can be commanded is dependent on motor speed and DC link voltage.
- **Gear Stress**

The torque commanded will not exceed that which will produce excess gear stress.
- **Horsepower Available**

The horsepower available will be estimated from the engine speed. Parasitic loads are taken into account. The torque will be limited so that the engine does not overload.

- **Jerk Limit**

The torque command will be slew-rate limited to prevent jerking motion.

- **Wheel Spin**

In the event that the inverters detect a wheel spin condition and reduce torque in the slipping wheel, the motor torque in the other wheel may be increased within the above constraints such that as much of the total desired torque as possible is maintained.

Retard Torque Control

The retard system converts braking torque from the wheel motors to energy dissipated in the resistor grid. The requested retard torque is based on the following three sources:

- **Retard Foot Pedal or Lever**

The maximum short time retard torque (at any speed, hence the constant torque level) will be scaled (linearly) by the retard foot pedal input (RPINHI) to produce the foot pedal retard torque call.

- **Overspeed**

While overspeed is active, the full available retard torque will be requested.

- **Retard Speed Control**

While RSC is active, the RSC retard torque call will be adjusted to control truck speed to the RSC set point. Retard speed control will not request any retard torque if RSC is not active.

The maximum torque call from the above three sources will be selected as the retard torque call. Retard torque limits are as follows:

- The retard torque call will be limited to the maximum torque level based on speed.
- The retard torque call will be limited to the maximum torque level available within the thermal constraints of the motors.
- The retard torque call will be limited as needed to prevent overvoltage on the DC link.
- While in retard, the minimum retard torque call will provide enough power to support at least one grid with 600 volts on the DC link. Retard will be dropped if the torque call falls below this value.
- At low speed, the available retard torque will be ramped to zero.

Wheel Slide Control

The inverters prevent wheel slide by limiting torque to maintain wheel speeds above preset limits. These preset limits are a function of truck speed and the allowable creep; additional compensation will be applied to provide for differences between wheel speeds during turns.

Resistor Grid Control

The first resistor grid (RG1) will always be engaged when retard is active since the grid blower motors are wired across it.

The second fixed resistor grids (controlled by RP2) will be engaged as needed to dissipate the energy produced in retard state.

Chopper Voltage Control

Chopper turn-on voltage will be set to give the motors as much of the retard envelope as possible (i.e., keep the voltage as close to the maximum value as possible) and to keep the DC link voltage at or below the maximum link voltage value.

EVENT DETECTION AND PROCESSING

The PSC contains very powerful troubleshooting software. The PSC software constantly monitors the AC drive system for any abnormalities (events).

Automatic self-tests are performed periodically on various parts of the system to ensure its integrity. Additionally, there are some elaborate tests which may be run by an electrician with the use of DID screens. Predictive analysis is used in some areas to report potential problems before they occur.

The troubleshooting system is composed of two parts:

- The PSC for detection, event logging, data storage and fault light indications.
- The TCI (or a PTU) for retrieval of stored event information, real time vehicle status, troubleshooting, etc.

The event detection function of the software is responsible for verifying the integrity of the PSC hardware and the systems to which the PSC interfaces by detecting an "event" (abnormal condition). The events fall into three detection categories:

Power-On Tests

Three power-on tests are executed once every time power is applied to the PSC. They are as follows:

- *CPU Card Checks* - Upon power-up, the PSC will confirm the integrity of its CPU card hardware before transferring execution control to the application program residing in its FLASH memory.
- *Battery-Backed RAM (BBRAM) Test/Adjustable Parameter Initialization* - A battery-backed RAM (BBRAM) check will be performed to check for BBRAM data integrity. If the check fails, all TCI/PTU-adjustable parameters will be initialized to their default values.
- *Inverter Powerup Tests* - The purpose of these tests is to verify that each inverter sub-system is functional:
 1. *Enabling Inverter Powerup Tests* - The power-up tests for a given inverter will be enabled if all of the following conditions are true:
 - a. The system is in Test state for the purpose of power-up.
 - b. The associated gate drive power converter has been enabled.
 - c. The engine is running.
 - d. Battery voltage is at least 25 VDC.
 - e. The inverter is requesting that the low voltage and/or high voltage powerup tests be performed.
 - f. The inverter has not been physically cut out of the system.
 - g. Active event restrictions do not preclude powering the DC link or running the inverter.
 2. *Low Voltage Test* - A given inverter will automatically perform its low voltage test if needed once inverter powerup testing is enabled per the above requirements. The PSC will declare the test failed and log an event if the test does not successfully complete within an expected time period.
 3. *High Voltage Test* - If the low voltage testing defined above is successful for a given inverter, the inverter will automatically perform its high voltage test if needed once there is sufficient power on the DC link. The PSC will declare the test failed and log an event if the test does not successfully complete within an expected time after the DC link is sufficiently powered.

- *DC Link Capacitance Test* - This test will run once every 24 hours when conditions allow, normally after a VI-test during the normal power-up sequence. This test can also be run from the DID panel to aid in troubleshooting. During test execution, engine speed is set to 1500 rpm and the DC link is charged to 120 VDC. The engine is then returned to idle while the DC link is allowed to discharge to 100 VDC. Total link capacitance is then calculated using the time it took to discharge.

If capacitance is getting low, but is still OK, event 70 is logged. If capacitance is below the minimum allowable level, event 71 is logged and the truck is restricted to 10 MPH. If the test is not able to be completed after numerous attempts, event 72 is logged, indicating a problem in the truck's ground detection circuit, and truck speed is limited to 10 mph.

Initiated Tests

These tests are performed when requested by maintenance personnel. The truck must be in the Test state for these tests to run.

- *Maintenance Tests* - The purpose of these tests is to facilitate verification of system installation and wiring, particularly the "digital" interfaces (relays, contactors, etc).
- *Self-Load Test* - Self-load testing is a means by which the truck's diesel engine can be checked for rated horsepower output.

Periodic Tests

These automatic tests are run continuously during the operation of the truck to verify certain equipment.

EVENT RESTRICTIONS

The PSC software will not override an event restriction as long as the "limp home" mode is not active. Transitions to restricted states will not be allowed. If the system is in a state which becomes restricted, it will transition down to the highest unrestricted state. The order of the states, from lowest to highest, is Startup/Shutdown, Rest, Test, Ready, Retard, Propel.

Transitions to the Test state or lower states in reaction to event restrictions will not be allowed until the truck is not moving. The "limp home" mode is a state which is entered when the truck has suffered a failure and is not able to continue normal operation, but is still capable of getting back to the maintenance area, or at least out of the way of other trucks.

Event Restrictions associated with a given event are listed in Table 1 earlier in this section.

EVENT LOGGING AND STORAGE

This software function is responsible for the recording of event information. There are two basic levels of event storage: event history buffer and data packs. The event history buffer provides a minimum set of information for a large number of events, while data packs provide extensive information for a limited number of events.

The following requirements apply to both data packs and the event history buffer:

- Fault information is maintained until overwritten; it is not cleared out following a reset. This allows the user to examine data associated with events that have been reset, as long as there have not been so many new events as to necessitate reuse of the storage space.
- If a given event is active (logged and not reset), logging of duplicate events (same event and sub-ID numbers) will not be allowed. If the event is reset and subsequently reoccurs, it may be logged again. Likewise, if an event reoccurs with a different sub-ID from the original occurrence, the event may be logged again.

Event History Buffer

Event history buffer is defined as a collection of event history records. A buffer contains 300 entries filled with event numbers occurring in chronological order. Also included in this buffer will be all the input and output values, time the event occurred, reset time, state information, etc, for each event. This buffer is filled continuously and overwritten (if necessary).

Limits (accept-limit) are placed on the amount of space which a given event code may consume. This prevents a frequently occurring event from using the memory space at the expense of a less frequent event. This data may be cleared (after downloading for troubleshooting) at each maintenance interval.

Data Packs

A data pack is defined as an extended collection of information relevant to a given event.

NOTE: The concepts of lockout, soft reset, and accept limit do not apply to data packs.

Thirty (30) data packs are stored with each containing 100 frames of real time snapshot data. Snapshot data is defined as a collection of key data parameter values for a single point in time). The purpose of each data pack is to show a little "movie" of what happened before and after a fault.

The time interval between snapshots is default to 50 ms, but each data pack may be programmed via the DID (or PTU) from 10 ms to 1 sec. (In multiples of 10 ms). The "TIME 0:00 frame #" at which the fault is logged is default to frame #60, but each data pack is programmable from 1 to 100.

In the above default cases, data is stored for 3 seconds (2.95 second actual) before the fault and 2 seconds after the fault.

A data pack status structure is assigned to each data pack plus any programmable settings. This status structure is used by the TCI (or PTU) to check for available data (event number, id, and status, should be set to zero if data pack is not frozen), as well as for control of the data packs.

If a data pack is unfrozen (not holding any particular fault data), it is continually updated each 100 frames, organized in a circular queue, with new real time snapshot data. When a fault occurs, the frame number at which the event occurred is used as a reference to mark the end of the data pack, and data is collected until the data pack is full. Only when the data pack is full will the event number, id and status be updated in the status structure.

All logic control variables are saved in battery backed RAM, in case a fault occurs and battery power is cycled before the data pack is filled with data (the software allows for proper recovery and then continues to fill the data pack). Maintenance personnel, by way of the DID (or PTU), can assign the data pack to hold only certain event numbers (for the case where it is desired to collect data on a particular fault).

However, in the default case, faults will be stored as they come until all data packs are frozen (holding fault data). When all data packs are frozen, the data pack with the fault that was RESET first (either automatically or by the DID/PTU), if any, will be unfrozen and will start storing new data in case a new fault occurs.

To Record and Save a Data Pack to a Disk

PSC:

1. With the PTU serial cable attached to the PSC port, type `c:\ACNMENU` and press {enter}.
2. Select "PTU TCI and PSC" and press {enter}
3. Type your name and press {enter}.
4. Type your password and press {enter}.
5. Cursor to "Special Operation" and press {enter}.
6. Cursor to "Event Data Menu" and press {enter}.
7. Cursor to "View Data Packs" and press {enter}.
8. Type FLTR number to be recorded and press {enter}.
9. Watch the lower right of the screen as 100 frames are recorded. Press the F2 key.
10. Cursor to "Record Screen" and press {enter}.
11. Assign a file name for the data pack.
12. Press {escape} until back to the DOS "C:>" prompt.
13. Insert a blank disk in the appropriate drive.
14. Type the following command:
`copy c:\geohvac\ptuaccurlf2data\filename`

NOTE: Insert the name assigned to the file in Step 11 in place of "filename" in the command in Step 14.

15. Press {enter} to copy the file to the disk.

TCI:

1. With the PTU serial cable attached to the TCI port, type `c:\ACNMENU` and press {enter}.
2. Select "PTU TCI and PSC and press {enter}
3. Type your name and press {enter}.
4. Type your password and press {enter}.
5. Cursor to "Special Operation" and press {enter}.
6. Cursor to "Event Data Menu" and press {enter}.
7. Cursor to "View Data Packs" and press {enter}.
8. Type PK number to be recorded and press {enter}.
9. Watch the lower right of the screen as 100 frames are recorded. Press the F2 key.
10. Cursor to "Record Screen" and press {enter}.
11. Assign a file name for the data pack.
12. Press {escape} until back to the DOS "C:>" prompt.
13. Insert a blank disk in the appropriate drive.
14. Type the following command: `copy c:\geoh-vac\ptuaccu\rf2data\filename`

NOTE: Insert the name assigned to the file in Step 11 in place of "filename" in the command in Step 14.

15. Press {enter} to copy the file to the disk.

Event Reset

There are two basic types of event resets: *soft* and *hard*. The difference between the soft and hard reset is that a soft reset only affects events that have not been locked out and a hard reset affects events regardless of lockout status.

Events will be reset:

- *On power-up* - A soft reset will be issued against all events at power-up.
- *By DID commands* - The TCI can issue both hard and soft resets.
- *By PTU commands* - The PTU can issue both hard and soft resets.

SERIAL DATA COMMUNICATIONS

The PSC system CPU card uses serial data busses to communicate with the TCI, the PTU, and the two inverter CPU cards.

PSC - TCI Communications Processing

This software function performs the processing necessary for the PSC to communicate with the TCI. The communication is comprised of periodic data and non-periodic data.

Periodic data is a predefined set of data which is used for transferring real time control information from the PSC to the TCI and from the TCI to the PSC at a fixed rate.

The non-periodic messages are used to transfer all background data. Background data consists of DID commands, remote monitor data, and download code.

Packets containing periodic data will be asynchronously (not initiated) transmitted from the PSC to the TCI and from the TCI to the PSC every 200 ms. The TCI initiates the transfer of non-periodic data.

The TCI and the PSC are interfaced using the General Electric Asynchronous Communications Protocol (ACP). ACP provides two general types of messages: *acknowledged* and *unacknowledged*. The acknowledged messages are used to transmit the background data. The unacknowledged messages are used to transmit the periodic data.

PSC - PTU Communications Processing

This software function performs the processing necessary for the PSC to communicate through an RS-232 serial link to the Portable Test Unit (PTU).

Inverter Communications Processing

This software function performs the processing necessary for the PSC system CPU card to communicate with both inverter CPU cards. The communication is through a high-speed serial link that is operated in a polled fashion with the system CPU card initiating communications to an inverter CPU card.

Every message transmitted across the serial link may contain two separate sections of information: *periodic data* and *acknowledged data*. The periodic data format is fixed and is used for transferring control information from the system CPU card to the inverter CPU card and vice versa. The acknowledged data format is used to transfer all background data. When large amounts of background data are to be transferred via the acknowledged data format, the originating CPU card will break the data down into smaller pieces and transmit each piece individually. All acknowledged data flows are initiated from the system CPU card with the inverter CPU card providing a response.

The system CPU card has one high-speed channel available for communications to the inverter CPU cards. This channel transfers periodic data across the serial link every 5 ms. This means that the periodic data to each inverter CPU card is updated every 10 ms. Each inverter responds to the data when the ID code in the periodic data matches the ID code of the specific inverter CPU card. The ID code is hard-wired in the card's backplane wiring.

OUTPUT PROCESSING

This software function processes all external outputs from the PSC. Refer to the G.E. publication *System Description* for a listing of the PSC outputs.

ABNORMAL CONDITIONS/OVERRIDING FUNCTIONS

Software functions given up to this point have assumed that the truck is operating under typical circumstances. The following information defines system operation under abnormal or exceptional circumstances. In the event of conflict between these functions and those given for normal operation, the following functions will take precedence.

Fast Start

A fast start software function is provided to address the case where the PSC is reset unexpectedly (power supply glitch, for example) while the system is running. Its purpose will be to regain control of the truck as quickly as possible.

Engine Shutdown/Engine Not Running

The engine must be running to enable the gate drives and to maintain power on the DC link. Typically, the PSC will be given advanced warning that the engine is about to be shut off. However, if the engine stalls or stops because of a mechanical malfunction, the system will most likely have no advance warning.

The system reaction to an engine not running condition will be the same as an event carrying a "no power" restriction except that no event will be recorded and no external reset to clear the condition will be required. The "no power" restriction will be automatically lifted as soon as the engine starts running.

If the system is given warning of an impending engine shutdown, the existing torque commands will be command to zero over a "long" ramp time (2 to 10 seconds). If no warning is given and the engine stops running, the existing torque commands will be command to zero over a "short" ramp time (0.1 to 0.5 second).

Limp Home Mode

The purpose of limp home mode is to address the situation where the truck has suffered a failure and is not able to continue normal operation but is still capable of “limping” (getting back to the maintenance area or at least out of the way of other trucks). The intent is that the limp home mode will be used by maintenance personnel operating the truck at low speeds with the truck unloaded. Maximum truck speed will be limited to a reduced value while in limp home mode.

If the TCI requests limp home mode, the state machine will ignore the restrictions associated with any fault for which limp home mode is possible.

The PSC will enter limp home mode if all of the following conditions are true:

- The truck is not moving.
- The TCI is requesting limp home mode.
- The PSC is in Ready or Test state and there is no initiated testing in progress.
- At least one inverter is functional.
- There are no events active for which limp home mode is not possible.
- If there are any events active for which an inverter must be turned off or cut out before limp home mode is allowed, those actions have been taken (inverter is turned off or cut out as required).

NOTE: The DID panel can be used to cut out an inverter. In some cases, certain DC link bus bars/ cables within the inverter also may need to be removed. The DID will prompt maintenance personnel if any of the above actions need to be accomplished.

The PSC will exit limp home mode if either of the following conditions occur:

- The TCI stops requesting limp home mode.
- An event occurs for which limp home mode is not possible.

PROPULSION SYSTEM COMPONENT ABBREVIATIONS & LOCATIONS

The Table 5 lists component abbreviations that are used in schematics and system description information. Refer to Figures 2-3 through 2-6 for the location of the components. A short description of the component's primary function is also listed.

Table 5: PROPULSION SYSTEM COMPONENTS DESCRIPTION			
	FIG. NO.	COMPONENT	FUNCTION
AFSE	2-4	Alternator Field Static Exciter Panel	Regulates current in the alternator field based on firing pulses from the PSC.
AFSER	2-4	Resistor	AFSE Battery boost command pull up resistor.
ALT		Alternator	Main alternator, propulsion and control system.
AMBTS	2-6	Ambient Temperature Sensor	Provides ambient air temperature input to the control group.
ANALOG I/O CARD		System analog input/output card	Provides signal conditioning for analog signals to and from the TCI and PSC.
BAROP	2-4	Barometric Pressure Sensor	Provides altitude input for control electronics.
BATFU1, 2	2-4	System Fuse	Provides overload protection for control equipment.
BATTSW		Battery Disconnect Switch	Connects and disconnects the 24 VDC truck batteries.
BDI	2-4	Battery Blocking Diode	Works in conjunction with BFC and BLFP to maintain battery voltage to CPU.
BFC	2-4	Battery Line Filter Capacitor	Additional capacitance for BLFP to prevent nuisance CPU resets.
BFCR	2-4	Battery Filter Resistor	Added to replace Battery line filter that was removed.
BM1, 2		Grid Blower Motors 1 and 2	DC motors driving blowers to provide cooling air for the retarding grids.
BM1I / BM2I	2-3	Current Sensing Modules	Monitors current flowing through grid blower motors #1 and #2.
CCF1, 2	2-3	DC Link Filter Capacitors	Absorbs and releases current to the DC link for the grid resistors when a current spike occurs.
CCLR1, 2	2-3	Capacitor Charge Resistor Panels 1 and 2	Connected across the DC link to provide a voltage attenuated sample of the DC link voltage to the Capacitor Charge Indicating lights.
CCL1, 2	2-4	Capacitor Charge Indicating Lights 1 and 2	Illuminated when 50 volts or more is present on the DC link (the DC bus connecting the Alternator output, Chopper Module/Resistor Grid circuits and traction Inverters).
CD1, 2	2-3	Chopper Diodes 1 and 2	Controls the DC voltage applied to the grids during retarding.
CF11, 22, 21, 22	2-3	DC Link Filter Capacitors	Absorbs and releases current to the DC link for the Traction Motors when a current spike occurs.
CGBM1, 2	2-3	Blower Motor Capacitors	Limit the rate of current increase when starting to optimize motor commutation.
CMAF	2-4	Alternator Field Current Sensing Module	Detects amount of current flowing through the Alternator field winding.
CMT	2-4	Alternator Tertiary Current Sensing Module	Detects amount of current flowing through the Alternator tertiary winding.
CM1, 2	2-3	Chopper IGBT Phase Module 1 and 2	Controls the DC voltage applied to the grids during retarding.

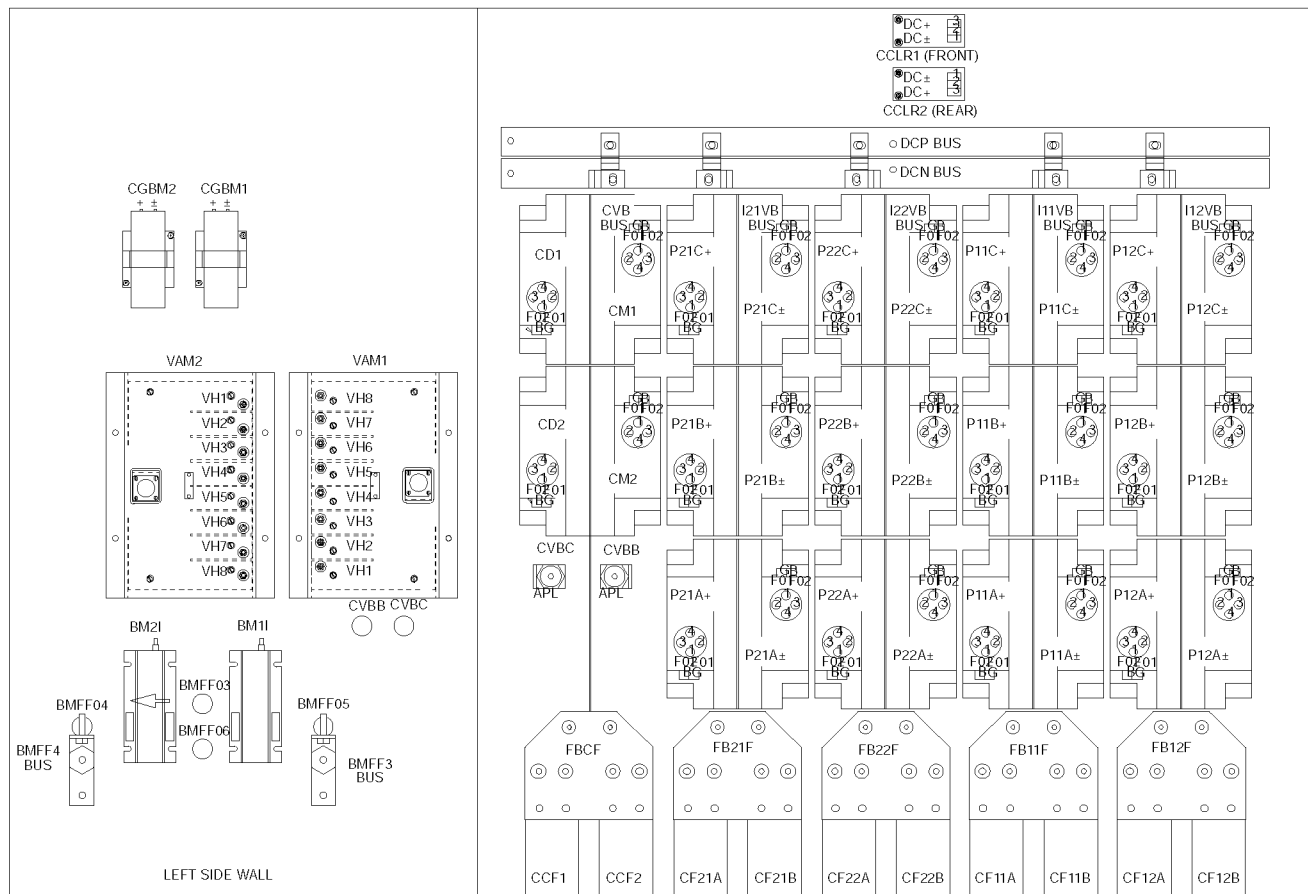
Table 5: PROPULSION SYSTEM COMPONENTS DESCRIPTION

	FIG. NO.	COMPONENT	FUNCTION
CM11A - 12C		Current Sensing Modules, Phase 1A, 1B and 1C	Detects amount of current flow through the A, B and C phases of Traction Motor 1.
CM21A - 22C		Current Sensing Modules, Phase 2A, 2B and 2C	Detects amount of current flow through the A, B and C phases of Traction Motor 2.
CPR	2-4	Control Power Relay	Picks up when the Key Switch and Control Power Switch are closed.
CPRD	2-4	Dual Diode Module	Allows two separate voltages to control the CPR coil.
CPRS	2-4	Control Power Relay Suppression Module	Suppresses voltage spike when CPR coil is de-energized.
CPS	2-4	Control Power Switch	Energizes CPR coil.
DCN BUS/DCP BUS	2-6	DC Link (-) and (+) Bus	The DC bus connects the Alternator output, Chopper Module/Resistor Grid circuits, and Traction inverters.
DID		Diagnostic Information Display	Provides maintenance personnel with the ability to monitor the operational status of certain truck systems and perform system diagnostic test.
DIGITAL I/O CARD		Digital Input/Output Card	Receives contactor, relay and switch feedback signals and provides drive signals to relays, contactors, indicator lamps, etc. Located in PSC and TCI.
FDR	2-6	Filter Discharge Resistor	Resistor divider network connected across the DC link, provides secondary discharge link for the DC link. Normal discharge is through RP1.
FIBER OPTIC ASSEMBLY		Fiber Optic Assembly	Provides voltage and electrical noise isolation for control and feedback signals between the PSC and Phase/ Chopper Modules.
FP	2-6	Filter Panel	Filters electrical noise on 3 phases of Alternator output.
GDPC1	2-4	Gate Driver Power Converter 1	Converts 19 to 95 VDC from the Gate Drive Power Supply to 25 kHz, 100 VRMS, square wave power to drive Inverter 1 IGBT Phase and Chopper Modules.
GDPC2	2-4	Gate Driver Power Converter 2	Converts 19 to 95 VDC from the Gate Drive Power Supply to 25 kHz, 100 VRMS, square wave power to drive Inverter 2 IGBT Phase and Chopper Modules.
GF	2-5	Alternator Field Contactor	Connects the AFSE to the Alternator field.
GFBR	2-4	Resistor	Provides a small load across the contactor feedbacks to help keep the contactors clean.
GFCO	2-4	Generator Field Contactor Cutout Switch	Disables Alternator output.
GFM1, 2		Gate Firing Module	Receives pulses from the Analog I/O card in the PSC, amplifies the pulses, and then splits the pulses to drive two SCR circuits in the AFSE. Located on AFSE panel.
GFR	2-5	Alternator Field Relay	Picks up with GF contactor and applies B+ to the AFSE (battery boost) during initial acceleration phase.
GFRS	2-5	Alternator Field Relay Coil Suppression Module	Suppresses voltage spikes when GF coil is de-energized.
GFS	2-5	Suppression Module	Suppresses voltage spikes in coil circuit when GF contactor is de-energized.
GRR	2-6	Ground Resistor Panel	Detects power circuit grounds.
GRR9, 10	2-4	Resistors	Used with GRR to detect power circuit grounds.

Table 5: PROPULSION SYSTEM COMPONENTS DESCRIPTION

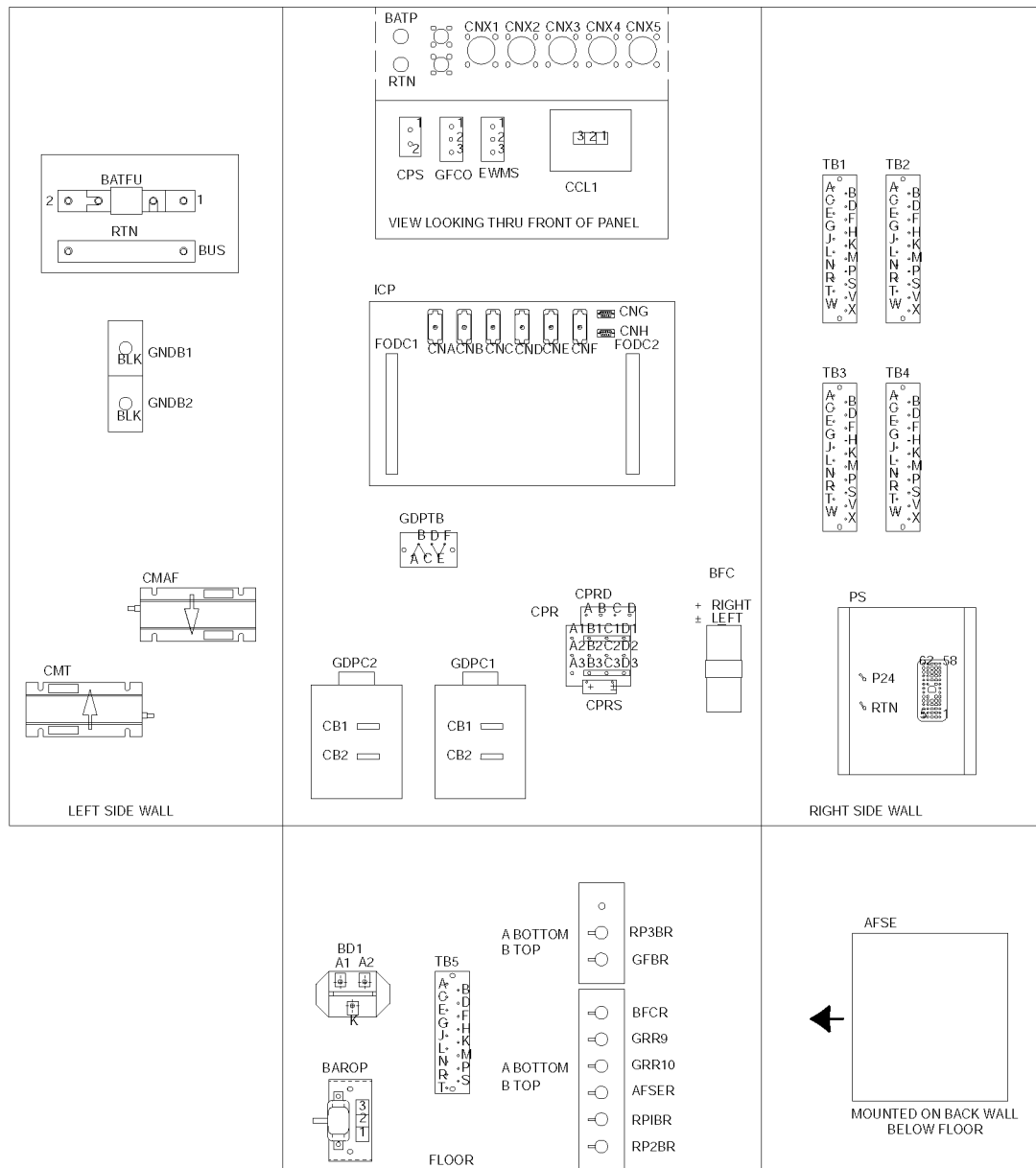
	FIG. NO.	COMPONENT	FUNCTION
ICP	2-4	Integrated Control Panel	The ICP is the main controller for the AC drive system. The ICP is composed of the PSC, TCI and inverter cards.
INV1 TMC CARD		Inverter 1 Central Processing Unit Card and Input/Output Card	Generates Phase Module turn-on/turn-off commands for the Inverter 1. Monitors voltages and currents from various areas for Inverter 1. Monitors Traction Motor 1 speed.
INV2 TMC CARD		Inverter 2 Central Processing Unit Card and Input/Output Card	Generates Phase Module turn-on/turn-off commands for the Inverter 2. Monitors voltages and currents from various areas for Inverter 2. Monitors Traction Motor 2 speed.
KEYSW		Key Switch	Connects battery voltage to CPR and control circuits when closed. (Located on instrument panel.)
LINK1	2-6	Link Current Sensing Module	Detects amount of current flow through the DC link.
L1, 2		Cabinet Lights	Provide interior cabinet illumination.
M1, 2		Motorized Wheels	Each Motorized Wheel consists of a Traction Motor and a Transmission Assembly. The 3-phase asynchronous Traction Motors convert electrical energy into mechanical energy. This mechanical energy is transmitted to the wheel hub through a double reduction gear train (Transmission).
P11A+, 11B+, 11C+ P12A+, 12B+, 12C+	2-3	IGBT Phase Modules	Provide positive driving voltages (PWM or square wave, depending on truck speed) for each of the three windings of Traction Motor 1.
P11A-, 11B-, 11C- P12A-, 12B-, 12C-	2-3	IGBT Phase Modules	Provide negative driving voltages (PWM or square wave, depending on truck speed) for each of the three windings of Traction Motor 1.
P21A+, 21B+, 21C+ P22A+, 22B+, 22C+	2-3	IGBT Phase Modules	Provide positive driving voltages (PWM or square wave, depending on truck speed) for each of the three windings of Traction Motor 2.
P21A-, 21B-, 21C- P22A-, 22B-, 22C-	2-3	IGBT Phase Modules	Provide negative driving voltages (PWM or square wave, depending on truck speed) for each of the three windings of Traction Motor 2.
PS	2-4	Power Supply	A DC to DC converter which provides regulated ± 24 VDC outputs from the unfiltered battery supply. Supplies power to PSC, TCI & LEMS.
PSC		Propulsion System Controller	The PSC is a part of the ICP, and is the main controller for the AC drive system. All propulsion and retarding functions are controlled by the PSC based on internally stored software instructions.
RDA, B, C	2-5	Rectifier Diode Panel	Converts Alternator 3-phase, AC voltage to DC voltage to power the two Inverters.
RG1A - 5C		Retard Grid Resistors	Dissipate power from the DC link during retarding, load box testing, and Inverter Filter Capacitor discharge operations.
RP1, 2	2-5	Retard Contactors 1and 2	When closed, connects Grid Resistors to the DC link during retarding, load box testing, and Inverter Filter discharge operations. <i>Note: Some trucks do not have RP3 installed.</i>
RP1S, RP2S	2-5	Suppression Modules	Suppresses voltage spikes in coil circuit when RP contactors are de-energized.
RP1BR \ RP2BR	2-4	Resistor	Provides a small load across the contactor feedbacks to help keep the contactors clean.

Table 5: PROPULSION SYSTEM COMPONENTS DESCRIPTION			
	FIG. NO.	COMPONENT	FUNCTION
R1	2-5	Battery Boost Resistor	Limits surge current in the Alternator field circuit when GFR contacts first close.
SS1, 2		Traction Motor Speed Sensors	Each speed sensor provides two output speed signals, proportional to the Traction Motor's rotor shaft speed.
SYS CPU Card		System Central Processing Unit Card	Provides control of propulsion and dynamic retarding functions, battery backed RAM, real-time clock, downloadable code storage, and an RS422 serial link.
TCI		Truck Control Interface	Is a part of the ICP Panel. Provides the main interface between the various truck systems, controls, and equipment and is used in conjunction with the DID by maintenance personnel.
TH1	2-5	Alternator Field Thyrite (Varistor)	Discharges the Alternator field when the AFSE is first turned off.
VAM1	2-3	Voltage Attenuation Module	Attenuates the three high voltage outputs applied to each phase winding of Traction Motor 1 to a level acceptable for use by the Analog I/O card in the ICP.
VAM2	2-3	Voltage Attenuation Module	Attenuates the three high voltage outputs applied to each phase winding of Traction Motor 2 to a level acceptable for use by the Analog I/O card in the ICP.
VAM3	2-6	Voltage Attenuation Module	Attenuates the high voltage outputs between the main alternator and the rectifier panel, and between the rectifier panel and the inverters to a level acceptable for use by the Analog I/O card in the ICP.
VAM4	2-5	Voltage Attenuation Module	Attenuates the high voltage outputs between the AFSE and the main alternator to a level acceptable for use by the Analog I/O card in the ICP.



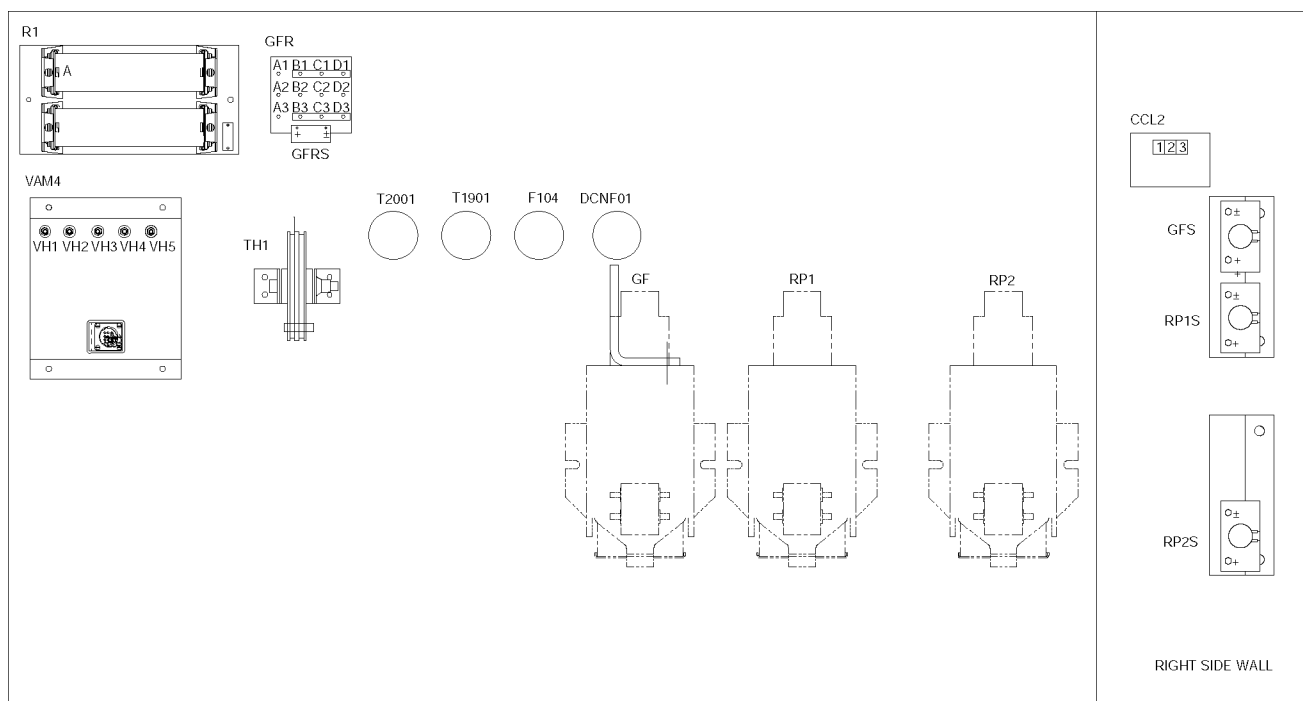
E020123

FIGURE 2-3. CONTROL CABINET COMPONENTS - HIGH VOLTAGE INVERTER AREA



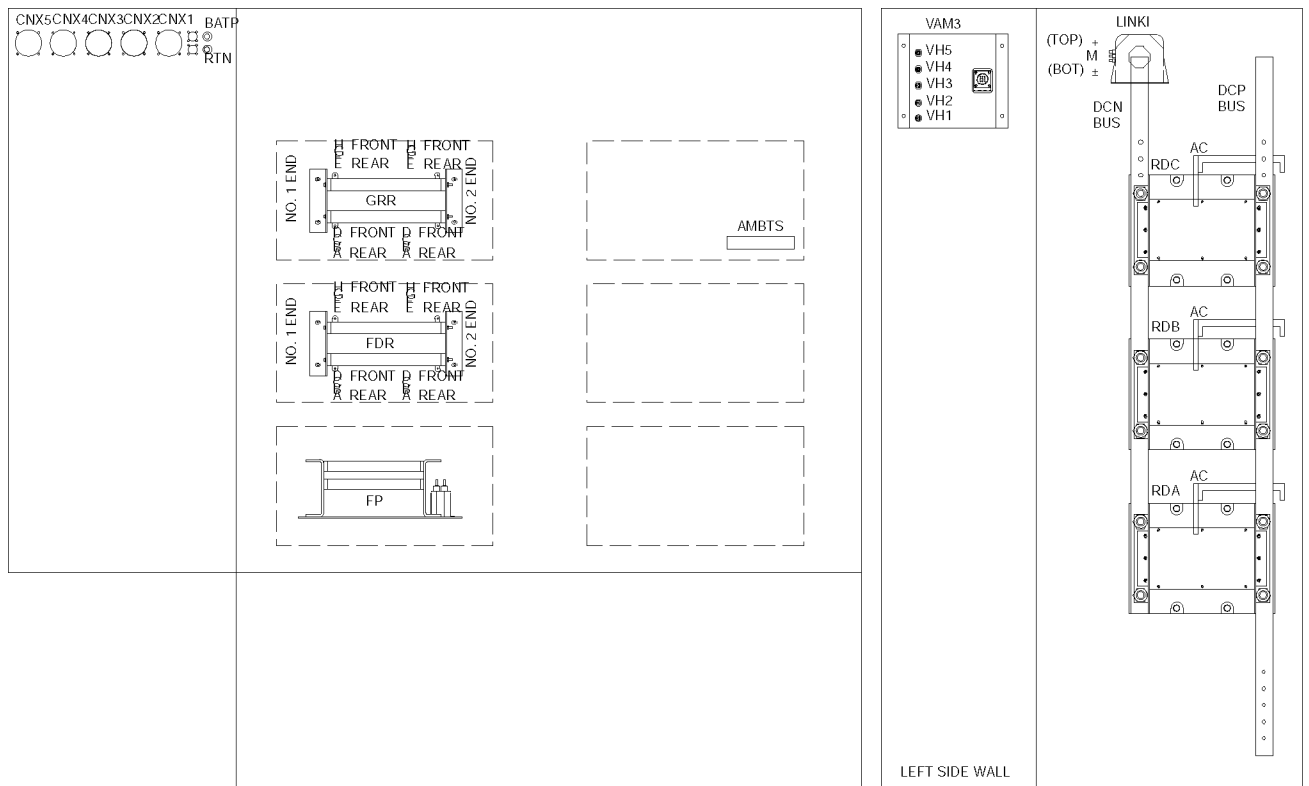
E020022

FIGURE 2-4. CONTROL CABINET COMPONENTS - LOW VOLTAGE CONTROL AREA



E020124

FIGURE 2-5. CONTROL CABINET COMPONENTS - CONTACTOR COMPARTMENT



E020125

FIGURE 2-6. CONTROL CABINET COMPONENTS - REAR CABINET VIEW

ELECTRONIC ACCELERATOR AND RETARD PEDALS

The accelerator pedal provides a signal to the Truck Control Interface (TCI) when the operator requests power. The retard pedal provides a signal to the Propulsion System Controller (PSC) when the operator requests retarding. The pedal signals are processed by the analog card in the respective panel for use by the system controllers to provide the desired mode of operation.

As the operator depresses the pedal, the internal potentiometer's wiper is rotated by a lever. The output voltage signal increases in proportion to the angle of depression of the pedal.

Repair and initial adjustment procedures are discussed in the following. Refer to AC Drive System Electrical Checkout Procedure for final calibration of the pedal potentiometer after installation in the truck.

Removal

NOTE: Repair procedures for the retard and accelerator pedal are identical. The retard pedal is mounted on the brake pedal. Refer to Section J for instructions for removing and installing the electronic pedal on the brake actuator.

NOTE: Note the routing and clamp location of the wiring harness. Proper wire routing is critical to prevent damage during operation after reinstallation.

1. Disconnect the pedal wiring harness from the truck harness connector.
2. Remove mounting cap screws, lockwashers and nuts, and remove the pedal assembly.

Installation

1. Install the pedal assembly using the mounting cap screws, lockwashers and nuts.
2. Connect the pedal wiring harness to the truck wiring harness.
3. Use the DID panel to calibrate the pedal potentiometer according to the instructions in the AC Drive System Electrical Checkout Procedure.

Disassembly

1. Remove the screws for cable clamps (1, Figure 2-7). The clamps can remain attached to wiring harness (2).
2. Remove the mounting screws and potentiometer (3).

Assembly

1. Position the potentiometer with the flat side toward the potentiometer cover and install it on the pedal shaft as follows:
 - a. Align the cutouts in the shaft with the potentiometer drive tangs.
 - b. Press the potentiometer onto the shaft until it bottoms against the housing.
2. Install the mounting screws.
3. Attach cable clamps (1) and tighten the screws securely.
4. Inspect the assembly and verify proper wiring clearance during operation of the pedal through the full range of travel.

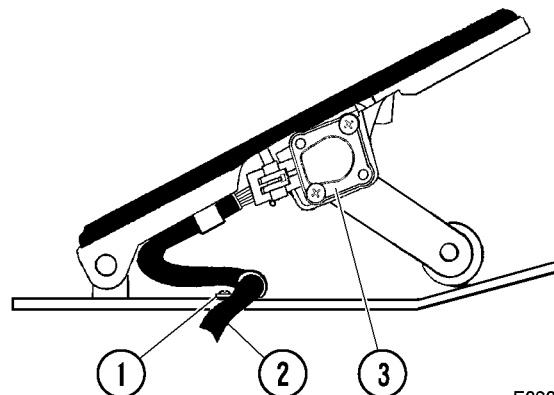


FIGURE 2-7. TYPICAL ELECTRONIC PEDAL

- | | |
|-------------------|------------------|
| 1. Cable Clamp | 3. Potentiometer |
| 2. Wiring Harness | |

SECTION E3

AC DRIVE SYSTEM ELECTRICAL CHECKOUT PROCEDURE

INDEX

AC DRIVE SYSTEM ELECTRICAL CHECKOUT PROCEDURE	E3-3
AC DRIVE SYSTEM MAINTENANCE	E3-3
NORMAL TRUCK SHUTDOWN PROCEDURE.....	E3-4
SHUTDOWN AFTER SYSTEM FAILURE	E3-4
SYSTEM CHECKOUT	E3-5
Battery and Control Circuit Checks - Battery Power OFF	E3-5
Battery and Control Circuit Checks - Battery Power ON	E3-8
Checks with Key Switch OFF	E3-9
Checks with Key Switch ON	E3-9
CPU Battery Checks.....	E3-9
MEMORY BACKUP BATTERY REPLACEMENT	E3-10
TCI PROGRAMMING.....	E3-10
PSC PROGRAMMING	E3-11
INVERTER PROGRAMMING	E3-11
CPU RESET	E3-11
PSC CHECKOUT	E3-12
PSC Digital Input Checks	E3-12
PSC Digital Output Checks.....	E3-16
TCI CHECKOUT.....	E3-17
Modular Mining Communication Port Check	E3-17
TCI Digital Input Checks.....	E3-17
TCI Digital Output Tests	E3-23
CALIBRATIONS	E3-25
Speedometer	E3-25
Accelerator Pedal, Retarder Pedal/Lever and RSC Dial	E3-25

ERASING EVENTS.....	E3-25
PSC.....	E3-25
TCI.....	E3-26
GATE DRIVER POWER CONVERTER TEST	E3-26
LOAD TESTING	E3-27
Preparation	E3-27
Alternator Speed Sensor Checks	E3-27
Battery Boost Check	E3-27
Brake Circuit Switch Checks	E3-28
Hoist & Steering Circuit Switch Checks.....	E3-28
Link Energized Checks	E3-28
Loadbox Test.....	E3-30
TROUBLESHOOTING	E3-34
PVM Optimum Load Curve Handshaking Troubleshooting.....	E3-34
Phase Module and Chopper Module Troubleshooting	E3-34
PHASE MODULE REPLACEMENT	E3-35
Phase Module Removal	E3-35
Phase Module Installation	E3-35

AC DRIVE SYSTEM ELECTRICAL CHECKOUT PROCEDURE

AC DRIVE SYSTEM MAINTENANCE



DANGEROUS VOLTAGE LEVELS ARE PRESENT WHEN THE ENGINE IS RUNNING AND CONTINUE TO EXIST AFTER SHUTDOWN IF THE REQUIRED SHUTDOWN PROCEDURES ARE NOT FOLLOWED. Before attempting repairs or working near propulsion system components, the following precautions and truck shutdown procedure must be followed:

- **DO NOT** step on or use any power cable as a handhold when the engine is running.
- **NEVER** open any electrical cabinet covers or touch the Retarding Grid elements until all shutdown procedures have been completed.
- **ALL** removal, repairs and installation of propulsion system electrical components, cables etc. must be performed by an electrical maintenance technician properly trained to service the system.
- **Power cables must be cleated in wood or other non-ferrous materials. Do not repair cable cleats by encircling the power cables with metal clamps or hardware. Always inspect power cable insulation prior to servicing the cables and prior to returning the truck to service. Discard cables with broken insulation.**
- **IN THE EVENT OF A PROPULSION SYSTEM MALFUNCTION, a qualified technician should inspect the truck and verify the propulsion system does not have dangerous voltage levels present before repairs are started.**



- *If weld repairs are required, the welding ground electrode should be attached as close as possible to the area to be welded. NEVER weld on the rear of the Electrical Control Cabinet or the retard grid exhaust air louvers. Power cables and wiring harnesses should be protected from weld spatter and heat.*
- *Prior to welding, disconnect Engine Control System (ECS) harnesses and ground wire (MTU engine). If equipped with DDEC or Komatsu engine, disconnect ECM harnesses. GE cards should be pulled forward far enough to disconnect card from backplane connector.*
- *Some power cable panels throughout the truck are made of aluminum or stainless steel. They must be repaired with the same material or the power cables may be damaged.*

After the truck is parked in position for the repairs, the truck must be shut down properly to ensure the safety of those working in the areas of the deck, electrical cabinet, traction motors, and retarding grids. The following procedures will ensure the electrical system is properly discharged before repairs are started.



If a problem occurs in the AC drive system that prevents use of normal shutdown procedures, ADDITIONAL PRECAUTIONS ARE NECESSARY to ensure that dangerous drive system voltages are not present when tests or repairs are performed.

NORMAL TRUCK SHUTDOWN PROCEDURE

1. Reduce the engine speed to idle. Place the directional control lever in PARK. Make sure that the parking brake applied indicator light is illuminated.
2. Place the rest switch in the ON position to put the AC drive system in the rest mode. Make sure that the rest mode indicator light is illuminated.
3. Turn the key switch to the OFF position. A timing sequence will be activated automatically to allow the engine to cool down before it is turned off.

If the engine does not shut down with the key switch after the timing sequence is done, use the engine shutdown switch on the operator cab center console. Pull this switch up until the engine stops.

4. With the key switch OFF and the engine stopped, wait at least 90 seconds. Make sure that the steering circuit is completely depressurized by turning the steering wheel back and forth several times. The front wheels should not turn when the hydraulic pressure is relieved. If the front wheels can still be turned, notify maintenance personnel.
5. Make sure that the link voltage light in the control cabinet is off. Notify maintenance personnel if the light remains illuminated for longer than five minutes after the engine is shut down.
6. Locate GF cutout switch (2, Figure 3-1) in the access panel on the left side of the main control cabinet. Place the switch in the CUTOUT position. This will prevent the alternator from re-energizing and creating system voltage until the switch is returned to its former position.

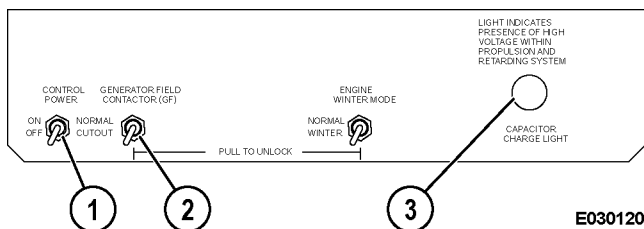


FIGURE 3-1. INFORMATION DISPLAY PANEL

1. Control Power Switch
2. GF Cutout Switch
3. Capacitor Charge Light



If there is any question whether the system has potential hazardous voltage present, return to the operator cab and perform the normal shutdown procedure. Normal operation of the drive system at shutdown should allow high voltages to be dissipated.

SHUTDOWN AFTER SYSTEM FAILURE

NOTE: In the event of a system failure, performing the following procedure will ensure that no hazardous voltages are present in the drive system.

1. Before shutting off the engine, verify the status of the drive system warning lights on the overhead display. Use the lamp check feature to verify proper lamp function.

NOTE: The link voltage light in the control cabinet is not lamp checked.



If any of the red drive system warning lights are on, DO NOT attempt to open any cabinets, disconnect any cables, or reach inside the control cabinet even after turning off the engine.

2. If all red drive system warning lights are off, move the directional selector lever to PARK, turn off the engine and chock the wheels.
3. After the engine has been off for at least five minutes, inspect the link voltage lights on the DID panel in cab. If all lights are off, the retarding grids, wheel motors, alternator, and power cables connecting these devices are safe to work on.



If the link voltage light continues to be illuminated after following the above steps, a fault has occurred.

Leave all cabinet doors in place. Do not touch the retarding grid elements. Do not disconnect any power cables or use them as hand or footholds. Notify the Komatsu factory representative or distributor immediately.

SYSTEM CHECKOUT

Test equipment that is required to fully test the AC drive system:

- One or two Portable Test Units (PTU) or laptop computers
- One digital multimeter
- Several jumper wires
- One analog VOM
- One 500 volt megger

The Portable Test Unit (PTU) is used to test, download and record system parameters on the TCI and PSC modules.

- The PTU is plugged into the DIAG1 port on the DID panel at the rear of the operator cab for monitoring the PSC module.
- The PTU is plugged into the DIAG3 port on the DID panel at the rear of the operator cab for monitoring the TCI module.

The TCI and PSC are programmed through the DB9 ports on the DID panel. The inverter cards are programmed through the DB9 ports on the ICP panel.

NOTE: If only one PTU is available, in some cases it will be necessary to switch between the PSC port and the TCI port to complete the test if it is necessary to monitor both during a test procedure. After the serial cable has been switched, exit to the Main Menu and the software will automatically switch to the menu for the connected panel.

Several different numbering methods or symbols are used in the following procedures to denote the operation to be performed:

1., 2., a., b., etc: Test preparation and instruction steps are preceded by a number or a letter.

- Procedures requiring visual checks, voltage measurements, etc, are preceded by this symbol.
- ◆ PTU keyboard entry steps are preceded by this symbol.

{escape}: When a keyboard key must be pressed, the key label is enclosed in braces.

- PTU screen display information is shown in this type font and preceded by this symbol.

NOTE: The following test procedures are applicable to Release 21 software. Procedures required for later software versions may vary. Contact the Komatsu distributor or factory representative for the current software version available.

Battery and Control Circuit Checks - Battery Power OFF



Make sure that the link voltage is drained down before servicing the propulsion system or performing tests.

1. Prepare for the following checks by performing the following procedure:
 - a. Turn both battery disconnect switches to the OFF position.
 - b. Disconnect circuit wire 21B from the starter solenoid.
 - c. Remove the 50 amp fuse (BATFU) from the left wall of the right side compartment of the control cabinet.
 - d. Disconnect the CN1 connector from the power supply on the right wall of the right side compartment of the control cabinet.
 - e. Open the ICP panel and slide the cards out far enough to disconnect them from the backplane.
 - f. Turn off all circuit breakers in the auxiliary control cabinet.
 - g. Make sure that the key switch is OFF, the 5 minute delay timer is OFF, and the rest switch is ON.
 - h. Turn off all lights and switches.

Battery Circuit Voltage Check:

2. In the auxiliary control cabinet, measure voltage to ground at each of the following circuits; 11, 11B1, 712 @ TB32, and 11ST @TB28.

☐ All voltages should be zero.

Resistance Checks, Low Voltage Circuits:

3. Measure the resistance from ground to the circuits listed in Table I. Stop and troubleshoot any direct short (0 ohms) to ground.

☐ All circuits should show some resistance as shown in Table I.

TABLE I. CIRCUIT RESISTANCE CHECKS
(All readings from circuit to ground)

CIRCUIT	LOCATION	APPROX. VALUE	NOTES
11B1	*	∞	Measure at the 12VDC insulator in the auxiliary control cabinet.
11	*	∞	Measure at the 24VDC insulator in the auxiliary control cabinet. All devices listed for the 11A circuit reading must be off.
15V	TB21	∞	
71GE	TB22	120 Ω	
439	TB25	∞	
10V	TB28	∞	
11SL	TB28	∞	Engine service lights must be turned off.
11ST	TB28	∞	
15PV	TB29	∞	
11S	TB30	∞	Ground level engine shutdown switch must be deactivated.
11A	TB30	∞	The following devices must be turned off: • Brake cabinet service light • Operator cab dome light • Hazard lights • Headlights • Ground level engine shutdown switch • Left and right side engine service lights
712	TB32	∞	Auxiliary control cabinet service lights must be turned off.
71	TB32	∞	
11KS	*	∞	Measure at the key switch.
12M	*	>10 Ω	Measure at AID Module terminal B-13.
12F	*	>200 Ω	Measure at AID Module terminal B-12.

Resistance Checks, Propulsion System Circuits:

4. Prepare for the following checks by performing the following additional procedure:
 - a. Make sure that all circuits are restored and the key switch is OFF.
 - b. Place the GF cutout switch, located on the right side of the control cabinet, in the CUTOUT position (down).
 - c. Disconnect the CCLR1 connector and the CCLR2 connector located in the center compartment of the control cabinet near the top of each vertical bus bar.
 - d. Disconnect the output plugs on the four VAM panels.
 - e. Remove the wires on the GNDB ground blocks located on the left wall of the right side compartment of the control cabinet. Ensure that the lugs on these wires are not touching one another after the wires are removed.
 - f. Remove the output plugs on both gate driver power converters (GDPC1 and GDPC2).
 - g. Remove the FAULTP02 wire on the GRR9 resistor.

DC Link Checks:

NOTE: Use an analog meter (VOM) to measure resistance in the following steps. The VOM must be on the Rx1 scale. Otherwise, the link capacitors will start charging and an accurate reading will not be possible.

5. Place the VOM positive lead on the DC(+) link bus and the VOM negative lead on a cabinet ground.
☐ Resistance should be 2 megohms or greater.
6. Place the VOM positive lead on the DC(-) link bus and the VOM negative lead on a cabinet ground.
☐ Resistance should be 2 megohms or greater.
7. Place the VOM positive lead on the DC(+) link bus and the VOM negative lead on the DC(-) link bus.
☐ Resistance should be approximately 1500 ohms.
8. Place the VOM positive lead on the DC(-) link bus and the VOM negative lead on the DC(+) link bus.
☐ Resistance should be approximately 6 ohms.

Chopper Modules (CM1, CM2):

9. With the VOM set on the Rx10,000 scale, measure the following at each chopper module in the control cabinet:
☐ GR(-) wire to ground resistance should be approximately 2 megohms or greater.
☐ GR(+) wire to ground resistance should be approximately 2 megohms or greater.

AFSE P1 Adjustment:

10. Connect an ohmmeter from the wiper of Pot P1 (cathode of ZD1) to Terminal E (GND) on the battery boost module.
☐ If necessary, adjust P1 to obtain an ohmmeter reading of 6000 ohms.
11. Restore the following circuits:
 - a. Reinsert all ICP panel cards.
 - b. Reconnect the ground wires at the GNDB ground blocks.
 - c. Reconnect the output plugs on the four VAM panels.
 - d. Reconnect the output plugs to both gate driver power converters (GDPC1 and GDPC2).
 - e. Reconnect the CCLR1 and CCLR2 connectors.
 - f. Reconnect the FAULTP02 wire to the GRR9 resistor.

GRR Wiring:

12. Measure the resistance between the FAULTP02 wire of the GRR9 resistor and ground.
☐ Resistance should be 60 ohms.
13. Measure the resistance between the FAULTP02 wire of the GRR9 resistor and the DC(-) link bus.
☐ Resistance should be approximately 1100 ohms.
14. Measure the resistance between the FAULTP02 wire of the GRR9 resistor and the DC(+) link bus.
☐ Resistance should be approximately 1500 ohms.
15. Reconnect the CN1 connector to the power supply. Remove any test equipment.
16. Activate the battery disconnect switches. Close all open circuit breakers.

**Battery and Control Circuit Checks -
Battery Power ON**



Make sure that the link voltage is drained down before servicing the propulsion system or performing tests.

1. Prepare for the following checks by performing the following procedure:
 - a. Remove the 50 amp fuse (BATFU) from the left wall of the right side compartment of the control cabinet.
 - b. Disconnect circuit wire 21B from the starter solenoid.
 - c. Plug in all the cards in the ICP panel. Verify that all the CN connectors are connected and control power switch (1, Figure 3-1) is OFF.
 - d. Verify that all circuit breakers are closed and the battery disconnect switches are deactivated.
 - e. Make sure that the key switch and the 5 minute delay timer are OFF.

Power Supply Check (PS):

2. Remove the CN1 connector on the power supply. Use an ohmmeter to check the harness side connector pins to ground. Refer to Table II for the resistance value at each pin.
3. Check for 1.4K ohms between TB3-K and TB3-L (LEM +24V to -24V power supply busses).
4. After resistance checks are complete, reconnect CN1 connector.

**TABLE II. POWER SUPPLY HARNESS
RESISTANCE CHECKS**

Pin	Ohms	Circuit
1	27	+5
2	27	+5
3	27	+5
4	27	+5
5	27	+5
6	27	+5
7	—	—
8	12K	+5
9	12K	+5
10	12K	+5
11	27	+5
12	0	return
13	0	return
14	0	return
15	0	return
16	0	return
17	0	return
18	—	—
19	0	return
20	0	return
21	0	return
22	—	—
23	770	+15
24	770	+15
25	770	+15
26	770	+15
27	—	—
28	210	+15
29	210	+15
30	0	return
31	0	return

Pin	Ohms	Circuit
32	0	return
33	0	return
34	0	return
35	0	return
36	0	return
37	0	return
38	1K	-15
39	1K	-15
40	1K	-15
41	1K	-15
42	0	return
43	—	—
44	0	return
45	0	return
46	860	+24
47	50	BP24
48	860	+24
49	—	—
50	—	—
51	—	—
52	—	—
53	—	—
54	0	return
55	0	return
56	1.4K	-24
57	8K	psstat
58	1.4K	-24
59	—	—
60	—	—
61	—	—
62	—	—

Checks with Key Switch OFF

5. With key switch OFF, verify 25VDC to ground minimum for the following circuits:

- ☐ 11
- ☐ 11s on TB30
- ☐ 11L on CB30
- ☐ 11A on TB30
- ☐ 11SL on TB28

NOTE: The voltage should be at least 25VDC. If the voltage is significantly low, check the battery circuits. If the voltage is slightly low, install a battery charger.

6. Check circuit 11B1 voltage to ground.

- ☐ The voltage should be approximately 12VDC.

7. Make sure that the CN1 connector on the power supply is connected. Install the BATFU fuse.

8. Activate the battery disconnect switches and turn on the ground level engine shutdown switch. Make sure all circuit breakers are closed.

10. Use a digital multimeter to check the polarity of the battery voltage at the BATFU fuse holder. Connect the positive lead to BATP and the negative lead to RTN.

- ☐ The voltage at BATP should be +24V.

Checks with Key Switch ON

1. Deactivate the battery disconnect switches. Turn the key switch ON.

2. Check the voltage of circuit 712 to ground.

- ☐ The voltage should be 25VDC.

3. Check the voltage of circuit 71CK to ground.

- ☐ The voltage should be 25VDC.

4. Turn control power switch (1, Figure 3-1) to ON. Verify that all five green lights in the power supply are lit.

5. Verify that the display on the DID panel is lit.

6. Check the voltage on the following circuits in the auxiliary control cabinet. Voltage at each should be at least 25VDC:

- ☐ 71 on TB32
- ☐ 71GE on TB22

7. Verify the specified voltage to ground at the following locations in the auxiliary control cabinet:

- ☐ +15VDC at the 15PV wire on TB29 (from TCI, supply to operator control pedals)
- ☐ +15VDC at the 15V wire on TB21 (from TCI, power to cab gauges)
- ☐ +15VDC at the 15SPD, 15RWS and 15LWS wires on TB21 (from TCI, power to front wheel speed sensors)
- ☐ Approximately 11VDC at the 10V wire on TB28
- ☐ +15VDC at the 15VL wire on TB32
- ☐ Approximately 14.8VDC at the 15SIM wire on TB32.

8. Measure voltage between circuit 72E (+) (TB24) and circuit 0 (-) (TB24-H) in the auxiliary control cabinet.

- ☐ The voltage should be 5VDC.

9. Install a jumper from 22F0 @ TB32 to ground.

- ☐ The voltage should change to 7VDC.

10. Remove the jumper to 22F0.

11. Verify 12VDC to ground for the following circuits:

- ☐ 65
- ☐ 67C
- ☐ 67R
- ☐ 67P

CPU Battery Checks

12. Turn the control power switch to OFF.

13. Connect a VOM across the screws securing the green plate on the edge of the PSC CPU card in the ICP panel. Connect the positive lead to top screw (2, Figure 3-2) and the negative lead to bottom screw (4).

- ☐ The value should be approximately 3.5 volts.

14. Connect a VOM across the screws securing the green plate on the edge of the TCI CPU card in the TCI panel. Connect the positive lead to top screw (2) and the negative lead to bottom screw (4).

- ☐ The value should be approximately 3.5 volts.

NOTE: If battery voltage is low in either battery check, refer to "Memory Backup Battery Replacement" instructions on the following page.

MEMORY BACKUP BATTERY REPLACEMENT

The replaceable memory backup battery on the CPU cards will require replacement if voltage is low when performing the CPU Battery Checks or if, during truck operation, an event code appears on the DID panel display as follows:

- ☐ Event Number 095 (BBRAM Battery Low)
- ☐ Event Number 633 (BBRAM Battery Failure)

To replace the battery on either card:

1. With control power OFF, remove the appropriate card and locate the green plate with the battery near the card edge. (See Figure 3-3.)
2. Have a replacement battery available for immediate installation. See your truck's Parts Book for the correct battery part number.

NOTE: To prevent data loss, the new battery must be installed within five minutes of removal of the old battery.

3. Remove both screws (2) and (4) that retain battery assembly (3) to the mounting bocks. Note the arrow direction (polarity) on the green plate before removal.
4. Remove the old battery and install the new battery. Make sure that it is positioned for proper polarity. Reinstall the screws.
5. Install the card in the appropriate panel slot.

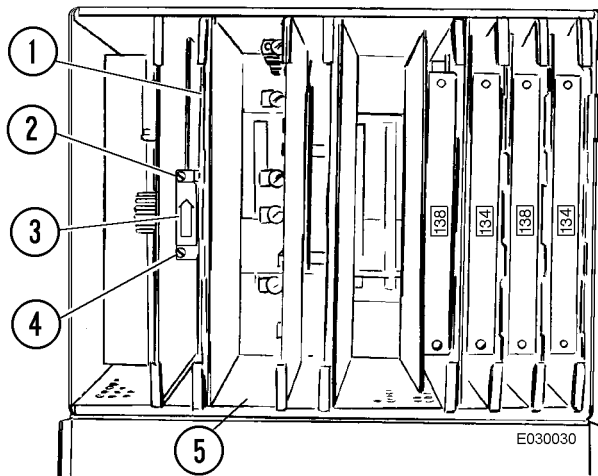


FIGURE 3-2. BATTERY LOCATION
(PSC Panel Shown)

- | | |
|-----------------------|-----------------------|
| 1. PSC CPU Card | 4. Negative (-) Screw |
| 2. Positive (+) Screw | 5. Panel Enclosure |
| 3. Battery Assembly | |

TCI PROGRAMMING



Make sure that the link voltage is drained down before servicing the propulsion system or performing tests.

1. Connect the serial communication cable from the PTU to the TCI port (DIAG3) on the DID panel located on the back wall of the operator cab.
2. Make sure that the directional control lever is in PARK and the rest switch is in the REST position.
3. Turn control power switch (1, Figure 3-1) and the key switch ON.

To program the TCI CPU card:

- ◆ Click START > Programs > GEOHVPTU_2.0 > AC TOOLS > wPTU AC v21.01
- ◆ Select "Normal" mode {enter}
- ◆ Type password "ok75e" {enter}
- ◆ Click "Program Panel".
- ◆ Select the GE Panel to download.
- ◆ To select the Configuration File, click "Browse", then "Up One Level", then select the file that matches your truck's wheels.
- ◆ Click "Open".
- ◆ Click "Begin Download".
- ◆ After the download is complete, click "Exit" when you see "Press exit to continue".

PSC PROGRAMMING



Make sure that the link voltage is drained down before servicing the propulsion system or performing tests.

1. Connect the serial communication cable from the PTU to the PSC port (DIAG1) on the DID panel located on the back wall of the operator cab.
2. Make sure that the directional control lever is in PARK and the rest switch is in the REST position.
3. Turn control power switch (1, Figure 3-1) and the key switch ON.

To program the PSC CPU card:

- ◆ Click START > Programs > GEOHVPTU_2.0 > AC TOOLS > wPTU AC v21.01
- ◆ Select "Normal" mode {enter}
- ◆ Type password "ok75e" {enter}
- ◆ Click "Program Panel".
- ◆ Select the GE Panel to download.
- ◆ To select the Configuration File, click "Browse", then "Up One Level", then select the file that matches your truck's wheels.
- ◆ Click "Open".
- ◆ Click "Begin Download".
- ◆ After the download is complete, click "Exit" when you see "Press exit to continue".

INVERTER PROGRAMMING



Make sure that the link voltage is drained down and the engine is not running before performing the following procedures.

1. Connect the serial communication cable from the PTU to the top ports on the ICP panel (CNG for inverters 11 and 12, CNH for inverters 21 and 22).
2. Turn GF cutout switch (2, Figure 3-1) to the CUTOUT position.

To program the inverters:

- ◆ Click START > Programs > GEOHVPTU_2.0 > AC TOOLS > wPTU AC v21.01
- ◆ Select "Normal" mode {enter}
- ◆ Type password "ok75e" {enter}
- ◆ Click "Program Panel".
- ◆ Select the GE Panel to download.
- ◆ Click "Begin Download".
- ◆ After the download is complete, click "Exit" when you see "Press exit to continue".

CPU RESET

After any programming is done, turn the control power switch to OFF. Wait for the green LEDs on the power supply turn off, then turn the control power switch to ON. This allows for synchronization of all CPU communication links.

PSC CHECKOUT

PSC Digital Input Checks

1. Open circuit breakers 1 and 2 for gate driver power converters 1 and 2.
2. Connect the serial communication cable from the PTU to the PSC port (DIAG1) on the DID panel located on the back wall of the operator cab.
3. Make sure that the directional control lever is in PARK and the rest switch is in the REST position.
4. Turn control power switch (1, Figure 3-1) and the key switch ON.

To check the PSC digital inputs:

- ◆ Click START > Programs > GEOHVPTU_2.0 > AC TOOLS > wPTU AC v21.01
- ◆ Select "Normal" mode {enter}
- ◆ Type password "ok75e" {enter}
- ◆ Under "Real Time", double-click "PSC Real Time Data".
- Verify that the analog values are similar to the example in Figure 3-3. Also, make sure that the correct truck ID is at the top of the screen and "COMMLINK" signal under "Modes" reads OK.
- The following normal power-on digital inputs should be highlighted: KEYSW, PSOK, CNX, CPSFB and CNFB.
- BRAKEON will be highlighted with wire 44R (TB26) jumpered to 712 (TB22). The wires do not have to be removed.
- GFNCO should be highlighted with the GF cutout switch in the NORMAL position.

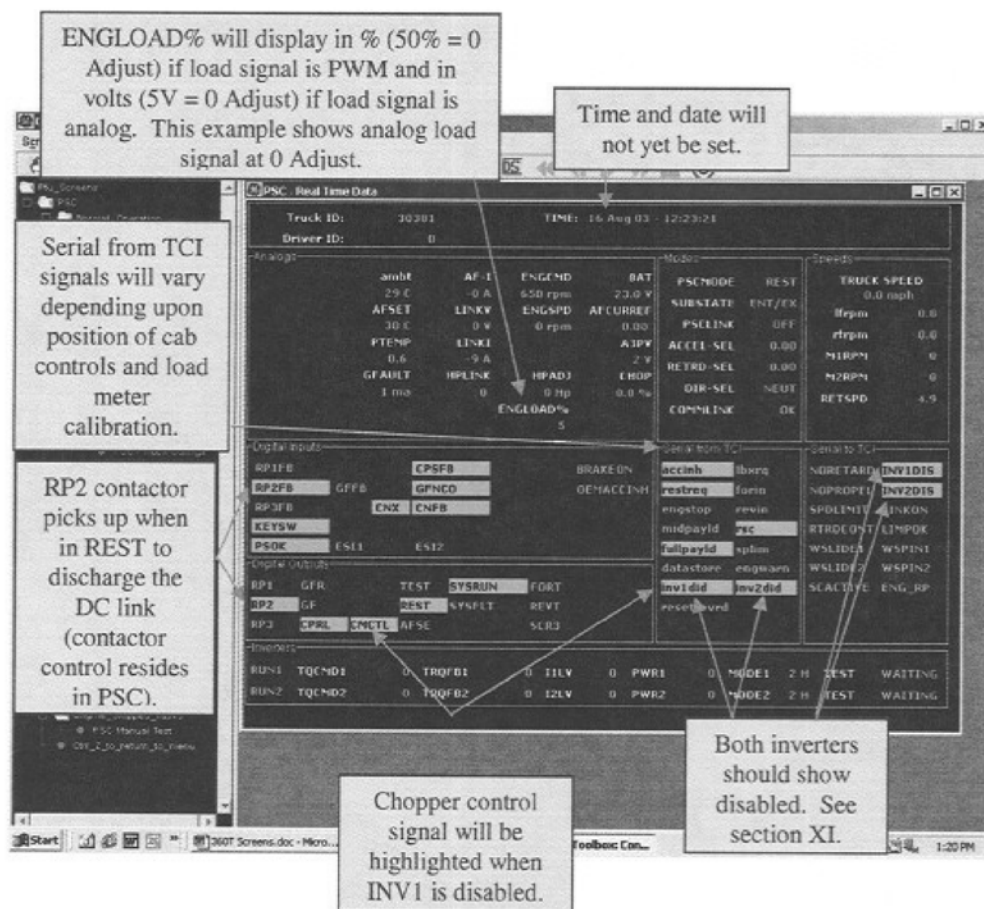


FIGURE 3-3. PSC REAL TIME DATA SCREEN

- ◆ Close the “PSC Real Time Data” screen, then double-click “PSC Serial Data”.
- Verify that the analog and digital values are similar to the example in Figure 3-4.

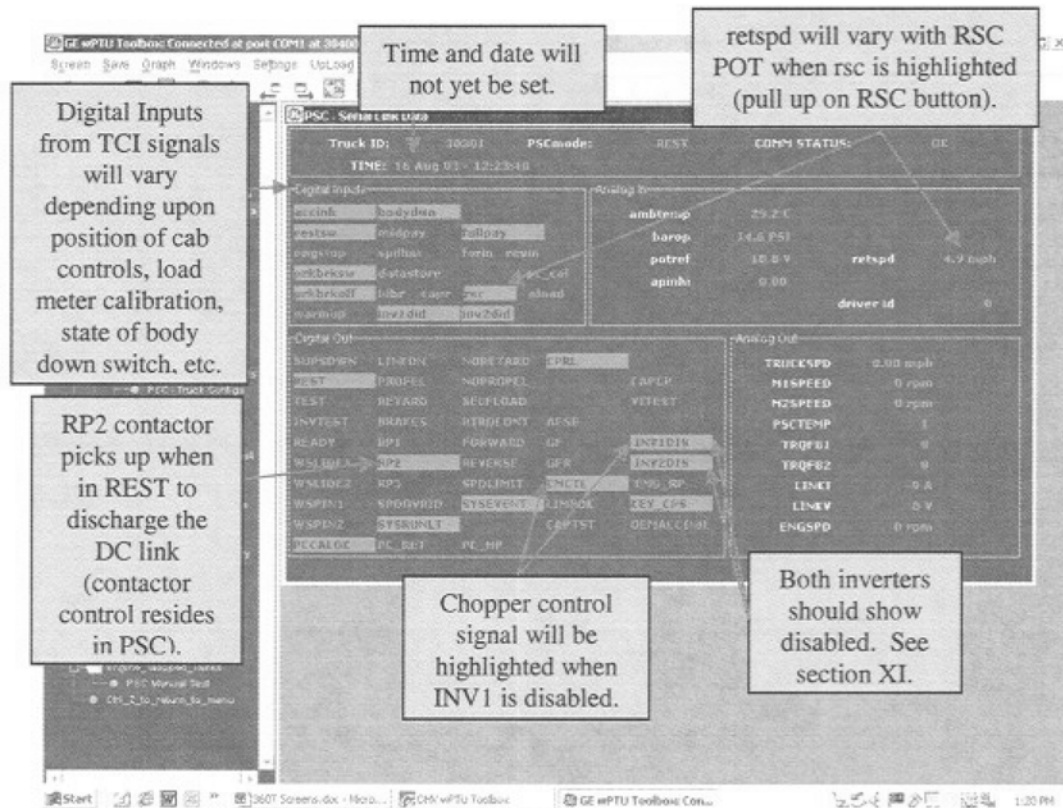


FIGURE 3-4. PSC SERIAL DATA SCREEN

- ◆ Close the “PSC Serial Data” screen, then double-click “PSC Analog Inputs”.
- Verify that the analog and digital values are similar to the example in Figure 3-5.
- With the retarder pedal not depressed, verify that the “RETARD PEDAL” signal is approximately 1.5 volts.
- With the retarder pedal fully depressed, verify that the “RETARD PEDAL” signal is approximately 8.5 volts.
- With the retarder lever fully up (OFF position), verify that the “RETARD LEVER” signal is approximately 0 volts.
- With the retarder lever fully down, verify that the “RETARD LEVER” signal is approximately 8.75 volts.
- The “ENGINE LOAD” signal should be either 50% if the PWM load signal is being used or 5.0 volts if the analog load signal is being used. This indicates a 0 HP adjust level with the engine off.

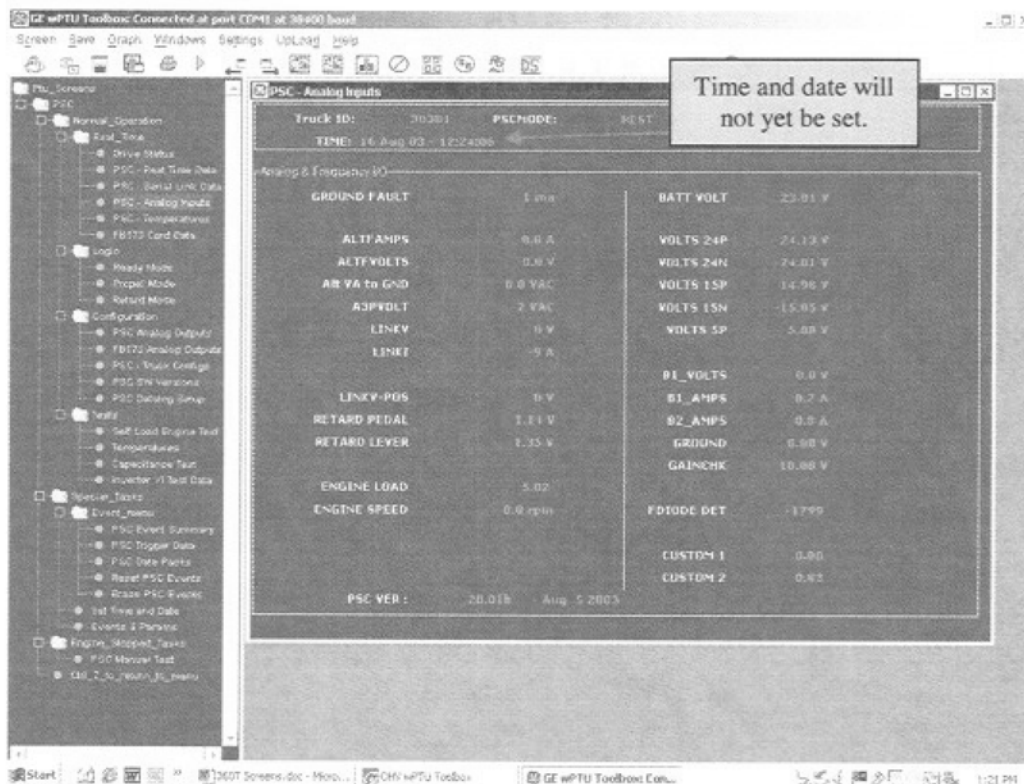


FIGURE 3-5. PSC ANALOG INPUTS SCREEN

- ◆ Close the “PSC Analog Inputs” screen, then double-click “PSC Temperatures”.
- Verify that the temperature values are similar to the example in Figure 3-5.

NOTE: All temperatures are calculated except for the **AFSE** and **AMBIENT TEMPERATURE** values.

- ◆ Close the “PSC Temperatures” screen.

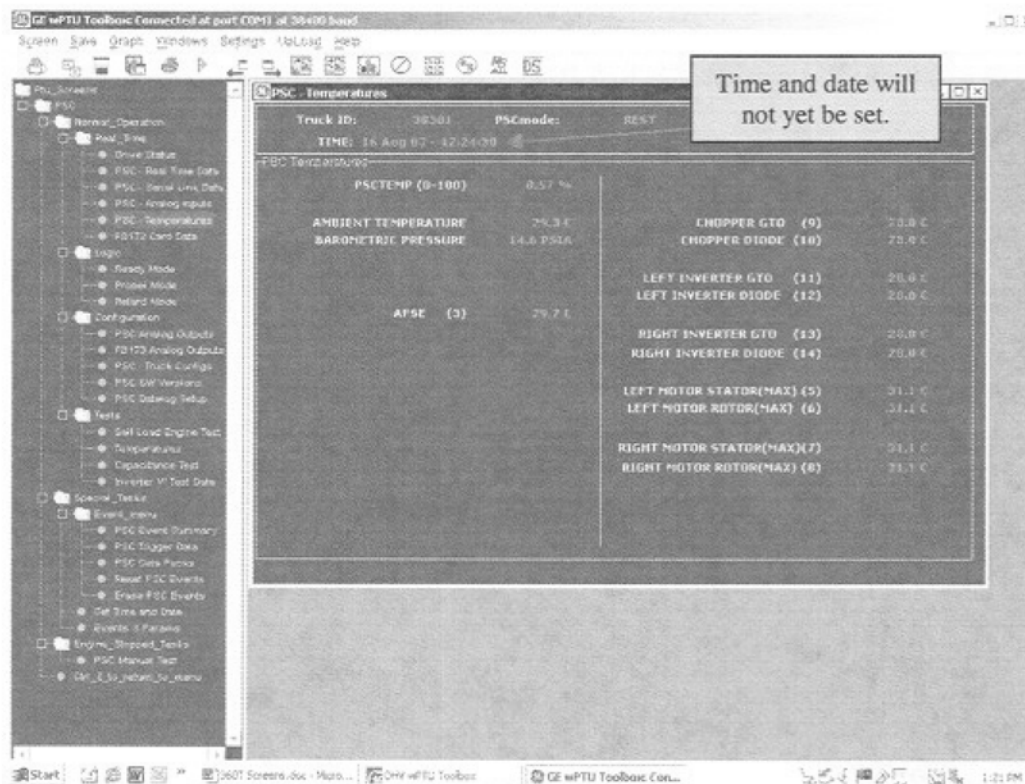


FIGURE 3-6. PSC TEMPERATURES SCREEN

PSC Digital Output Checks

1. Open circuit breakers 1 and 2 for gate driver power converters 1 and 2.
2. Connect the serial communication cable from the PTU to the PSC port (DIAG1) on the DID panel located on the back wall of the operator cab.
3. Make sure that the directional control lever is in PARK and the rest switch is in the REST position.
4. Turn control power switch (1, Figure 3-1) and the key switch ON.

To check the PSC digital outputs:

- ◆ Click START > Programs > GEOHVPTU_2.0 > AC TOOLS > wPTU AC v21.01
- ◆ Select "Normal" mode {enter}
- ◆ Type password "ok75e" {enter}
- ◆ Click "LOGIN to wPTU Toolbox".
- ◆ Under "Engine Stopped Task", double-click "PSC Manual Test".
- ◆ The "PSC Manual Test" screen looks similar to the "PSC Real Time Data" screen in Figure 3-3. Clicking the buttons in the "Digital Outputs" field will toggle the output on and off. See Table III.

NOTE: For Steps 1 through 5, remove the contactor arc chutes and observe the contacts while they are closed to ensure that the tips are mating properly. These contactors are interlocked so they will not close with the arc chutes removed. For testing, the interlock can be pushed inward to allow contact closure with the arc chute removed.

NOTE: Do not check CMCTL at this time.

TABLE III. PSC DIGITAL OUTPUT CHECKS

STEP	OUTPUT	DESCRIPTION	DEVICE CHECKOUT
1	RP1	RP1 Contactor	Verify that RP1 picks up and RP1FB is highlighted.
2	RP2	RP2 Contactor	Verify that RP2 picks up and RP2FB is highlighted.
3	RP3	RP3 Contactor (If installed)	If installed, verify that RP3 picks up and RP3FB is highlighted.
4	GFR	GFR Contactor	Verify that the GFR relay picks up.
5	GF	GF Contactor	Verify that the GF contactor picks up and GFFB is highlighted on the PTU. The GF Cutout Switch must be in the NORMAL (up) position to check.
6	GF	GF Cutout Switch Safety Check	Move the GF Cutout Switch to the CUTOFF (down) position. Verify that the GF contactor does not pick up and GFFB is not highlighted on the PTU. Return the GF Cutout Switch to the NORMAL (up) position.
7	CPRL	Control Power Relay	With CPRL highlighted, turn off the Control Power Switch and verify that control power is not lost. Turn the switch back on.
8	AFSE	Alternator Field Static Exciter	With AFSE highlighted, verify 24 volts to ground on the "+25" terminal on the AFSE terminal board.
9	FORT	Forward Travel Direction	Verify that circuit 72FD changes from 24VDC to 0VDC when FORT is activated.
10	REVT	Reverse Travel Direction	Verify that circuit 79RD changes from 24VDC to 0VDC when REVT is activated.

TCI CHECKOUT



Make sure that the link voltage is drained down before servicing the propulsion system or performing tests.

Modular Mining Communication Port Check

1. Connect the serial communication cable from the PTU to the Modular Mining port on the DID panel located on the back wall of the operator cab.
2. Make sure that the directional control lever is in PARK and the rest switch is in the REST position.
3. Turn control power switch (1, Figure 3-1) and the key switch ON.

The PTU baud rate defaults to 38400. However, the Modular Mining port is at 9600 baud. Therefore, before testing the Modular Mining port, the PTU baud rate must be changed to 9600.

- ◆ Click START > Programs > GEOHVPTU_2.0 > AC TOOLS > wPTU AC v21.01
- ◆ Select "Normal" mode {enter}
- ◆ Click "Options".
- ◆ Remove "38400" and add "9600" to the column "Selected Baud Sequence". Click "OK".
- The GE wPTU Toolbox Login Screen Connection Status window should now show "Connected to AC TCI 360T-DIGBT at 9600 Baud on COM1". This verifies the port communication.
- If the Connection Status window shows "Connection to target failed", exit the program and restart the PTU.

TCI Digital Input Checks

1. Connect the serial communication cable from the PTU to the TCI port (DIAG3) on the DID panel located on the back wall of the operator cab.
2. Make sure that the directional control lever is in PARK and the rest switch is in the REST position.
3. Turn control power switch (1, Figure 3-1) and the key switch ON.

To check the TCI digital inputs:

- ◆ Click START > Programs > GEOHVPTU_2.0 > AC TOOLS > wPTU AC v21.01
- ◆ Select "Normal" mode {enter}
- ◆ Type password "ok75e" {enter}
- ◆ Click "LOGIN to wPTU Toolbox".
- ◆ Double-click "TCI Real Time Data".
- Verify that the analog values are similar to the example in Figure 3-7. Also, make sure that the "COMMLINK" signal under "Modes" reads OK.
- Test the inputs in the "Digital Inputs" field as described in Table IV.

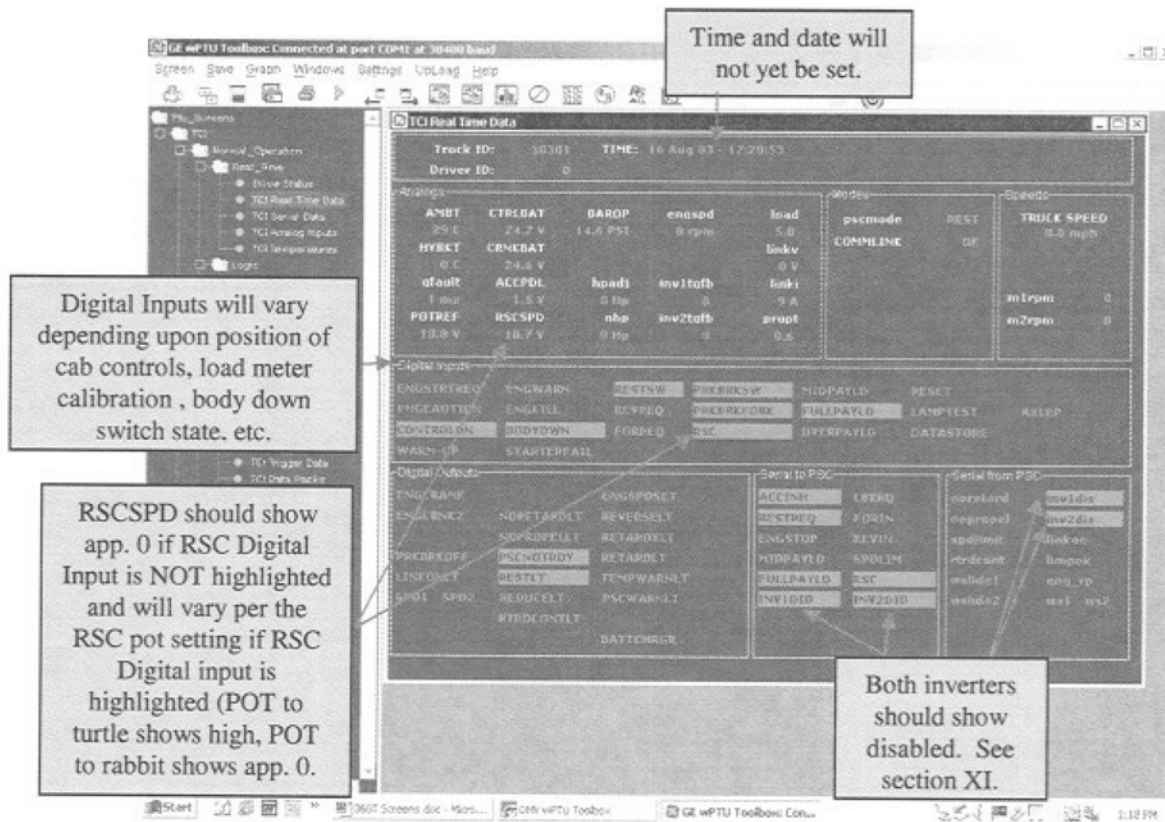


FIGURE 3-7. TCI REAL TIME DATA SCREEN

TABLE IV. TCI DIGITAL INPUT CHECKS

STEP	INPUT	DESCRIPTION
1	ENGSTRTREQ	Engine Start Request - DO NOT CHECK AT THIS TIME.
2	ENGCAUTION	Check Engine Caution Lamp - Jumper circuit 419M @ TB30 to ground to illuminate the lamp in the overhead panel. ENGCAUTION on the PTU should be highlighted.
3	CONTROLON WARM-UP	Control Power ON & Engine Warm-Up - DO NOT CHECK AT THIS TIME.
4	ENGWARN	Engine Service Light - Jumper 528A @ TB32 to ground to illuminate the blue indicator lamp on the back of the center console.
5	ENGINILL	Engine Shutdown Switch - Pull up on switch on center console. Verify that circuit 439 @ TB25 changes from 24VDC to 0VDC. Push down the switch to reset the system.
6	BODYDWN	Body Up Switch (activated when body is down) - Place a washer on body-up switch. Verify that circuits 63L & 71F change from 0VDC to 24VDC and the lamp in the overhead panel is OFF.
7	RESTSW	Rest Switch - Move the Rest Switch to the REST position. Verify that the internal lamp on the Rest Switch illuminates when in the REST position.
8	REVREQ	Reverse Request - Move the selector lever to the REVERSE position. The parking brake will release, circuit 52PBO will be 24VDC, and circuit 52CS will be 0VDC.
9	FORREQ	Forward Request - Move the selector lever to the FORWARD position. The parking brake will release, circuit 52PBO will be 24VDC, and circuit 52CS will be 0VDC.
10	NEUREQ	Neutral Request - Move the selector lever to the NEUTRAL position. The parking brake will release, circuit 52PBO will be 24VDC, and circuit 52CS will be 0VDC.
11	PRKBRKSW	Parking Brake Switch - Move the selector lever to the PARK position. PRKBRKSW on the PTU should be highlighted. (NOTE: The parking brake solenoid is controlled by the Interface Module. See Interface Module Checkout in Section D12 to fully test this function.)
12	RSC	Retard Speed Control Switch - Pull up the switch on the center console.
13	MIDPAYLD	Truck at 70% Payload - Jumper 73MS @TB25 to ground. (NOTE: RB2 circuit breakers must be OFF if installed.)
14	FULLPAYLD	Truck Fully Loaded - Jumper 73LS @TB25 to ground. (NOTE: RB2 circuit breakers must be OFF if installed.)
15	OVERPAYLD	Truck Overloaded - Jumper circuit 72IP@TB29 to circuit 712 @ TB32.
16	RESET	Override/Fault Reset Switch - Push the switch on the center console.
17	LAMPTEST	Lamp Test Switch - Push the switch on the dash panel. All warning/status lights in the overhead panel should illuminate except for the bottom two rows, the backup horn sounds, and the retard lights at the rear of the truck and on top of the cab turn on.
18	DATASTORE	Data Store Switch - Push the switch on the back of the center console.
19	AXLEP	Axle Pressure Switch - Jumper across the air pressure switch in the rear axle housing.

- ◆ Close the “TCI Real Time Data” screen, then double-click “TCI Serial Data”.
- Verify that the analog and digital values are similar to the example in Figure 3-8.

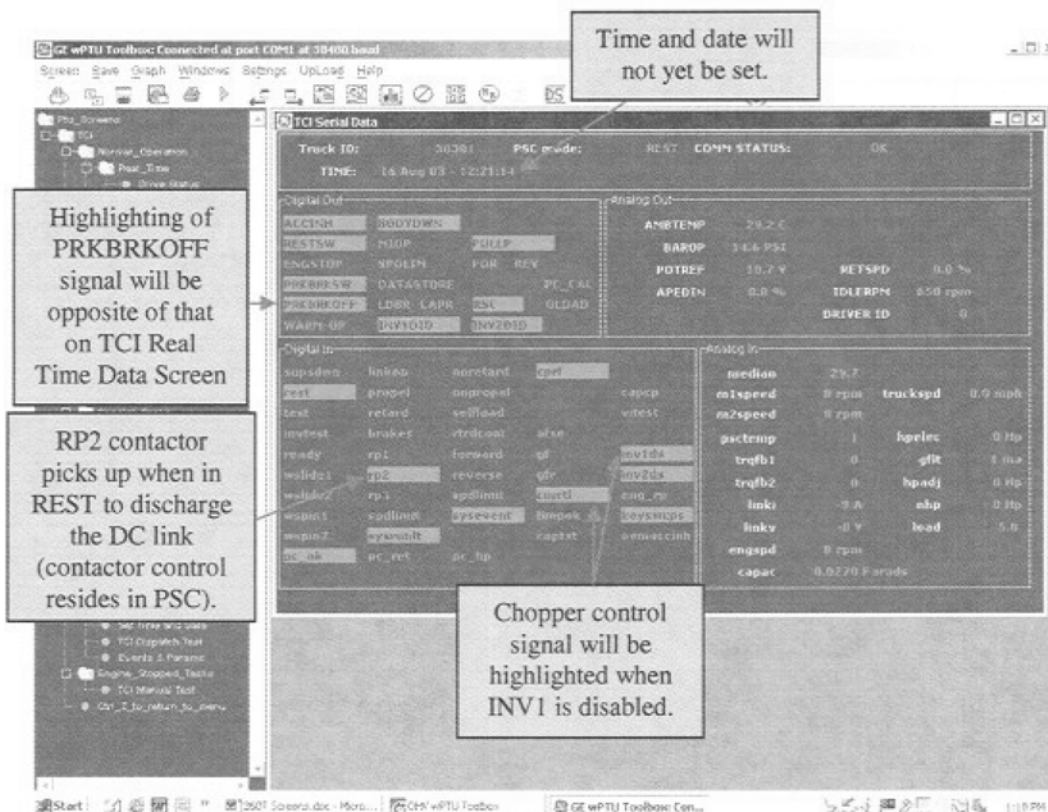


FIGURE 3-8. TCI SERIAL DATA SCREEN

- ◆ Close the “TCI Serial Data” screen, then double-click “TCI Analog Inputs”.
- Verify that the analog values are similar to the example in Figure 3-9.
- With the accelerator pedal not depressed, verify that the “ACCEL PEDAL” signal is approximately 1.5 volts.
- With the accelerator pedal fully depressed, verify that the “ACCEL PEDAL” signal is approximately 8.5 volts.
- With the RSC switch up (OFF position) and the RSC dial fully counterclockwise, verify that the “RSC POT” signal is approximately 10.7 volts.
- With the RSC switch up (OFF position) and the RSC dial fully clockwise, verify that the “RSC POT” signal is approximately 0 volts.

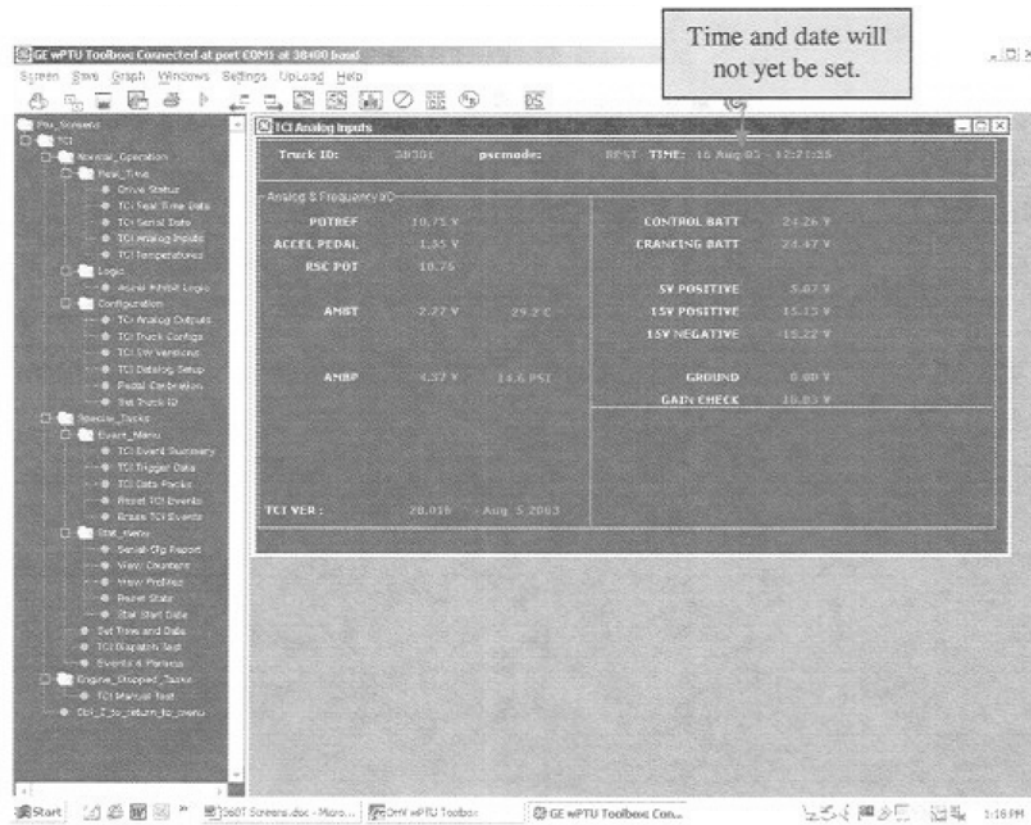


FIGURE 3-9. TCI ANALOG INPUTS SCREEN

- ◆ Close the “TCI Analog Inputs” screen, then double-click “TCI Temperatures”.
- Verify that the temperature values are similar to the example in Figure 3-10.

NOTE: All temperatures are calculated except for the AFSE and AMBIENT TEMPERATURE values.

- ◆ Close the “TCI Temperatures” screen.

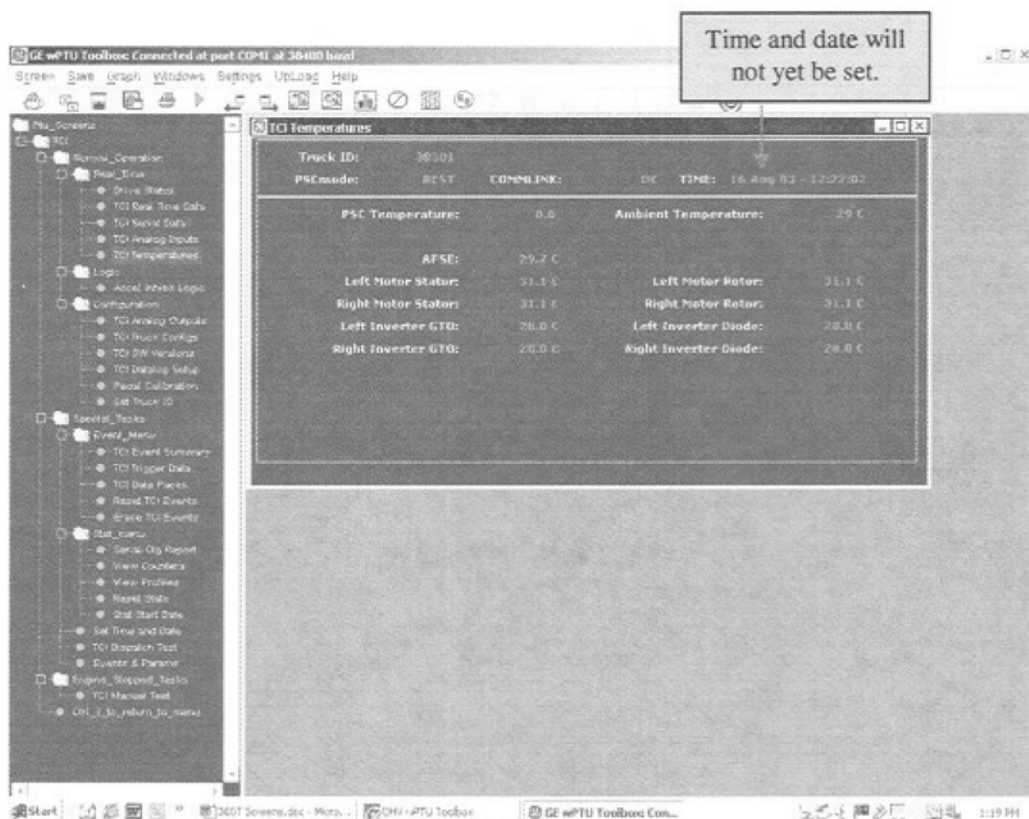


FIGURE 3-10. TCI TEMPERATURES SCREEN

TCI Digital Output Tests

1. Connect the serial communication cable from the PTU to the TCI port (DIAG3) on the DID panel located on the back wall of the operator cab.
2. Make sure that the directional control lever is in PARK and the rest switch is in the REST position.
3. Turn control power switch (1, Figure 3-1) and the key switch ON.

NOTE: The lamp test switch for the overhead panel will not activate the GE propulsion system lamps when the PTU is in the Manual Test mode.

To check the PSC digital outputs:

- ◆ Click START > Programs > GEOHVPTU_2.0 > AC TOOLS > wPTU AC v21.01
- ◆ Select "Normal" mode {enter}
- ◆ Type password "ok75e" {enter}
- ◆ Click "LOGIN to wPTU Toolbox".
- ◆ Under "Engine Stopped Task", double-click "TCI Manual Test".
- ◆ The "TCI Manual Test" screen looks similar to the "TCI Real Time Data" screen in Figure 3-7. Clicking the buttons in the "Digital Outputs" field will toggle the output on and off. See Table V.

TABLE V. TCI DIGITAL OUTPUT CHECKS

STEP	INPUT	DESCRIPTION	LOCATION (Fig. 3-11)
1	BATSEPC	Battery Separate Relay - Measure 24 VDC from circuit 21BSR @ TB28 to ground. With BATSEPC highlighted on the PTU, press {enter}. Verify 0 VDC.	
2	LINKONLT	Link Energized Light on the back of the center console will illuminate.	
3	SPD1	not used	
	SPD2	not used	
4	NORETARD	No Retard/Propel Light will illuminate.	A5
5	NOPROPEL	No Propel Light will illuminate.	A6
6	PSCNOTRDY	Propulsion System Not Ready Light will illuminate.	C6
7	RESTIT	Propulsion System at Rest Light will illuminate.	B6
8	REDUCELT	Propulsion System at Reduced Level Light will illuminate.	D6
9	RTRDCON	Retard System at Continuous Level Light will illuminate.	E6
10	BATTCHGR	Battery Charger System Failure Light will illuminate. (NOTE: This function is controlled by the Interface Module. See Interface Module Checkout in Section D12 to fully test this function.)	E5
11	ENGSPDSET	not used	
12	REVERSELT	Backup horn and backup lights will activate.	B4
13	RETARDXLT	Retard light on top of the cab and at rear of truck will turn on.	
14	RETARDLT	Dynamic Retarding Applied Light will illuminate.	D3
15	TEMPWARN	Propulsion System Temperature Light will illuminate.	C5
16	PSCWARNLT	Propulsion System Light will illuminate.	B5
17	HYDBHOTLT	Hydraulic Brake Oil Hot Light - cannot be checked (NOTE: This function is controlled by the Interface Module. See Interface Module Checkout in Section D12 to test this function.)	D5
18	ENGCRANK	Engine Crank Signal - See the following procedure to test this function.	

To check the Engine Crank Signal:

NOTE: Before checking ENGCRANK, verify that the 21B wires are removed from the starter solenoid relay in the battery control box. Also, make sure that the engine oil is at the proper level.

1. Change the PTU from the "Engine Stopped Tasks" screen to the "TCI Real Time Data" screen.

2. Turn the key switch to START.

Trucks without prelube system:

- ☐ Measure 24VDC to ground on circuit 21A @ TB25 and circuit 21B @ TB31.

Trucks with prelube system:

- ☐ Measure 24VDC to ground on circuit 21A @ TB25.
- ☐ After the prelube system has reached the proper oil pressure, measure 24VDC to ground on circuit 21PT @ TB28 to ground.
- ☐ After circuit 21PT is 24VDC, circuits 21ST and 21B will measure 24VDC to the starter solenoid.
- ☐ Verify that ENGSTRTREQ, ENGCRANK, ENGCRNK2, and BATSEPC are highlighted on the PTU when circuits 21A, 21PT and 21B are 24VDC.

3. Turn the key switch to ON.

4. Move the directional control lever to FORWARD.

5. Turn the key switch to START.

- ☐ Circuit 21A should remain 0VDC.

6. Release the key switch.

7. Move the directional control lever to NEUTRAL.

8. Turn key switch to START position.

- ☐ Circuit 21A should remain 0VDC.

9. Release the key switch.

10. Move the directional control lever to PARK.

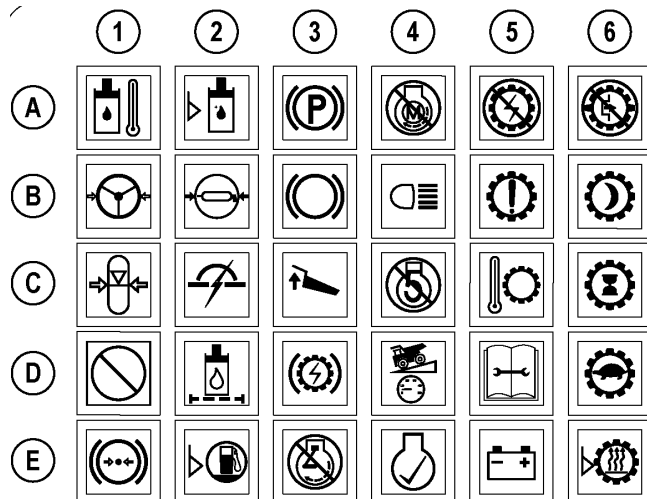


FIGURE 3-11. STATUS/WARNING LIGHTS

Row/Column	Indicator Description	Color
A1*	High Hydraulic Oil Temperature	Red
B1*	Low Steering Pressure	Red
C1	Low Accumulator Precharge	Red
D1	Spare	
E1	Low Brake Pressure	Red
A2*	Low Hydraulic Tank Oil Level	Red
B2*	Low Auto Lube Pressure	Amber
C2*	Circuit Breaker Tripped	Amber
D2*	Hydraulic Oil Filter Restricted	Amber
E2*	Low Fuel	Amber
A3*	Parking Brake Applied	Amber
B3*	Service Brake Applied	Amber
C3*	Body Up	Amber
D3*	Dynamic Retarding Applied	Amber
E3	Stop Engine	Red
A4*	Starter Failure	Amber
B4*	Manual Backup Lights	Amber
C4*	5 Minute Shutdown Timer	Amber
D4*	Retard Speed Control	Amber
E4*	Cheack Engine	Amber
A5	No Propel/Retard	Red
B5	Propulsion System Warning	Amber
C5	Propulsion System Temperature	Amber
D5	Maintenance Monitor	Red
E5	Battery Charger System Failure	Red
A6	No Propel	Red
B6	Propulsion System at Rest	Amber
C6*	Propulsion System Not Ready	Amber
D6*	Propulsion System at Reduced Level	Amber
E6*	Retard System at Continuous Level	Amber

CALIBRATIONS

The following procedures are used to calibrate the retarder and accelerator pedals, retarder lever, and the hydraulic brake temperature and propel system temperature gauges and the speedometer for the software. If any of the above components require replacement during truck servicing or troubleshooting procedures, the new or rebuilt component must be recalibrated using the applicable procedure before the truck is returned to service.

Speedometer

The speedometer can be calibrated by using the DID panel at the back of the operator cab.

1. On the DID panel, press the function keys
F4 - Menu > F1 - Test Menu > F4 - Speedometer.
 2. Adjust the speedometer to read 32 kph (20 mph).
 3. Enter "40" on the DID panel keypad.
- ☐ Verify that the speedometer reads 64 kph (40 mph).

Accelerator Pedal, Retarder Pedal/Lever and RSC Dial

The pedals and retarder lever can be calibrated by using the DID panel at the back of the operator cab. Press the function keys F4 - Menu > F4 - Truck Cfg > F2 - Begin, then follow the instructions on the screen.

The pedals, retarder lever and RSC dial can also be calibrated by using the PTU as follows:

1. Connect the serial communication cable from the PTU to the PSC port (DIAG1) on the DID panel located on the back wall of the operator cab.
 2. Make sure that the directional control lever is in PARK and the rest switch is in the REST position.
 3. Turn control power switch (1, Figure 3-1) and the key switch ON.
- ◆ Click START > Programs > GEOHVPTU_2.0 > AC TOOLS > wPTU AC v21.01
 - ◆ Select "Normal" mode {enter}
 - ◆ Type password "ok75e" {enter}
 - ◆ Under "Real Time", double-click "PSC Real Time Data".
 - ☐ Verify the values in Table VI.

TABLE VI. CALIBRATION VALUES

INPUT	DESCRIPTION
ACCEL-SEL	0.00 - accelerator pedal released
	1.00 - accelerator pedal fully depressed
RETRD-SEL	0.00 - retarder pedal released and retarder lever fully up
	1.00 - retarder pedal fully depressed and retarder lever fully up
	1.00 - retarder pedal released and retarder lever fully down
RETSPD	5 - RSC dial pulled up and turned fully counterclockwise
	34 - RSC dial pulled up and turned fully clockwise

ERASING EVENTS

PSC

1. Connect the serial communication cable from the PTU to the PSC port (DIAG1) on the DID panel located on the back wall of the operator cab.
 2. Make sure that the directional control lever is in PARK and the rest switch is in the REST position.
 3. Turn control power switch (1, Figure 3-1) and the key switch ON.
- ◆ Click START > Programs > GEOHVPTU_2.0 > AC TOOLS > wPTU AC v21.01
 - ◆ Select "Normal" mode {enter}
 - ◆ Type password "ok75e" {enter}
 - ◆ Click "LOGIN to wPTU Toolbox".
 - ◆ Under the "Special Tasks" heading, double-click "Erase PSC Events".
 - ◆ Click "YES".
 - ◆ Double-click "PSC Event Summary".
 - ☐ Only two events should be listed and active: Event 91 (Inverter 1 Cutout) and Event 92 (Inverter 2 Cutout). Investigate any other events that are listed.

TCI

1. Connect the serial communication cable from the PTU to the TCI port (DIAG3) on the DID panel located on the back wall of the operator cab.
2. Make sure that the directional control lever is in PARK and the rest switch is in the REST position.
3. Turn control power switch (1, Figure 3-1) and the key switch ON.
 - ◆ Click START > Programs > GEOHVPTU_2.0 > AC TOOLS > wPTU AC v21.01
 - ◆ Select "Normal" mode {enter}
 - ◆ Type password "ok75e" {enter}
 - ◆ Click "LOGIN to wPTU Toolbox".
 - ◆ Under the "Special Tasks" heading, double-click "Erase TCI Events".
 - ◆ Click "YES".
 - ◆ Double-click "TCI Event Summary".
- ☐ No events should be listed. Investigate any events that are listed.
7. Close circuit breaker CB2 on GDPC1 in the right side compartment of the control cabinet.
8. Use an analog meter to check the voltage between the pins on the P12A+ round connector.
 - ☐ There should be 90 - 100VDC.
9. Open circuit breaker CB2 on GDPC1. Reconnect the round connector to P12A+ and close circuit breaker CB2 again.
10. Carefully remove the gray plug on top of each P12 phase module. Without looking directly into the plug hole in each phase module, verify that a red light is present. Insert the gray plugs.
11. Carefully remove the gray plug on top of chopper module CM2. Without looking directly into the plug hole, verify that a red light is present. Insert the gray plug.
12. Close circuit breaker CB1 on GDPC2 in the right side compartment of the control cabinet.
13. Use an analog meter to check the voltage between the pins on the P21A+ round connector.
 - ☐ There should be 90 - 100VDC.
14. Open circuit breaker CB1 on GDPC2. Reconnect the round connector to P21A+ and close circuit breaker CB1 again.
15. Carefully remove the gray plug on top of each P21 phase module. Without looking directly into the plug hole in each phase module, verify that a red light is present. Insert the gray plugs.
16. Close circuit breaker CB2 on GDPC2 in the right side compartment of the control cabinet.
17. Use an analog meter to check the voltage between the pins on the P22A+ round connector.
 - ☐ There should be 90 - 100VDC.
18. Open circuit breaker CB2 on GDPC2. Reconnect the round connector to P22A+ and close circuit breaker CB2 again.
19. Carefully remove the gray plug on top of each P22 phase module. Without looking directly into the plug hole in each phase module, verify that a red light is present. Insert the gray plugs.

GATE DRIVER POWER CONVERTER TEST



Never look directly into the fiber optic light. Eye damage could result.

1. Disconnect the round connector on top of phase modules P11A+, P12A+, P21A+ and P22A+.
2. Close circuit breaker CB1 on GDPC1 in the right side compartment of the control cabinet.
3. Use an analog meter to check the voltage between the pins on the P11A+ round connector.
 - ☐ There should be 90 - 100VDC.
4. Open circuit breaker CB1 on GDPC1. Reconnect the round connector to P11A+ and close circuit breaker CB1 again.
5. Carefully remove the gray plug on top of each P11 phase module. Without looking directly into the plug hole in each phase module, verify that a red light is present. Insert the gray plugs.
6. Carefully remove the gray plug on top of chopper module CM1. Without looking directly into the plug hole, verify that a red light is present. Insert the gray plug.

LOAD TESTING

Preparation

1. Ensure that the wheels are chocked and the directional control lever is in PARK.
2. Ensure that all blower motor and alternator brushes are installed correctly.
3. Install locks on the contactor box door and left side compartment door of the control cabinet. The right side compartment will be accessed.
4. Use the DID panel to cutout both inverters:
 - a. Press F4 - MENU > F3 - Inv Cutout > F1 - Inv #1 > F4 - Toggle.
 - ☐ The display will show "Inverter #1 = cut-out".
 - b. Press F5 - Return > F2 - Inv #2 > F4 - Toggle.
 - ☐ The display will show "Inverter #2 = cut-out".
 - c. Press F5 - Return > F5 - Return to return to the main DID panel display.
5. Turn the rest switch ON.
6. Move GF cutout switch (2, Figure 3-1) to the CUTOFF (down) position.
7. Ensure that circuit breakers CB1 and CB2 on both gate driver power converters are closed.
8. Reconnect the 21B wires to the starter solenoids.

Alternator Speed Sensor Checks

1. Connect an AC voltmeter to circuits 74X (TB22) and 74Z (TB22).
2. Ensure that the GF cutout switch is in the CUTOFF (down) position and the rest switch is ON.
 - ☐ Verify approximately 4VAC on the meter.
 - ☐ Verify that the tachometer in the operator cab reads approximately 700 RPM.
4. Remove the voltmeter.
5. Ensure engine speed control by varying the position of the accelerator pedal.
6. To check the PSC alternator speed feedback, connect the serial communication cable from the PTU to the PSC port (DIAG1) on the DID panel located on the back wall of the operator cab.
7. Make sure that the directional control lever is in PARK and the rest switch is in the REST position.
8. Turn control power switch (1, Figure 3-1) and the key switch ON.

- ◆ Click START > Programs > GEOHVPTU_2.0 > AC TOOLS > wPTU AC v21.01
 - ◆ Select "Normal" mode {enter}
 - ◆ Type password "ok75e" {enter}
 - ◆ Under "Real Time", double-click "PSC Real Time Data".
 - ☐ Verify that ENGSPD in the "Analog" field shows the correct alternator speed value.
9. Leave the PTU connected to the PSC port for further checks. Use the emergency stop switch on the center console of the operator cab to stop the engine.

Battery Boost Check



The battery boost check must be performed exactly as described in the following procedure. Failure to do so may result in serious injury.

The contactors in the control cabinet with the R1 resistor may be energized while the engine is running. DANGEROUS VOLTAGES ARE PRESENT INSIDE THE CONTROL CABINET.

NOTE: The engine must be OFF during initial setup.

1. Turn the rest switch ON.
2. Move GF cutout switch (2, Figure 3-1) to the CUTOFF (down) position.
3. Verify that all link voltage lights are OFF.
4. Connect a voltmeter across resistor R1 located in the right side compartment of the control cabinet.
 - a. Connect the positive lead to BAT
 - b. Connect the negative lead to F101.
5. Start the engine.
6. Move the GF cutout switch to the NORMAL (up) position.
7. Turn the rest switch OFF.
 - ☐ The voltmeter will momentarily show a reading of approximately 18VAC, then drop to zero.
8. Turn the rest switch ON.
10. Move the GF cutout switch to the CUTOFF (down) position.
11. Use the emergency stop switch on the center console of the operator cab to stop the engine.
12. Verify that all link voltage lights are OFF. Remove the voltmeter.

Brake Circuit Switch Checks

1. Turn the rest switch ON.
2. Start engine and allow engine to warm up for approximately 10 minutes.
 - ☐ Verify that all status/warning lights in the overhead panel are off except Parking Brake Applied (A3, Figure 3-11), Propulsion System at Rest (B6) and Propulsion System Not Ready (C6).
3. Turn the wheel brake lock switch ON.
4. Short circuit 33T to ground. This is for the brake lock degradation switch located in brake cabinet.
 - ☐ Note that when the wheel brake lock is applied, the service brake lights on the truck are active and the service brake light indicator on the overhead panel is lit.
5. Connect the serial communication cable from the PTU to the PSC port (DIAG1) on the DID panel located on the back wall of the operator cab.
 - ◆ Click START > Programs > GEOHVPTU_2.0 > AC TOOLS > wPTU AC v21.01
 - ◆ Select "Normal" mode {enter}
 - ◆ Type password "ok75e" {enter}
 - ◆ Click "LOGIN to wPTU Toolbox".
 - ◆ Under "Real Time", double-click "PSC Real Time Data".
- ☐ Verify that BRAKEON is highlighted when the wheel brake lock is applied and the engine is running.
6. Turn the wheel brake lock switch OFF.
7. In the brake cabinet, short circuit 33 on the brake pressure switch to ground.
 - ☐ The low brake pressure light on the overhead panel and the low brake pressure buzzer should activate.
8. On the LH frame rail, short circuit 33F at the steering pressure switch on the bleeddown manifold to ground.
 - ☐ The low brake pressure light, low steering pressure light and low brake pressure buzzer should activate.

Hoist & Steering Circuit Switch Checks

9. On the inner side of the fuel tank, short circuit 39 on hoist circuit hydraulic filters bypass indicator switch to ground.
 - ☐ The hydraulic oil filter light in the overhead panel should illuminate.
10. At the steering circuit hydraulic filter, short circuit 39 on the filter bypass indicator switch to ground.
 - ☐ The hydraulic oil filter light in the overhead panel should illuminate.
11. Short circuit 51A at the nitrogen precharge pressure switches on the top of the steering accumulators to ground.
 - ☐ The low accumulator precharge indicator light is activated. This light stays on even when the short is removed.
12. Use emergency shutdown switch on the center console to shut off the engine. Do not turn the key switch OFF.
 - ☐ The low accumulator precharge light should remain on and the brakes and steering pressure should remain charged.
13. Turn the key switch OFF.
 - ☐ Verify that the steering pressure bleeds down.

Link Energized Checks

1. Start the engine.
2. Move the GF cutout switch to the NORMAL (up) position.
3. Turn the rest switch OFF.
 - ◆ Click START > Programs > GEOHVPTU_2.0 > AC TOOLS > wPTU AC v21.01
 - ◆ Select "Normal" mode {enter}
 - ◆ Type password "ok75e" {enter}
 - ◆ Click "LOGIN to wPTU Toolbox".
 - ◆ Under "Real Time", double-click "PSC Real Time Data".
- ☐ Verify that capacitor charge light (3, Figure 3-1) and the link energized indicator light on the rear of the center console are lit.
- ☐ Verify that LINKV and both inverter link voltages (I1LV & I2LV) are approximately 700 volts.
- ☐ Verify the other values and highlighted functions are similar on the various PSC screens in Figures 3-12 through 3-14.

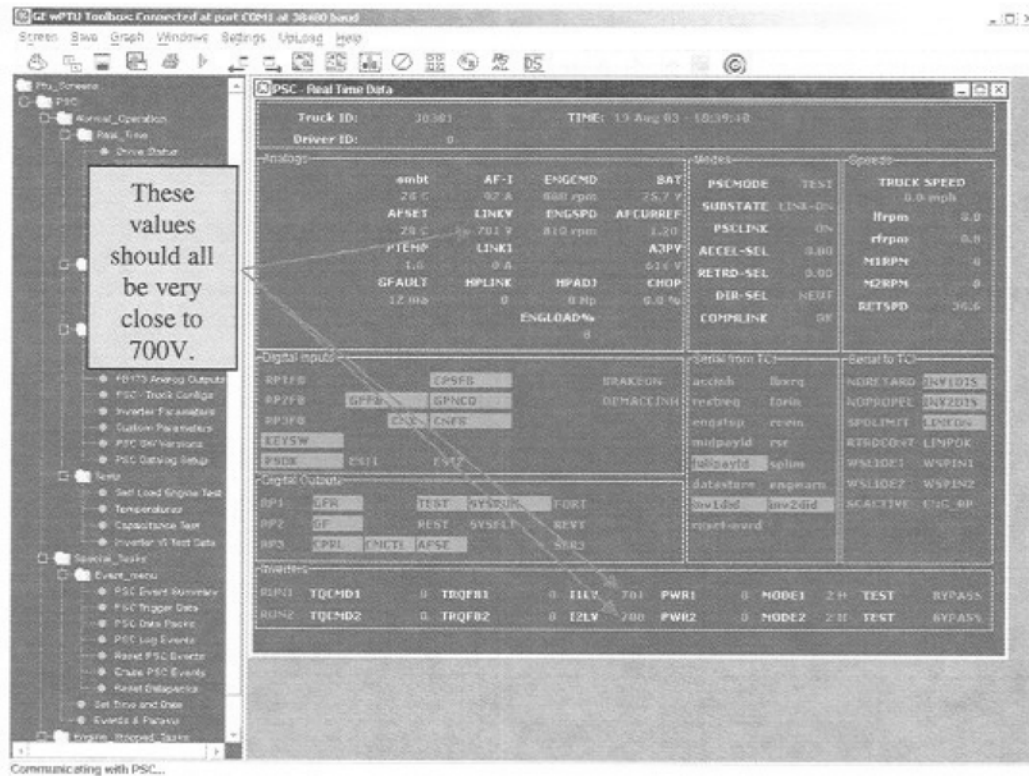


FIGURE 3-12. PSC REAL TIME DATA SCREEN

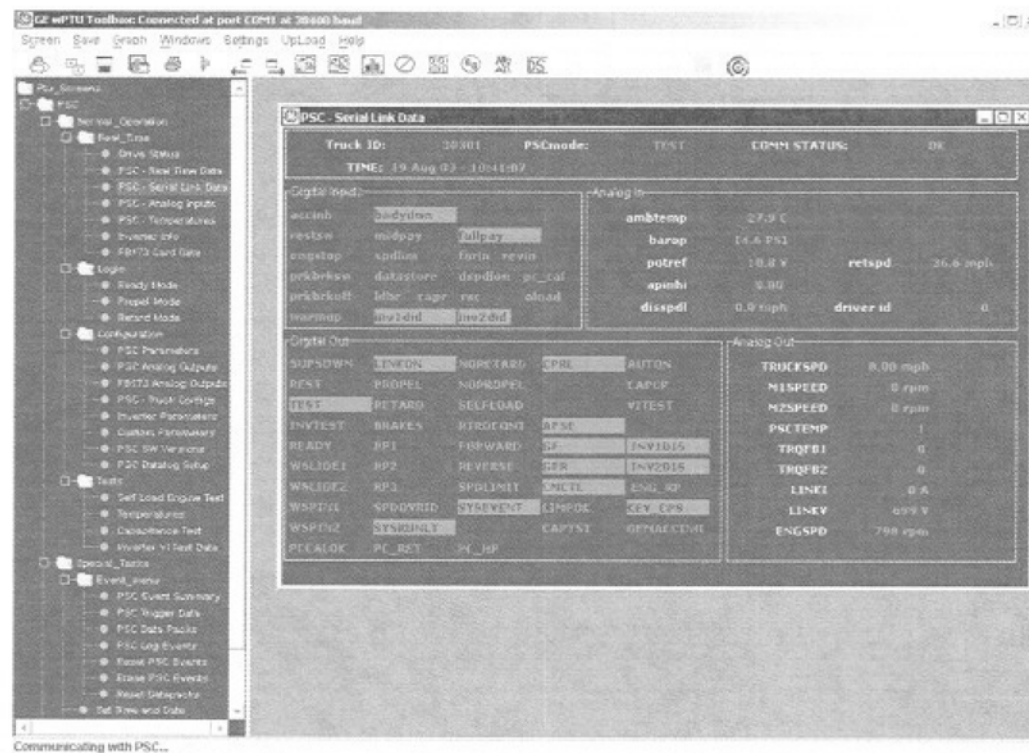


FIGURE 3-13. PSC SERIAL DATA SCREEN

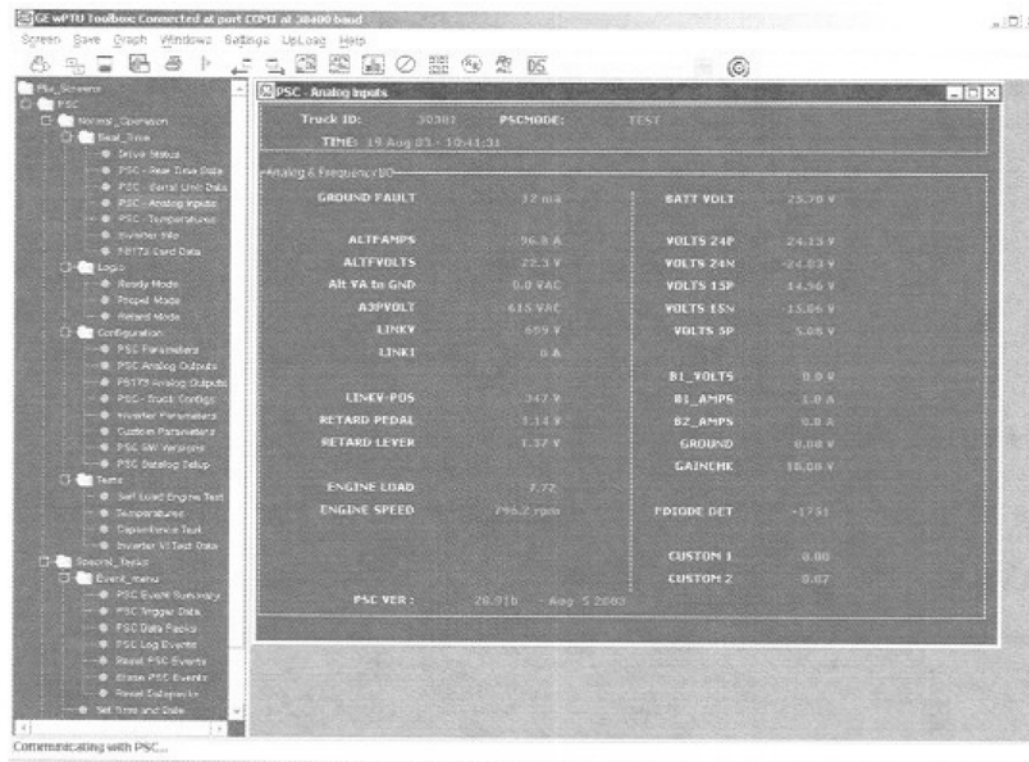


FIGURE 3-14. PSC ANALOG INPUTS SCREEN

4. Make sure that the directional control lever is in PARK and the rest switch is in the REST position.
5. Turn control power switch (1, Figure 3-1) and the key switch ON.
6. Connect the serial communication cable from the PTU to the TCI port (DIAG3) on the DID panel located on the back wall of the operator cab.
 - ◆ Click START > Programs > GEOHVPTU_2.0 > AC TOOLS > wPTU AC v21.01
 - ◆ Select "Normal" mode {enter}
 - ◆ Type password "ok75e" {enter}
 - ◆ Click "LOGIN to wPTU Toolbox".
 - ◆ Under "Real Time", double-click "TCI Real Time Data".
- Verify that the analog values and highlighted functions are similar to the TCI Real Time Data screen in Figure 3-15.
7. Exit the TCI Real Time Data screen. Leave the engine running and the PTU connected to the TCI port for the loadbox test.

Loadbox Test



Verify that the control cabinet doors are closed and locked before performing the following tests. DANGEROUS VOLTAGES ARE PRESENT INSIDE THE CONTROL CABINET WHEN THE ENGINE IS RUNNING.

1. Jumper fan clutch control circuit 22FO @ TB32 to ground to lock the fan in full on condition.
 2. With the engine running, move the GF cutout switch to the NORMAL (up) position.
 3. Turn the rest switch OFF.
- ◆ Under "Test", double-click "Self Load Engine Test".
 - ◆ Click "Enter LDBX".
 - Verify that the values are similar to the initial Self Load Engine Test screen in Figure 3-16.

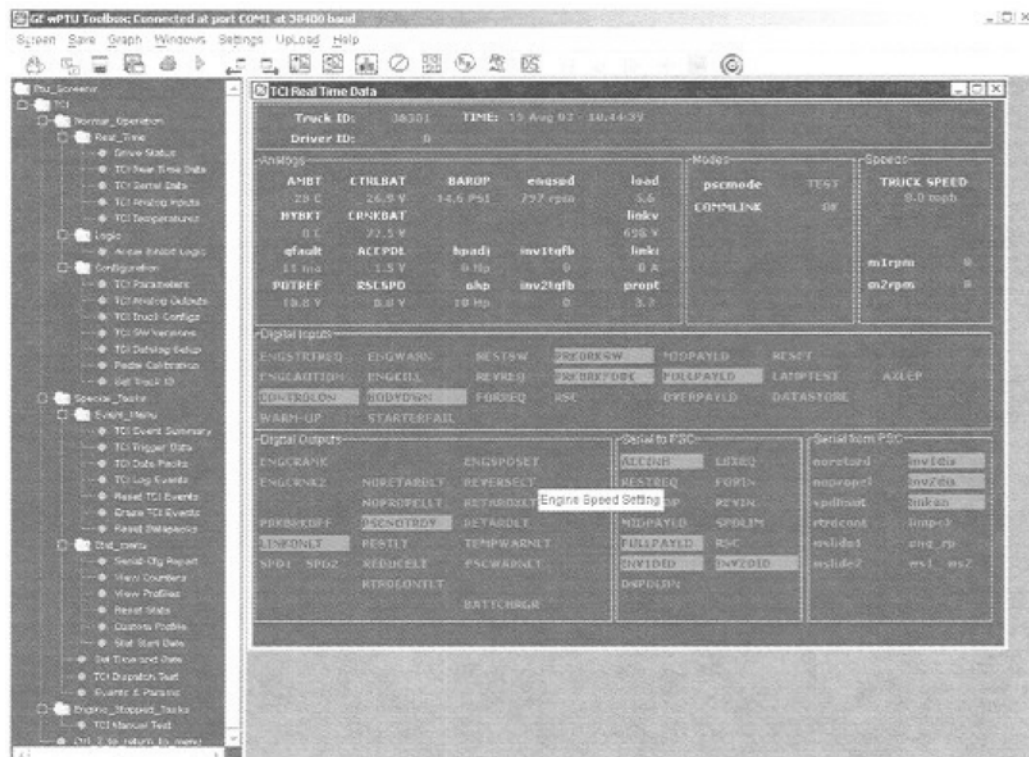


FIGURE 3-15. TCI REAL TIME DATA SCREEN

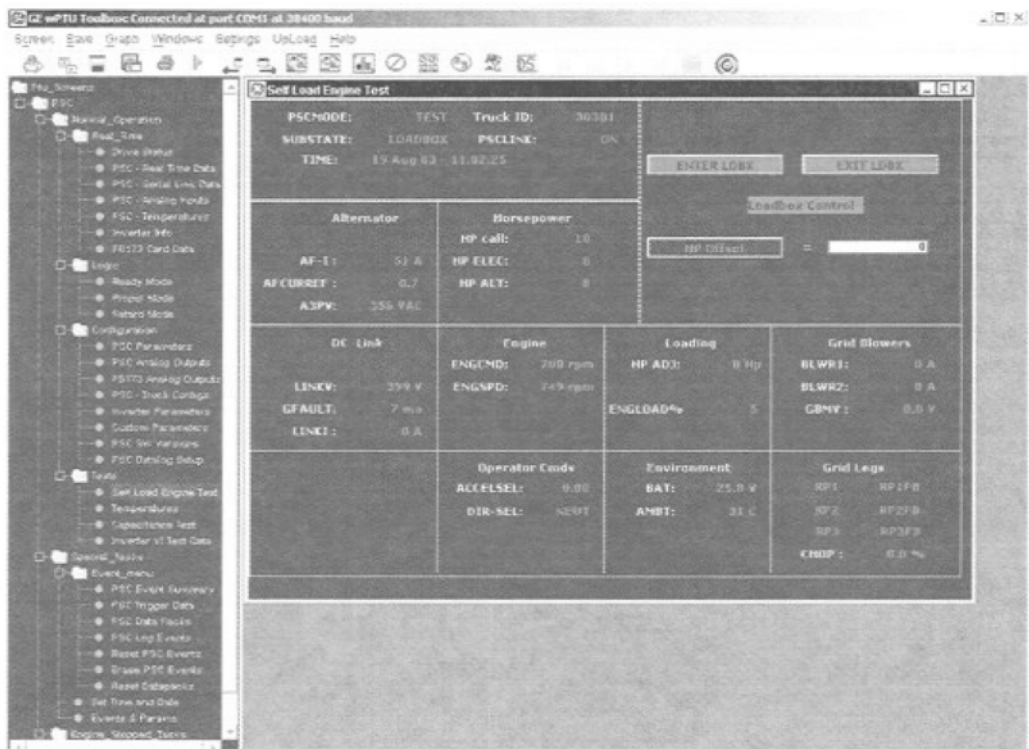


FIGURE 3-16. INITIAL SELF LOAD ENGINE TEST SCREEN

⚠ WARNING

In the following step, exhaust air from grid vents may be very hot.

4. Put the directional control lever in NEUTRAL and depress the accelerator pedal to just pick up contactor RP1. This should occur at approximately 1150 rpm.
- ☐ Check for hot air flow from both front sections of the retarding grid.
- ☐ Check for air flow from the rectifier air exhausts on the back of the control cabinet.
- ☐ On the PTU, verify that the values for BLWR1 and BLWR2 are balanced but opposite polarity.

NOTE: If the HPADJ value is fixed at zero and the ENGLOAD% value is fixed at 50% (or 5.0V if using an analog load signal), it is an indication that the PWM engine load signal is not getting to the PSC. Check for Event 63 (Engine Load Signal) on the DID panel. Refer to Troubleshooting for more information.

5. Depress the accelerator pedal to pick up contactors RP1 and RP2. This should occur at approximately 1375 rpm.
6. Let up on the accelerator pedal just enough so that contactor RP2 drops out but contactor RP1 is still picked up. This should occur at approximately 1375 rpm.

NOTE: Some trucks are not equipped with contactor RP3.

7. Depress the accelerator pedal to pick up contactors RP1, RP2 and RP3. This should occur at approximately 1550 rpm.
8. Warm up the engine until the engine coolant temperature stabilizes. Then fully depress the accelerator pedal to pickup all the RP contactors.
- ☐ The CHOP value on the PTU should be be 25% at approximately 1900 rpm.
9. Record the PTU screen while viewing the screen during full load.
- ◆ Under the "Save" menu, select "Single Snapshot", then click "Save".
- ◆ To view the recorded screen, under the "View" menu, select "Screen Relay", the highlight the file and click "Open".

10. Note the ENGLOAD value on the screen.

- ☐ If the value is 5 volts during load testing, loading is satisfactory.
- ☐ If the value is below 5 volts, the electrical system needs to remove horsepower loading. This is an indication of a weak engine.
- ☐ If the value is above 5 volts, the electrical system needs to load the engine more. This is an indication of a strong engine.

11. The Self Load Engine Test screen should be recorded and the values compared to values that are calculated to account for parasitic losses at the elevation of the test site and ambient temperature during testing as follows:

- a. Output horsepower should be 2700 HP $\pm$ 5% @ 1900 +10/-15 rpm.
- b. Requested rpm from GE must be 1900 rpm.
- c. Refer to Figure 3-17 for parasitic losses curve.
- ☐ Read the parasitic losses from the graph based on ambient temperature and altitude.
- ☐ Add the value on the graph to the delivered HP to GE and compare that to the "-5%" value at the rpm rated tolerance (i.e. 2612 HP + value from graph = corrected HP).

Manual Offset HP Output Adjustment:

12. If it is necessary to troubleshoot HP problems, use the following procedure:
 - ◆ With loadbox initiated, enter a + or - offset value in the "HP Offset" field.
 - ◆ Click the "HP Offset" box.
13. Perform the load test again.
 - ◆ Return the offset to 0.0
 - ◆ Click the "HP Offset" box.
 - ◆ Click "EXIT LDBX" to exit the Self Load Engine Test screen.
14. Allow the engine to cool down until the engine temperature and pressure gauges show normal operating values.
15. Turn the rest switch ON.
16. Turn the key switch OFF. Allow approximately 90 seconds for the steering accumulators to bleed down.
17. Remove the jumper from the fan clutch control circuit
18. Record all data to create a truck record for future comparison.

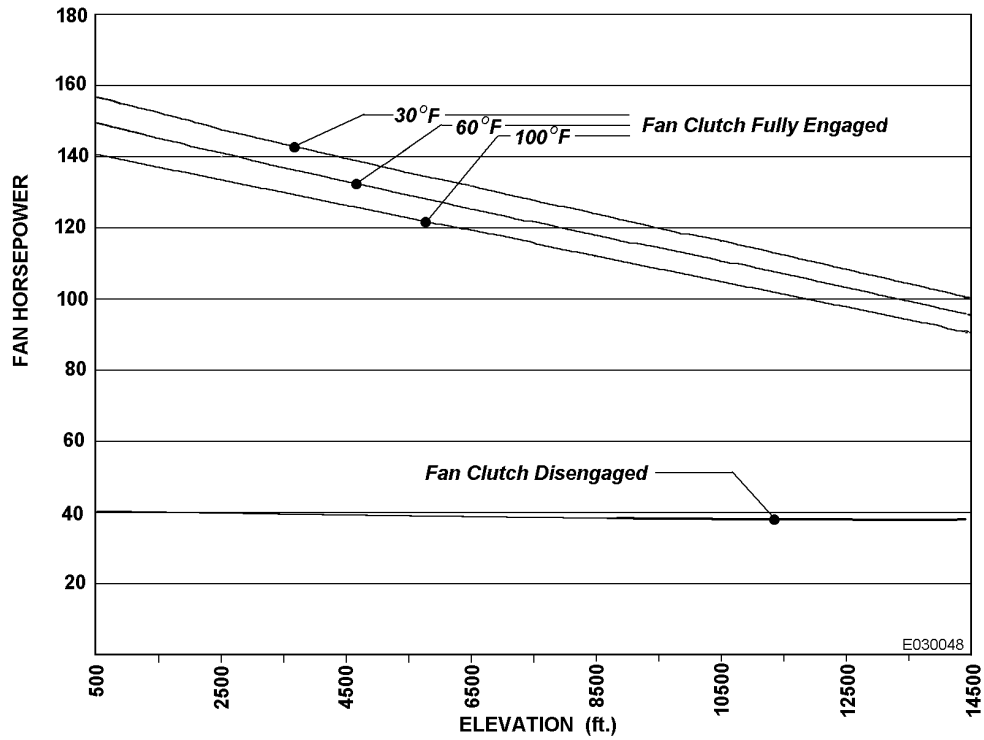


FIGURE 3-17. TOTAL PARASITIC LOSS AT FULL POWER
Komatsu SSDA16V160, 2700 GHP, ECS 8 Blade, 78" dia. 5.3" PW @ 798 RPM

TROUBLESHOOTING

PVM Optimum Load Curve Handshaking Troubleshooting

NOTE: A value of less than 0.5VDC or more than 9.5VDC on circuit 72E indicates a failure.

1. With the engine off, key switch ON and control power switch ON, measure the voltage between the 72E (+) lead and the 72R (-) lead.

□ The voltage should be 5.0VDC.

- a. Jumper circuit 22FO to ground and verify voltage on 72E to 72R changes to 7.0VDC.
- b. If the voltage is 0VDC, verify that the connections to the PVM are correct and that circuits 439 and 11SL (connected to CN P382 positions 5 and 40) are 24VDC.

2. With the engine running and under load, and the key switch and control power switch ON, check the voltage at 72E (+) to 72R (-).

□ The voltage should be 5.0VDC.

- a. Check the PVM diagnostic connector P381.
 - b. Verify that the voltage between position A to B is 8 to 11VDC. A reading of 0VDC indicates that the 1939 transmission line failed. Check 1939 wiring.
 - c. Verify the voltage between position C to B is 8 to 11VDC. A reading of 0VDC indicates that the PVM has failed only if the voltage from position A to B is correct and the filtering circuit is correct. Check the filtering circuit resistors and capacitors connected to P383 positions 12 and 20 and P382 position 33 mounted on diode board DB1.
3. If both Step 1 and 2 are 0VDC, then circuit 439 or 11SL or both are incorrect.

⚠ DANGER

Allow adequate time for link voltage to drain down before opening the control cabinet to perform the following checks or repairs.

Phase Module and Chopper Module Troubleshooting

1. To troubleshoot a phase module or chopper module, stop the engine and turn the rest switch OFF. Connect the serial communication cable from the PTU to the PSC panel and access the PSC Manual Test screen.
2. Click the appropriate GD1E or GD2E signal in the "Digital Output" field and turn it ON. (GD1E turns on all Inverter 1 phase modules and chopper module 1. GD2E turns on all Inverter 2 phase modules and chopper module 2.)
3. Disconnect the gray fiber optic cable on the phase module or chopper module that is being checked.

⚠ WARNING

In the following step, DO NOT look directly at the red light. Eye damage could result.

4. If a red light is visible out of the gray receptacle on the gate driver module, the phase module or chopper module is OK.
5. If a red light is not visible, disconnect the round power supply harness from the gate driver module.
6. Check the AC voltage in the two pins in the harness. There should be 100 VAC square wave on the harness. The actual reading on the VOM will depend on the meter and how it is designed to measure AC voltage. Most meters read less than 100 volts. Normally, there will either be proper voltage on the harness or no voltage at all.
7. If there is no voltage, troubleshoot the appropriate gate driver power converter or the harness. See "Gate Driver Power Converter Test" earlier in this section.
8. If there is voltage, reconnect the harness and disconnect the gate lead on the "G" terminal.
10. If there is a red light visible with the gate lead disconnected, there is a short and the phase module or chopper module must be replaced.
11. If a red light is not visible with the gate lead disconnected, the gate driver module is faulty and must be replaced. All the gate driver sections for phase modules and chopper modules are alike and interchangeable. The red-covered and white-covered gate driver modules are interchangeable where mounted by the six cap screws to the cooling tubes of the phase module or chopper module.

PHASE MODULE REPLACEMENT

Phase Module Removal

1. Place the control power switch in the OFF position and the GF cutout switch in the CUTOFF position.
2. Use a VOM to ensure that there is no voltage present between the (+) and (-) DC links and ground.
3. Disconnect the fiber optic cables and the round plug at the top of the phase module. Tuck the removed cables under the loom to protect the cables when the module is pulled out.
4. Remove the mounting hardware that secures the phase module to the vertical bus bar. Note the length of the bolts for proper reinstallation.
5. Remove the mounting hardware that secures the two fuses.
6. Mark each phase module so that it will be reinstalled in its original location.

NOTE: Each phase module weighs 29.5 kg (65 lb).

7. Support the phase module and remove the two nuts and washers that secure the phase module to the control cabinet.
8. Slide the phase module forward by the extended mounting arms and remove it from the control cabinet. **Do not pull on the gate card cover.**

Phase Module Installation

1. Inspect the rear cooling air sealing gasket. Replace it if damaged.
2. Return the phase module to its original location.
3. Install the two mounting bolts and washers that secure the phase module to the control cabinet. Tighten the bolts to **64 N·m (47 ft lb)**.
4. Install the mounting hardware that secures the two fuses. Tighten the bolts to **19 N·m (14 ft lb)**.
5. Install the mounting hardware that secures the phase module to the vertical bus bar. Tighten the bolts to **26 N·m (19 ft lb)**.
6. Reconnect the fiber optic cables and the round plug at the top of the phase module.
7. Place the GF cutout switch in the NORMAL position and the control power switch in the ON position.

NOTE: For removal of other control cabinet components, refer to the GE service manual.

NOTES

SECTION E5

TROUBLESHOOTING FAULT CODES (A001-A146)

INDEX

Fault Code A001: Left front suspension pressure sensor signal is high.	E5-3
Fault Code A002: Left front suspension pressure sensor signal is low	E5-4
Fault Code A003: Right front suspension pressure sensor signal is high	E5-5
Fault Code A004: Right front suspension pressure sensor signal is low	E5-6
Fault Code A005: Left rear suspension pressure sensor signal is high	E5-7
Fault Code A006: Left rear suspension pressure sensor signal is low.	E5-8
Fault Code A007: Right rear suspension pressure sensor signal is high.	E5-9
Fault Code A008: Right rear suspension pressure sensor signal is low	E5-10
Fault Code A009: Incline sensor signal is high.	E5-11
Fault Code A010: Incline sensor signal is low	E5-12
Fault Code A011: Payload meter speed sensor signal has failed	E5-13
Fault Code A013: Body up switch has failed	E5-14
Fault Code A014: Payload meter checksum computation has failed.	E5-15
Fault Code A016: Payload meter write to flash memory has failed	E5-16
Fault Code A017: Payload meter flash memory read has failed	E5-17
Fault Code A018: Right rear flat suspension cylinder warning.	E5-18
Fault Code A019: Left rear flat suspension cylinder warning	E5-20
Fault Code A022: Carryback load is excessive	E5-22
Fault Code A026: Payload meter user select switch has failed	E5-24
Fault Code A027: Payload meter user clear switch has failed	E5-25
Fault Code A101: High pressure detected across an hydraulic pump filter	E5-26
Fault Code A105: Fuel level sensor is shorted to ground, indicating a false high fuel level. . .	E5-28
Fault Code A107: GE has generated a propel system caution	E5-30
Fault Code A108: GE has generated a propel system temperature caution	E5-32
Fault Code A109: GE has generated a propel system reduced level signal	E5-34
Fault Code A115: Low steering precharge pressure is detected	E5-36
Fault Code A123: GE has generated a reduced retarding caution.	E5-38
Fault Code A124: GE has generated a no propel / no retard warning	E5-40
Fault Code A125: GE has generated a no propel warning.	E5-42
Fault Code A126: Oil level in the hydraulic tank is low.	E5-44
Fault Code A127: IM furnished +5 volt output for sensors is low	E5-46
Fault Code A128: IM furnished +5 volt output for sensors is high	E5-48
Fault Code A139, A310: Low fuel level warning.	E5-50
Fault Code A145: Hydraulic temperature sensors cause advance of engine rpm to advance level 2 for cooling of hydraulic oil.	E5-52
Fault Code A146: Hydraulic temperature sensors cause posting of this fault	E5-54

NOTES

Fault Code A001: Left front suspension pressure sensor signal is high

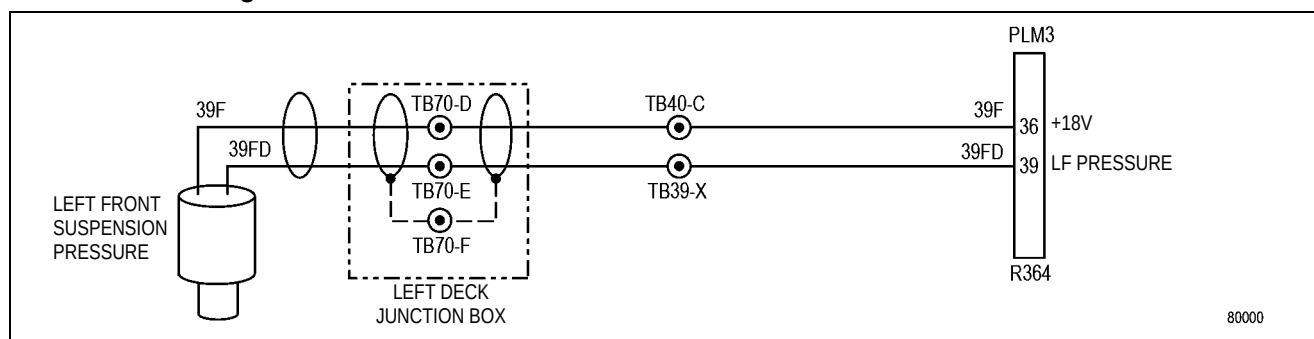
Operator Action	None
Fault Code	A001
Description	Left front suspension pressure sensor signal is high.
Fault Conditions	Sets if pressure signal is out of range high (sensor current over 22 ma). Resets if reading returns to normal.
Operator Alerting System Response	PLM Display: -----
Resulting Problem(s)	Bad payload computation.
Related Information	

Table

1. This fault is generated by PLM in response to a problem in the sensor circuit. This sensor circuit may have a related fault that can be used to resolve the problem. Refer to the Troubleshooting Instructions for the active parameter fault(s).
2. If there are no active parameter fault codes, then, based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items.
3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.

Parameter	Expected State and/or Related Fault(s)
Left Front Pressure Sensor (PLM3 36,39)	Sensor current > 22 ma: failed high Sensor current < 2 ma: failed low Sensor current > 2 ma but less than 22 ma: valid readings Fault(s): A002

Related circuit diagram



Fault Code A002: Left front suspension pressure sensor signal is low

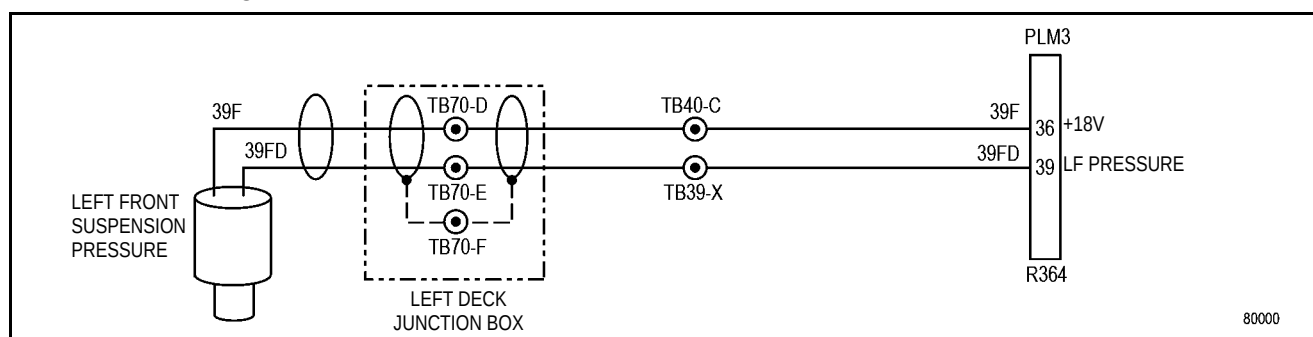
Operator Action	None
Fault Code	A002
Description	Left front suspension pressure sensor signal is low.
Fault Conditions	Sets if pressure signal is out of range low (sensor current less than 2 ma). Resets if reading returns to normal.
Operator Alerting System Response	PLM Display: -----
Resulting Problem(s)	Bad payload computation.
Related Information	

Table

1. This fault is generated by PLM in response to a problem in the sensor circuit. This sensor circuit may have a related fault that can be used to resolve the problem. Refer to the Troubleshooting Instructions for the active parameter fault(s).
2. If there are no active parameter fault codes, then, based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items.
3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.

Parameter	Expected State and/or Related Fault(s)
Left Front Pressure Sensor (PLM3 36,39)	Sensor current > 22 ma: failed high Sensor current < 2 ma: failed low Sensor current > 2 ma but less than 22 ma: valid readings Fault(s): A001

Related circuit diagram



Fault Code A003: Right front suspension pressure sensor signal is high

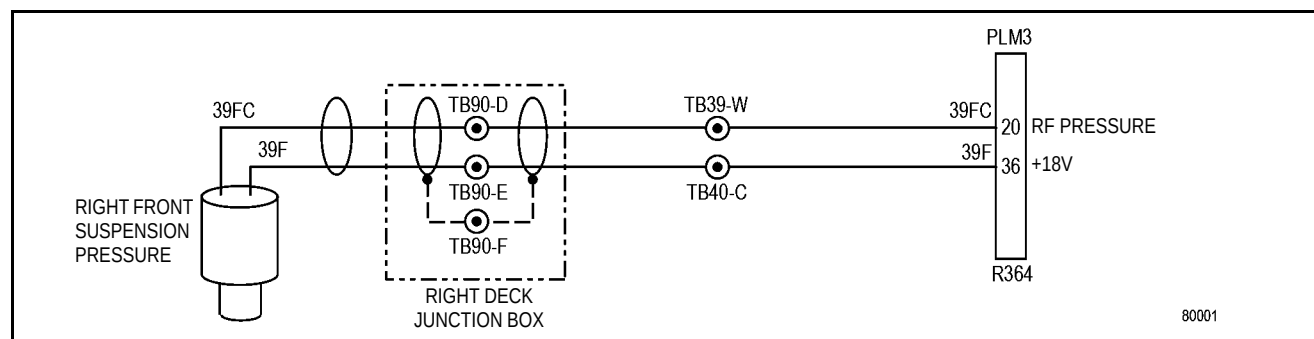
Operator Action	None
Fault Code	A003
Description	Right front suspension pressure sensor signal is high.
Fault Conditions	Sets if pressure signal is out of range high (sensor current over 22 ma). Resets if reading returns to normal.
Operator Alerting System Response	PLM Display: -----
Resulting Problem(s)	Bad payload computation.
Related Information	

Table

1. This fault is generated by PLM in response to a problem in the sensor circuit. This sensor circuit may have a related fault that can be used to resolve the problem. Refer to the Troubleshooting Instructions for the active parameter fault(s).
2. If there are no active parameter fault codes, then, based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items.
3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.

Parameter	Expected State and/or Related Fault(s)
Right Front Pressure Sensor (PLM3 36,20)	Sensor current > 22 ma: failed high Sensor current < 2 ma: failed low Sensor current > 2 ma but less than 22 ma: valid readings Fault(s): A004

Related circuit diagram



Fault Code A004: Right front suspension pressure sensor signal is low

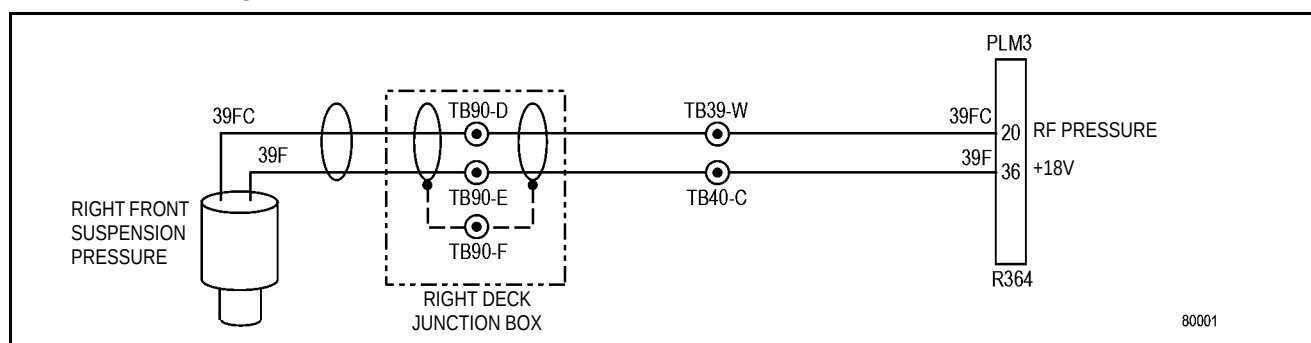
Operator Action	None
Fault Code	A004
Description	Right front suspension pressure sensor signal is low.
Fault Conditions	Sets if pressure signal is out of range low (sensor current less than 2 ma). Resets if reading returns to normal.
Operator Alerting System Response	PLM Display: -----
Resulting Problem(s)	Bad payload computation.
Related Information	

Table

1. This fault is generated by PLM in response to a problem in the sensor circuit. This sensor circuit may have a related fault that can be used to resolve the problem. Refer to the Troubleshooting Instructions for the active parameter fault(s).
2. If there are no active parameter fault codes, then, based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items.
3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.

Parameter	Expected State and/or Related Fault(s)
Right Front Pressure Sensor (PLM3 36,20)	Sensor current > 22 ma: failed high Sensor current < 2 ma: failed low Sensor current > 2 ma but less than 22 ma: valid readings Fault(s): A003

Related circuit diagram



Fault Code A005: Left rear suspension pressure sensor signal is high

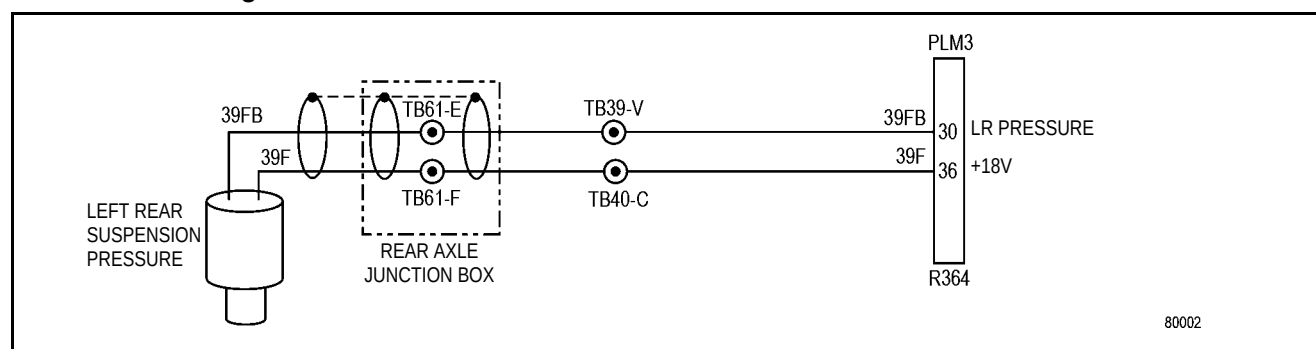
Operator Action	None
Fault Code	A005
Description	Left rear suspension pressure sensor signal is high.
Fault Conditions	Sets if pressure signal is out of range high (sensor current over 22 ma). Resets if reading returns to normal.
Operator Alerting System Response	PLM Display: -----
Resulting Problem(s)	Bad payload computation.
Related Information	

Table

1. This fault is generated by PLM in response to a problem in the sensor circuit. This sensor circuit may have a related fault that can be used to resolve the problem. Refer to the Troubleshooting Instructions for the active parameter fault(s).
2. If there are no active parameter fault codes, then, based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items.
3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.

Parameter	Expected State and/or Related Fault(s)
Left Rear Pressure Sensor (PLM3 36,30)	Sensor current > 22 ma: failed high Sensor current < 2 ma: failed low Sensor current > 2 ma but less than 22 ma: valid readings Fault(s): A006

Related circuit diagram



Fault Code A006: Left rear suspension pressure sensor signal is low

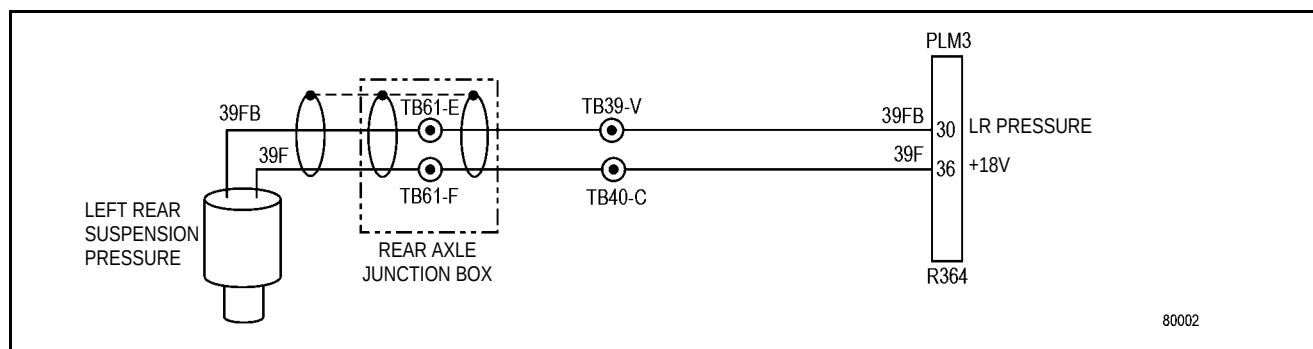
Operator Action	None
Fault Code	A006
Description	Left rear suspension pressure sensor signal is low.
Fault Conditions	Sets if pressure signal is out of range low (sensor current less than 2 ma). Resets if reading returns to normal.
Operator Alerting System Response	PLM Display: -----
Resulting Problem(s)	Bad payload computation.
Related Information	

Table

1. This fault is generated by PLM in response to a problem in the sensor circuit. This sensor circuit may have a related fault that can be used to resolve the problem. Refer to the Troubleshooting Instructions for the active parameter fault(s).
2. If there are no active parameter fault codes, then, based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items.
3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.

Parameter	Expected State and/or Related Fault(s)
Left Rear Pressure Sensor (PLM3 36,30)	Sensor current > 22 ma: failed high Sensor current < 2 ma: failed low Sensor current > 2 ma but less than 22 ma: valid readings Fault(s): A005

Related circuit diagram



Fault Code A007: Right rear suspension pressure sensor signal is high

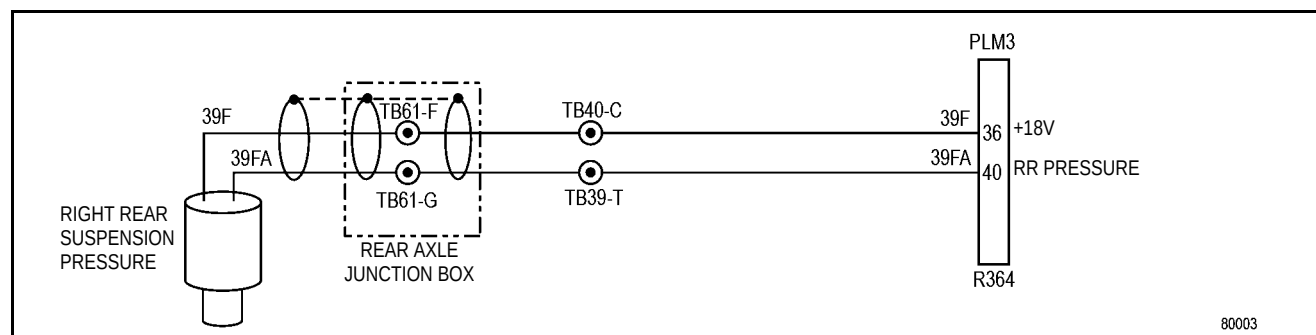
Operator Action	None
Fault Code	A007
Description	Right rear suspension pressure sensor signal is high.
Fault Conditions	Sets if pressure signal is out of range high (sensor current over 22 ma). Resets if reading returns to normal.
Operator Alerting System Response	PLM Display: -----
Resulting Problem(s)	Bad payload computation.
Related Information	

Table

1. This fault is generated by PLM in response to a problem in the sensor circuit. This sensor circuit may have a related fault that can be used to resolve the problem. Refer to the Troubleshooting Instructions for the active parameter fault(s).
2. If there are no active parameter fault codes, then, based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items.
3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.

Parameter	Expected State and/or Related Fault(s)
Right Rear Pressure Sensor (PLM3 36,40)	Sensor current > 22 ma: failed high Sensor current < 2 ma: failed low Sensor current > 2 ma but less than 22 ma: valid readings Fault(s): A008

Related circuit diagram



Fault Code A008: Right rear suspension pressure sensor signal is low

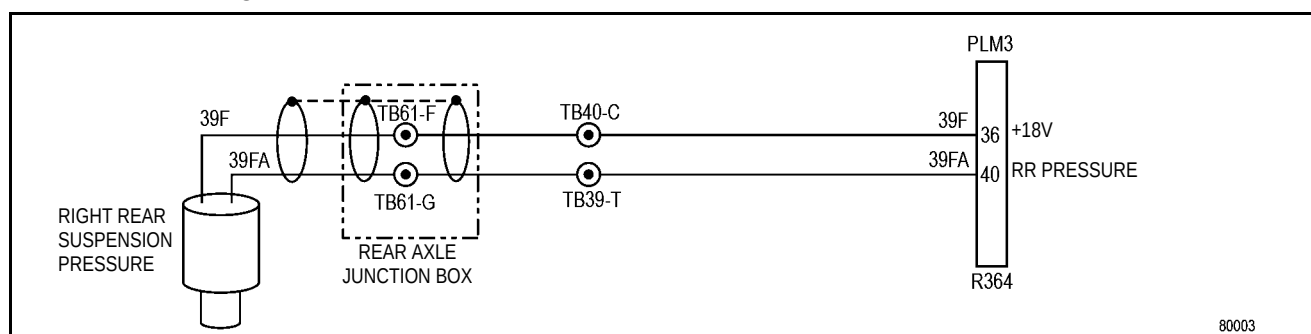
Operator Action	None
Fault Code	A008
Description	Right rear suspension pressure sensor signal is low.
Fault Conditions	Sets if pressure signal is out of range low (sensor current less than 2 ma). Resets if reading returns to normal.
Operator Alerting System Response	PLM Display: -----
Resulting Problem(s)	Bad payload computation.
Related Information	

Table

1. This fault is generated by PLM in response to a problem in the sensor circuit. This sensor circuit may have a related fault that can be used to resolve the problem. Refer to the Troubleshooting Instructions for the active parameter fault(s).
2. If there are no active parameter fault codes, then, based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items.
3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.

Parameter	Expected State and/or Related Fault(s)
Right Rear Pressure Sensor (PLM3 36,40)	Sensor current > 22 ma: failed high Sensor current < 2 ma: failed low Sensor current > 2 ma but less than 22 ma: valid readings Fault(s): A007

Related circuit diagram



Fault Code A009: Incline sensor signal is high

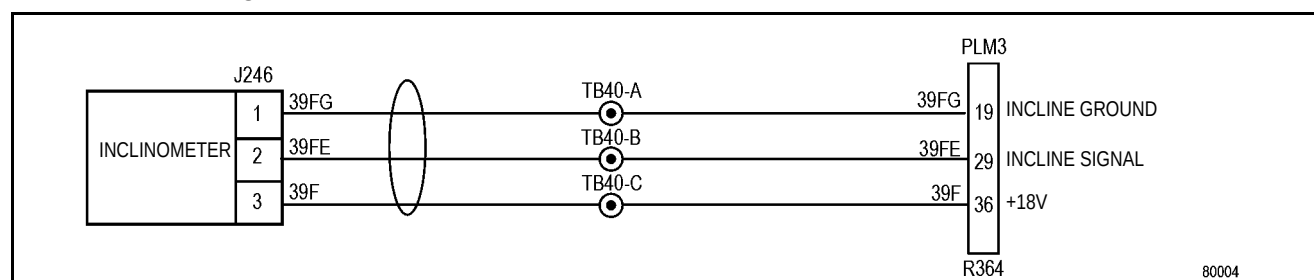
Operator Action	None
Fault Code	A009
Description	Incline sensor signal is high.
Fault Conditions	Sets if incline signal is out of range high (sensor voltage less than 0.565 volts). Resets if reading returns to normal.
Operator Alerting System Response	PLM Display: -----
Resulting Problem(s)	Bad payload computation.
Related Information	

Table

1. This fault is generated by PLM in response to a problem in the sensor circuit. This sensor circuit may have a related fault that can be used to resolve the problem. Refer to the Troubleshooting Instructions for the active parameter fault(s).
2. If there are no active parameter fault codes, then, based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items.
3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.

Parameter	Expected State and/or Related Fault(s)
Incline Sensor (PLM3 36,29,19)	Sensor voltage < 0.565: failed high Sensor voltage > 5.08: failed low Sensor voltage > 0.565 but < 5.08: valid readings Fault(s): A010

Related circuit diagram



Fault Code A010: Incline sensor signal is low

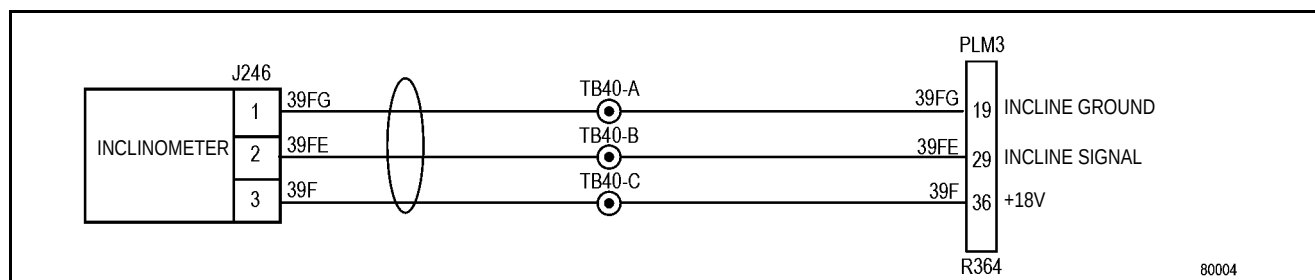
Operator Action	None
Fault Code	A010
Description	Incline sensor signal is low.
Fault Conditions	Sets if incline signal is out of range low (sensor voltage greater than 5.08 volts). Resets if reading returns to normal.
Operator Alerting System Response	PLM Display: -----
Resulting Problem(s)	Bad payload computation.
Related Information	

Table

1. This fault is generated by PLM in response to a problem in the sensor circuit. This sensor circuit may have a related fault that can be used to resolve the problem. Refer to the Troubleshooting Instructions for the active parameter fault(s).
2. If there are no active parameter fault codes, then, based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items.
3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.

Parameter	Expected State and/or Related Fault(s)
Incline Sensor (PLM3 36,29,19)	Sensor voltage < 0.565: failed high Sensor voltage > 5.08: failed low Sensor voltage > 0.565 but < 5.08: valid readings Fault(s): A009

Related circuit diagram



Fault Code A011: Payload meter speed sensor signal has failed

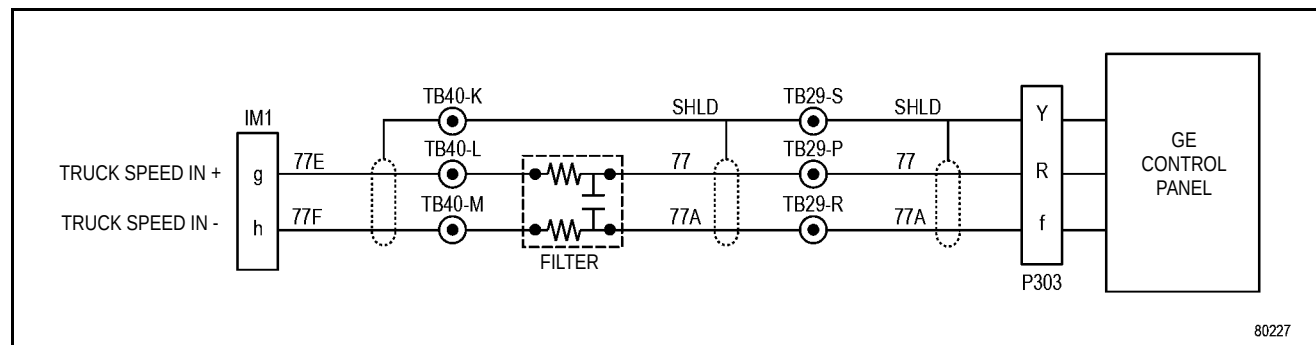
Operator Action	None
Fault Code	A011
Description	Payload meter speed sensor signal has failed.
Fault Conditions	Sets when payload meter declares a speed sensor fault. Resets when payload meter resets the speed sensor fault.
Operator Alerting System Response	PLM Display: -----
Resulting Problem(s)	Payload and haul cycle data is bad.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. The primary correction for this fault is to correct any external wiring or replace the GE signal source.
2. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items.
3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.

Parameter	Expected State and/or Related Fault(s)
Truck Speed Signal (PLM3 15, 25)	7 Hz: truck is not moving Greater than 7 Hz: truck is moving Fault(s): A212

Related circuit diagram



Fault Code A013: Body up switch has failed

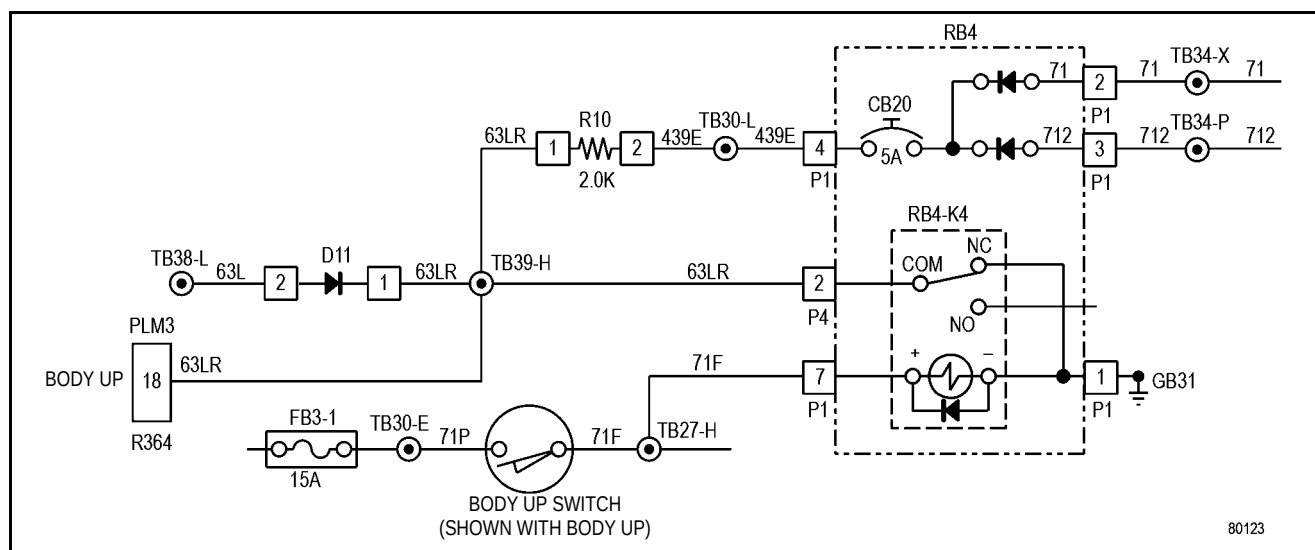
Operator Action	None
Fault Code	A013
Description	Body up switch has failed.
Fault Conditions	The switch no longer responds to payload cycles.
Operator Alerting System Response	PLM Display: -----
Resulting Problem(s)	Payload and haul cycle data is bad.
Related Information	

Table

1. The primary corrective measure for this fault is to change / correct the body up switch and wiring to PLM. No other faults are available to troubleshoot this problem.

Parameter	Expected State and/or Related Fault(s)
Body Up Switch Input (PLM 18)	0: Body is up 1: Body is down

Related circuit diagram



Fault Code A014: Payload meter checksum computation has failed

Operator Action	None
Fault Code	A014
Description	Payload meter checksum computation has failed.
Fault Conditions	Payload meter reports a checksum failure.
Operator Alerting System Response	PLM Display: -----
Resulting Problem(s)	Payload and haul cycle data is bad.
Related Information	

Table
1. This fault is totally contained within the payload meter.

Related circuit diagram

None

Fault Code A016: Payload meter write to flash memory has failed

Operator Action	None
Fault Code	A016
Description	Payload meter write to flash memory has failed.
Fault Conditions	Payload meter reports a failure in write to flash memory.
Operator Alerting System Response	PLM Display: -----
Resulting Problem(s)	Payload and haul cycle data is bad.
Related Information	

Table
1. This fault is totally contained within the payload meter.

Related circuit diagram

None

Fault Code A017: Payload meter flash memory read has failed

Operator Action	None
Fault Code	A017
Description	Payload meter flash memory read has failed.
Fault Conditions	Payload meter reports a failure of flash memory read.
Operator Alerting System Response	PLM Display: -----
Resulting Problem(s)	Payload and haul cycle data is bad.
Related Information	

Table
1. This fault is totally contained within the payload meter.

Related circuit diagram

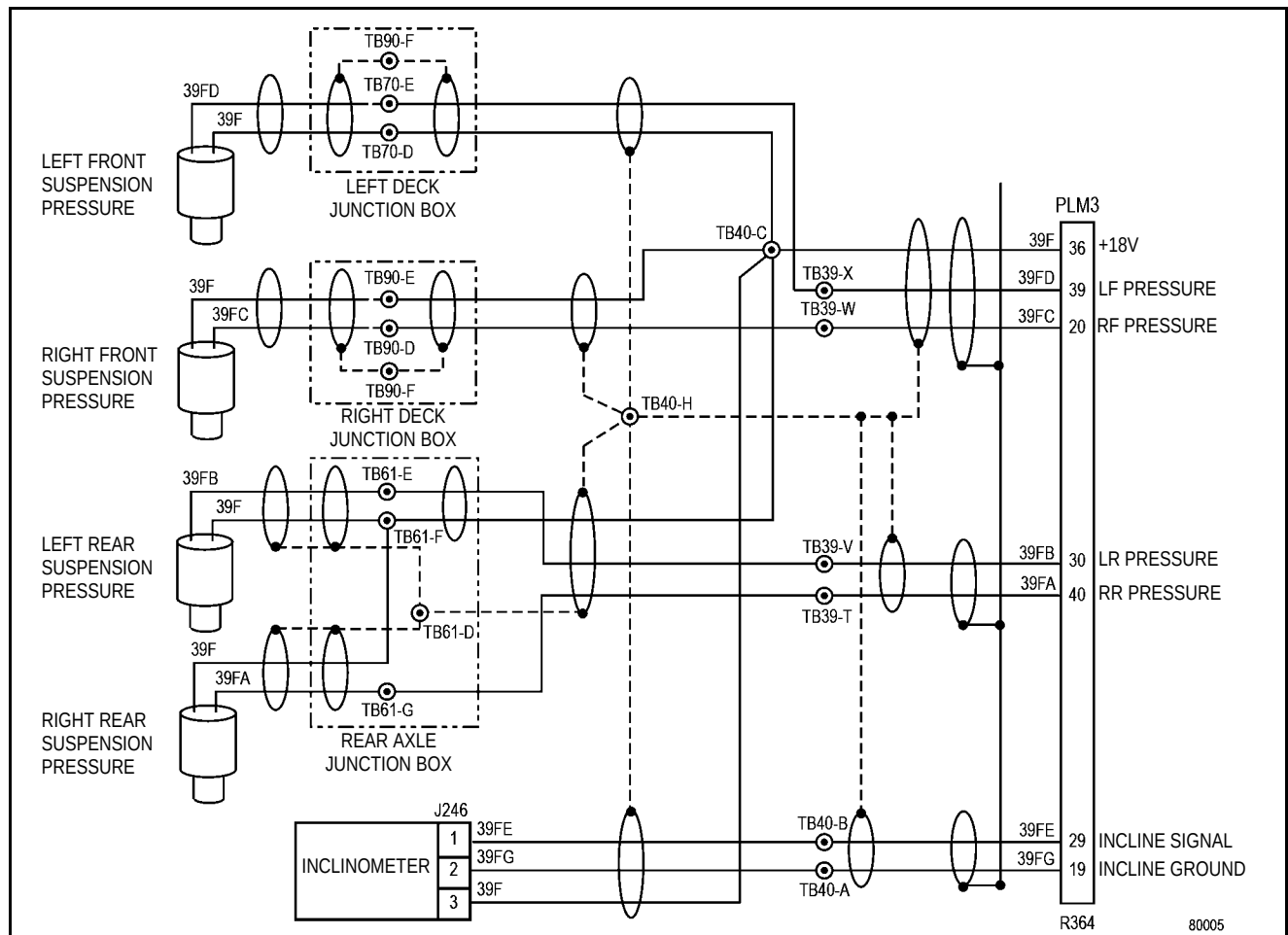
None

Fault Code A018: Right rear flat suspension cylinder warning

Operator Action	None
Fault Code	A018
Description	Right rear flat suspension cylinder warning.
Fault Conditions	Sets if PLMIII detects a flat right rear suspension cylinder. Resets if conditions clears.
Operator Alerting System Response	PLM Display: -----
Resulting Problem(s)	Potential damage to suspension and frame.
Related Information	

Table	
1. The logic that produces this fault is intended to identify flat suspension cylinders. The primary correction is to recharge / repair the cylinder. If the cylinder is not defective, use the following procedure to correct PLMIII. 2. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 3. If there are no active parameter fault codes, then, based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 4. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Right Rear Pressure Sensor (PLM3 36,40)	Sensor current > 22 ma: failed high Sensor current < 2 ma: failed low Sensor current > 2 ma but less than 22 ma: valid readings Fault(s): A007, A008
Left Rear Pressure Sensor (PLM3 36,30)	Sensor current > 22 ma: failed high Sensor current < 2 ma: failed low Sensor current > 2 ma but less than 22 ma: valid readings Fault(s): A005, A006
Right Front Pressure Sensor (PLM3 36,20)	Sensor current > 22 ma: failed high Sensor current < 2 ma: failed low Sensor current > 2 ma but less than 22 ma: valid readings Fault(s): A003, A004
Left Front Pressure Sensor (PLM3 36,39)	Sensor current > 22 ma: failed high Sensor current < 2 ma: failed low Sensor current > 2 ma but less than 22 ma: valid readings Fault(s): A001, A002
Incline Sensor (PLM3 36,29,19)	Sensor voltage < 0.565: failed high Sensor voltage > 5.08: failed low Sensor voltage > 0.565 but < 5.08: valid readings Fault(s): A009, A010

Related circuit diagram

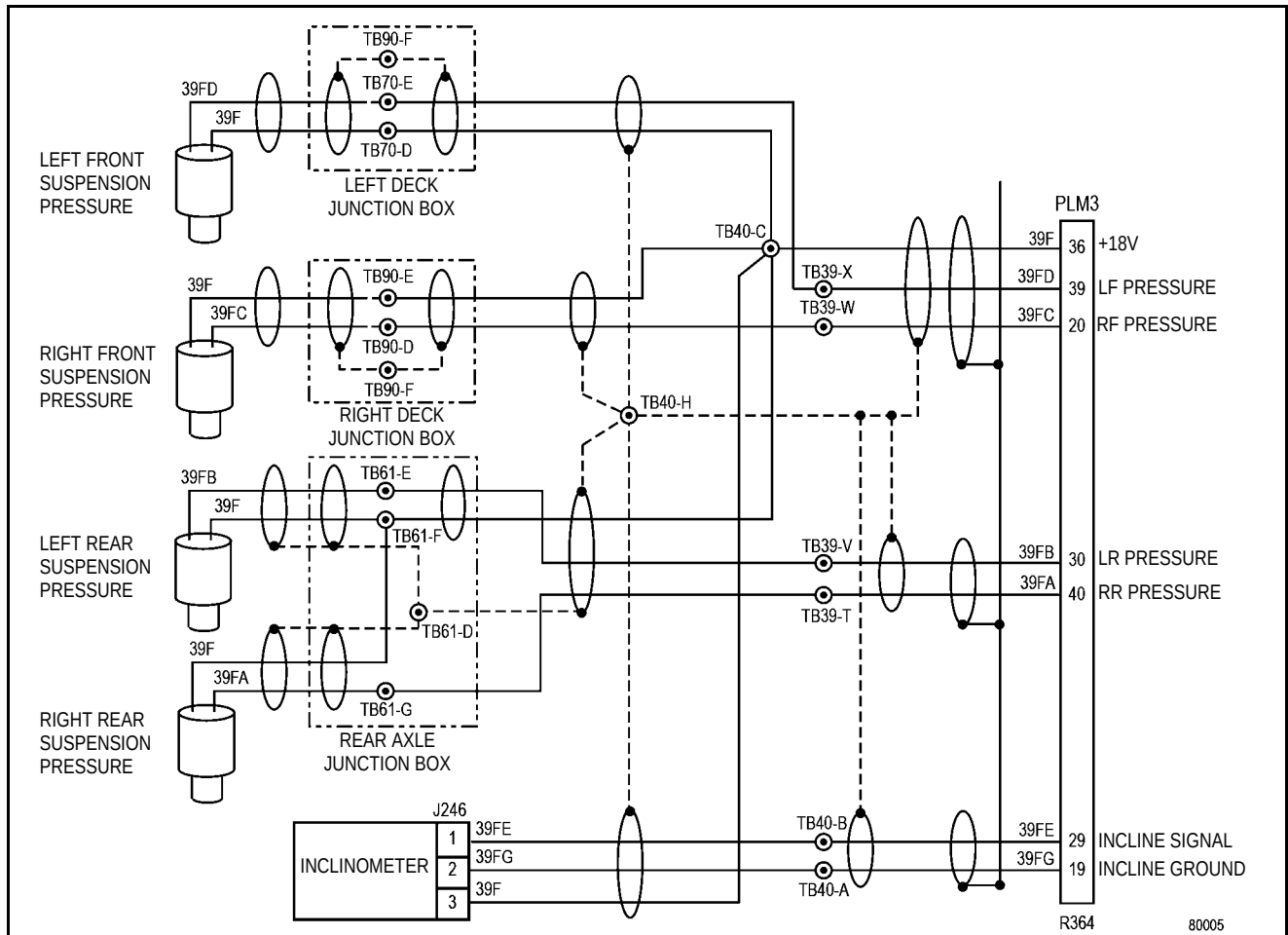


Fault Code A019: Left rear flat suspension cylinder warning

Operator Action	None
Fault Code	A019
Description	Left rear flat suspension cylinder warning.
Fault Conditions	Sets if PLMIII detects a flat left rear suspension cylinder. Resets if conditions clears.
Operator Alerting System Response	PLM Display: -----
Resulting Problem(s)	Potential damage to suspension and frame.
Related Information	

Table	
1. The logic that produces this fault is intended to identify flat suspension cylinders. The primary correction is to recharge / repair the cylinder. If the cylinder is not defective, use the following procedure to correct PLMIII. 2. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 3. If there are no active parameter fault codes, then, based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 4. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Left Rear Pressure Sensor (PLM3 36,30)	Sensor current > 22 ma: failed high Sensor current < 2 ma: failed low Sensor current > 2 ma but less than 22 ma: valid readings Fault(s): A005, A006
Right Rear Pressure Sensor (PLM3 36,40)	Sensor current > 22 ma: failed high Sensor current < 2 ma: failed low Sensor current > 2 ma but less than 22 ma: valid readings Fault(s): A007, A008
Left Front Pressure Sensor (PLM3 36,39)	Sensor current > 22 ma: failed high Sensor current < 2 ma: failed low Sensor current > 2 ma but less than 22 ma: valid readings Fault(s): A001, A002
Right Front Pressure Sensor (PLM3 36,20)	Sensor current > 22 ma: failed high Sensor current < 2 ma: failed low Sensor current > 2 ma but less than 22 ma: valid readings Fault(s): A003, A004
Incline Sensor (PLM3 36,29,19)	Sensor voltage < 0.565: failed high Sensor voltage > 5.08: failed low Sensor voltage > 0.565 but < 5.08: valid readings Fault(s): A009, A010

Related circuit diagram

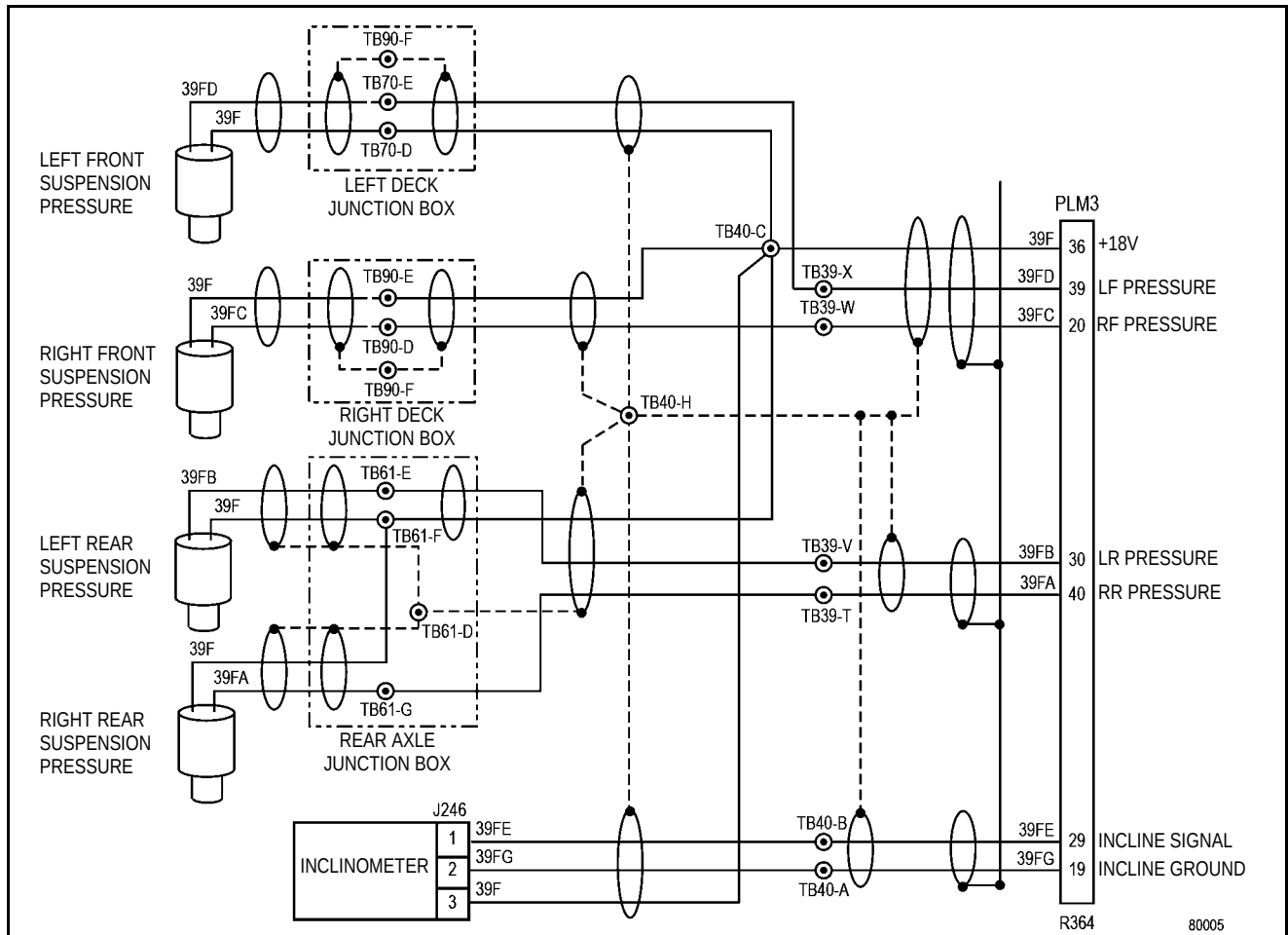


Fault Code A022: Carryback load is excessive

Operator Action	None
Fault Code	A022
Description	Carryback load is excessive.
Fault Conditions	Sets if PLMIII detects excessive load remaining in the bed after dumping is complete. Resets if conditions clears.
Operator Alerting System Response	PLM Display: -----
Resulting Problem(s)	Reduced production.
Related Information	

Table	
1. The logic that produces this fault is intended to identify excessive carryback load. This is a KOMTRAX Plus recording only. No display announces the fault. If excessive carryback does not exist, troubleshoot the payload system. 2. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 3. If there are no active parameter fault codes, then, based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 4. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Left Rear Pressure Sensor (PLM3 36,30)	Sensor current > 22 ma: failed high Sensor current < 2 ma: failed low Sensor current > 2 ma but less than 22 ma: valid readings Fault(s): A005, A006
Right Rear Pressure Sensor (PLM3 36,40)	Sensor current > 22 ma: failed high Sensor current < 2 ma: failed low Sensor current > 2 ma but less than 22 ma: valid readings Fault(s): A007, A008
Left Front Pressure Sensor (PLM3 36,39)	Sensor current > 22 ma: failed high Sensor current < 2 ma: failed low Sensor current > 2 ma but less than 22 ma: valid readings Fault(s): A001, A002
Right Front Pressure Sensor (PLM3 36,20)	Sensor current > 22 ma: failed high Sensor current < 2 ma: failed low Sensor current > 2 ma but less than 22 ma: valid readings Fault(s): A003, A004
Incline Sensor (PLM3 36,29,19)	Sensor voltage < 0.565: failed high Sensor voltage > 5.08: failed low Sensor voltage > 0.565 but < 5.08: valid readings Fault(s): A009, A010

Related circuit diagram

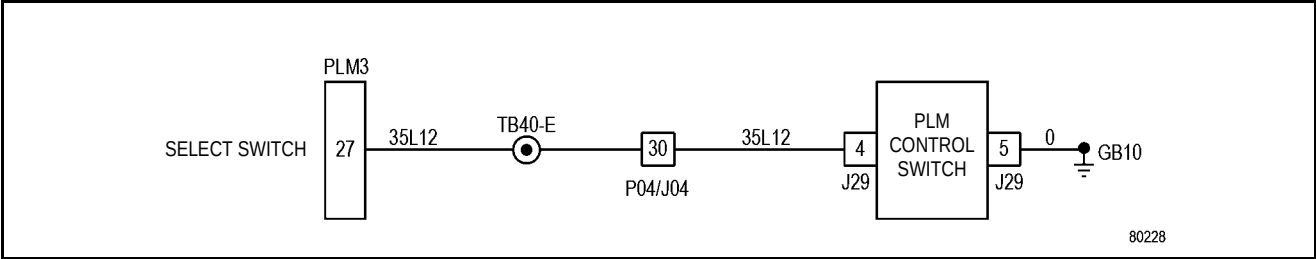


Fault Code A026: Payload meter user select switch has failed

Operator Action	None
Fault Code	A026
Description	Payload meter user select switch has failed.
Fault Conditions	Sets if the payload meter detects a switch failure.
Operator Alerting System Response	None
Resulting Problem(s)	Difficulty selecting payload data on the display.
Related Information	

Table	
1. The primary corrective measure for this fault is to change / correct the user select switch and wiring to PLM. No other faults are available to troubleshoot this problem.	
Parameter	Expected State and/or Related Fault(s)
User Select Switch (PLM3 27)	0: Select switch in SELECT position 1: Select switch in OFF position

Related circuit diagram

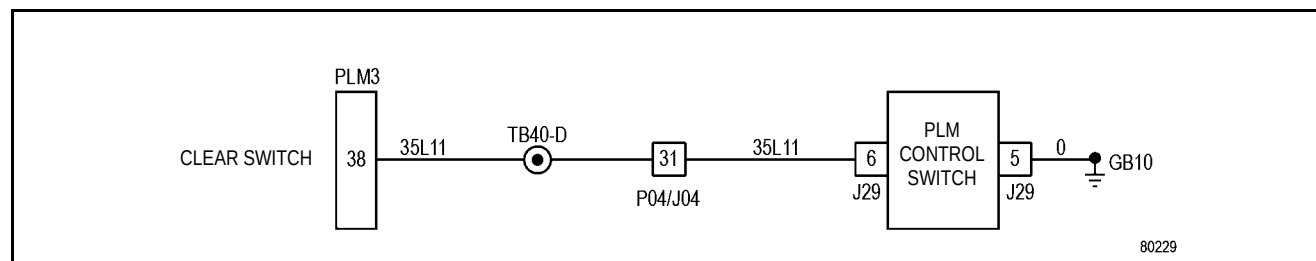


Fault Code A027: Payload meter user clear switch has failed

Operator Action	None
Fault Code	A027
Description	Payload meter user clear switch has failed.
Fault Conditions	Sets if the payload meter detects a switch failure.
Operator Alerting System Response	None
Resulting Problem(s)	Difficulty clearing payload data on the display.
Related Information	

Table	
1. The primary corrective measure for this fault is to change / correct the user clear switch and wiring to PLM. No other faults are available to troubleshoot this problem.	
Parameter	Expected State and/or Related Fault(s)
User Clear Switch (PLM3 38)	0: Select switch in CLEAR position 1: Select switch in OFF position

Related circuit diagram



Fault Code A101: High pressure detected across a hydraulic pump filter

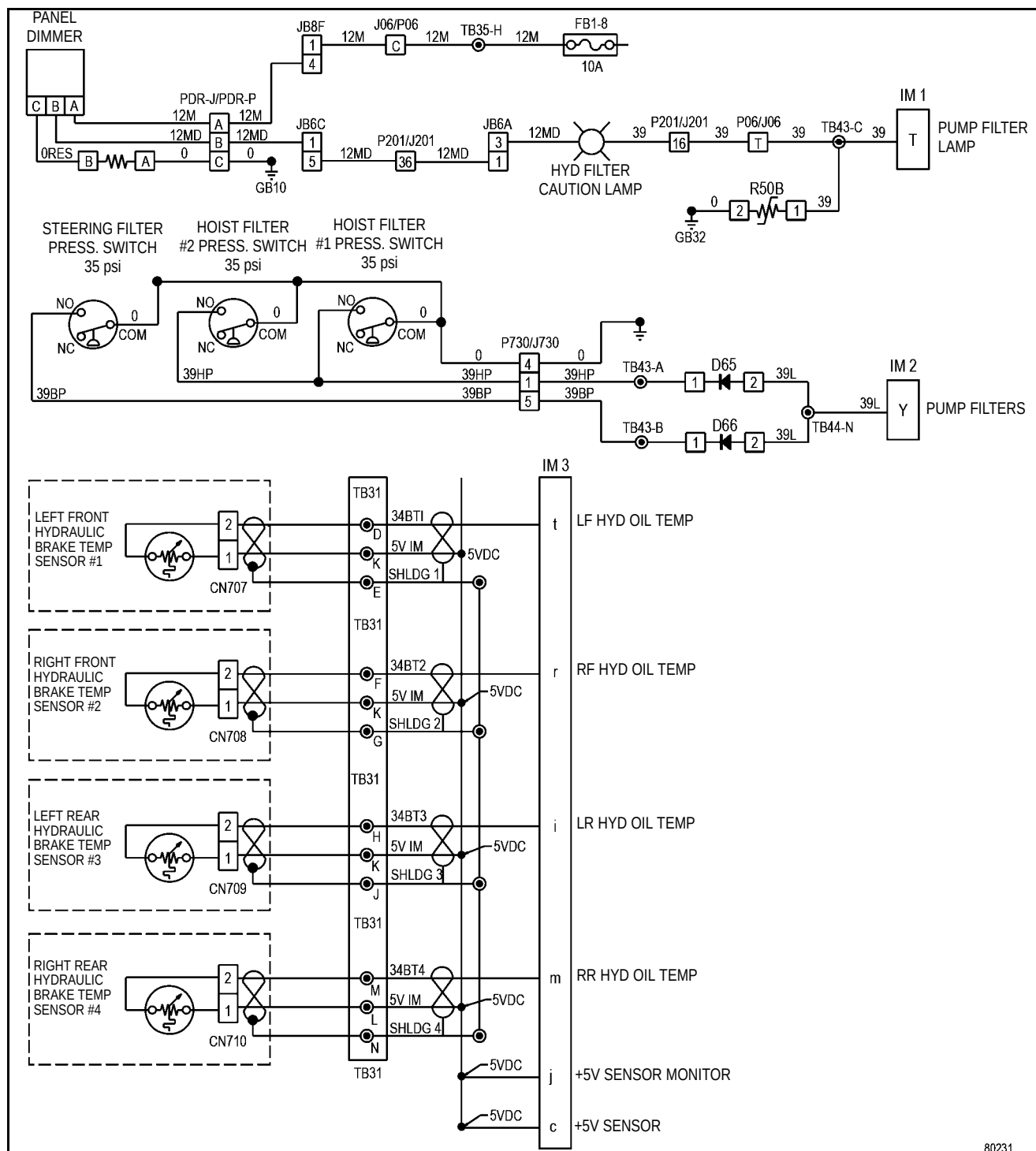
Operator Action	Go To Shop Now
Fault Code	A101
Description	High pressure detected across an hydraulic pump filter.
Fault Conditions	Sets if switch is on, engine is running, and oil temperature at any of four positions is above 50 °C (122 °F) for 10 seconds. Resets if any of these not true for 10 seconds.
Operator Alerting System Response	Amber, dimmable "Hydraulic Oil Filter" Lamp No Buzzer Left Lamp Module Row 4, Column 2
Resulting Problem(s)	Hydraulic oil filters will bypass oil without filtration. Continuing operation may damage hydraulic system components.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. This fault is generated by parallel switch inputs to IM from the hydraulic oil filter switches. Any of the three switches can close and cause the warning. The normal corrective measure for this fault is to change one or more hydraulic filters. 2. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 3. If there are no active parameter fault codes, then, based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 4. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Pump Filter Switches (IM2 Y)	0: No hydraulic filter switches open 1: One or more hydraulic filter switches open Fault(s): A273
Hydraulic Oil Temperature - Left Rear (IM3 i)	0.016 Volt to 4.89 Volt: Good Readings < 0.016 Volt or > 4.89 Volt: Defective Sensor or Circuit Fault(s): A166, A170
Hydraulic Oil Temperature - Right Rear (IM3 m)	0.016 Volt to 4.89 Volt: Good Readings < 0.016 Volt or > 4.89 Volt: Defective Sensor or Circuit Fault(s): A167, A171
Hydraulic Oil Temperature - Left Front (IM3 t)	0.016 Volt to 4.89 Volt: Good Readings < 0.016 Volt or > 4.89 Volt: Defective Sensor or Circuit Fault(s): A168, A172
Hydraulic Oil Temperature - Right Front (IM3 r)	0.016 Volt to 4.89 Volt: Good Readings < 0.016 Volt or > 4.89 Volt: Defective Sensor or Circuit Fault(s): A169, A173
Engine Speed [RPM] (CAN/J1939) (IM1 q,r,s)	Less than 300 RPM: engine is not running Greater than 300 RPM for 4 seconds: engine is running Fault(s): A184

Pump Filter Switches Lamp (IM1 T)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A361
--------------------------------------	--

Related circuit diagram



80231

Fault Code A105: Fuel level sensor is shorted to ground, indicating a false high fuel level

Operator Action	Go To Shop Now
Fault Code	A105
Description	Fuel level sensor is shorted to ground, indicating a false high fuel level.
Fault Conditions	Sets if fuel level indication is at 110% for 5 seconds with keyswitch on, and battery voltage does not drop below 18, with engine speed below 600 rpm. Resets if fuel level indication is at 104.2% for 3 seconds.
Operator Alerting System Response	Amber "Low Fuel Level" Lamp Fuel Gauge Buzzer producing bursts of sound every 15 minutes Left Lamp Module Row 5, Column 2
Resulting Problem(s)	False fuel gauge readings. If ignored, possible shutdown of engine while on the haul road.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. The primary correction for this fault is to correct or replace the fuel level sensor and wiring. 2. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 3. If there are no active parameter fault codes, then, based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 4. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Fuel Level Sensor (IM3 g)	2.8 to 6.2 Volts: Good Readings at 24 Volt Battery (varies with battery voltage) < 2.8 or > 6.2 Volts: Defective Sensor or Circuit at 24 Volt Battery (varies with battery voltage) Fault(s): A158
Battery Voltage, 24 Volt (IM1 A)	> 18 Volt: Good Reading < 18 Volt: Low battery voltage (while cranking) blocks this fault code. Fault(s): A153, A154, A155, A250
Keyswitch (IM3 G)	0: keyswitch is off and the truck is not moving and the engine is not running 1: keyswitch is on or the truck is moving (GE power not yet off) Fault(s): A240
Engine Speed [RPM] (CAN/J1939) (IM1 q,r,s)	Less than 600 RPM: combined with low voltage means cranking is in process and fault is blocked Greater than 600 RPM: engine is running normally Fault(s): A184

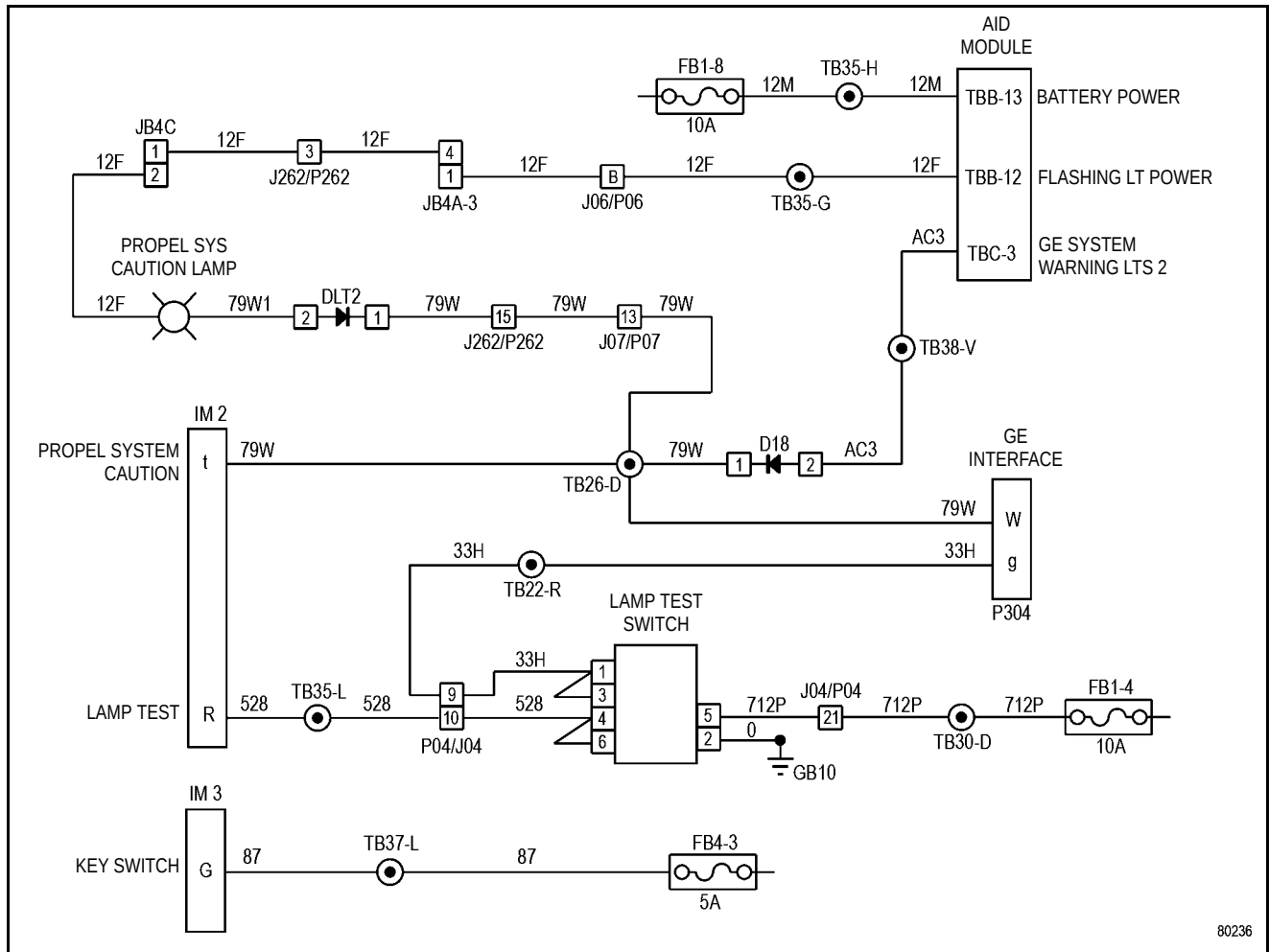
Wiring diagram for the fuel system of a 1977-1978 Chevrolet. The diagram shows the battery power (IM1) connected to a 10A fuse (FB2-4) and a 5A fuse (FB1-3). The 10A fuse feeds the fuel gauge (J04/P04) and the fuel level sensor (IM3). The 5A fuse feeds the fuel level sender (P730/J730) and the key switch (IM3). The fuel gauge is grounded to the chassis (GB10). The fuel level sender is grounded to the chassis (P730/J730). The fuel level sensor is grounded to the chassis (TB32-D). The key switch is grounded to the chassis (TB37-L).

Fault Code A107: GE has generated a propel system caution

Operator Action	Max Speed Limited
Fault Code	A107
Description	GE has generated a propel system caution.
Fault Conditions	Sets if the GE controller generates a propel system caution event with keyswitch on and lamp test off. Resets if conditions change.
Operator Alerting System Response	Amber flashing "Propel System Caution" Lamp No buzzer Right Lamp Module Row 2, Column 1
Resulting Problem(s)	Propel system may not permit the truck to remain at full performance. Shutdown of propel system may occur if severity increases.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault. A laptop running GE's PTU software may be required to resolve this fault.

Table	
1. This fault is generated by GE driving an input pin of IM to ground. 2. It may be necessary to use GE's PTU software to determine the actual cause of the propel system distress. 3. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 4. If there are no active parameter fault codes, then, based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 5. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Propel System Caution (IM2 t)	0: GE has generated a propel system caution 1: no propel system caution exists
Keyswitch (IM3 G)	0: keyswitch is off and the truck is not moving and the engine is not running 1: keyswitch is on or the truck is moving (GE power not yet off) Fault(s): A240
Lamp Test Switch (IM2 R)	0: lamp test switch is off - normal position 1: lamp test switch is in test position

Related circuit diagram



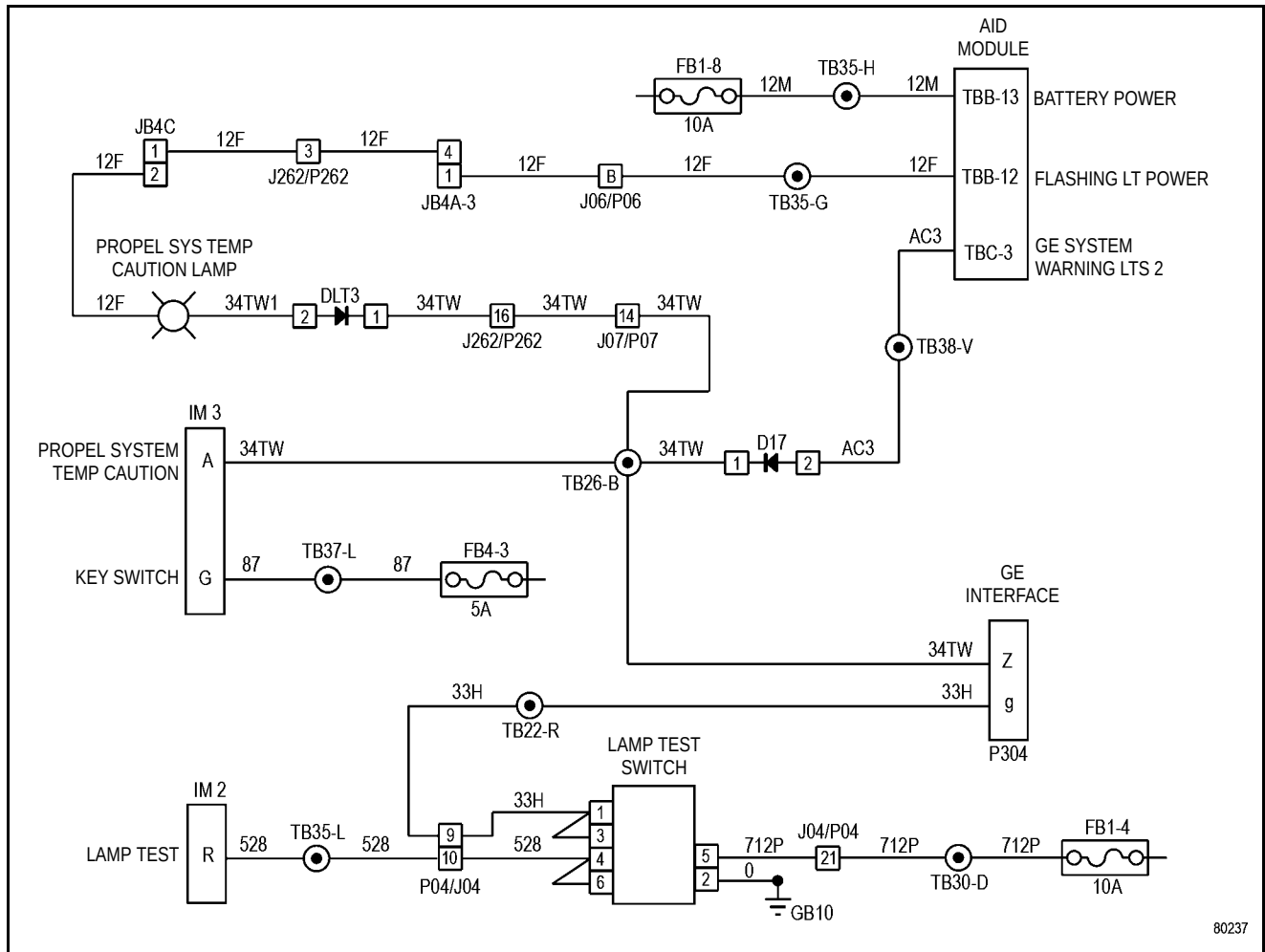
80236

Fault Code A108: GE has generated a propel system temperature caution

Operator Action	None
Fault Code	A108
Description	GE has generated a propel system temperature caution.
Fault Conditions	Sets if the GE controller generates a propel system caution event with keyswitch on and lamp test off. Resets if conditions change.
Operator Alerting System Response	Amber flashing "Propel System Temperature Caution" Lamp No buzzer Right Lamp Module Row 3, Column 1
Resulting Problem(s)	Propel system may not permit the truck to remain at full performance. Shutdown of propel system may occur if severity increases.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault. A laptop running GE's PTU software may be required to resolve this fault.

Table	
1. This fault is generated by GE driving an input pin of IM to ground. 2. It may be necessary to use GE's PTU software to determine the actual cause of the propel system distress. 3. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 4. If there are no active parameter fault codes, then, based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 5. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Propel System Temperature Caution (IM3 A)	0: GE has generated a propel system temperature caution 1: no propel system caution exists
Keyswitch (IM3 G)	0: keyswitch is off and the truck is not moving and the engine is not running 1: keyswitch is on or the truck is moving (GE power not yet off) Fault(s): A240
Lamp Test Switch (IM2 R)	0: lamp test switch is off - normal position 1: lamp test switch is in test position

Related circuit diagram

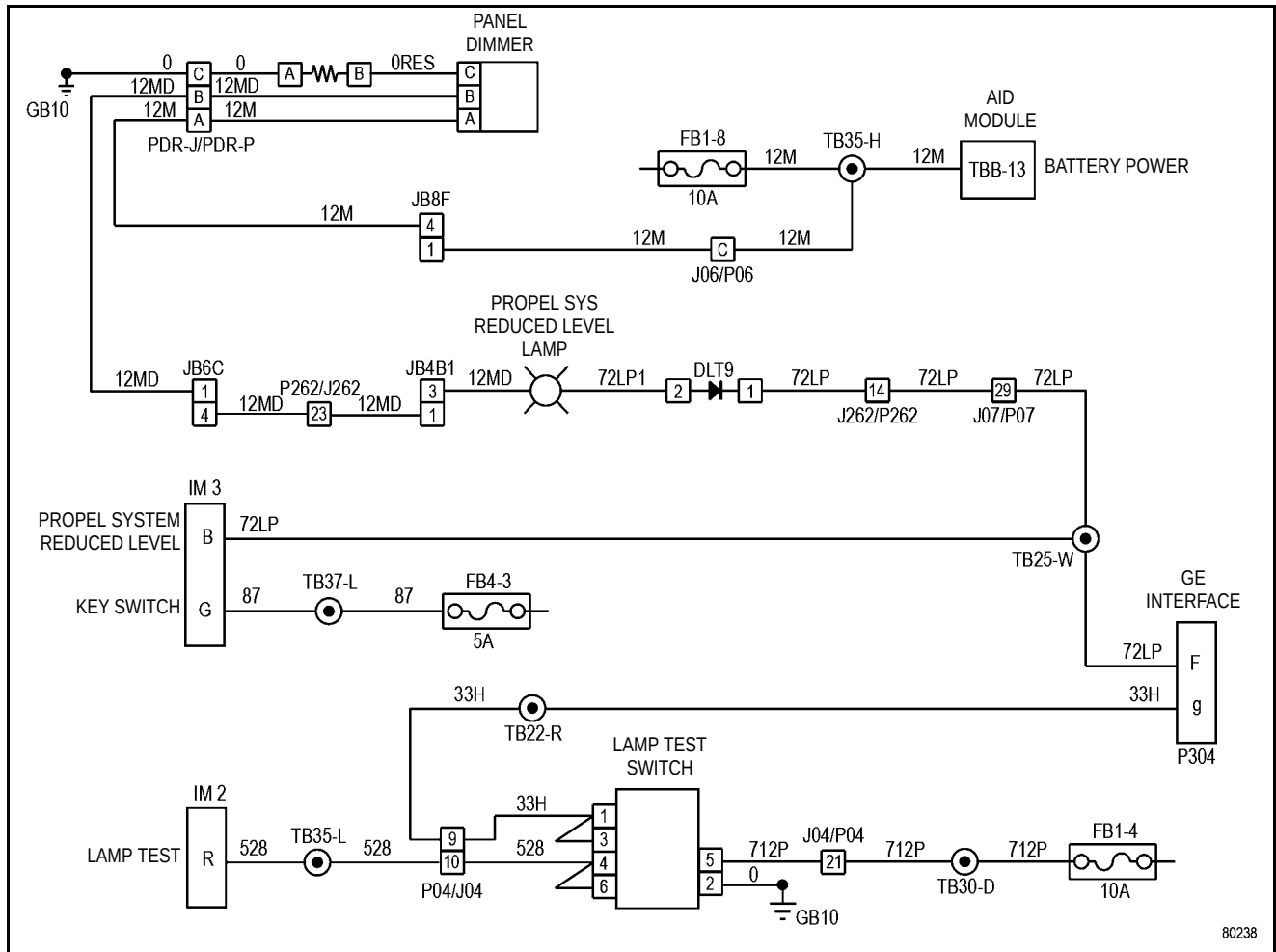


Fault Code A109: GE has generated a propel system reduced level signal

Operator Action	Max Speed Limited
Fault Code	A109
Description	GE has generated a propel system reduced level signal.
Fault Conditions	Sets if the GE controller generates a propel system reduced level signal with key-switch on and lamp test off. Resets if conditions change.
Operator Alerting System Response	Amber flashing "Propel System Reduced Level" Lamp No buzzer Right Lamp Module Row 4, Column 2
Resulting Problem(s)	Propel system will not permit the truck to remain at full performance. Shutdown of propel system may occur if severity increases
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault. A laptop running GE's PTU software may be required to resolve this fault.

Table	
1. This fault is generated by GE driving an input pin of IM to ground. 2. It may be necessary to use GE's PTU software to determine the actual cause of the propel system distress. 3. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 4. If there are no active parameter fault codes, then, based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 5. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Propel System Reduced Level (IM3 B)	0: GE has generated a propel system temperature caution 1: no propel system caution exists
Keyswitch (IM3 G)	0: keyswitch is off and the truck is not moving and the engine is not running 1: keyswitch is on or the truck is moving (GE power not yet off) Fault(s): A240
Lamp Test Switch (IM2 R)	0: lamp test switch is off - normal position 1: lamp test switch is in test position

Related circuit diagram



Fault Code A115: Low steering precharge pressure is detected

Operator Action	Stop; Park
Fault Code	A115
Description	Low steering precharge pressure is detected.
Fault Conditions	Sets and latches if low steering precharge switch operates for two seconds with key-switch on and "Steering Precharge Mask" not on except when operated by lamp test. The "Steering Precharge Mask" prevents warnings that might occur from the beginning of steering bleeddown until five minutes after bleeddown is complete or timed out or keyswitch is turned back on. Resets when steering bleeddown comes on. Mask prevents new operations for five minutes.
Operator Alerting System Response	Red flashing "Low Accumulator Precharge" Lamp Buzzer Left Lamp Module Row 3, Column 1
Resulting Problem(s)	Low steering accumulator precharge will reduce the amount of stored energy in the steering system.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. The primary correction for this fault is to adjust the nitrogen charge in the steering accumulators. 2. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 3. If there are no active parameter fault codes, then, based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 4. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Low Steering Precharge Pressure Switch (IM2 W)	0: Normal reading - full precharge pressure 1: Steering precharge pressure is low
Keyswitch (IM3 G)	0: keyswitch is off and the truck is not moving and the engine is not running 1: keyswitch is on or the truck is moving (GE power not yet off) Fault(s): A240
Lamp Test Switch (IM2 R)	0: lamp test switch is off - normal position 1: lamp test switch is in test position

Fault Code A123: GE has generated a reduced retarding caution

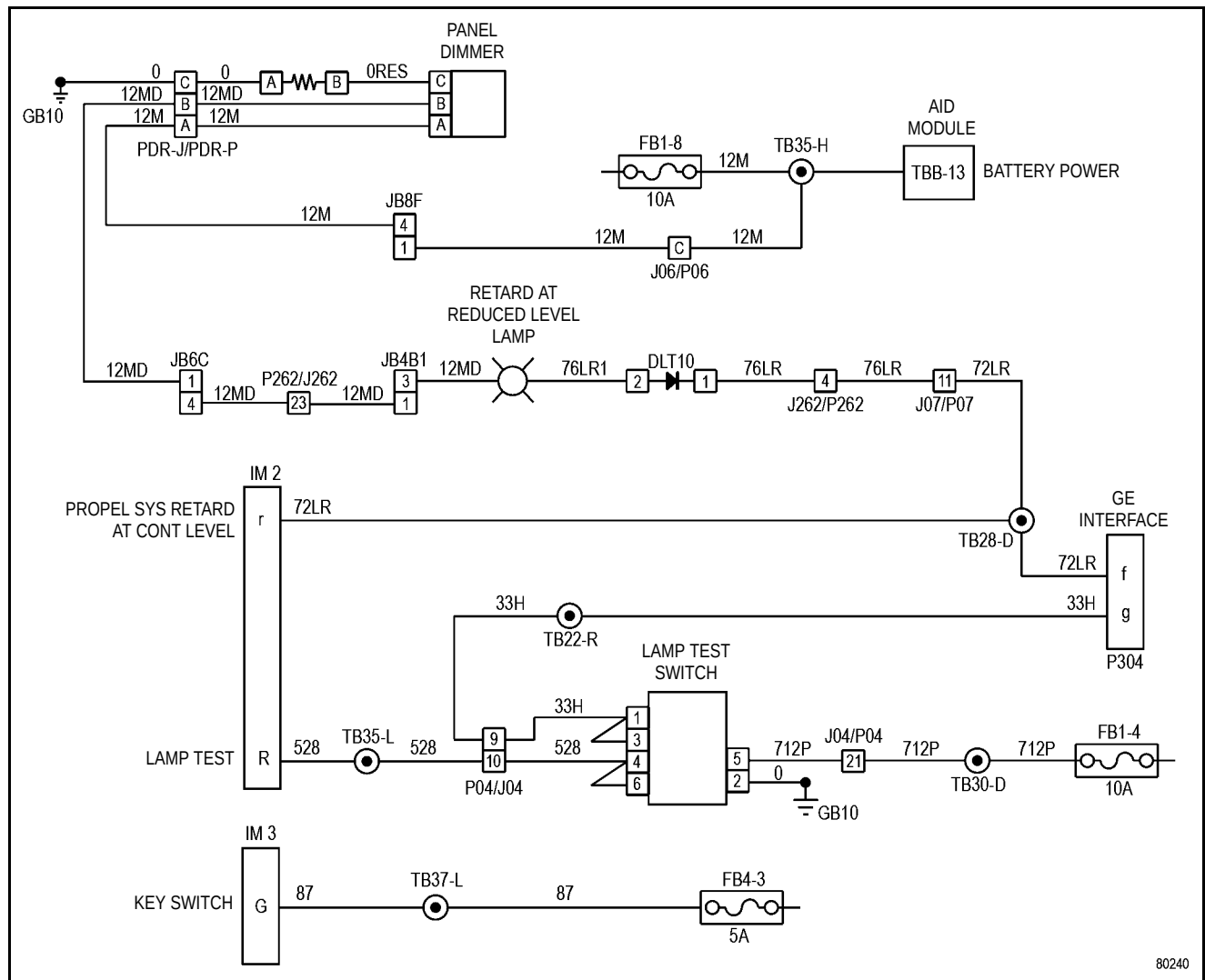
Operator Action	Slow Down Hill
Fault Code	A123
Description	GE has generated a reduced retarding caution.
Fault Conditions	Sets if the GE controller generates a reduced retarding event with the keyswitch on and lamp test switch off for 0.4 seconds. Resets if GE removes the reduced retarding event.
Operator Alerting System Response	Amber, dimmable "Retard at Reduced Level" Lamp No Buzzer Right Lamp Module Row 5, Column 2
Resulting Problem(s)	Retarding will be at somewhat reduced level, but the truck will still be driveable.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault. A laptop running GE's PTU software may be required to resolve this fault.

Table

1. The primary correction for this fault is to descend the hill at a slower speed so that retarding capacity is not exceeded.
2. This fault is generated by GE driving an input pin of IM to ground.
3. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s).
4. If there are no active parameter fault codes, then, based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items.
5. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.

Parameter	Expected State and/or Related Fault(s)
Reduced Retarding (IM2 R)	0: GE has generated a reduced retarding caution 1: no reduced retarding caution exists
Keyswitch (IM3 G)	0: keyswitch is off and the truck is not moving and the engine is not running 1: keyswitch is on or the truck is moving (GE power not yet off) Fault(s): A240
Lamp Test Switch (IM2 R)	0: lamp test switch is off - normal position 1: lamp test switch is in test position

Related circuit diagram



80240

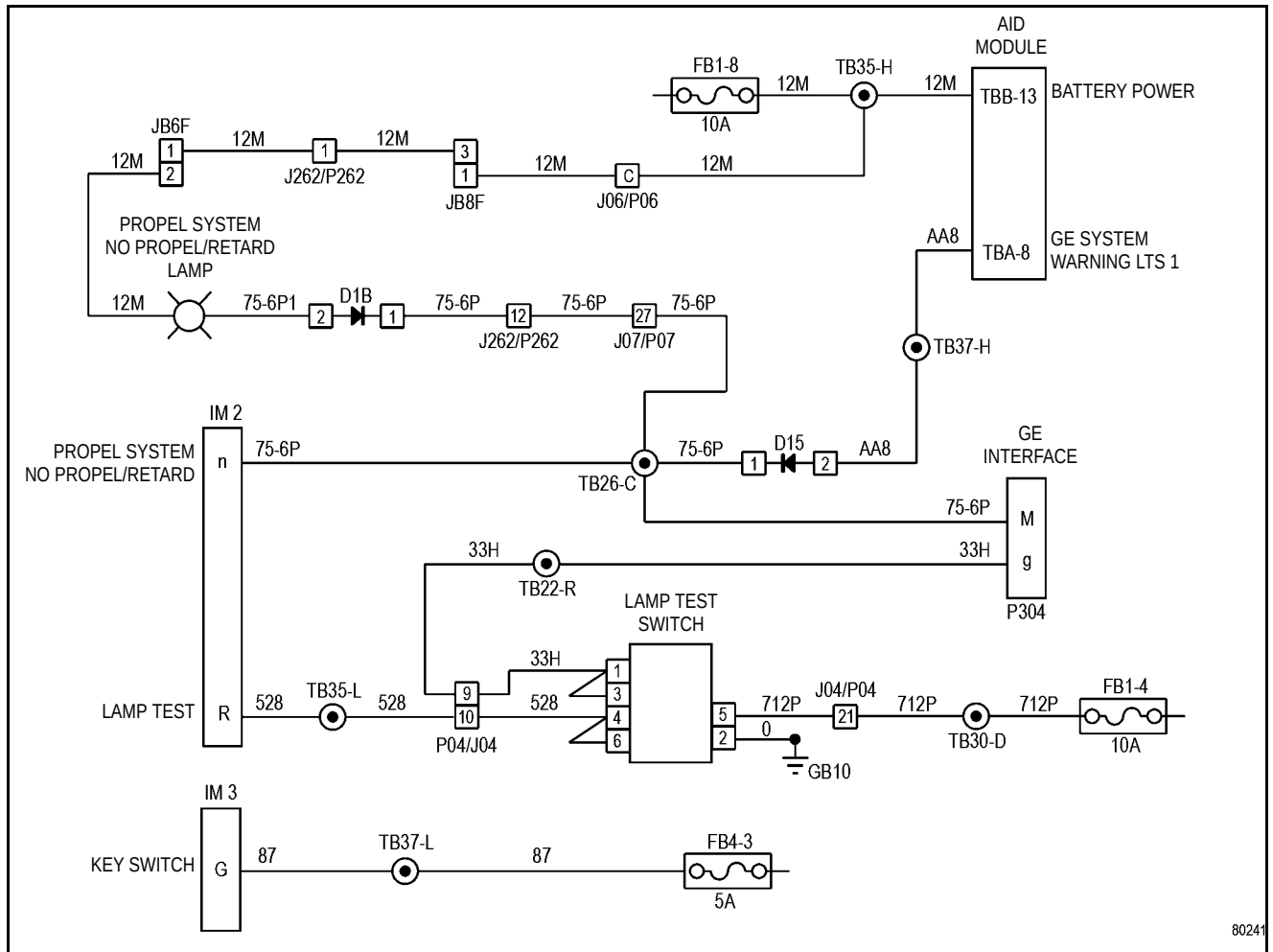
Fault Code A124: GE has generated a no propel / no retard warning

Operator Action	Stop; Park
Fault Code	A124
Description	GE has generated a no propel / no retard warning.
Fault Conditions	Sets 0.25 sec after GE controller generates a no propel / no retard event with key-switch on and lamp test off. Resets 2.0 sec after GE removes the no propel / no retard event.
Operator Alerting System Response	Red "No Propel / Retard" Lamp Buzzer Right Lamp Module Row 1, Column 1
Resulting Problem(s)	The drive system is inoperable in both propel and retard.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault. A laptop running GE's PTU software may be required to resolve this fault.

Table

1. This fault is generated by GE driving an input pin of IM to ground. 2. It may be necessary to use GE's PTU software to determine the actual cause of the propel system distress. 3. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 4. If there are no active parameter fault codes, then, based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 5. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
No Propel / Retard (IM2 n)	0: GE has generated a no propel / retard warning 1: the no propel / retard warning doesn't exist
Keyswitch (IM3 G)	0: keyswitch is off and the truck is not moving and the engine is not running 1: keyswitch is on or the truck is moving (GE power not yet off) Fault(s): A240
Lamp Test Switch (IM2 R)	0: lamp test switch is off - normal position 1: lamp test switch is in test position

Related circuit diagram

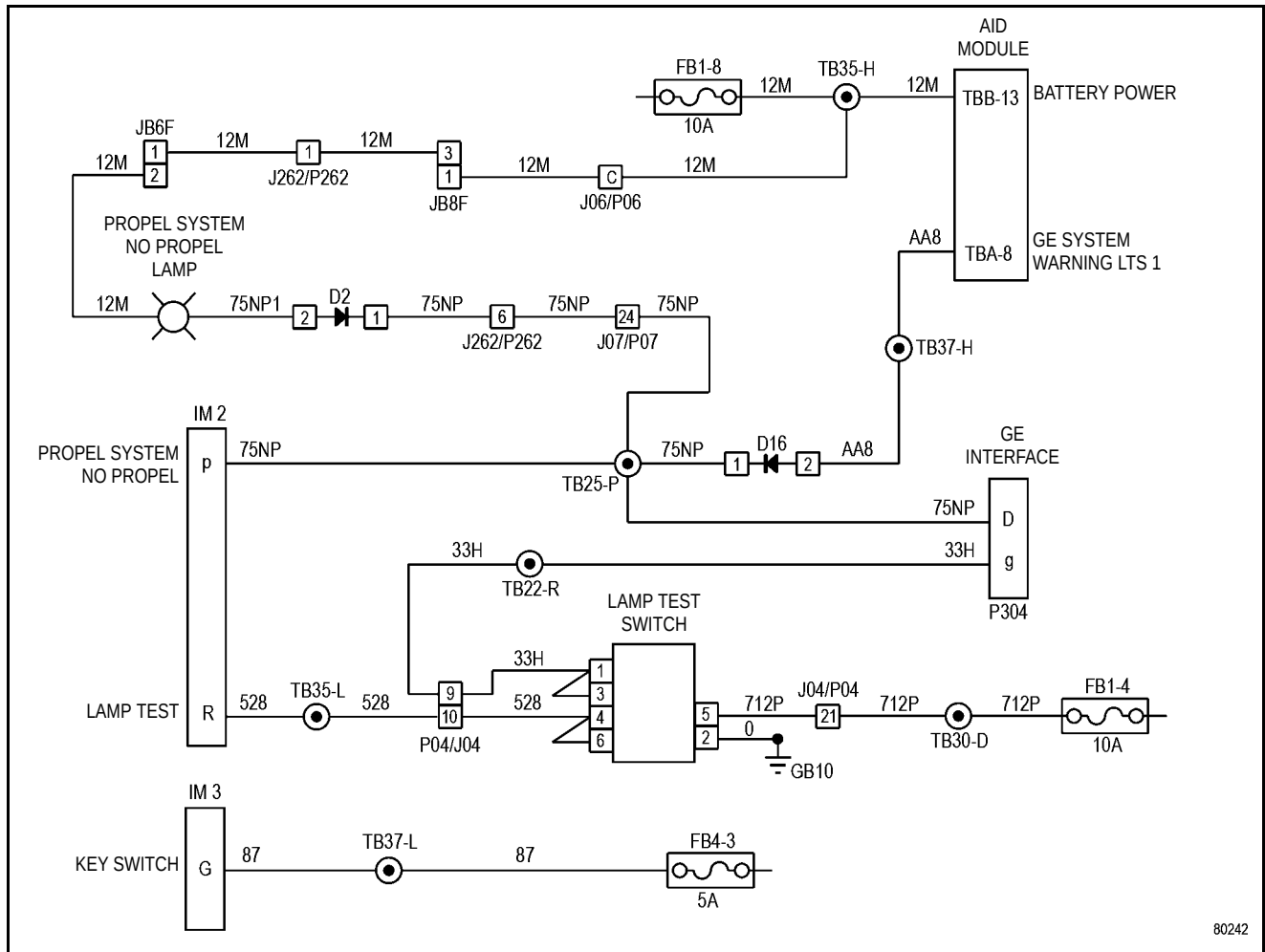


Fault Code A125: GE has generated a no propel warning

Operator Action	Stop; Park
Fault Code	A125
Description	GE has generated a no propel warning.
Fault Conditions	Sets 0.25 sec after GE controller generates a no propel event with keyswitch on and lamp test switch off. Resets 2.0 sec after GE removes the no propel event.
Operator Alerting System Response	Red "No Propel / Retard" Lamp Buzzer Right Lamp Module Row 1, Colum 2
Resulting Problem(s)	The drive system is inoperable in propel.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault. A laptop running GE's PTU software may be required to resolve this fault.

Table	
1. This fault is generated by GE driving an input pin of IM to ground. 2. It may be necessary to use GE's PTU software to determine the actual cause of the propel system distress. 3. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 4. If there are no active parameter fault codes, then, based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 5. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
No Propel (IM2 p)	0: GE has generated a no propel warning 1: the no propel warning doesn't exist
Keyswitch (IM3 G)	0: keyswitch is off and the truck is not moving and the engine is not running 1: keyswitch is on or the truck is moving (GE power not yet off) Fault(s): A240
Lamp Test Switch (IM2 R)	0: lamp test switch is off - normal position 1: lamp test switch is in test position

Related circuit diagram



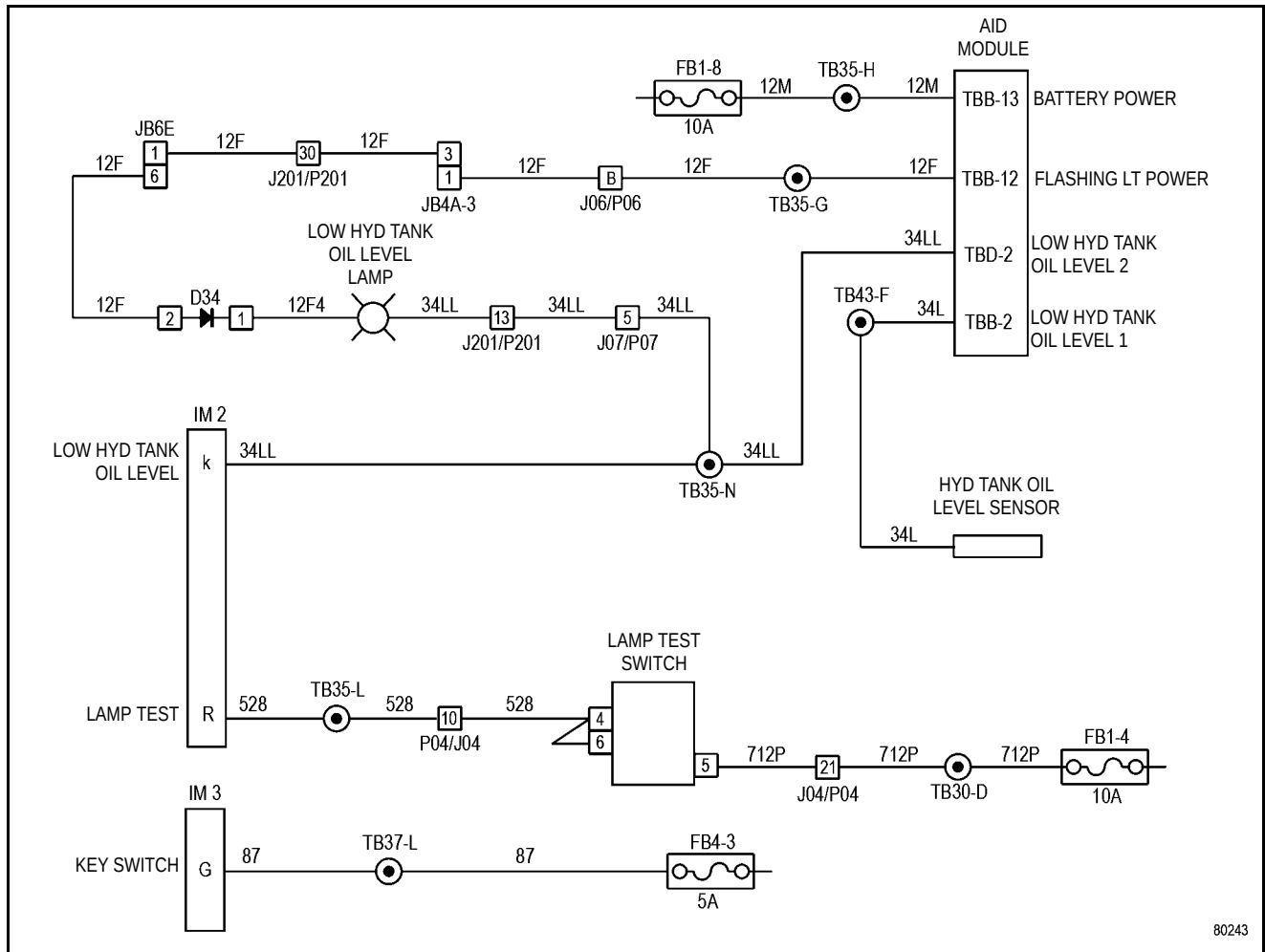
80242

Fault Code A126: Oil level in the hydraulic tank is low

Operator Action	Stop; Park; Power Down; Check
Fault Code	A126
Description	Oil level in the hydraulic tank is low.
Fault Conditions	Sets if AID sensing reports oil level to be low with keyswitch on and lamp test off. Resets if oil level is restored or if steering bleeddown comes on.
Operator Alerting System Response	Red flashing "Low Hydraulic Tank Level" Lamp Buzzer Left Lamp Module Row 1, Column 2
Resulting Problem(s)	Low hydraulic oil level can cause damage to equipment and failure to operate properly. Steering and brakes may deteriorate in performance.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table	
1. This primary correction for this fault is to add oil to the hydraulic system and check for hydraulic leaks. 2. If oil is sufficient, check the sensor and sensor wiring. 3. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 4. If there are no active parameter fault codes, then, based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 5. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Hydraulic Oil Level Sensor (IM2 k)	0: Oil level is normal 1: Oil level is low
Keyswitch (IM3 G)	0: keyswitch is off and the truck is not moving and the engine is not running 1: keyswitch is on or the truck is moving (GE power not yet off) Fault(s): A240
Lamp Test Switch (IM2 R)	0: lamp test switch is off - normal position 1: lamp test switch is in test position

Related circuit diagram

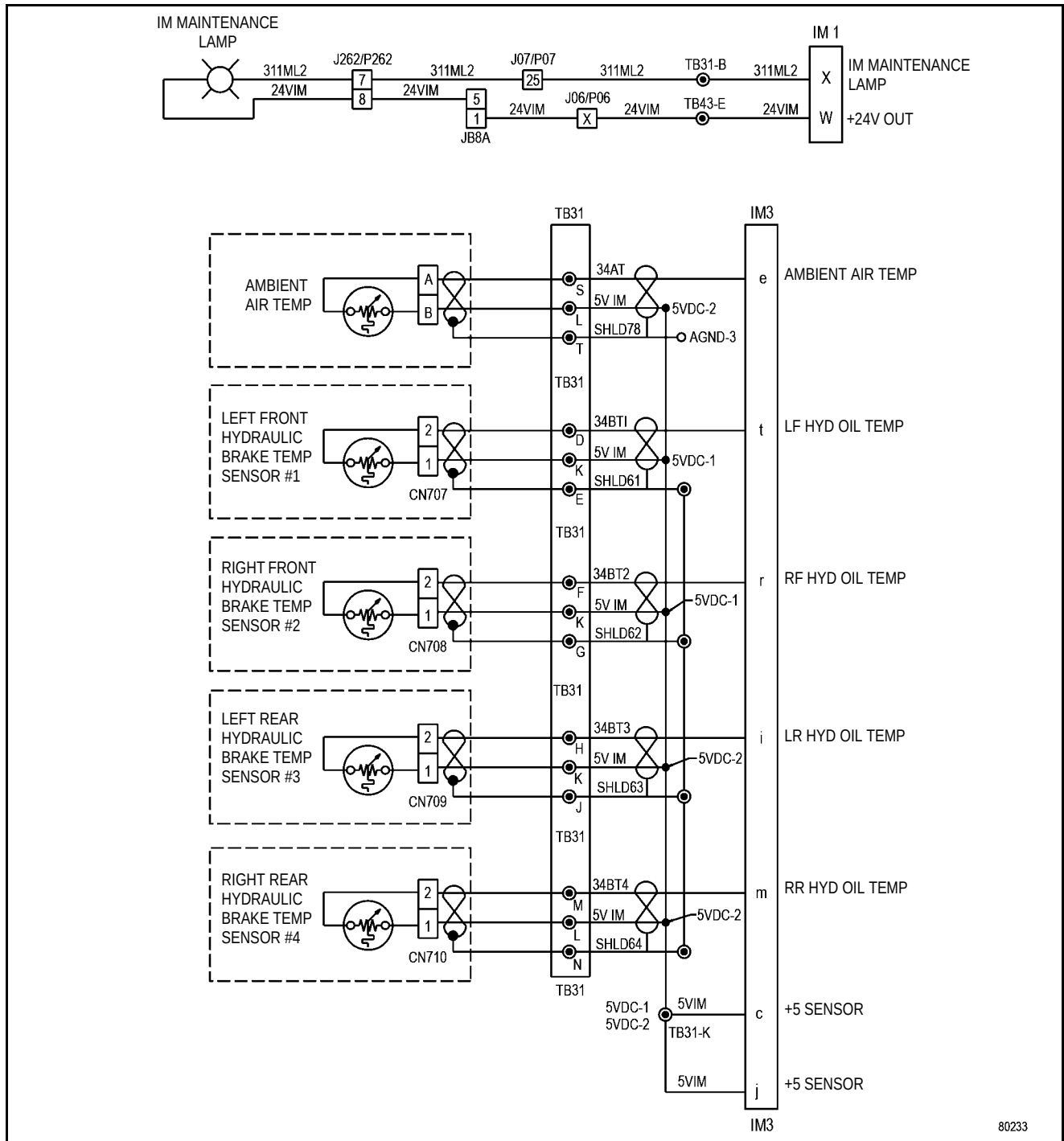


Fault Code A127: IM furnished +5 volt output for sensors is low

Operator Action	Go To Shop Now
Fault Code	A127
Description	IM furnished +5 volt output for sensors is low.
Fault Conditions	Sets when sensor +5V input drops below 4.52 volts for two seconds. Resets when sensor +5V input recovers to 4.66 volts for two seconds.
Operator Alerting System Response	Amber "IM Maintenance" Lamp No Buzzer Right Lamp Module Row 5, Column 3
Resulting Problem(s)	All active temperature sensors using the +5 volt supply will report low readings. Warnings of high oil temperature will be compromised. Equipment damage may result. Ambient temperature will also be reported low.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table	
1. Check external wiring and sensors to determine if problem is outside of IM. 2. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 3. If there are no active parameter fault codes, then, based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 4. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Sensor +5V Analog Input (IM3 j)	> 4.52 Volts: Normal voltage < 4.52 Volts: Voltage low Fault(s): A128
Sensor +5V Output (IM3 c)	> 4.52 Volts: Normal voltage < 4.52 Volts: Voltage low Fault(s): A128
IM Maintenance Lamp (IM1 X)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A363

Related circuit diagram



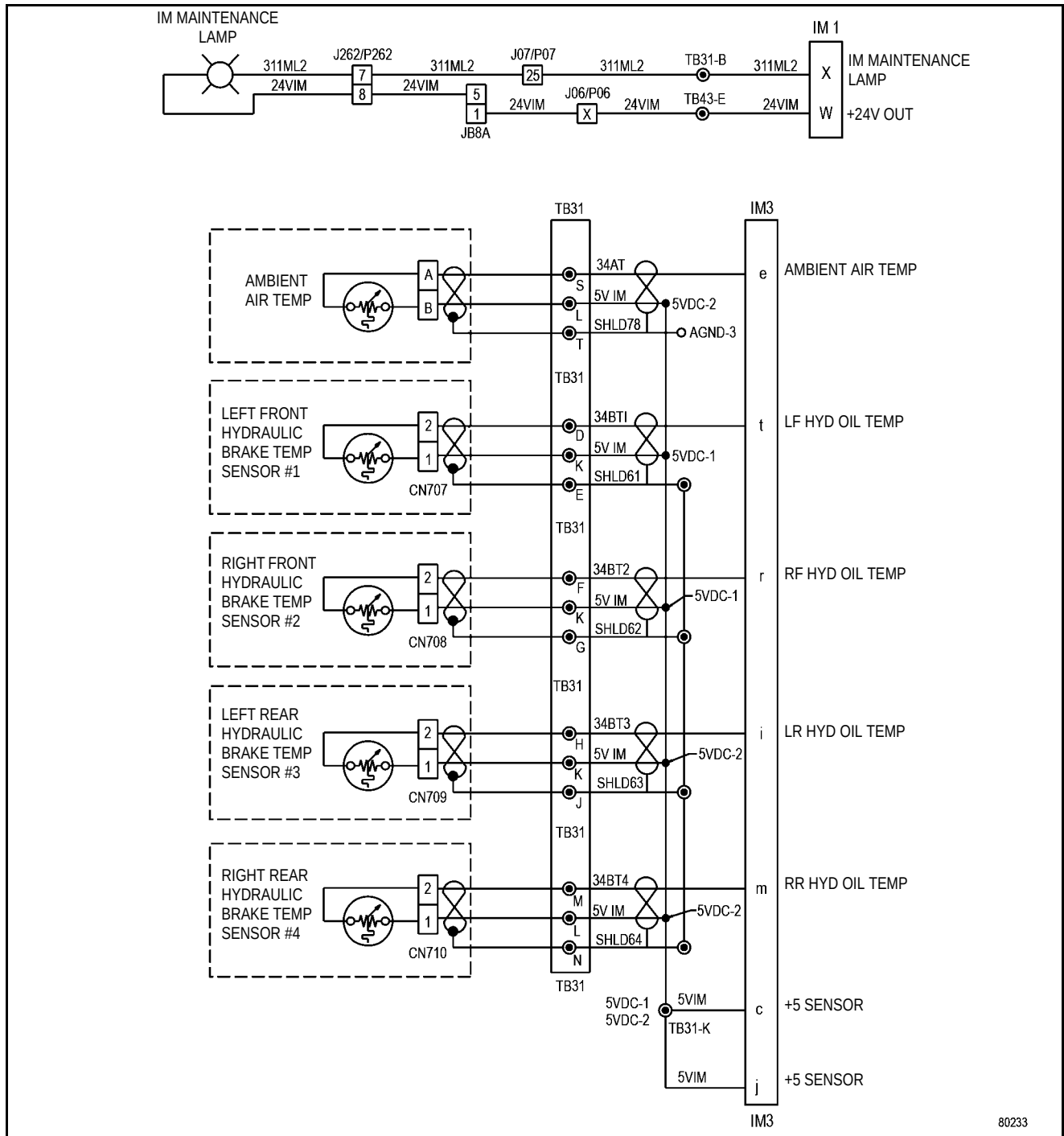
Fault Code A128: IM furnished +5 volt output for sensors is high

Operator Action	None
Fault Code	A128
Description	IM furnished +5 volt output for sensors is high.
Fault Conditions	Sets when sensor +5V input drops below 4.52 volts for two seconds. Resets when sensor +5V input recovers to 4.66 volts for two seconds.
Operator Alerting System Response	Amber "IM Maintenance" Lamp No Buzzer Right Lamp Module Row 5, Column 3
Resulting Problem(s)	All five temperature sensors using the +5 volt supply will report high readings. Warnings of high oil temperature will occur at lower than normal temperatures. Ambient temperature will be reported high.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. Check external wiring and sensors to determine if problem is outside of IM. 2. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 3. If there are no active parameter fault codes, then, based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 4. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Sensor +5V Analog Input (IM3 j)	< 5.51 Volts: Normal voltage > 5.51 Volts: Voltage high Fault(s): A127
Sensor +5V Output (IM3 c)	< 5.51 Volts: Normal voltage > 5.51 Volts: Voltage high Fault(s): A127
IM Maintenance Lamp (IM1 X)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A363

Related circuit diagram



Fault Code A139, A310: Low fuel level warning

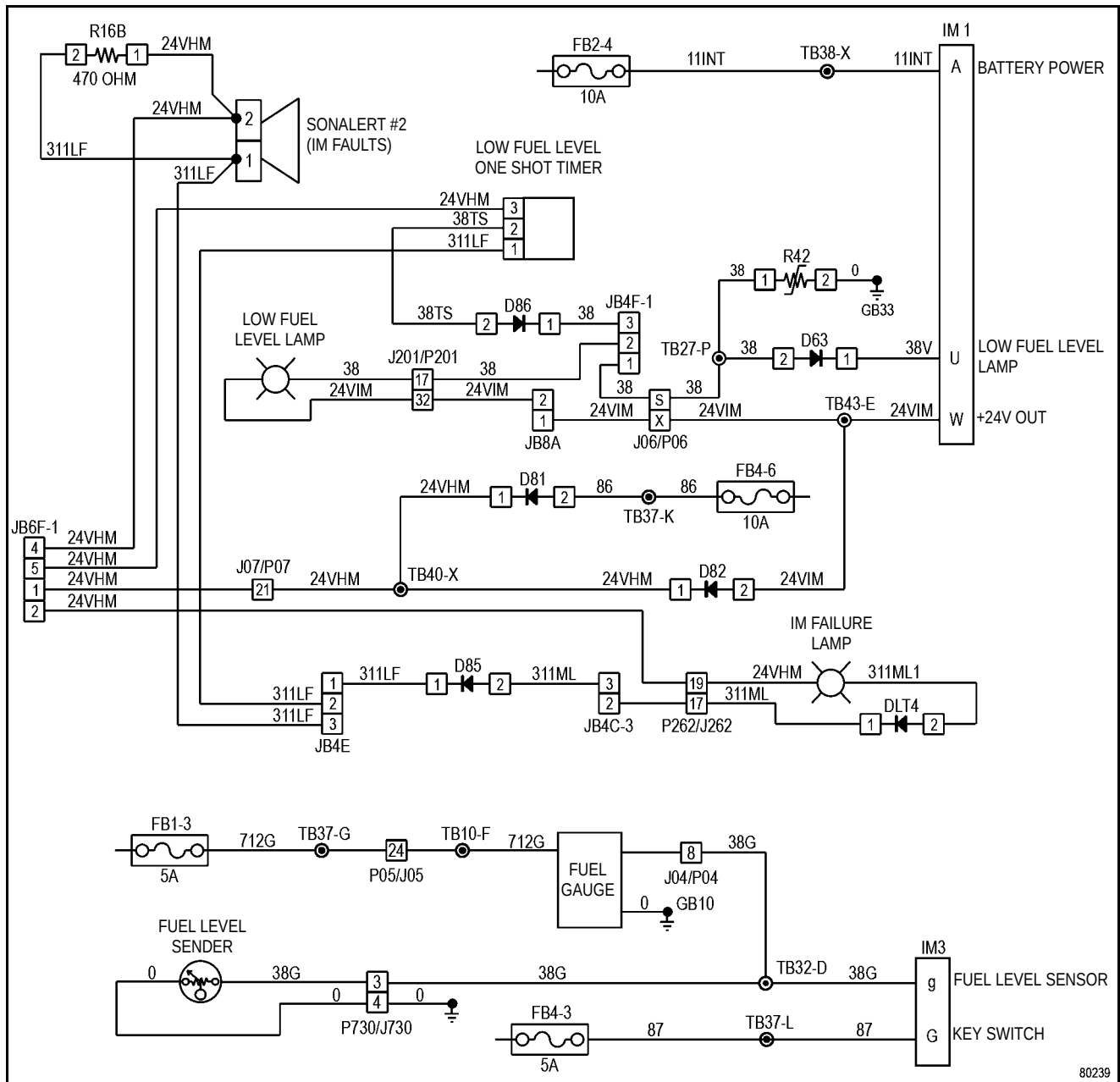
Operator Action	Go To Shop Now
Fault Code	A139, A310
Description	A139 is a low fuel level warning for storage by KOMTRAX Plus and for use by A310. A310 is a low fuel warning driver that activates display elements in response to A139.
Fault Conditions	A139 sets after 15 seconds of fuel level readings below 10% of full if the keyswitch is on, low voltage due to cranking is not sensed, and the fuel level sensor is not high (false indication of no fuel). A139 resets after 1 minute if fuel level recovers to 15% of full. A310 sets if A139 comes on or the fuel sensor is low or high. It flashes twice, then stays on for 15 minutes.
Operator Alerting System Response	Fuel Gauge Amber "Low Fuel Level" Lamp flashes twice then stays on for 15 minutes Buzzer makes two bursts of sound then stays off for 15 minutes Amber "IM Maintenance" Lamp flashes with the buzzer Left Lamp Module Row 5, Column 2
Resulting Problem(s)	Possible running out of fuel on haul road. Possible engine damage.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. The primary correction for this fault is to add fuel to the tank. 2. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 3. If there are no active parameter fault codes, then, based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 4. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Fuel Level Sensor (IM3 g)	2.8 to 6.2 Volts: Good Readings at 24 Volt Battery (varies with battery voltage) < 2.8 or > 6.2 Volts: Defective Sensor or Circuit at 24 Volt Battery (varies with battery voltage) Fault(s): A105, A158
Battery Voltage, 24 Volt (IM1 A)	> 18 Volt: Good Reading < 18 Volt: Low battery voltage (while cranking) blocks this fault code. Fault(s): A153, A154, A155
Keyswitch (IM3 G)	0: keyswitch is off and the truck is not moving and the engine is not running 1: keyswitch is on or the truck is moving (GE power not yet off) Fault(s): A240
Engine Speed [RPM] (CAN/J1939) (IM1 q,r,s)	Less than 600 RPM: combined with low voltage means cranking is in process and fault is blocked Greater than 300 RPM for 4 seconds: engine is running Fault(s): A184

Low Fuel Lamp (IM1 U)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A362
----------------------------------	---

Related circuit diagram



Fault Code A145: Hydraulic temperature sensors cause advance of engine rpm to advance level 2 for cooling of hydraulic oil

Operator Action	None
Fault Code	A145
Description	Hydraulic temperature sensors cause advance of engine rpm to advance level 2 for cooling of hydraulic oil.
Fault Conditions	Sets (advances engine rpm) after 5 seconds at 105° C after filtering out data from sensors failed high. Resets (returns engine to normal) after 5 seconds at 100° C.
Operator Alerting System Response	None. This event is recorded by KOMTRAX Plus only.
Resulting Problem(s)	If this control is non-functional, excessive oil temperatures can cause oil and equipment damage.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 2. If there are no active parameter fault codes, then, based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Brake Cooling RPM Advance 1 (IM1 L)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected. No problem Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot 0: brake cooling rpm advance to advance idle 1: brake cooling rpm advance to normal idle Fault(s): A355
Hydraulic Oil Temperature - Left Rear (IM3 i)	0.016 Volt to 4.89 Volt: Good Readings < 0.016 Volt or > 4.89 Volt: Defective Sensor or Circuit Fault(s): A166, A170
Hydraulic Oil Temperature - Right Rear (IM3 m)	0.016 Volt to 4.89 Volt: Good Readings < 0.016 Volt or > 4.89 Volt: Defective Sensor or Circuit Fault(s): A167, A171
Hydraulic Oil Temperature - Left Front (IM3 t)	0.016 Volt to 4.89 Volt: Good Readings < 0.016 Volt or > 4.89 Volt: Defective Sensor or Circuit Fault(s): A168, A172
Hydraulic Oil Temperature - Right Front (IM3 r)	0.016 Volt to 4.89 Volt: Good Readings < 0.016 Volt or > 4.89 Volt: Defective Sensor or Circuit Fault(s): A169, A173

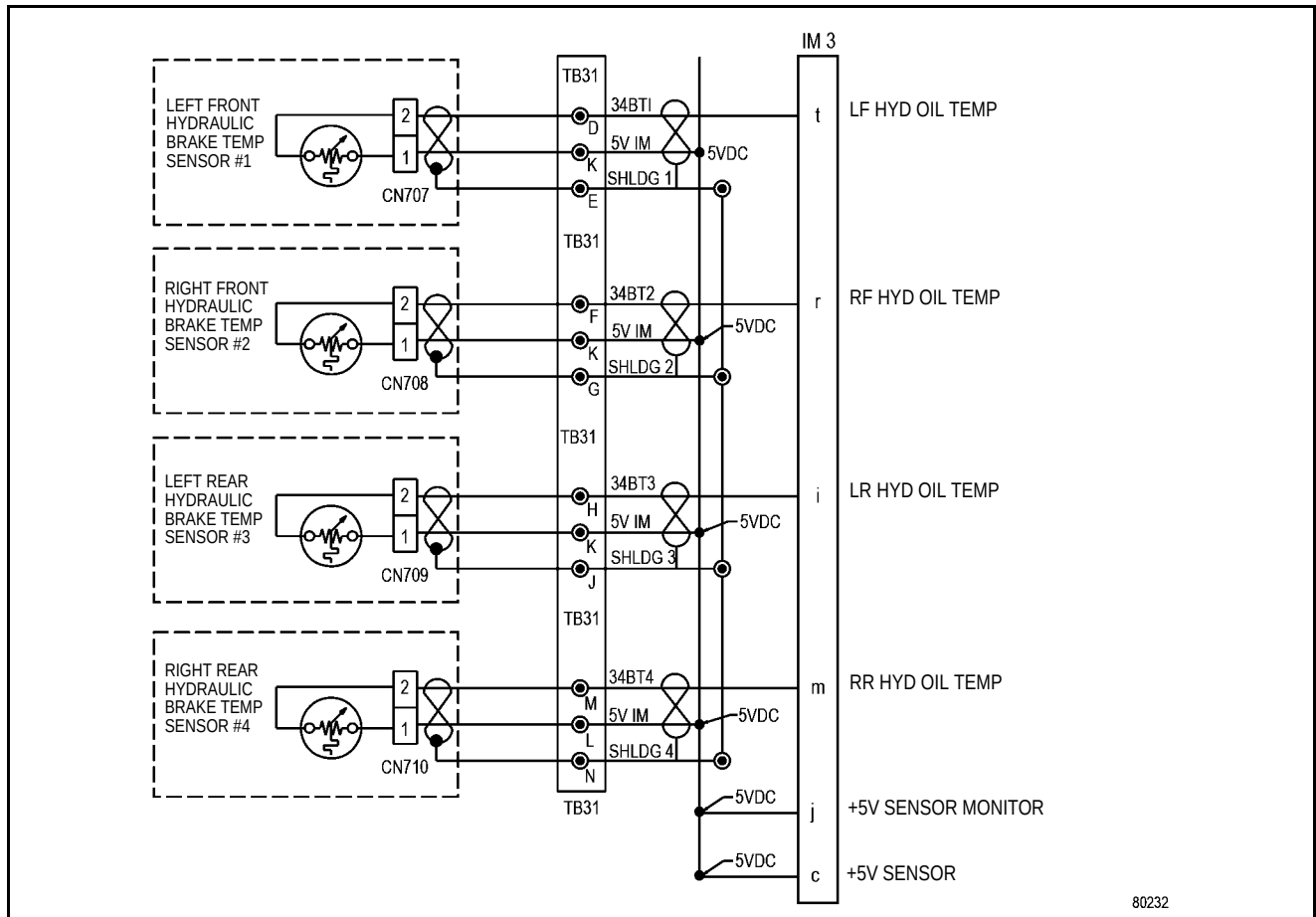
Fault Code A146: Hydraulic temperature sensors cause posting of this fault

Operator Action	None
Fault Code	A146
Description	Hydraulic temperature sensors cause posting of this fault.
Fault Conditions	Sets after 5 seconds at 120° C after filtering out data from sensors failed high. This fault is stored in KOMTRAX Plus only and does not affect engine rpm because it is already at maximum (level 2). Resets after 5 seconds at 108° C.
Operator Alerting System Response	None. This event is recorded by KOMTRAX Plus only.
Resulting Problem(s)	If this control is non-functional, excessive oil temperatures can cause oil and equipment damage.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 2. If there are no active parameter fault codes, then, based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Hydraulic Oil Temperature - Left Rear (IM3 i)	0.016 Volt to 4.89 Volt: Good Readings < 0.016 Volt or > 4.89 Volt: Defective Sensor or Circuit Fault(s): A166, A170
Hydraulic Oil Temperature - Right Rear (IM3 m)	0.016 Volt to 4.89 Volt: Good Readings < 0.016 Volt or > 4.89 Volt: Defective Sensor or Circuit Fault(s): A167, A171
Hydraulic Oil Temperature - Left Front (IM3 t)	0.016 Volt to 4.89 Volt: Good Readings < 0.016 Volt or > 4.89 Volt: Defective Sensor or Circuit Fault(s): A168, A172
Hydraulic Oil Temperature - Right Front (IM3 r)	0.016 Volt to 4.89 Volt: Good Readings < 0.016 Volt or > 4.89 Volt: Defective Sensor or Circuit Fault(s): A169, A173

Related circuit diagram



NOTES

SECTION E5

TROUBLESHOOTING FAULT CODES (A152-A216)

INDEX

Fault Code A152: Starter failure	E5-2
Fault Code A153: Battery voltage is low with truck in operation.	E5-4
Fault Code A154: Battery charging voltage is excessive	E5-6
Fault Code A155: Battery charging voltage is low	E5-8
Fault Code A158: Fuel level sensor is open or shorted high, indicating a false low fuel level . .	E5-10
Fault Code A166: Left rear hydraulic oil temperature sensor is low.	E5-12
Fault Code A167: Right rear hydraulic oil temperature sensor is low.	E5-14
Fault Code A168: Left front hydraulic oil temperature sensor is low	E5-16
Fault Code A169: Right front hydraulic oil temperature sensor is low	E5-18
Fault Code A170: Left rear hydraulic oil temperature sensor is high	E5-20
Fault Code A171: Right rear hydraulic oil temperature sensor is high	E5-21
Fault Code A172: Left front hydraulic oil temperature sensor is high.	E5-22
Fault Code A173: Right front hydraulic oil temperature sensor is high	E5-23
Fault Code A184: The J1939 data link is not connected	E5-24
Fault Code A190: Auto lube control has detected an incomplete lube cycle	E5-26
Fault Code A194: Left front hydraulic oil temperature is high.	E5-28
Fault Code A195: Right front hydraulic oil temperature is high	E5-30
Fault Code A196: Left rear hydraulic oil temperature is high	E5-32
Fault Code A197: Right rear hydraulic oil temperature is high.	E5-34
Fault Code A198: Hoist pressure 1 sensor is high.	E5-36
Fault Code A199: Hoist pressure 2 sensor is high.	E5-37
Fault Code A200: Steering pressure sensor is high.	E5-38
Fault Code A201: Brake pressure sensor is high.	E5-39
Fault Code A202: Hoist pressure 1 sensor is low.	E5-40
Fault Code A203: Hoist pressure 2 sensor is low.	E5-41
Fault Code A204: Steering pressure sensor is low.	E5-42
Fault Code A205: Brake pressure sensor is low	E5-43
Fault Code A206: Ambient temperature sensor is high	E5-44
Fault Code A207: Ambient temperature sensor is low	E5-45
Fault Code A212: Bad truck speed signal	E5-46
Fault Code A213: Parking brake should have applied but is detected as not having applied . .	E5-48
Fault Code A214: Parking brake should have released but is detected as not having released	E5-52
Fault Code A215: Brake auto apply relay circuit is defective	E5-56
Fault Code A216: An open or short to ground has been detected in the parking brake command valve circuit	E5-58

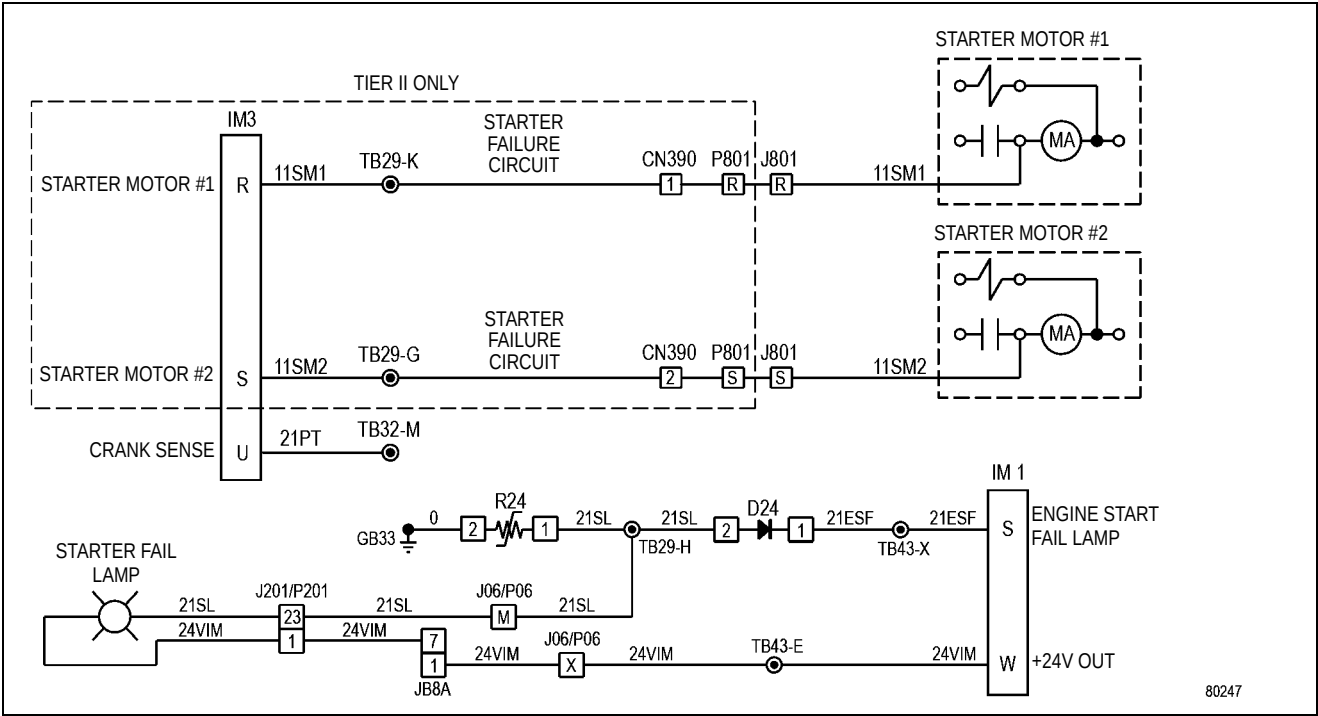
Fault Code A152: Starter failure

Operator Action	None
Fault Code	A152
Description	Starter has failed.
Fault Conditions	Sets if either starter motor is not energized for 2 seconds when cranking is attempted. Resets only when steering bleed occurs.
Operator Alerting System Response	Amber "Engine Start Fail" Lamp No Buzzer Left Lamp Module Row 1, Column 4
Resulting Problem(s)	One good starter may be destroyed or engine may not start at all.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 2. If there are no active parameter fault codes, then, based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Starter Motor 1 Energized (IM3 R)	0: starter motor 1 not energized 1: starter motor 1 energized Fault(s): A275
Starter Motor 2 Energized (IM3 S)	0: starter motor 2 not energized 1: starter motor 2 energized Fault(s): A275
Crank Sense (IM3 U)	0: no cranking in process 1: cranking in process
Engine Start Fail Lamp (IM1 S)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A360

Related circuit diagram



Fault Code A153: Battery voltage is low with truck in operation

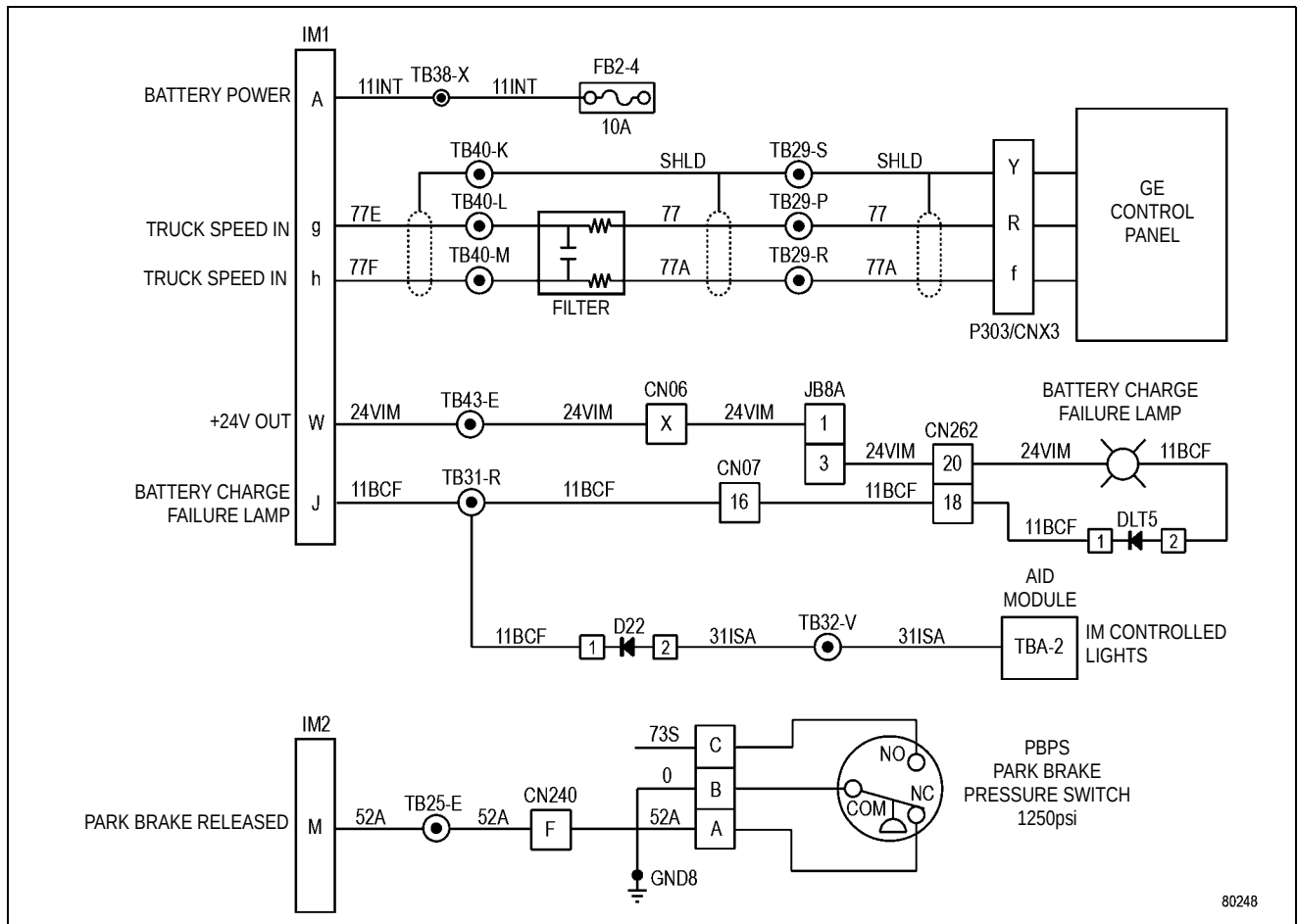
Operator Action	Stop; Park
Fault Code	A153
Description	Battery voltage is low with the truck in operation.
Fault Conditions	Sets if battery voltage with the engine running or truck moving is below 23 volts for 5 seconds. Truck moving sets if truck speed is above 0 or if a bad speed signal fault is active and the parking brake is not set. Resets if voltage recovers to 25.5 volts for 5 seconds or when steering bleed begins at shutdown.
Operator Alerting System Response	Red "Battery Charge System Failure" Lamp Buzzer Right Lamp Module Row 5, Column1
Resulting Problem(s)	Voltage may continue to drop and cause improper operation of the many electrical controls on the truck.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. The primary correction for this fault is to charge the batteries. 2. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s) 3. If there are no active parameter fault codes, then, based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 4. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Battery Voltage, 24 Volt (IM1 A)	< 23.0 volts: Too low to continue operation. > 23.0 volts: OK to continue operation. Fault(s): A250, A154, A155
Engine Speed [RPM] (CAN/J1939) (IM1 q,r,s)	Less than 300 RPM: engine is not running Greater than 300 RPM for 4 seconds: engine is running Fault(s): A184
Truck Speed [kph] (IM1 g,h)	0: truck is not moving Greater than 0: truck is moving Fault(s): A212
Park Brake Released (IM2 M)	0: parking brake pressure switch indicating low pressure and an applied parking brake 1: parking brake pressure switch indicating higher pressure and an unapplied parking brake Fault(s): A213

Battery Charge System Failure Lamp (IM1 J)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A353
--	--

Related circuit diagram

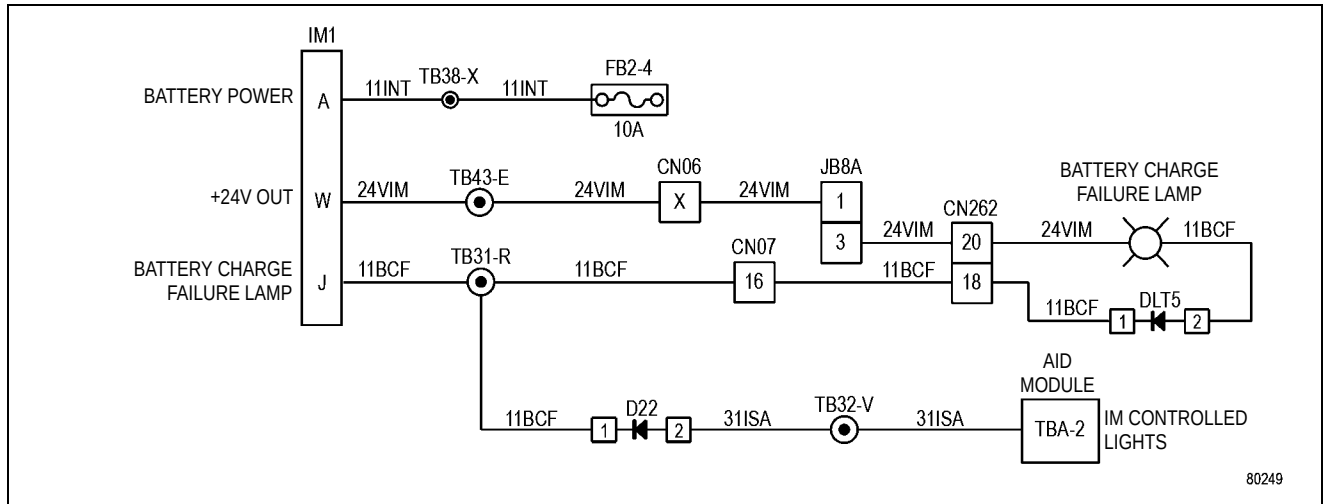


Fault Code A154: Battery charging voltage is excessive

Operator Action	Stop; Park; Power Down
Fault Code	A154
Description	Battery charging voltage is excessive.
Fault Conditions	Sets if above 32 volts for 5 seconds. Resets if below 27.5 volts for 5 seconds.
Operator Alerting System Response	Red "Battery Charge System Failure" Lamp Buzzer Right Lamp Module Row 5, Column1
Resulting Problem(s)	Sustained excessive charging voltage may burn out electrical and electronic components.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table	
1. The primary correction for this fault is to check / replace the battery charger. 2. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 3. If there are no active parameter fault codes, then, based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 4. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Battery Voltage, 24 Volt (IM1 A)	< 23.0 volts: Too low to continue operation. > 23.0 volts: OK to continue operation. Fault(s): A153, A155
Battery Charge System Failure Lamp (IM1 J)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A353

Related circuit diagram

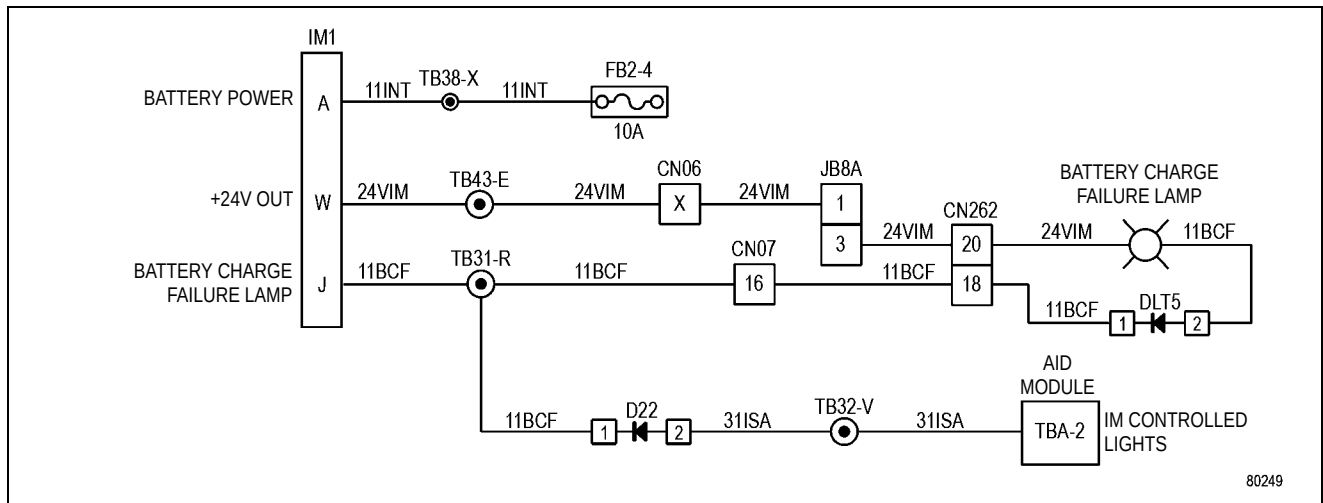


Fault Code A155: Battery charging voltage is low

Operator Action	Go To Shop Now
Fault Code	A155
Description	Battery charging voltage is low.
Fault Conditions	Sets if below 24.5 volts for 5 seconds with engine above 1400 rpm. Resets if above 26.0 volts for 5 seconds or at steering bleed.
Operator Alerting System Response	Red "Battery Charge System Failure" Lamp Buzzer Right Lamp Module Row 5, Column1
Resulting Problem(s)	If defective battery charging alternators are not replaced, batteries may become discharged and electrical and electronic equipment will stop working.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table	
1. The primary correction for this fault is to check / replace the battery charger. 2. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 3. If there are no active parameter fault codes, then, based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 4. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Battery Voltage, 24 Volt (IM1 A)	> 32.0 volts: Too high to continue operation. < 32.0 volts: OK to continue operation. Fault(s): A153, A154
Battery Charge System Failure Lamp (IM1 J)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A353

Related circuit diagram



Fault Code A158: Fuel level sensor is open or shorted high, indicating a false low fuel level

Operator Action	Go To Shop Now
Fault Code	A158
Description	Fuel level sensor is open or shorted high, indicating a false low fuel level.
Fault Conditions	Sets if fuel level indication is at -18.5% for 3 seconds, and battery voltage does not drop below 18, with engine speed below 600 rpm. Resets if fuel level indication is above -12.5% for 3 seconds.
Operator Alerting System Response	Amber "Low Fuel Level" Lamp Fuel Gauge Buzzer producing bursts of sound every 15 minutes Left Lamp Module Row 5, Column 2
Resulting Problem(s)	Fuel readings will show an empty tank, whether there is fuel in the tank or not.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. The primary correction for this fault is to correct or replace the fuel level sensor and wiring. 2. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 3. If there are no active parameter fault codes, then, based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 4. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Fuel Level Sensor (IM3 g)	2.8 to 6.2 Volts: Good Readings at 24 Volt Battery (varies with battery voltage) < 2.8 or > 6.2 Volts: Defective Sensor or Circuit at 24 Volt Battery (varies with battery voltage) Fault(s): A158
Battery Voltage, 24 Volt (IM1 A)	> 18 Volt: Good Reading < 18 Volt: Low battery voltage (while cranking) blocks this fault code. Fault(s): A153, A154, A155, A250
Keyswitch (IM3 G)	0: keyswitch is off and the truck is not moving and the engine is not running 1: keyswitch is on or the truck is moving (GE power not yet off) Fault(s): A240
Engine Speed [RPM] (CAN/J1939) (IM1 q,r,s)	Less than 600 RPM: combined with low voltage means cranking is in process and fault is blocked Greater than 600 RPM: engine is running normally Fault(s): A184

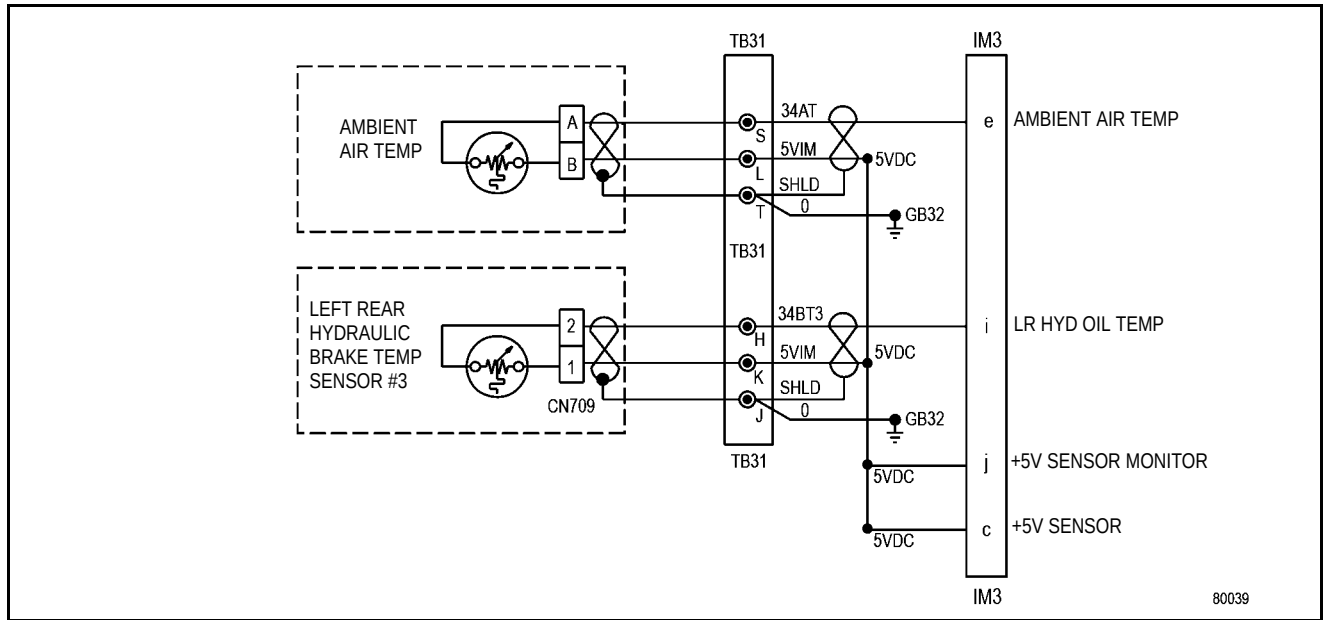
Wiring diagram for the fuel system of a 1977-1978 Chevrolet. The diagram shows the battery power (IM1) connected to a 10A fuse (FB2-4) and a 5A fuse (FB1-3). The 10A fuse feeds the fuel gauge (J04/P04) and the fuel level sensor (IM3). The 5A fuse feeds the fuel level sender (P730/J730) and the key switch (IM3). The fuel gauge is grounded to the chassis (GB10). The fuel level sender is grounded to the chassis (P730/J730). The fuel level sensor is grounded to the chassis (TB32-D). The key switch is grounded to the chassis (TB37-L).

Fault Code A166: Left rear hydraulic oil temperature sensor is low

Operator Action	None
Fault Code	A166
Description	Left rear hydraulic oil temperature sensor is low.
Fault Conditions	Sets if temperature reading drops to -51°C (.016 Volts) for 3 seconds or remains 10°C below ambient temperature for 15 minutes after the engine runs. Resets if temperature reading rises to -46°C (.032 Volt) for 3 seconds and recovers to within 10°C of ambient temperature.
Operator Alerting System Response	Red "Hydraulic Oil Temperature High" Lamp Hydraulic Oil Temperature Gauge Buzzer Left Lamp Module Row 1, Column 1
Resulting Problem(s)	Loss of monitoring of the left rear brake temperature begins a compromise of the brake system temperature monitoring that can only worsen if other sensors are lost.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table	
1. The primary correction for this fault is to correct any external wiring or replace the sensor. 2. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 3. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 4. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Hydraulic Oil Temperature - Left Rear (IM3 i)	0.016 Volt to 4.89 Volt: Good Readings <0.016 Volt or >4.89 Volt: Defective Sensor or Circuit Fault(s): A170
Ambient Temperature (IM3 e)	0.016 Volt to 4.89 Volt: Good Readings <0.016 Volt or >4.89 Volt: Defective Sensor or Circuit Fault(s): A206, A207
Engine Speed [RPM] (CAN/J1939) (IM1 q,r,s)	Less than 300 RPM: engine is not running Greater than 300 RPM for 4 seconds: engine is running Fault(s): A184

Related circuit diagram



Fault Code A167: Right rear hydraulic oil temperature sensor is low

Operator Action	None
Fault Code	A167
Description	Right rear hydraulic oil temperature sensor is low.
Fault Conditions	Sets if temperature reading drops to -51°C (.016 Volts) for 3 seconds or remains 10°C below ambient temperature for 15 minutes after the engine runs. Resets if temperature reading rises to -46°C (.032 Volt) for 3 seconds and recovers to within 10°C of ambient temperature.
Operator Alerting System Response	Red "Hydraulic Oil Temperature High" Lamp Hydraulic Oil Temperature Gauge Buzzer Left Lamp Module Row 1, Column 1
Resulting Problem(s)	Loss of monitoring of the right rear brake temperature begins a compromise of the brake system temperature monitoring that can only worsen if other sensors are lost.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table	
1. The primary correction for this fault is to correct any external wiring or replace the sensor. 2. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 3. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 4. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Hydraulic Oil Temperature - Right Rear (IM3 m)	0.016 Volt to 4.89 Volt: Good Readings <0.016 Volt or >4.89 Volt: Defective Sensor or Circuit Fault(s): A171
Ambient Temperature (IM3 e)	0.016 Volt to 4.89 Volt: Good Readings <0.016 Volt or >4.89 Volt: Defective Sensor or Circuit Fault(s): A206, A207
Engine Speed [RPM] (CAN/J1939) (IM1 q,r,s)	Less than 300 RPM: engine is not running Greater than 300 RPM for 4 seconds: engine is running Fault(s): A184

The diagram illustrates the wiring for two temperature sensors connected to the IM3 module. The sensors are represented by dashed boxes on the left:

- AMBIENT AIR TEMP:** A sensor with terminals A and B. Terminal A is connected to TB31 pin 34AT (S). Terminal B is connected to TB31 pin 5VIM (L).
- RIGHT REAR HYDRAULIC BRAKE TEMP SENSOR #4:** A sensor with terminals 1 and 2. Terminal 1 is connected to TB31 pin 5VIM (L). Terminal 2 is connected to TB31 pin 34BT4 (M).

The TB31 connector has the following pins and connections:

- Pin 34AT (S):** Connected to IM3 pin e (AMBIENT AIR TEMP).
- Pin 5VIM (L):** Connected to IM3 pin m (RR HYD OIL TEMP).
- Pin 34BT4 (M):** Connected to IM3 pin j (+5V SENSOR MONITOR).
- Pin 5VIM (L):** Connected to IM3 pin c (+5V SENSOR).

Power and ground connections are shown on the right side of the IM3 module:

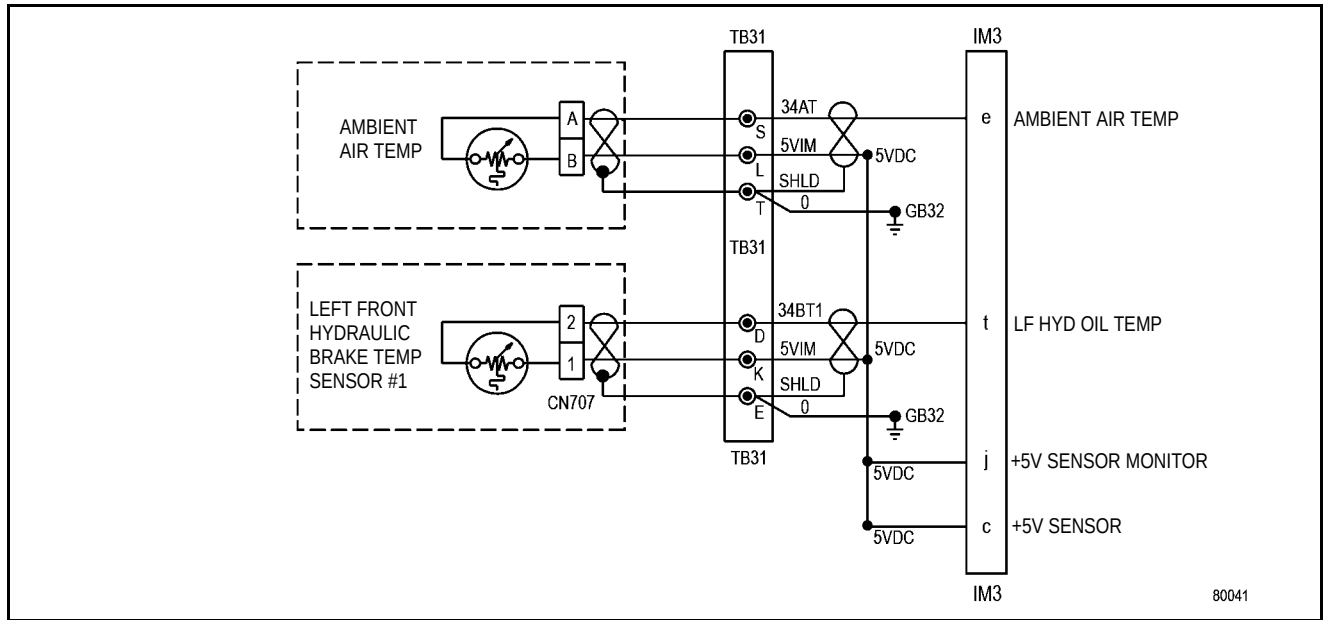
- 5VDC:** Two 5VDC sources are connected to the IM3 module. One is connected to TB31 pin SHLD (0) and the other to TB31 pin SHLD (0).
- GB32:** Ground connections are shown for TB31 pin SHLD (0) and TB31 pin SHLD (0).

Fault Code A168: Left front hydraulic oil temperature sensor is low

Operator Action	None
Fault Code	A168
Description	Left front hydraulic oil temperature sensor is low.
Fault Conditions	Sets if temperature reading drops to -51°C (.016 Volts) for 3 seconds or remains 10°C below ambient temperature for 15 minutes after the engine runs. Resets if temperature reading rises to -46°C (.032 Volt) for 3 seconds and recovers to within 10°C of ambient temperature.
Operator Alerting System Response	Red "Hydraulic Oil Temperature High" Lamp Hydraulic Oil Temperature Gauge Buzzer Left Lamp Module Row 1, Column 1
Resulting Problem(s)	Loss of monitoring of the left front brake temperature begins a compromise of the brake system temperature monitoring that can only worsen if other sensors are lost.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table	
1. The primary correction for this fault is to correct any external wiring or replace the sensor. 2. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 3. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 4. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Hydraulic Oil Temperature - Left Front (IM3 t)	0.016 Volt to 4.89 Volt: Good Readings <0.016 Volt or >4.89 Volt: Defective Sensor or Circuit Fault(s): A172
Ambient Temperature (IM3 e)	0.016 Volt to 4.89 Volt: Good Readings <0.016 Volt or >4.89 Volt: Defective Sensor or Circuit Fault(s): A206, A207
Engine Speed [RPM] (CAN/J1939) (IM1 q,r,s)	Less than 300 RPM: engine is not running Greater than 300 RPM for 4 seconds: engine is running Fault(s): A184

Related circuit diagram

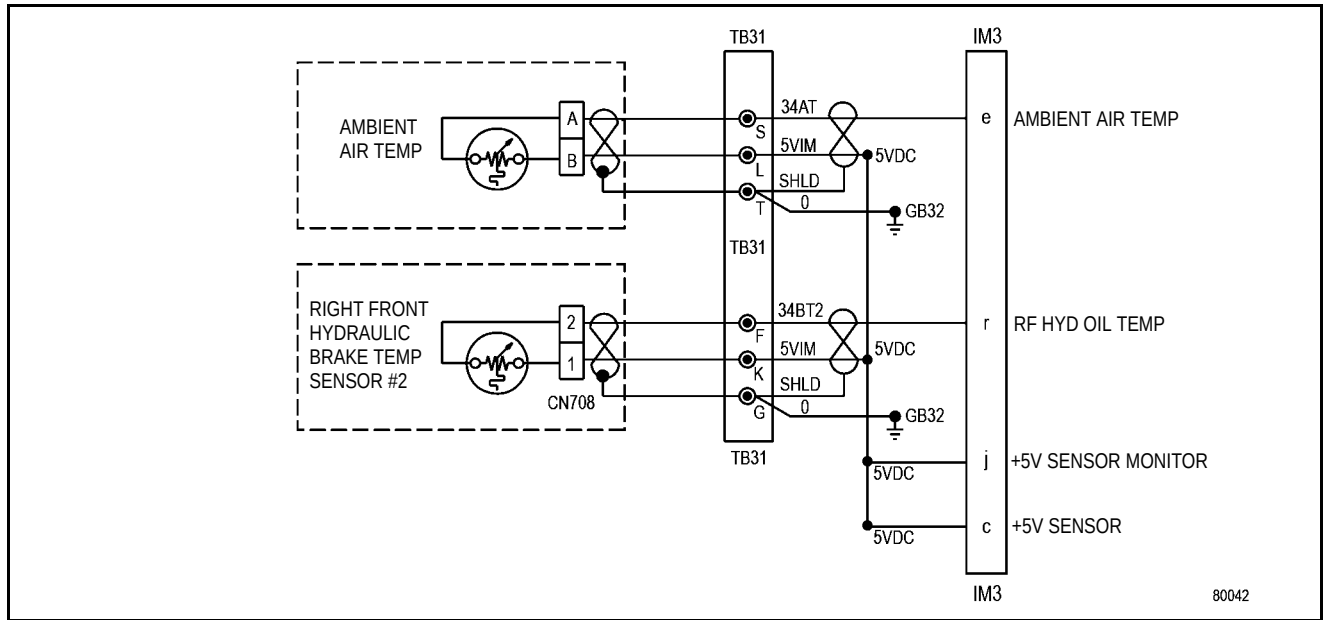


Fault Code A169: Right front hydraulic oil temperature sensor is low

Operator Action	None
Fault Code	A169
Description	Right front hydraulic oil temperature sensor is low.
Fault Conditions	Sets if temperature reading drops to -51°C (.016 Volts) for 3 seconds or remains 10°C below ambient temperature for 15 minutes after the engine runs. Resets if temperature reading rises to -46°C (.032 Volt) for 3 seconds and recovers to within 10°C of ambient temperature.
Operator Alerting System Response	Red "Hydraulic Oil Temperature High" Lamp Hydraulic Oil Temperature Gauge Buzzer Left Lamp Module Row 1, Column 1
Resulting Problem(s)	Loss of monitoring of the right front brake temperature begins a compromise of the brake system temperature monitoring that can only worsen if other sensors are lost.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table	
1. The primary correction for this fault is to correct any external wiring or replace the sensor. 2. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 3. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 4. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Hydraulic Oil Temperature - Right Front (IM3 r)	0.016 Volt to 4.89 Volt: Good Readings <0.016 Volt or >4.89 Volt: Defective Sensor or Circuit Fault(s): A173
Ambient Temperature (IM3 e)	0.016 Volt to 4.89 Volt: Good Readings <0.016 Volt or >4.89 Volt: Defective Sensor or Circuit Fault(s): A206, A207
Engine Speed [RPM] (CAN/J1939) (IM1 q,r,s)	Less than 300 RPM: engine is not running Greater than 300 RPM for 4 seconds: engine is running Fault(s): A184

Related circuit diagram



80042

Fault Code A170: Left rear hydraulic oil temperature sensor is high

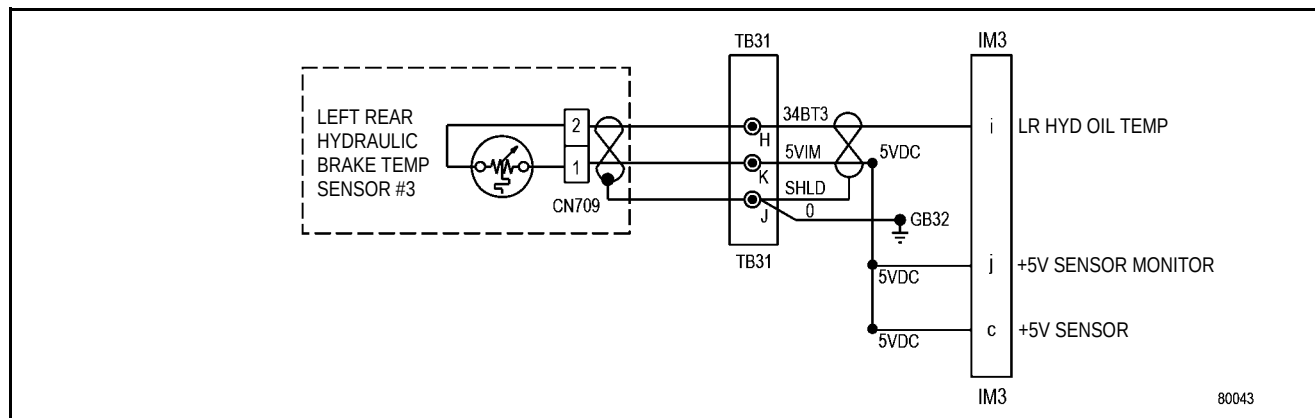
Operator Action	None
Fault Code	A170
Description	Left rear hydraulic oil temperature sensor is high.
Fault Conditions	Sets if temperature reading rises to 211°C (4.89 Volts) for 3 seconds. Resets if temperature reading drops to 188°C (4.77 Volt) for 3 seconds.
Operator Alerting System Response	Red "Hydraulic Oil Temperature High" Lamp Hydraulic Oil Temperature Gauge Buzzer Left Lamp Module Row 1, Column 1
Resulting Problem(s)	Loss of monitoring of the left rear brake temperature begins a compromise of the brake system temperature monitoring that can only worsen if other sensors are lost.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. The primary correction for this fault is to correct any external wiring or replace the sensor.
2. This fault's logic contains only one parameter.

Parameter	Expected State and/or Related Fault(s)
Hydraulic Oil Temperature - Left Rear (IM3 i)	0.016 Volt to 4.89 Volt: Good Readings <0.016 Volt or >4.89 Volt: Defective Sensor or Circuit Fault(s): A166

Related circuit diagram



Fault Code A171: Right rear hydraulic oil temperature sensor is high

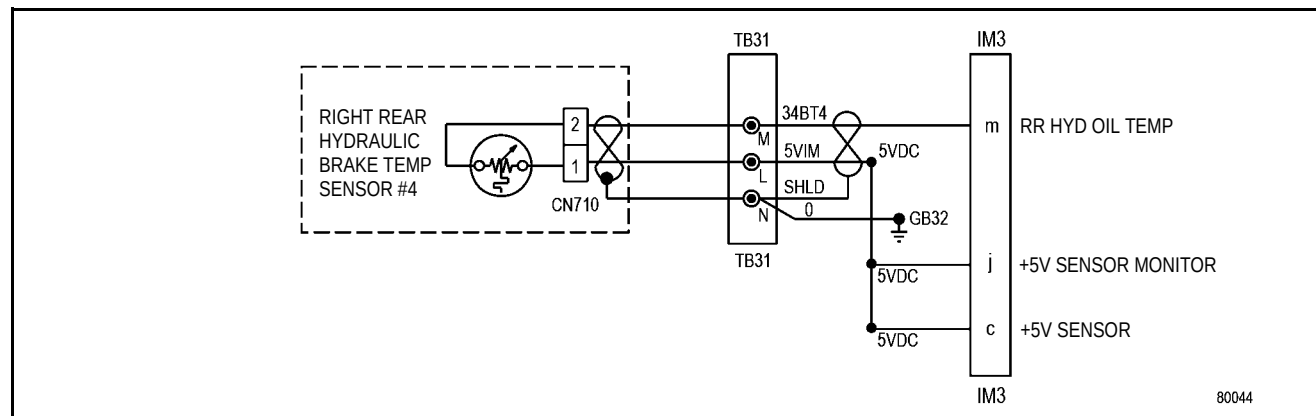
Operator Action	None
Fault Code	A171
Description	Right rear hydraulic oil temperature sensor is high.
Fault Conditions	Sets if temperature reading rises to 211°C (4.89 Volts) for 3 seconds. Resets if temperature reading drops to 188°C (4.77 Volt) for 3 seconds.
Operator Alerting System Response	Red "Hydraulic Oil Temperature High" Lamp Hydraulic Oil Temperature Gauge Buzzer Left Lamp Module Row 1, Column 1
Resulting Problem(s)	Loss of monitoring of the right rear brake temperature begins a compromise of the brake system temperature monitoring that can only worsen if other sensors are lost.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. The primary correction for this fault is to correct any external wiring or replace the sensor.
2. This fault's logic contains only one parameter.

Parameter	Expected State and/or Related Fault(s)
Hydraulic Oil Temperature - Right Rear (IM3 m)	0.016 Volt to 4.89 Volt: Good Readings <0.016 Volt or >4.89 Volt: Defective Sensor or Circuit Fault(s): A167

Related circuit diagram



Fault Code A172: Left front hydraulic oil temperature sensor is high

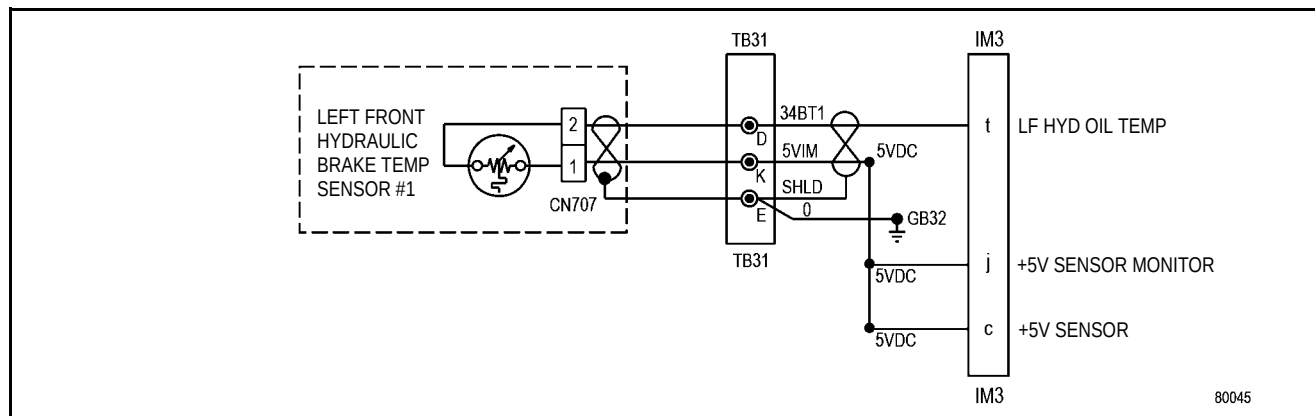
Operator Action	None
Fault Code	A172
Description	Left front hydraulic oil temperature sensor is high.
Fault Conditions	Sets if temperature reading rises to 211°C (4.89 Volts) for 3 seconds. Resets if temperature reading drops to 188°C (4.77 Volt) for 3 seconds.
Operator Alerting System Response	Red "Hydraulic Oil Temperature High" Lamp Hydraulic Oil Temperature Gauge Buzzer Left Lamp Module Row 1, Column 1
Resulting Problem(s)	Loss of monitoring of the left front brake temperature begins a compromise of the brake system temperature monitoring that can only worsen if other sensors are lost.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. The primary correction for this fault is to correct any external wiring or replace the sensor.
2. This fault's logic contains only one parameter.

Parameter	Expected State and/or Related Fault(s)
Hydraulic Oil Temperature - Left Front (IM3 t)	0.016 Volt to 4.89 Volt: Good Readings <0.016 Volt or >4.89 Volt: Defective Sensor or Circuit Fault(s): A168

Related circuit diagram



Fault Code A173: Right front hydraulic oil temperature sensor is high

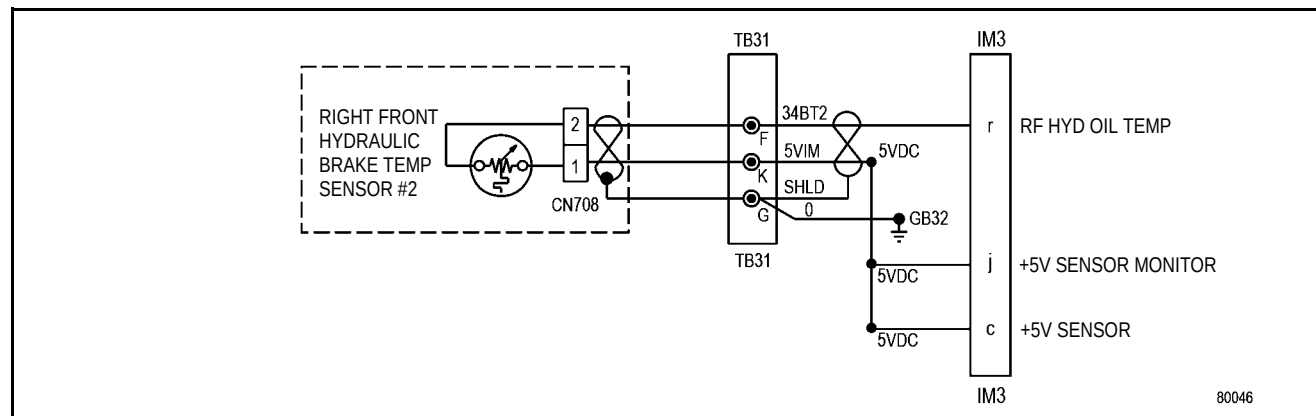
Operator Action	None
Fault Code	A173
Description	Right front hydraulic oil temperature sensor is high.
Fault Conditions	Sets if temperature reading rises to 211°C (4.89 Volts) for 3 seconds. Resets if temperature reading drops to 188°C (4.77 Volt) for 3 seconds.
Operator Alerting System Response	Red "Hydraulic Oil Temperature High" Lamp Hydraulic Oil Temperature Gauge Buzzer Left Lamp Module Row 1, Column 1
Resulting Problem(s)	Loss of monitoring of the right front brake temperature begins a compromise of the brake system temperature monitoring that can only worsen if other sensors are lost.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. The primary correction for this fault is to correct any external wiring or replace the sensor.
2. This fault's logic contains only one parameter.

Parameter	Expected State and/or Related Fault(s)
Hydraulic Oil Temperature - Right Front (IM3 r)	0.016 Volt to 4.89 Volt: Good Readings <0.016 Volt or >4.89 Volt: Defective Sensor or Circuit Fault(s): A169

Related circuit diagram

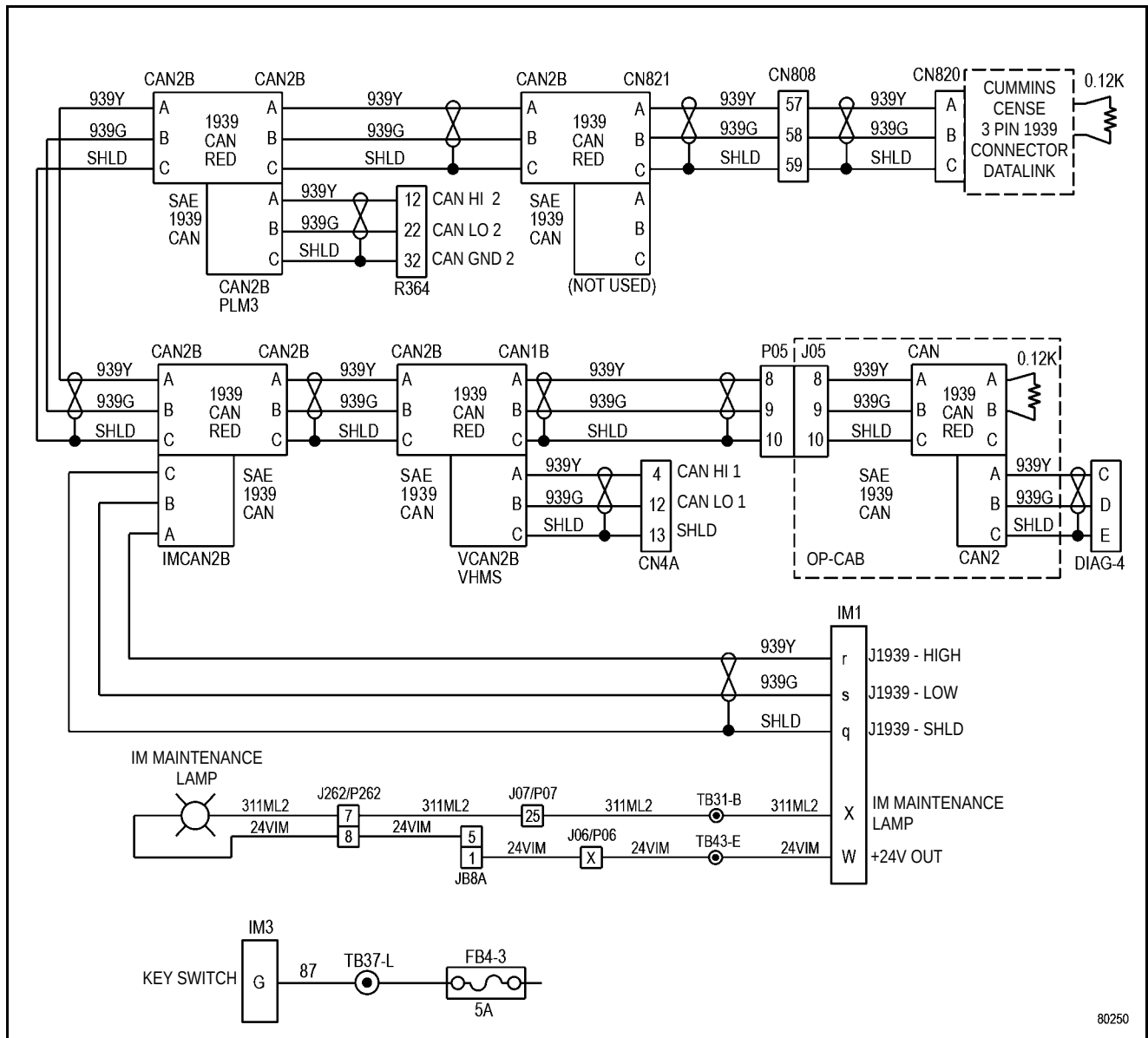


Fault Code A184: J1939 data link is not connected

Operator Action	Stop; Park; Power Down
Fault Code	A184
Description	J1939 data link is not connected.
Fault Conditions	Sets if no data received from the engine for 10 seconds after an initial 11 second delay at key on. Resets immediately when communication is established.
Operator Alerting System Response	Amber "IM Maintenance" Lamp No Buzzer Right Lamp Module Row 5, Column 3
Resulting Problem(s)	Some engine data is compromised.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table	
1. The primary correction for this fault is to correct any J1939 wiring faults. 2. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 3. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 4. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
CAN J1939 Data Link (IM1 q,r,s)	No communication for 10 seconds if keyswitch has been on for 11 seconds: J1939 not connected. Successful read of PGN 61444: J1939 is connected.
Keyswitch (IM3 G)	0: keyswitch is off and the truck is not moving and the engine is not running 1: keyswitch is on or the truck is moving or the engine is running <i>NOTE: For this fault, the keyswitch function is delayed for 11 seconds internally after initial turn on</i> Fault(s): A240
IM Maintenance Lamp (IM1 X)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A363

Related circuit diagram



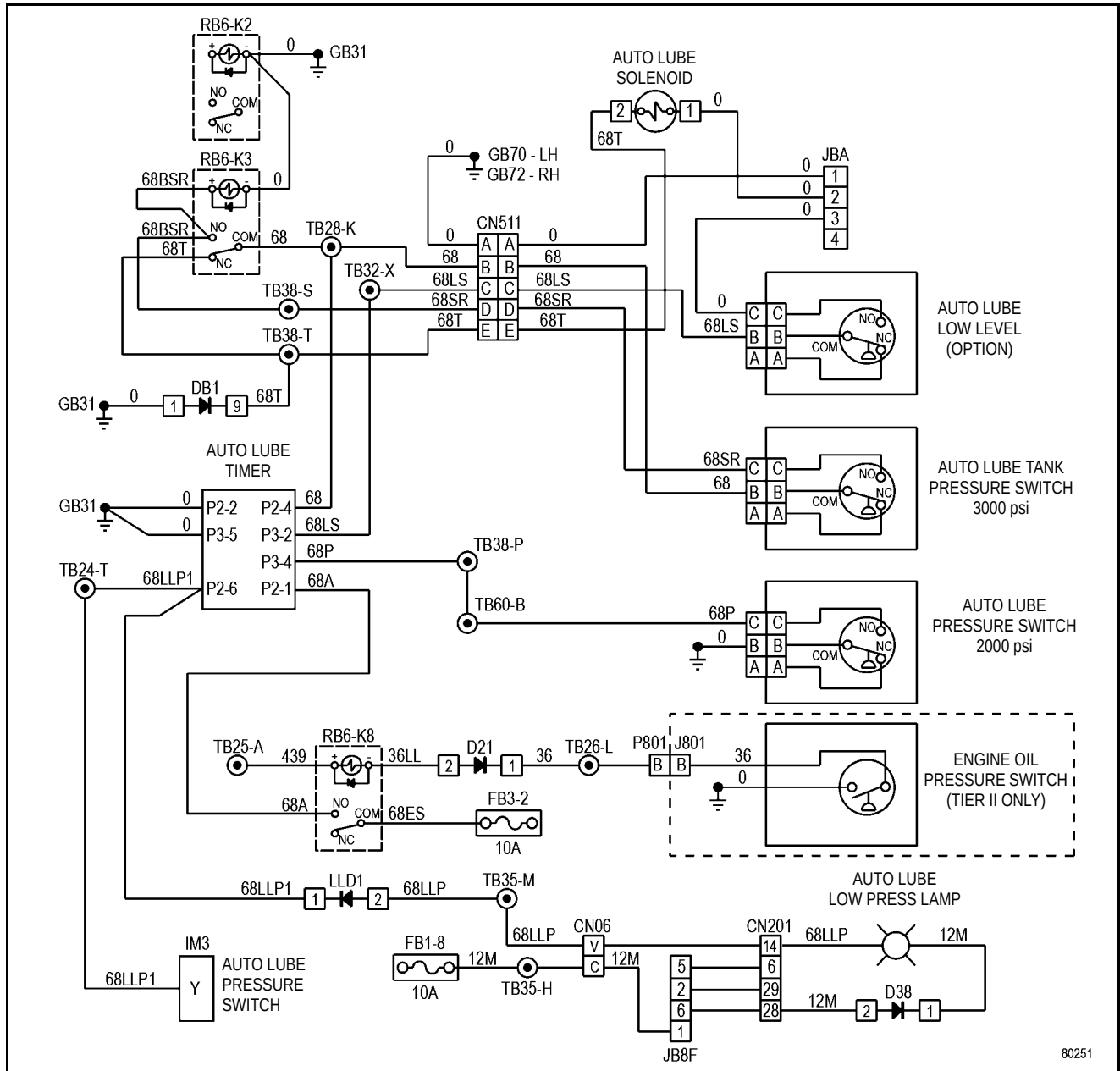
80250

Fault Code A190: Auto lube control has detected an incomplete lube cycle

Operator Action	None
Fault Code	A190
Description	Auto lube control has detected an incomplete lube cycle.
Fault Conditions	Sets if lubricating pressure is low. Lubrication only proceeds when engine is running.
Operator Alerting System Response	Red "Auto Lube Low Pressure" Lamp No Buzzer Left Lamp Module Row 2, Column 2
Resulting Problem(s)	Insufficient lube may result and, if left uncorrected, could lead to equipment damage.
Related Information	

Table	
1. This fault's logic contains no additional parameters. 2. The primary correction for this fault is to add grease or correct circuitry or wiring that is causing the incomplete lube cycle.	
Parameter	Expected State and/or Related Fault(s)
Auto Lube Switch (IM3 Y)	0: Low pressure and incomplete lube cycle 1: Pressure and lube cycling OK

Related circuit diagram

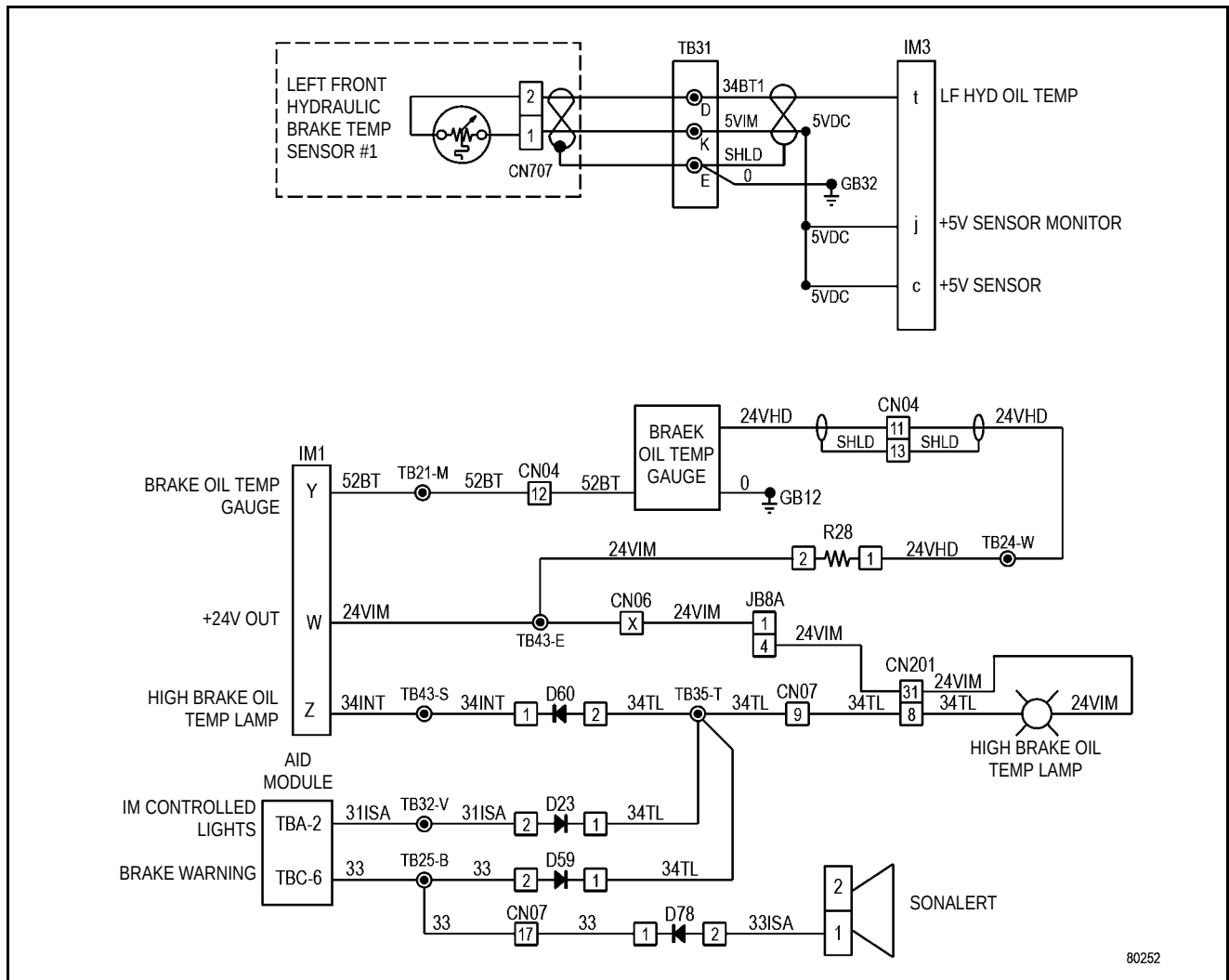


Fault Code A194: Left front hydraulic oil temperature is high

Operator Action	Stop; Park; Run Engine
Fault Code	A194
Description	Left front hydraulic oil temperature is high.
Fault Conditions	Sets at 120°C (4.13 volts) after 5 seconds. Resets at 103°C (3.78 volts) after 5 seconds or at steering bleed.
Operator Alerting System Response	Red "Hydraulic Oil Temperature High" Lamp Hydraulic Oil Temperature Gauge Buzzer Left Lamp Module Row 1, Column 1
Resulting Problem(s)	Excessively hot oil can cause equipment damage and even reduce service brake effectiveness.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table	
1. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 2. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Hydraulic Oil Temperature - Left Front (IM3 t)	0.016 Volt to 4.89 Volt: Good Readings <0.016 Volt or >4.89 Volt: Defective Sensor or Circuit Fault(s): A168, A172
Steering Bleed Valve	0: steering bleed valve is off 1: steering bleed valve is on
High Brake Oil Temperature Lamp (IM1 Z)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A365
Hydraulic Oil Temperature Gauge (IM1 Y)	Status - Open Load: Unexpected: Troubleshoot Status - Shorted to Ground: Unexpected: Troubleshoot Status - Overload: Unexpected: Troubleshoot Approximate Calibration: Start Green Band: 60°C Start Red Band: 120°C Fault(s): A364

Related circuit diagram

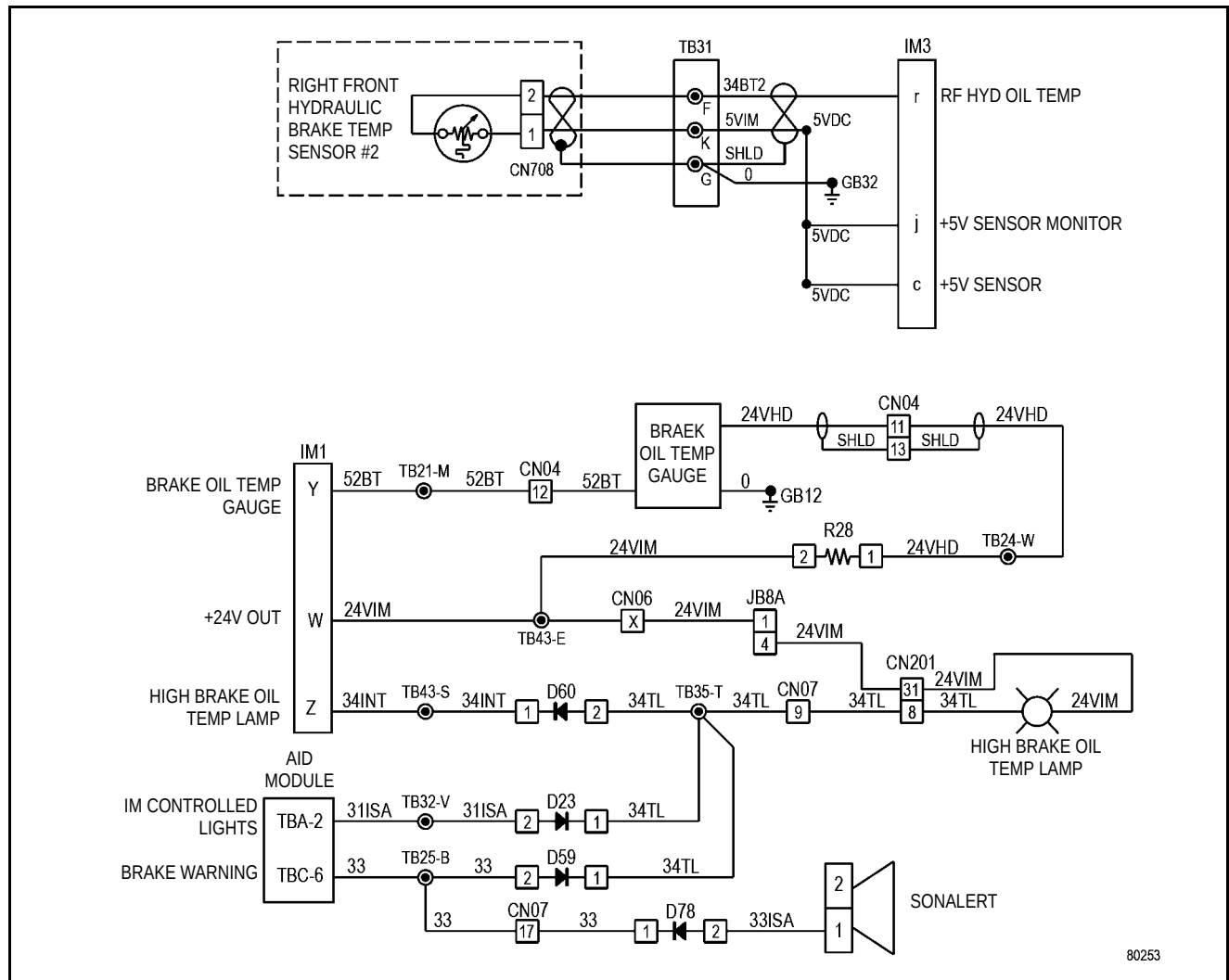


Fault Code A195: Right front hydraulic oil temperature is high

Operator Action	Stop; Park; Run Engine
Fault Code	A195
Description	Right front hydraulic oil temperature is high.
Fault Conditions	Sets at 120°C (4.13 volts) after 5 seconds. Resets at 103°C (3.78 volts) after 5 seconds or at steering bleed.
Operator Alerting System Response	Red "Hydraulic Oil Temperature High" Lamp Hydraulic Oil Temperature Gauge Buzzer Left Lamp Module Row 1, Column 1
Resulting Problem(s)	Excessively hot oil can cause equipment damage and even reduce service brake effectiveness.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table	
1. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 2. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Hydraulic Oil Temperature - Right Front (IM3 r)	0.016 Volt to 4.89 Volt: Good Readings <0.016 Volt or >4.89 Volt: Defective Sensor or Circuit Fault(s): A169, A173
Steering Bleed Valve	0: steering bleed valve is off 1: steering bleed valve is on
High Brake Oil Temperature Lamp (IM1 Z)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A365
Hydraulic Oil Temperature Gauge (IM1 Y)	Status - Open Load: Unexpected: Troubleshoot Status - Shorted to Ground: Unexpected: Troubleshoot Status - Overload: Unexpected: Troubleshoot Approximate Calibration: Start Green Band: 60°C Start Red Band: 120°C Fault(s): A364

Related circuit diagram

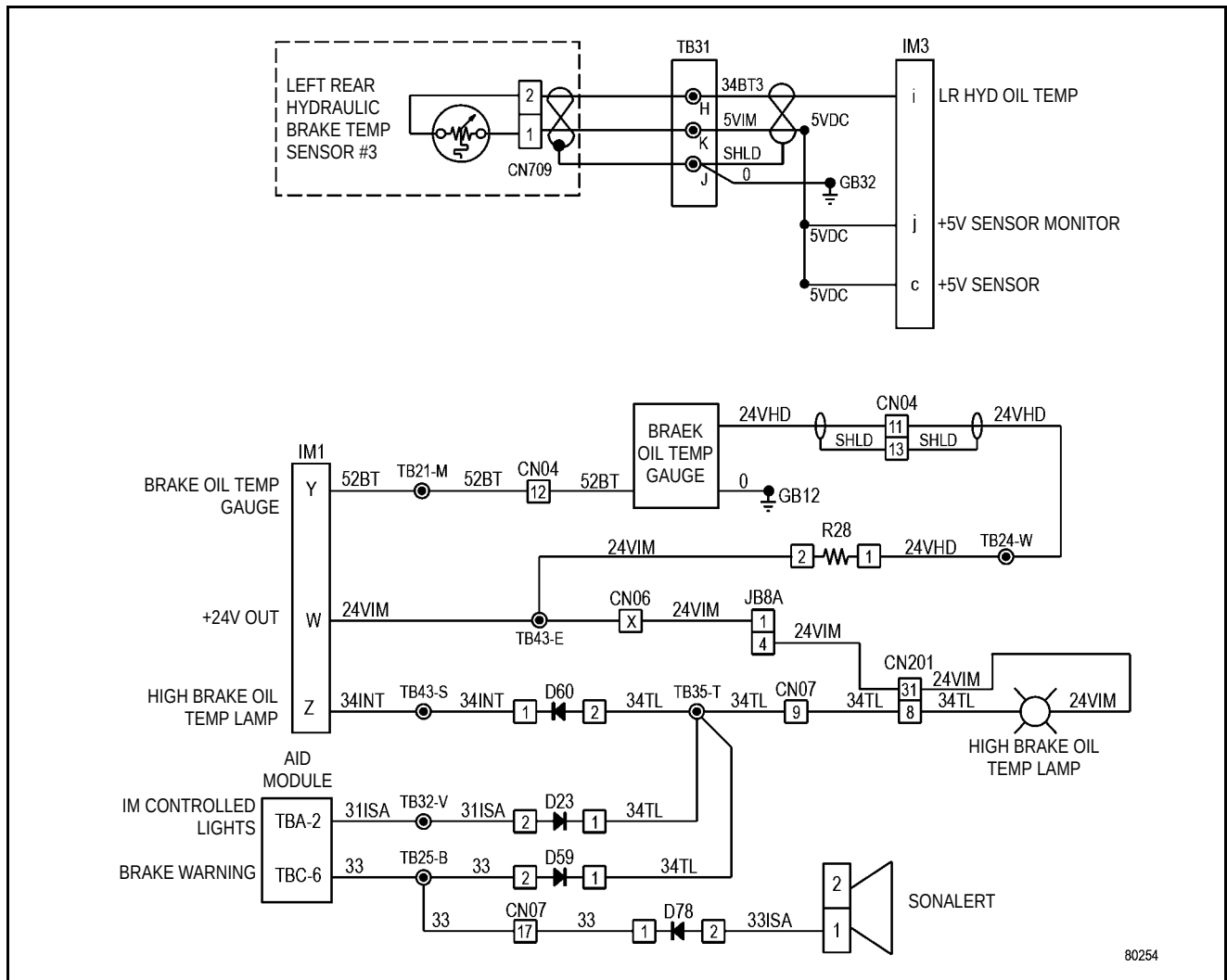


Fault Code A196: Left rear hydraulic oil temperature is high

Operator Action	Stop; Park; Run Engine
Fault Code	A196
Description	Left rear hydraulic oil temperature is high.
Fault Conditions	Sets at 120°C (4.13 volts) after 5 seconds. Resets at 103°C (3.78 volts) after 5 seconds or at steering bleed.
Operator Alerting System Response	Red "Hydraulic Oil Temperature High" Lamp Hydraulic Oil Temperature Gauge Buzzer Left Lamp Module Row 1, Column 1
Resulting Problem(s)	Excessively hot oil can cause equipment damage and even reduce service brake effectiveness.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table	
1. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 2. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Hydraulic Oil Temperature - Left Rear (IM3 i)	0.016 Volt to 4.89 Volt: Good Readings <0.016 Volt or >4.89 Volt: Defective Sensor or Circuit Fault(s): A166, A170
Steering Bleed Valve	0: steering bleed valve is off 1: steering bleed valve is on
High Brake Oil Temperature Lamp (IM1 Z)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A365
Hydraulic Oil Temperature Gauge (IM1 Y)	Status - Open Load: Unexpected: Troubleshoot Status - Shorted to Ground: Unexpected: Troubleshoot Status - Overload: Unexpected: Troubleshoot Approximate Calibration: Start Green Band: 60°C Start Red Band: 120°C Fault(s): A364

Related circuit diagram



Fault Code A197: Right rear hydraulic oil temperature is high

Operator Action	Stop; Park; Run Engine
Fault Code	A197
Description	Right rear hydraulic oil temperature is high.
Fault Conditions	Sets at 120°C (4.13 volts) after 5 seconds. Resets at 103°C (3.78 volts) after 5 seconds or at steering bleed.
Operator Alerting System Response	Red "Hydraulic Oil Temperature High" Lamp Hydraulic Oil Temperature Gauge Buzzer Left Lamp Module Row 1, Column 1
Resulting Problem(s)	Excessively hot oil can cause equipment damage and even reduce service brake effectiveness.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table	
1. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 2. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Hydraulic Oil Temperature - Right Rear (IM3 m)	0.016 Volt to 4.89 Volt: Good Readings <0.016 Volt or >4.89 Volt: Defective Sensor or Circuit Fault(s): A167, A171
Steering Bleed Valve	0: steering bleed valve is off 1: steering bleed valve is on
High Brake Oil Temperature Lamp (IM1 Z)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A365
Hydraulic Oil Temperature Gauge (IM1 Y)	Status - Open Load: Unexpected: Troubleshoot Status - Shorted to Ground: Unexpected: Troubleshoot Status - Overload: Unexpected: Troubleshoot Approximate Calibration: Start Green Band: 60°C Start Red Band: 120°C Fault(s): A364

[illegible]

Fault Code A198: Hoist pressure 1 sensor is high

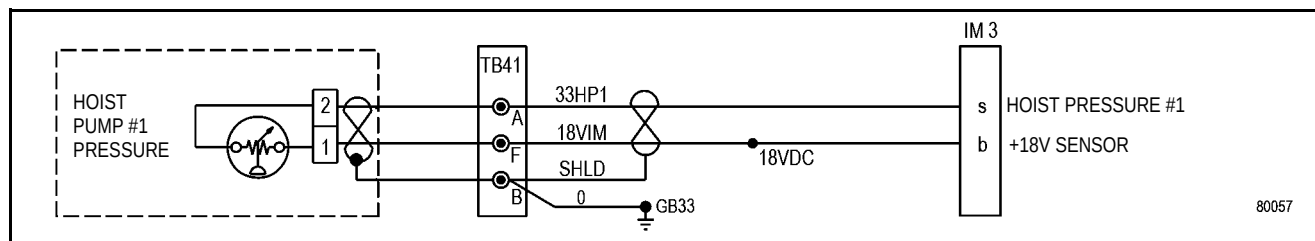
Operator Action	None
Fault Code	A198
Description	Hoist pressure 1 sensor is high.
Fault Conditions	Sets at 4025 psi (20.1mA) for 5 seconds. Resets at 3650 pi (18.6 mA) for 5 seconds.
Operator Alerting System Response	None
Resulting Problem(s)	Monitoring of hoist pressures in KOMTRAX Plus will be compromised.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. The primary correction for this fault is to correct any external wiring or replace the sensor.
2. This fault's logic contains only one parameter.

Parameter	Expected State and/or Related Fault(s)
Hoist Pressure 1 Sensor (IM3 s)	2.4 mA to 20.1 mA: good readings Less than 2.4 mA or more than 20.1 mA: Defective sensor or circuit Fault(s): A202

Related circuit diagram

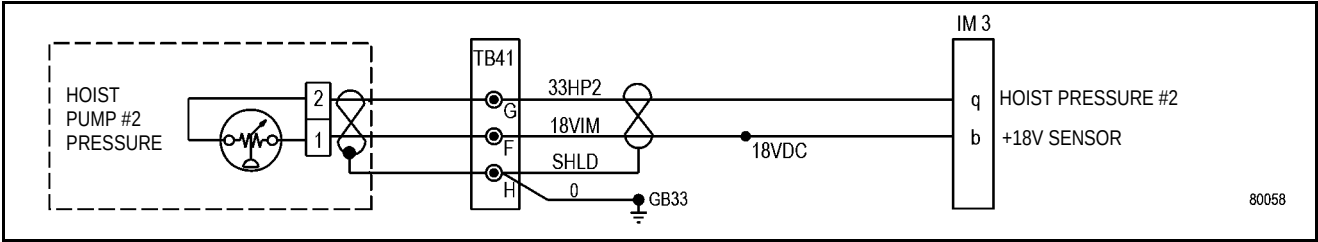


Fault Code A199: Hoist pressure 2 sensor is high

Operator Action	None
Fault Code	A199
Description	Hoist pressure 2 sensor is high.
Fault Conditions	Sets at 4025 psi (20.1mA) for 5 seconds. Resets at 3650 pi (18.6 mA) for 5 seconds.
Operator Alerting System Response	None
Resulting Problem(s)	Monitoring of hoist pressures in KOMTRAX Plus will be compromised.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table	
1. The primary correction for this fault is to correct any external wiring or replace the sensor. 2. This fault's logic contains only one parameter.	
Parameter	Expected State and/or Related Fault(s)
Hoist Pressure 2 Sensor (IM3 q)	2.4 mA to 20.1 mA: good readings Less than 2.4 mA or more than 20.1 mA: Defective sensor or circuit Fault(s): A203

Related circuit diagram

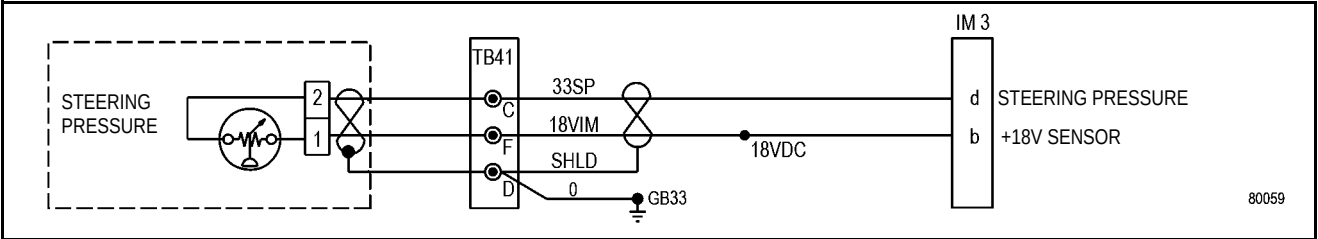


Fault Code A200: Steering pressure sensor is high

Operator Action	None
Fault Code	A200
Description	Steering pressure sensor is high.
Fault Conditions	Sets at 4025 psi (20.1mA) for 5 seconds. Resets at 3650 pi (18.6 mA) for 5 seconds.
Operator Alerting System Response	None
Resulting Problem(s)	Monitoring of steering pressure in KOMTRAX Plus will be compromised.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table	
1. The primary correction for this fault is to correct any external wiring or replace the sensor. 2. This fault's logic contains only one parameter.	
Parameter	Expected State and/or Related Fault(s)
Steering Pressure Sensor (IM3 d)	2.4 mA to 20.1 mA: good readings Less than 2.4 mA or more than 20.1 mA: Defective sensor or circuit Fault(s): A204

Related circuit diagram



Fault Code A201: Brake pressure sensor is high

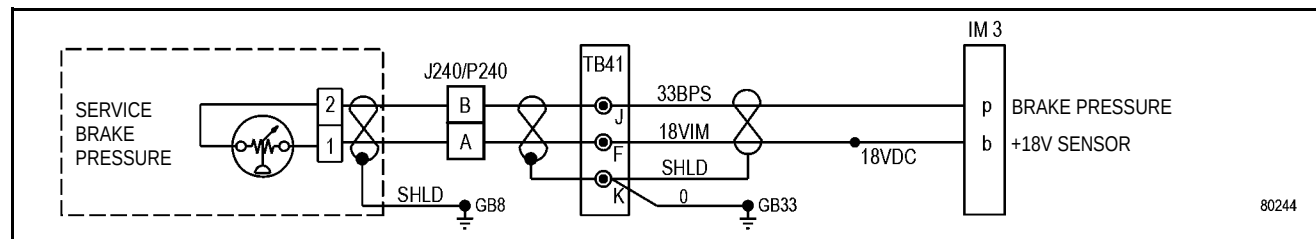
Operator Action	None
Fault Code	A201
Description	Brake pressure sensor is high.
Fault Conditions	Sets at 4025 psi (20.1mA) for 5 seconds. Resets at 3650 pi (18.6 mA) for 5 seconds.
Operator Alerting System Response	None
Resulting Problem(s)	Monitoring of service brake system for driver input as well as KOMTRAX Plus will be compromised.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. The primary correction for this fault is to correct any external wiring or replace the sensor.
2. This fault's logic contains only one parameter.

Parameter	Expected State and/or Related Fault(s)
Brake Pressure Sensor (IM3 p)	2.4 mA to 20.1 mA: good readings Less than 2.4 mA or more than 20.1 mA: Defective sensor or circuit Fault(s): A205

Related circuit diagram



Fault Code A202: Hoist pressure 1 sensor is low

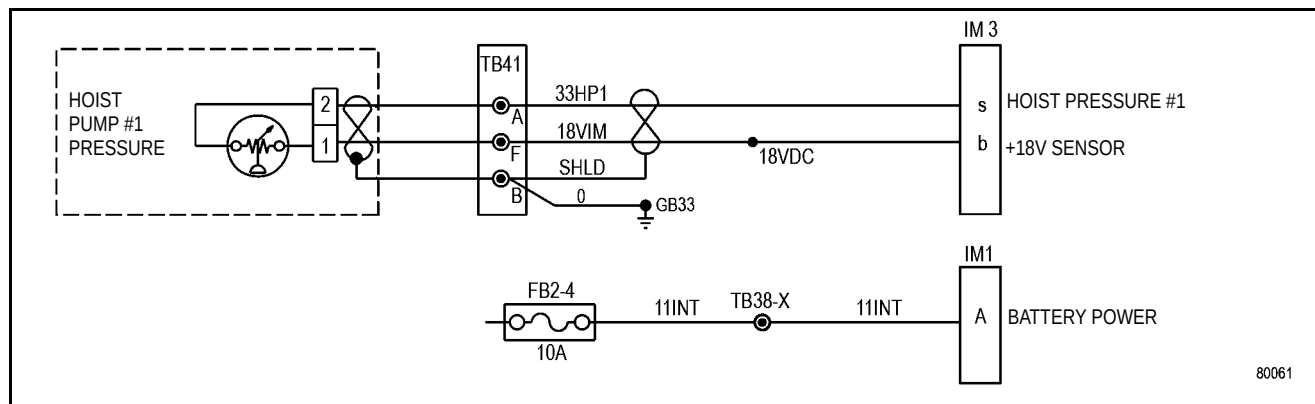
Operator Action	None
Fault Code	A202
Description	Hoist pressure 1 sensor is low.
Fault Conditions	Sets at -401 psi (2.4 mA) for 5 seconds with cranking state not sensed (<600 engine rpm and <18 battery volts). Resets at -206 psi (3.2 mA)
Operator Alerting System Response	None
Resulting Problem(s)	Monitoring of hoist pressures in KOMTRAX Plus will be compromised.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. The primary correction for this fault is to correct any external wiring or replace the sensor.
2. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s).
3. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items.
4. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.

Parameter	Expected State and/or Related Fault(s)
Hoist Pressure 1 Sensor (IM3 s)	2.4 mA to 20.1 mA: good readings Less than 2.4 mA or more than 20.1 mA: Defective sensor or circuit Fault(s): A198
Battery Voltage, 24 Volt (IM1 A)	>18 Volt: Good Reading <18 Volt: Low battery voltage (while cranking) blocks this fault.
Engine Speed [RPM] (CAN/J1939) (IM1 q,r,s)	Less than 600 RPM: combined with low voltage means cranking is in process and fault is blocked Greater than 600 RPM: engine is running normally Fault(s): A184

Related circuit diagram



Fault Code A203: Hoist pressure 2 sensor is low

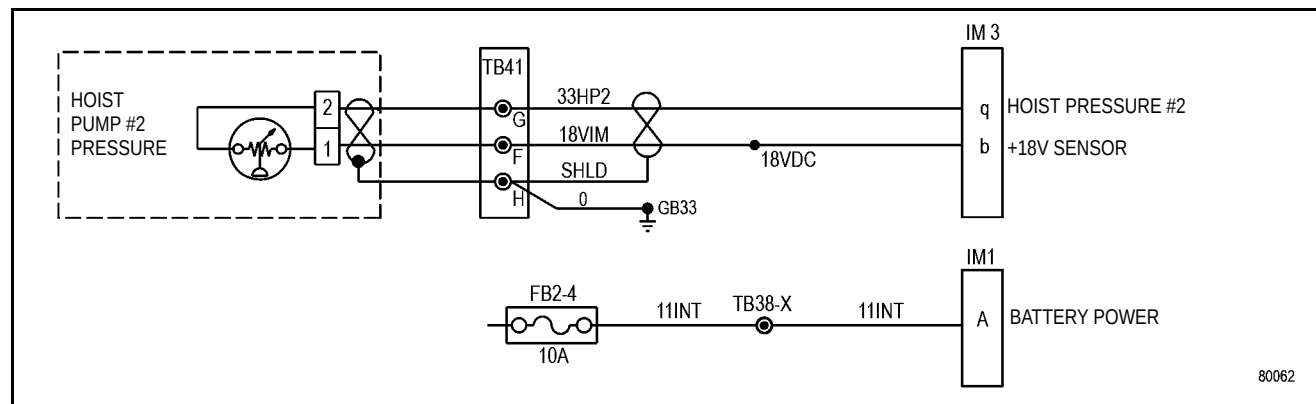
Operator Action	None
Fault Code	A203
Description	Hoist pressure 2 sensor is low.
Fault Conditions	Sets at -401 psi (2.4 mA) for 5 seconds with cranking state not sensed (<600 engine rpm and <18 battery volts). Resets at -206 psi (3.2 mA)
Operator Alerting System Response	None
Resulting Problem(s)	Monitoring of hoist pressures in KOMTRAX Plus will be compromised.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. The primary correction for this fault is to correct any external wiring or replace the sensor.
2. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s).
3. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items.
4. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.

Parameter	Expected State and/or Related Fault(s)
Hoist Pressure 2 Sensor (IM3 q)	2.4 mA to 20.1 mA: good readings Less than 2.4 mA or more than 20.1 mA: Defective sensor or circuit Fault(s): A199
Battery Voltage, 24 Volt (IM1 A)	>18 Volt: Good Reading <18 Volt: Low battery voltage (while cranking) blocks this fault.
Engine Speed [RPM] (CAN/J1939) (IM1 q,r,s)	Less than 600 RPM: combined with low voltage means cranking is in process and fault is blocked Greater than 600 RPM: engine is running normally Fault(s): A184

Related circuit diagram



Fault Code A204: Steering pressure sensor is low

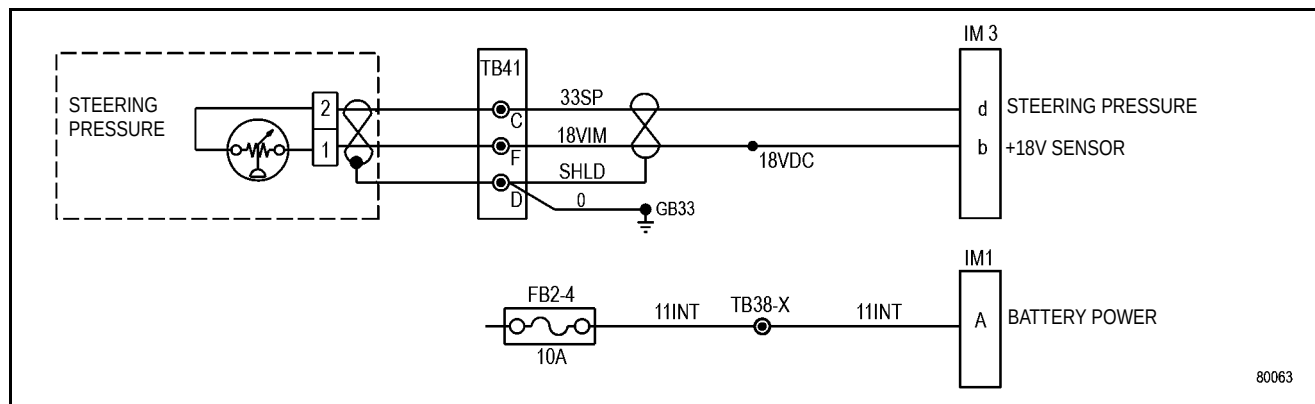
Operator Action	None
Fault Code	A204
Description	Steering pressure sensor is low.
Fault Conditions	Sets at -401 psi (2.4 mA) for 5 seconds with cranking state not sensed (<600 engine rpm and <18 battery volts). Resets at -206 psi (3.2 mA)
Operator Alerting System Response	None
Resulting Problem(s)	Monitoring of steering pressure in KOMTRAX Plus will be compromised.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. The primary correction for this fault is to correct any external wiring or replace the sensor.
2. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s).
3. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items.
4. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.

Parameter	Expected State and/or Related Fault(s)
Steering Pressure Sensor (IM3 d)	2.4 mA to 20.1 mA: good readings Less than 2.4 mA or more than 20.1 mA: Defective sensor or circuit Fault(s): A200
Battery Voltage, 24 Volt (IM1 A)	>18 Volt: Good Reading <18 Volt: Low battery voltage (while cranking) blocks this fault.
Engine Speed [RPM] (CAN/J1939) (IM1 q,r,s)	Less than 600 RPM: combined with low voltage means cranking is in process and fault is blocked Greater than 600 RPM: engine is running normally Fault(s): A184

Related circuit diagram



Fault Code A205: Brake pressure sensor is low

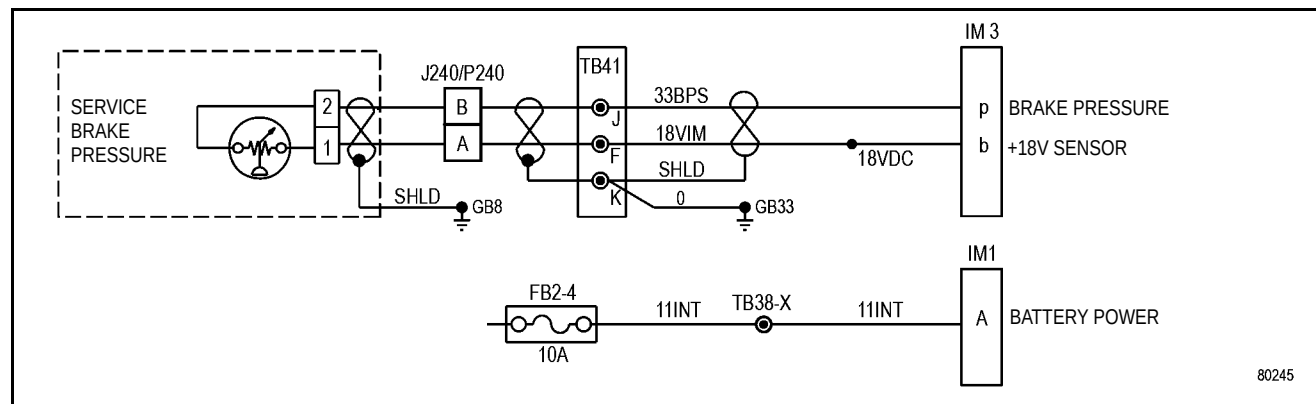
Operator Action	None
Fault Code	A205
Description	Brake pressure sensor is low.
Fault Conditions	Sets at -401 psi (2.4 mA) for 5 seconds with cranking state not sensed (<600 engine rpm and <18 battery volts). Resets at -206 psi (3.2 mA)
Operator Alerting System Response	None
Resulting Problem(s)	Monitoring of brake pressure in KOMTRAX Plus will be compromised.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. The primary correction for this fault is to correct any external wiring or replace the sensor.
2. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s).
3. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items.
4. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.

Parameter	Expected State and/or Related Fault(s)
Brake Pressure Sensor (IM3 p)	2.4 mA to 20.1 mA: good readings Less than 2.4 mA or more than 20.1 mA: Defective sensor or circuit Fault(s): A201
Battery Voltage, 24 Volt (IM1 A)	>18 Volt: Good Reading <18 Volt: Low battery voltage (while cranking) blocks this fault.
Engine Speed [RPM]	Less than 600 RPM: combined with low voltage means cranking is in process and fault is blocked Greater than 600 RPM: engine is running normally Fault(s): A184

Related circuit diagram

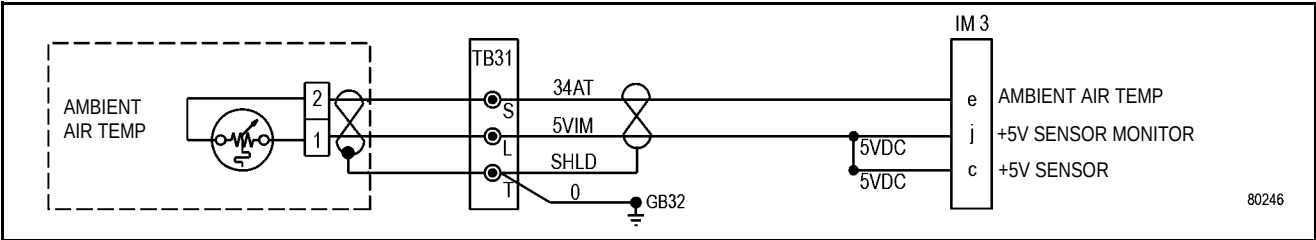


Fault Code A206: Ambient temperature sensor is high

Operator Action	None
Fault Code	A206
Description	Ambient temperature sensor is high.
Fault Conditions	Sets at 133° C (4.89V) for 3 seconds. Resets at 107° C (4.77V) for 3 seconds.
Operator Alerting System Response	None
Resulting Problem(s)	Monitoring of temperatures will be compromised without ambient temperature information.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table	
1. The primary correction for this fault is to correct any external wiring or replace the sensor. 2. This fault's logic contains only one parameter.	
Parameter	Expected State and/or Related Fault(s)
Ambient Temperature Sensor (IM3 e)	0.016 Volt to 4.89 Volt: Good Readings <0.016 Volt or >4.89 Volt: Defective Sensor or Circuit Fault(s): A207

Related circuit diagram

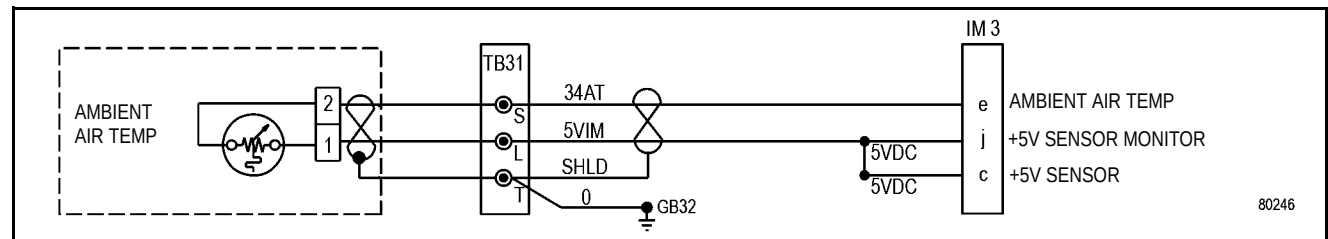


Fault Code A207: Ambient temperature sensor is low

Operator Action	None
Fault Code	A207
Description	Ambient temperature sensor is low.
Fault Conditions	Sets at -62° C (0.016V) for 3 seconds. Resets at -59° C (0.032V) for 3 seconds.
Operator Alerting System Response	None
Resulting Problem(s)	Monitoring of temperatures will be compromised without ambient information.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table	
1. The primary correction for this fault is to correct any external wiring or replace the sensor. 2. This fault's logic contains only one parameter.	
Parameter	Expected State and/or Related Fault(s)
Ambient Temperature Sensor (IM3 e)	0.016 Volt to 4.89 Volt: Good Readings <0.016 Volt or >4.89 Volt: Defective Sensor or Circuit Fault(s): A206

Related circuit diagram



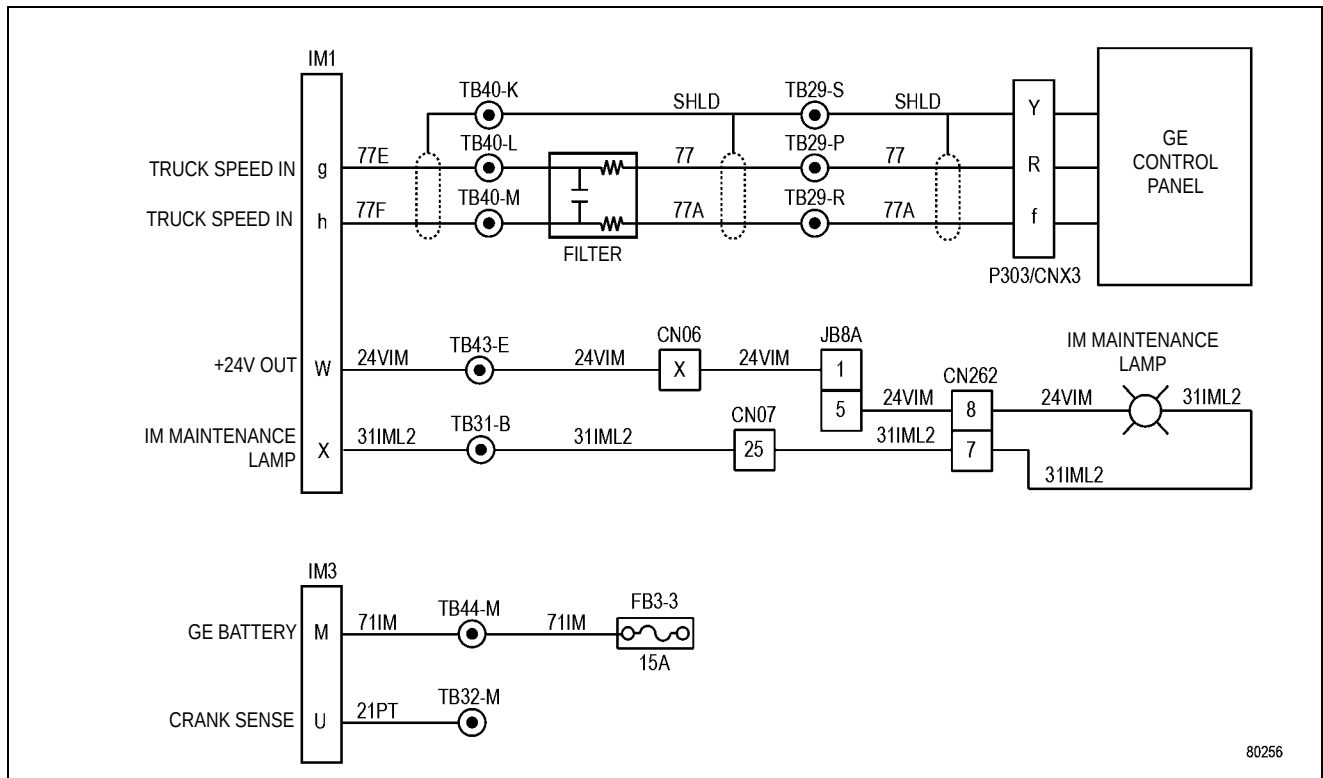
Fault Code A212: Bad truck speed signal

Operator Action	Go To Shop Now
Fault Code	A212
Description	Bad truck speed signal
Fault Conditions	Sets if truck speed timeout (frequency input does not produce 5 periods within 1 second) persists for 12 seconds with GE powered up and engine not being cranked. Resets if timeouts clear for 3 seconds.
Operator Alerting System Response	Amber "IM Maintenance" Lamp No Buzzer Right Lamp Module Row 5, Column 3
Resulting Problem(s)	Loss of truck speed signal caused loss of protection from mis-application of the parking brake and loss of speedometer for the driver.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. The primary correction for this fault is to correct any external wiring or replace the GE signal source. 2. This fault's logic contains more than one parameter. 3. Since there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 4. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Truck Speed [kph] (IM1 g,h)	0: truck is not moving Greater than 0: truck is moving
Drive System Control Power (IM3 M)	0: power to GE control is off 1: power to GE control is on
Crank Sense (IM3 U)	0: Engine not being cranked 1: Engine is being cranked
IM Maintenance Lamp (IM1 X)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A363

Related circuit diagram



Fault Code A213: Parking brake should have applied but is detected as not having applied

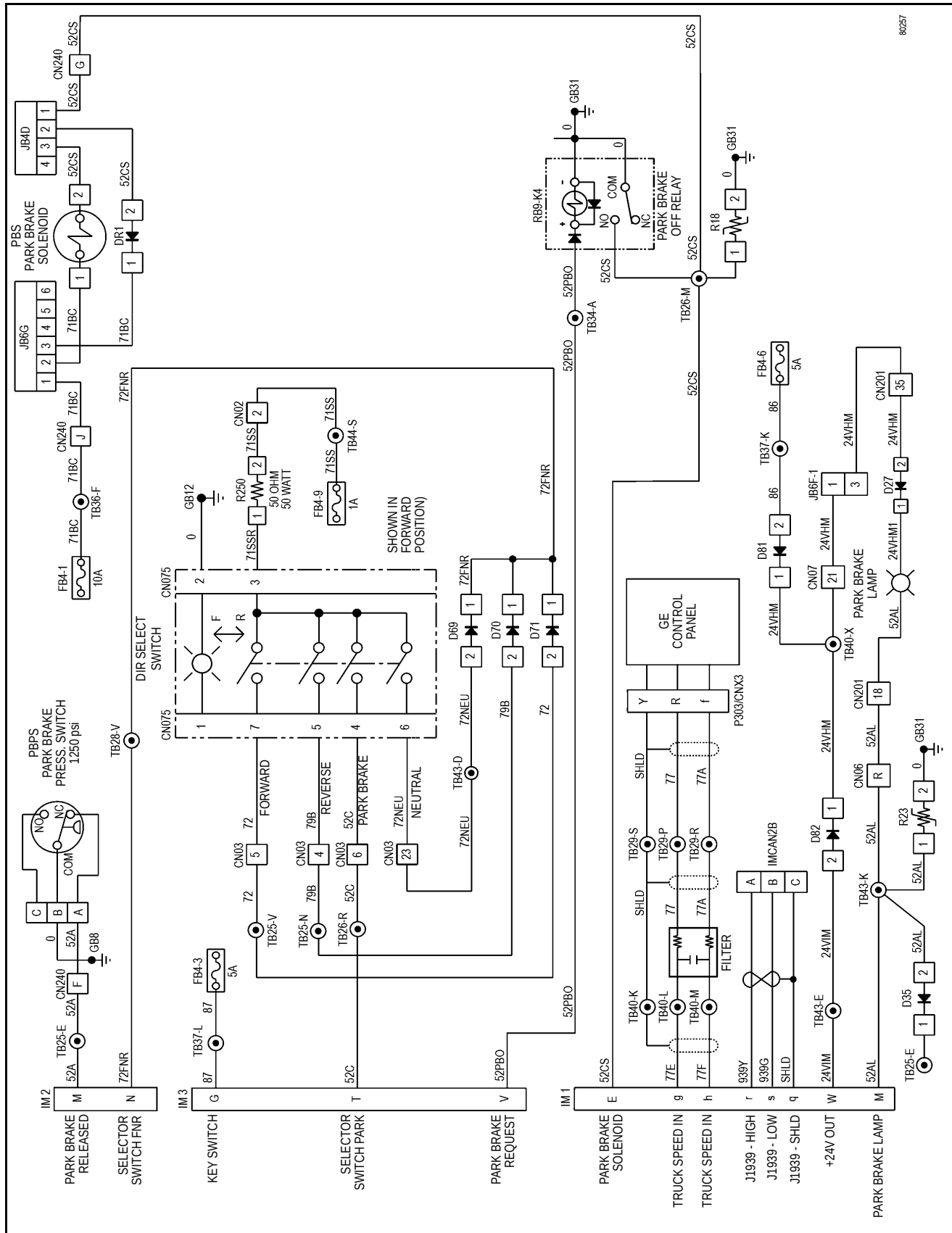
Operator Action	Secure Vehicle
Fault Code	A213
Description	Parking brake should have applied but is detected as not having applied.
Fault Conditions	Sets if park brake does not apply within 3 seconds if the shifter is not in the forward, neutral, or reverse positions or the engine oil pressure is below the setpoint of the engine oil pressure switch and either the speed of the truck has been 0.8kph or less for 1 second or the service brakes are applied, or after the engine has been off for 15 seconds with the J1939 link working, or within 2 seconds after the key switch has been turned off, the drive system control power is off, the engine is not running, and the truck is not moving. Park brake set is defined as closure of the park brake set pressure switch and opening of the park brake release pressure switch. Resets after 3 seconds if the park brake applies, or if the park brake request is cancelled and the shifter is not in the park position and the shifter is in the forward, neutral, or reverse positions and the engine is running.
Operator Alerting System Response	Amber flashing "Parking Brake" Lamp No Buzzer Left Lamp Module Row 1, Column 3
Resulting Problem(s)	The parking brake may be non-functional. Engine cranking may be prevented. Smart Timed Engine Shutdown may be prevented.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 2. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Selector Switch (FNR) (IM2 N)	0: shifter is not in forward, neutral, or reverse 1: shifter is in forward, neutral, or reverse Fault(s): A271, A303
Park Brake Released (IM2 M)	0: parking brake pressure switch indicating low pressure and an applied parking brake 1: parking brake pressure switch indicating high pressure and a released parking brake
Selector Switch (Park) (IM3 T)	0: shifter is not in park 1: shifter is in park Fault(s): A271, A303

Park Brake Request (IM3 V)	0: shifter is not in the forward, neutral, or reverse positions or the engine oil pressure is below the setpoint of the engine oil pressure switch 1: shifter is in the forward, neutral, or reverse positions and the engine oil pressure is above the setpoint of the engine oil pressure switch Fault(s): A264
Engine Speed [RPM] (CAN/J1939) (IM1 q,r,s)	0: engine is not running Greater than 0: engine is running Fault(s): A184
Truck Speed [kph] (IM1 g,h)	0: truck is not moving Greater than 0: truck is moving Fault(s): A212
CAN J1939 connection (IM1 q,r,s)	Fault(s): A184
Keyswitch (IM3 G)	0: keyswitch is off and the truck is not moving and the engine is not running 1: keyswitch is on or the truck is moving or the engine is running Fault(s): A240
Park Brake Solenoid (IM1 E)	Status - Open Load: Unexpected. Troubleshoot. Status - Normal: Expected. No Problem. Status - Shorted to Ground: Expected if shifter is in the forward, neutral, or reverse positions and the engine oil pressure is above the setpoint of the engine oil pressure switch. Otherwise unexpected and must troubleshoot. Status - Overload: Unexpected. Troubleshoot. 0: shifter is not in the forward, neutral, or reverse positions or the engine oil pressure is below the setpoint of the engine oil pressure switch and either the speed of the truck has been 0.8kph or less for 1 second or the service brakes are applied. 1: shifter is in the forward, neutral, or reverse positions and the engine oil pressure is above the setpoint of the engine oil pressure switch. Fault(s): A216, A351
Parking Brake Lamp (IM1 M)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A356

Related circuit diagram



NOTES

Fault Code A214: Parking brake should have released but is detected as not having released

Operator Action	Stop; Park
Fault Code	A214
Description	Parking brake should have released but is detected as not having released.
Fault Conditions	Sets if the engine has been running 90 seconds or more with the keyswitch on and the park brake does not release in response to the selector switch being moved out of park or in response to truck speeds over 0.8 kph for 3 seconds. Park brake release is defined as the closure of the park brake release pressure switch and the opening of the park brake set pressure switch. Resets after 3 seconds if the park brake releases, or engine stops running, or key-switch and propel control power are both turned off, or selector switch is moved to park position.
Operator Alerting System Response	Amber "Parking Brake" Lamp (not flashing) No Buzzer Left Lamp Module Row 1, Column 3
Resulting Problem(s)	Operation of the truck may be prevented or the parking brake may be damaged.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 2. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Selector Switch (Park) (IM3 T)	0: shifter is not in park 1: shifter is in park Fault(s): A271, A303
Park Brake Released (IM2 M)	0: parking brake pressure switch indicating high pressure and a released parking brake 1: parking brake pressure switch indicating lower pressure and an applied parking brake
Park Brake Request (IM3 V)	0: shifter is not in the forward, neutral, or reverse positions or the engine oil pressure is below the setpoint of the engine oil pressure switch 1: shifter is in the forward, neutral, or reverse positions and the engine oil pressure is above the setpoint of the engine oil pressure switch Fault(s): A264
Engine Speed [RPM] (CAN/J1939) (IM1 q,r,s)	0: engine is not running Greater than 0: engine is running Fault(s): A184

Truck Speed [kph] (IM1 g,h)	0: truck is not moving Greater than 0: truck is moving Fault(s): A212
CAN J1939 connection (IM1 q,r,s)	Fault(s): A184
Keyswitch (IM3 G)	0: keyswitch is off and the truck is not moving and the engine is not running 1: keyswitch is on or the truck is moving or the engine is running Fault(s): A240
Park Brake Solenoid (IM1 E)	Status - Open Load: Unexpected. Troubleshoot. Status - Normal: Expected. No Problem. Status - Shorted to Ground: Expected if shifter is in the forward, neutral, or reverse positions and the engine oil pressure is above the setpoint of the engine oil pressure switch. Otherwise unexpected and must troubleshoot. Status - Overload: Unexpected. Troubleshoot. 0: shifter is not in the forward, neutral, or reverse positions or the engine oil pressure is below the setpoint of the engine oil pressure switch and either the speed of the truck has been 0.8kph or less for 1 second or the service brakes are applied 1: shifter is in the forward, neutral, or reverse positions and the engine oil pressure is above the setpoint of the engine oil pressure switch Fault(s): A216, A351

IM2
M N
PARK BRAKE RELEASED
SELECTOR SWITCH FNR
72FNR

IM3
G T V
KEY SWITCH
PARK BRAKE
SELECTOR SWITCH PARK
PARK BRAKE REQUEST
72FNR 52C 52PBO

IM1
E g h r s q
PARK BRAKE SOLENOID
TRUCK SPEED IN
TRUCK SPEED IN
J1939 - HIGH
J1939 - LOW
J1939 - SHLD

Wiring Details:

- Park Brake Solenoid:** Connected to IM2 M (52A) and IM1 E (52CS). The solenoid is labeled "PARK BRAKE SOLENOID" and "PARK BRAKE RELEASED".
- Park Brake Pressure Switch:** Connected to IM2 N (52A) and IM1 E (52CS). The switch is labeled "PARK BRAKE PRESS. SWITCH 1250 PSI".
- Control Panels:** The diagram shows connections to a "GE CONTROL PANEL" and a "P303/CNX3" panel. The GE panel has terminals Y, R, and f. The P303 panel has terminals SHLD, 77, 77A, and 77B.
- Relays and Switches:** The diagram includes a "PARK BRAKE OFF RELAY" and a "PARK BRAKE ON RELAY". The off relay is connected to IM1 E (52CS) and IM1 g (77E). The on relay is connected to IM1 E (52CS) and IM1 h (77F).
- Diodes and Fuses:** The diagram includes diodes D69, D70, and D71, and fuses F4-1 (10A) and F4-3 (5A).
- Grounding:** The diagram shows multiple ground points, including G88, G831, and G832.

Legend for IM1:

Terminal	Label
E	52CS
g	77E
h	77F
r	939Y
s	939G
q	SHLD

NOTES

Fault Code A215: Brake auto apply relay circuit is defective

Operator Action	Go To Shop
Fault Code	A215
Description	Brake auto apply relay circuit is defective.
Fault Conditions	Sets if the output driver circuit detects an open circuit or short to ground when the key-switch is on and the park brake is requested, or sets if the output driver circuit does not detect an open circuit with the key switch on and the park brake not requested. Resets only at power down.
Operator Alerting System Response	Amber "IM Maintenance" Lamp No Buzzer Right Lamp Module Row 5, Column 3
Resulting Problem(s)	Parking brake may not be protected in an application while rolling, or service brake may not release and permit movement.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 2. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Park Brake Request (IM3 V)	0: shifter is not in the forward, neutral, or reverse positions or the engine oil pressure is below the setpoint of the engine oil pressure switch 1: shifter is in the forward, neutral, or reverse positions and the engine oil pressure is above the setpoint of the engine oil pressure switch Fault(s): A264, A317
Keyswitch (IM3 G)	0: keyswitch is off and the truck is not moving and the engine is not running 1: keyswitch is on or the truck is moving (GE power not yet off) Fault(s): A240
Auto Brake Apply Solenoid (IM1 R)	Status - Open Load: Unexpected if park brake request and keyswitch are on. But expected if park brake request is off or keyswitch is off. Troubleshoot if inconsistent. Status - Normal: Expected if park brake request and keyswitch are on. Troubleshoot if status is normal with park brake request off or keyswitch off. Status - Shorted to ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: normal operation with park brake either on or off. This valve operates only transiently when the park brake is turned on. 1: brake auto apply valve operates for 1.5 seconds after the park brake is turned on if the truck is stopped. If moving, it will not operate until truck speed has dropped to 0.8 kph. Fault(s): A359

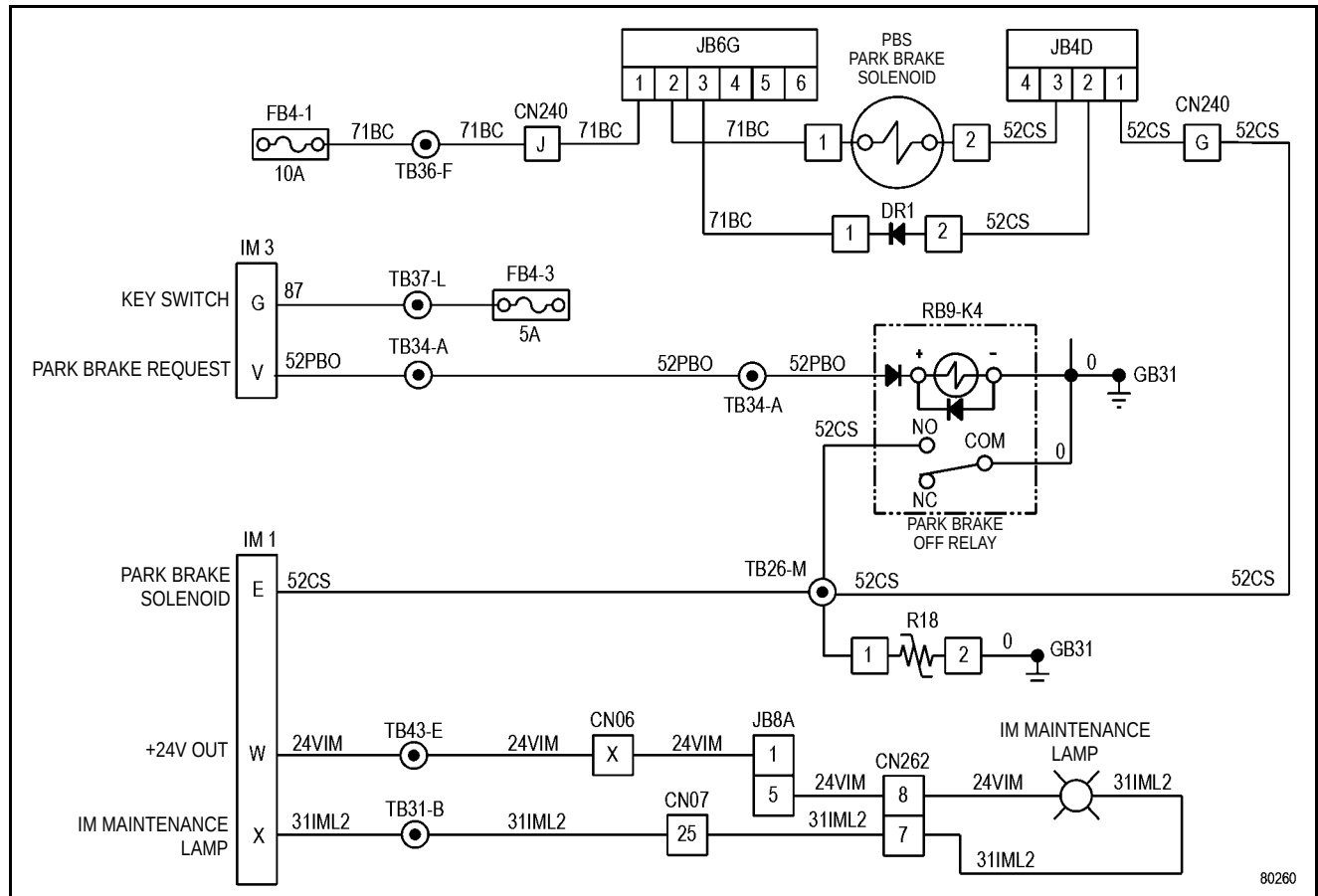
Fault Code A216: An open or short to ground has been detected in the parking brake command valve circuit

Operator Action	Stop; Park
Fault Code	A216
Description	An open or short to ground has been detected in the parking brake command valve circuit.
Fault Conditions	Sets if valve circuit is open or shorted to ground with keyswitch and park brake request on, or if valve circuit is open without park brake request being on. Resets if any conditions change.
Operator Alerting System Response	Amber "IM Maintenance" Lamp No Buzzer Right Lamp Module Row 5, Column 3
Resulting Problem(s)	Parking brake may not apply or release properly, causing equipment and roll-away concerns.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table	
1. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 2. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Park Brake Solenoid (IM1 E)	Status - Open Load: Unexpected. Troubleshoot. Status - Normal: Expected. No Problem. Status - Shorted to Ground: Expected if Park Brake Request is in the 24 volt condition. Otherwise unexpected and must troubleshoot. Status - Overload: Unexpected. Troubleshoot. 0: Park Brake Request Input is in the low voltage (request) condition 1: Park Brake Request Input is in the high voltage (not requested) condition Fault(s): A351
Park Brake Request (IM3 V)	0: shifter is not in the forward, neutral, or reverse positions or the engine oil pressure is below the setpoint of the engine oil pressure switch 1: shifter is in the forward, neutral, or reverse positions and the engine oil pressure is above the setpoint of the engine oil pressure switch Fault(s): A264
Keyswitch (IM3 G)	0: keyswitch is off and the truck is not moving and the engine is not running 1: keyswitch is on or the truck is moving or the engine is running Fault(s): A240

IM Maintenance Lamp (IM1 X)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A363
--------------------------------	--

Related circuit diagram



NOTES

SECTION E5

TROUBLESHOOTING FAULT CODES (A223-A292)

INDEX

Fault Code A223: Excessive engine cranking has occurred or a jump start has been attempted	E5-2
Fault Code A230: Parking brake has been requested while truck is still moving	E5-4
Fault Code A236: Steering accumulator has not properly bled down after 90 seconds	E5-6
Fault Code A240: The keyswitch input to the interface module is open	E5-8
Fault Code A246: Payload meter reports truck overload	E5-9
Fault Code A249: Red warning lamp output used to drive the steering bleed fault lamp is shorted	E5-10
Fault Code A250: Battery voltage is low with the truck parked	E5-12
Fault Code A252: Start enable output circuit is either open or shorted to ground	E5-14
Fault Code A253: Steering bleed circuit is not open while running	E5-16
Fault Code A256: Red warning lamp output used to drive the steering bleed fault lamp is open	E5-18
Fault Code A257: Payload CAN/RPC is not connected	E5-20
Fault Code A258: Steering accumulator bleed pressure switch circuit is defective	E5-22
Fault Code A260: Parking brake failure	E5-24
Fault Code A262: Steering bleed circuit is not open during shutdown	E5-26
Fault Code A263: Steering bleed circuit is shorted to ground	E5-28
Fault Code A264: Parking brake relay circuit is defective	E5-30
Fault Code A265: Service brake failure	E5-32
Fault Code A270: Brake lock switch power supply is not on when required	E5-36
Fault Code A272: Brake lock switch power supply is not off when required	E5-38
Fault Code A273: A fault has been detected in the hoist or steering pump filter pressure switch circuit	E5-42
Fault Code A274: A brake setting fault has been detected	E5-44
Fault Code A275: A starter has been detected as engaged without a cranking attempt	E5-46
Fault Code A276: The drive system data link is not connected	E5-48
Fault Code A282: The number of excessive cranking counts and jump starts without the engine running has reached 7	E5-50
Fault Code A283: An engine shutdown delay was aborted because the parking brake was not set	E5-52
Fault Code A284: An engine shutdown delay was aborted because the secondary shutdown switch was operated	E5-54
Fault Code A285: The parking brake was not set when the key switch was turned off	E5-56
Fault Code A286: A fault was detected in the shutdown delay relay circuit	E5-58
Fault Code A292: The shutdown delay relay has remained on after the latched key switch circuit is off	E5-60

Fault Code A223: Excessive engine cranking has occurred or a jump start has been attempted

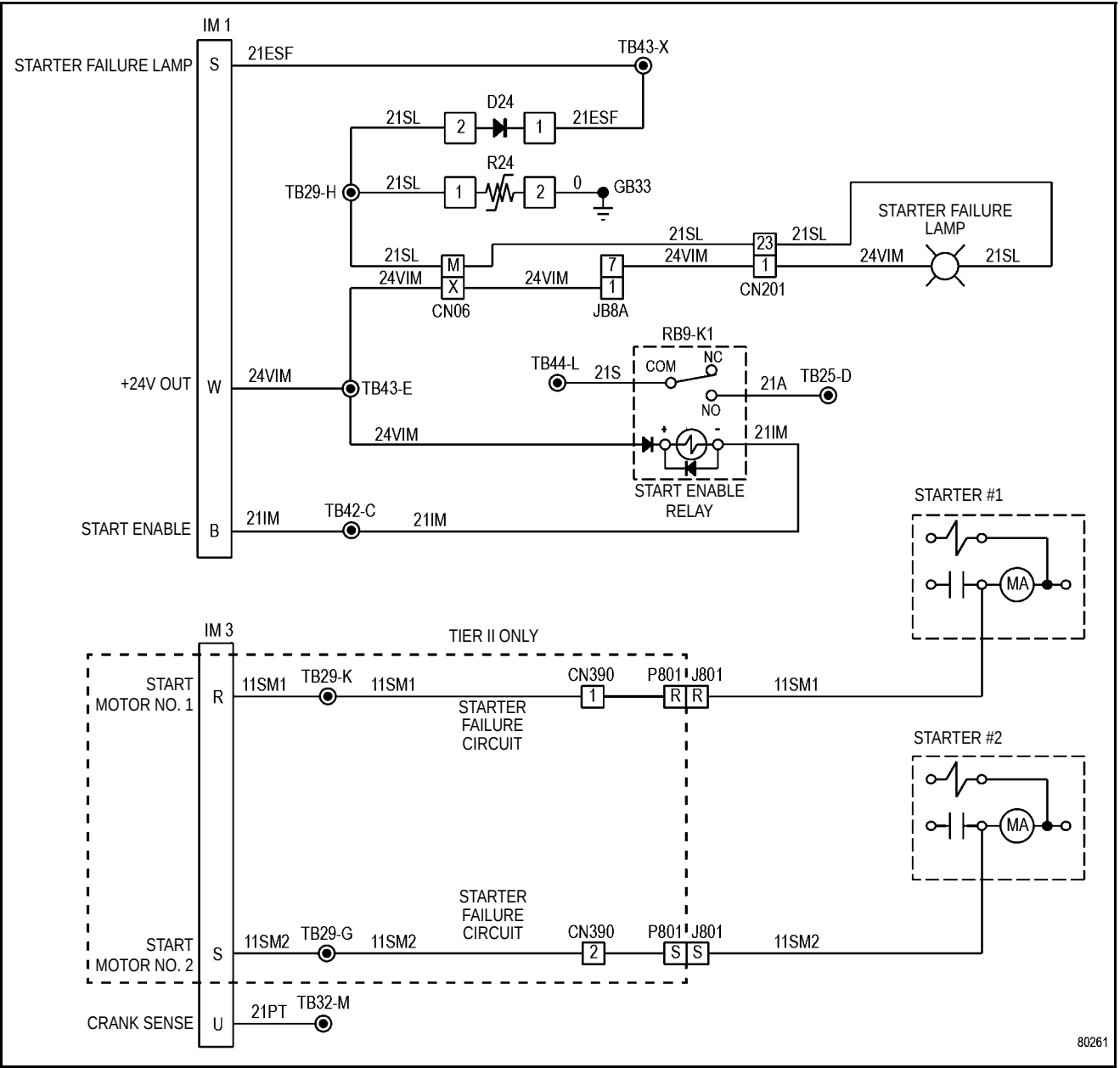
Operator Action	Wait 120 sec, Retry
Fault Code	A223
Description	Excessive engine cranking has occurred or a jump start has been attempted.
Fault Conditions	Sets after 30 seconds of continuous cranking or if either starter motor is energized without the start enable circuit (jump start). Resets after 120 seconds of no cranking.
Operator Alerting System Response	Amber "Engine Start Fail" Lamp No Buzzer Left Lamp Module Row 1, Column 4
Resulting Problem(s)	Starters can be damaged by excessive engine cranking.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. The primary correction for this fault is not to excessively crank the engine. Starting problems must be corrected, or starter failures will result. 2. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 3. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 4. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Crank Sense (IM3 U)	0: Engine not being cranked 1: Engine is being cranked
Starter Motor 1 Energized (IM3 R)	0: starter motor 1 not energized 1: starter motor 1 energized Fault(s): A152, A275, A316
Starter Motor 2 Energized (IM3 S)	0: starter motor 2 not energized 1: starter motor 2 energized Fault(s): A152, A275, A316
Start Enable (IM 1B)	Status - Open Load: Unexpected. Troubleshoot. Status - Normal: Expected. No problem. Status - Shorted to Ground: Unexpected. Troubleshoot. Status - Overload: Unexpected. Troubleshoot. 0: One of several interlocking situations exist to prevent cranking (excessive cranking history, selector switch in wrong position, engine red light, J1939 not ok, engine speed either not 0 to begin cranking or over 400 rpm while cranking). 1. No interlocking situations exist to prevent cranking. Fault(s): A252

Engine Start Fail Lamp (IM1 S)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A360
-----------------------------------	--

Related circuit diagram



Fault Code A230: Parking brake has been requested while truck still moving

Operator Action	Move shift lever out of PARK
Fault Code	A230
Description	Parking brake has been requested while truck is still moving.
Fault Conditions	Sets if parking brake is requested (selector switch moved from forward, neutral, reverse) while moving 0.8 kph and with parking brake released. Resets if any condition changes.
Operator Alerting System Response	None
Resulting Problem(s)	Parking brake could be damaged if protective circuitry did not work.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. The primary correction for this fault is just not to move the selector switch while the truck is in motion. 2. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 3. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 4. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Selector Switch (FNR) (IM2 N)	0: shifter is not in forward, neutral, or reverse 1: shifter is in forward, neutral, or reverse Fault(s): A271, A303
Truck Speed [kph] (IM1 g,h)	Less than 0.8 kph: truck is not moving Greater than 0.8 kph: truck is moving Fault(s): A212
Park Brake Released (IM2 M)	0: parking brake pressure switch indicating high pressure and a released parking brake 1: parking brake pressure switch indicating low pressure and an unreleased parking brake

[illegible]

Fault Code A236: Steering accumulator has not properly bled down after 90 seconds

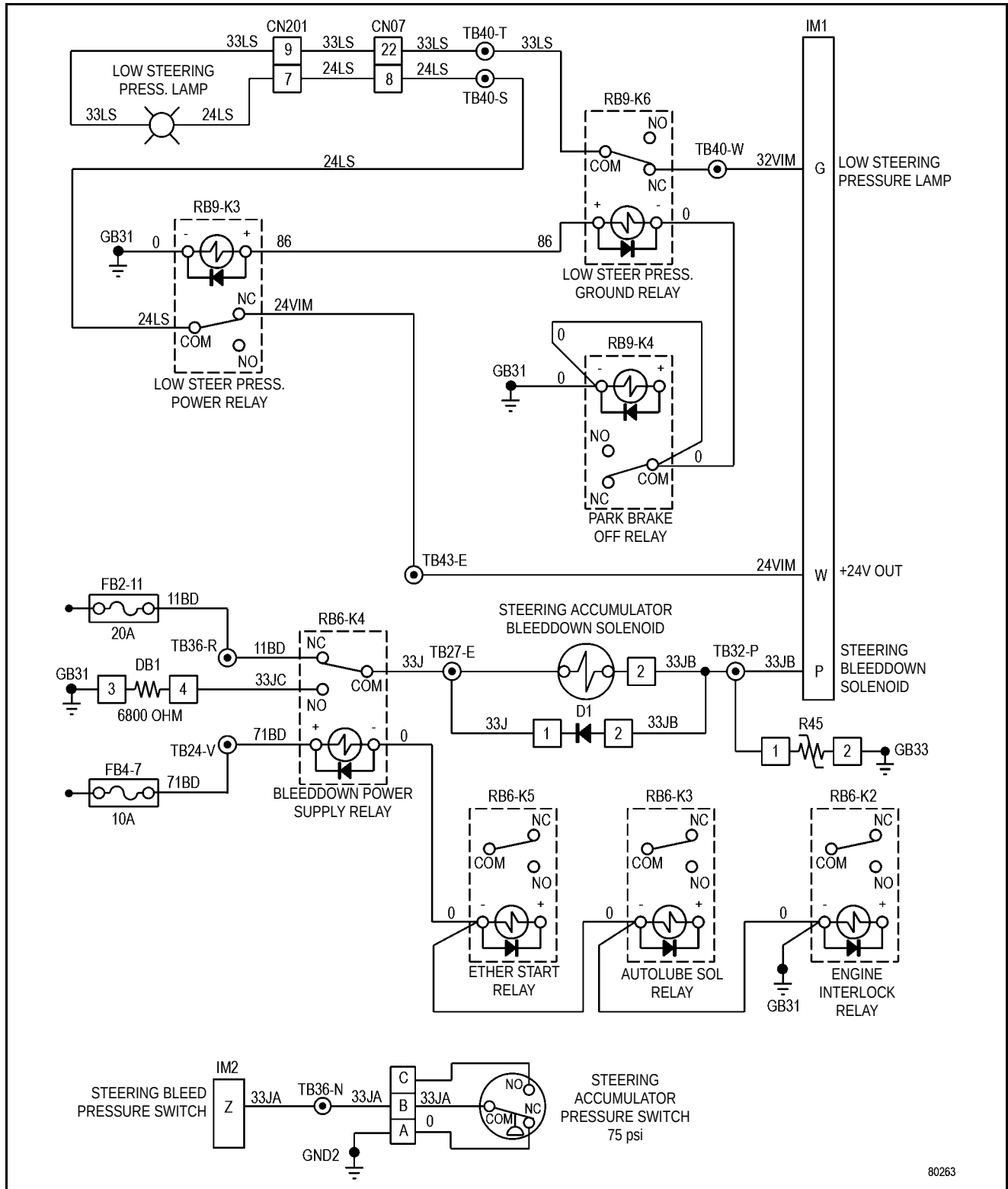
Operator Action	Park, do not steer
Fault Code	A236
Description	Steering accumulator has not properly bled down after 90 seconds.
Fault Conditions	Sets if the steering accumulator bleed pressure switch does not indicate that bleed down has been completed within 90 seconds. Resets either at power down, or if keyswitch is turned back on, thereby turning off the steering bleed valve.
Operator Alerting System Response	Red "Low Steering Pressure" Lamp after keyswitch is turned off (not flashing) No Buzzer Left Lamp Module Row 2, Column 1
Resulting Problem(s)	Failure to bleed the steering accumulator can create dangerous situations under the truck.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s).
2. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items.
3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.

Parameter	Expected State and/or Related Fault(s)
Steering Bleed Valve (IM1 P)	Status - Open Load: Expected with key switch on. No problem. Otherwise, unexpected. Must troubleshoot. Status - Normal: Expected only with key off and steering bleed in process. Troubleshoot if found with key on. Status - Shorted to Ground: Unexpected. Troubleshoot. Status - Overload: Unexpected. Troubleshoot. 0: steering bleed valve is off. This is expected in normal running operation. 1: steering bleed valve is on. This is expected after key off initiates steering bleed operation. Fault(s): A253, A358, A262, A363
Steering Accum Bleed Pressure Sw (IM2 Z)	0: Accumulator is bled. 1: Accumulator is not completely bled. Fault(s): A258, A280
Steering Bleed Fault Lamp (IM1 G)	Status - Open Load with Keyswitch On: Normal: Expected Status - Open Load with Keyswitch Off: Unexpected, Troubleshoot Status - Normal with Keyswitch On::Unexpected, Troubleshoot Status - Normal with Keyswitch Off: Normal, Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground)

Related circuit diagram



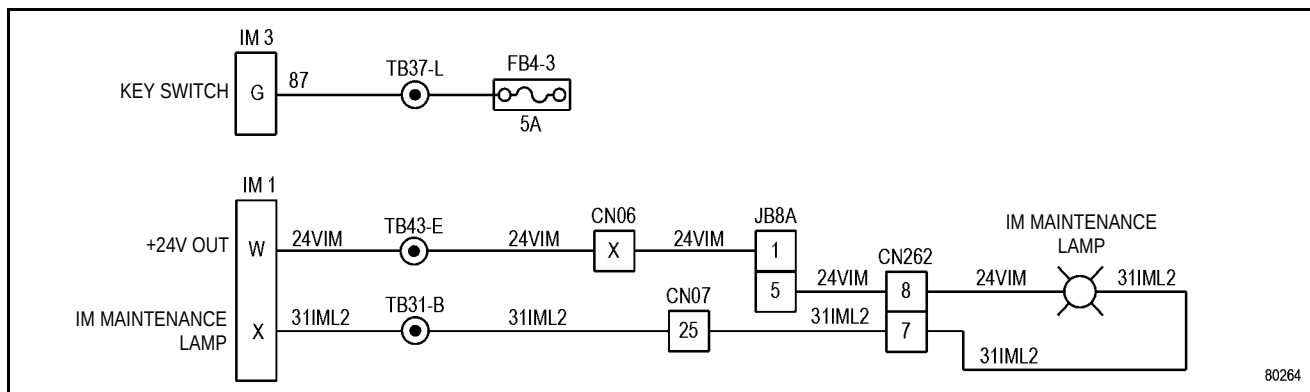
Fault Code A240: The keyswitch input to the interface module is open

Operator Action	Stop, Park, Repair
Fault Code	A240
Description	The keyswitch input to the interface module is open.
Fault Conditions	Sets if both keyswitch input and steering bleed valve are missing for 15 seconds. Resets if either one turns on.
Operator Alerting System Response	Amber "IM Maintenance" Lamp No Buzzer Right Lamp Module Row 5, Column 3
Resulting Problem(s)	Many warnings will be non-functional. Some controls will not function as normal. Steering will bleed as soon as truck comes to a complete stop and the engine stops running without the keyswitch being turned off.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. The primary correction for this fault is to correct the wiring from the keyswitch to the Interface Module. 2. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 3. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 4. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Keyswitch (IM3 G)	0: keyswitch is off and the truck is not moving and the engine is not running 1: keyswitch is on or the truck is moving or the engine is running
Steering Bleed Valve	Status - Open Load: Expected with key switch on. No problem. Otherwise, unexpected. Must troubleshoot. Status - Normal: Expected only with key off and steering bleed in process. Troubleshoot if found with key on. Status - Shorted to Ground: Unexpected. Troubleshoot. Status - Overload: Unexpected. Troubleshoot. 0: steering bleed valve is off. This is expected in normal running operation. 1: steering bleed valve is on. This is expected after key off initiates steering bleed operation. Fault(s): A253, A262, A263, A358
IM Maintenance Lamp (IM1 X)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A363

Related circuit diagram



Fault Code A246: Payload meter reports truck overload

Operator Action	Speed Limit
Fault Code	A246
Description	Payload meter reports truck overload.
Fault Conditions	Sets if PLM3 reports an overloaded truck. Resets if PLM3 clears the overloaded truck report.
Operator Alerting System Response	PLM3 reading only
Resulting Problem(s)	Truck could be damaged by operating in overload condition.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table	
1. This fault is entirely contained within PLM3 (payload meter). The panel and communication to it must be working in order for IM to be aware of the fault. Normal corrective measures would be to correct the loading process. 2. Any faults within the payload meter system that might affect its accuracy are developed and reported by PLM3.	
Parameter	Expected State and/or Related Fault(s)

Related circuit diagram

None.

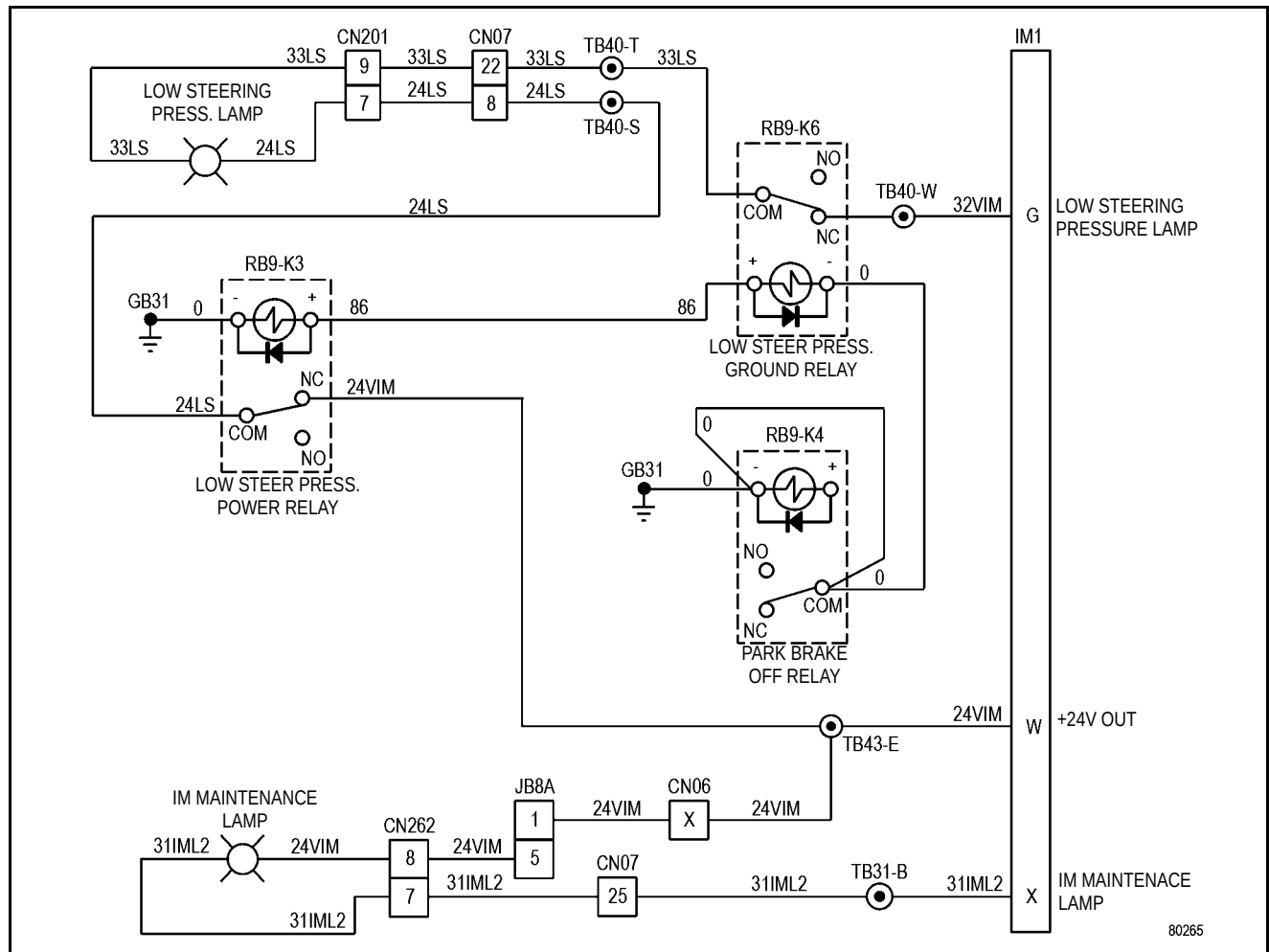
Fault Code A249: Red warning lamp output used to drive the steering bleed fault lamp is shorted

Operator Action	Go to Shop
Fault Code	A249
Description	Red warning lamp output used to drive the steering bleed fault lamp is shorted.
Fault Conditions	Sets if the red warning lamp feedback voltage is high for 400 ms with the red warning lamp on. Resets at power down.
Operator Alerting System Response	Amber "IM Maintenance" Lamp No Buzzer Right Lamp Module Row 5, Column 3
Resulting Problem(s)	Warnings will be audible only.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table	
1. Check the wiring and connections between the IM and low steering/bleed fault lamp. The red warning lamp feedback connection is made internally. 2. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 3. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 4. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Red Warning Lamp (IM1 G)	0: red lamp is not on 1: red lamp is on Fault(s): A256
Red Warning Lamp Feedback (IM1 G)	Above 3.75 volts for 400 ms with the lamp on: shorted Less than 3.75 volts for 400 ms with the lamp on: normal
Steering Bleed Fault Lamp (IM1 G)	Status - Open Load with Keyswitch On: Normal: Expected Status - Open Load with Keyswitch Off: Unexpected, Troubleshoot Status - Normal with Keyswitch On::Unexpected, Troubleshoot Status - Normal with Keyswitch Off: Normal, Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground)

IM Maintenance Lamp (IM1 X)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A363
--------------------------------	--

Related circuit diagram



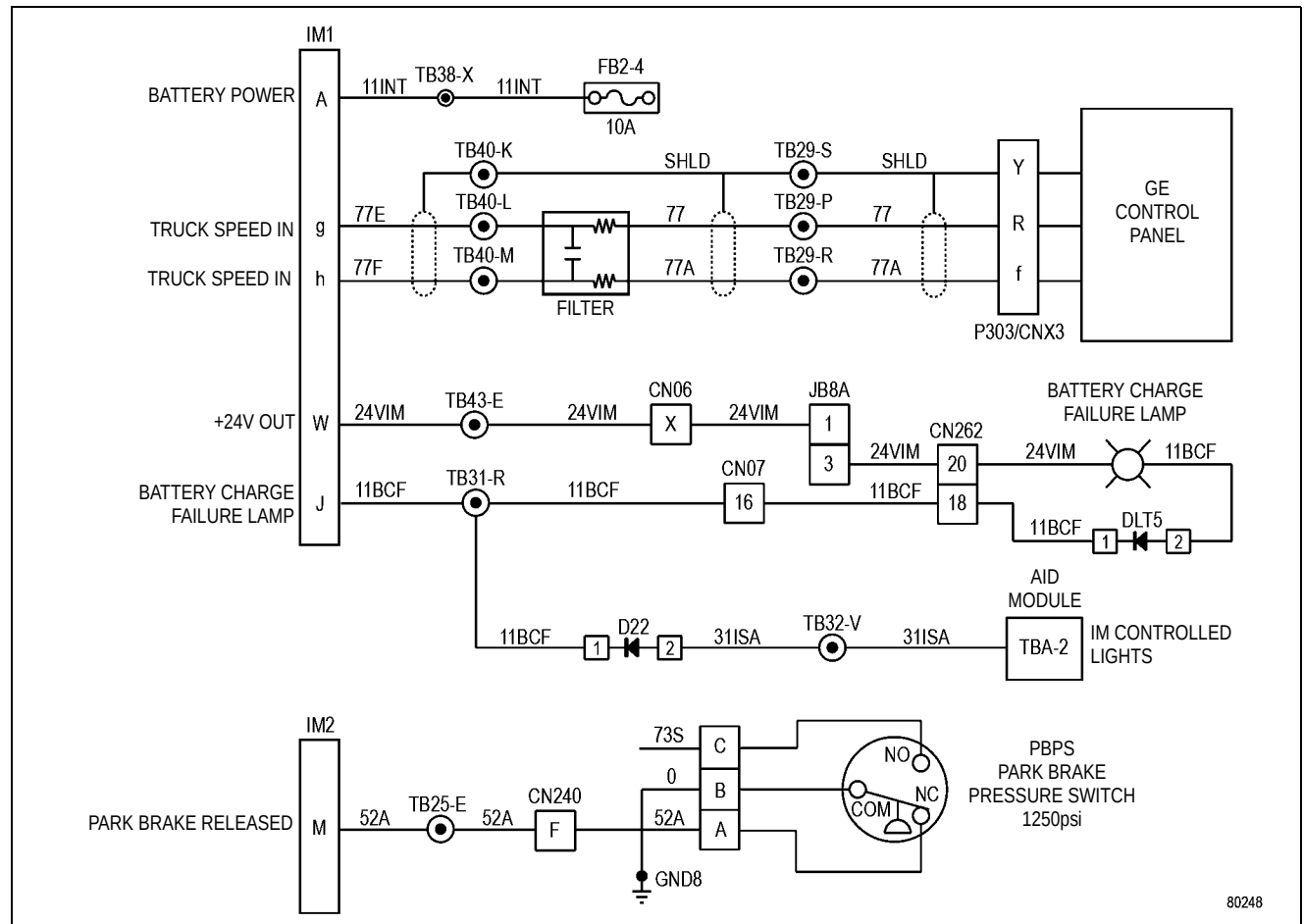
Fault Code A250: Battery voltage is low with the truck parked

Operator Action	Charge Batteries
Fault Code	A250
Description	Battery voltage is low with the truck parked.
Fault Conditions	Sets if battery voltage with the engine not running, truck not moving, crank state not sensed, and not in steering bleed is below 23 volts for 5 seconds. Engine not running is less than 300 rpm. Truck not moving is truck speed equal to zero or if truck speed is bad, park brake is set. Crank state sensed is battery less than 18 volts for 1 sec with engine less than 600 rpm. Park brake set is defined as closure of the park brake set pressure switch and opening of the park brake release pressure switch. Resets if voltage recovers to 25.5 volts.
Operator Alerting System Response	Red "Battery Charging System Failure" Lamp Buzzer Right Lamp Module Row 5, Column 1
Resulting Problem(s)	Voltage may continue to drop and cause improper operation of the many electrical controls on the truck.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table	
1. The primary correction for this fault is to charge the batteries. 2. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 3. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 4. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Battery Voltage, 24 Volt (IM1 A)	<23.0 volts: Too low to continue operation. >23.0 volts: OK to continue operation. <18 volts: cranking is sensed. Fault(s): A153, A154, A155
Engine Speed [RPM] (CAN/J1939) (IM1 q,r,s)	Less than 600 RPM: engine not being cranked Less than 300 RPM: engine is not running Greater than 300 RPM for 4 seconds: engine is running Fault(s): A184
Truck Speed [kph] (IM1 g,h)	0: truck is not moving Greater than 0: truck is moving Fault(s): A212

Park Brake Released (IM2 M)	0: parking brake pressure switch indicating high pressure and a released parking brake 1: parking brake pressure switch indicating low pressure and an applied parking brake
Steering Bleed Valve	0: steering bleed valve is off 1: steering bleed valve is on
Battery Charging Failure Lamp (IM1 J)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A353

Related circuit diagram



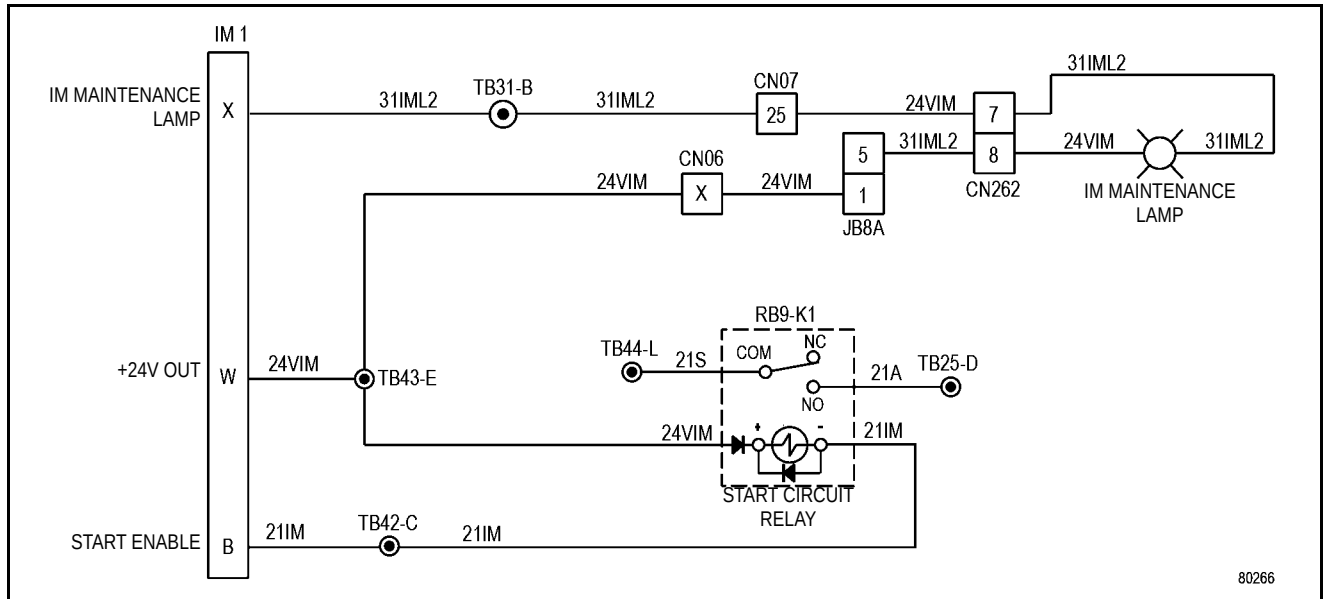
Fault Code A252: Start enable output circuit is either open or shorted to ground

Operator Action	None
Fault Code	A252
Description	Start enable output circuit is either open or shorted to ground.
Fault Conditions	Sets if the start enable relay circuit is open or short to ground for 2 seconds. Resets only at power down.
Operator Alerting System Response	Amber "IM Maintenance" Lamp No Buzzer Right Lamp Module Row 5, Column 3
Resulting Problem(s)	Starting will either not be possible, or will lack the protections that IM provides for the start enable circuit (excessive cranking, selector switch position, J1939 OK, and engine speed 0 to start and not above 400 rpm at finish).
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 2. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Start Enable (IM 1B)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected. No problem. Status - Shorted to Ground: Unexpected. Troubleshoot. Status - Overload: Unexpected. Troubleshoot. 0: One of several interlocking situations exist to prevent cranking (excessive cranking history, selector switch in wrong position, engine red light, J1939 not ok, engine speed either not 0 to begin cranking or over 400 rpm while cranking). 1. No interlocking situations exist to prevent cranking. Fault(s): A350
Steering Bleed Valve	0: steering bleed valve is off 1: steering bleed valve is on
IM Maintenance Lamp (IM1 X)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A363

Related circuit diagram



Fault Code A253: Steering bleed circuit is not open while running

Operator Action	None
Fault Code	A253
Description	Steering bleed circuit is not open while running.
Fault Conditions	Sets if the steering bleed circuit is not open for 2 seconds, beginning 11 seconds after the keyswitch is turned on. Resets only at power down.
Operator Alerting System Response	Amber "IM Maintenance" Lamp No Buzzer Right Lamp Module Row 5, Column 3
Resulting Problem(s)	Steering bleed down will be possible during normal running if IM or some of its inputs should also malfunction.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 2. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Steering Bleed Valve (IM1 P)	Status - Open Load: Expected with key switch on. No problem. Otherwise, unexpected. Must troubleshoot. Status - Normal: Expected only with key off and steering bleed in process. Troubleshoot if found with key on. Status - Shorted to Ground: Unexpected. Troubleshoot. Status - Overload: Unexpected. Troubleshoot. 0: steering bleed valve is off. This is expected in normal running operation. 1: steering bleed valve is on. This is expected after key off initiates steering bleed operation. Fault(s): A262, A263, A358
Keyswitch (IM3 G)	0: keyswitch is off and the truck is not moving and the engine is not running 1: keyswitch is on or the truck is moving or the engine is running Fault(s): A240
IM Maintenance Lamp (IM1 X)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A363

[illegible]

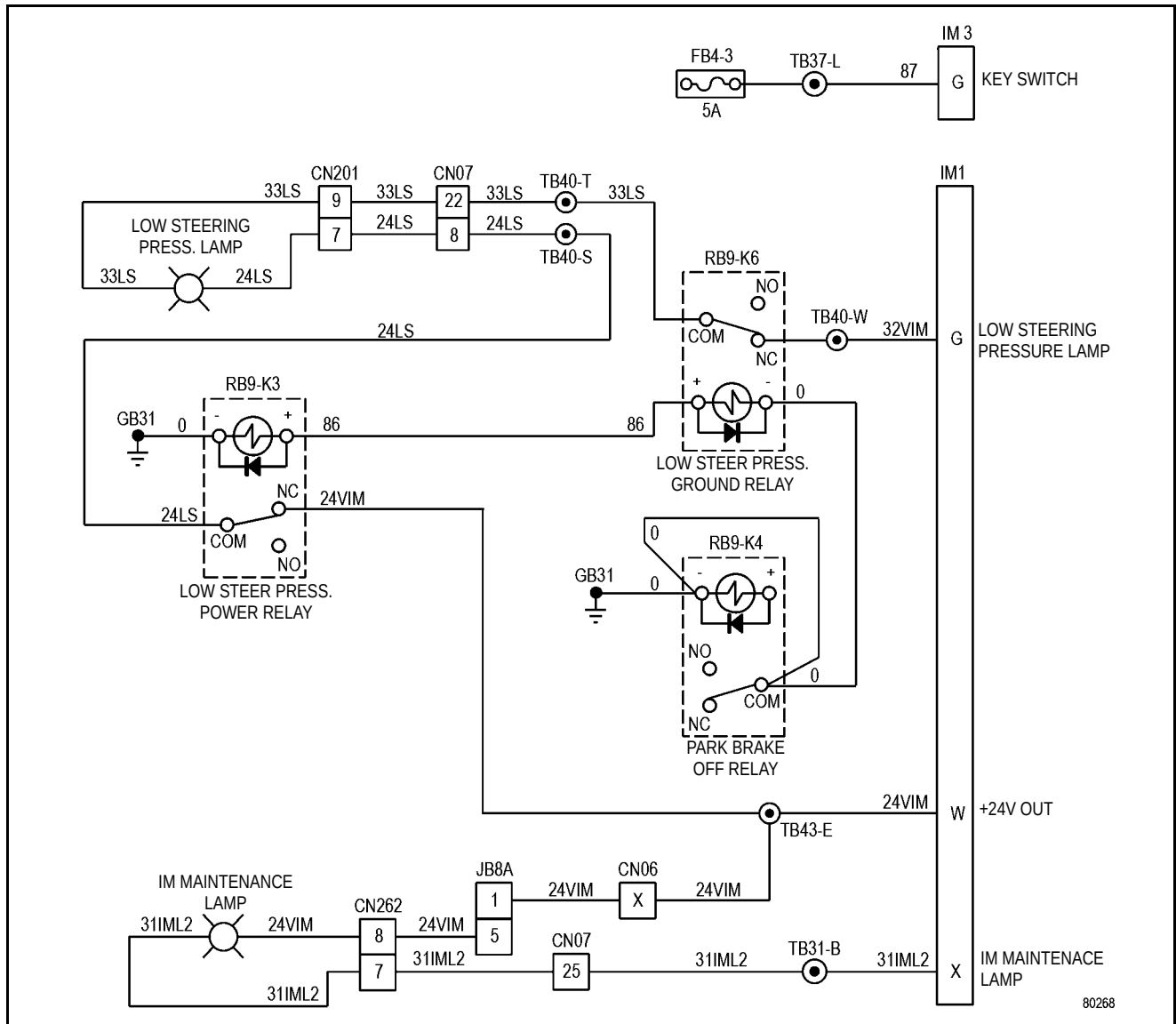
Fault Code A256: Red warning lamp output used to drive the steering bleed fault lamp is open

Operator Action	Go to Shop
Fault Code	A256
Description	Red warning lamp output used to drive the steering bleed fault lamp is open.
Fault Conditions	Sets if red lamp stays open for 1 second with steering bleed off. (The red lamp is open if feedback voltage stays low for 400 ms with red lamp off.) Resets if red lamp returns to normal or steering bleed operates for 1 sec. (The red lamp is normal if feedback voltage goes high for 400 ms with red lamp off.)
Operator Alerting System Response	Amber "IM Maintenance" Lamp No Buzzer Right Lamp Module Row 5, Column 3
Resulting Problem(s)	Warnings will be audible only.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table	
1. Check the wiring and connections between the IM and low steering/bleed fault lamp. The red warning lamp feedback connection is made internally. 2. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 3. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 4. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Red Warning Lamp (IM1 G)	0: red lamp is not on 1: red lamp is on Fault(s): A249
Red Warning Lamp Feedback (IM1 G)	Above 3.75 volts for 400 ms with the lamp on: shorted Less than 3.75 volts for 400 ms with the lamp on: normal
Steering Bleed Fault Lamp (IM1 G)	Status - Open Load with Keyswitch On: Normal: Expected Status - Open Load with Keyswitch Off: Unexpected, Troubleshoot Status - Normal with Keyswitch On::Unexpected, Troubleshoot Status - Normal with Keyswitch Off: Normal, Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground)
Keyswitch (IM3 G)	0: keyswitch is off and the truck is not moving and the engine is not running 1: keyswitch is on or the truck is moving or the engine is running Fault(s): A240

IM Maintenance Lamp (IM1 X)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A363
--------------------------------	--

Related circuit diagram



Fault Code A257: Payload CAN/RPC is not connected

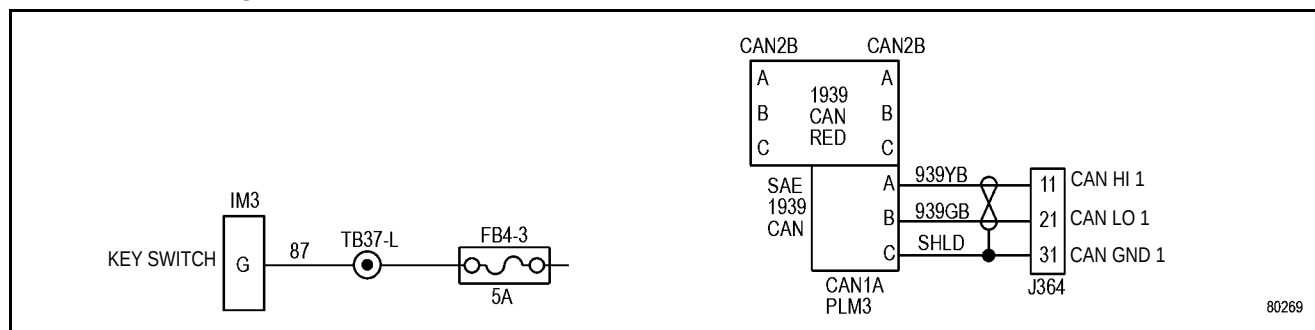
Operator Action	None
Fault Code	A257
Description	Payload CAN/RPC is not connected.
Fault Conditions	Sets if payload has not been updated for 10 seconds and keyswitch has been on at least 11 seconds since power up. Resets when payload is updated.
Operator Alerting System Response	None
Resulting Problem(s)	Payload and overload data is not properly recorded.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. The primary correction for this fault is to correct any external wiring or replace the display or replace the Interface Module.
2. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s).
3. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items.
4. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.

Parameter	Expected State and/or Related Fault(s)
CAN/RPC (IM1 j,k,i)	No communication for 10 seconds if keyswitch has been on for 11 seconds: CAN/RPC not connected. Successful read of payload: J1939 is connected.
Keyswitch (IM3 G)	0: keyswitch is off and the truck is not moving and the engine is not running 1: keyswitch is on or the truck is moving or the engine is running Fault(s): A240

Related circuit diagram



NOTES

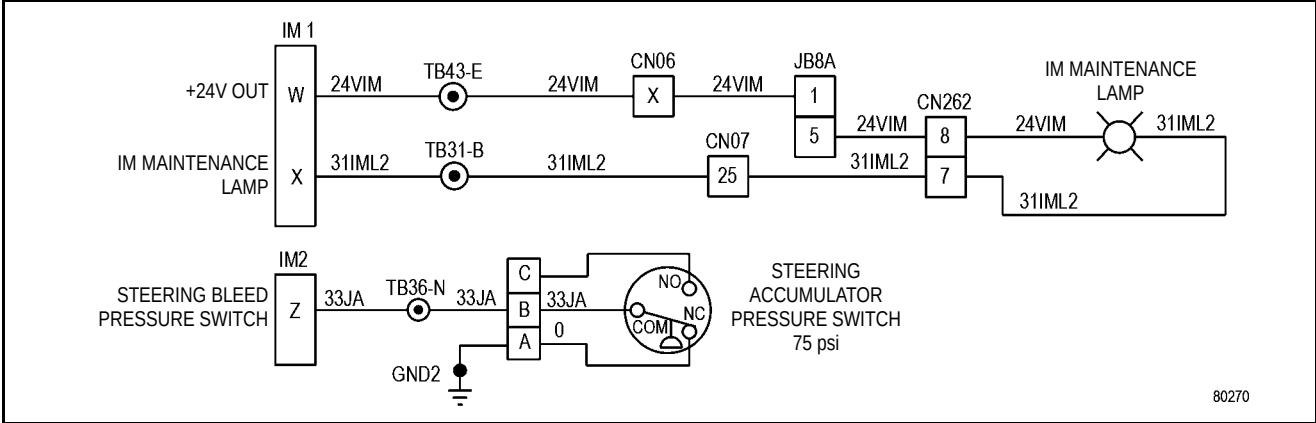
Fault Code A258: Steering accumulator bleed pressure switch circuit is defective

Operator Action	Go to Shop
Fault Code	A258
Description	Steering accumulator bleed pressure switch circuit is defective.
Fault Conditions	Sets if switch input continues to indicate steering accumulator bled after 90 seconds of engine running above 300 rpm. Resets at power down.
Operator Alerting System Response	Amber "IM Maintenance" Lamp No Buzzer Right Lamp Module Row 5, Column 3
Resulting Problem(s)	Indication of steering bleed is unreliable.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 2. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Steering Accum Bleed Pressure Sw (IM2 Z)	0: Accumulator is bled. 1: Accumulator is not completely bled. Fault(s): A280
Engine Speed [RPM] (CAN/J1939) (IM1 q,r,s)	0: engine is not running Greater than 300 rpm: engine is running Fault(s): A184
IM Maintenance Lamp (IM1 X)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A363

Related circuit diagram



Fault Code A260: Parking brake failure

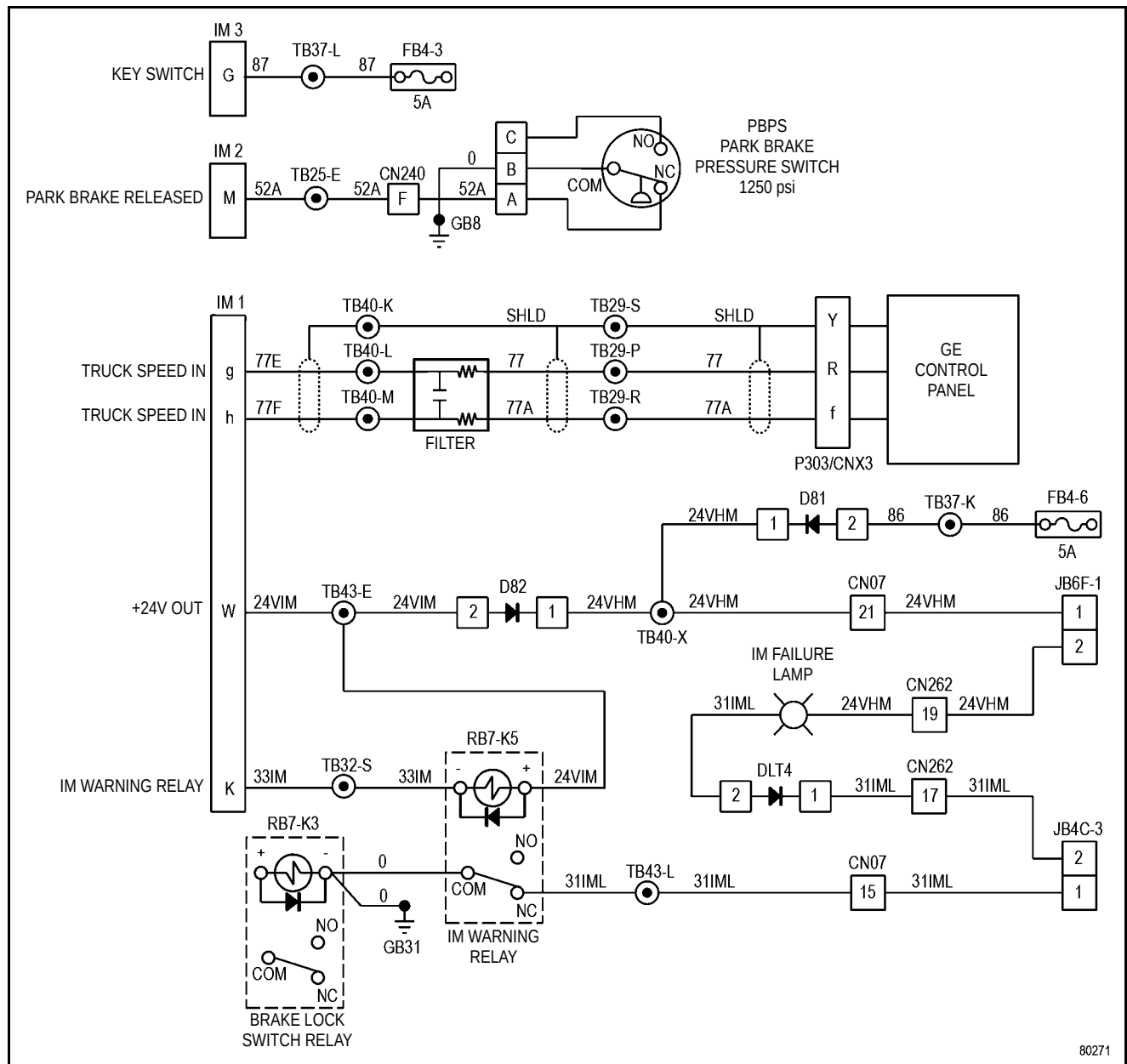
Operator Action	Secure Vehicle
Fault Code	A260
Description	Parking brake failure (assumed if truck rolls after application of parking brake or if parking brake is applied while moving)
Fault Conditions	Two cases are tested: After truck is stopped completely (defined as park brake set and truck not moving for 1.5 seconds and reset only by park brake release), and truck rolling (defined as not stopped completely). Truck Rolling After Stop: Sets if truck rolls 2 kph for 1.5 seconds after the truck is stopped completely. This case resets if conditions change. Parking Brake Application While Moving: Sets if truck's speed is 0.8 kph or over for 1.5 seconds with parking brake not released (parking brake released is defined as parking brake release switch being on and parking brake set switch being off). This case resets only at power down.
Operator Alerting System Response	Red "System Failure" Lamp No Buzzer Right Lamp Module Row 4, Column 5
Resulting Problem(s)	Parking brake may not hold truck.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 2. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Park Brake Released (IM2 M)	0: parking brake pressure switch indicating high pressure and a released parking brake (NOTE: this input is ignored if keyswitch is off). 1: parking brake pressure switch indicating lower pressure and an applied parking brake Fault(s): A213
Truck Speed [kph] (IM1 g,h)	Less than 2.0 kph: truck is not moving in case: Truck Rolling After Stop Less than 0.8 kph: truck is not moving in case: Park Brake Applied While Moving Greater than 2.0 kph: truck is moving in case: Truck Rolling After Stop Greater than 0.8 kph: truck is moving in case: Park Brake Applied While Moving Fault(s): A212
Keyswitch (IM3 G)	0: keyswitch is off and the truck is not moving and the engine is not running 1: keyswitch is on Fault(s): A240

IM Warn Relay (IM1 K)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Relay Off (IM output at B+) - Lamp is on 1: Relay On (IM output at ground) - Lamp is off Fault(s): A354
-----------------------	---

Related circuit diagram



Fault Code A262: Steering bleed valve circuit open during shutdown

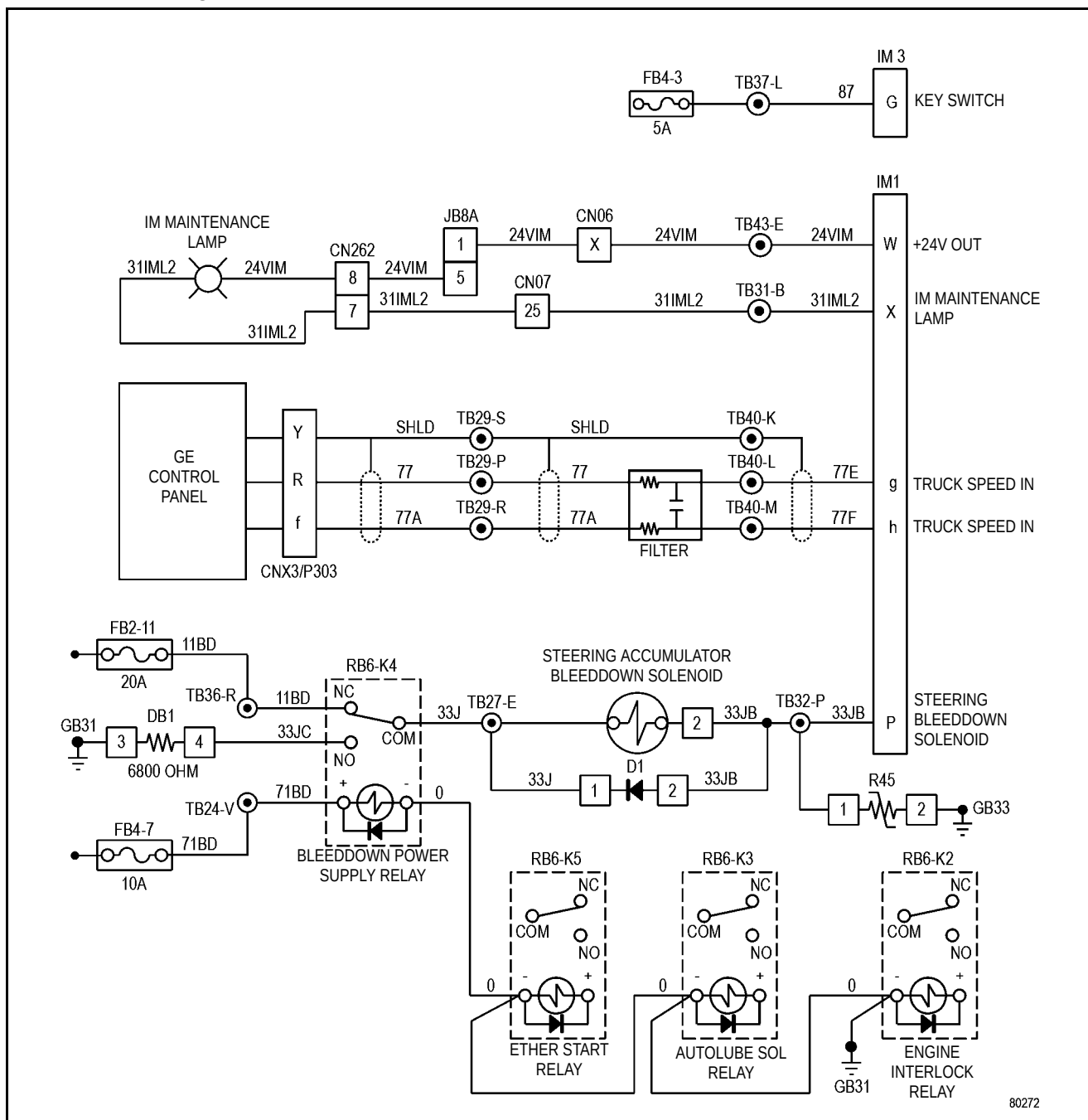
Operator Action	Go to Shop
Fault Code	A262
Description	Steering bleed valve circuit open during shutdown.
Fault Conditions	Sets if steering bleed valve circuit is open for 1 second when the key switch has been turned off, the engine no longer runs, and the truck is stopped (drive system control power turned off). Energization of the steering bleed valve is delayed for 2 seconds to provide a valid test. Resets at power down.
Operator Alerting System Response	Amber "IM Maintenance" Lamp No Buzzer Right Lamp Module Row 5, Column 3
Resulting Problem(s)	Steering will not bleed down properly if circuit is open.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 2. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Steering Bleed Valve (IM1 P)	Status - Open Load: Expected with key switch on. No problem. Otherwise, unexpected, must troubleshoot. Status - Normal: Expected only with key off and steering bleed in process. Troubleshoot if found with key on. Status - Shorted to Ground: Unexpected. Troubleshoot. Status - Overload: Unexpected. Troubleshoot. 0: steering bleed valve is off. This is expected in normal running operation. 1: steering bleed valve is on. This is expected after key off initiates steering bleed operation. Fault(s): A253, A263, A358
Keyswitch (IM3 G)	0: keyswitch is off and the truck is not moving and the engine is not running 1: keyswitch is on or the truck is moving or the engine is running Fault(s): A240
Engine Speed [RPM] (CAN/J1939) (IM1 q,r,s)	0: engine is not running Greater than 0: engine is running Fault(s): A184
Truck Speed [kph] (IM1 g,h)	0: truck is not moving Greater than 0: truck is moving Fault(s): A212

IM Maintenance Lamp (IM1 X)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A363
--------------------------------	--

Related circuit diagram

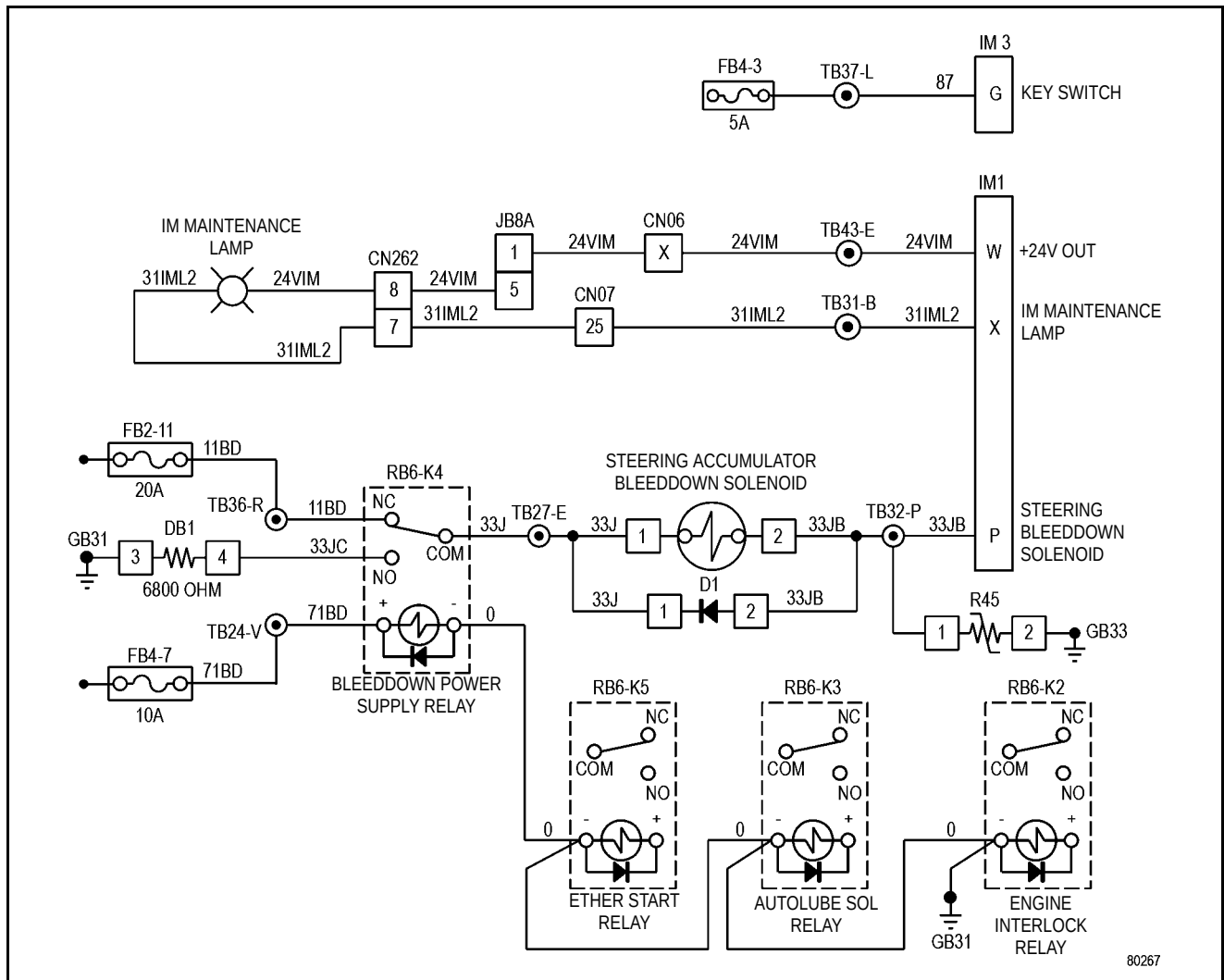


Fault Code A263: Steering bleed circuit is shorted to ground

Operator Action	Go to Shop
Fault Code	A263
Description	Steering bleed circuit is shorted to ground.
Fault Conditions	Sets if the steering bleed circuit is shorted to ground for 2 seconds, beginning 11 seconds after the keyswitch is turned on. Resets only at power down.
Operator Alerting System Response	Amber "IM Maintenance" Lamp No Buzzer Right Lamp Module Row 5, Column 3
Resulting Problem(s)	Steering may bleed all the time and will not be controllable by IM.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table	
1. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 2. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Steering Bleed Valve (IM1 P)	Status - Open Load: Expected with key switch on. No problem. Otherwise, unexpected. Must troubleshoot. Status - Normal: Expected only with key off and steering bleed in process. Troubleshoot if found with key on. Status - Shorted to Ground: Unexpected. Troubleshoot. Status - Overload: Unexpected. Troubleshoot. 0: steering bleed valve is off. This is expected in normal running operation. 1: steering bleed valve is on. This is expected after key off initiates steering bleed operation. Fault(s): A262, A263, A358
Keyswitch (IM3 G)	0: keyswitch is off and the truck is not moving and the engine is not running 1: keyswitch is on or the truck is moving or the engine is running Fault(s): A240
IM Maintenance Lamp (IM1 X)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A363

Related circuit diagram



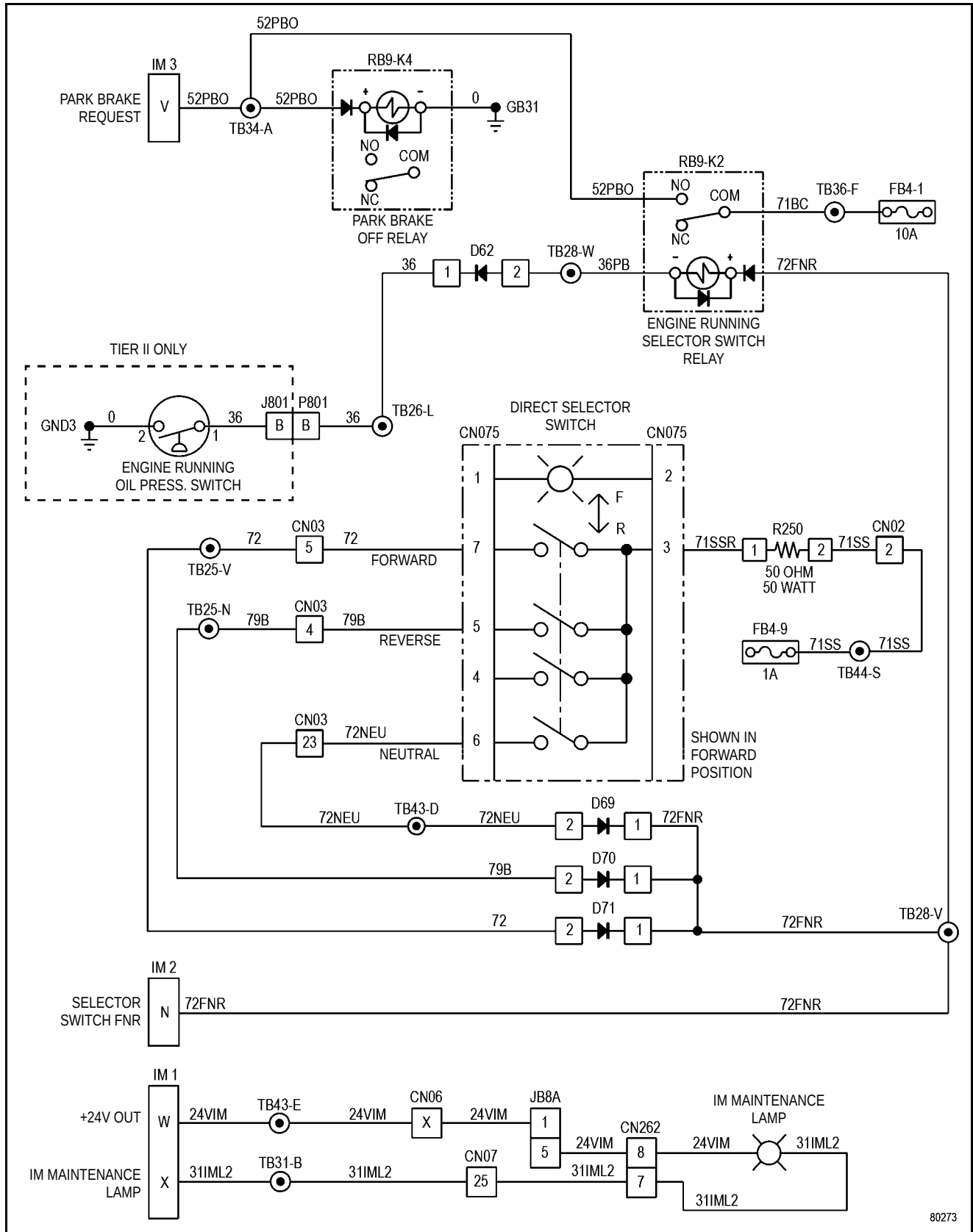
Fault Code A264: Parking brake relay circuit is defective

Operator Action	None
Fault Code	A264
Description	Parking brake relay circuit is defective.
Fault Conditions	Sets if parking brake request is not on when selector lever is not in forward, neutral, or reverse, or engine has not been running for 15 seconds with the J1939 data link to the engine OK. Sets if parking brake request is on when selector lever is in forward, neutral, or reverse and engine has been running for 15 seconds. Resets only at power down.
Operator Alerting System Response	Amber "IM Maintenance" Lamp No Buzzer Right Lamp Module Row 5, Column 3
Resulting Problem(s)	Parking brake may not set or release when expected.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 2. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Park Brake Request (IM 3V)	0: Park Brake Set is Requested (shifter is not in the forward, neutral, or reverse positions or the engine oil pressure is below the setpoint of the engine oil pressure switch) 1: Park Brake Release is Requested (shifter is in the forward, neutral, or reverse positions and the engine oil pressure is above the setpoint of the engine oil pressure switch) Fault(s): A213, A214
Selector Switch (FNR) (IM 2N)	0: shifter is not in forward, neutral, or reverse 1: shifter is in forward, neutral, or reverse Fault(s): A271, A303
Engine Speed [RPM] (CAN/J1939) (IM1 q,r,s)	0: engine is not running Greater than 0: engine is running Fault(s): A184
IM Maintenance Lamp (IM1 X)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A363

Related circuit diagram



80273

Fault Code A265: Service brake failure

Operator Action	Stop: Park
Fault Code	A265
Description	A service brake failure exists. Operation of either brake lock or brake pedal does not produce brake light switch operation.
Fault Conditions	Sets if brake lock is on or if front brake pressure is above 250 psi for 2 seconds and the service brake pressure switch does not turn on if the engine has been running for 90 seconds. Resets if conditions change for 2 seconds.
Operator Alerting System Response	Amber, dimmable "Service Brake" Lamp flashes Left Lamp Module Row 2, Column 3 Red "System Failure" Lamp No Buzzer Right Lamp Module Row 4, Column 5
Resulting Problem(s)	Service brake may not respond to brake lock switch or pedal. Brake light switch and brake lights may not respond to brake application.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 2. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Brake Lock Input (IM2 i)	0: brake lock valve is off 1: brake lock valve is on
Brake Pressure (kPa) (IM3 b,p)	Less than 1724 kPa (250 psi): front brake not applied Greater than 1724 kPa (250 psi): front brake applied Fault(s): A201, A205
Service Brake Pressure Switch (IM3 C)	0: Service brake not applied 1: Service brake applied
Engine Speed [RPM] (CAN/J1939) (IM1 q,r,s)	Less than 400 rpm: engine is not running Greater than 400 rpm for 15 seconds: engine is running Fault(s): A184

IM Warn Relay (IM1 K)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Relay Off (IM output at B+) - Lamp is on 1: Relay On (IM output at ground) - Lamp is off Fault(s): A354
Service Brake Lamp (IM1 N)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A357

[illegible]

NOTES

Fault Code A270: Brake lock switch power supply is not on when required

Operator Action	Go to shop now
Fault Code	A270
Description	Brake lock switch power supply is not on when required
Fault Conditions	Sets if brake lock switch power supply is off when not in steering bleed, truck speed above 1 kph and park brake not set, or with park brake not requested and the engine running 400 rpm for 15 seconds and the delayed shutdown relay not on. Park brake set is defined as closure of the park brake set pressure switch and opening of the park brake release pressure switch. Resets at beginning of steering bleed.
Operator Alerting System Response	Amber "IM Maintenance" Lamp No Buzzer Right Lamp Module Row 5, Column 3
Resulting Problem(s)	Missing brake lock power supply makes it impossible to use the brake lock in a normal manner for loading a dumping operations.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table	
1. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). 2. Since there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Brake Lock Switch Power Supply (IM3 L)	0: no power available for brake lock switch 1: power is available for brake lock switch Fault(s): A272
Truck Speed [kph] (IM1 g,h)	Less than 1 kph: truck is not moving Greater than 1 kph: truck is moving Fault(s): A212
Engine Speed [RPM] (CAN/J1939) (IM1 q,r,s)	Less than 400 rpm: engine is not running Greater than 400 rpm for 15 seconds: engine is running Fault(s): A184
IM Maintenance Lamp (IM1 X)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A363
Park Brake Released (IM 2M)	0: parking brake pressure switch indicating high pressure and a released parking brake 1: parking brake pressure switch indicating low pressure and an applied parking brake Fault(s): A214

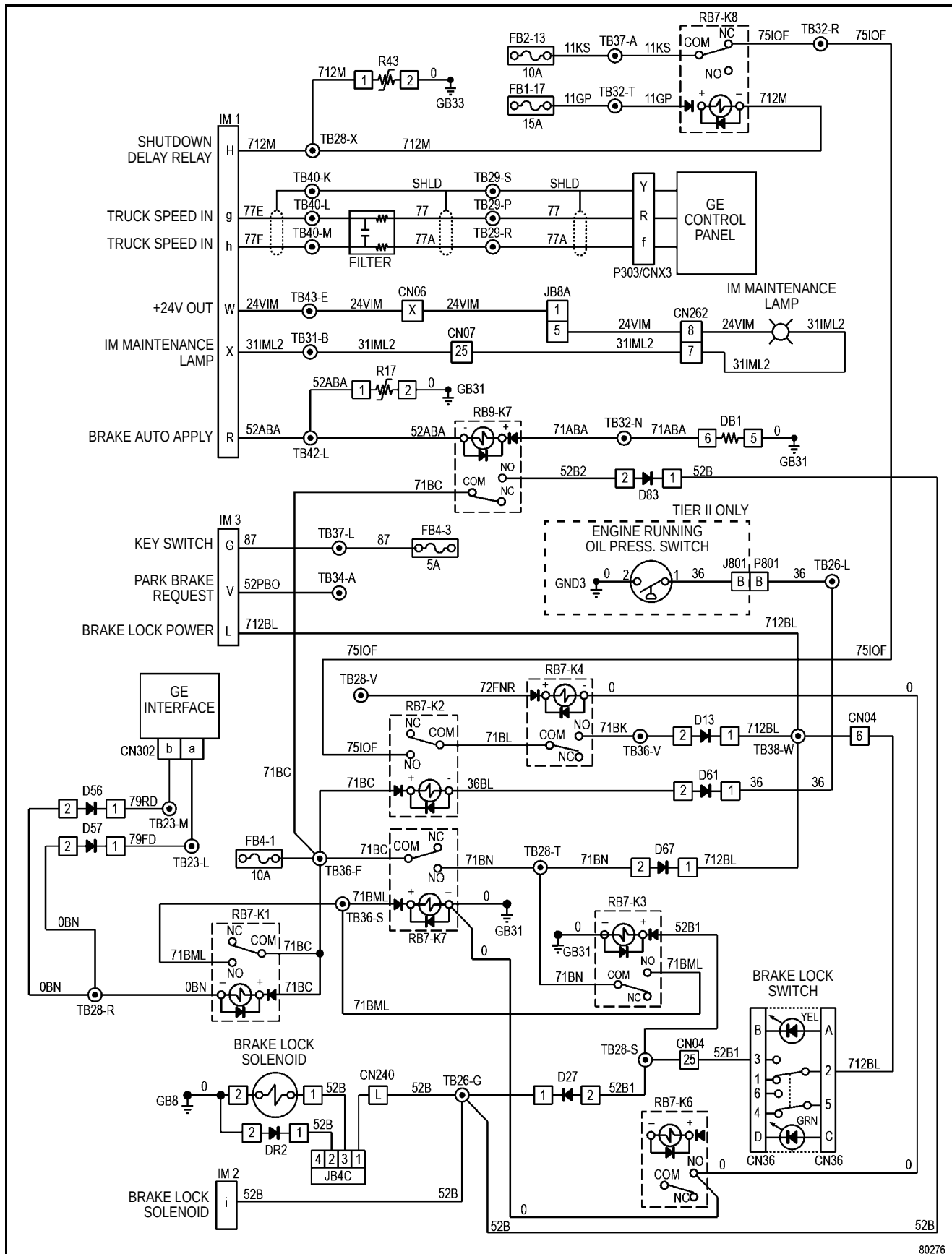
Fault Code A272: Brake lock switch power supply is not off when required

Operator Action	Go to shop now
Fault Code	A272
Description	Brake lock switch power supply not off when required.
Fault Conditions	Sets if brake lock switch power supply is not off when brake lock is off, and truck speed is less than 0.5 kph and park brake request is on or engine speed has been 0 for 15 seconds and J1939 data link is OK, or the shutdown relay is on or the keyswitch is off, for 3 seconds. Resets at power down. Alternatively sets if brake lock switch power supply is not off when brake lock is on and brake auto apply is not on and truck speed is less than 0.5 kph and park brake request is on or engine speed has been 0 for 15 seconds and J1939 data link is OK, or the shutdown relay is on or the keyswitch is off, for 1 second. Resets in 1 second if conditions change.
Operator Alerting System Response	Amber "IM Maintenance" Lamp No Buzzer Right Lamp Module Row 5, Column 3
Resulting Problem(s)	Defective brake lock power supply circuitry makes it impossible to use the brake lock in a normal manner for loading and dumping operations.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table	
1. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). 2. Since there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Brake Lock Switch Power Supply (IM3 L)	0: no power available for brake lock switch 1: power is available for brake lock switch Fault(s): A270
Brake Lock (IM2 i)	0: brake lock not on 1: brake lock is on
Auto Brake Apply Relay (IM1 R)	Status - Open Load: Unexpected if park brake request and keyswitch are on. But expected if park brake request is off or keyswitch is off. Troubleshoot if inconsistent. Status - Normal: Expected if park brake request and keyswitch are on. Troubleshoot if status is normal with park brake request off or keyswitch off. Status - Shorted to ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: normal operation with park brake either on or off. This valve operates only transiently when the park brake is turned on. 1: brake auto apply valve operates for 1.5 seconds after the park brake is turned on if the truck is stopped. If moving, it will not operate until truck speed has dropped to 0.8 kph. Fault(s): A359

IM Maintenance Lamp (IM1 X)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A363
Truck Speed [kph] (IM1 g,h)	Less than 1 kph: truck is not moving Greater than 1 kph: truck is moving Fault(s): A212
Engine Speed [RPM] (CAN/J1939) (IM1 q,r,s)	Less than 400 rpm: engine is not running Greater than 400 rpm for 15 seconds: engine is running Fault(s): A184
Park Brake Request (IM3 V)	0: shifter is not in the forward, neutral, or reverse positions or the engine oil pressure is below the setpoint of the engine oil pressure switch 1: shifter is in the forward, neutral, or reverse positions and the engine oil pressure is above the setpoint of the engine oil pressure switch Fault(s): A264
Keyswitch (IM3 G)	0: keyswitch is off and the truck is not moving and the engine is not running 1: keyswitch is on or the truck is moving or the engine is running Fault(s): A240
Shutdown Delay Relay (IM1 H)	0: Turns off when conditions clear or engine speed drops to zero. 1: Turns on when Keyswitch, Direct is off, Cummins Shutdown Delay is on, Park brake is set, Selector Switch is in park, but not J1939 Not Connected, not High Batt Charge Voltage, not Starter Stuck On, not Active Engine Red Light, and engine speed above 300. Fault(s): A286, A292

Related circuit diagram



80276

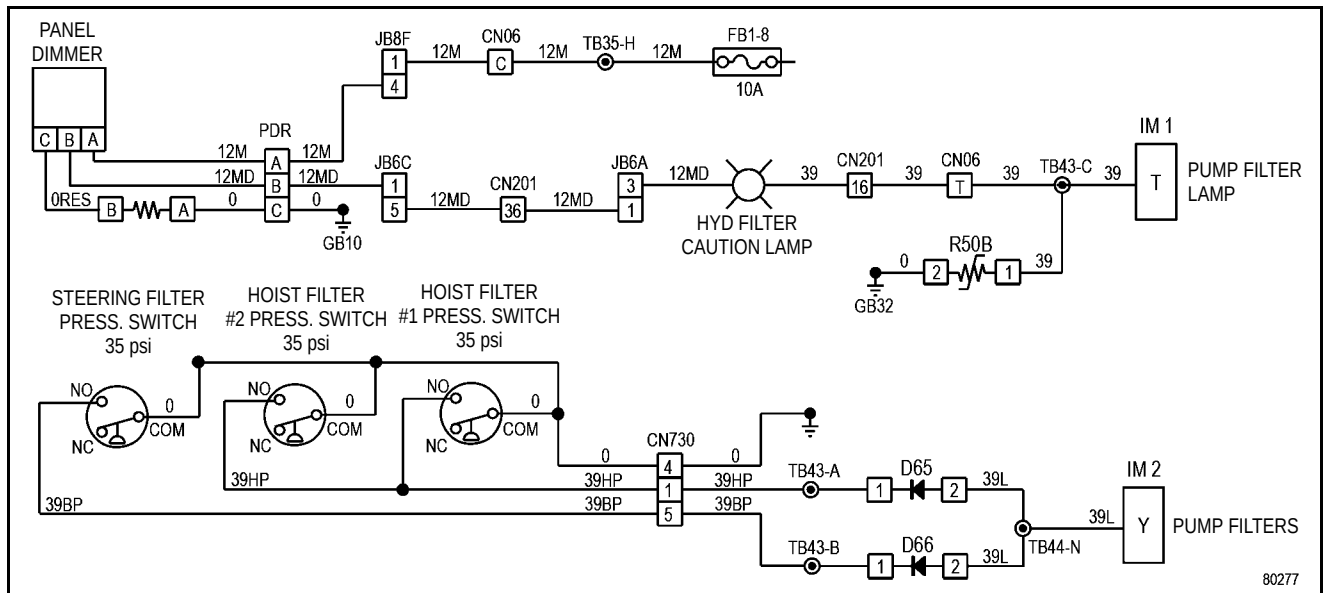
NOTES

Fault Code A273: A fault has been detected in the hoist or steering pump filter pressure switch circuit

Operator Action	None
Fault Code	A273
Description	A fault has been detected in the hoist or steering pump filter pressure switch circuit.
Fault Conditions	Sets if the pump filter pressure switch circuit indicates high pressure across the filters for 2 seconds with the engine not running at least 300 rpm. Resets if pump filter pressure switch circuit indicates normal pressure across the filters or the engine begins to run.
Operator Alerting System Response	Amber "Hydraulic Oil Filter" Lamp Dimmable Left Lamp Module Row 4, Column 2
Resulting Problem(s)	Hydraulic oil filters might plug up and stop filtering without warning.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table	
1. This fault's logic contains more than one parameter, but no additional fault codes are available for them. 2. Since there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Pump Filter Switches (IM2 Y)	0: No hydraulic filter switches open 1: One or more hydraulic filter switches open Fault(s): A101
Engine Speed [RPM] (CAN/J1939) (IM1 q,r,s)	Less than 400 rpm: engine is not running Greater than 400 rpm for 15 seconds: engine is running Fault(s): A184
Pump Filter Switches Lamp (IM1 T)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A361

Related circuit diagram



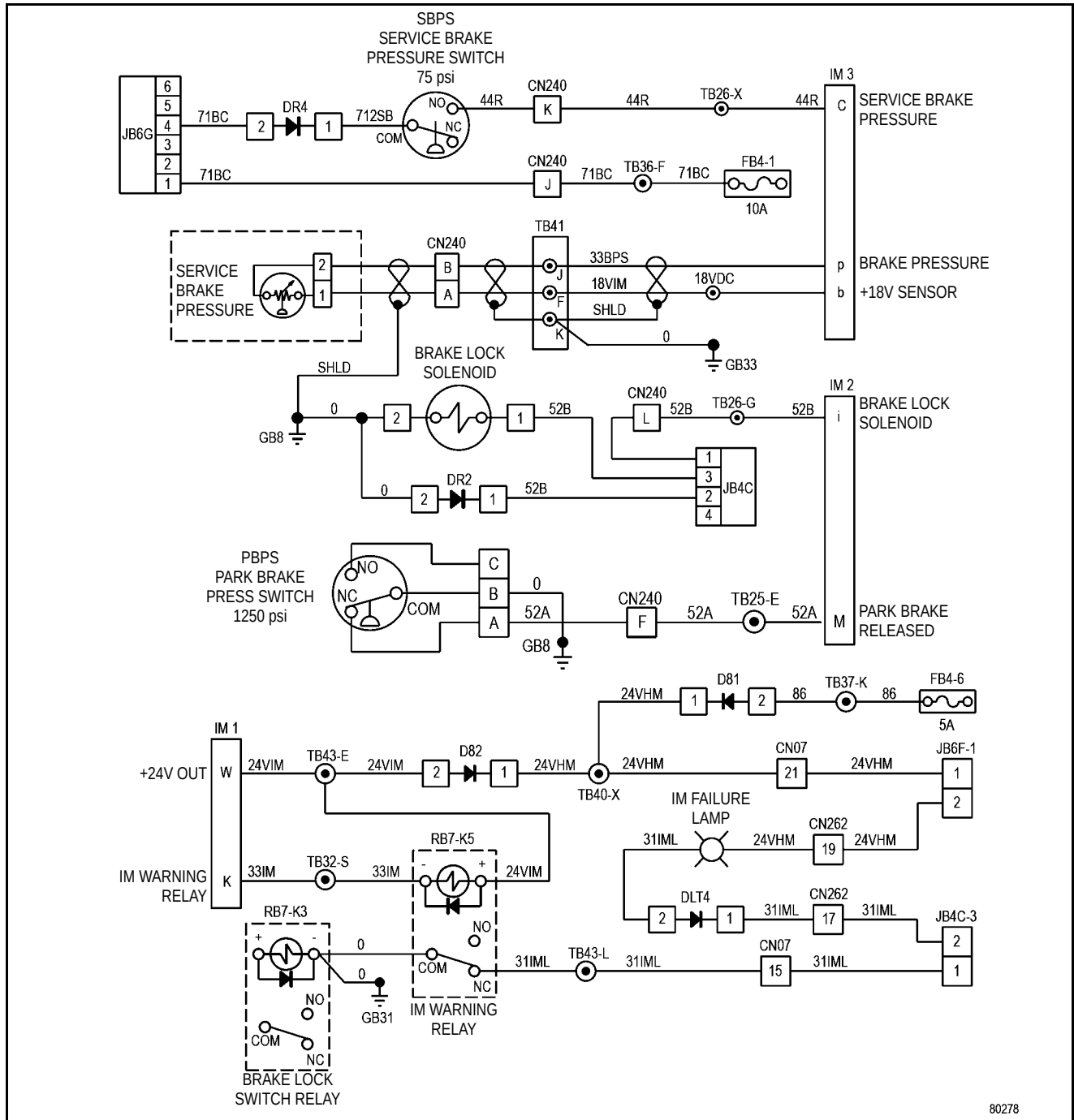
Fault Code A274: A brake setting fault has been detected

Operator Action	Secure vehicle
Fault Code	A274
Description	A brake setting fault has been detected.
Fault Conditions	Sets if both park brake and brake lock are set and brake lock has not been set by brake auto apply for 5 seconds. Park brake set is defined as closure of the park brake set pressure switch and opening of the park brake release pressure switch. Resets after 5 seconds if conditions change.
Operator Alerting System Response	Red "System Failure" Lamp No Buzzer Right Lamp Module Row 4, Column 5
Resulting Problem(s)	Improperly controlled brakes are dangerous to equipment as well as operator.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). 2. Since there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Park Brake Released (IM2 M)	0: parking brake pressure switch indicating high pressure and a released parking brake 1: parking brake pressure switch indicating low pressure and an applied parking brake
Service Brake Pressure Switch (IM3 C)	0: Service brake not applied 1: Service brake applied
Brake Pressure (kPa) (IM3 b,p)	Less than 1724 kPa (250 psi): front brake not applied Greater than 1724 kPa (250 psi): front brake applied Fault(s): A201, A205
Brake Lock (IM2 i)	0: brake lock not on 1: brake lock is on
IM Warn Relay (IM1 K)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Relay Off (IM output at B+) - Lamp is on 1: Relay On (IM output at ground) - Lamp is off Fault(s): A354

Related circuit diagram



80278

Fault Code A275: A starter has been detected as engaged without a cranking attempt

Operator Action	Stop, park, power down, check
Fault Code	A275
Description	A starter has been detected as engaged without a cranking attempt.
Fault Conditions	Sets if either starter is engaged for 2 seconds without crank sense being on and while not in steering bleed. Resets after 2 seconds if conditions change.
Operator Alerting System Response	Amber "Engine Start Fail" Lamp No Buzzer Left Lamp Module Row 1, Column 4 Red "System Failure" Lamp No Buzzer Right Lamp Module Row 4, Column 5
Resulting Problem(s)	Starters may be destroyed by continued operation when stuck on.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

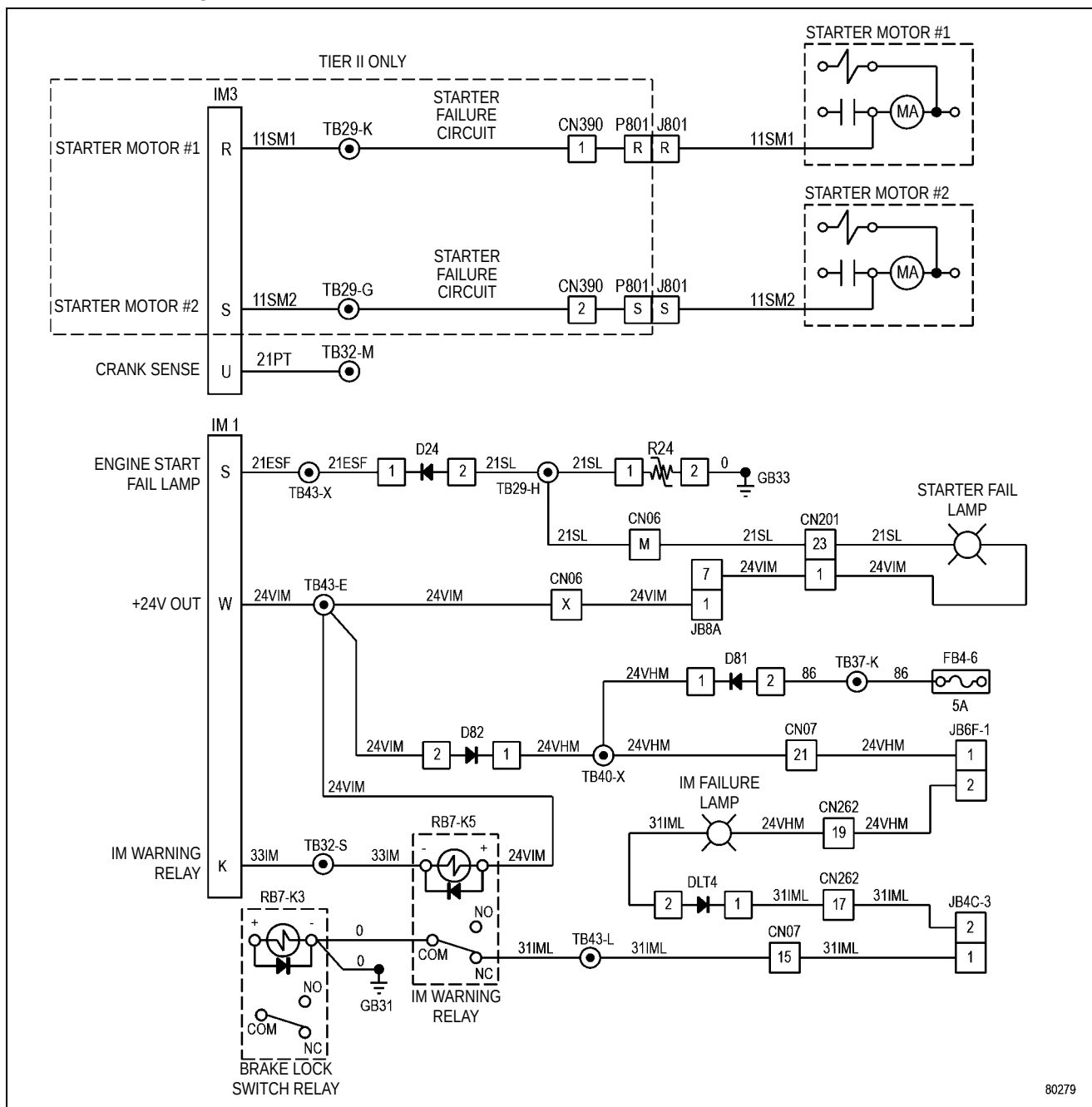
Table

1. The normal corrective measure for this fault is to check the starter motors and engagement to the flywheel.
2. This fault's logic contains more than one parameter, each of which may have its own related fault code(s).
3. Since there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items.
4. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.

Parameter	Expected State and/or Related Fault(s)
Starter Motor 1 Energized (IM3 R)	0: starter motor 1 not energized 1: starter motor 1 energized Fault(s): A152, A316
Starter Motor 2 Energized (IM3 S)	0: starter motor 2 not energized 1: starter motor 2 energized Fault(s): A152, A316
Crank Sense (IM3 U)	0: no cranking in process 1: cranking in process
Steering Bleed Valve	0: steering bleed valve is off 1: steering bleed valve is on
Engine Start Fail Lamp (IM1 S)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A360

IM Warn Relay (IM1 K)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Relay Off (IM output at B+) - Lamp is on 1: Relay On (IM output at ground) - Lamp is off Fault(s): A354
-----------------------	---

Related circuit diagram



Fault Code A276: The drive system data link is not connected

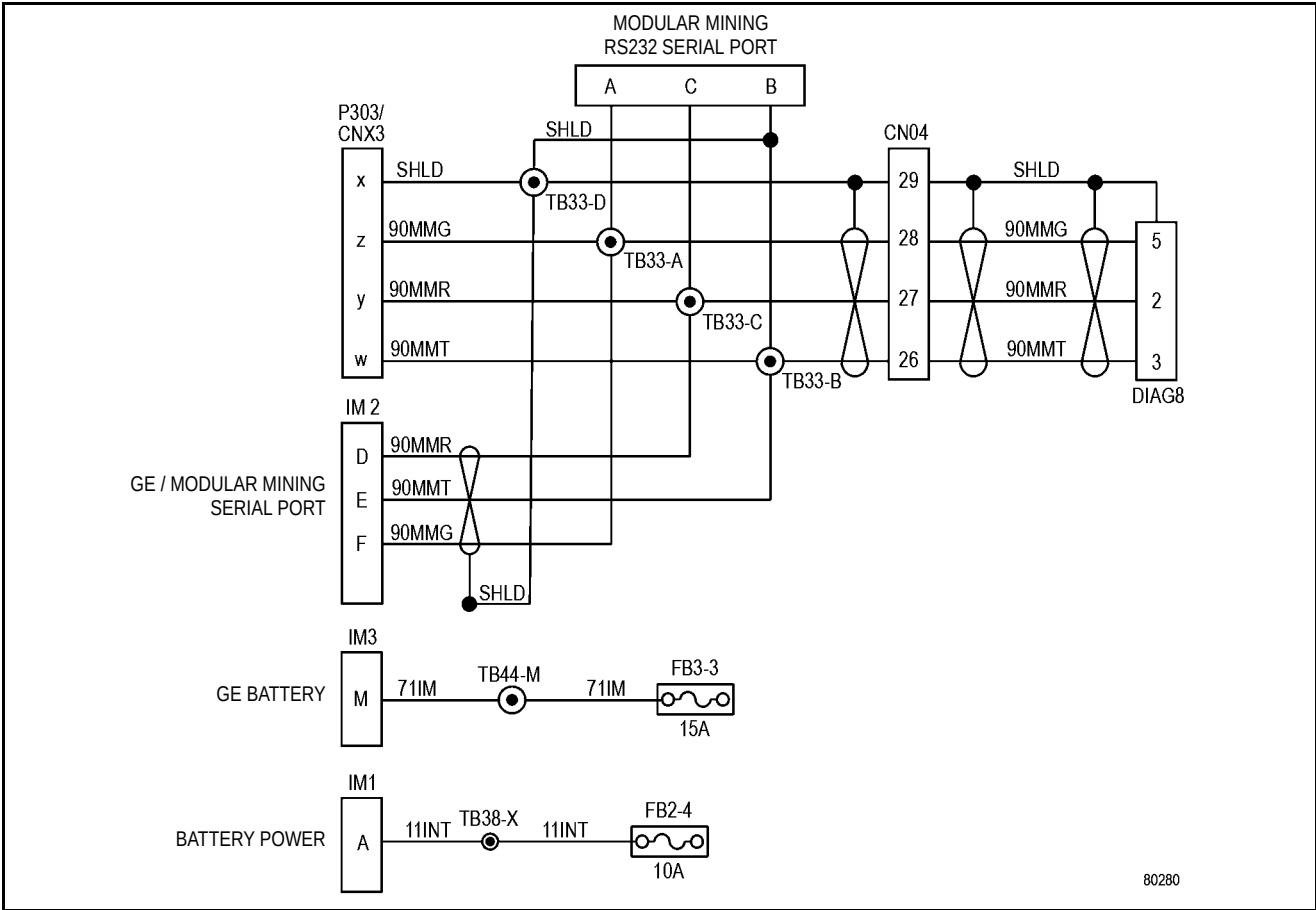
Operator Action	None
Fault Code	A276
Description	The drive system data link is not connected.
Fault Conditions	Sets after 60 seconds of no communication if battery voltage is above 23V and GE serial has been previously detected. Resets 5 seconds after communication is detected.
Operator Alerting System Response	None
Resulting Problem(s)	Details of warnings not available to drivers or KOMTRAX Plus.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. This fault's logic contains more than one parameter, each of which may have its own related fault code(s).
2. Since there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items.
3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.

Parameter	Expected State and/or Related Fault(s)
Drive System Data Link (IM2 D,E,F)	Should operate 20 seconds after initial power up as long as GE control power is on.
Battery Voltage, 24 Volt (IM1 A)	> 23 Volt: Good Reading < 23 Volt: Low Reading Fault(s): A153, A154, A155, A250
Drive System Control Power (IM3 M)	0: power to GE control is off 1: power to GE control is on

Related circuit diagram



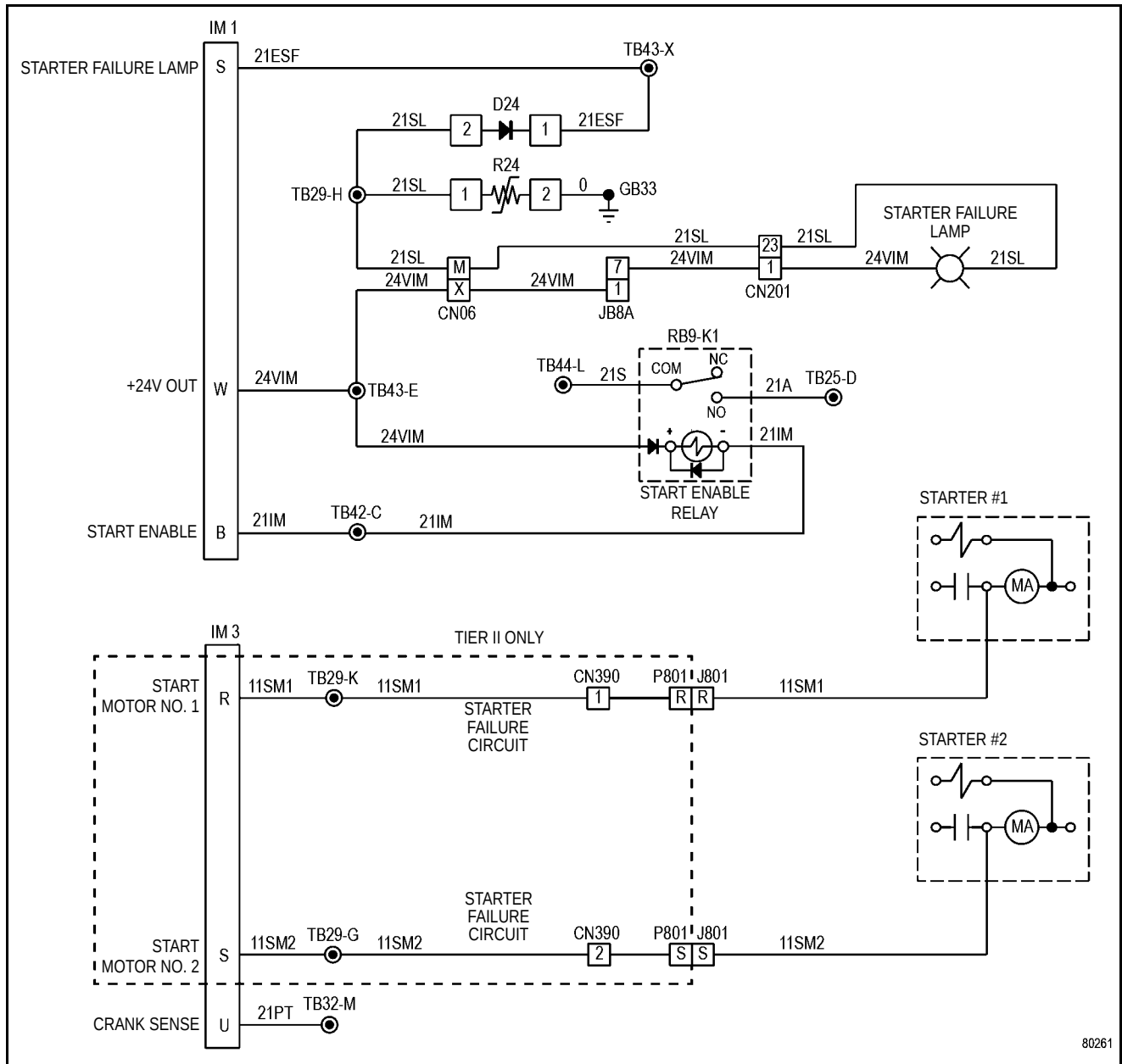
Fault Code A282: The number of excessive cranking counts and jump starts without the engine running has reached 7

Operator Action	Stop, park, power down, check
Fault Code	A282
Description	The number of excessive cranking counts and jump starts without the engine running has reached 7.
Fault Conditions	Count is increased every time cranking reaches 30 seconds continuous and every time a jump start is attempted (engagement of either starter without start enable). A count of 7 disables further start attempts until the power is shut off.
Operator Alerting System Response	Amber "Engine Start Fail" Lamp No Buzzer Left Lamp Module Row 1, Column 4" Will not crank again without shutting down keyswitch power to reset.
Resulting Problem(s)	Excessive cranking destroys starters.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table	
1. The normal corrective measure for this fault is to check and correct engine and/or starting system deficiencies. 2. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). 3. Since there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 4. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Crank Sense (IM3 U)	0: no cranking in process 1: cranking in process
Engine Speed [RPM] (CAN/J1939) (IM1 q,r,s)	Less than 400 rpm: engine is not running Greater than 400 rpm for 15 seconds: engine is running Fault(s): A184
Start Enable (IM1 B)	0: One of several interlocking situations exists to prevent cranking (excessive cranking history, selector switch in wrong position, engine red light, J1939 not ok, engine speed either not 0 to begin cranking or over 400 rpm while cranking). 1: No interlocking situations exist to prevent cranking. Fault(s): A252, A350
Starter Motor 1 Energized (IM3 R)	0: starter motor 1 not energized 1: starter motor 1 energized Fault(s): A152, A223, A275, A316
Starter Motor 2 Energized (IM3 S)	0: starter motor 2 not energized 1: starter motor 2 energized Fault(s): A152, A223, A275, A316

Engine Start Fail Lamp (IM1 S)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A360
-----------------------------------	--

Related circuit diagram

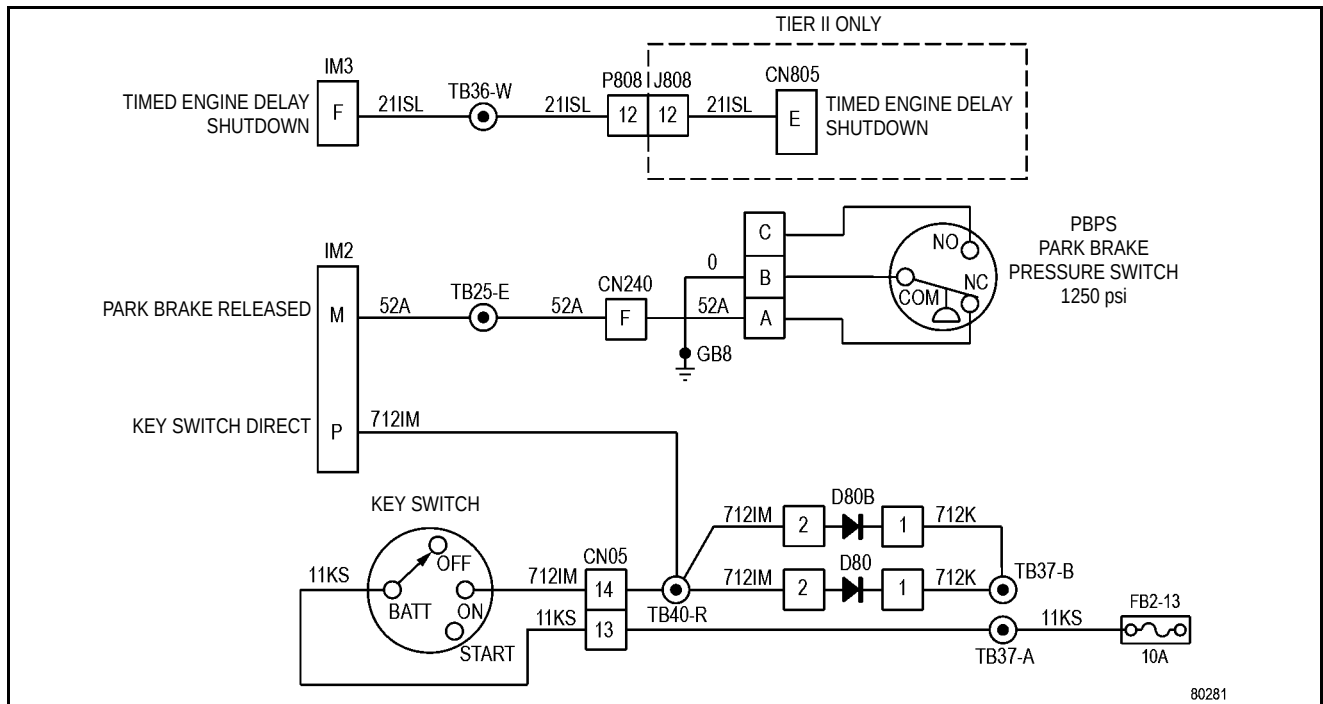


Fault Code A283: An engine shutdown delay was aborted because the parking brake was not set

Operator Action	None
Fault Code	A283
Description	An engine shutdown delay was aborted because the parking brake was not set.
Fault Conditions	Sets if key switch is turned off and engine shutdown delay is on and engine is running but parking brake set is not on. Parking brake set is defined as opening of the parking brake release pressure switch. Resets if key switch is turned on, and engine shutdown delay is off, and engine is running.
Operator Alerting System Response	None
Resulting Problem(s)	The engine can suffer damage when shut down immediately rather than in a controlled delay.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table	
1. The normal corrective measure for this fault is to set the parking brake before turning off the key switch. 2. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). 3. Since there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 4. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Keyswitch, Direct (IM2 P)	0: keyswitch is off 1: keyswitch is on Fault(s): A240
Engine Shutdown Delay (IM3 F)	0: no shutdown delay 1: engine shutdown delay signal is on Fault(s): A284
Parking Brake Released (IM2 M)	0: parking brake pressure switch indicating high pressure and a released parking brake 1: parking brake pressure switch indicating low pressure and an applied parking brake Fault(s): A214
Engine Speed [RPM] (CAN/J1939) (IM1 q,r,s)	Less than 400 rpm: engine is not running Greater than 400 rpm for 15 seconds: engine is running Fault(s): A184

Related circuit diagram



Fault Code A284: An engine shutdown delay was aborted because the secondary shutdown switch was operated

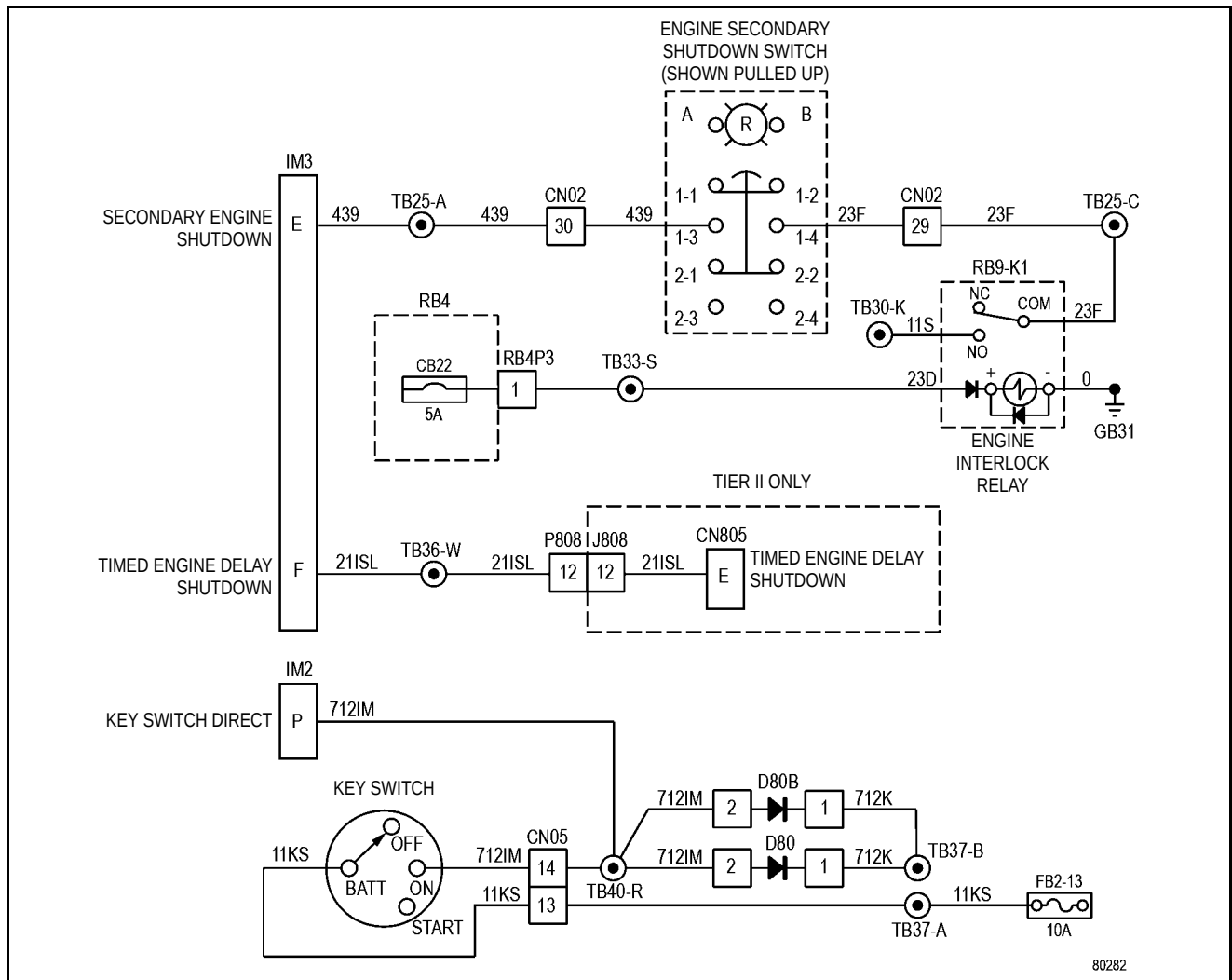
Operator Action	None
Fault Code	A284
Description	An engine shutdown delay was aborted because the secondary shutdown switch was operated.
Fault Conditions	Sets if keyswitch is turned off and engine shutdown delay is on and engine is running but secondary engine shutdown switch is operated. Resets if key switch is turned back on, engine shutdown delay is off, and engine is running.
Operator Alerting System Response	None
Resulting Problem(s)	The engine can suffer damage when shut down immediately rather than in a controlled delay.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. The normal corrective measure for this fault is to shut down the engine with the keyswitch.
2. This fault's logic contains more than one parameter, each of which may have its own related fault code(s).
3. Since there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items.
4. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.

Parameter	Expected State and/or Related Fault(s)
Secondary Engine Shutdown Switch (IM3 E)	0: engine being shut down by secondary switch 1: engine not being shut down by secondary switch
Keyswitch, Direct (IM2 P)	0: keyswitch is off 1: keyswitch is on Fault(s): A240
Engine Shutdown Delay (IM3 F)	0: no shutdown delay 1: 1: engine shutdown delay signal is on Fault(s): A283
Engine Speed [RPM] (CAN/J1939) (IM1 q,r,s)	Less than 400 rpm: engine is not running Greater than 400 rpm for 15 seconds: engine is running Fault(s): A184

Related circuit diagram



Fault Code A285: The parking brake was not set when the key switch was turned off

Operator Action	Set parking brake
Fault Code	A285
Description	The parking brake was not set when the key switch was turned off.
Fault Conditions	Sets if parking brake is not already set when keyswitch is turned off. Parking brake set is defined as closure of the parking brake set pressure switch and opening of the parking brake release pressure switch. Resets if parking brake is set or if key switch is turned back on.
Operator Alerting System Response	Red "System Failure" Lamp No Buzzer Right Lamp Module Row 4, Column 5
Resulting Problem(s)	Truck could roll away if parking brake is not set at shutdown.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. The normal corrective measure for this fault is to set the parking brake before turning off the key switch. 2. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). 3. Since there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 4. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Keyswitch, Direct (IM2 P)	0: keyswitch is off 1: keyswitch is on Fault(s): A240
Parking Brake Released (IM 2M)	0: parking brake pressure switch indicating high pressure and a released parking brake 1: parking brake pressure switch indicating low pressure and an applied parking brake Fault(s): A214
IM Warn Relay (IM1 K)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Relay Off (IM output at B+) - Lamp is on 1: Relay On (IM output at ground) - Lamp is off Fault(s): A354

The diagram illustrates the electrical system for the IM (Intermittent Motor) and related components. Key elements include:

- Park Brake Released:** A switch (IM2) that controls the IM1 motor through a 52A fuse (TB25-E) and a 52A fuse (CN240).
- Key Switch Direct:** A switch (P) that controls the IM1 motor through a 712IM fuse (CN05).
- Key Switch:** A switch with positions for OFF, BATT, ON, and START. It controls the IM1 motor through a 11KS fuse (CN05).
- IM1:** The main intermittent motor, controlled by the Park Brake Released and Key Switch Direct switches.
- IM2:** A switch that controls the IM1 motor through a 52A fuse (TB25-E) and a 52A fuse (CN240).
- IM Warning Relay:** A relay (RB7-K3) that controls the IM1 motor through a 33IM fuse (TB32-S).
- Brake Lock Switch Relay:** A relay (RB7-K5) that controls the IM1 motor through a 33IM fuse (TB32-S).
- IM Failure Lamp:** A lamp (31IML) that indicates a failure in the IM system.
- Relays and Switches:** Various relays and switches (e.g., TB43-E, TB43-L, TB40-X, TB40-R, TB37-K, TB37-A, TB37-B, TB25-E, CN240, CN05, CN07, CN262, D81, D82, D80B, D80, DLT4, RB7-K3, RB7-K5, IM WARNING RELAY, BRAKE LOCK SWITCH RELAY, IM FAILURE LAMP) are used to control the IM system.
- Fuses:** Various fuses (e.g., 52A, 712IM, 11KS, 33IM, 31IML, 5A, 10A) are used to protect the circuit.
- Grounds:** Grounds (e.g., GB8, GB31) are used to complete the circuit.

Fault Code A286: A fault was detected in the shutdown delay relay circuit

Operator Action	None
Fault Code	A286
Description	A fault was detected in the shutdown delay relay circuit.
Fault Conditions	Sets if an open circuit or short to ground is detected in the relay driver circuit when the keyswitch is on. Resets only at power down.
Operator Alerting System Response	Amber "IM Maintenance" Lamp No Buzzer Right Lamp Module Row 5, Column 3
Resulting Problem(s)	A malfunctioning shutdown delay circuit could cause damage to the engine.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

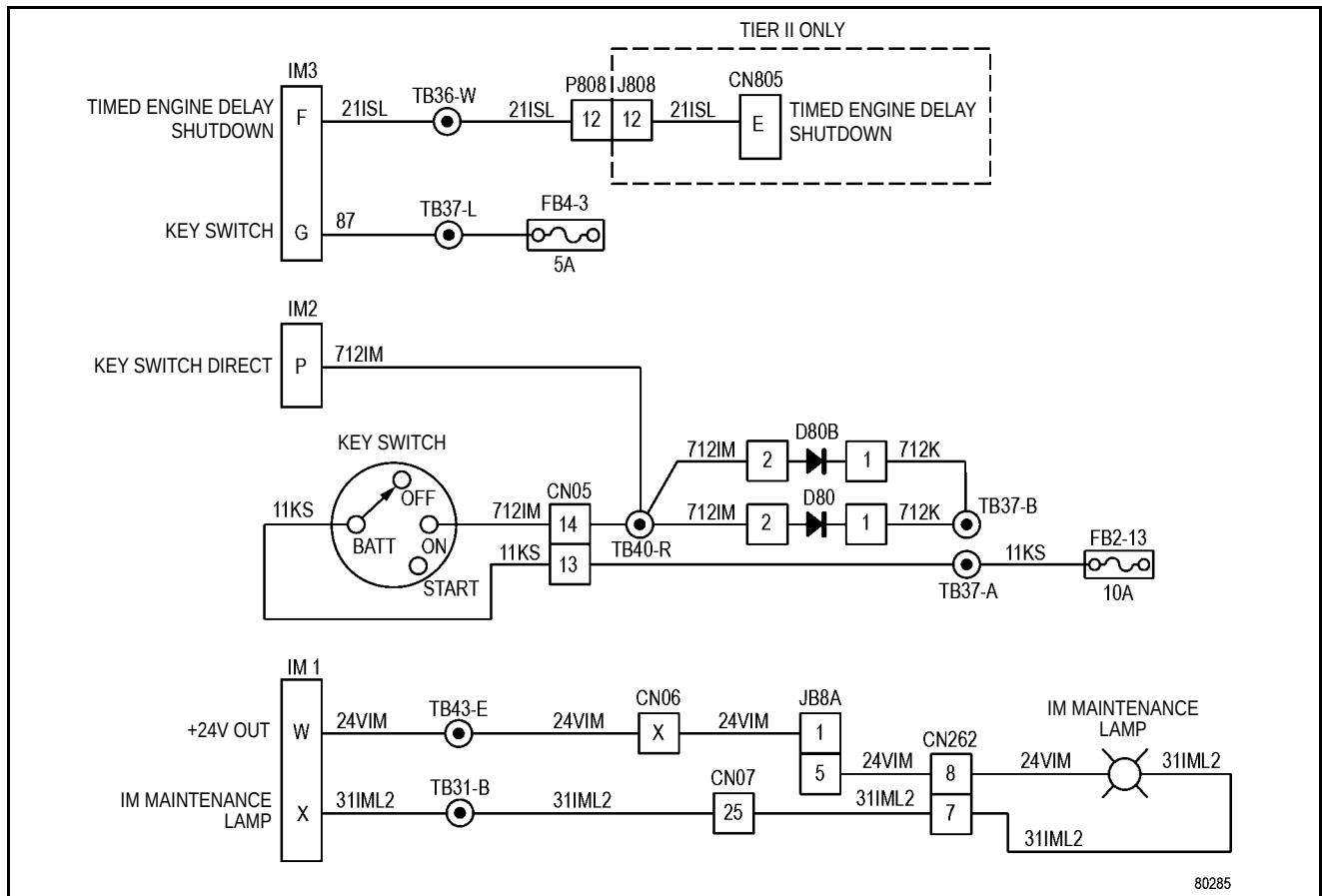
1. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 2. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Shutdown Delay Relay (IM1 H)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected. No problem. Status - Shorted to Ground: Unexpected. Troubleshoot. Status - Overload: Unexpected. Troubleshoot. 0: Turns off when conditions clear or engine speed drops to zero. 1: Turns on when Keyswitch, Direct is off, engine shutdown delay is on, park brake is set, selector switch is in Park, but not J1939 Not Connected, not High Batt Charge Voltage, not Starter Stuck On, not Active Engine Red Light, and engine speed above 300. Fault(s): A292
Keyswitch, Direct (IM2 P)	0: keyswitch is off 1: keyswitch is on Fault(s): A240
IM Maintenance Lamp (IM1 X)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A363

Fault Code A292: The shutdown delay relay has remained on after the latched key switch circuit is off

Operator Action	None
Fault Code	A292
Description	The shutdown delay relay has remained on after the latched key switch circuit is off.
Fault Conditions	Sets if the shutdown relay is still on after the latched key switch circuit (GE latches the circuit until the truck and engine have stopped completely) is off. Resets if key switch is turned back on or shutdown delay relay turns off.
Operator Alerting System Response	Amber "IM Maintenance" Lamp No Buzzer Right Lamp Module Row 5, Column 3
Resulting Problem(s)	A malfunctioning shutdown delay circuit could cause damage to the engine.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table	
1. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 2. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Keyswitch, Direct (IM2 P)	0: keyswitch is off 1: keyswitch is on Fault(s): A240
Engine Shutdown Delay (IM3 F)	0: no shutdown delay 1: engine shutdown delay signal is on Fault(s): A284
Keyswitch (IM3 G)	0: keyswitch is off and the truck is not moving and the engine is not running 1: keyswitch is on or the truck is moving (GE power not yet off) Fault(s): A240
IM Maintenance Lamp (IM1 X)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A363

Related circuit diagram



NOTES

SECTION E5

TROUBLESHOOTING FAULT CODES (A303-A365)

INDEX

Fault Code A303: Shifter is defective	E5-2
Fault Code A306: Red warning lamp output used to drive the steering bleed fault lamp is not open when it should be	E5-4
Fault Code A309: No brakes applied when expected	E5-6
Fault Code A311: Brake lock switch is on when it should not be	E5-10
Fault Code A312: DC-DC converter 12 volt circuit sensing is producing low readings	E5-12
Fault Code A313: DC-DC converter 12 volt circuit sensing is producing high readings.	E5-13
Fault Code A315: DC-DC converter 12 volt circuit is low.	E5-14
Fault Code A316: Starter engagement has been attempted with engine running	E5-16
Fault Code A317: Operation of brake auto apply valve without a detected response	E5-18
Fault Code A318: Unexpected power loss to interface module	E5-21
Fault Code A328: Drive system not powered up	E5-22
Fault Code A350: Overload on output 1B.	E5-24
Fault Code A351: Overload on output 1E.	E5-26
Fault Code A352: Overload on output 1H	E5-28
Fault Code A353: Overload on output 1J.	E5-30
Fault Code A354: Overload on output 1K.	E5-32
Fault Code A355: Overload on output 1L.	E5-34
Fault Code A356: Overload on output 1M	E5-36
Fault Code A357: Overload on output 1N	E5-38
Fault Code A358: Overload on output 1P.	E5-40
Fault Code A359: Overload on output 1R	E5-42
Fault Code A360: Overload on output 1S.	E5-44
Fault Code A361: Overload on output 1T.	E5-46
Fault Code A362: Overload on output 1U	E5-48
Fault Code A363: Overload on output 1X.	E5-50
Fault Code A364: Overload on output 1Y.	E5-52
Fault Code A365: Overload on output 1Z.	E5-54

Fault Code A303: Shifter is defective

Operator Action	Stop, Park, Power Down
Fault Code	A303
Description	Shifter is defective.
Fault Conditions	Sets if Selector Switch (FNR) and Selector Switch (Park) operate simultaneously for 1 second. Resets if conditions change for 1 second.
Operator Alerting System Response	Red "System Failure" Lamp No Buzzer Right Lamp Module Row 4, Column 5
Resulting Problem(s)	Control of truck and parking brake will be defective.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 2. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Selector Switch (Park) (IM3 T)	0: shifter is not in park 1: shifter is in park Fault(s): A271
Selector Switch (FNR) (IM2 N)	0: shifter is not in forward, neutral, or reverse 1: shifter is in forward, neutral, or reverse Fault(s): A271
IM Warn Relay (IM1 K)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Relay Off (IM output at B+) - Lamp is on 1: Relay On (IM output at ground) - Lamp is off Fault(s): A354

[illegible]

Fault Code A306: Red warning lamp output used to drive the steering bleed fault lamp is not open when it should be

Operator Action	Go to Shop
Fault Code	A306
Description	Red warning lamp output used to drive the steering bleed fault lamp is not open when it should be.
Fault Conditions	Sets if the red warning lamp circuit is not open with keyswitch on for 2 seconds. Resets if circuit changes for 2 seconds.
Operator Alerting System Response	Amber "IM Maintenance" Lamp No Buzzer Right Lamp Module Row 5, Column 3
Resulting Problem(s)	The red warning lamp will not properly function as a low steering pressure lamp if not open circuit when the keyswitch is on.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 2. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Red Warning Lamp (IM1 G)	0: red lamp is not on 1: red lamp is on Fault(s): A256
Red Warning Lamp Feedback (IM1 G)	Above 3.75 volts for 400 ms with the lamp on: shorted Less than 3.75 volts for 400 ms with the lamp on: normal
Steering Bleed Fault Lamp (IM1 G)	Status - Open Load with Keyswitch On: Normal: Expected Status - Open Load with Keyswitch Off: Unexpected, Troubleshoot Status - Normal with Keyswitch On::Unexpected, Troubleshoot Status - Normal with Keyswitch Off: Normal, Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground)
Keyswitch (IM3 G)	0: keyswitch is off and the truck is not moving and the engine is not running 1: keyswitch is on or the truck is moving (GE power not yet off) Fault(s): A240

Fault Code A309: No brakes applied when expected

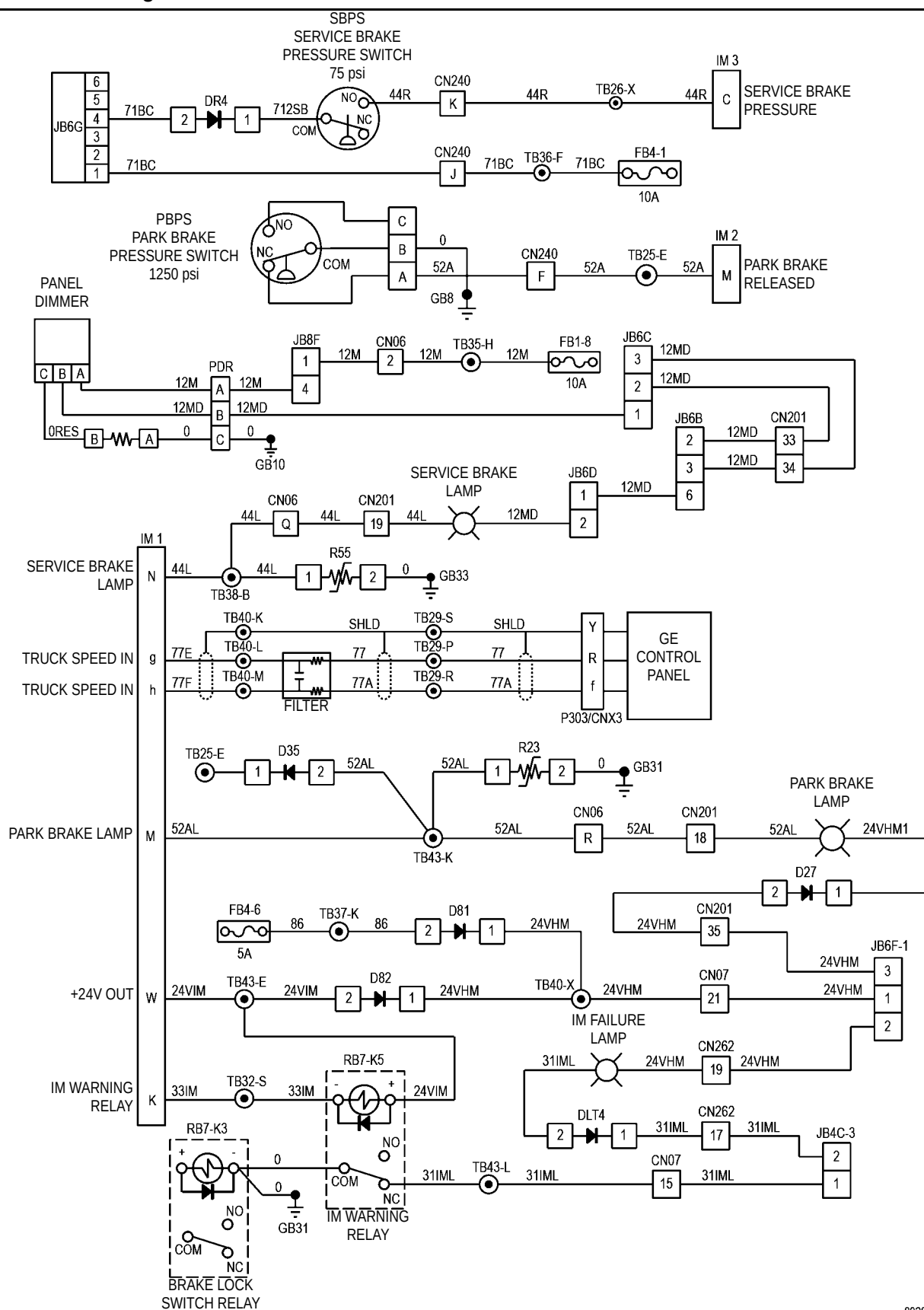
Operator Action	Apply brake or travel
Fault Code	A309
Description	No brakes applied when expected.
Fault Conditions	Sets if truck speed is 0, park brake is not set, service brake is not set, and engine is running for 2 seconds. Park brake set is defined as opening of the park brake release pressure switch. Resets if conditions change for 2 seconds.
Operator Alerting System Response	Amber, dimmable "Service Brake" Lamp flashes Left Lamp Module Row 2, Column 3 Amber "Park Brake" Lamp flashes Left Lamp Module Row 1, Column 3 Red "System Failure" Lamp No Buzzer Right Lamp Module Row 4, Column 5
Resulting Problem(s)	No brakes set may let truck roll away.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 2. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Park Brake Released (IM2 M)	0: parking brake pressure switch indicating high pressure and a released parking brake 1: parking brake pressure switch indicating low pressure and an applied parking brake Fault(s): A214
Service Brake Pressure Switch (IM3 C)	0: Service brake not applied 1: Service brake applied Fault(s): A265, A274, A317
Truck Speed [kph] (IM1 g,h)	0: truck is not moving Greater than 0: truck is moving Fault(s): A212
Engine Speed [RPM]	Less than 300 RPM: engine is not running Greater than 300 RPM for 4 seconds: engine is running Fault(s): A184

Service Brake Lamp (IM1 N)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A357
Park Brake Lamp (IM1 M)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A356
IM Warn Relay (IM1 K)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Relay Off (IM output at B+) - Lamp is on 1: Relay On (IM output at ground) - Lamp is off Fault(s): A354

Related circuit diagram



NOTES

Fault Code A311: Brake lock switch is on when it should not be

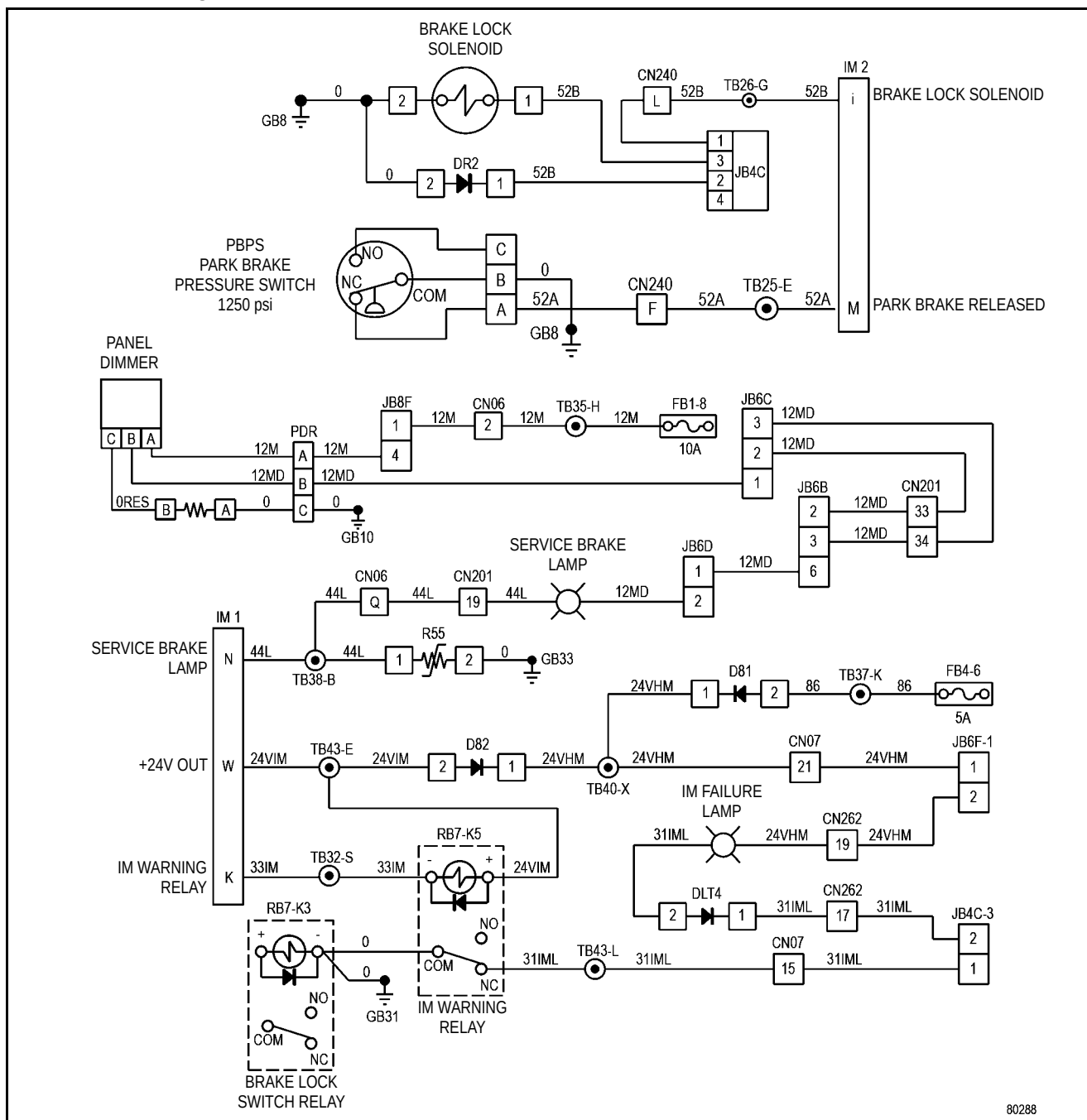
Operator Action	Turn Off Brake Lock Switch
Fault Code	A311
Description	Brake lock switch is on when it should not be.
Fault Conditions	Sets if parking brake is set and brake lock is on for 2 seconds. Parking brake set is defined as opening of the park brake release pressure switch. Resets when conditions clear for 2 seconds.
Operator Alerting System Response	Amber, dimmable "Service Brake" Lamp flashes Left Lamp Module Row 2, Column 3 Red "System Failure" Lamp No Buzzer Right Lamp Module Row 4, Column 5
Resulting Problem(s)	The roll away protection of the braking systems could be defeated by improper operation.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 2. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Park Brake Released (IM2 M)	0: parking brake pressure switch indicating high pressure and a released parking brake 1: parking brake pressure switch indicating low pressure and an applied parking brake Fault(s): A214
Brake Lock (IM2 i)	0: brake lock is not on 1: brake lock is on Fault(s): A272
Service Brake Lamp (IM1 N)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A357

IM Warn Relay (IM1 K)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Relay Off (IM output at B+) - Lamp is on 1: Relay On (IM output at ground) - Lamp is off Fault(s): A354
-----------------------	---

Related circuit diagram

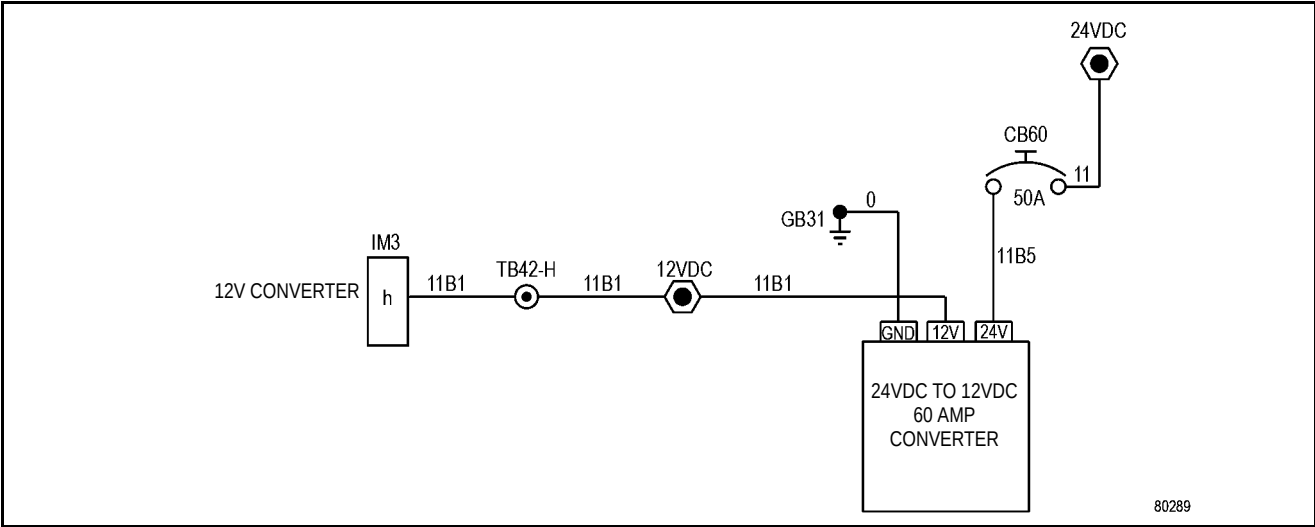


Fault Code A312: DC-DC converter 12 volt circuit sensing is producing low readings

Operator Action	None
Fault Code	A312
Description	The DC-DC converter 12 volt circuit sensing is producing low readings.
Fault Conditions	Sets if voltage at 12 volt input drops to 0.2 volts for 3 seconds. Resets if voltage at 12 volt input rises to 0.8 volts for 3 seconds.
Operator Alerting System Response	None
Resulting Problem(s)	Monitoring is lost for the 12 volt output of the DC-DC Converter
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table	
1. The primary correction for this fault is to correct any external wiring to the 12 volt input.	
Parameter	Expected State and/or Related Fault(s)
DC-DC Converter, 12 Volt (IM3 h)	0.2 volts to 16.1 volts: Normal measurement range. <0.2 or >16.1: Bad sensing circuit or external connection. Fault(s): A315

Related circuit diagram

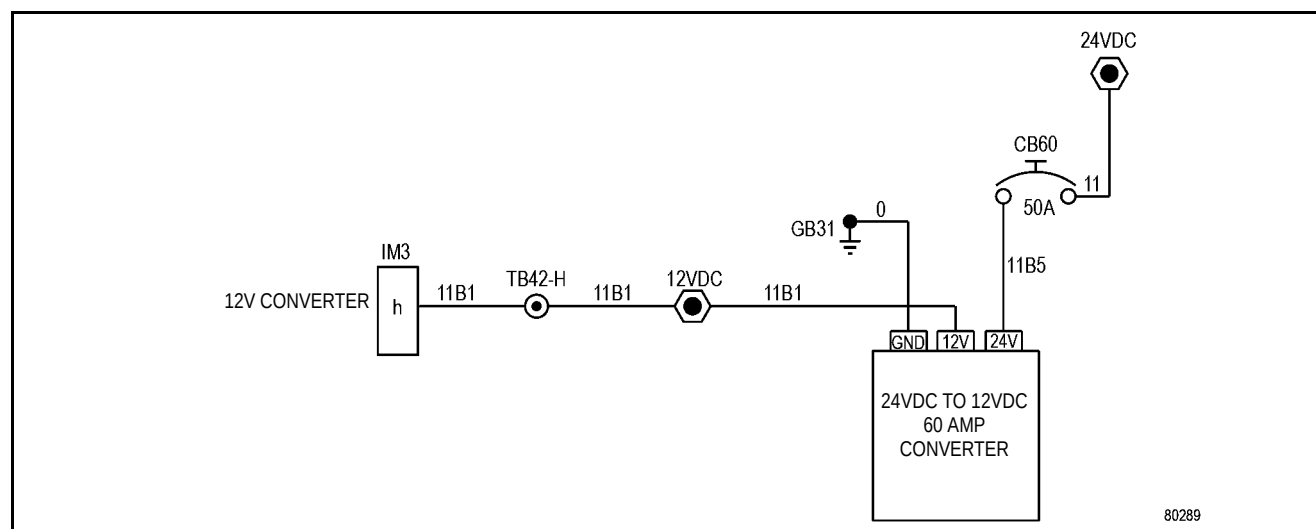


Fault Code A313: DC-DC converter 12 volt circuit sensing is producing high readings

Operator Action	None
Fault Code	A313
Description	The DC-DC converter 12 volt circuit sensing is producing high readings.
Fault Conditions	Sets if voltage at 12 volt input rises to 16.1 volts for 3 seconds. Resets if voltage at 12 volt input drops to 15.5 volts for 3 seconds.
Operator Alerting System Response	None
Resulting Problem(s)	Monitoring is lost for the 12 volt battery tap circuit.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table	
1. The primary correction for this fault is to correct any external wiring to the 12 volt input.	
Parameter	Expected State and/or Related Fault(s)
DC-DC Converter, 12 Volt (IM3 h)	0.2 volts to 16.1 volts: Normal measurement range. <0.2 or >16.1: Bad sensing circuit or external connection. Fault(s): A314

Related circuit diagram



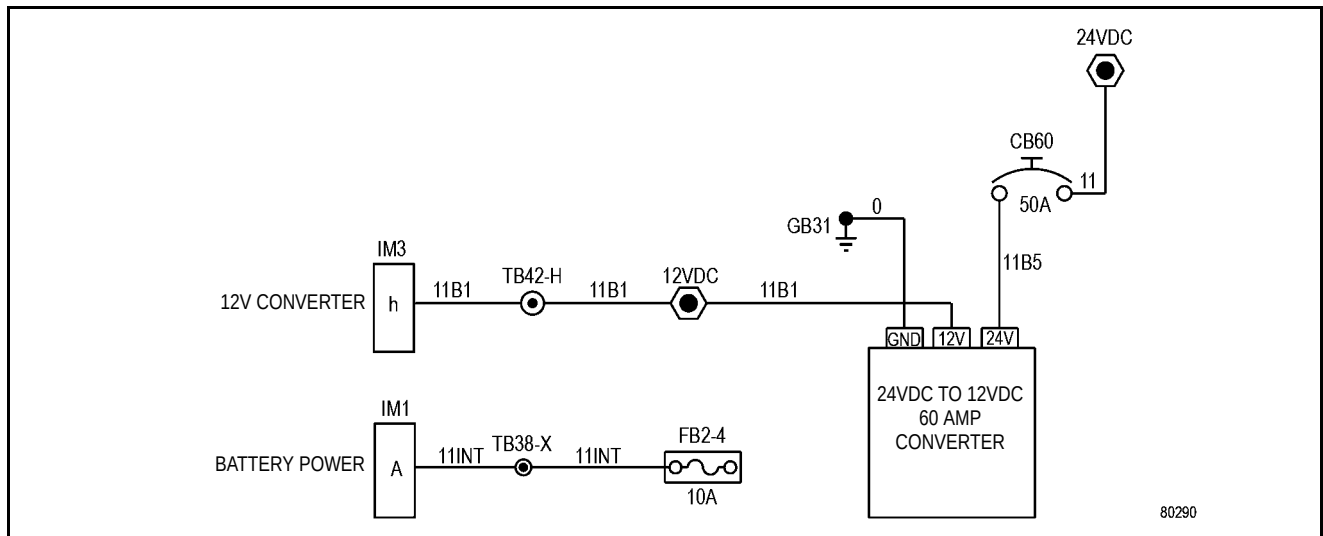
Fault Code A315: DC-DC converter 12 volt circuit is low

Operator Action	None
Fault Code	A315
Description	The DC-DC converter 12 volt circuit is low.
Fault Conditions	When engine is running and battery volts are above 22, sets if voltage is below 12.5. If battery voltage is below 22, sets if voltage is more than 1.5 volts lower than battery voltage divided by 2. Resets at power down only.
Operator Alerting System Response	None
Resulting Problem(s)	The 12 Volt DC-DC converter is putting out low voltage. Correction is required to continue good performance in the 12 volt loads.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. The primary correction for this fault is to eliminate any overload, or replace the converter. 2. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 3. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 4. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
DC-DC Converter, 12 Volt (IM3 h)	<12.5 volts when battery voltage >22 volts: Voltage is low. < (Batt Voltage / 2) - 1.5 volts : Voltage is low. Otherwise: Voltage is normal. Fault(s): A312
Battery Voltage, 24 Volt (IM1 A)	>22 volts: 12 volt circuit should have normal readings <22 volts: 12 volt circuit might have abnormal readings
Engine Speed [RPM]	Less than 300 RPM: engine is not running Greater than 300 RPM for 4 seconds: engine is running Fault(s): A184

Related circuit diagram



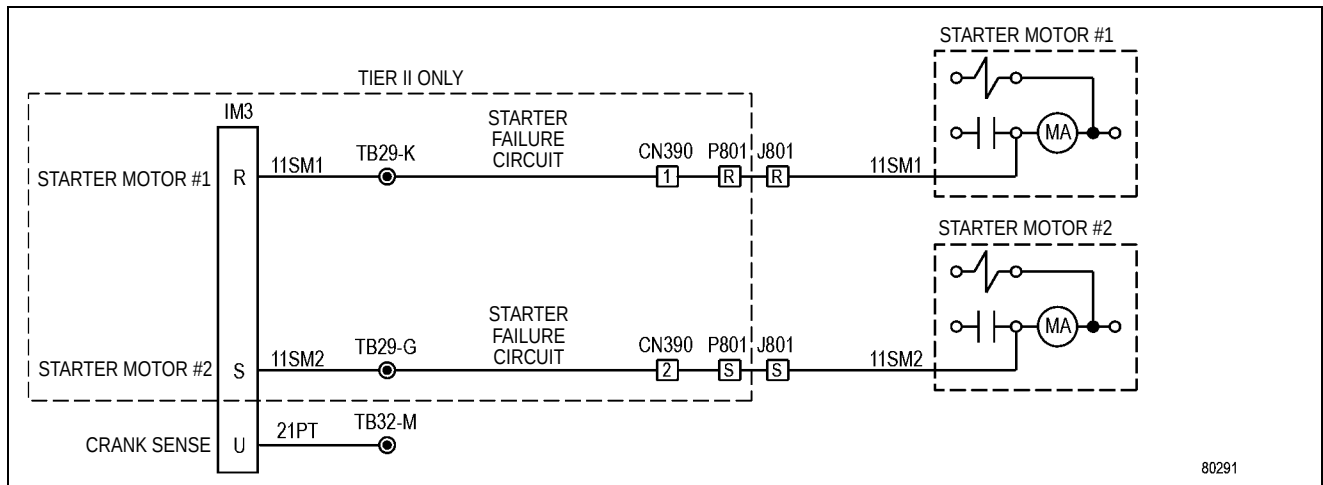
Fault Code A316: Starter engagement has been attempted with engine running

Operator Action	Do not crank
Fault Code	A316
Description	Starter engagement has been attempted with engine running.
Fault Conditions	If not steering bleed, sets after 2 seconds if either starter engages with engine speed above 500 rpm with crank sense on. Resets after 2 seconds if conditions change.
Operator Alerting System Response	None
Resulting Problem(s)	Damage to starter and engine flywheel may occur.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. The normal corrective measure for this fault is to not engage the starter with the engine already running. 2. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). 3. Since there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 4. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Starter Motor 1 Energized (IM3 R)	0: starter motor 1 not energized 1: starter motor 1 energized Fault(s): A152, A275
Starter Motor 2 Energized (IM3 S)	0: starter motor 2 not energized 1: starter motor 2 energized Fault(s): A152, A275
Engine Speed [RPM]	Less than 500 rpm: engine is not running Greater than 500 rpm: engine is running Fault(s): A184
Crank Sense (IM3 U)	0: no cranking in process 1: cranking in process
Steering Bleed Valve	0: steering bleed valve is off 1: steering bleed valve is on

Related circuit diagram



Fault Code A317: Operation of brake auto apply valve without a detected response

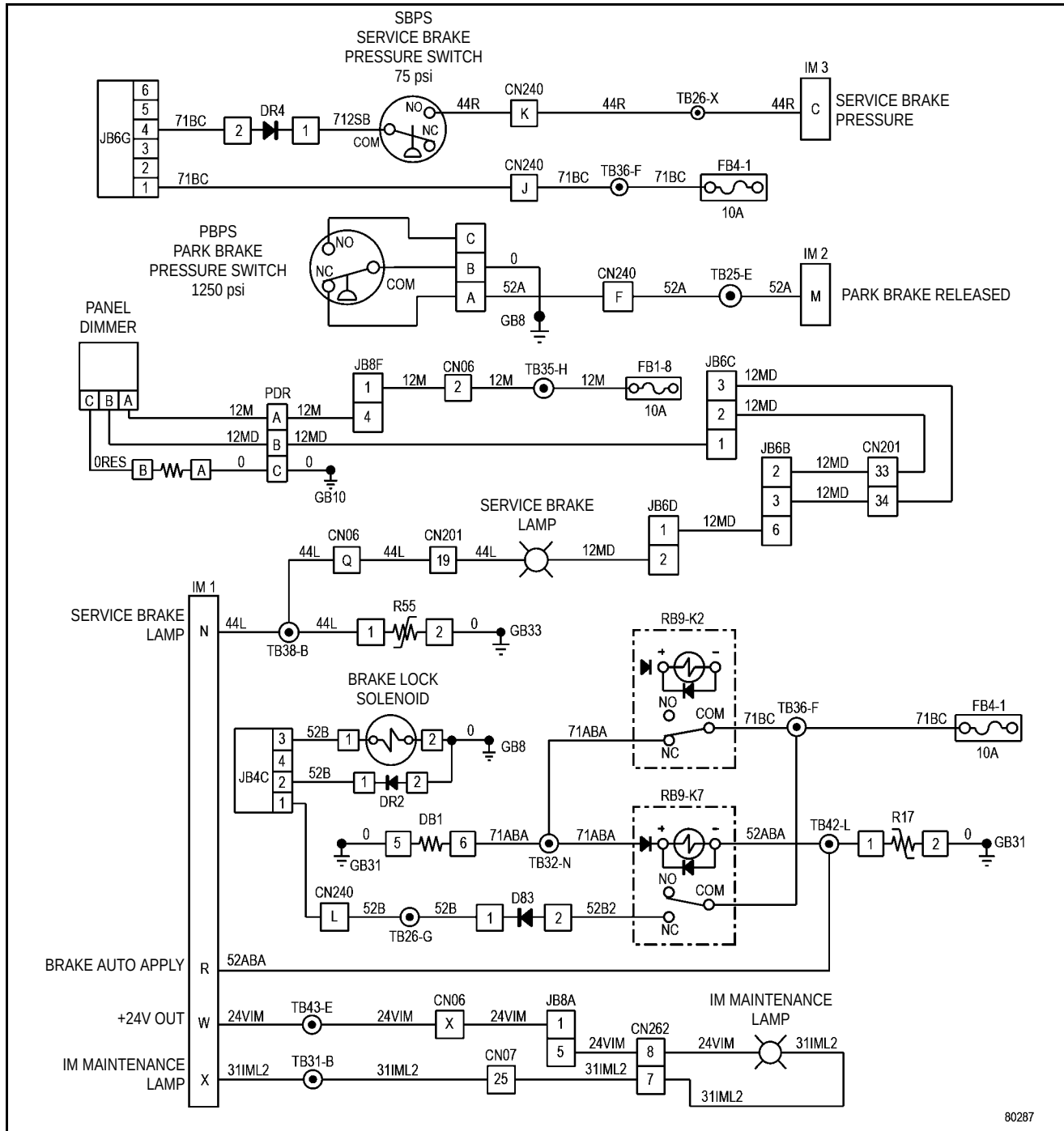
Operator Action	Go to shop now
Fault Code	A317
Description	Operation of brake auto apply valve without a detected response.
Fault Conditions	Sets if operation of brake auto apply valve for 1 second (with keyswitch on and after engine has been running for 90 seconds) does not cause the service brake pressure switch to operate. Parking brake release is defined as the opening of the parking brake release pressure switch. Resets when parking brake is released for 5 seconds.
Operator Alerting System Response	Amber, dimmable "Service Brake" Lamp flashes Left Lamp Module Row 2, Column 3 Amber "IM Maintenance" Lamp No Buzzer Right Lamp Module Row 5, Column 3
Resulting Problem(s)	Failure of brake auto apply circuit could cause damage to parking brake.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 2. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Auto Brake Apply Solenoid (IM1 R)	Status - Open Load: Unexpected if park brake request and keyswitch are on. But expected if park brake request is off or keyswitch is off. Troubleshoot if inconsistent. Status - Normal: Expected if park brake request and keyswitch are on. Troubleshoot if status is normal with park brake request off or keyswitch off. Status - Shorted to ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: normal operation with park brake either on or off. This valve operates only transiently when the park brake is turned on. 1: brake auto apply valve operates for 1.5 seconds after the park brake is turned on if the truck is stopped. If moving, it will not operate until truck speed has dropped to 0.8 kph. Fault(s): A359
Service Brake Pressure Switch (IM3 C)	0: Service brake not applied 1: Service brake applied

Park Brake Released (IM2 M)	0: parking brake pressure switch indicating high pressure and a released parking brake 1: parking brake pressure switch indicating low pressure and an applied parking brake Fault(s): A214
IM Maintenance Lamp (IM1 X)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A363
Service Brake Lamp (IM1 N)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A357

Related circuit diagram



Fault Code A318: Unexpected power loss to interface module

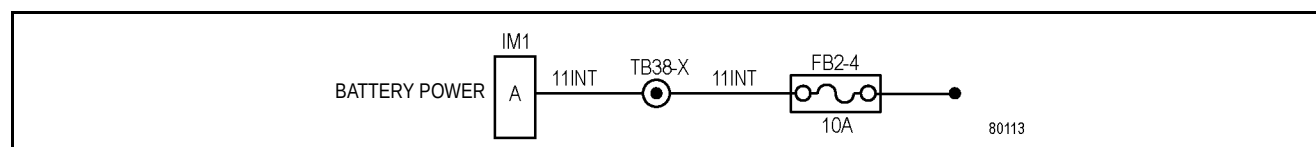
Operator Action	None
Fault Code	A318
Description	Unexpected power loss to interface module.
Fault Conditions	Sets if open file markers are discovered at power up. Resets if no open file markers are discovered at power up.
Operator Alerting System Response	None
Resulting Problem(s)	Data may be compromised by irregular shutdowns.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. The primary corrective action for this fault is to keep the Interface Module working and shutting itself down in a regular manner.
2. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s).
3. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items.
4. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.

Parameter	Expected State and/or Related Fault(s)
Open file markers	Stored in IM FLASH
Battery Voltage, 24 Volt (IM1 A)	>22 volts: 12 volt circuit should have normal readings <22 volts: 12 volt circuit might have abnormal readings

Related circuit diagram

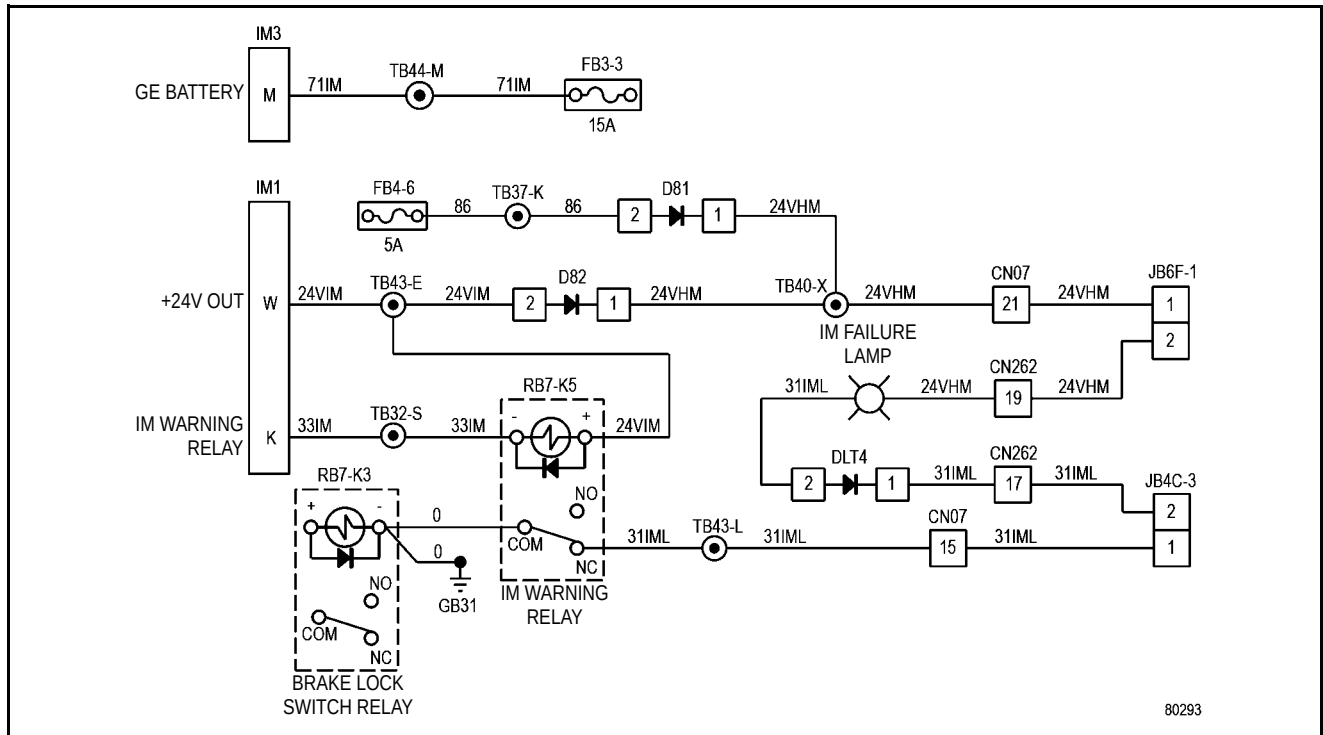


Fault Code A328: Drive system not powered up

Operator Action	Stop; Park
Fault Code	A328
Description	Drive system not powered up.
Fault Conditions	Sets if drive system does not power up within 30 seconds after engine is running. Resets if conditions change for 2 seconds.
Operator Alerting System Response	Red "System Failure" Lamp No Buzzer Right Lamp Module Row 4, Column 5
Resulting Problem(s)	Drive system can neither retard nor propel without control power.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table	
1. Check wiring from GE control power to the interface module. 2. Check control power wiring, relay, and drive circuitry from GE. 3. This fault's logic contains more than one parameter, each of which may have its own related fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 4. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 5. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Engine Speed [RPM]	Less than 500 rpm: engine is not running Greater than 500 rpm: engine is running Fault(s): A184
Drive System Control Power (IM3 M)	0: Control power is off 1: Control power is on
IM Warn Relay (IM1 K)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Relay Off (IM output at B+) - Lamp is on 1: Relay On (IM output at ground) - Lamp is off Fault(s): A354

Related circuit diagram



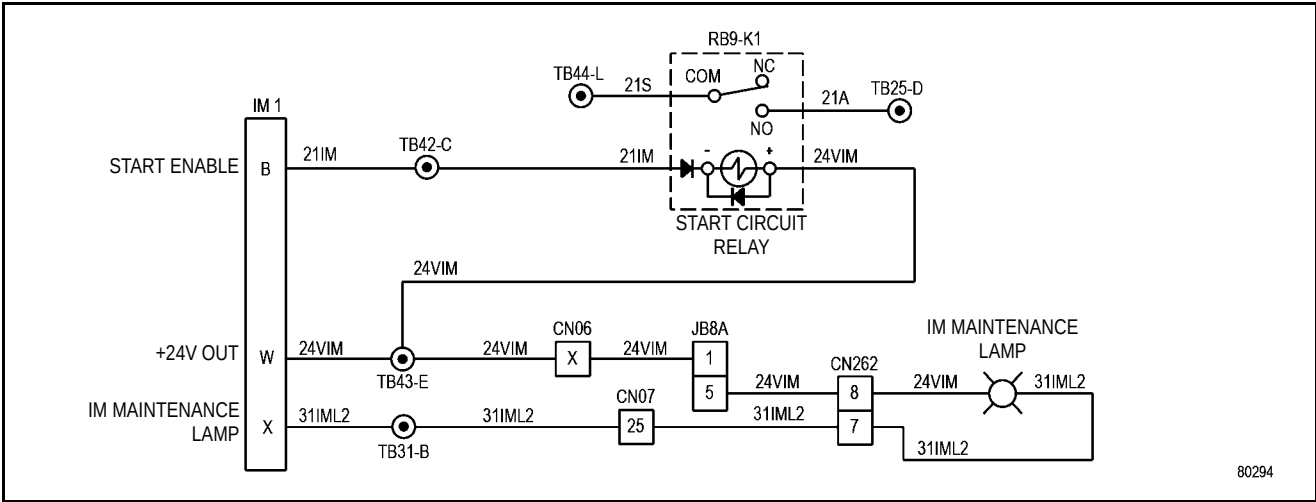
Fault Code A350: Overload on output 1B

Operator Action	None
Fault Code	A350
Description	Overload on output 1B.
Fault Conditions	Sets if driver chip detects overcurrent or over temp on output 1B. Output is turned off when overload is detected. Resets at power down.
Operator Alerting System Response	Amber "IM Maintenance" Lamp No Buzzer Right Lamp Module Row 5, Column 3
Resulting Problem(s)	Starting circuit is disabled.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. This fault's logic contains just one parameter, which may have additional fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 2. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Start Enable (IM1 B)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected. No problem. Status - Shorted to Ground: Unexpected. Troubleshoot. Status - Overload: Unexpected. Troubleshoot. 0: One of several interlocking situations exist to prevent cranking (excessive cranking history, selector switch in wrong position, engine red light, J1939 not ok, engine speed either not 0 to begin cranking or over 400 rpm while cranking). 1: No interlocking situations exist to prevent cranking. Fault(s): A252
IM Maintenance Lamp (IM1 X)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A363

Related circuit diagram



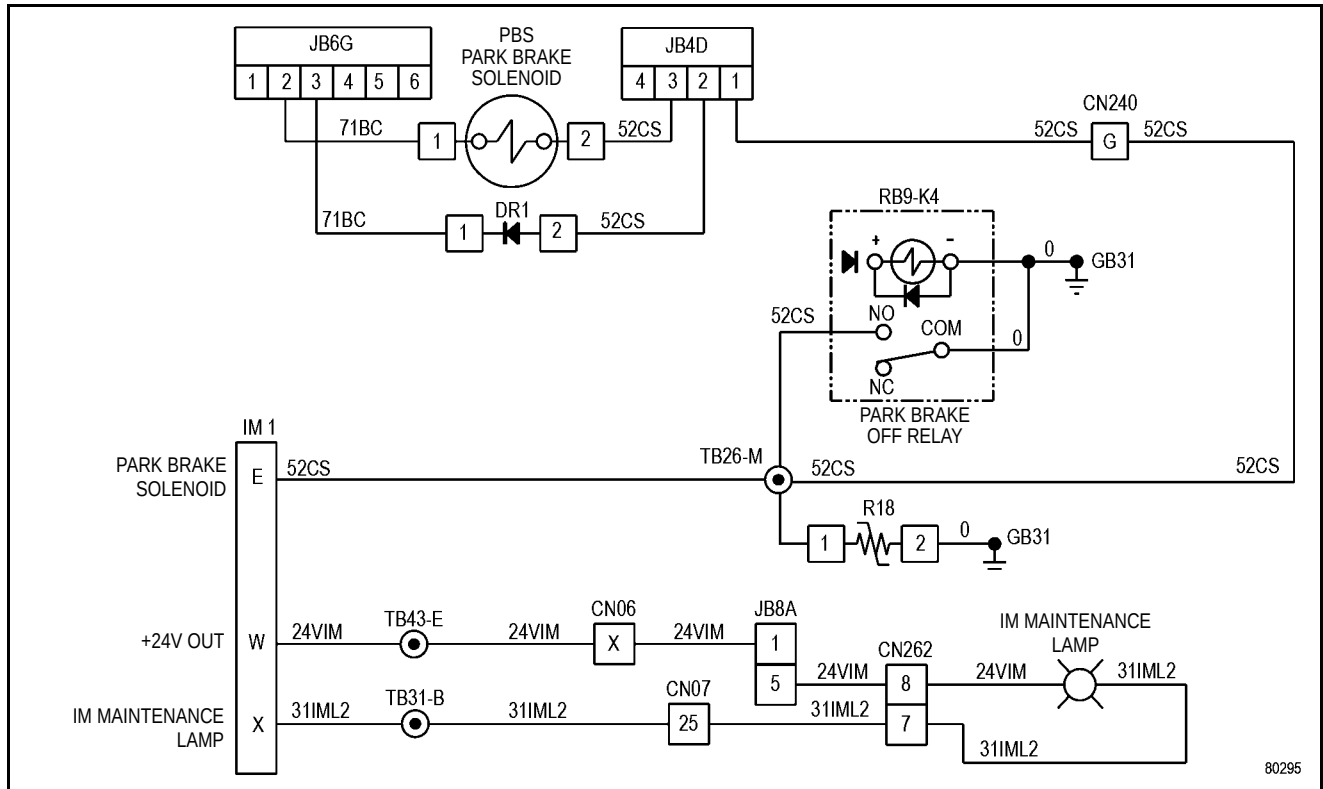
Fault Code A351: Overload on output 1E

Operator Action	Go to shop now
Fault Code	A351
Description	Overload on output 1E.
Fault Conditions	Sets if driver chip detects overcurrent or over temp on output 1E. Output is turned off when overload is detected. Resets at power down.
Operator Alerting System Response	Amber "IM Maintenance" Lamp No Buzzer Right Lamp Module Row 5, Column 3
Resulting Problem(s)	Parking brake solenoid valve circuit is disabled (parking brake will not release).
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. This fault's logic contains just one parameter, which may have additional fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 2. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Park Brake Solenoid (IM1 E)	Status - Open Load: Unexpected. Troubleshoot. Status - Normal: Expected. No Problem. Status - Shorted to Ground: Expected if Park Brake Request is in the 24 volt condition. Otherwise unexpected and must troubleshoot. Status - Overload: Unexpected. Troubleshoot. 0: Park Brake Request Input is in the low voltage (request) condition 1: Park Brake Request Input is in the high voltage (not requested) condition Fault(s): A214, A216
IM Maintenance Lamp (IM1 X)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A363

Related circuit diagram



Fault Code A352: Overload on output 1H

Operator Action	None
Fault Code	A352
Description	Overload on output 1H.
Fault Conditions	Sets if driver chip detects overcurrent or over temp on output 1H. Output is turned off when overload is detected. Resets at power down.
Operator Alerting System Response	Amber "IM Maintenance" Lamp No Buzzer Right Lamp Module Row 5, Column 3
Resulting Problem(s)	Shutdown delay (5 Minute Idle Latch or Shutdown Delay Relay) circuit is disabled.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

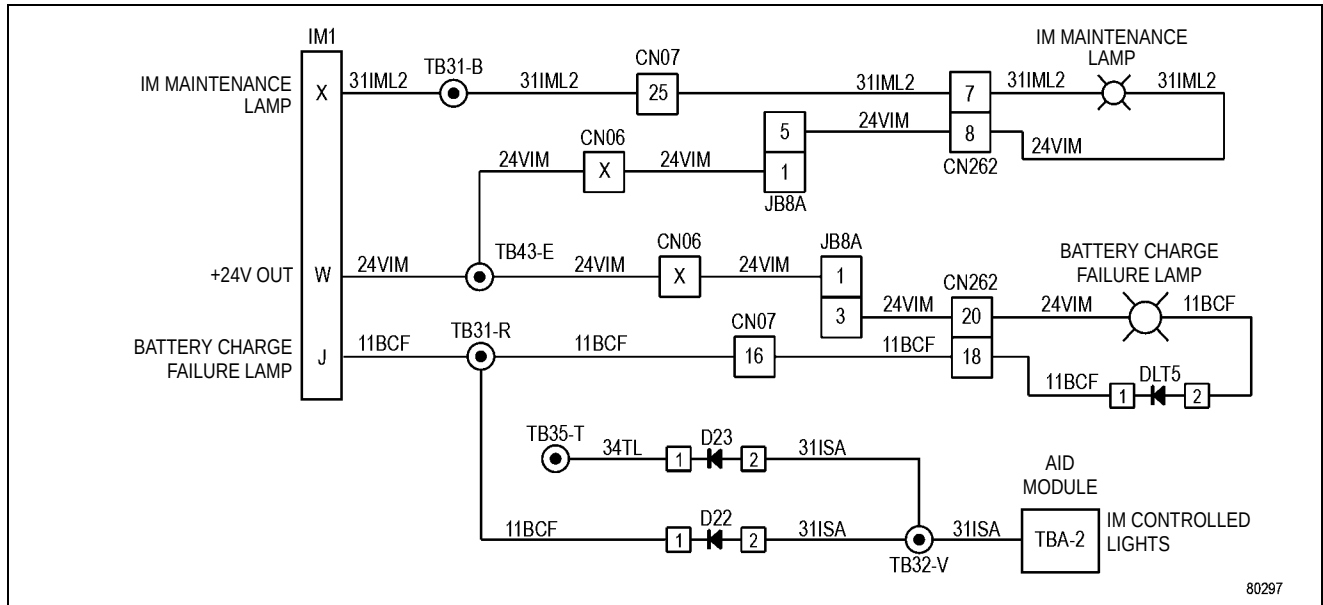
1. This fault's logic contains just one parameter, which may have additional fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 2. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Shutdown Delay Relay (IM1 H)	Status - Open Load: Unexpected. Troubleshoot. Status - Normal: Expected. No problem. Status - Shorted to Ground: Unexpected. Troubleshoot. Status - Overload: Unexpected. Troubleshoot. 0: Turns off when conditions clear or engine speed drops to zero. 1: Turns on when Keyswitch, Direct is off, Cummins Shutdown Delay is on, Park brake is set, Selector Switch is in park, but not J1939 Not Connected, not High Batt Charge Voltage, not Starter Stuck On, not Active Engine Red Light, and engine speed above 300. Fault(s): A286, A292
IM Maintenance Lamp (IM1 X)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A363

Fault Code A353: Overload on output 1J

Operator Action	None
Fault Code	A353
Description	Overload on output 1J.
Fault Conditions	Sets if driver chip detects overcurrent or over temp on output 1J. Output is turned off when overload is detected. Resets at power down.
Operator Alerting System Response	Amber "IM Maintenance" Lamp No Buzzer Right Lamp Module Row 5, Column 3
Resulting Problem(s)	No warning by Battery Charging Failure lamp is provided.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table	
1. This fault's logic contains just one parameter. There are no additional fault codes to investigate. 2. Any spare wires connected to this output should be removed to prevent ultimate failure of the driver chip.	
Parameter	Expected State and/or Related Fault(s)
Battery Charge Failure Lamp (IM1 J)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground)
IM Maintenance Lamp (IM1 X)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A363

Related circuit diagram

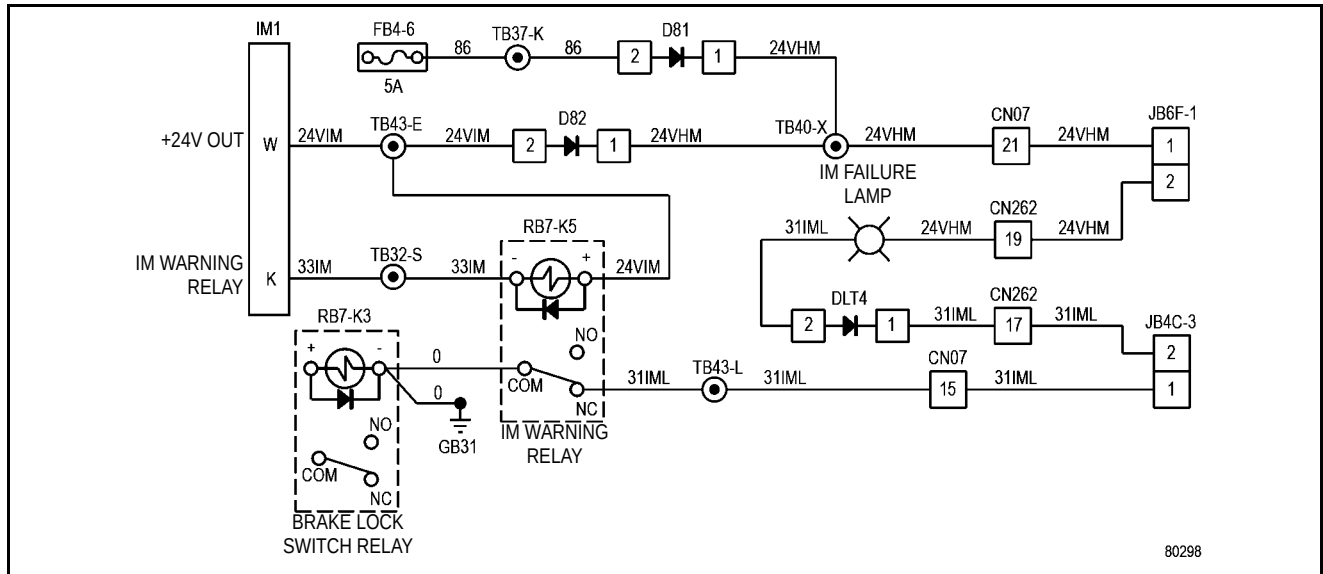


Fault Code A354: Overload on output 1K

Operator Action	None
Fault Code	A354
Description	Overload on output 1K.
Fault Conditions	Sets if driver chip detects overcurrent or over temp on output 1K. Output is turned off when overload is detected. Resets at power down.
Operator Alerting System Response	Red "System Failure" Lamp No Buzzer Right Lamp Module Row 4, Column 5
Resulting Problem(s)	Faults normally associated with the System Failure lamp will not be available, but the lamp itself will remain on.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table	
1. This fault's logic contains just one parameter, which may have additional fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 2. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
IM Warn Relay (IM1 K)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot 0: Relay Off (IM output at B+) - Lamp is on 1: Relay On (IM output at ground) - Lamp is off

Related circuit diagram



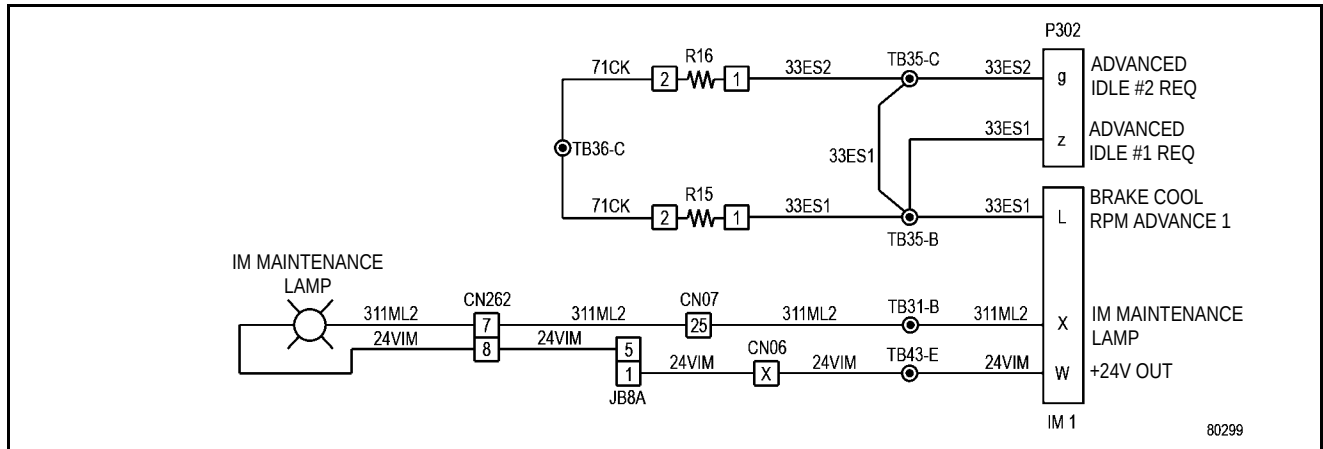
Fault Code A355: Overload on output 1L

Operator Action	None
Fault Code	A355
Description	Overload on output 1L.
Fault Conditions	Sets if driver chip detects overcurrent or over temp on output 1L. Output is turned off when overload is detected. Resets at power down.
Operator Alerting System Response	Amber "IM Maintenance" Lamp No Buzzer Right Lamp Module Row 5, Column 3
Resulting Problem(s)	Unused outputs with short circuits on them might damage the driver chip if not corrected.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. This fault's logic contains just one parameter, which may have additional fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 2. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Brake Cooling RPM Advance 1: (IM1 L)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected. No problem. Status - Shorted to Ground: Unexpected. Troubleshoot. Status - Overload: Unexpected. Troubleshoot. 0: Off. 1: On. Fault(s): A145
IM Maintenance Lamp (IM1 X)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A363

Related circuit diagram



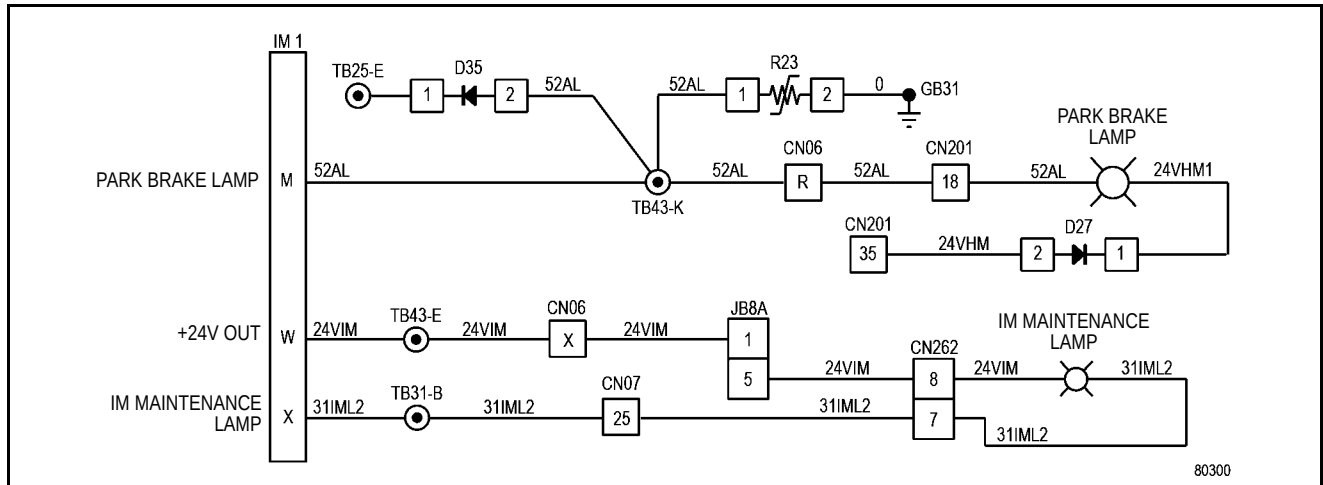
Fault Code A356: Overload on output 1M

Operator Action	Go to shop now
Fault Code	A356
Description	Overload on output 1M.
Fault Conditions	Sets if driver chip detects overcurrent or over temp on output 1M. Output is turned off when overload is detected. Resets at power down.
Operator Alerting System Response	Amber "IM Maintenance" Lamp No Buzzer Right Lamp Module Row 5, Column 3
Resulting Problem(s)	Operation of the parking brake will not be indicated by the parking brake lamp.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. This fault's logic contains just one parameter, which may have additional fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 2. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Park Brake Lamp (IM1 M)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground)
IM Maintenance Lamp (IM1 X)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A363

Related circuit diagram



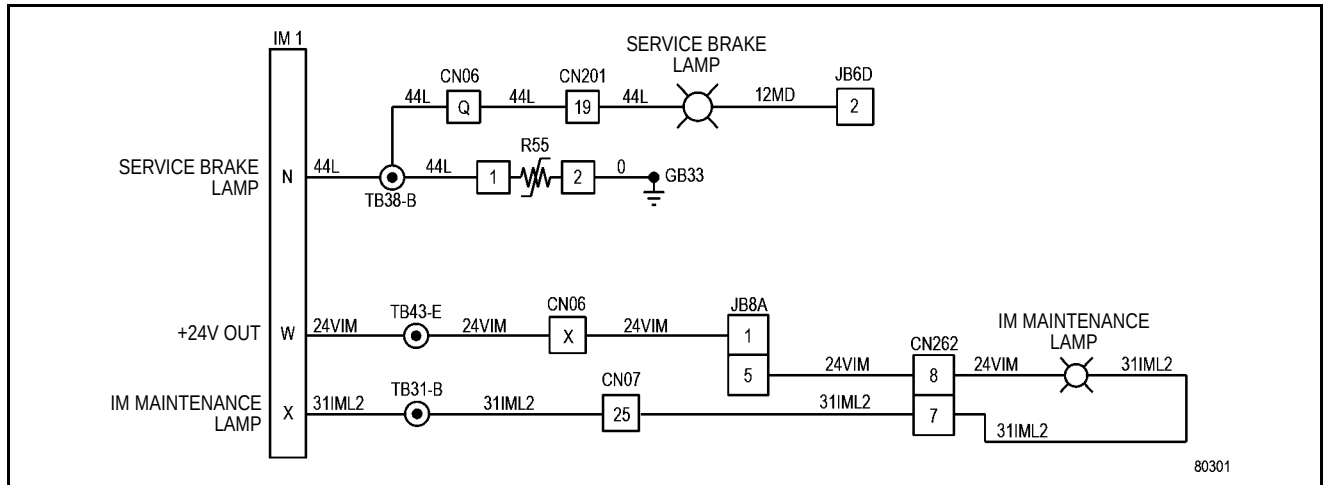
Fault Code A357: Overload on output 1N

Operator Action	None
Fault Code	A357
Description	Overload on output 1N.
Fault Conditions	Sets if driver chip detects overcurrent or over temp on output 1N. Output is turned off when overload is detected. Resets at power down.
Operator Alerting System Response	Amber "IM Maintenance" Lamp No Buzzer Right Lamp Module Row 5, Column 3
Resulting Problem(s)	Operation of the service brake will not be annunciated by the service brake lamp.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. This fault's logic contains just one parameter, which may have additional fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 2. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Service Brake Lamp (IM1 N)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground)
IM Maintenance Lamp (IM1 X)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A363

Related circuit diagram

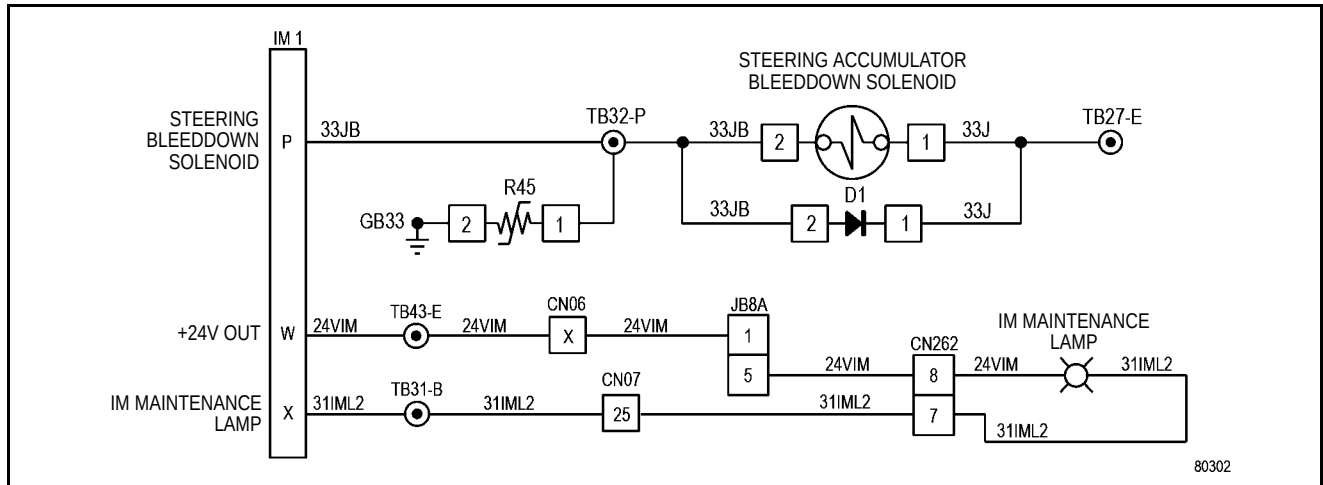


Fault Code A358: Overload on output 1P

Operator Action	Go to shop now
Fault Code	A358
Description	Overload on output 1P.
Fault Conditions	Sets if driver chip detects overcurrent or over temp on output 1P. Output is turned off when overload is detected. Resets at power down.
Operator Alerting System Response	Amber "IM Maintenance" Lamp No Buzzer Right Lamp Module Row 5, Column 3
Resulting Problem(s)	Normal steering bleiddown after shutdown will not occur.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table	
1. This fault's logic contains just one parameter, which may have additional fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 2. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Steering Bleed Valve (IM1 P)	Status - Open Load: Expected with key switch on. No problem. Otherwise, unexpected. Must troubleshoot. Status - Normal: Expected only with key off and steering bleed in process. Troubleshoot if found with key on. Status - Shorted to Ground: Unexpected. Troubleshoot. Status - Overload: Unexpected. Troubleshoot. 0: steering bleed valve is off. This is expected in normal running operation. 1: steering bleed valve is on. This is expected after key off initiates steering bleed operation. Fault(s): A235, A236, A253, A262, A263
IM Maintenance Lamp (IM1 X)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A363

Related circuit diagram



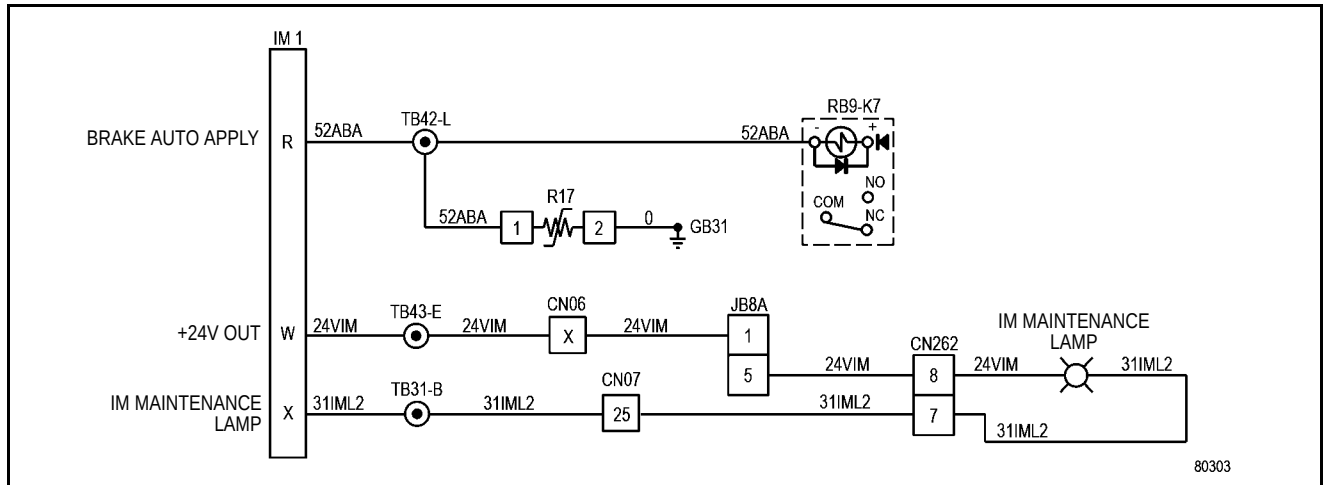
Fault Code A359: Overload on output 1R

Operator Action	Go to shop now
Fault Code	A359
Description	Overload on output 1R.
Fault Conditions	Sets if driver chip detects overcurrent or over temp on output 1R. Output is turned off when overload is detected. Resets at power down.
Operator Alerting System Response	Amber "IM Maintenance" Lamp No Buzzer Right Lamp Module Row 5, Column 3
Resulting Problem(s)	Unused outputs with short circuits on them might damage the driver chip if not corrected.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. This fault's logic contains just one parameter, which may have additional fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 2. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Auto Brake Apply Relay (IM1 R)	Status - Open Load: Unexpected if park brake request and keyswitch are on. But expected if park brake request is off or keyswitch is off. Troubleshoot if inconsistent. Status - Normal: Expected if park brake request and keyswitch are on. Troubleshoot if status is normal with park brake request off or keyswitch off. Status - Shorted to ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: normal operation with park brake either on or off. This valve operates only transiently when the park brake is turned on. 1: brake auto apply valve operates for 1.5 seconds after the park brake is turned on if the truck is stopped. If moving, it will not operate until truck speed has dropped to 0.8 kph. Fault(s): A359
IM Maintenance Lamp (IM1 X)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A363

Related circuit diagram



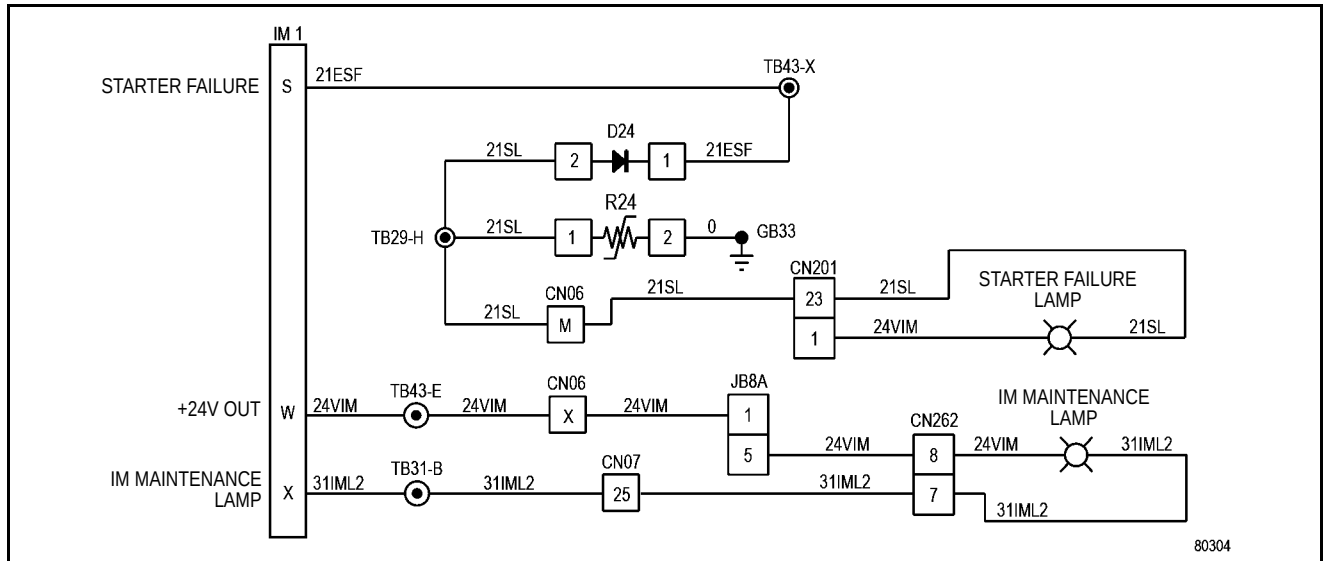
Fault Code A360: Overload on output 1S

Operator Action	None
Fault Code	A360
Description	Overload on output 1S.
Fault Conditions	Sets if driver chip detects overcurrent or over temp on output 1S. Output is turned off when overload is detected. Resets at power down.
Operator Alerting System Response	Amber "IM Maintenance" Lamp No Buzzer Right Lamp Module Row 5, Column 3
Resulting Problem(s)	Unused outputs with short circuits on them might damage the driver chip if not corrected.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. This fault's logic contains just one parameter, which may have additional fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 2. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Engine Start Fail Lamp (IM1 S)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground)
IM Maintenance Lamp (IM1 X)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A363

Related circuit diagram

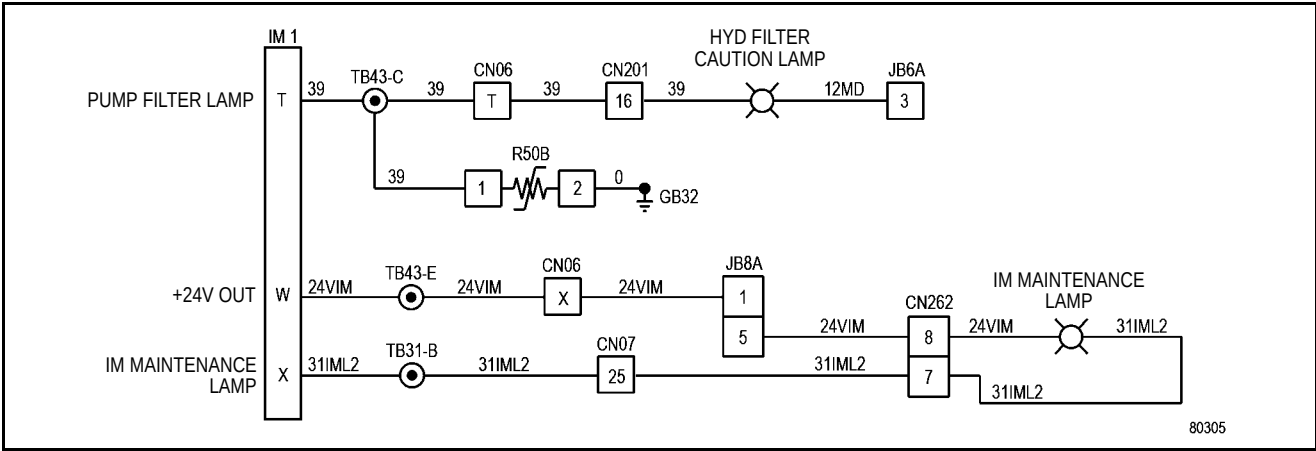


Fault Code A361: Overload on output 1T

Operator Action	None
Fault Code	A361
Description	Overload on output 1T
Fault Conditions	Sets if driver chip detects overcurrent or over temp on output 1T. Output is turned off when overload is detected. Resets at power down.
Operator Alerting System Response	Amber "IM Maintenance" Lamp No Buzzer Right Lamp Module Row 5, Column 3
Resulting Problem(s)	Hydraulic filter problems will not be indicated by the Pump Filter lamp.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table	
1. This fault's logic contains more than one parameter, which may have additional fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 2. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Pump Filter Switches Lamp (IM1 T)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground)
IM Maintenance Lamp (IM1 X)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A363

Related circuit diagram



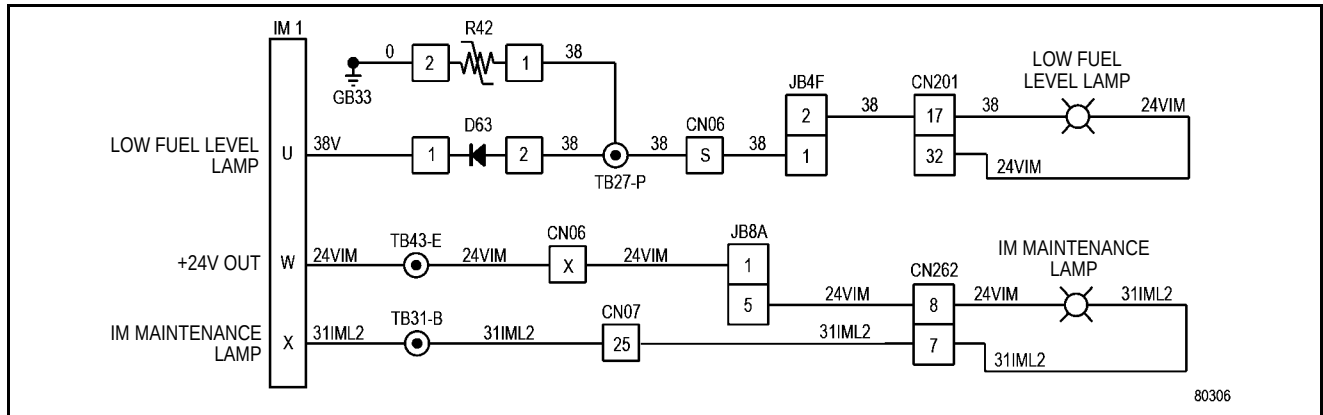
Fault Code A362: Overload on output 1U

Operator Action	None
Fault Code	A362
Description	Overload on output 1U
Fault Conditions	Sets if driver chip detects overcurrent or over temp on output 1U. Output is turned off when overload is detected. Resets at power down.
Operator Alerting System Response	Amber "IM Maintenance" Lamp No Buzzer Right Lamp Module Row 5, Column 3
Resulting Problem(s)	Low fuel level lamp will not operate.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. This fault's logic contains more than one parameter, which may have additional fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 2. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Low Fuel Level Lamp (IM1 U)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground)
IM Maintenance Lamp (IM1 X)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A363

Related circuit diagram



Fault Code A363: Overload on output 1X

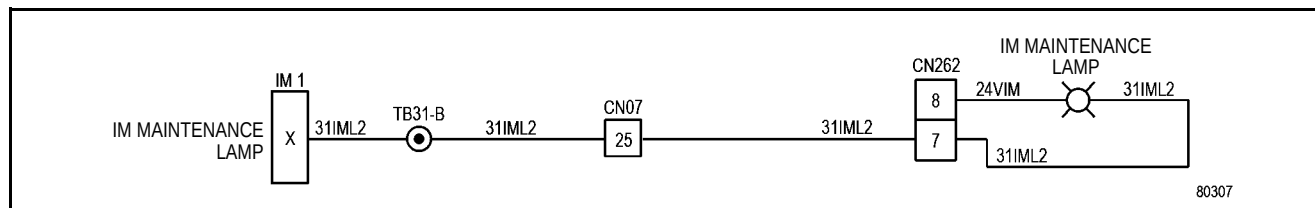
Operator Action	None
Fault Code	A363
Description	Overload on output 1X.
Fault Conditions	Sets if driver chip detects overcurrent or over temp on output 1X. Output is turned off when overload is detected. Resets at power down.
Operator Alerting System Response	None
Resulting Problem(s)	IM Maintenance lamp will not operate.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table

1. This fault's logic contains just one parameter. This output normally drives the IM Maintenance Lamp. No other information is available.

Parameter	Expected State and/or Related Fault(s)
IM Maintenance Lamp (IM1 X)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground)

Related circuit diagram



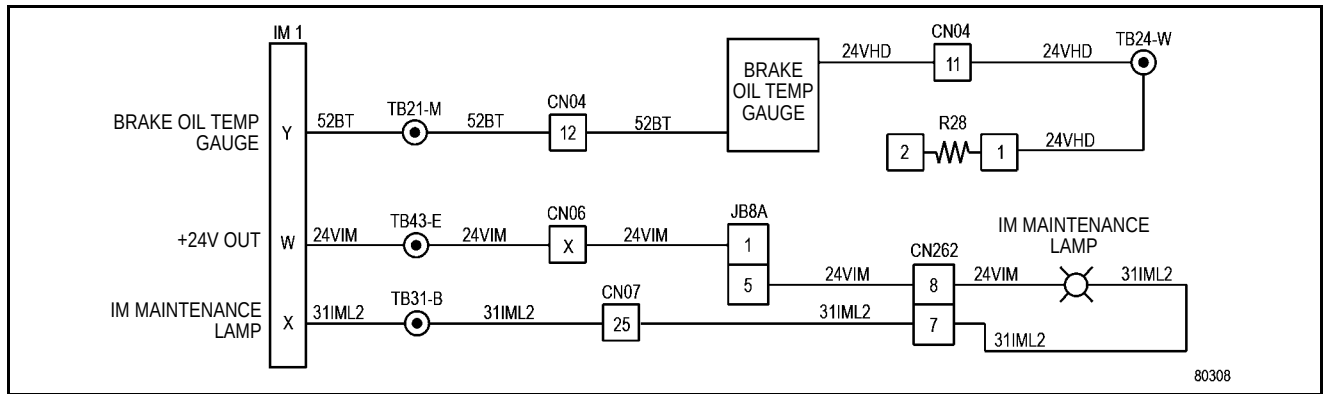
NOTES

Fault Code A364: Overload on output 1Y

Operator Action	None
Fault Code	A364
Description	Overload on output 1Y
Fault Conditions	Sets if driver chip detects overcurrent or over temp on output 1Y. Output is turned off when overload is detected.
Operator Alerting System Response	Amber "IM Maintenance" Lamp No Buzzer Right Lamp Module Row 5, Column 3
Resulting Problem(s)	Hydraulic oil temperature gauge will not operate.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table	
1. This fault's logic contains more than one parameter, which may have additional fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 2. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
Hydraulic Oil Temp Gauge (IM1 Y)	Status - Open Load: Unexpected: Troubleshoot Status - Shorted to Ground: Unexpected: Troubleshoot Status - Overload: Unexpected: Troubleshoot Approximate Calibration: Start Green Band: 60 C Start Red Band: 120 C
IM Maintenance Lamp (IM1 X)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A363

Related circuit diagram

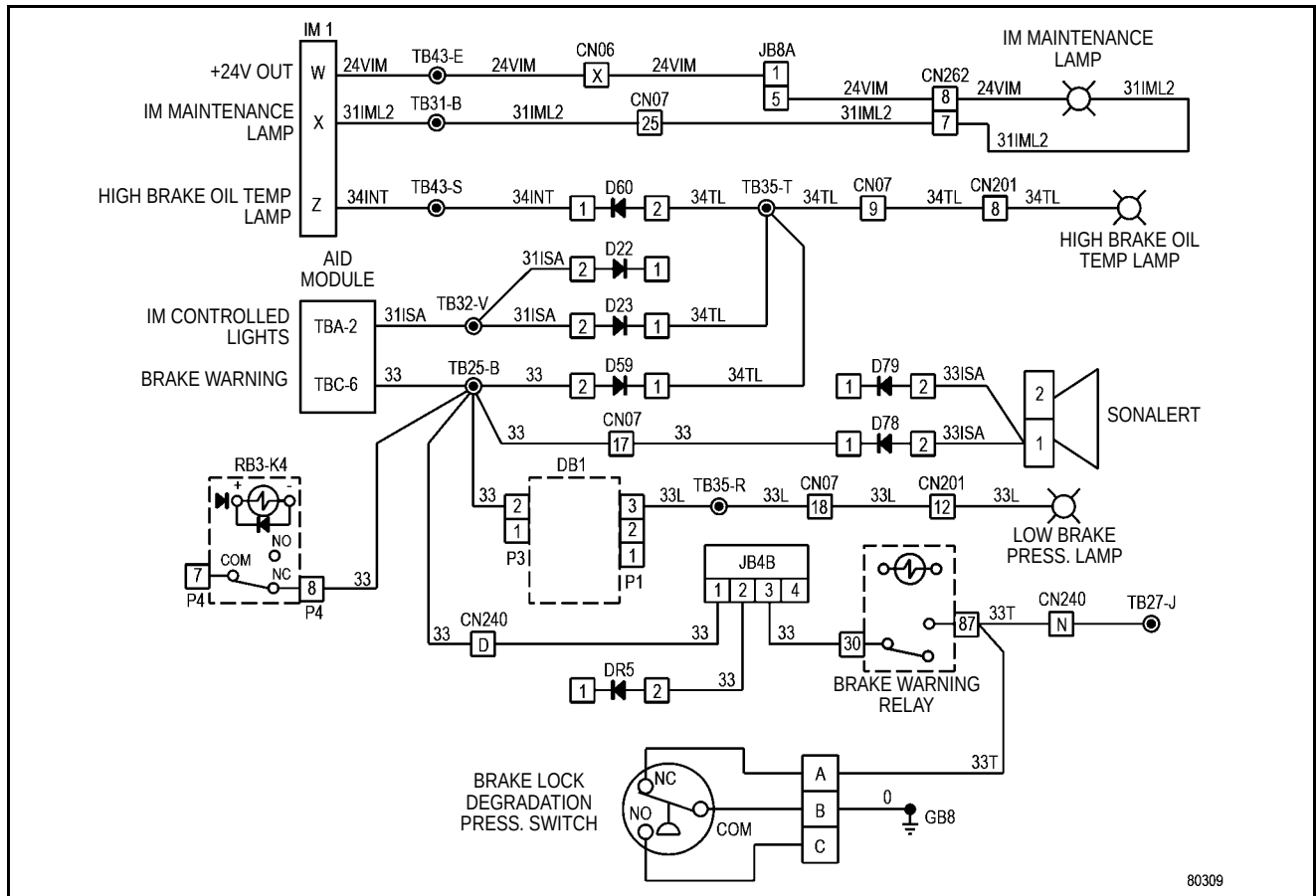


Fault Code A365: Overload on output 1Z

Operator Action	None
Fault Code	A365
Description	Overload on output 1Z
Fault Conditions	Sets if driver chip detects overcurrent or over temp on output 1Z. Output is turned off when overload is detected.
Operator Alerting System Response	Amber "IM Maintenance" Lamp No Buzzer Right Lamp Module Row 5, Column 3
Resulting Problem(s)	Unused outputs with short circuits on them might damage the driver chip if not corrected.
Related Information	A laptop running "IM Realtime Data Monitor" software may be required to resolve this fault.

Table	
1. This fault's logic contains more than one parameter, which may have additional fault code(s). This fault may be resolved by resolving the parameter(s) active fault code(s). Refer to the Troubleshooting Instructions for the active parameter fault(s). 2. If there are no active parameter fault codes, then based on the truck's setup while this fault is active, determine which of the parameters in the Table is not shown in its expected state on the IM Realtime Data Monitor program. Refer to the schematic to identify which item(s) may be causing the parameter(s) to be in the unexpected state. Troubleshoot these items. 3. If this fault is not currently active or if no parameters are currently in the unexpected state and the malfunction is still unresolved, then check each of the parameters in the Table for proper functionality. Troubleshoot all item(s) related to the parameter(s) which are found to be malfunctioning.	
Parameter	Expected State and/or Related Fault(s)
High Brake Oil Temp Lamp (IM1 Z)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground)
IM Maintenance Lamp (IM1 X)	Status - Open Load: Unexpected. Troubleshoot Status - Normal: Expected Status - Shorted to Ground: Unexpected. Troubleshoot Status - Overload: Unexpected. Troubleshoot. 0: Lamp Off (IM output at B+) 1: Lamp On (IM output at ground) Fault(s): A363

Related circuit diagram



80309

NOTES

SECTION G
REAR AXLE, SPINDLES AND WHEELS
INDEX

TIRES AND RIMS	G2-1
FRONT WHEEL HUB AND SPINDLE.....	G3-1
REAR AXLE MOUNTING	G4-1
REAR AXLE AND WHEEL MOTOR	G5-1

NOTES

SECTION G2

TIRES AND RIMS

INDEX

TIRES AND RIMS	G2-3
WHEEL STUD MAINTENANCE	G2-4
FRONT TIRES AND RIMS	G2-5
Removal	G2-5
Installation	G2-6
REAR TIRES AND RIMS	G2-7
Removal	G2-7
Installation	G2-8
RIM	G2-9
Tire Removal	G2-9
Tire Installation	G2-9

NOTES

TIRES AND RIMS

The truck tires should be inspected and tire pressure checked with an accurate pressure gauge before each working shift. Tire pressure will vary according to manufacturer and local working conditions. Consult the tire manufacturer for recommended tire pressure.

Insure valve caps are securely applied to valve stems. The caps protect valves from dirt build up and damage. DO NOT bleed air from tires which are hot due to operation; under such circumstances, it is normal for pressure to increase in the tire due to expansion.

A bent or damaged rim which does not support the bead properly may cause abnormal strain on the tire resulting in tire damage. If a tire becomes deeply cut, it should be removed and repaired. Neglected cuts cause many tire problems; water, sand, dirt and other foreign materials work into the tire through a cut, eventually causing tread or ply separation.

Tires should be stored indoors if possible. If stored outdoors, cover tires with tarpaulin to keep out dirt, water and other foreign materials. Long exposure to the sun will cause ozone cracks. Storage should be in a cool, dry, dark, draft free location. Tires should be stored vertically. If they must be laid on their sides for a short period, avoid distortion by stacking no more than three tires on top of one another. Avoid contact with oil, grease and other petroleum products.

Before storing used tires, clean thoroughly and inspect for damage. Repair as necessary. When a truck is placed in storage, it should be blocked to remove the weight from the tires. If a stored truck cannot be blocked, check air pressure and inspect tires twice a month for proper inflation pressure.

⚠ WARNING

DO NOT weld or apply heat on the rim assembly with the tire mounted on the rim. Remaining gases inside the tire may ignite causing explosion of tire and rim.

DO NOT go near a tire if a brake or wheel motor has experienced a fire until the tire has cooled.

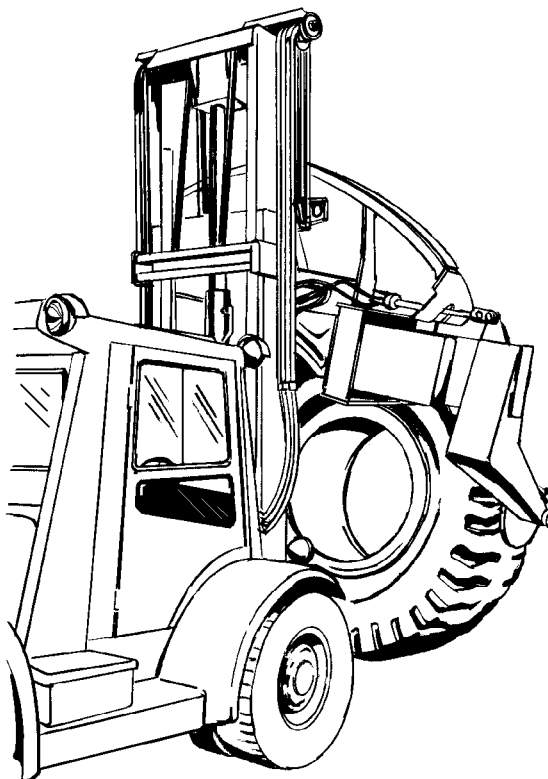
⚠ DANGER

When inflating tires always use a safety cage. Never inflate a tire until the locking is securely in place. Do not stand in front of or over the locking during inflation procedures. Never overinflate a tire. Refer to tire manufacturer's recommendations.

Always keep personnel away from a wheel and tire assembly when it is being removed or installed.

The tire and rim weigh approximately 7 165 kg (15,800 lb). Make sure that tire handling equipment is capable of lifting and maneuvering the load.

Due to the size and weight of the tire and rim assemblies, special handling equipment, such as a modified fork lift called a "tire handler" as shown in Figure 2-1, is desirable. Consult local tire vendors for sources of equipment designed especially to remove, repair, and install large off-highway truck tires.



91573A

FIGURE 2-1. TYPICAL TIRE HANDLER

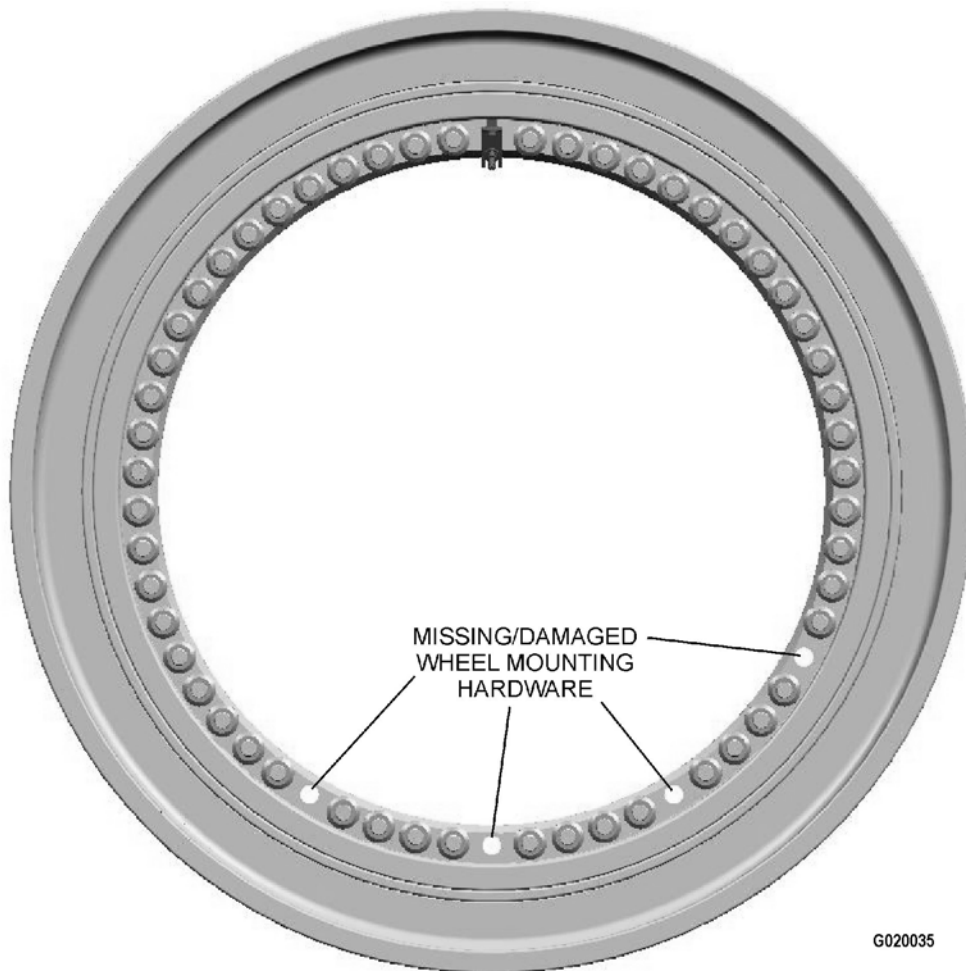
WHEEL STUD MAINTENANCE

The wheel mounting hardware used on some trucks may no longer be able to maintain the specified clamping force (tightening torque). Inadequate clamping force may lead to broken studs, stripped threads, and/or failure of the stud knurl which would allow the stud to rotate when the nut is loosened or tightened. Stud replacement is necessary based on the following guidelines:

- The maximum allowable number of missing or damaged studs in any one bolt circle is four.
- The position of missing or damaged studs must not be in consecutive locations. In addition, there must be a minimum of four properly functioning studs between each missing stud. Refer to Figure 2-2.

CAUTION

While replacing missing or damaged studs, the condition of all other stud threads should be inspected. If minor corrosion or dirt is found in the threads, wire brush thoroughly. If damaged (knicked or highly corroded) threads are found, the stud(s) must be replaced.



G020035

FIGURE 2-2. MISSING/DAMAGED WHEEL STUD ALLOWANCE

FRONT TIRES AND RIMS

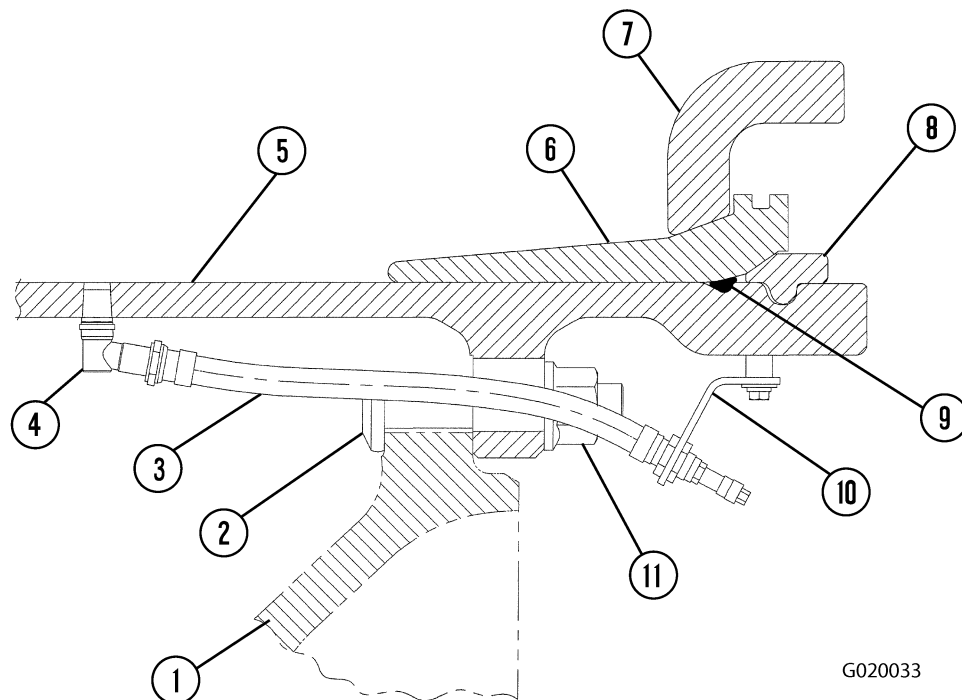
Removal

1. Apply parking brake and block rear wheels to prevent movement of truck.
2. Following normal shutdown procedures, place the rest switch in the ON position, shut down the engine and verify that the link voltage lights are OFF. Allow at least 90 seconds for the accumulators to bleed down. Turn the steering wheel to ensure that no pressure remains. As a safety precaution, bleed down the brake accumulators.
3. Place a jack under the spindle or frame at the front cross tube.
4. Raise the front end of the truck until the tire clears the ground. Block up the truck securely under the frame.
5. Inspect the hydraulic brake lines for damage or leaking fittings.
6. Grip the tire and wheel assembly with tire a handler. Remove nuts (11, Figure 2-3) that secure the wheel assembly.
7. Be careful not to damage the inflation hose during tire removal. Move the wheel assembly away from the wheel hub and into a clean work area.

WARNING

Do not attempt to disassemble wheel assembly until all air pressure is bled off.

Always keep personnel away from a wheel assembly when it is being removed or installed.



G020033

FIGURE 2-3. FRONT WHEEL HUB AND RIM ASSEMBLY

- | | | |
|------------------------|-------------------|-------------------|
| 1. Wheel Hub | 5. Rim | 9. O-Ring |
| 2. Stud | 6. Bead Seat Band | 10. Clamp Bracket |
| 3. Tire Inflation Hose | 7. Side Flange | 11. Flanged Nut |
| 4. Swivel Connector | 8. Lock Ring | |

Installation

NOTE: Remove all dirt and rust from mating parts before installing wheel assembly.

1. Grip the wheel assembly with the tire handler and align the tire inflation hose and wheel hub (1, Figure 2-3). Position the rim onto the wheel hub studs.



Grease containing molybdenum disulphide must never be used on wheel mounting hardware. Use of this type of grease on wheel mounting hardware may result in wheel mounting studs stretching beyond their elastic limit, making them susceptible to breakage.

2. Lubricate all stud threads and nut seating flanges with a lithium based grease that does not contain molybdenum disulphide. Install and tighten the nuts in the following sequence:
 - a. Install six nuts at the 12 o'clock and six nuts at the 6 o'clock positions. Tighten each nut to **2 325 ± 136 N·m (1,715 ± 100 ft lb)**.
 - b. Install three nuts directly below the 3 o'clock and three nuts directly above the 9 o'clock positions. Tighten the nuts to **2 325 ± 136 N·m (1,715 ± 100 ft lb)**.
 - c. Install three nuts directly above the 3 o'clock and three nuts directly below the 9 o'clock positions. Tighten these nuts to **2 325 ± 136 N·m (1,715 ± 100 ft lb)**.
 - d. Install the remaining nuts and torque in a clockwise direction to **2 325 ± 136 N·m (1,715 ± 100 ft lb)**.
 - e. Retighten all nuts in a clockwise direction to **2 325 ± 136 N·m (1,715 ± 100 ft lb)**.
3. Remove the blocking and lower the jack.
4. Operate the truck for one load and retighten the wheel nuts to the specified torque.

REAR TIRES AND RIMS

Removal

1. Park the truck on level ground and block the front wheels. Position a jack under the rear suspension mounting plates as shown in Figure 2-4.

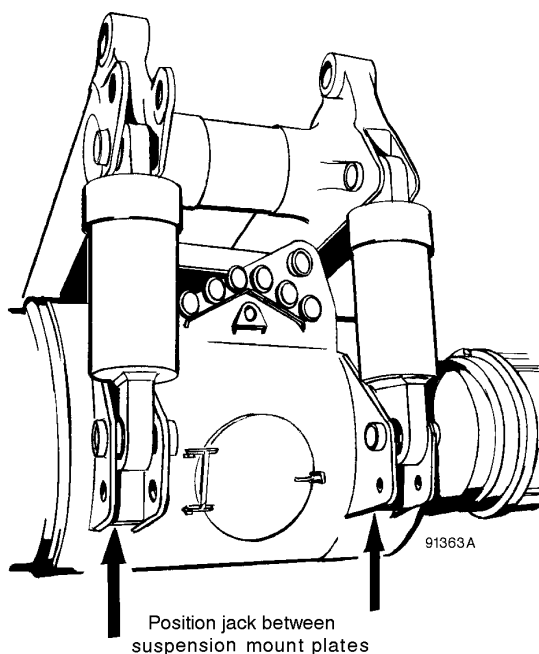


FIGURE 2-4. REAR AXLE JACK LOCATION

2. Raise the rear axle housing until the tires clear the ground. Securely block up the rear axle housing near the wheel motor mounting flange.
3. If the rear inner tire is to be removed, remove the inner tire inflation hose at the clamp on the outer wheel and disconnect it from inner extension (7, Figure 2-6).
4. Grip outer wheel and tire with tire handler arms as shown in Figure 2-5. Remove flanged nuts (10, Figure 2-6) from adaptor ring (9) securing outer rim to wheel motor hub.
5. Pull straight out on the outer wheel assembly and remove it.
6. If inner wheel removal is necessary, disconnect inner tire inflation extension (7) and remove it.

7. Position a tire handler to grip the inner wheel. Remove flanged nuts (6).
8. Pull straight out to remove the tire from the wheel hub.

⚠ IMPORTANT ⚠

If the wheel motor is to be removed from the truck for service, install approximately eight flanged nuts with appropriate spacers in place of the outer wheel adaptor ring and nuts removed in Step 4. This will provide additional support for the wheel motor transmission during removal and transportation.

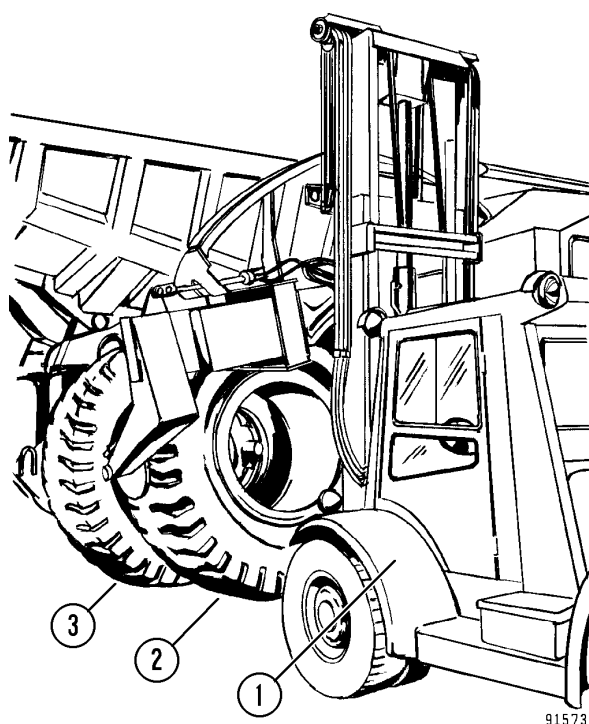
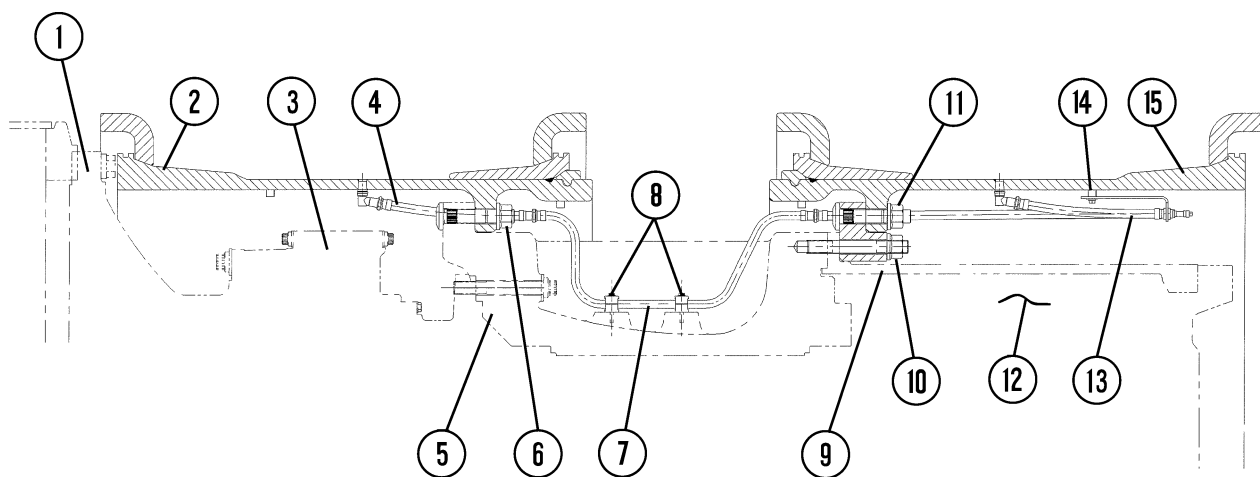


FIGURE 2-5. REAR TIRE REMOVAL

- | | |
|--------------------|--------------------|
| 1. Tire Handler | 3. Inner Rear Tire |
| 2. Outer Rear Tire | |



G020034

FIGURE 2-6. REAR WHEEL ASSEMBLY

- | | | |
|--------------------------------|--------------------|------------------------------|
| 1. Wheel Motor Mounting Flange | 6. Flanged Nut | 11. Flanged Nut |
| 2. Inner Wheel Rim | 7. Inner Extension | 12. Wheel Motor Transmission |
| 3. Disc Brake Assembly | 8. Clamps | 13. Extension |
| 4. Extension | 9. Adapter Ring | 14. Bracket |
| 5. Wheel Hub | 10. Flanged Nut | 15. Outer Wheel Rim |

Installation



Always keep personnel away from a wheel assembly when it is being removed and installed.

NOTE: Clean all mating surfaces and check the stud threads before installing the wheel assemblies.

1. If either wheel motor has been removed, bleed the disc brakes before installing the rear tires. Refer to Section J, Wet Disc Brake Assembly - Wet Disc Brake Bleeding Procedure.
2. Grip the inner wheel assembly with a tire handler and install it onto wheel hub (5, Figure 2-6). Carefully align tire inflation extension line (4) for mating with inner extension (7).
3. Lubricate all stud threads and nut seating flanges with lithium based grease. Install and tighten the nuts in the following sequence:
 - a. Install six nuts at the 12 o'clock and six nuts at the 6 o'clock positions. Tighten each nut to **2 325 ± 136 N·m (1,715 ± 100 ft lb)**.
 - b. Install three nuts directly below the 3 o'clock and three nuts directly above the 9 o'clock positions. Tighten these nuts to **2 325 ± 136 N·m (1,715 ± 100 ft lb)**.
 - c. Install three nuts directly above the 3 o'clock and three nuts directly below the 9 o'clock positions. Tighten these nuts to **2 325 ± 136 N·m (1,715 ± 100 ft lb)**.
 - d. Install the remaining nuts and tighten in a clockwise direction to **2 325 ± 136 N·m (1,715 ± 100 ft lb)**.
 - e. Retighten all nuts in a clockwise direction to the required **2 325 ± 136 N·m (1,715 ± 100 ft lb)**.
4. If adapter ring (9) requires installation, install the adapter onto the wheel hub/wheel motor. Lubricate the studs and nut flanges with lithium based grease.
5. Using the procedures in Step 3, install flanged nuts (10) in the sequence described and tighten to the same torque value as the inner wheel.

6. Grip the outer wheel assembly with a tire handler and position it onto the wheel motor hub.

NOTE: Position the outer dual wheel to align the tire valve bracket with the inner wheel inflation line.

7. Using the procedures in Step 3, install flanged nuts (11) in the sequence described and tighten to the same torque value as the inner wheel.
8. Secure the inner and outer dual tire inflation lines to bracket (14) on the outer rim.
9. Remove the blocks from under the truck and lower the truck to the ground. Operate the truck for one load and retighten outer wheel nuts (11) and adapter flange nuts (10) to **2 325 ± 136 N·m (1,715 ± 100 ft lb)**.

NOTE: Inner flanged nuts (5 & 7) and studs should be inspected for breakage or missing nuts during scheduled maintenance checks by inserting a mirror between the rear tires.

RIM

Tire Removal



DO NOT weld or apply heat on the rim assembly with the tire mounted on the rim. Resulting gases inside the tire may ignite, causing an explosion.

When inflating tires always use a safety cage. Never inflate a tire until the lockring is securely in place. Do not stand in front of or over the lockring during inflation procedures. Never overinflate a tire. Refer to tire manufacturers recommendations.

1. Place the tire and wheel assembly in a safety cage and discharge all air pressure from the tire.
2. Attach a hydraulic bead breaker to the rim by slipping the jaws of the frame assembly over the outer edge of flange (7, Figure 2-7). Make sure that the jaws of the frame are as near to bead seat band (6) as possible.
3. Following the tool manufacturer's instructions, move the tire bead in far enough to permit placing a wedge between the tire and the flange at the side of the tool.

4. Repeat this procedure at locations approximately 90° from the first application. Continue this procedure until the tire bead is free from the rim.
5. After the bead is broken loose, insert the flat of a tire tool in the beading notch on lockring (8). Pry the lockring up and out of the groove on the rim.
6. Pry in on bead seat band (6) until O-ring (9) is exposed. Remove the O-ring.
7. Remove bead seat band (6) from rim (5) and remove flange (7).
8. Reposition wheel assembly and repeat removal procedure on opposite side of tire. Remove the tire from the rim.

Tire Installation

1. Before mounting the tire to the rim, remove all dirt and rust from the rim parts, particularly the O-ring groove and bead seats. Also touch up all metal parts with anti-rust paint to prevent bare metal from being exposed to the weather.

NOTE: Do not allow paint, rust or other contamination to cover the mating faces of lock ring (8, Figure 2-7) and rim (5).



Make sure that proper rim parts are used for reassembly. Use of incompatible parts may not properly secure the assembly, resulting in violently flying parts upon inflation.

2. If the tire inflation hose and hardware were removed, reinstall them in the rim. Position the hose assembly for proper routing.
3. Install the inner flange on the rim. Coat the beads of the tire with tire mounting soap solution.



Prying against tire bead may cause damage to tire bead and will cause air leaks.

4. Position the tire over the rim and work the tire on as far as possible without prying against the beads.

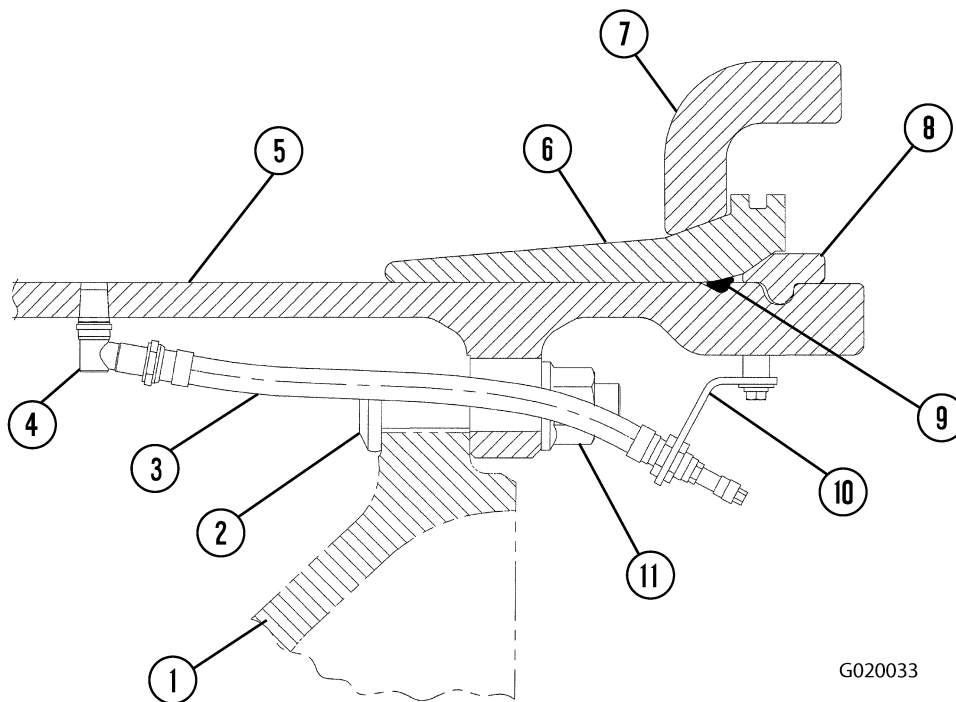


FIGURE 2-7. FRONT WHEEL HUB AND RIM ASSEMBLY

- | | | |
|------------------------|-------------------|-------------------|
| 1. Wheel Hub | 5. Rim | 9. O-Ring |
| 2. Stud | 6. Bead Seat Band | 10. Clamp Bracket |
| 3. Tire Inflation Hose | 7. Side Flange | 11. Flanged Nut |
| 4. Swivel Connector | 8. Lock Ring | |

5. Install outer flange (7, Figure 2-7) in position and install bead seat band (6). Push in on the bead seat band to expose the O-ring groove in the rim.

6. Lubricate new O-ring (9) with soap solution and install it in the groove of the rim.

7. Install lockring (8) and tap it into place with a lead hammer. The lockring lug must fit into the slot of the rim.

9. If the beads of the tire and O-ring do not seat within one minute, raise the tire slightly and tap the bead seat band. This will help the air pressure to push the tire bead out into position.

10. As soon as the seating has been accomplished, install the valve core and inflate the tire to the recommended tire pressure.

⚠ DANGER

Use a safety cage whenever possible. Stand to one side as tire is being inflated. Never start inflating unless the lockring is securely in place. DO NOT stand in front of or over the lockring when inflating.

8. With the tire inside a safety cage, remove the valve core from the valve stem and inflate the tire to seat the beads of the tire and the O-ring as specified by the tire manufacturer.

SECTION G3

FRONT WHEEL HUB AND SPINDLE

INDEX

FRONT WHEEL HUB AND SPINDLE	G3-3
WHEEL HUB AND SPINDLE ASSEMBLY	G3-3
Removal	G3-3
Spindle Removal Procedure (off of the truck)	G3-6
Installation	G3-8
Disassembly	G3-9
Cleaning and Inspection	G3-11
Assembly	G3-11
Wheel Bearing Adjustment	G3-12
Brake Installation	G3-13
Seal Assembly Gap Check	G3-14
WHEEL SPEED SENSOR TESTING	G3-15
STEERING CYLINDERS	G3-16
Spherical Bearing Wear Limits	G3-16
Removal	G3-17
Installation	G3-17
Bearing Replacement	G3-17
TIE ROD	G3-19
Removal	G3-19
Installation	G3-19
Disassembly	G3-20
Assembly	G3-20
Toe-In Adjustment	G3-21
TIE ROD INSPECTION AND TORQUE PROCEDURE	G3-22

NOTES

FRONT WHEEL HUB AND SPINDLE

WHEEL HUB AND SPINDLE ASSEMBLY

The following instructions will cover the complete removal, installation, disassembly, assembly and bearing adjustment of the front wheel hub and spindle. If only brake service is to be performed, refer to Section J, Brake Circuit.

⚠ WARNING

Do not loosen or disconnect any hydraulic line or component until the engine is stopped, the key switch is OFF for 90 seconds, and the brake accumulators are depressurized.

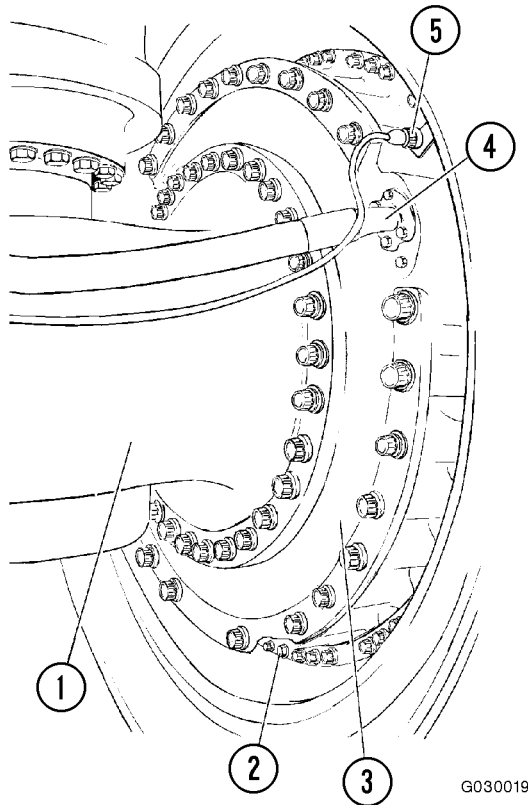


FIGURE 3-1. FRONT WHEEL & SPINDLE INSTALLATION

- | | |
|-----------------------|-----------------------|
| 1. Spindle | 4. Brake Cooling Line |
| 2. Drain Plug | 5. Speed Sensor |
| 3. Disc Brake Housing | Connector |

Removal

1. Shut down the truck. Refer to "Normal Engine Shutdown Procedure" in Section A3, General Safety & Operating Instructions.
2. Depressurize the brake accumulators by using the bleed valves on the brake manifold.
3. Activate the battery disconnect switches.
4. Remove the front tire and rim assembly. Refer to "Front Tire and Rim - Removal" in Section G2.
5. Close the hydraulic pump shutoff valves.
6. Disconnect speed sensor connector(s) (5, Figure 3-1). Tie the cables back away from the spindle to prevent damage during spindle removal.

NOTE: The left wheel has two speed sensors installed.

7. Disconnect brake cooling oil lines (4) at the inlet and outlet ports on the brake housing. Plug the hoses and ports to help prevent contamination.
8. Disconnect the brake apply line. Cap the hoses and ports to prevent contamination.
9. Remove drain plug (2) and drain the oil from the brake housing and hub bearings into a suitable container.
10. Remove the lubrication lines from the tie rod and steering cylinder.
11. Disconnect the steering cylinder and the tie rod from the spindle being removed. Refer to "Steering Cylinders and Tie Rod - Removal" in this section.

12. Position a fork lift under the wheel hub and spindle assembly as shown in Figure 3-2. The weight of the spindle assembly is approximately 6 652 kg (14,665 lb). Ensure the lifting device can handle the load safely. Ensure the spindle and hub are securely fastened to the lifting device to prevent tipping when it is free from the suspension.
13. Remove capscrews and washers (1, Figure 3-2) that secure retainer plate (2) to the spindle structure and suspension. To prevent thread damage, loosen the capscrews in a circular pattern in torque increments of **678 N·m (500 ft lb)**. Remove the retainer plate.
14. Remove 16 steering arm capscrews (5, Figure 3-2) as follows:
 - a. Refer to Figure 3-4. Remove only the capscrews that are designated by an "X".
 - b. To prevent thread damage, remove these capscrews using a circular pattern in torque increments of **678 N·m (500 ft lb)**. DO NOT attempt to remove each capscrew in one sequence.

CAUTION

DO NOT remove the cap screws in one step with an air wrench. Revolve around the circular pattern at least 3 times, gradually loosening the cap screws during each revolution.

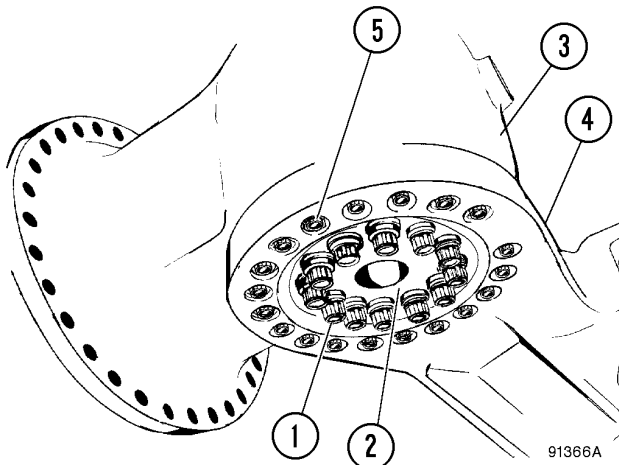


FIGURE 3-2. STEERING ARM ASSEMBLY

- | | |
|------------------------|-----------------|
| 1. Capscrews & Washers | 4. Steering Arm |
| 2. Retainer Plate | 5. Capscrews |
| 3. Spindle | |

15. Run the proper size tap into the threaded holes to ensure clean, good quality threads.

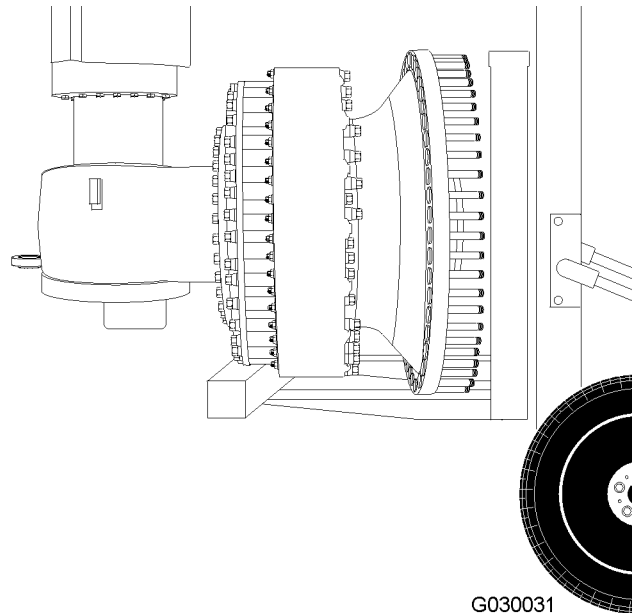


FIGURE 3-3. TYPICAL LIFTING DEVICE (FORKLIFT)

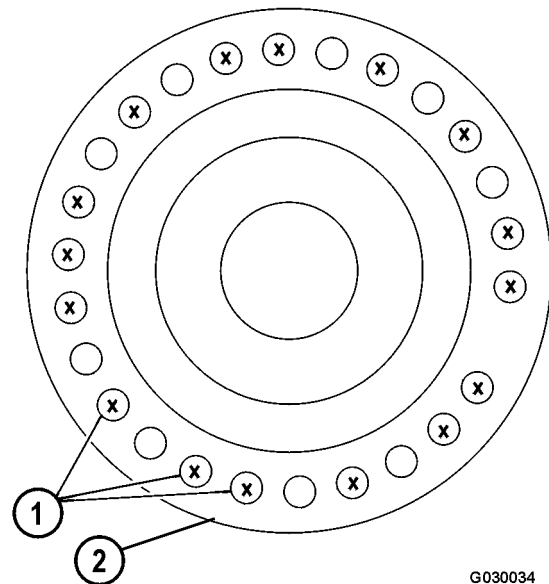


FIGURE 3-4. SPINDLE REMOVAL TOOL

- | | |
|-----------------------------------|-------------------------------|
| 1. Capscrew Insertion Holes ("X") | 2. Tool Structure Holes ("X") |
|-----------------------------------|-------------------------------|

NOTE: Older pusher tools (5, Figure 3-5) may have been made with a round tube (shown as a dotted line) welded on one side. The tool can be installed upside down and used with a hydraulic ram as shown.

Refer to Section M8, Special Tools, for dimensions for fabricating the spindle pusher tool and spacers (6, Figure 3-5). Special cap screws and washers are also required.

The hydraulic ram must be a low profile hydraulic ram (pancake jack) to fit in this area. Most hydraulic rams that will fit in this area are rated at 150 tons.

16. Locate 16 cap screws (1, Figure 3-5) and washers (2) as specified below:

Cap Screw (P/N KC7095). 1 1/4 x 8 in.

Hardened Washer (P/N WA0366) 1 1/4 in.

17. Lubricate the cap screw threads and flat washers with chassis grease.

18. Install pusher tool (5) as shown in Figure 3-5 using the only three cap screws (1), three washers (2) and three spacers (6). The three cap screws must be evenly spaced. Tighten the three cap screws to **68 N·m (50 ft lb)**.

NOTE: Verify minimum thread engagement is 41.1 mm (1.62 in.) on pusher cap screws when inserted.

19. Ensure hydraulic ram (4) is completely retracted. Install hydraulic ram with reaction plate (3) on top of the hydraulic ram.

NOTE: If there is a large gap between the reaction plate and the suspension, it may be necessary to add steel spacers below the hydraulic ram.



Do not apply pressure to the hydraulic ram until all 16 cap screws are installed.

20. Install the remaining 13 cap screws and tighten them to **68 N·m (50 ft lb)**. This is to ensure all cap screws will pull evenly when pressure is applied by the hydraulic ram.



The maximum force that can be applied during the spindle removal process is not to exceed 701 000 kgf (1,545,600 lb).

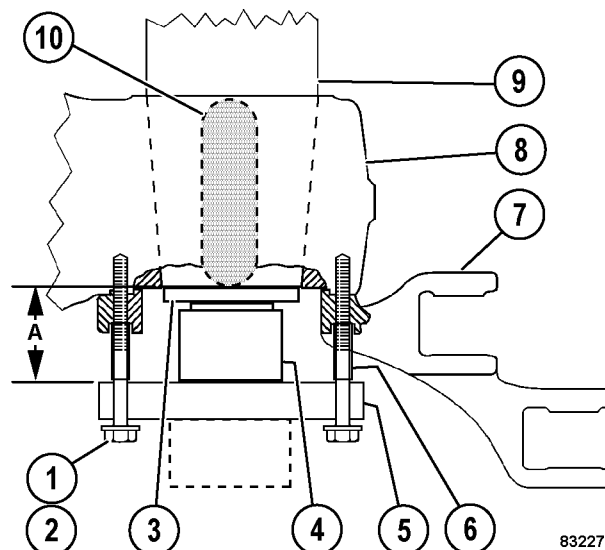


FIGURE 3-5. PULLER TOOL WITH RAM

- | | |
|-----------------------------|----------------------|
| 1. Cap Screw (KC7095) | 5. Pusher Tool |
| 2. Hardened Washer (WA0366) | 6. Spacer |
| 3. Reaction Plate | 7. Steering Arm |
| 4. Hydraulic Ram | 8. Spindle |
| | 9. Suspension Piston |
| | 10. Area to heat |

WARNING

Heavy components and high forces are involved in this procedure. Use caution at all times when applying force to these parts. Sudden release of the spindle could cause components to move forcefully and unexpectedly.

21. Start applying pressure to the hydraulic ram to separate the parts. If the specified maximum force of the hydraulic ram is reached and the tapered parts have not separated, slowly and uniformly apply heat to the spindle as shown (10, Figure 3-5). Heat must be applied in two locations 180 degrees apart. Allow heat to penetrate into the spindle. Reapply heat as required. **Do not exceed 454 °C (850 °F) anywhere on the spindle.**

CAUTION

Heating the spindle in excess of 454° C (850° F) may cause serious damage to the spindle.

22. Use heat as specified in the previous step and a large hammer to carefully tap the top surface of the spindle until the spindle breaks free.

NOTE: If the spindle does not separate from the suspension, the spindle and suspension must be removed from the truck as an assembly. Then use the "Spindle Removal Procedure (off of the truck)" procedure to separate the spindle from the suspension.

23. After separation, lower the wheel hub and spindle assembly away from suspension piston rod. Be careful during removal to prevent damage to the suspension piston rod taper and the tapered spindle bore.
24. Move the spindle and hub assembly to a clean work area for repair.

Spindle Removal Procedure (off of the truck)

If the hydraulic ram method (with heat) did not separate the spindle from the suspension, then the suspension and spindle must be removed from the truck as an assembly. With the suspension on the ground, the pusher tool can be used to separate the spindle from the suspension.

WARNING

Heavy structures and high forces are involved in this procedure. Use caution at all times when applying force to these parts. Sudden release of the spindle could cause components to move forcefully and unexpectedly.

Refer to Section M8, Special Tools for dimensions for fabricating the spindle pusher tool and the cap screws and washers required. Multiple flatwashers may be required under the cap screws to be effective.

NOTE: Hardened flat washers must be used under the pusher cap screws to prevent galling. Lubricate cap screw threads and washers with a lubricant such as chassis lube.

1. Position the suspension and spindle assembly on the work floor as shown in Figure 3-6. The combined weight of the spindle, hub and brake assembly and front suspension is approximately 9 375 kg (20,665 lb). Ensure the lifting device can handle the load safely.
2. The wheel hub must be supported with an overhead hoist. The spindle, hub and brake assembly weighs approximately 6 652 kg (14,665 lb). Ensure the lifting device can handle the load safely.

NOTE: If the spindle is turned so the wheel hub is resting on the ground, the spindle must still be attached to an overhead hoist.

3. Install pusher tool (2, Figure 3-6) using cap screws and washers (1) specified below:

Cap Screw (P/N KC7095) 1 1/4 x 8 in.

Hardened Washer (P/N WA0366) 1 1/4 in.

NOTE: Verify minimum thread engagement is 41.1 mm (1.62 in.) on pusher cap screws when inserted.

4. After the pusher tool has been installed, progressively increase the torque on the cap screws in a circular pattern until the tapered piston breaks loose, or until the maximum specified torque on the cap screws of **2 142 N·m (1,580 ft lb)** is reached.
5. If the specified torque is reached and the parts have not separated, slightly loosen the cap screws and slowly and uniformly apply heat to the spindle as shown (3, Figure 3-6). Heat must be applied in two locations 180 degrees apart. Allow heat to penetrate into the spindle. Reapply heat as required. **Do not exceed 454 °C (850 °F) anywhere on the spindle.**

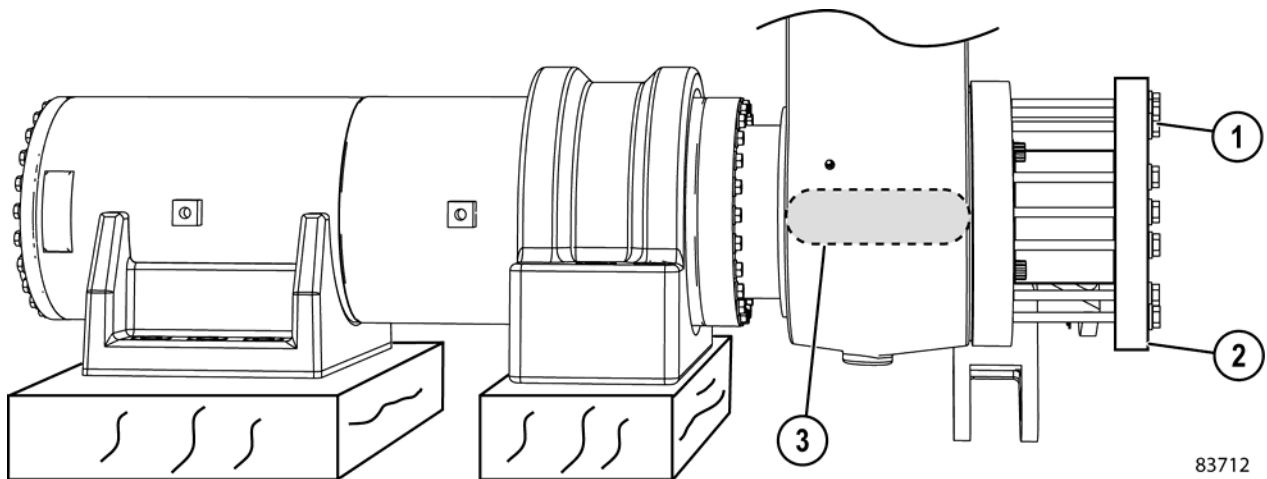


Heating the spindle in excess of 454° C (850° F) may cause serious damage to the spindle.

6. Tighten the cap screws again to the maximum specified torque as described in Step 4.
7. Using a large hammer and heat at the specified locations, carefully tap on the top surface of the spindle until the piston breaks free.

NOTE: In extreme cases, it may be necessary to remove additional steering arm retaining cap screws and use additional pusher cap screws to apply more force.

8. After separation, use the lifting device to move the spindle assembly to clean work area for repair.



83712

FIGURE 3-6. PUSHER TOOL INSTALLATION

- | | |
|-----------------------|-----------------|
| 1. Cap Screw & Washer | 3. Area to heat |
| 2. Pusher Tool | |

Installation

1. Clean the spindle bore and suspension rod taper so that they are free of rust, dirt, etc.
2. Lubricate the spindle bore and suspension rod taper with multi-purpose grease Number 2 with a minimum of 5% molybdenum disulphide.



Use of anti-seize compounds that contain copper are prohibited from use on spindle bores and rod tapers. These lubricants can cause severe damage. Only use multi-purpose grease Number 2 with a minimum of 5% molybdenum disulphide.

3. Position the spindle and wheel hub assembly on a fork lift or similar lifting device as shown in Figure 3-3.
4. Raise the spindle and wheel hub assembly into position.
5. Lubricate capscrews (1, Figure 3-2) and (5) on the threads and seats with lithium based chassis grease (multi-purpose, EP NLGI).
6. Secure the spindle to the suspension using retainer plate (2) and capscrews (1). Tighten the capscrews using the following procedure:
 - a. Tighten the capscrews uniformly to **678 N·m (500 ft lb)**.
 - b. Continue to tighten the capscrews in increments of **339 N·m (250 ft lb)** to obtain a final torque of **2 705 ± 135 N·m (1,995 ± 100 ft lb)**.
7. If removed, install the steering arm. Tighten the capscrews to **2 705 ± 135 N·m (1,995 ± 100 ft lb)**.
8. Install the steering cylinder into the steering arm. Use EH4638 sleeve alignment tool (included in tool group) to position the spacers and bearing during pin insertion. Tighten the pin retaining nut to **1 017 ± 102 N·m (750 ± 75 ft lb)**.
9. Use EH4638 alignment tool to install the tie rod on the steering arm. Tighten the pin retaining nut to **1 017 ± 102 N·m (750 ± 75 ft lb)**.
10. Connect the lubrication lines.
11. Install the brake system cooling hoses using new O-rings in the flange fittings. Install the brake apply line. Ensure that drain plug (2, Figure 3-1) is installed.
12. Reconnect the speed sensor cables.
13. Install the wheel and tire as described in "Front Tires and Rims - Installation" in Section G2.
14. Ensure that the hydraulic pump shut-off valves are open.
15. Start the engine. Move the hoist control lever to the FLOAT position to allow hydraulic oil to circulate through the brake cooling system and fill the front wheel hub and disc brake assembly with oil.
16. Bleed the air from the brake apply lines as described in "Wet Disc Brake Bleeding Procedure" in Section J5.
17. Shut off the engine. Check the hydraulic tank oil level and refill if necessary.
18. Inspect the brake assembly and all hose connections for leaks.

Disassembly

1. Remove the wheel hub and spindle as covered in the Removal procedure before proceeding to Step 2. Remove any dirt and mud from the assembly.
2. To aid in the complete disassembly of the wheel hub and spindle assembly, support the assembly in a vertical (hub cover up) position using a fabricated spindle stand.
3. Install six 0.50 in. -13 x 0.75 in. socket head capscrews (34, Figure 3-7) through the disc brake back plate into the mating holes in the seal carrier. Tighten securely.

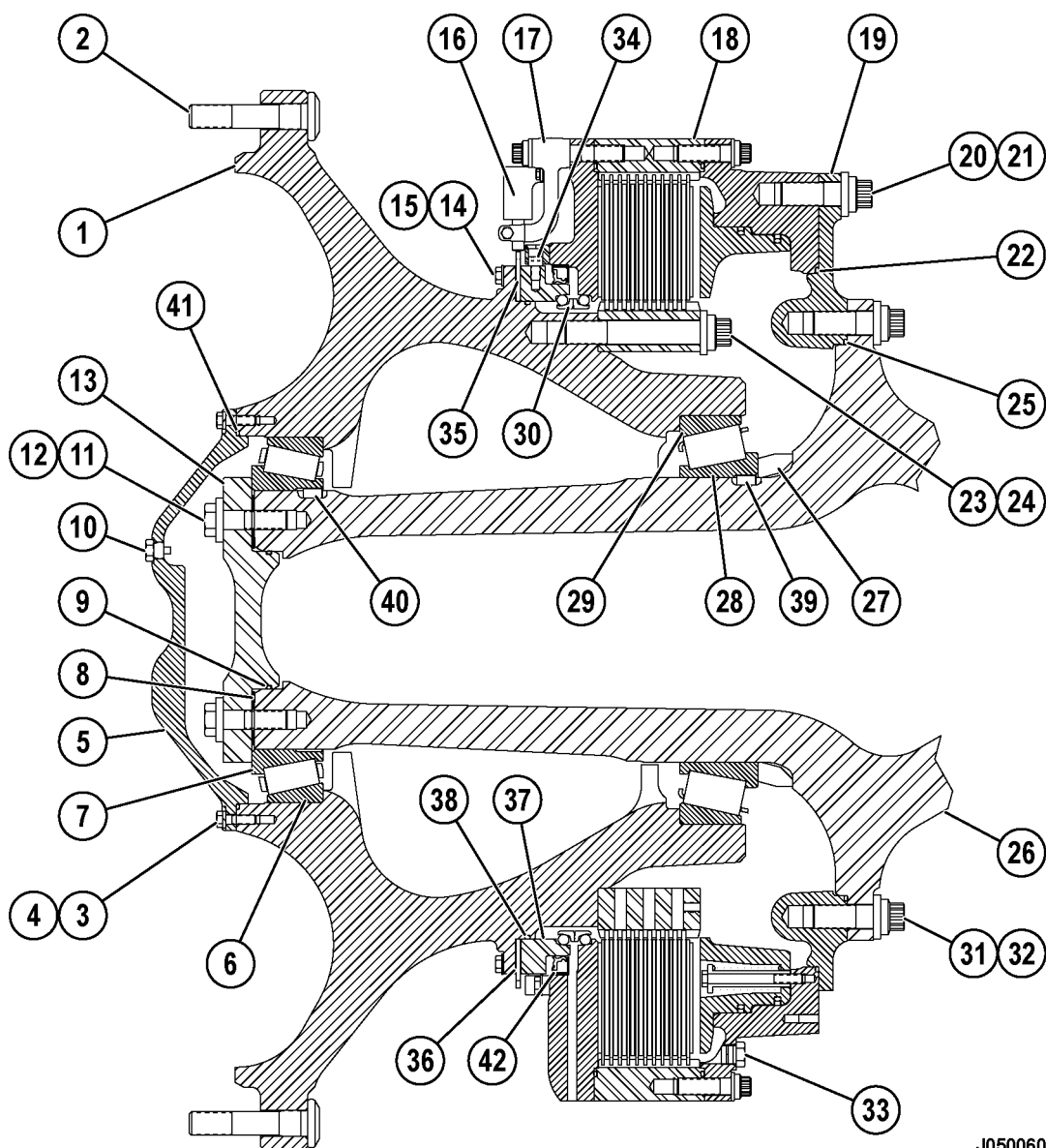
NOTE: The capscrews installed in Step 3 will secure the seal carrier and face seal assembly to the brake housing during brake removal. DO NOT rotate the wheel hub.

4. Disconnect the speed sensor cables. Loosen the sensor clamping capscrews and remove speed sensors (16).
5. Remove capscrews (20) and hardened flat washers (21) that secure brake adapter (19) to the brake housing.
6. Remove the capscrews and flat washers that secure cover (5). Remove the cover and discard O-ring seal (41).
7. Remove capscrews (11) and hardened flat washers (12).
8. Remove bearing retainer (13), O-ring (9), and shims (8). Discard the O-ring.

9. Attach a lifting device to the wheel hub/brake assembly and carefully lift it straight up and off the spindle.
10. Remove outer bearing cone (7) and retainer pin (40).
11. Rotate the hub vertically 180°, and place it on blocking to prevent damage to wheel studs and machined surfaces.
12. Remove capscrews (14) and flat washers (15) that secure seal carrier (37), sensor gear (35), and shims (36) to the wheel hub.
13. Remove capscrews (23) and hardened flat washers (24) that secure the brake assembly inner gear to the wheel hub.
14. Attach lifting eyes and an overhead hoist to the brake assembly. Carefully lift the assembly off the hub.

Refer to Section J for brake assembly rebuild instructions.

15. Remove and discard seal carrier O-ring (38).
16. Remove speed sensor gear (35) and shims (36).
17. If bearings require replacement, press cups (6) and (29) from the wheel hub.
18. Remove capscrews (31) and hardened flat washers (32). Remove brake adapter (19) and discard O-rings (22) and (25).
19. Remove inner bearing cone (28) and retainer pin (39).
20. Remove bearing spacer (27).



J050060

FIGURE 3-7. FRONT WHEEL SPINDLE, HUB & BRAKE ASSEMBLY

- | | | |
|--------------------------|--------------------------|--------------------------|
| 1. Wheel Hub | 15. Flat Washer | 29. Inner Bearing Cup |
| 2. Stud | 16. Speed Sensor | 30. Seal Assembly |
| 3. Capscrew | 17. Sensor Bracket | 31. Capscrew |
| 4. Washer | 18. Disc Brake Assembly | 32. Hardened Flat Washer |
| 5. Cover | 19. Brake Adapter | 33. Drain Plug |
| 6. Outer Bearing Cup | 20. Capscrew | 34. Socket Head Capscrew |
| 7. Outer Bearing Cone | 21. Hardened Washer | 35. Speed Sensor Gear |
| 8. Shims | 22. O-Ring | 36. Shims |
| 9. O-Ring | 23. Capscrew | 37. Seal Carrier |
| 10. Magnetic Plug | 24. Hardened Flat Washer | 38. O-Ring |
| 11. Capscrew | 25. O-Ring | 39. Bearing Retainer Pin |
| 12. Hardened Flat Washer | 26. Spindle | 40. Bearing Retainer Pin |
| 13. Bearing Retainer | 27. Bearing Spacer | 41. O-Ring |
| 14. Capscrew | 28. Inner Bearing Cone | 42. Oil Seal |

Cleaning and Inspection

1. Clean all metal parts in fresh cleaning solvent.
2. Remove and clean magnetic plug (10, Figure 3-7).
3. Inspect wheel hub studs (2). Replace if damaged or broken.

NOTE: If new studs are installed, coat the hole in the hub and serrated portion of the stud with an anti-seize compound prior to installation. DO NOT coat threads.

4. Inspect the tapped hole threads. Re-tap if necessary.
5. Inspect the bearing seating surfaces in the hub and on the spindle. Inspect the bearing spacer. Use a stone to carefully dress high spots that may interfere with re-assembly.
6. Inspect all other machined surfaces for damage.
7. Always use new O-ring seals during assembly.

Assembly

IMPORTANT

All mating surfaces of the wheel and brake components must be clean and dry during assembly. No thread lubricant is permitted on these surfaces.

1. Position the spindle vertically (hub end up).
2. Check bearing cones (7) and (28, Figure 3-7) for a slip fit on spindle (26).
3. Check the threads in the bearing retainer holes in the end of the spindle to ensure that the capscrews will thread freely. If not, re-tap the threads.
4. Install bearing cups (6) and (29) in wheel hub (1) as follows:
 - a. Preshrink the cups by packing them in dry ice or by placing them in a deep-freeze unit.

NOTE: Do not cool below -54°C (-65°F).

- b. Install the cups in the wheel hub bores.
- c. After the cups have warmed to ambient temperature, press the cups tight against the hub shoulder as follows:

Inner cup (29) - Apply **20 860 kg (23 tons)** force.

Outer cup (6) - Apply **19 050 kg (21 tons)** force.

5. Install spacer (27). If necessary, tap lightly to seat the spacer against the spindle. The spacer must fit tightly against the spindle shoulder.
6. Install pin (39) in the pin groove and install inner bearing cone (28) over the pin and against the spacer. The cone is a loose fit on the spindle.
7. Install wheel hub (1) onto spindle (26). Install outer pin (40) and outer bearing cone (7).

NOTE: To ensure bearing lubrication during initial operation, lightly lubricate the bearings with clean hydraulic oil.

IMPORTANT

To facilitate the bearing adjustment procedure, DO NOT install the brake assembly at this time.

Wheel Bearing Adjustment

8. Install bearing retainer (13, Figure 3-7) onto the spindle without O-ring (9) and shims (8). The thickness of the retainer is etched on the surface of the retainer. This side should be facing outward. Use four equally-spaced capscrews in order to secure the retainer. Two capscrews must be positioned adjacent to the 13 mm (0.50 in.) diameter access holes in the retainer.

9. Seat the wheel bearings using the following procedure:

a. Tighten the four capscrews to the proper torque, then rotate the wheel hub at least three full revolutions.

Standard bearings **108 N·m (80 ft lb)**

Heavy duty bearings . . . **135 N·m (100 ft lb)**

b. Tighten the four capscrews to the proper torque, then rotate the wheel hub at least three full revolutions.

Standard bearings **217 N·m (160 ft lb)**

Heavy duty bearings . . . **271 N·m (200 ft lb)**

c. Repeat Step b until the torque is maintained.

d. Loosen the four capscrews and rotate the hub at least three revolutions.

e. Tighten the four capscrews to the proper torque, then rotate the wheel hub at least three full revolutions.

Standard bearings **81 N·m (60 ft lb)**

Heavy duty bearings **54 N·m (40 ft lb)**

f. Tighten the four capscrews to the proper torque, then rotate the wheel hub at least three full revolutions.

Standard bearings **136 N·m (100 ft lb)**

Heavy duty bearings **95 N·m (70 ft lb)**

g. Repeat Step f until the torque is maintained.

10. Use a depth micrometer to measure and record the distance between the face of bearing retainer (13) and spindle (26) through each of the two 13 mm (0.50 in.) diameter access holes.

11. Add the two dimensions measured in the Step 10. Divide the sum by 2 to obtain the average depth.

12. Subtract the retainer plate thickness, which is etched on the surface of bearing retainer (13), from the result calculated in Step 11.

13. Assemble a shim pack that equals the dimension calculated in Step 12 within 0.0254 mm (0.001 in.).

NOTE: Measure the shims individually for accuracy. The above procedure results in a shim pack which will provide a nominal 0.508 mm (0.020 in.) preload for standard bearings, or 0.305 mm (0.012 in.) preload for heavy duty bearings, after assembly.

Example:

Step 12 (average depth) 1.416 in.

Step 13 subtract etched dim. -1.375 in.

Step 14 Required shim pack = 0.041 in.

NOTE: After a shim pack has been determined, the shim pack, spindle, wheel hub, spacer, and bearings are now an interdependent group. If any of the parts are replaced, the shim pack is no longer valid and a new pack must be calculated.

14. Remove bearing retainer (13), outer cone (7), retainer pin (40), and wheel hub (1) from spindle (26) for installation of the brake assembly.

Brake Installation

15. Check the integrity of the seals before installing the brakes on the truck.
 - a. Cap the cooling oil ports and pressurize the brake to a maximum of 103 kPa (15 psi).
 - b. Close off the air supply and monitor the interior brake pressure for 15 minutes.
 - c. If the air pressure decreases rapidly, a displaced or damaged brake seal is indicated. Disassemble the brake and reinstall or replace the seal.
16. Position the wheel hub vertically on blocks with the inner bearing bore at the top.
17. Assemble the brake to the hub as follows:
 - a. Install nine 0.51 mm (0.020 in.) thick shims (36, Figure 3-7) on top of the wheel hub flange at each of the six gear/seal carrier mounting locations. Shims may be installed on either side of speed sensor gear (35) for speed sensor alignment.
 - b. Install speed sensor gear (35).
 - c. Install new O-ring (38) in the groove in the hub. Ensure that the O-ring is not twisted. Lubricate the O-ring.
 - d. Install lifting eyes on the brake assembly and attach it to an overhead hoist.
 - e. Lower the brake assembly onto the hub while aligning the hub and seal carrier mounting holes.

NOTE: Do not remove the shipping bars until the inner gear ring of the brake assembly is attached to the hub.

- f. Install capscrews (14) and flat washers (15) from under hub flange. Tighten the capscrews to the standard torque.
 - g. Align the brake hub holes with the wheel hub and install hardened flat washers (24) and capscrews (23). Tighten the capscrews to **2 705 ± 135 N·m (1,995 ± 100 ft lb)**.
 - h. Remove the shipping bars from the brake assembly.

18. Install new O-ring (25) to brake adapter (19). Place the assembly into position on the spindle.
 19. Install hardened flat washers (32) and capscrews (31). Tighten capscrews to **2 705 ± 135 N·m (1,995 ± 100 ft lb)**.

IMPORTANT

Socket head capscrews (34) are installed to position the seal carrier on the brake back plate after assembly and prior to installation on the hub/spindle. In the next step, it may be necessary to loosen these capscrews slightly to align the brake assembly holes with the brake adapter.

NOTE: The following instructions must be performed carefully to prevent damage to brake adaptor O-ring seal (22) during assembly.

20. Prior to installing the hub and brake assembly on the spindle, install four alignment studs on the brake adapter in place of capscrews (20). Space the studs 90° apart to ensure that the brake assembly is properly aligned during the following steps.
 - a. Install new O-ring (22) onto the brake adapter shoulder. Ensure that the O-ring is not twisted and is properly seated.
 - b. Lubricate the O-ring with petroleum jelly or chassis grease.

21. Attach an overhead hoist to the wheel hub and brake assembly. Rotate the hub 180° in preparation for lowering it onto the spindle.

NOTE: The hoist must be rigid enough to prevent "springing" or "jerking" as the hub and brake assembly is lowered into position.

22. Lift the hub over the spindle and slowly lower while aligning the brake mounting holes with the alignment studs.
 - a. When the brake piston housing is approximately 13 mm (0.50 in.) from the adaptor, install four capscrews (20) and flat washers (21) spaced evenly in between the alignment studs.
 - b. Carefully pull the brake and wheel hub assembly into position by tightening the capscrews evenly and in small increments.
 - c. DO NOT allow the assembly to "cock" (angle). If the assembly is cocked, lift the assembly and inspect O-ring (22) for damage. Replace the O-ring if necessary, and repeat the procedure.
23. After the brake assembly is properly positioned on the adapter and the four capscrews are snug, remove the alignment studs.
24. Remove socket head capscrews (34) that secure seal carrier (37) to brake assembly (18).
25. Install remaining brake adapter/brake assembly mounting capscrews (20) and hardened flat washers (21). Tighten the capscrews to **2 705 ± 135 N·m (1,995 ± 100 ft lb)**.
26. Install bearing retainer pin (40) in the groove in the spindle. Align and slide outer bearing cone (7) over the pin. Lubricate the bearing with clean hydraulic oil.
27. Install shim pack (8), retainer (13), capscrews (11), and flat washers (12). Tighten the capscrews alternately in several successive increments while rotating the hub to a final torque of **1 017 ± 102 N·m (750 ± 75 ft lb)**.
28. Install new O-ring (41) on cover (5). Install the cover, capscrews, and flat washers. Tighten the capscrews to the standard torque.

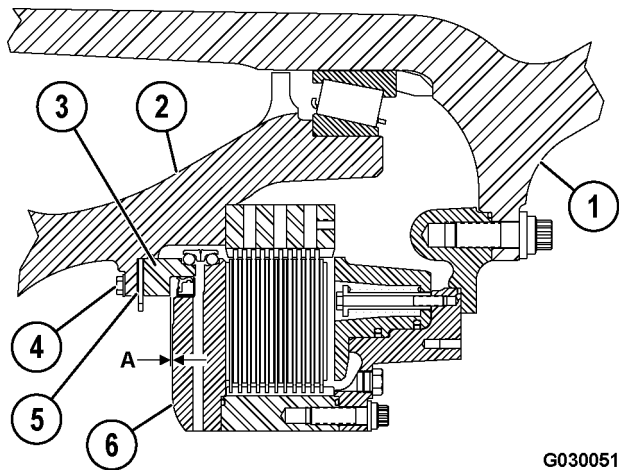
Seal Assembly Gap Check

After the assembly of the wheel and brake is complete, the gap between seal carrier (3, Figure 3-8) and back plate (6) must be measured and adjusted if necessary.

The ideal gap is 4.81 mm (0.189 in). If necessary, use shims to maintain a gap of 4.56 - 5.06 mm (0.179 - 0.199 in.).

29. Measure the seal gap as follows:

- a. Measure gap (dimension "A", Figure 3-8) at three equally-spaced places and record the results.
 - b. Add the three dimensions and divide the result by 3 to obtain the average gap width.
30. If the average gap width is not within the allowable range, loosen seal carrier capscrews (4) and add shims (5) as required to reduce the gap, or remove shims to increase the gap. **The quantity and thickness of shims at each of the six locations must be equal.**
 31. Re-tighten the seal carrier capscrews to the standard torque and measure the seal gap as described in Step 29. If necessary, repeat Step 30 until the proper gap is maintained.



G030051

FIGURE 3-8. MEASURING SEAL GAP

- | | |
|-----------------|---------------|
| 1. Spindle | 4. Capscrews |
| 2. Wheel Hub | 5. Shims |
| 3. Seal Carrier | 6. Back Plate |

32. Install speed sensor(s) (16, Figure 3-7) in support bracket(s) (17). Adjust the sensor as follows:

- a. Rotate the hub to position the center line of a gear tooth directly under the sensor tip.
- b. Turn in the sensor until the tip contacts the gear tooth. Then turn out the sensor until the flats of the sensor housing are perpendicular to the gear tooth motion (see Figure 3-9). Clearance between the sensor tip and gear tooth must be 2.0 ± 0.1 mm (0.079 ± 0.004 in.).
- c. Lock the sensor in place. Rotate the hub 180° to check whether there is enough sensor clearance.

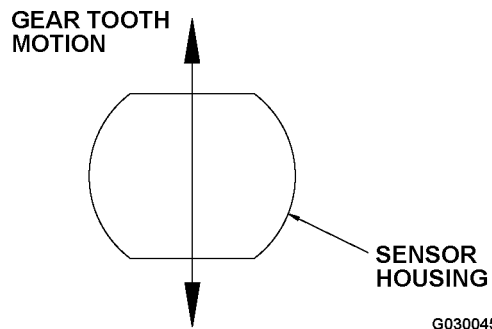


FIGURE 3-9. SPEED SENSOR ADJUSTMENT

33. Install the speed sensor cables.
34. Install the hub and spindle assembly according to the Installation instructions.

WHEEL SPEED SENSOR TESTING

NOTE: The wheel speed sensor can be tested while on the truck or bench tested. Testing the speed sensor requires power in either instance.

1. Connect a ground wire to pin C (black wire) on the speed sensor connector. This is TB21, circuit 0LF/0RF in the auxiliary control cabinet.
2. Connect 15 ± 1 VDC to pin A (red wire) on the speed sensor connector. This is TB21, circuit 15LWS/15RWS in the auxiliary control cabinet.
3. Connect the positive lead of a digital voltmeter to pin B (white wire) on the speed sensor connector. This is TB21, circuit 33LF/33RF in the auxiliary control cabinet.
4. Connect the common lead of the digital voltmeter to pin C (black wire) on the speed sensor connector. This is TB21, circuit 0LF/0RF in the auxiliary control cabinet.
5. Pass a ferrous metal target near the tip of the sensing tip on the speed sensor. A 6.4 mm (0.25 in.) diameter chisel or punch will work.

The sensor is a non-directional sensor. This means the ferrous metal target can pass by the sensor in any direction and the sensor will generate a voltage. If the target is moved toward the sensing tip and stopped in front of the tip, the voltmeter will display 15V. If the target passes by the sensing tip quickly, because of response time, the meter will likely only display a positive voltage, and not necessarily 15V.

6. If voltage is detected, the sensor is good. If voltage is not detected as the target passes by the sensing tip, the speed sensor is faulty.
7. If the speed sensor is faulty, replace the speed sensor.

STEERING CYLINDERS

The steering cylinders and tie rod mounting arrangements are similar. The removal, installation, and bearing wear limits are applicable to both.

Spherical Bearing Wear Limits

For optimum steering performance, it is necessary to determine the condition of the spherical bearings on the steering linkage components. Ball diameter, new dimensions, and maximum allowable wear specifications are listed in Table 1. Bearings that exceed the maximum wear limits must be replaced.

If premature wear of the bearings is evident, check the automatic lubrication system to ensure that the proper amount of lubrication is being received at the joint(s) in question. If lubrication is done manually, ensure that a sufficient amount of grease is being applied on a regular basis. Refer to Section P2, Lubrication and Service, for information on proper lubrication intervals.

It is also important to ensure that the steering linkage components are tightened to the proper torque. Use the torque specifications for steering linkage components listed in this section.



Use extreme caution when performing maintenance on any vehicle with an active steering system. Serious injury or death can result from contact with moving parts. Always keep a safe distance from crush points.

Table 1: STEERING SPHERICAL BEARING WEAR SPECIFICATIONS	
Spherical Bearing Ball Diameter (New)	109.47 mm (4.31 in.)
Maximum Allowable Wear	1.09 mm (0.043 in.)

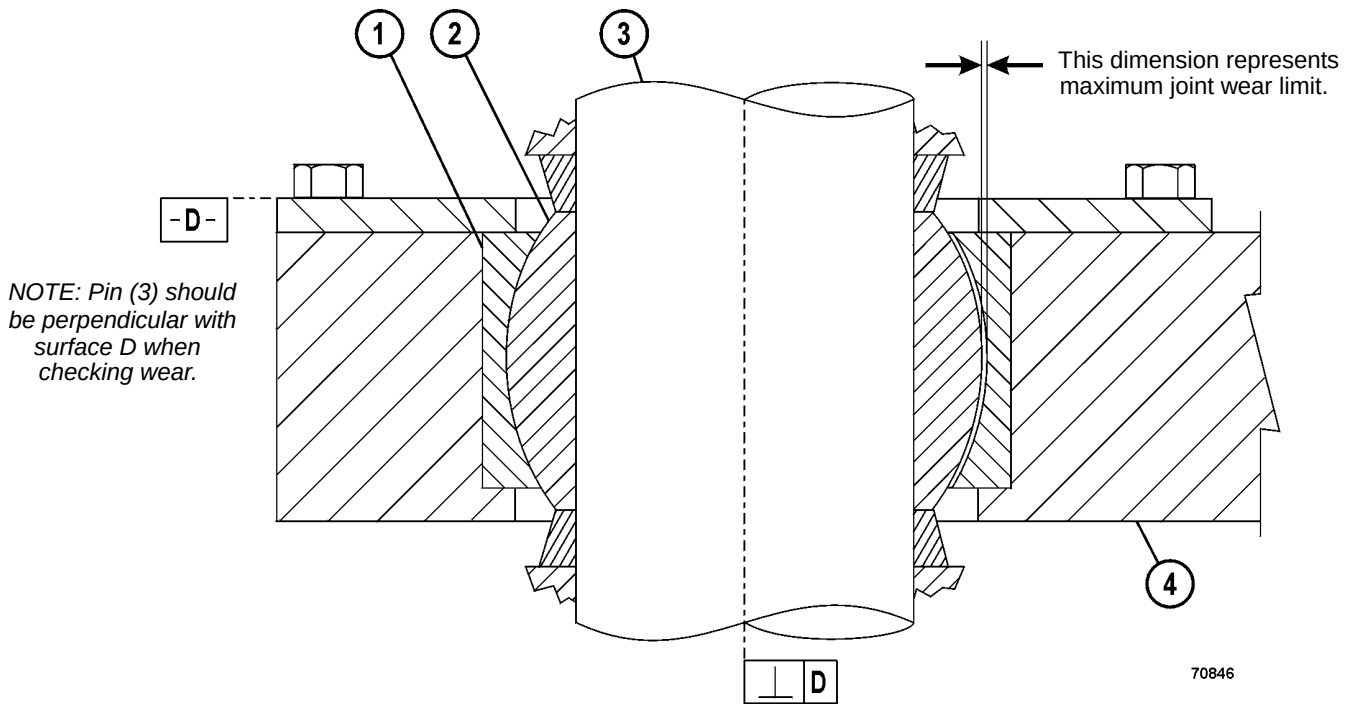


FIGURE 3-10. SPHERICAL BEARING WEAR LIMITS

1. Outer Race
2. Ball
3. Pin
4. Bearing Housing

Removal

1. With engine shut down and key switch OFF, allow at least 90 seconds for the accumulator to bleed down. Turn the steering wheel to ensure no hydraulic pressure is present.
2. Block front and back of rear wheels to prevent truck movement.
3. Disconnect hydraulic lines at the steering cylinders. Plug all line connections and cylinder ports to prevent contamination of hydraulic system.
4. Remove locknuts (12, Figure 3-11), retainers (11), and cap screws (1) from both ends of steering cylinder.
5. The steering cylinder weighs approximately 126 kg (278 lb). Attach a suitable lifting device that can safely handle the load safely to the steering cylinder and take up the slack.
6. Remove pins (2) from each end of the cylinder and move them to a clean work area. Bearing spacers (4) will be free when the pin is removed. **Ensure the bearing spacers do not drop out and become damaged when removing the pin.**
7. Use the lifting device to lift the cylinder from the truck.
8. Remove seals (3), spacers (4) and washer (10).

Installation

NOTE: Use sleeve alignment tool EH4638 (included in the tool group) to hold the bearing spacers and spherical bearings in position when the pins are inserted during assembly.

1. The steering cylinder weighs approximately 126 kg (278 lb). Use a suitable lifting device to lift the cylinder into position on the truck.

NOTE: Ensure anti-rotation retainer (7, Figure 3-11) is installed facing downwards.

2. Align bearing spacers (4), seals (3) and the barrel end of the steering cylinder with pin bores on truck frame.



Always install pin retaining capscrews from the top with the locknut on the bottom side at the steering arm.

3. Install pin (2), cap screw (1) and retainer (11) and secure with locknut (12).
4. Align bearing spacers (4), seals (3), washer (10) and rod end with pin bores on steering arm.
5. Install pin (2), cap screw (1) and retainer (11) and secure with locknut (12).
6. Tighten both locknuts (12) to **1 017 ± 102 N·m (750 ± 75 ft lb)**.
7. Connect the grease lines to their respective ports. Operate the steering and check for leaks and proper operation.

Bearing Replacement (Steering Cylinder and Tie Rod)

1. Remove cap screws (8, Figure 3-11) and lockwashers (9). Remove bearing retainer (7).
2. Press bearing (5) out of the bore in the steering cylinder or tie rod end.
3. Press a new bearing into the bore.
4. Install anti-rotation retainers (7) with cap screws and lockwashers. Tighten the cap screws to standard torque.